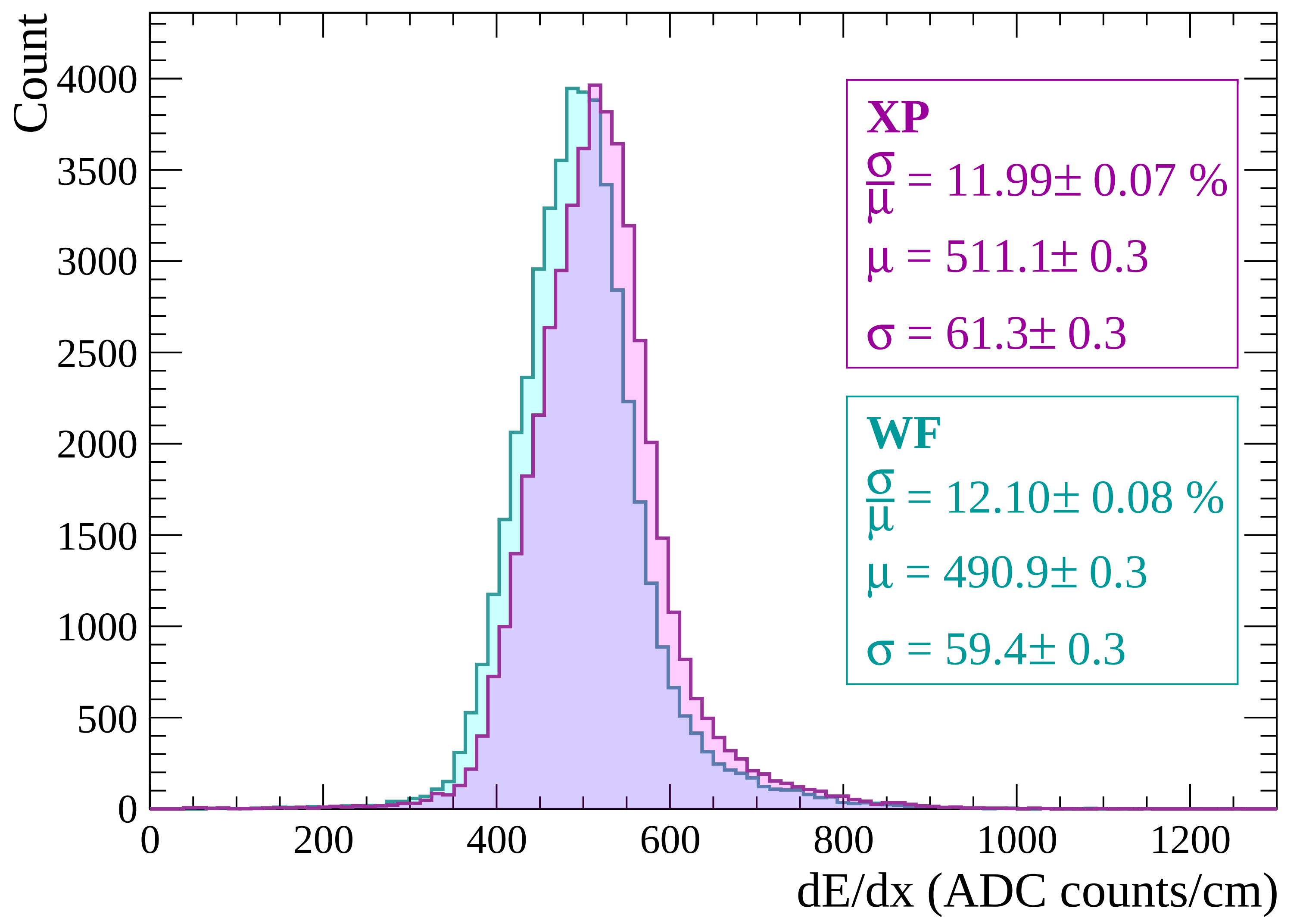
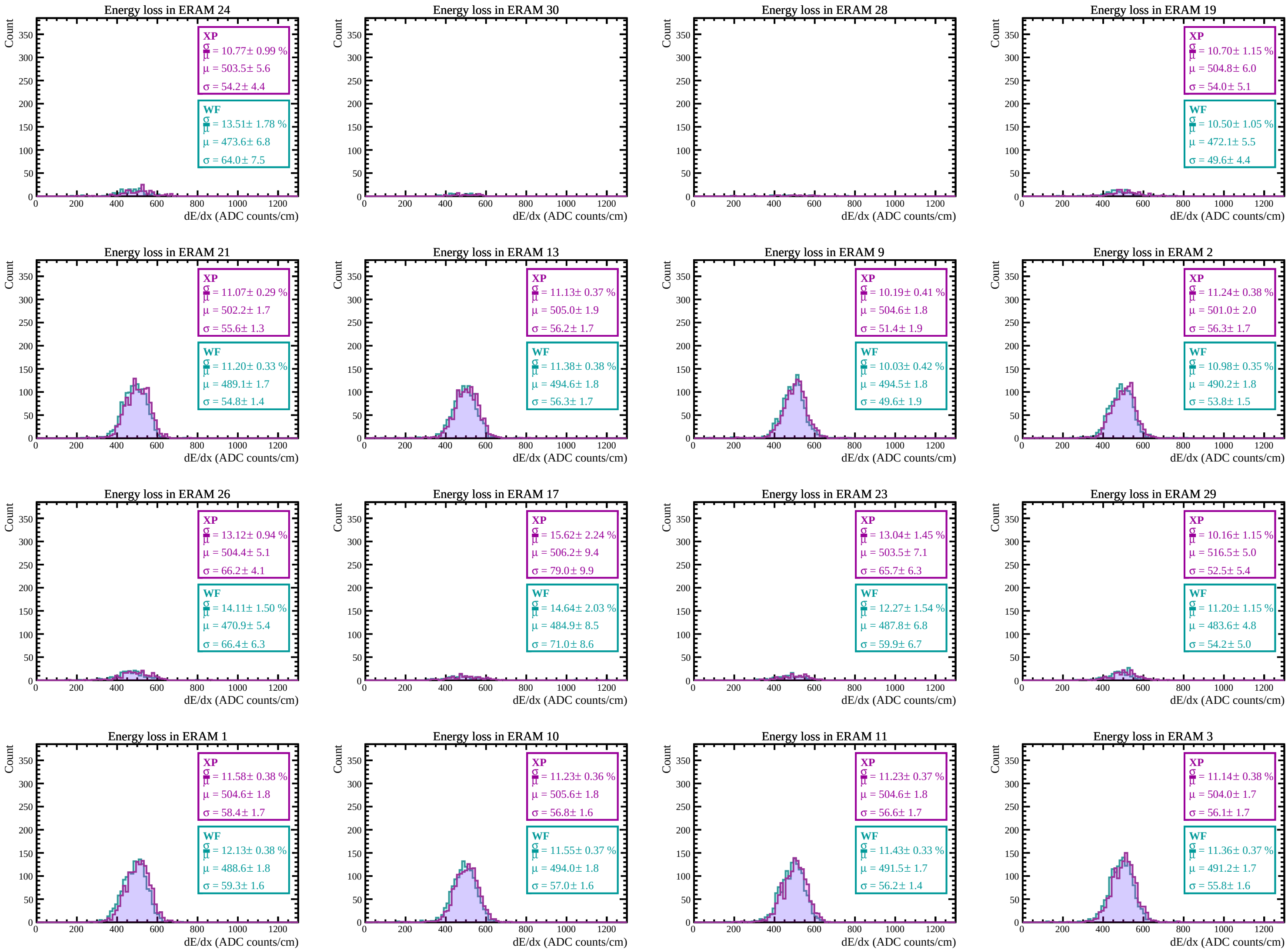
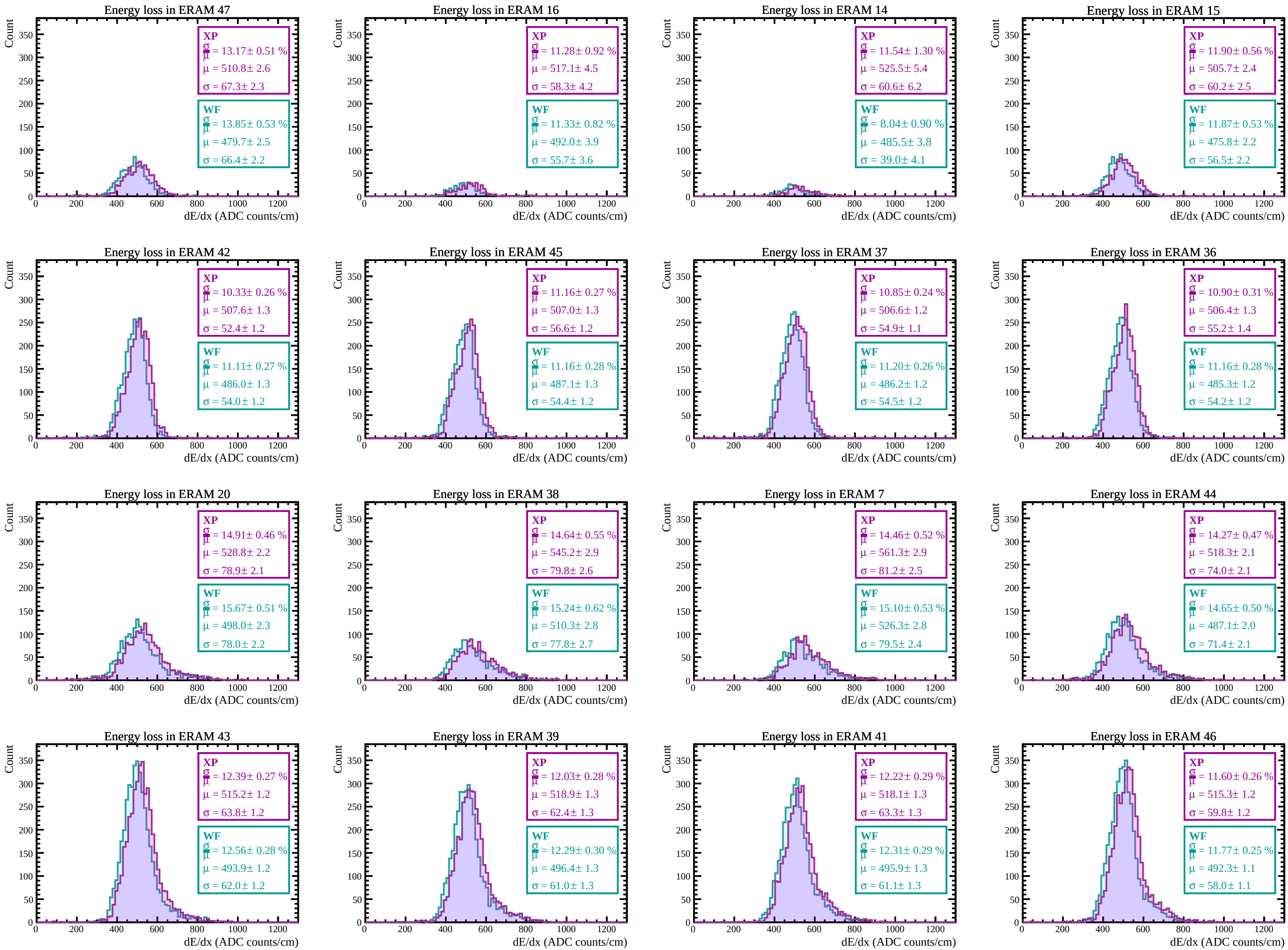


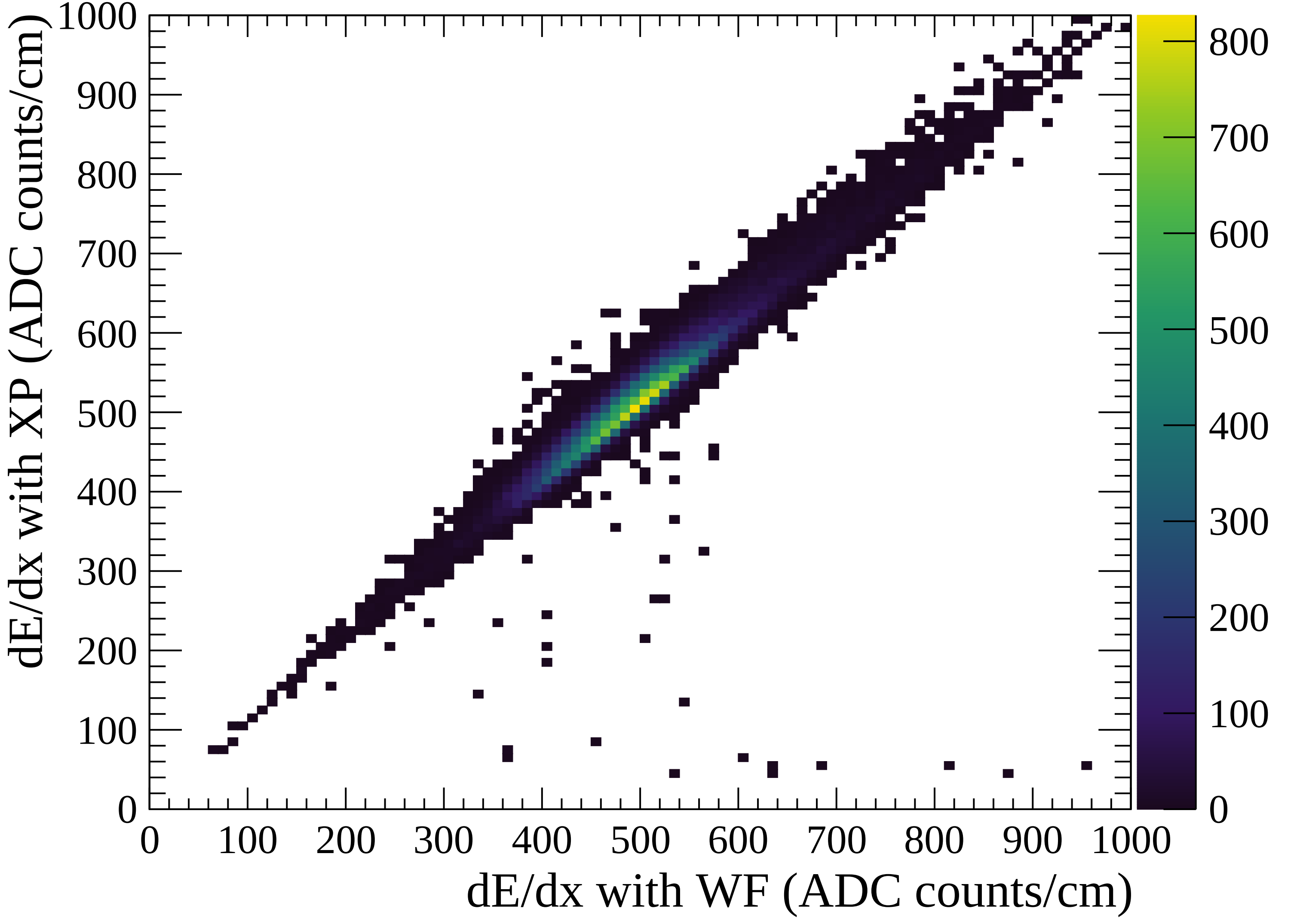
Count
4000
3500
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.99± 0.07 %
μ = 511.1± 0.3
σ = 61.3± 0.3
WF
σ/μ = 12.10± 0.08 %
μ = 490.9± 0.3
σ = 59.4± 0.3



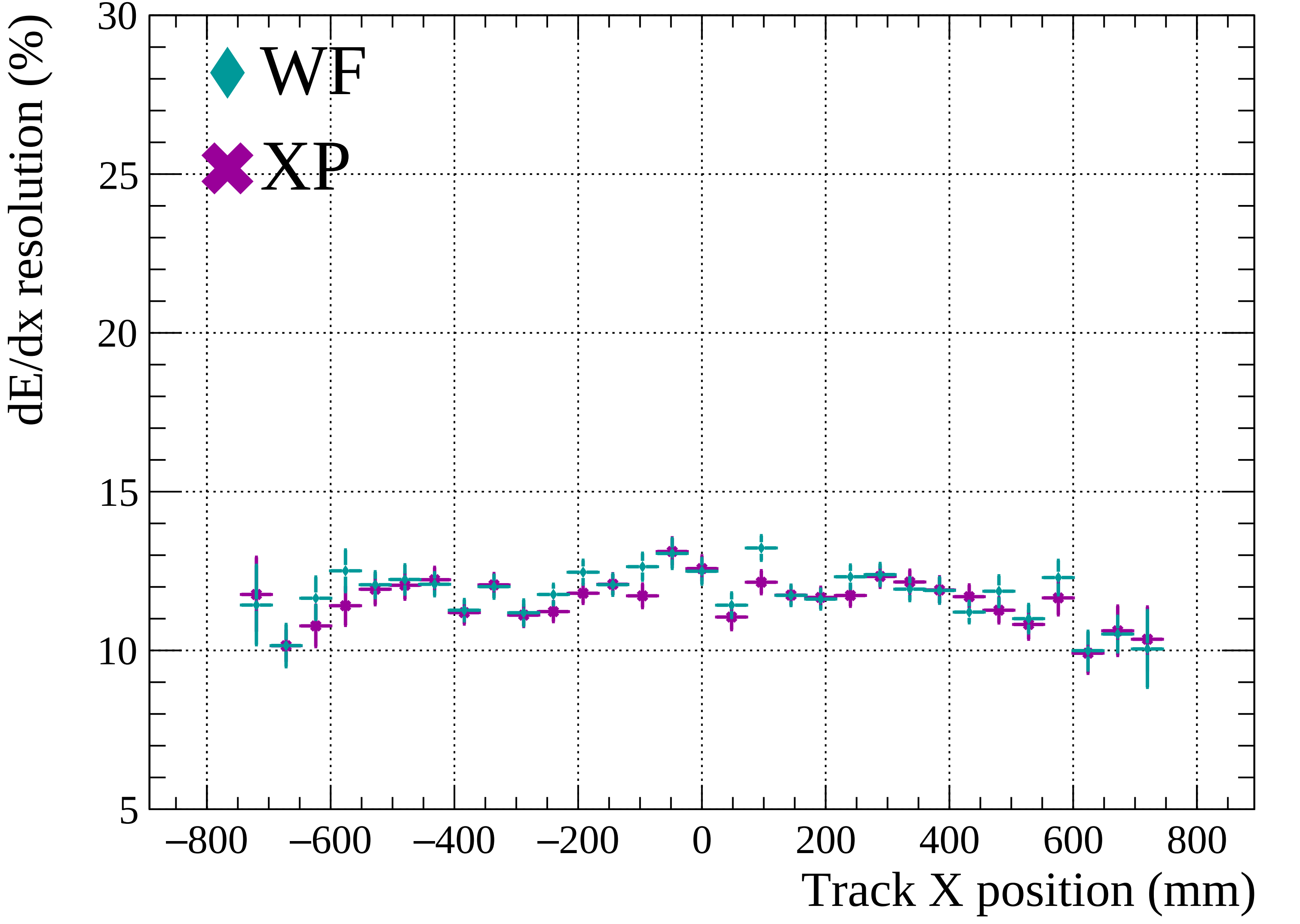
Energy loss in ERAM 24
XP
σ/μ = 10.77± 0.99 %
μ = 503.5± 5.6
σ = 54.2± 4.4
WF
σ/μ = 13.51± 1.78 %
μ = 473.6± 6.8
σ = 64.0± 7.5
Energy loss in ERAM 30
Energy loss in ERAM 28
Energy loss in ERAM 19
XP
σ/μ = 10.70± 1.15 %
μ = 504.8± 6.0
σ = 54.0± 5.1
WF
σ/μ = 10.50± 1.05 %
μ = 472.1± 5.5
σ = 49.6± 4.4
Energy loss in ERAM 21
XP
σ/μ = 11.07± 0.29 %
μ = 502.2± 1.7
σ = 55.6± 1.3
WF
σ/μ = 11.20± 0.33 %
μ = 489.1± 1.7
σ = 54.8± 1.4
Energy loss in ERAM 13
XP
σ/μ = 11.13± 0.37 %
μ = 505.0± 1.9
σ = 56.2± 1.7
WF
σ/μ = 11.38± 0.38 %
μ = 494.6± 1.8
σ = 56.3± 1.7
Energy loss in ERAM 9
XP
σ/μ = 10.19± 0.41 %
μ = 504.6± 1.8
σ = 51.4± 1.9
WF
σ/μ = 10.03± 0.42 %
μ = 494.5± 1.8
σ = 49.6± 1.9
Energy loss in ERAM 2
XP
σ/μ = 11.24± 0.38 %
μ = 501.0± 2.0
σ = 56.3± 1.7
WF
σ/μ = 10.98± 0.35 %
μ = 490.2± 1.8
σ = 53.8± 1.5
Energy loss in ERAM 26
XP
σ/μ = 13.12± 0.94 %
μ = 504.4± 5.1
σ = 66.2± 4.1
WF
σ/μ = 14.11± 1.50 %
μ = 470.9± 5.4
σ = 66.4± 6.3
Energy loss in ERAM 17
XP
σ/μ = 15.62± 2.24 %
μ = 506.2± 9.4
σ = 79.0± 9.9
WF
σ/μ = 14.64± 2.03 %
μ = 484.9± 8.5
σ = 71.0± 8.6
Energy loss in ERAM 23
XP
σ/μ = 13.04± 1.45 %
μ = 503.5± 7.1
σ = 65.7± 6.3
WF
σ/μ = 12.27± 1.54 %
μ = 487.8± 6.8
σ = 59.9± 6.7
Energy loss in ERAM 29
XP
σ/μ = 10.16± 1.15 %
μ = 516.5± 5.0
σ = 52.5± 5.4
WF
σ/μ = 11.20± 1.15 %
μ = 483.6± 4.8
σ = 54.2± 5.0
Energy loss in ERAM 1
XP
σ/μ = 11.58± 0.38 %
μ = 504.6± 1.8
σ = 58.4± 1.7
WF
σ/μ = 12.13± 0.38 %
μ = 488.6± 1.8
σ = 59.3± 1.6
Energy loss in ERAM 10
XP
σ/μ = 11.23± 0.36 %
μ = 505.6± 1.8
σ = 56.8± 1.6
WF
σ/μ = 11.55± 0.37 %
μ = 494.0± 1.8
σ = 57.0± 1.6
Energy loss in ERAM 11
XP
σ/μ = 11.23± 0.37 %
μ = 504.6± 1.8
σ = 56.6± 1.7
WF
σ/μ = 11.43± 0.33 %
μ = 491.5± 1.7
σ = 56.2± 1.4
Energy loss in ERAM 3
XP
σ/μ = 11.14± 0.38 %
μ = 504.0± 1.7
σ = 56.1± 1.7
WF
σ/μ = 11.36± 0.37 %
μ = 491.2± 1.7
σ = 55.8± 1.6
Count
dE/dx (ADC counts/cm)



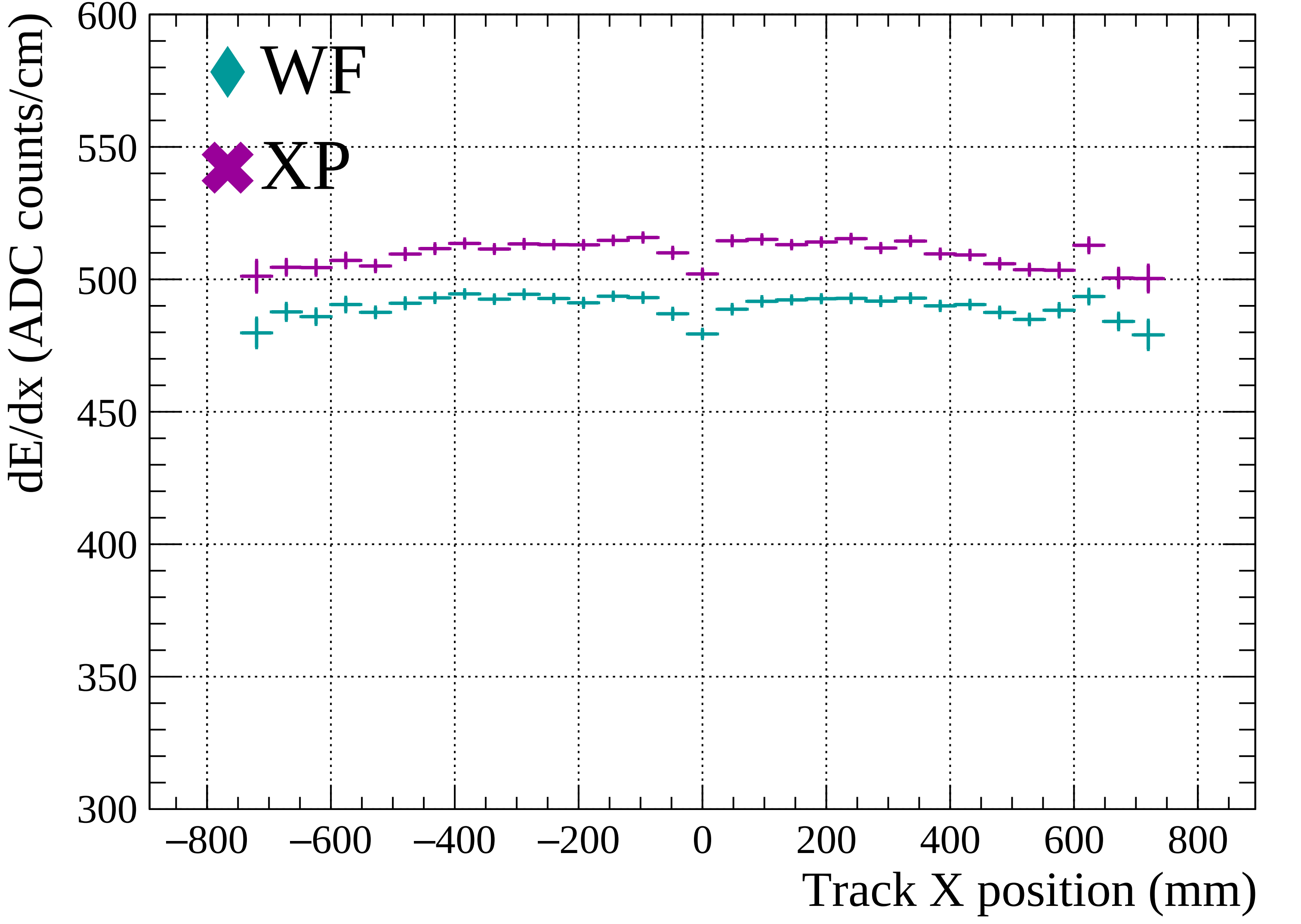
Energy loss in ERAM 47
XP
σ/μ = 13.17± 0.51 %
μ = 510.8± 2.6
σ = 67.3± 2.3
WF
σ/μ = 13.85± 0.53 %
μ = 479.7± 2.5
σ = 66.4± 2.2
Energy loss in ERAM 16
XP
σ/μ = 11.28± 0.92 %
μ = 517.1± 4.5
σ = 58.3± 4.2
WF
σ/μ = 11.33± 0.82 %
μ = 492.0± 3.9
σ = 55.7± 3.6
Energy loss in ERAM 14
XP
σ/μ = 11.54± 1.30 %
μ = 525.5± 5.4
σ = 60.6± 6.2
WF
σ/μ = 8.04± 0.90 %
μ = 485.5± 3.8
σ = 39.0± 4.1
Energy loss in ERAM 15
XP
σ/μ = 11.90± 0.56 %
μ = 505.7± 2.4
σ = 60.2± 2.5
WF
σ/μ = 11.87± 0.53 %
μ = 475.8± 2.2
σ = 56.5± 2.2
Energy loss in ERAM 42
XP
σ/μ = 10.33± 0.26 %
μ = 507.6± 1.3
σ = 52.4± 1.2
WF
σ/μ = 11.11± 0.27 %
μ = 486.0± 1.3
σ = 54.0± 1.2
Energy loss in ERAM 45
XP
σ/μ = 11.16± 0.27 %
μ = 507.0± 1.3
σ = 56.6± 1.2
WF
σ/μ = 11.16± 0.28 %
μ = 487.1± 1.3
σ = 54.4± 1.2
Energy loss in ERAM 37
XP
σ/μ = 10.85± 0.24 %
μ = 506.6± 1.2
σ = 54.9± 1.1
WF
σ/μ = 11.20± 0.26 %
μ = 486.2± 1.2
σ = 54.5± 1.2
Energy loss in ERAM 36
XP
σ/μ = 10.90± 0.31 %
μ = 506.4± 1.3
σ = 55.2± 1.4
WF
σ/μ = 11.16± 0.28 %
μ = 485.3± 1.2
σ = 54.2± 1.2
Energy loss in ERAM 20
XP
σ/μ = 14.91± 0.46 %
μ = 528.8± 2.2
σ = 78.9± 2.1
WF
σ/μ = 15.67± 0.51 %
μ = 498.0± 2.3
σ = 78.0± 2.2
Energy loss in ERAM 38
XP
σ/μ = 14.64± 0.55 %
μ = 545.2± 2.9
σ = 79.8± 2.6
WF
σ/μ = 15.24± 0.62 %
μ = 510.3± 2.8
σ = 77.8± 2.7
Energy loss in ERAM 7
XP
σ/μ = 14.46± 0.52 %
μ = 561.3± 2.9
σ = 81.2± 2.5
WF
σ/μ = 15.10± 0.53 %
μ = 526.3± 2.8
σ = 79.5± 2.4
Energy loss in ERAM 44
XP
σ/μ = 14.27± 0.47 %
μ = 518.3± 2.1
σ = 74.0± 2.1
WF
σ/μ = 14.65± 0.50 %
μ = 487.1± 2.0
σ = 71.4± 2.1
Energy loss in ERAM 43
XP
σ/μ = 12.39± 0.27 %
μ = 515.2± 1.2
σ = 63.8± 1.2
WF
σ/μ = 12.56± 0.28 %
μ = 493.9± 1.2
σ = 62.0± 1.2
Energy loss in ERAM 39
XP
σ/μ = 12.03± 0.28 %
μ = 518.9± 1.3
σ = 62.4± 1.3
WF
σ/μ = 12.29± 0.30 %
μ = 496.4± 1.3
σ = 61.0± 1.3
Energy loss in ERAM 41
XP
σ/μ = 12.22± 0.29 %
μ = 518.1± 1.3
σ = 63.3± 1.3
WF
σ/μ = 12.31± 0.29 %
μ = 495.9± 1.3
σ = 61.1± 1.3
Energy loss in ERAM 46
XP
σ/μ = 11.60± 0.26 %
μ = 515.3± 1.2
σ = 59.8± 1.2
WF
σ/μ = 11.77± 0.25 %
μ = 492.3± 1.1
σ = 58.0± 1.1
Count
dE/dx (ADC counts/cm)



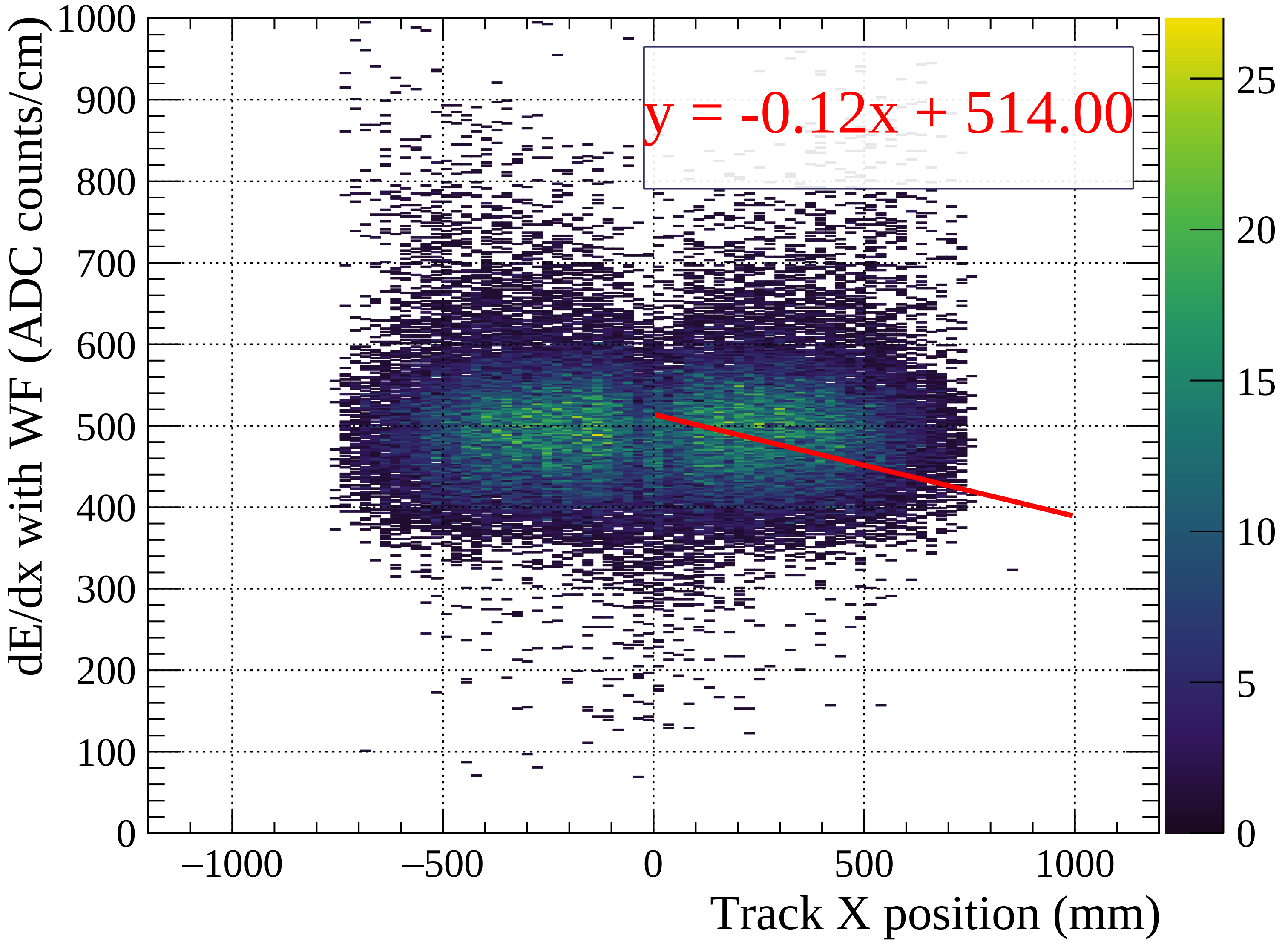
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000

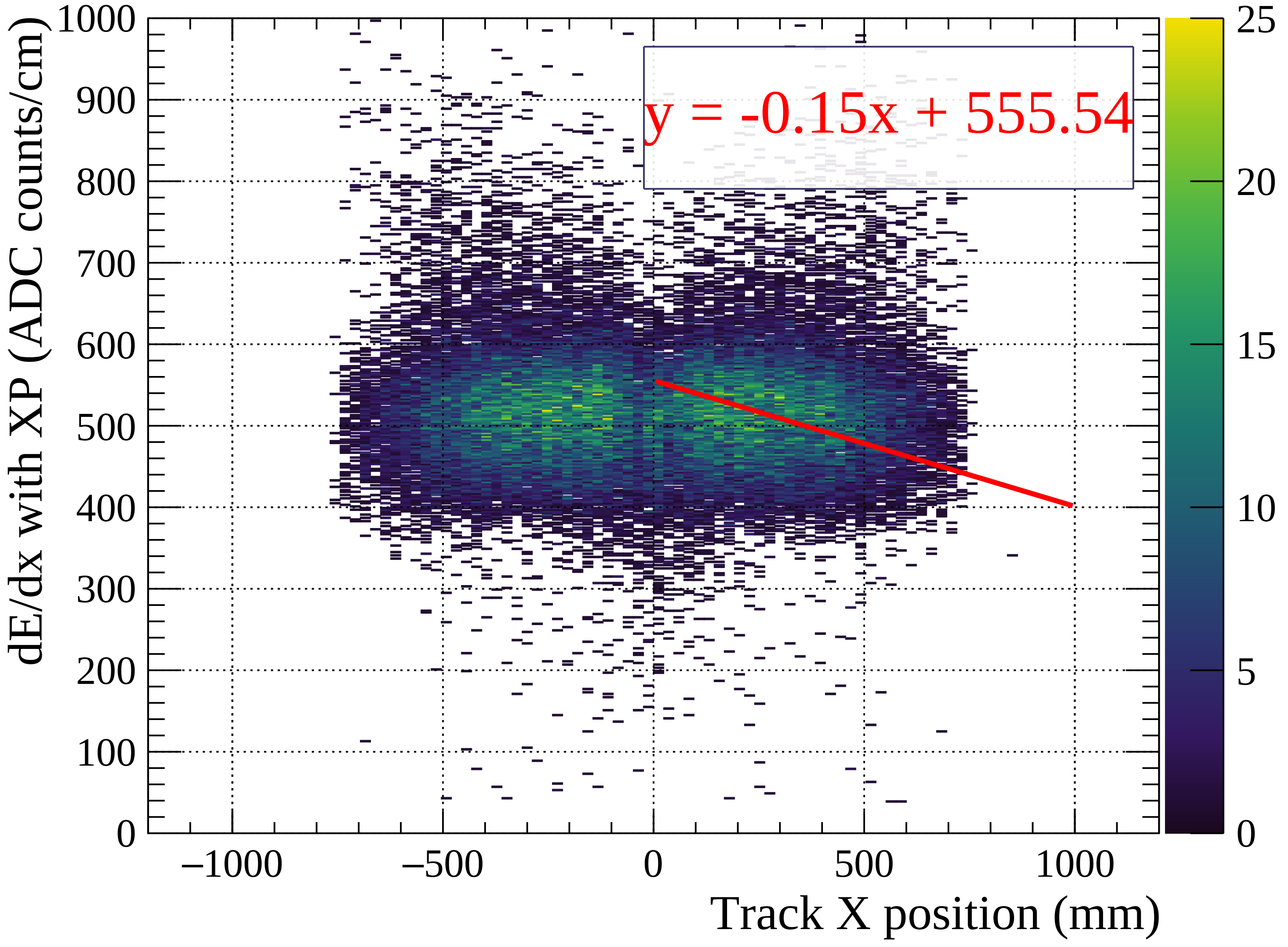


WF
XP
dE/dx resolution (%)
30
25
20
15
10
5
−800
−600
−400
−200
0
200
400
600
800
Track X position (mm)

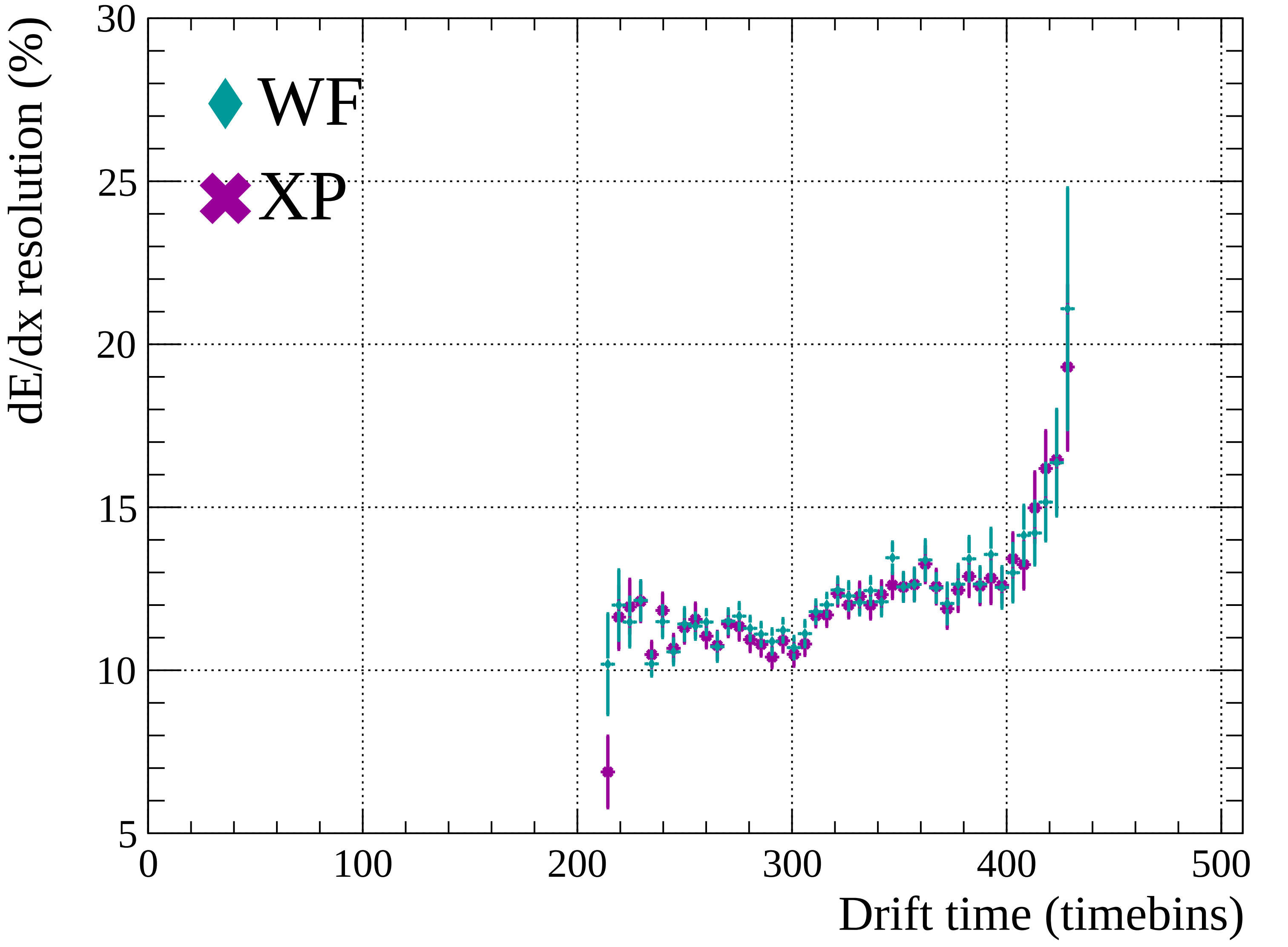


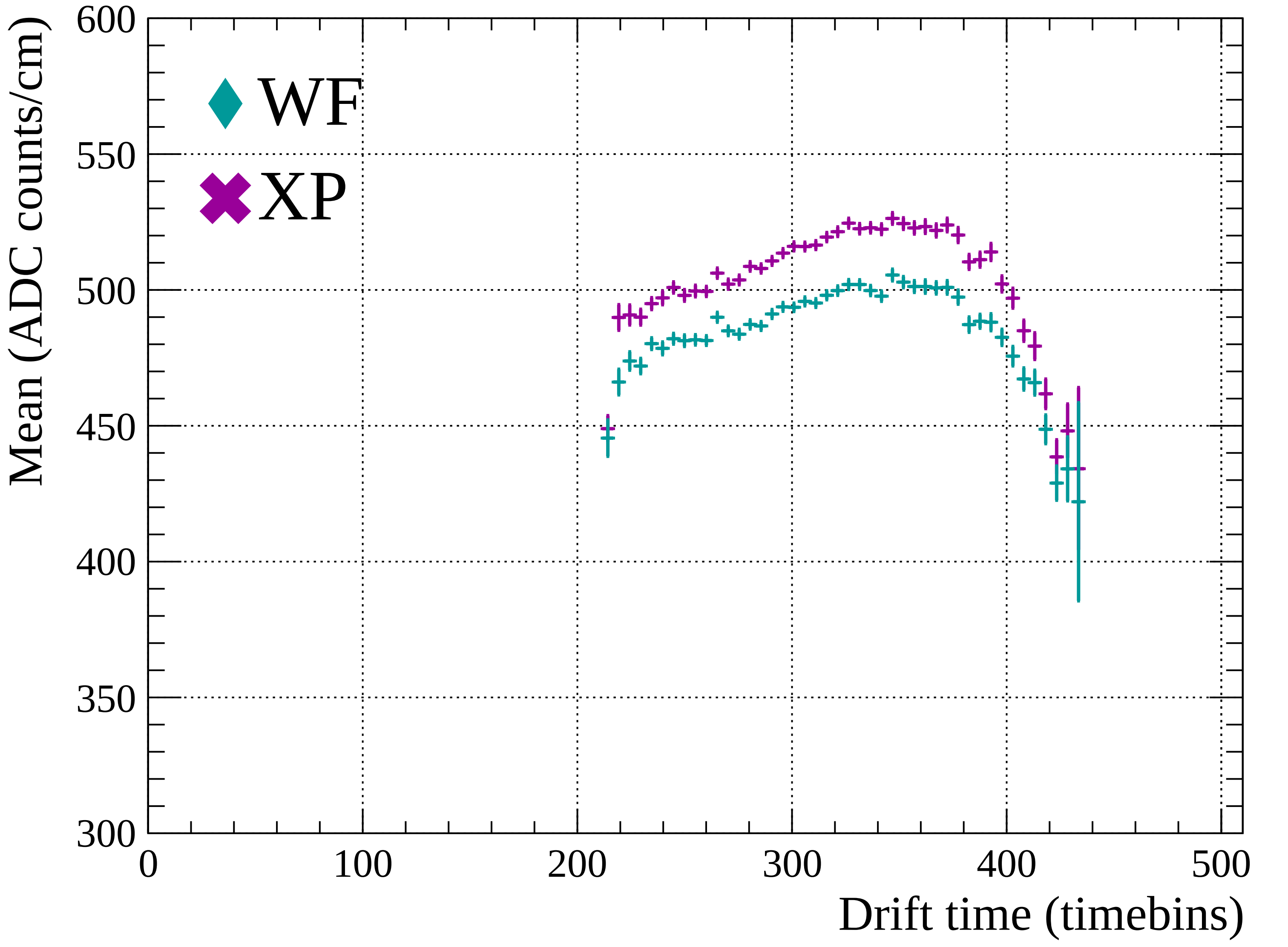
WF
XP
dE/dx (ADC counts/cm)
Track X position (mm)
600
550
500
450
400
350
300
−800
−600
−400
−200
0
200
400
600
800


y = -0.12x + 514.00
dE/dx with WF (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
5
10
15
20
25

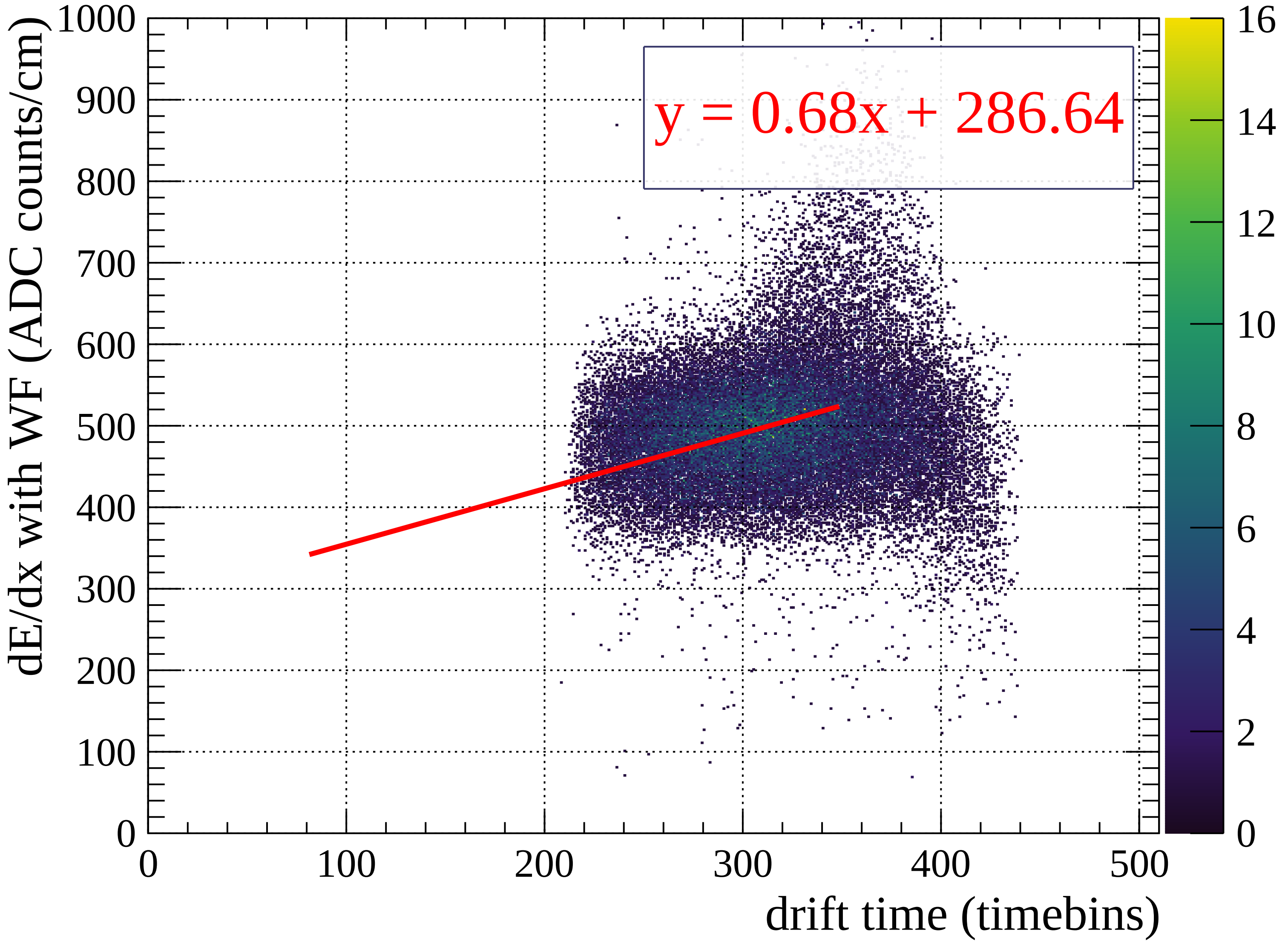


y = -0.15x + 555.54
dE/dx with XP (ADC counts/cm)
Track X position (mm)
1000
900
800
700
600
500
400
300
200
100
0
−1000
−500
0
500
1000
25
20
15
10
5
0

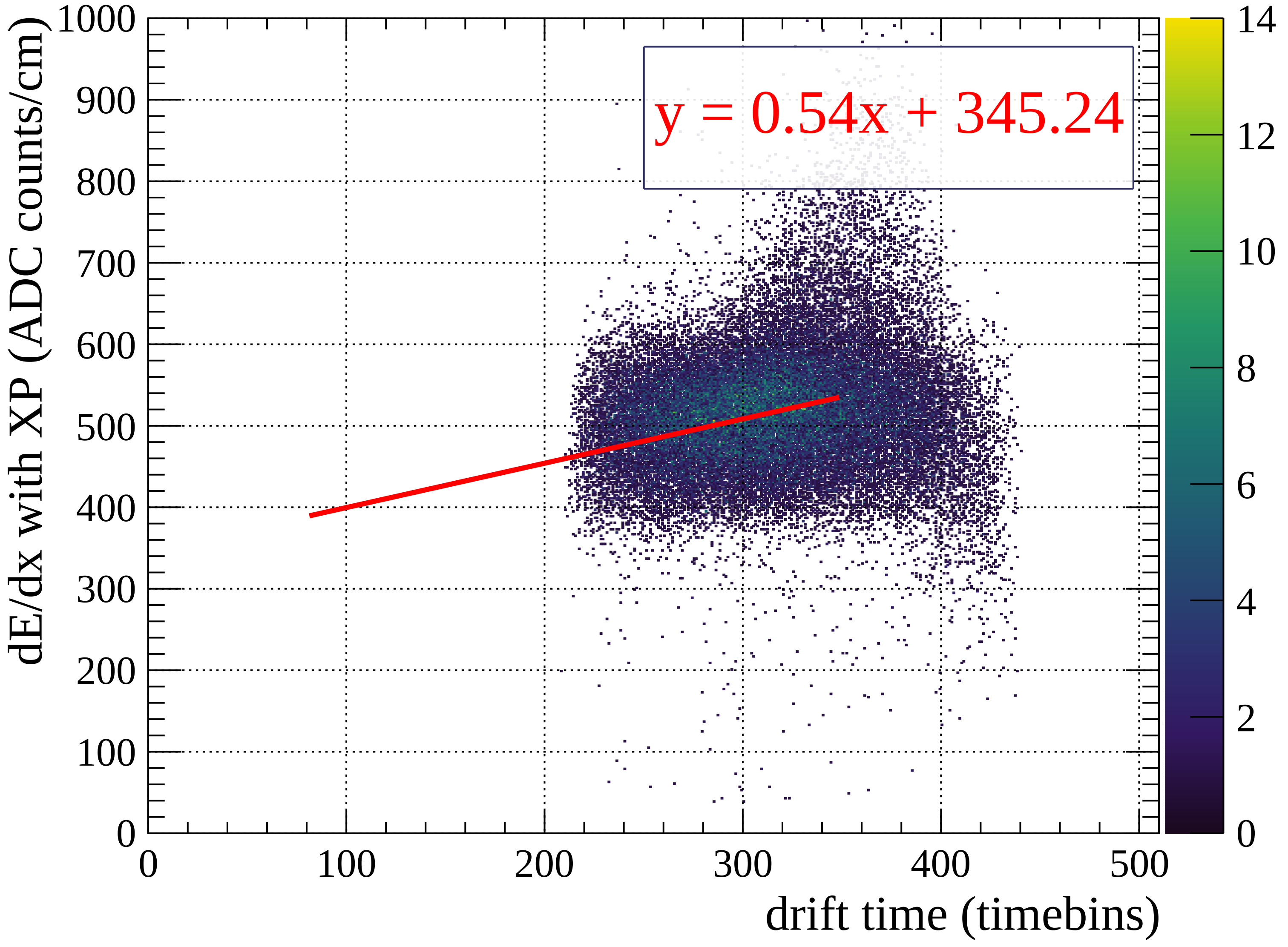

WF
XP
dE/dx resolution (%)
Drift time (timebins)
30
25
20
15
10
5
0
100
200
300
400
500

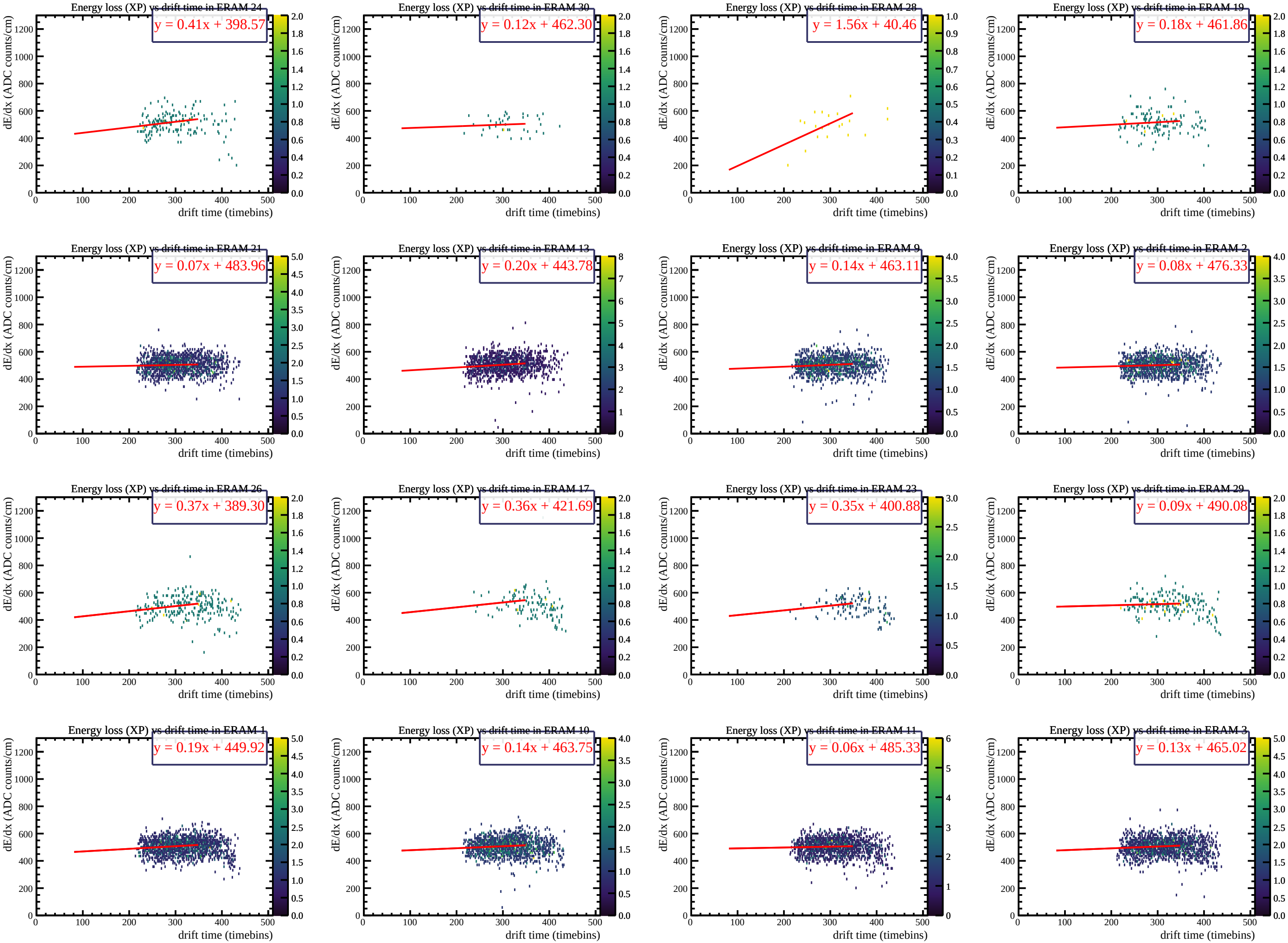


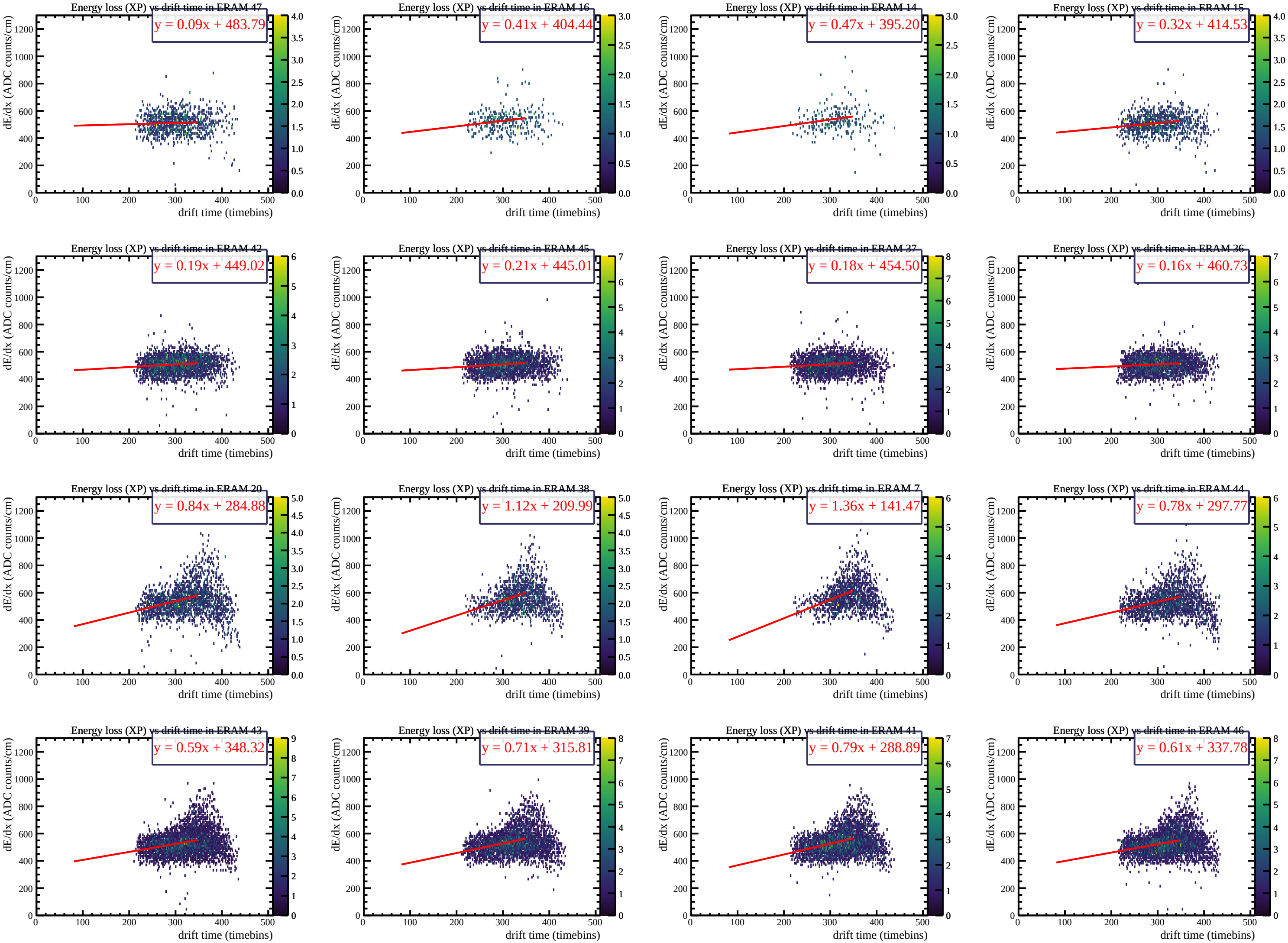
WF
XP
Mean (ADC counts/cm)
Drift time (timebins)
600
550
500
450
400
350
300
0
100
200
300
400
500

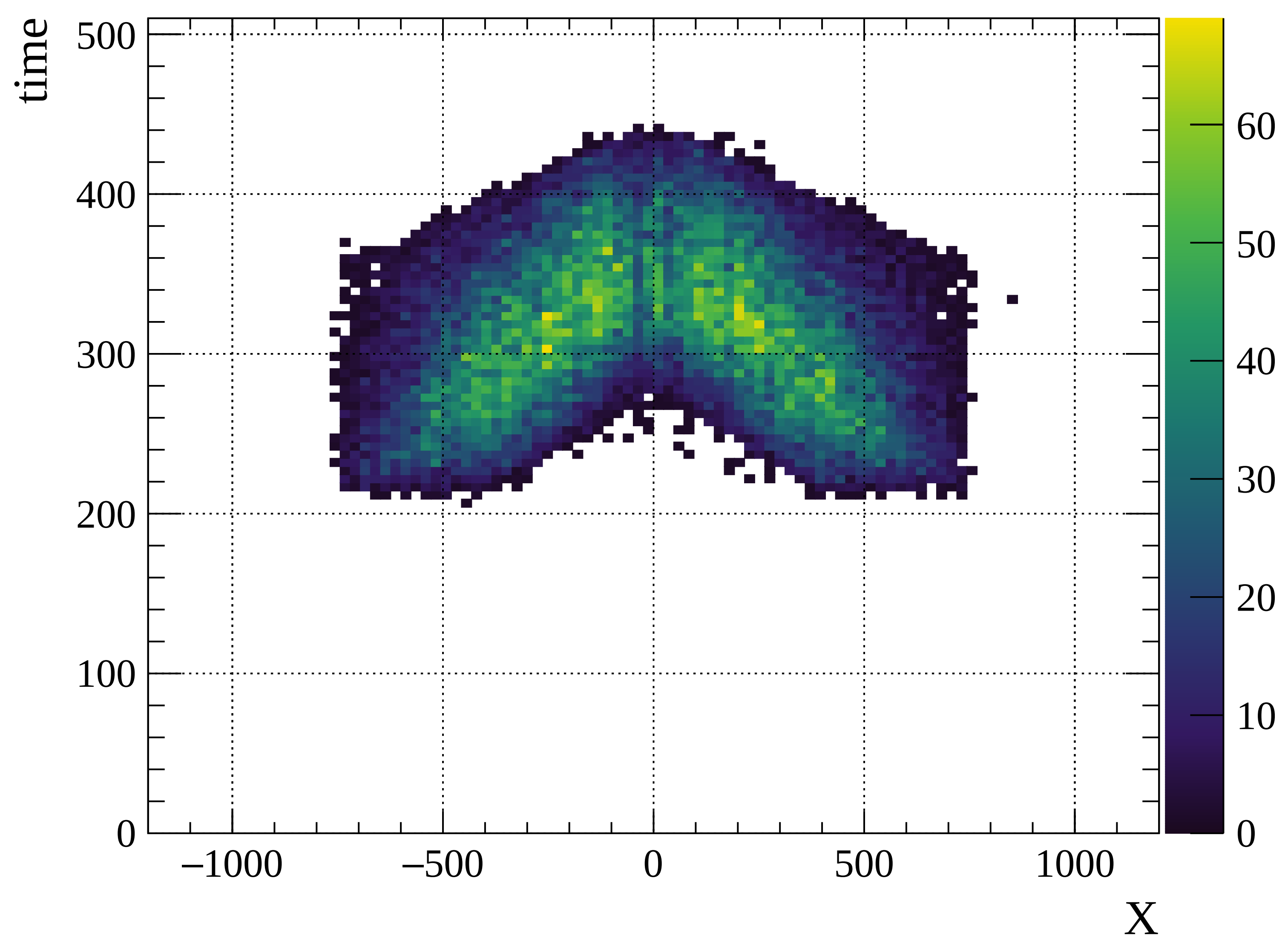


y = 0.68x + 286.64
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
2
4
6
8
10
12
14
16

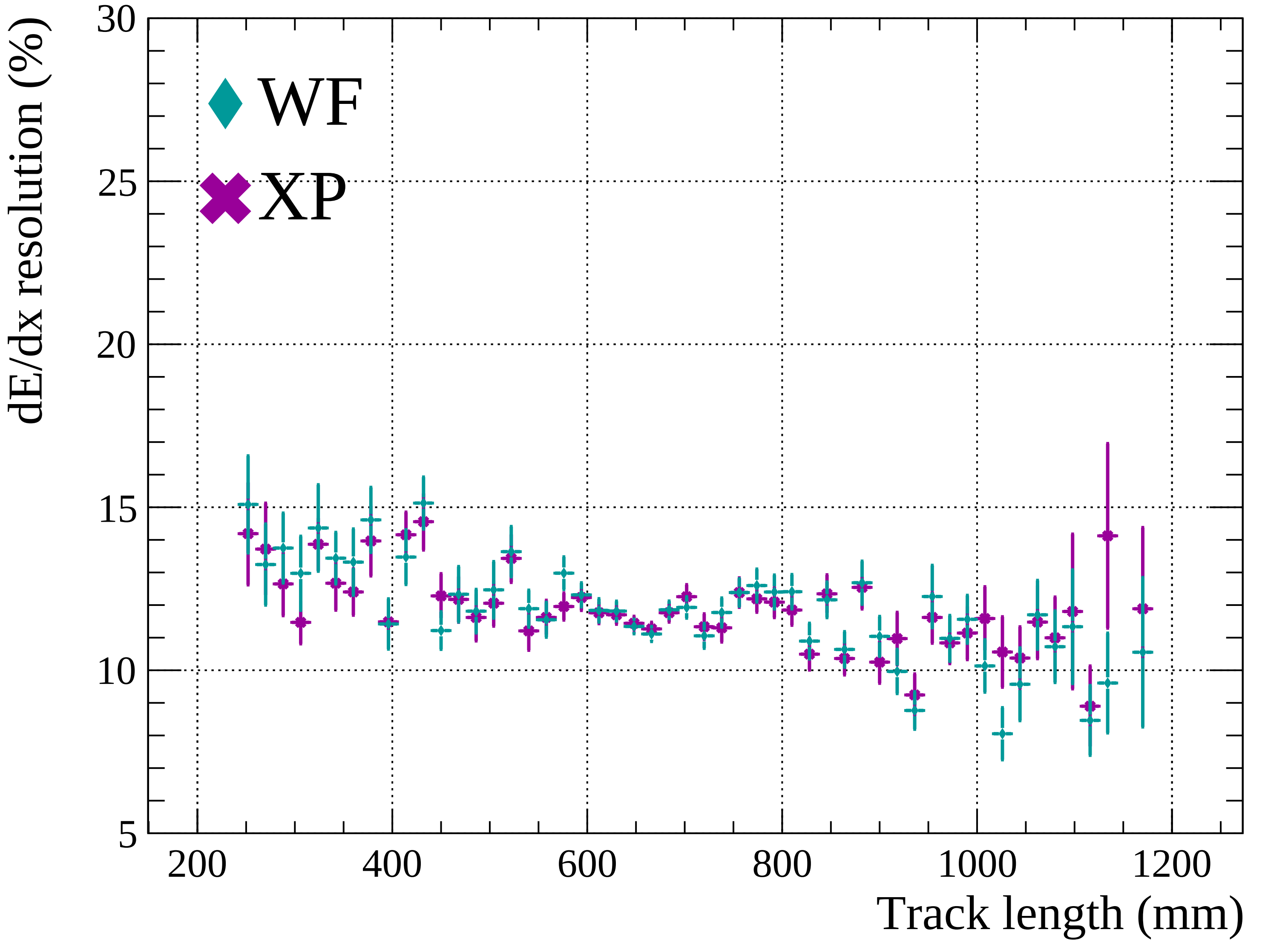

y = 0.54x + 345.24
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
2
4
6
8
10
12
14

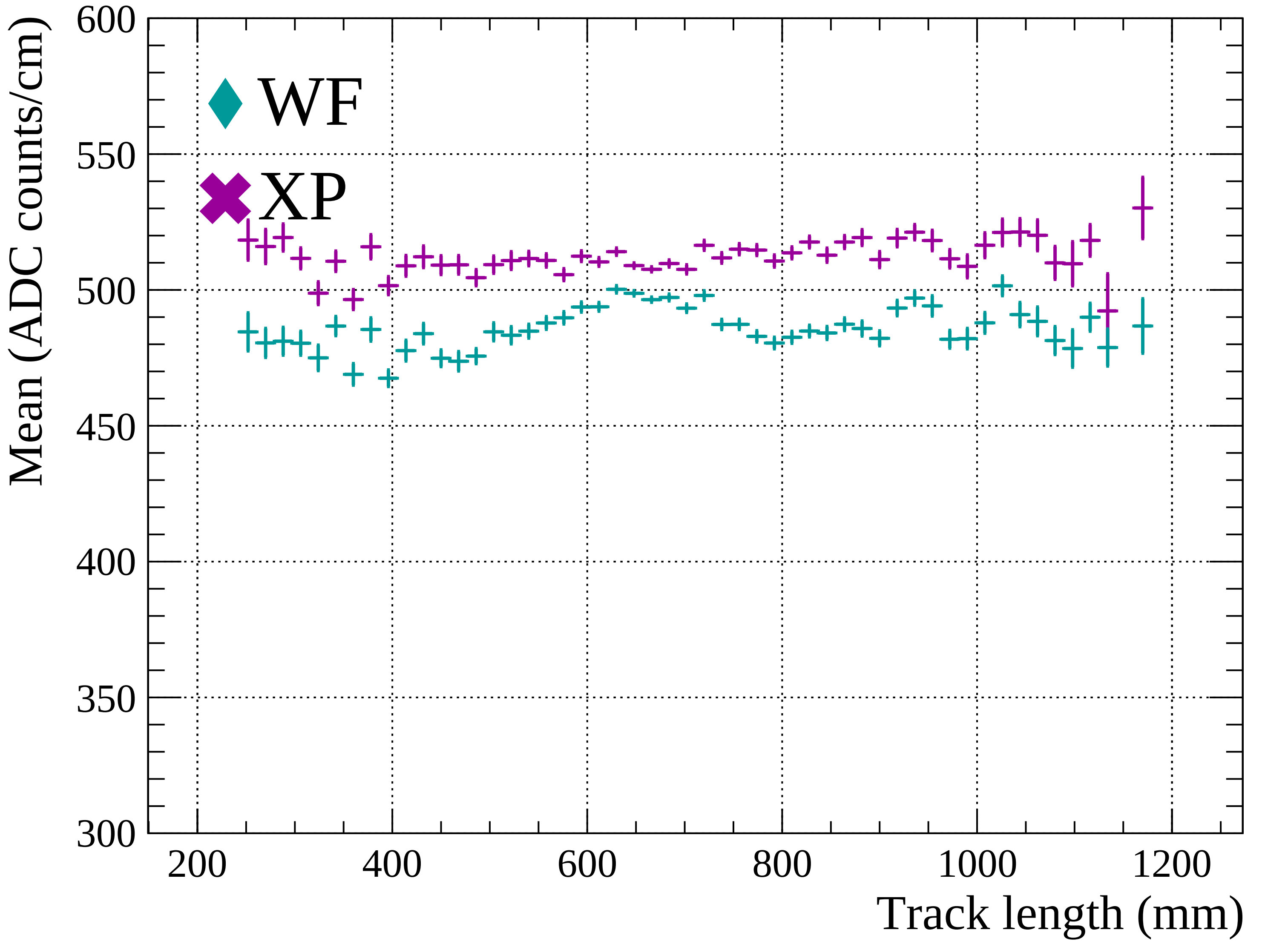

Energy loss (XP) vs drift time in ERAM 24
y = 0.41x + 398.57
Energy loss (XP) vs drift time in ERAM 30
y = 0.12x + 462.30
Energy loss (XP) vs drift time in ERAM 28
y = 1.56x + 40.46
Energy loss (XP) vs drift time in ERAM 19
y = 0.18x + 461.86
Energy loss (XP) vs drift time in ERAM 21
y = 0.07x + 483.96
Energy loss (XP) vs drift time in ERAM 13
y = 0.20x + 443.78
Energy loss (XP) vs drift time in ERAM 9
y = 0.14x + 463.11
Energy loss (XP) vs drift time in ERAM 2
y = 0.08x + 476.33
Energy loss (XP) vs drift time in ERAM 26
y = 0.37x + 389.30
Energy loss (XP) vs drift time in ERAM 17
y = 0.36x + 421.69
Energy loss (XP) vs drift time in ERAM 23
y = 0.35x + 400.88
Energy loss (XP) vs drift time in ERAM 29
y = 0.09x + 490.08
Energy loss (XP) vs drift time in ERAM 1
y = 0.19x + 449.92
Energy loss (XP) vs drift time in ERAM 10
y = 0.14x + 463.75
Energy loss (XP) vs drift time in ERAM 11
y = 0.06x + 485.33
Energy loss (XP) vs drift time in ERAM 3
y = 0.13x + 465.02
dE/dx (ADC counts/cm)
drift time (timebins)


Energy loss (XP) vs drift time in ERAM 47
y = 0.09x + 483.79
Energy loss (XP) vs drift time in ERAM 16
y = 0.41x + 404.44
Energy loss (XP) vs drift time in ERAM 14
y = 0.47x + 395.20
Energy loss (XP) vs drift time in ERAM 15
y = 0.32x + 414.53
Energy loss (XP) vs drift time in ERAM 42
y = 0.19x + 449.02
Energy loss (XP) vs drift time in ERAM 45
y = 0.21x + 445.01
Energy loss (XP) vs drift time in ERAM 37
y = 0.18x + 454.50
Energy loss (XP) vs drift time in ERAM 36
y = 0.16x + 460.73
Energy loss (XP) vs drift time in ERAM 20
y = 0.84x + 284.88
Energy loss (XP) vs drift time in ERAM 38
y = 1.12x + 209.99
Energy loss (XP) vs drift time in ERAM 7
y = 1.36x + 141.47
Energy loss (XP) vs drift time in ERAM 44
y = 0.78x + 297.77
Energy loss (XP) vs drift time in ERAM 43
y = 0.59x + 348.32
Energy loss (XP) vs drift time in ERAM 39
y = 0.71x + 315.81
Energy loss (XP) vs drift time in ERAM 41
y = 0.79x + 288.89
Energy loss (XP) vs drift time in ERAM 46
y = 0.61x + 337.78
dE/dx (ADC counts/cm)
drift time (timebins)

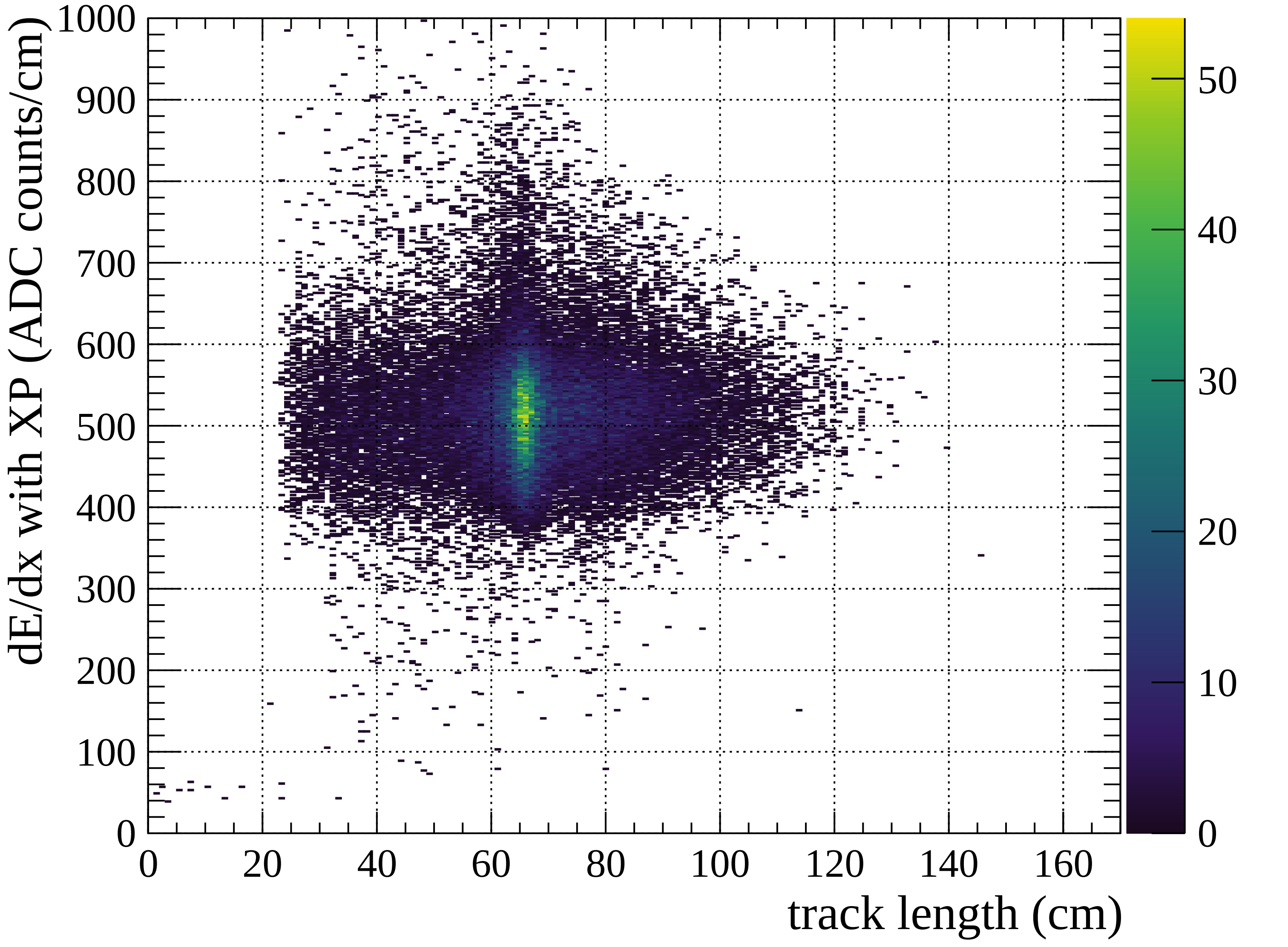


time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
60
50
40
30
20
10
0

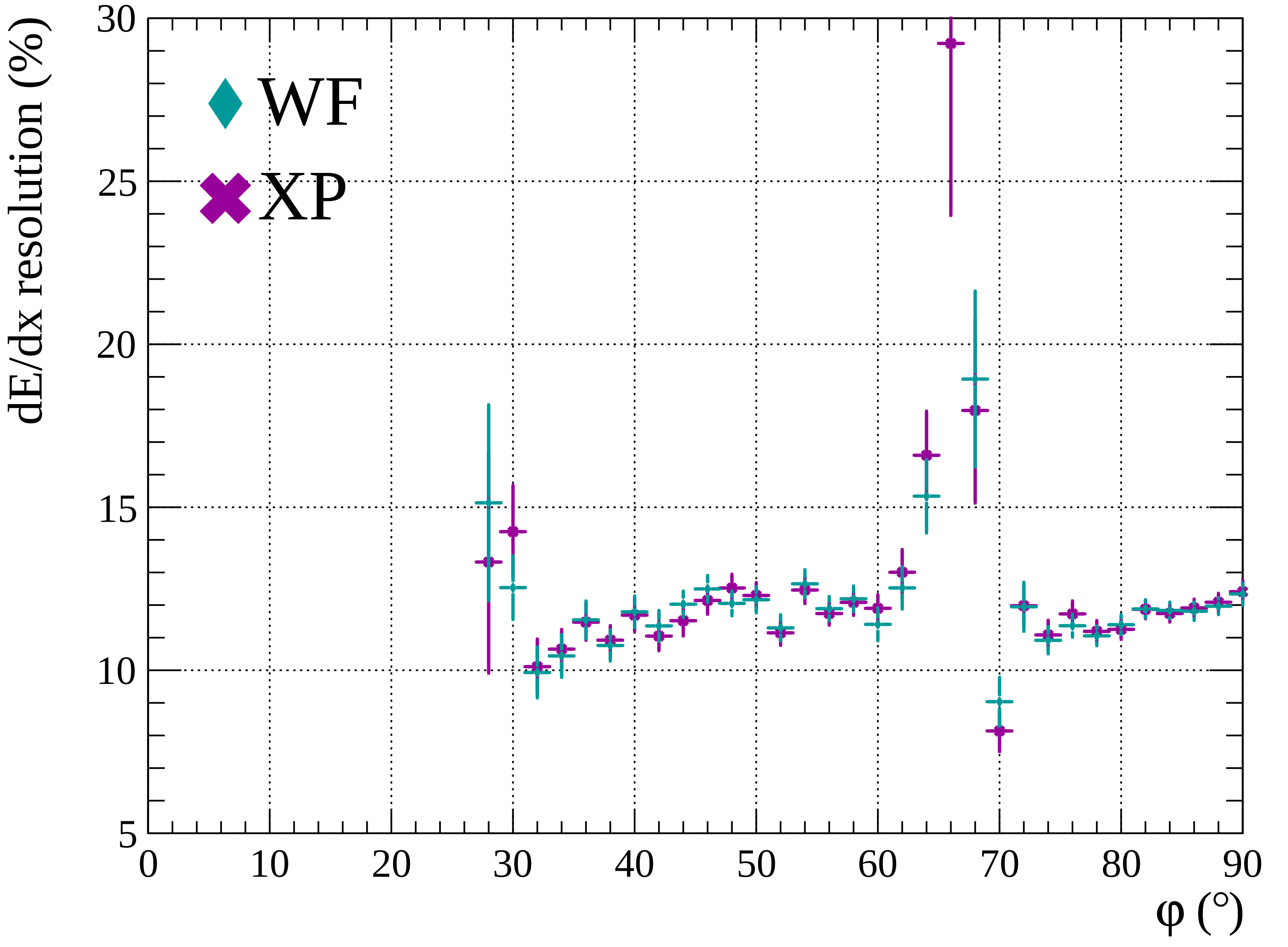

WF
XP
dE/dx resolution (%)
Track length (mm)
5
10
15
20
25
30
200
400
600
800
1000
1200



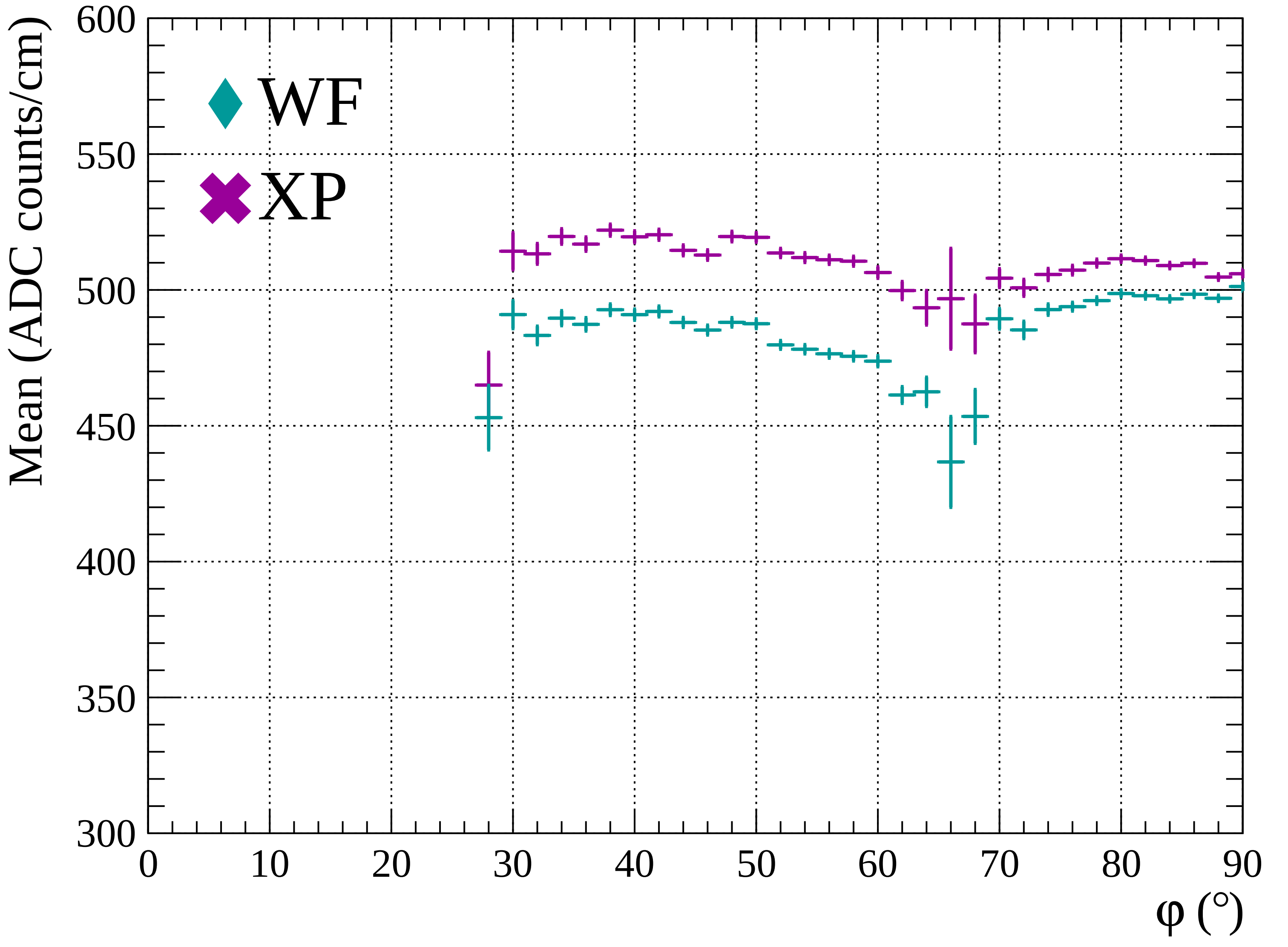
WF
XP
Mean (ADC counts/cm)
Track length (mm)
600
550
500
450
400
350
300
200
400
600
800
1000
1200



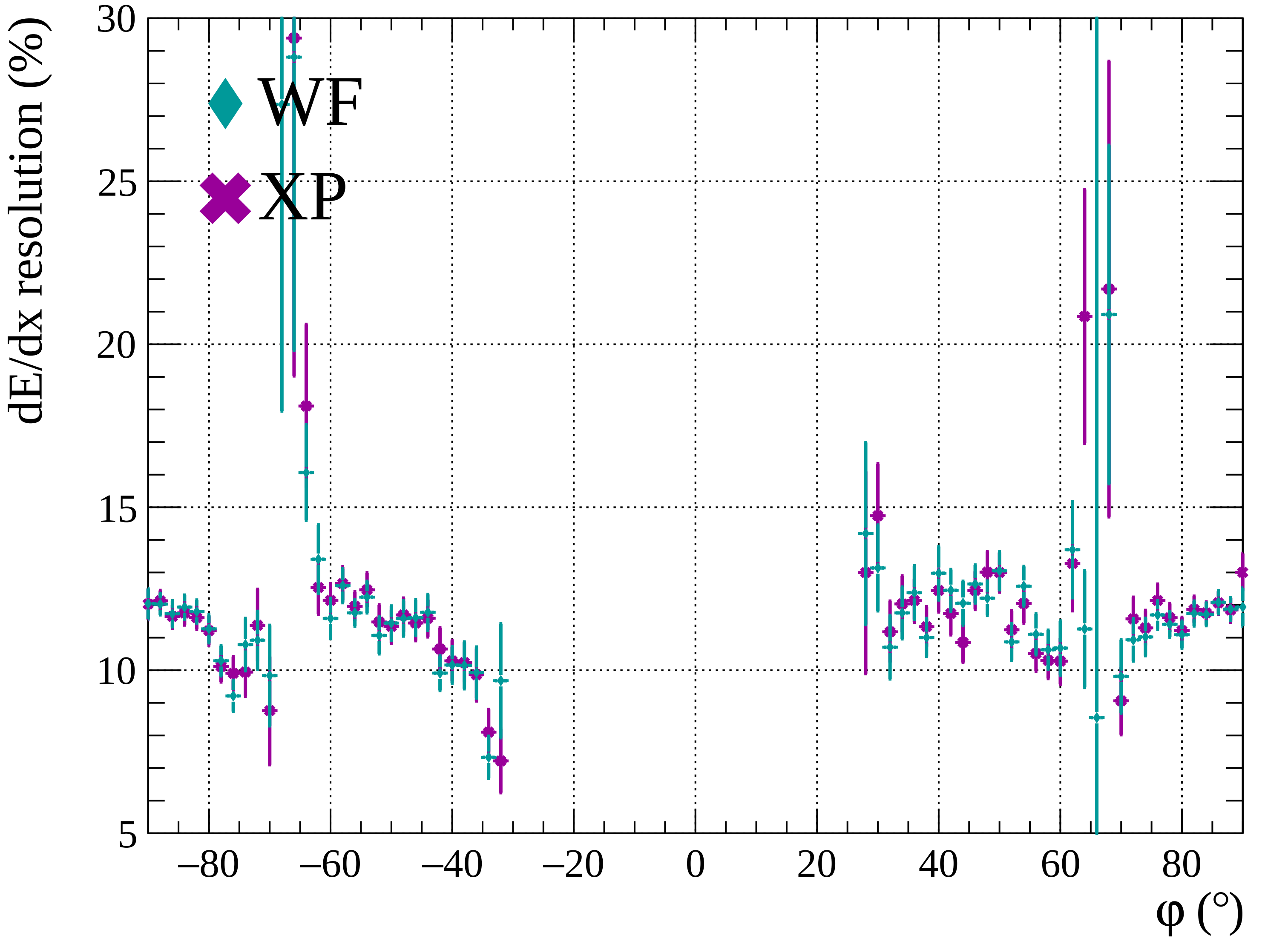
dE/dx with XP (ADC counts/cm)
track length (cm)
0
100
200
300
400
500
600
700
800
900
1000
0
20
40
60
80
100
120
140
160
0
10
20
30
40
50



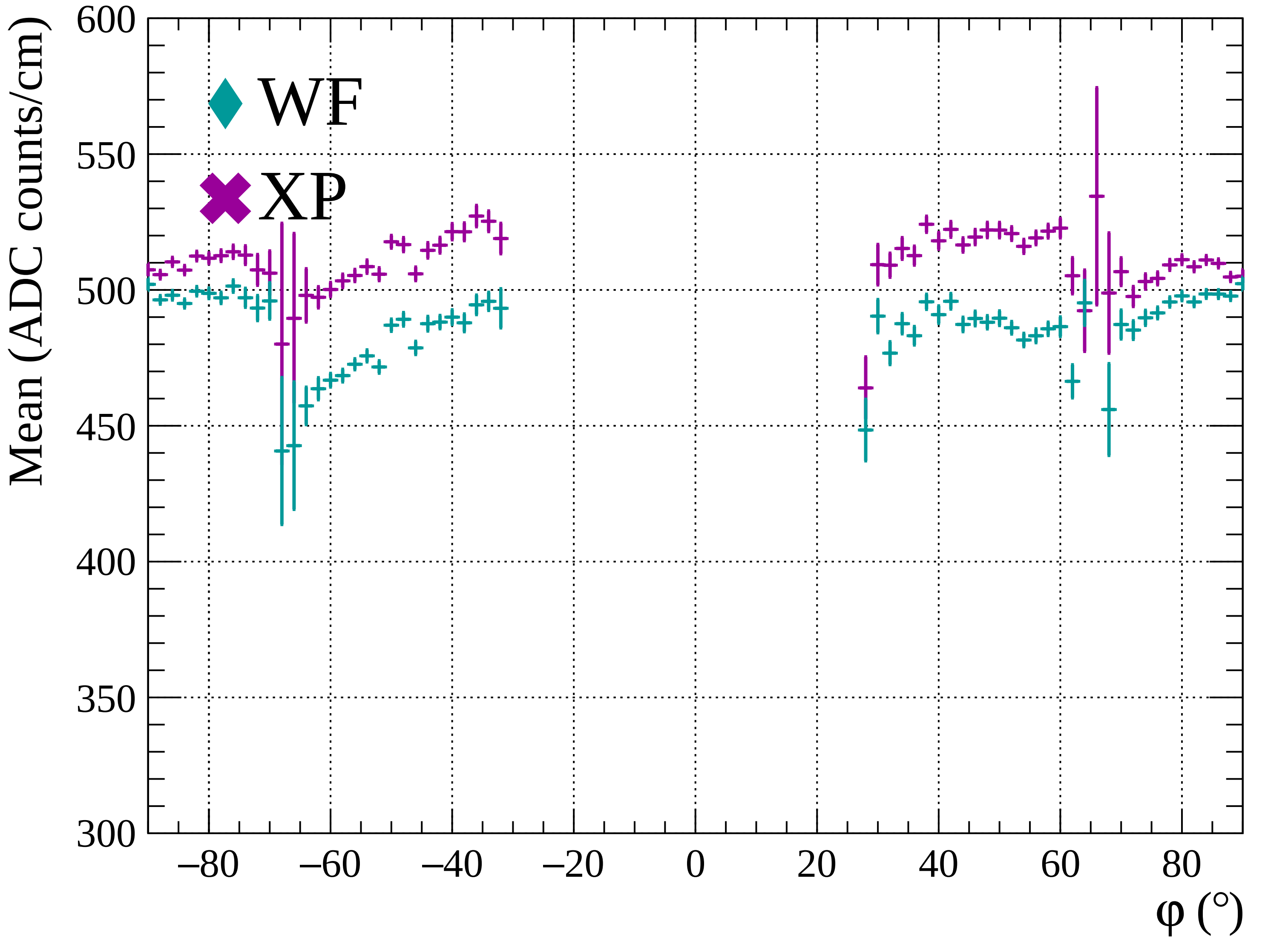
dE/dx resolution (%)
WF
XP
φ (°)
30
25
20
15
10
5
0
10
20
30
40
50
60
70
80
90



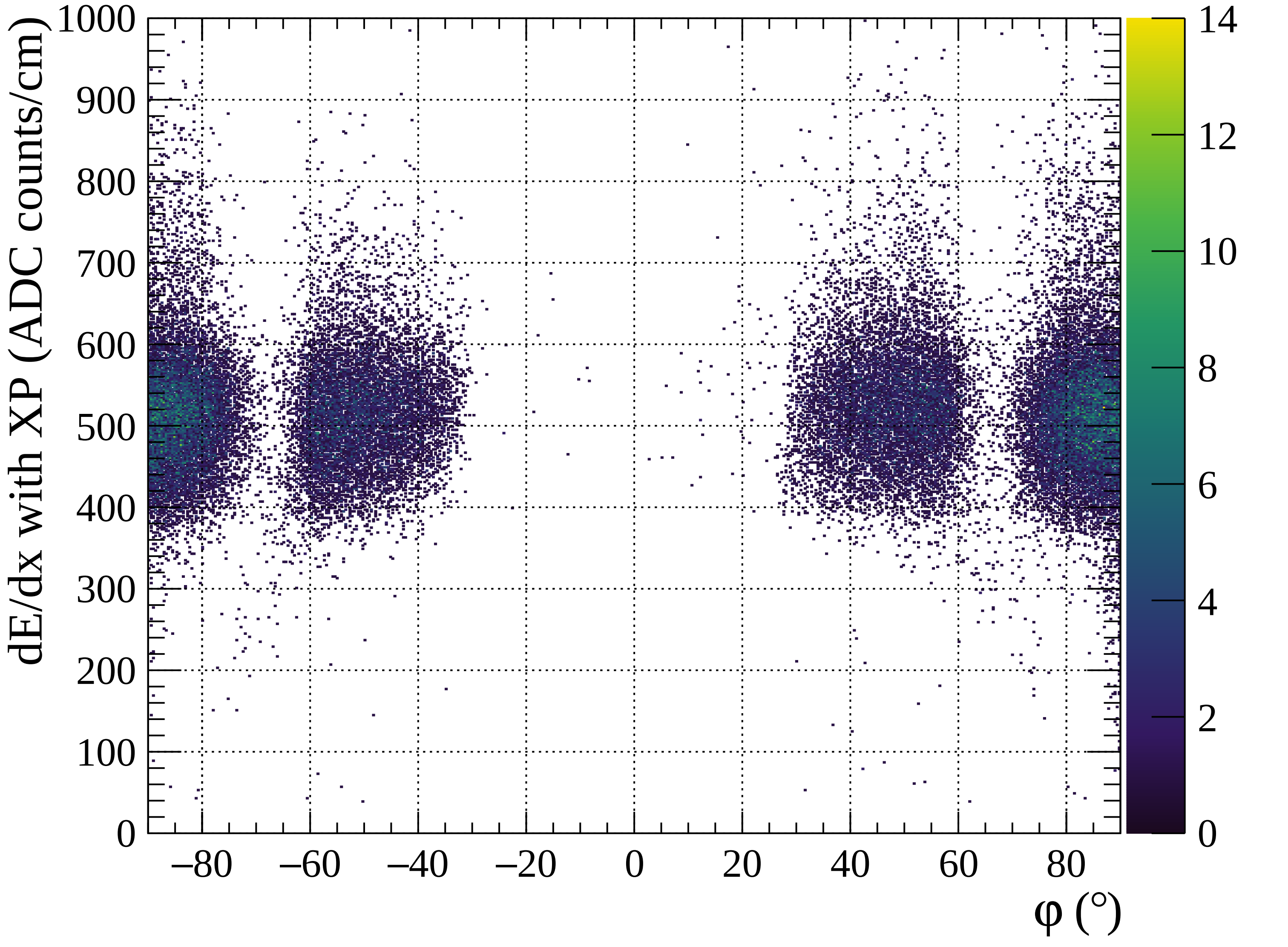
WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90
φ (°)



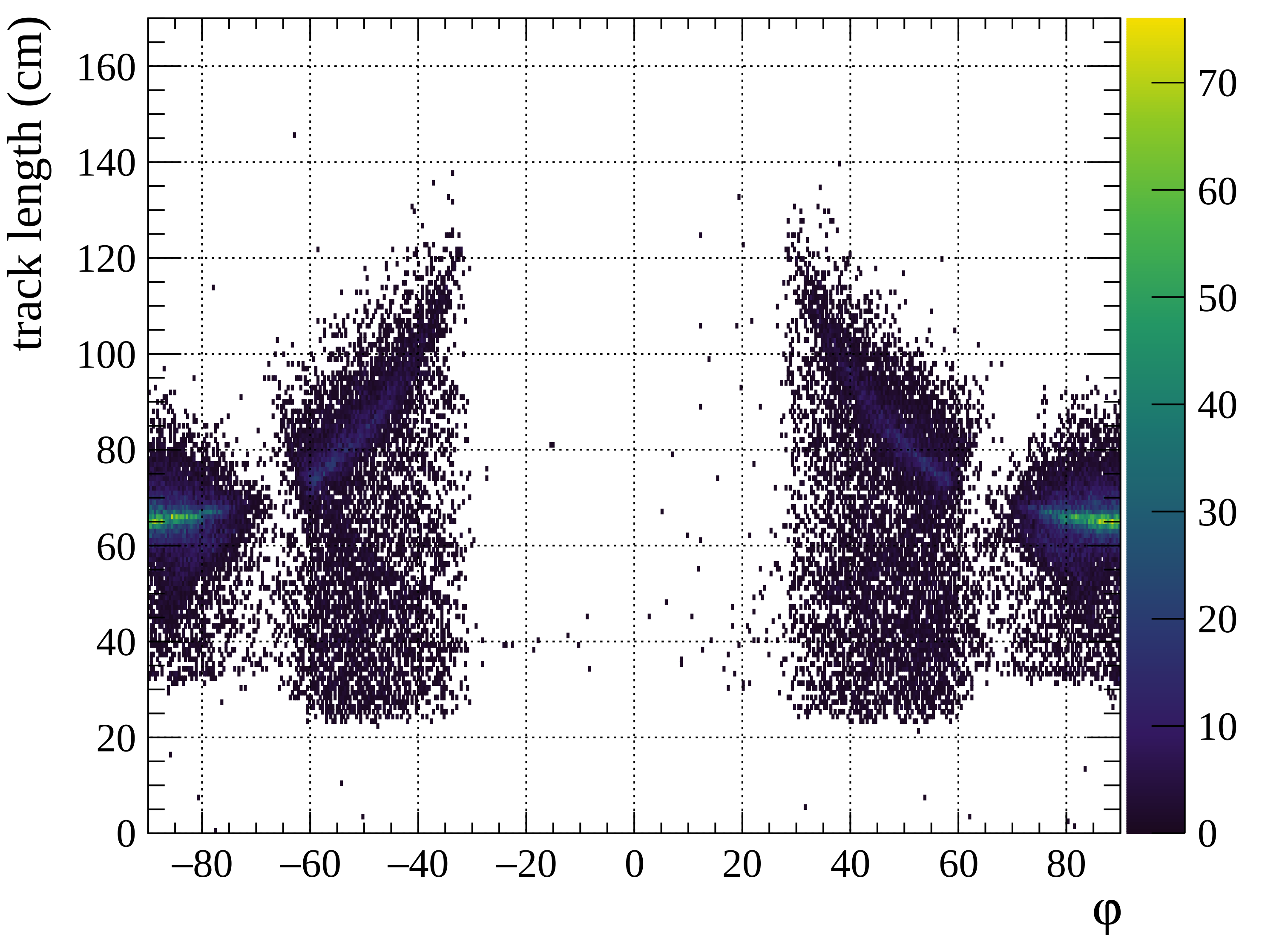
dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
−80
−60
−40
−20
0
20
40
60
80
φ (°)



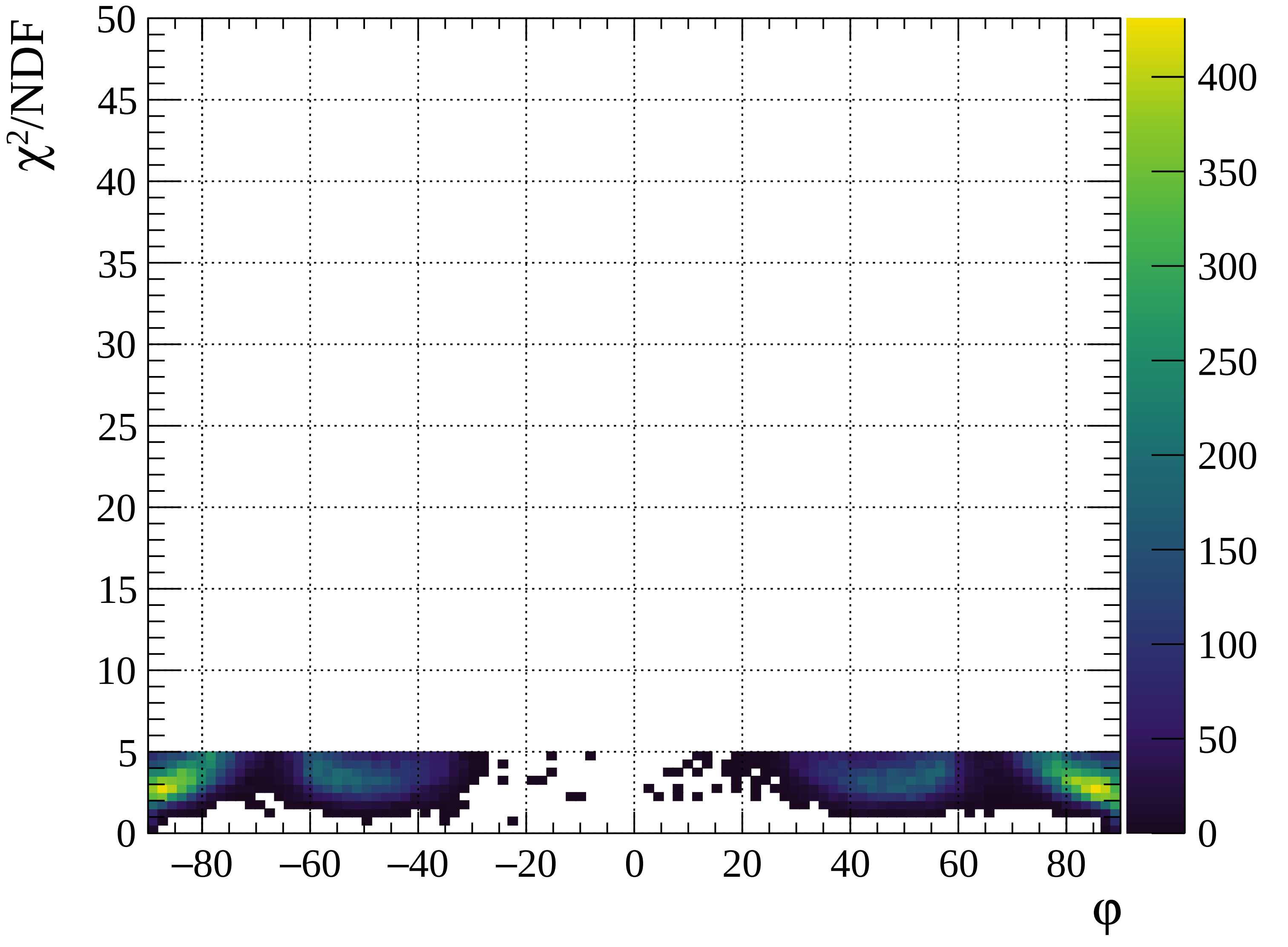
WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80
φ (°)



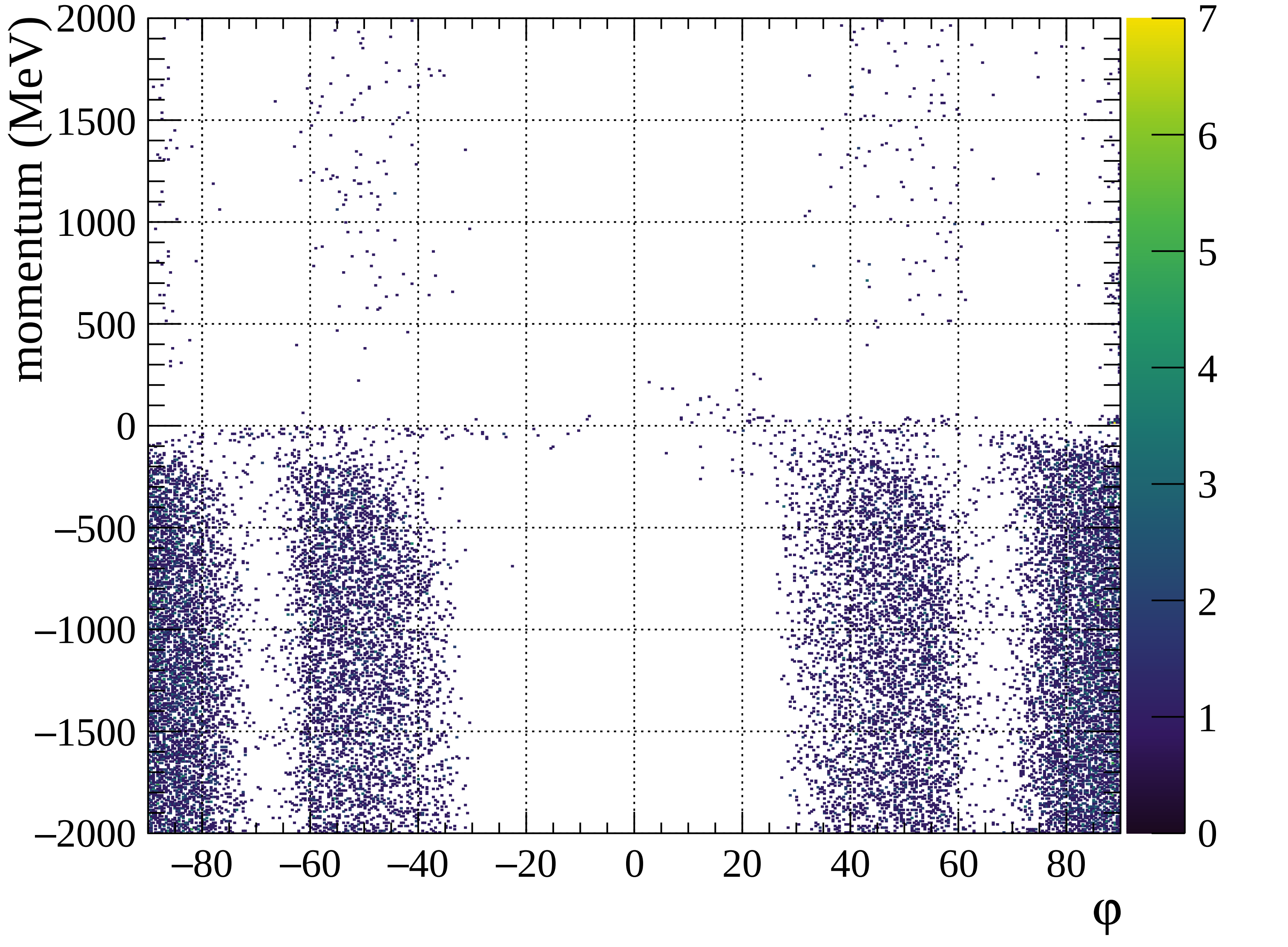
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
φ (°)
14
12
10
8
6
4
2
0

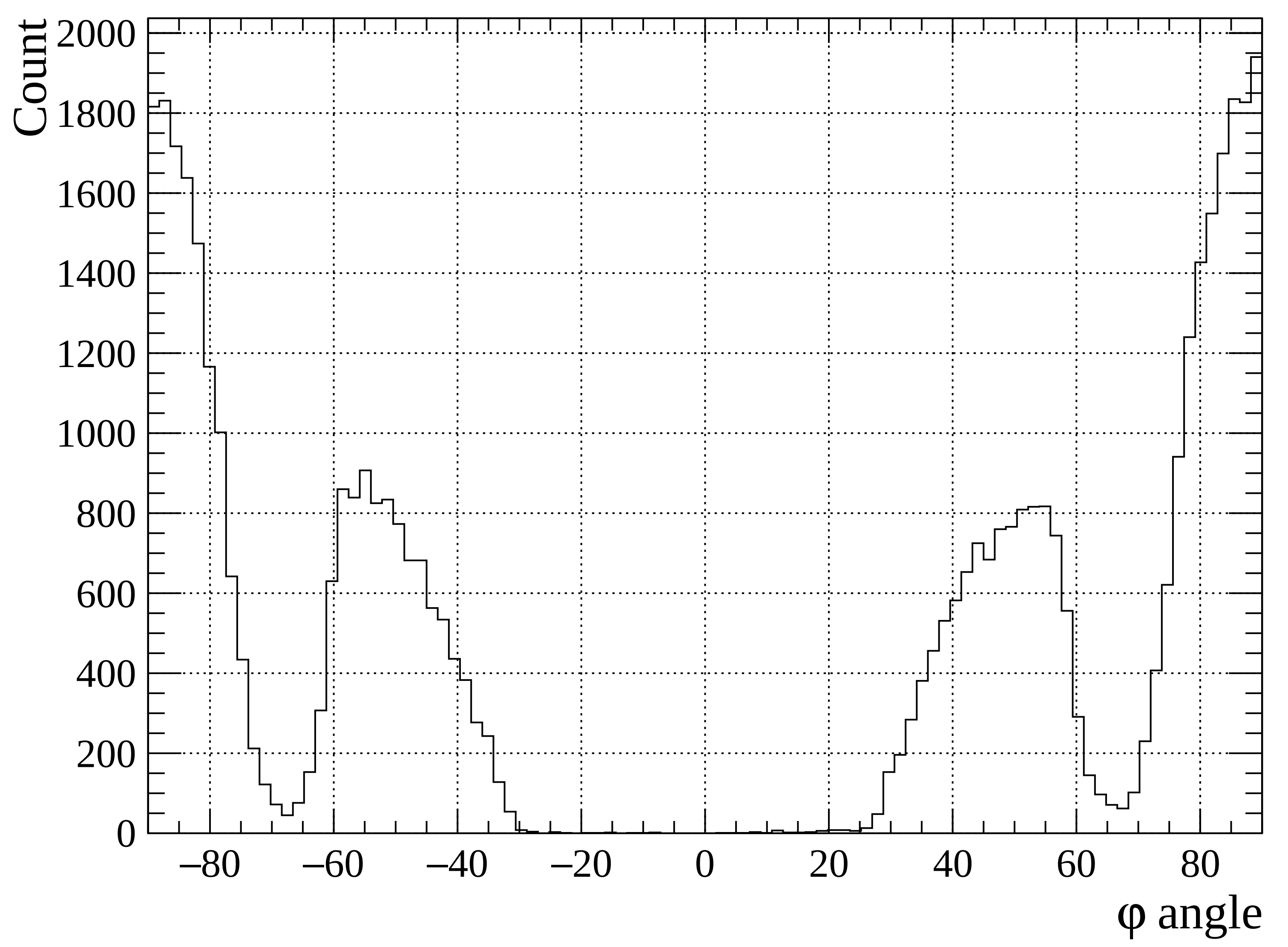


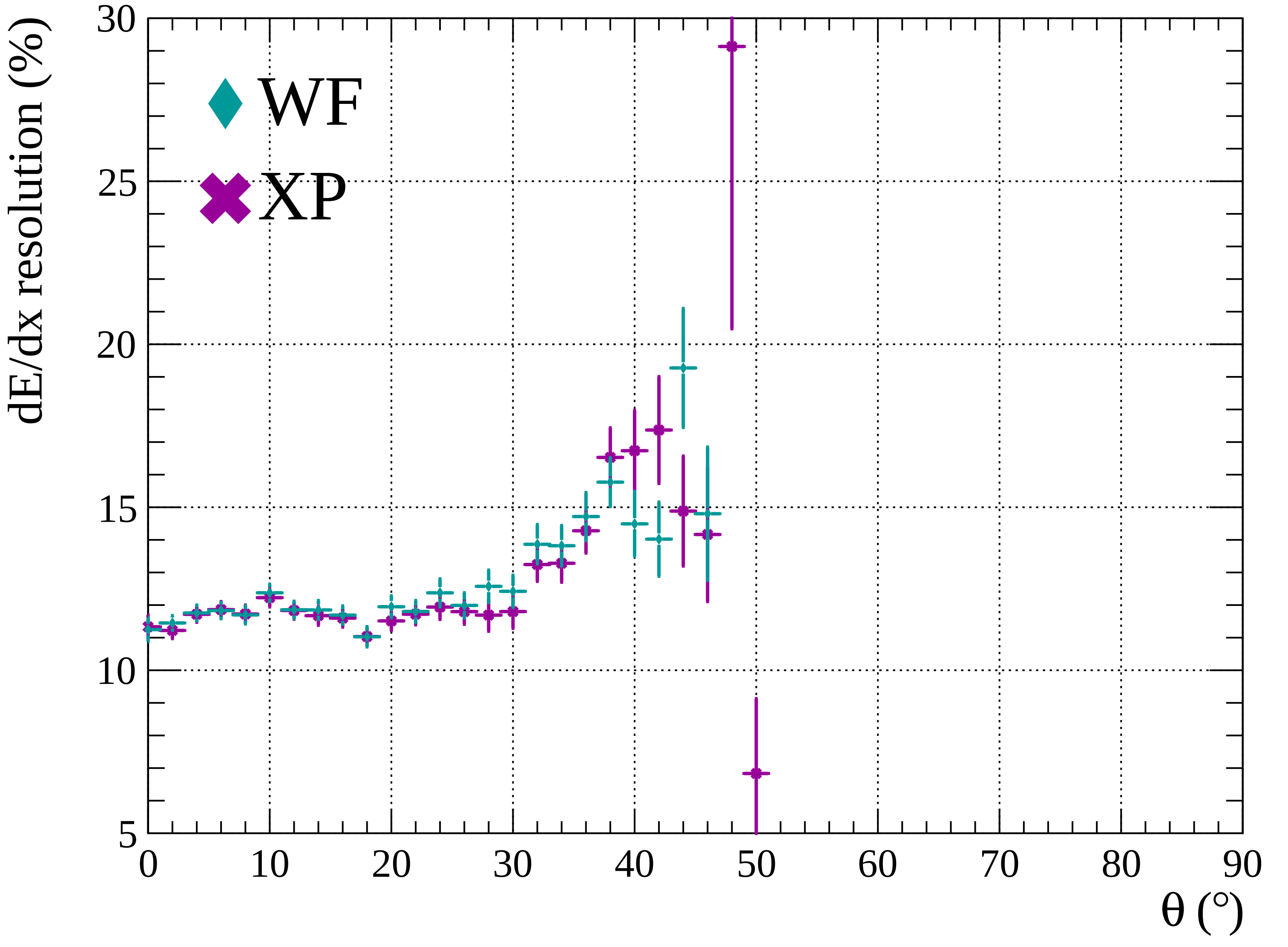
track length (cm)
φ
0
20
40
60
80
100
120
140
160
−80
−60
−40
−20
0
20
40
60
80
0
10
20
30
40
50
60
70

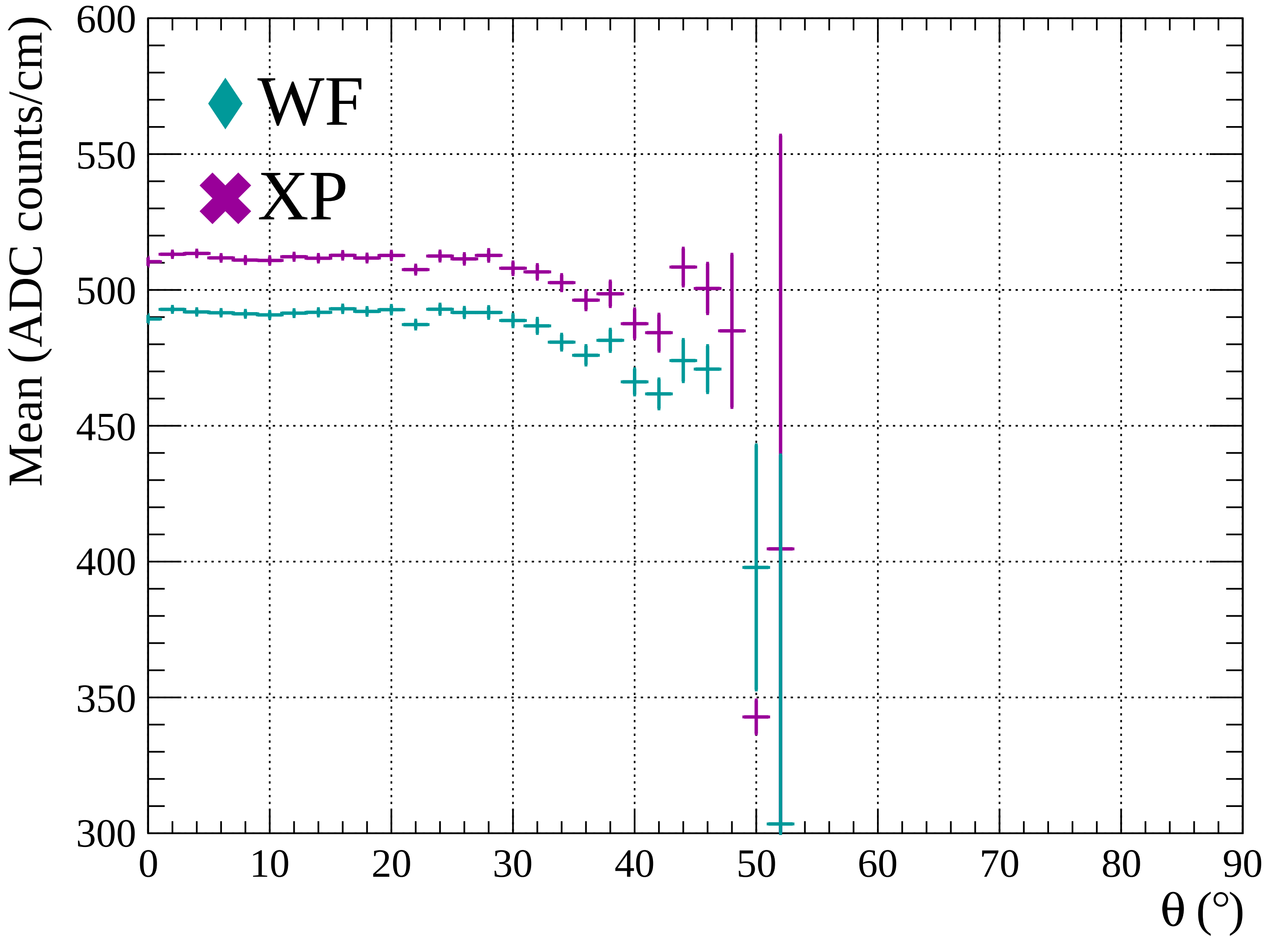


χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ
400
350
300
250
200
150
100
50
0

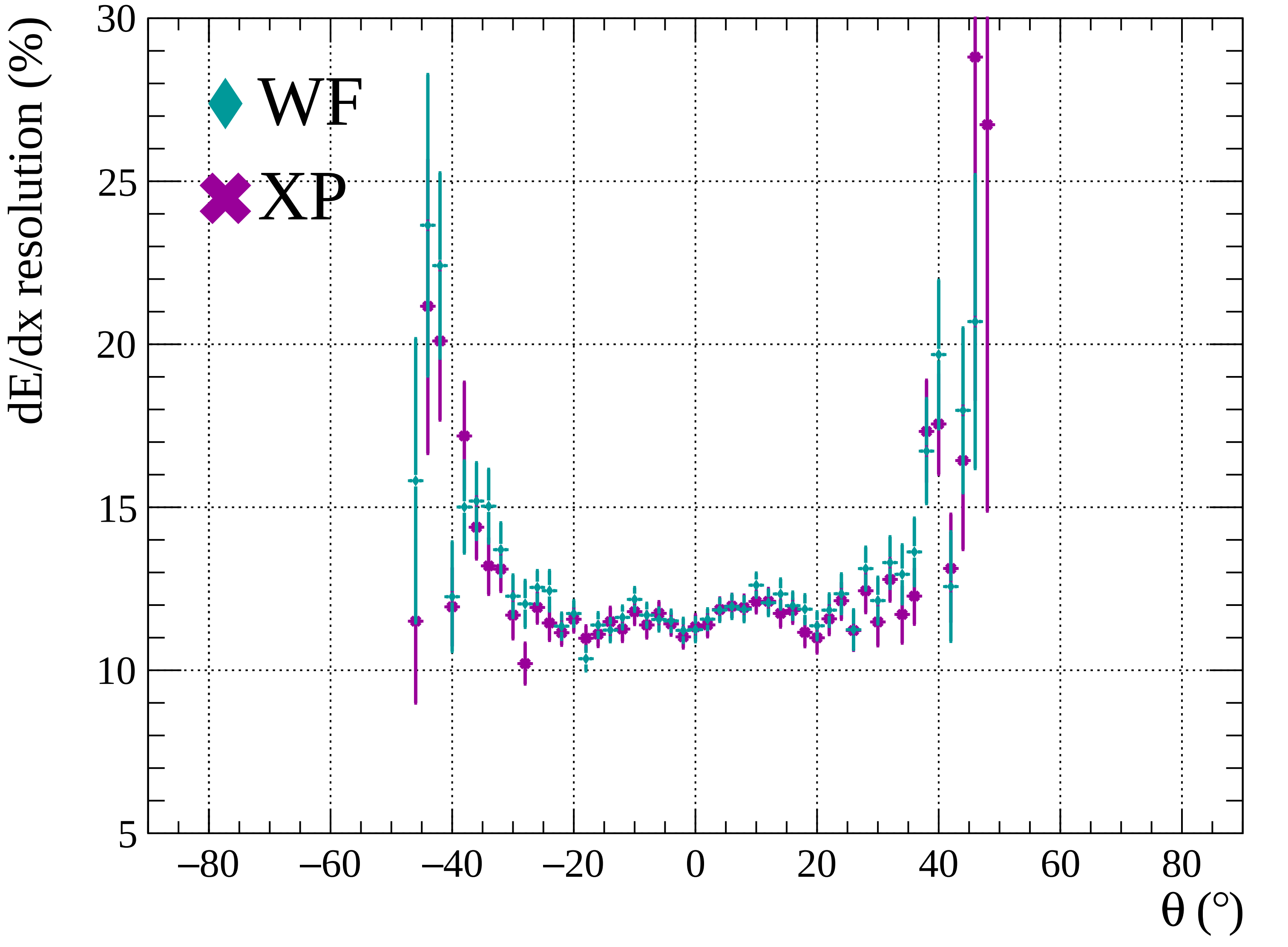

momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
7
6
5
4
3
2
1
0


Count
2000
1800
1600
1400
1200
1000
800
600
400
200
0
−80
−60
−40
−20
0
20
40
60
80
φ angle

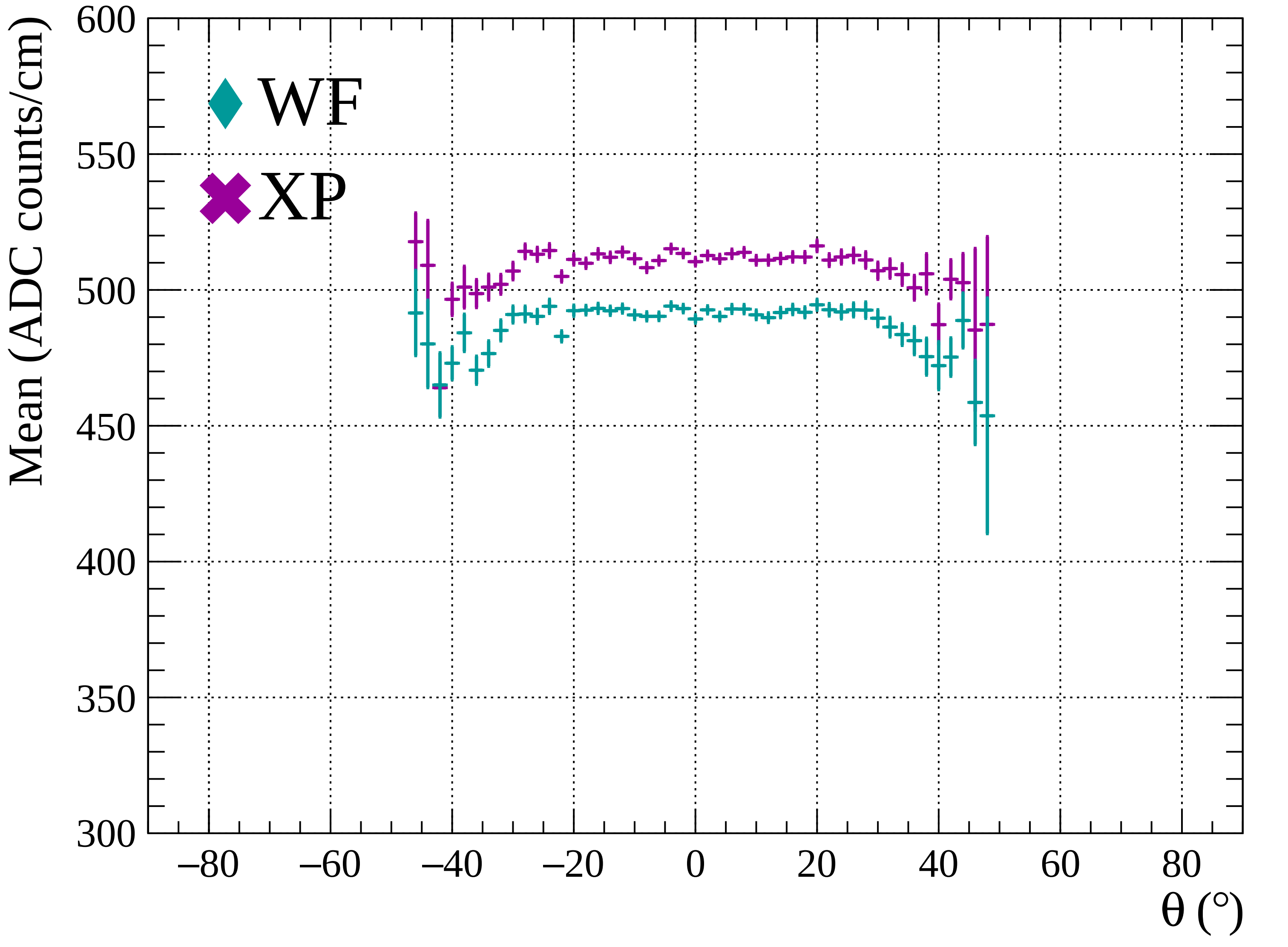

dE/dx resolution (%)
θ (°)
WF
XP
30
25
20
15
10
5
0
10
20
30
40
50
60
70
80
90



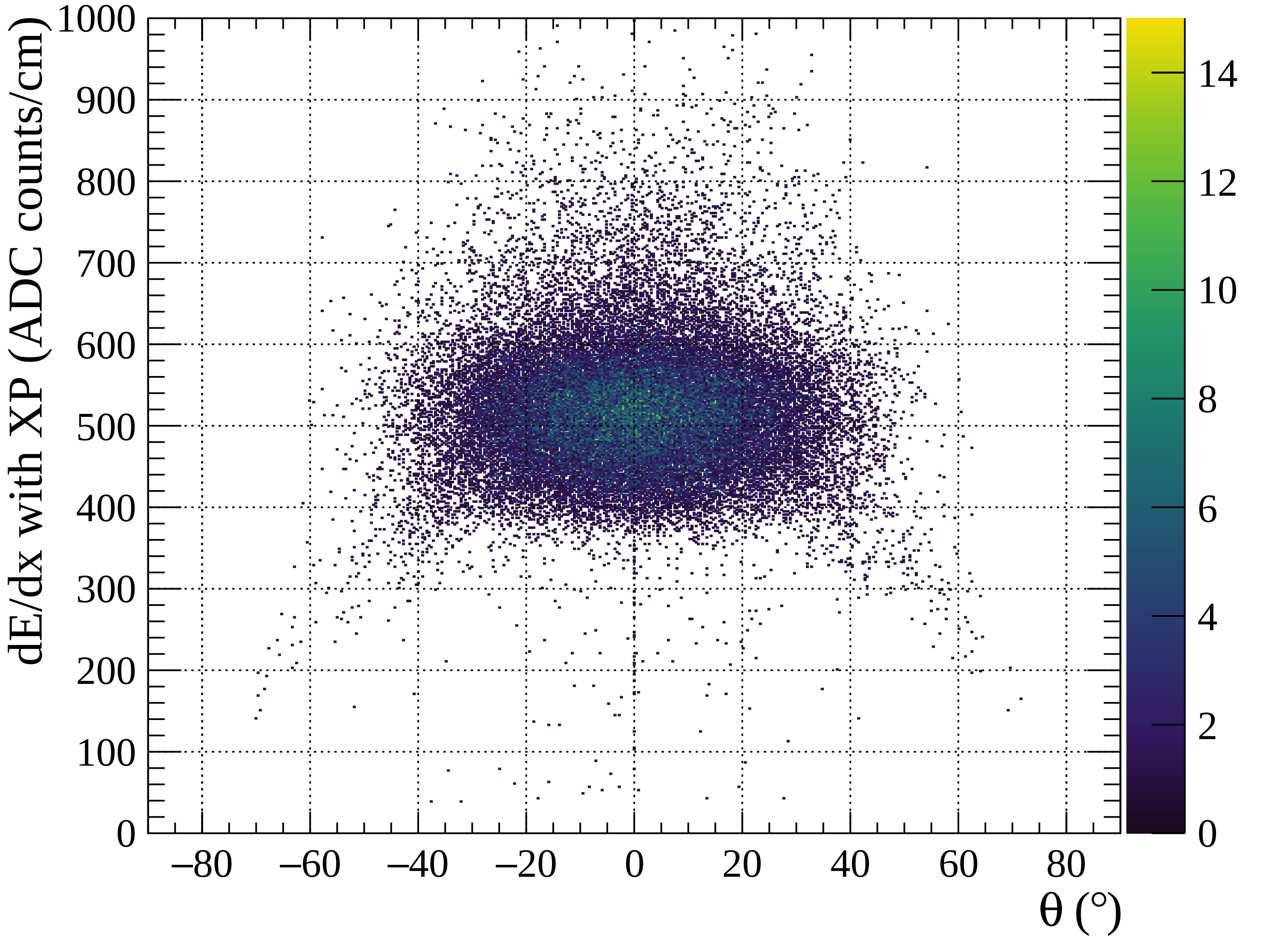
WF
XP
Mean (ADC counts/cm)
θ (°)
300
350
400
450
500
550
600
0
10
20
30
40
50
60
70
80
90



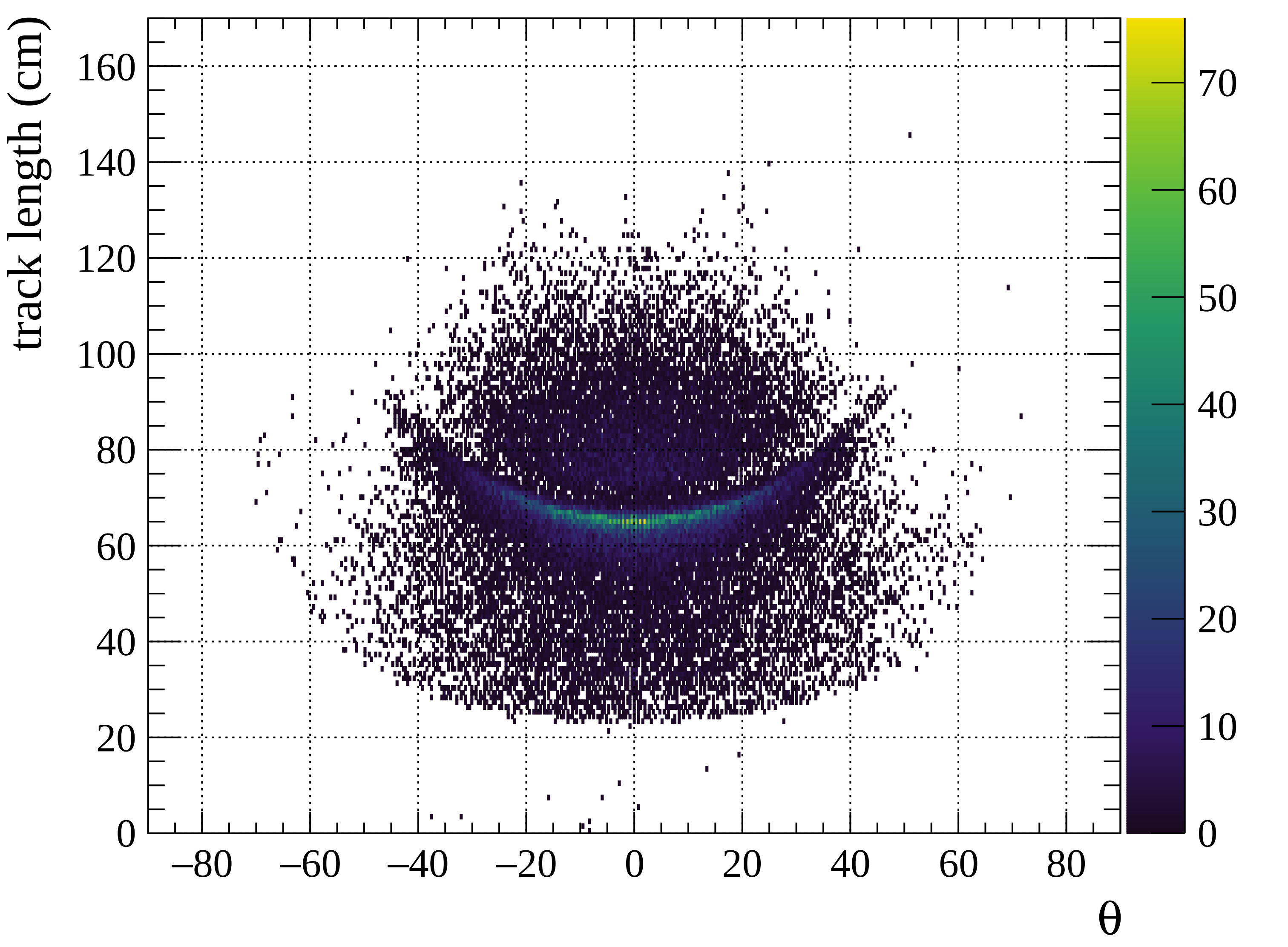
dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
−80
−60
−40
−20
0
20
40
60
80
θ (°)

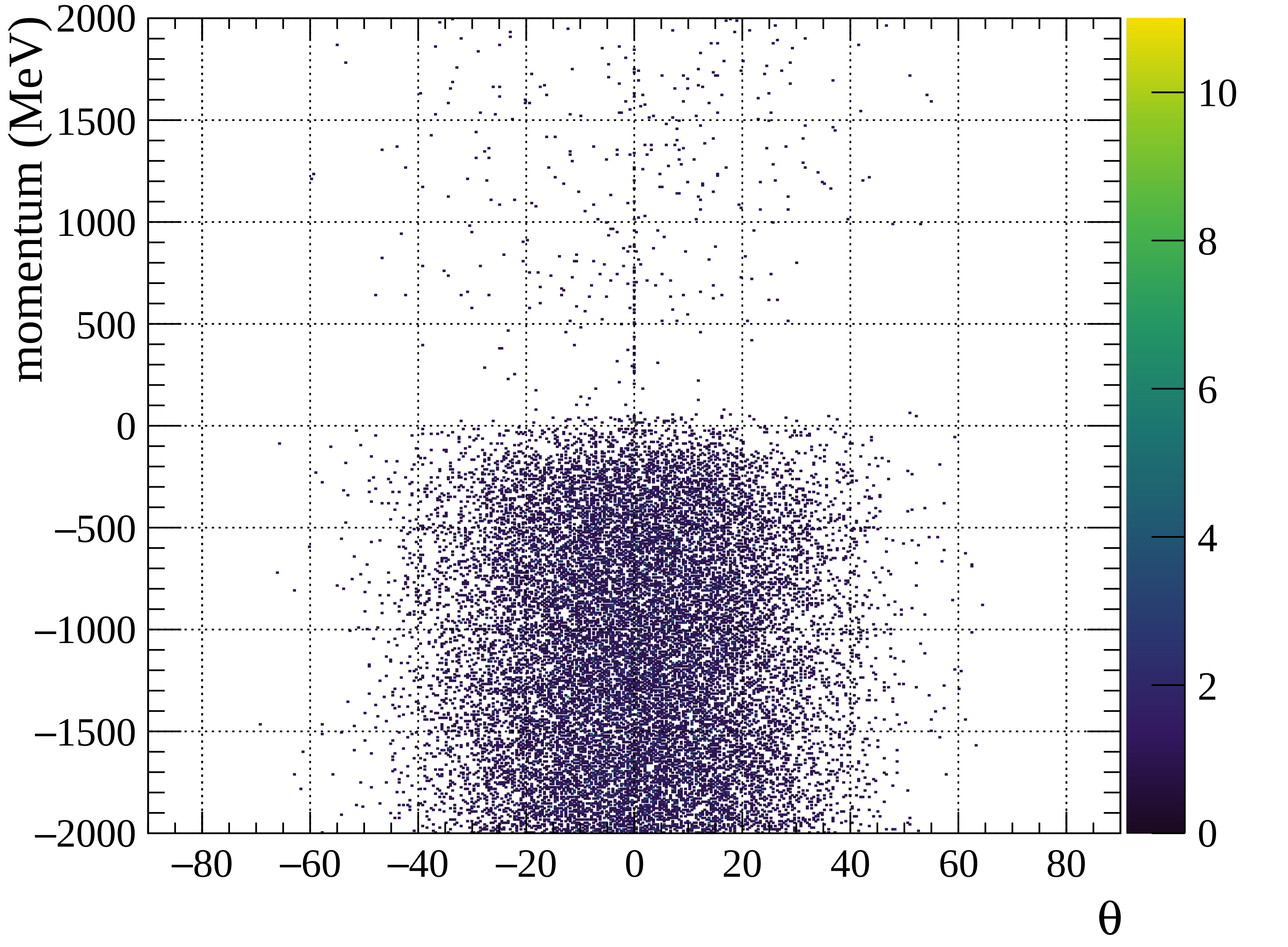


Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
−80
−60
−40
−20
0
20
40
60
80
θ (°)

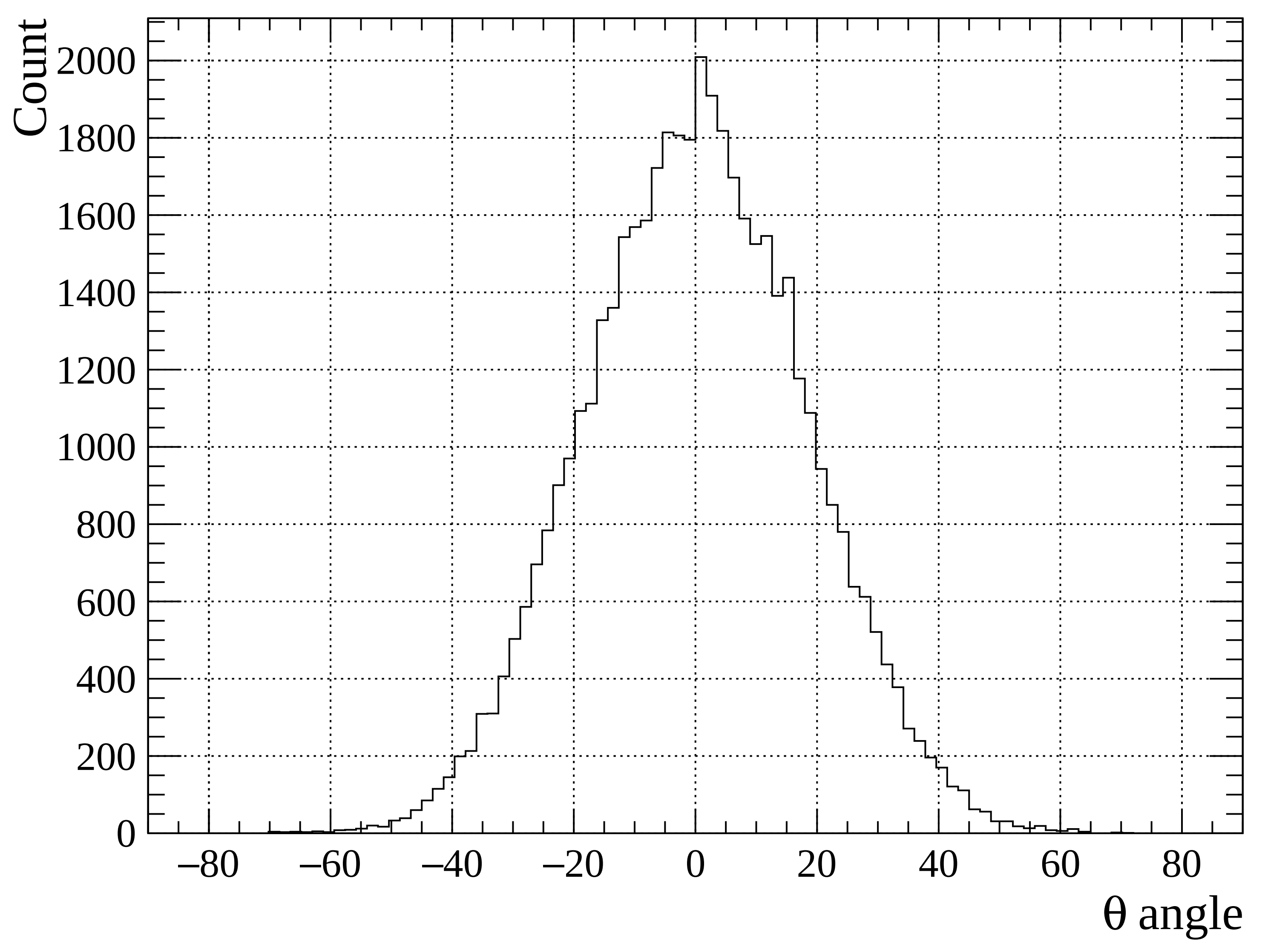


dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
14
12
10
8
6
4
2
0

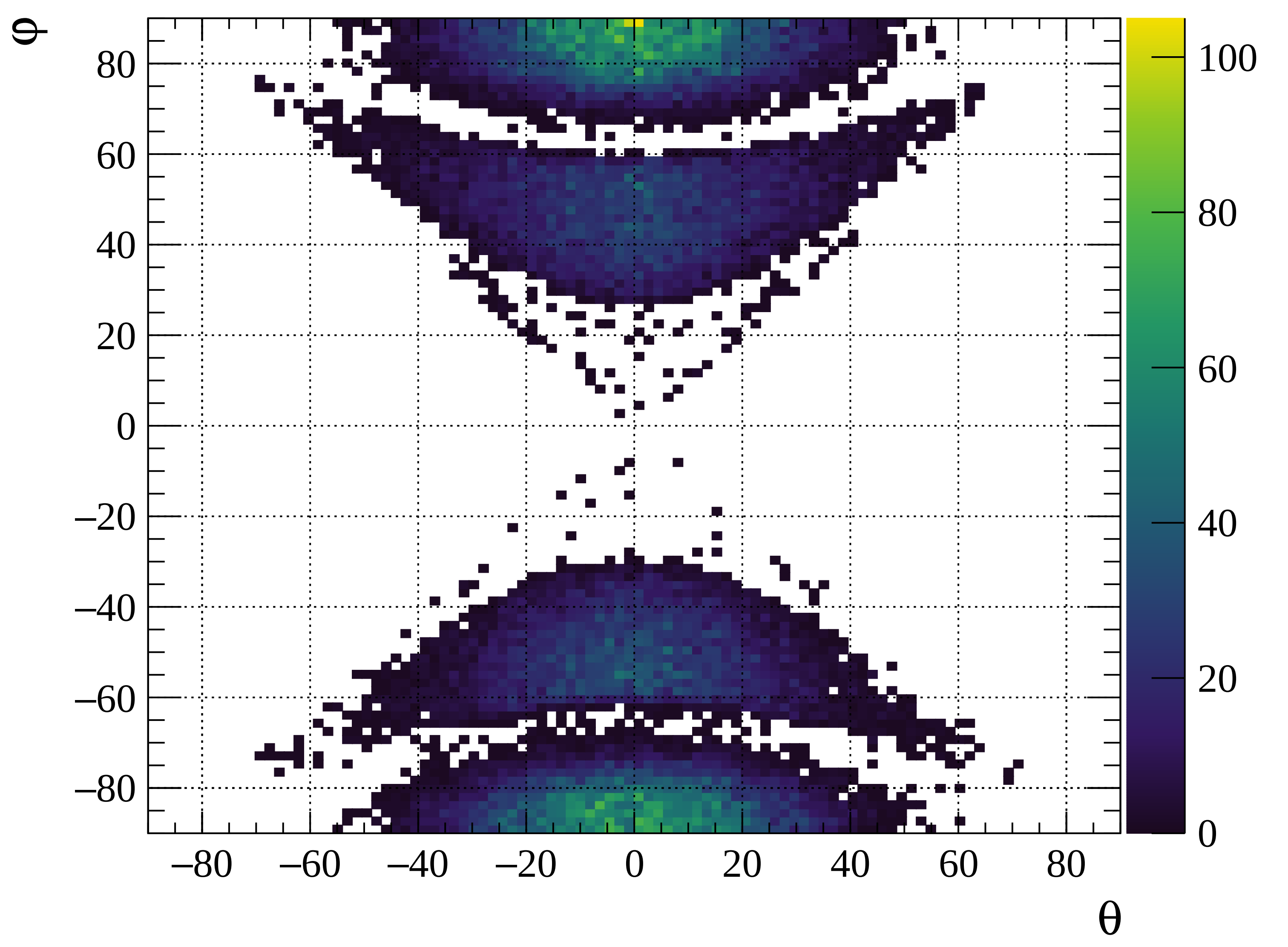

track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
θ
70
60
50
40
30
20
10
0



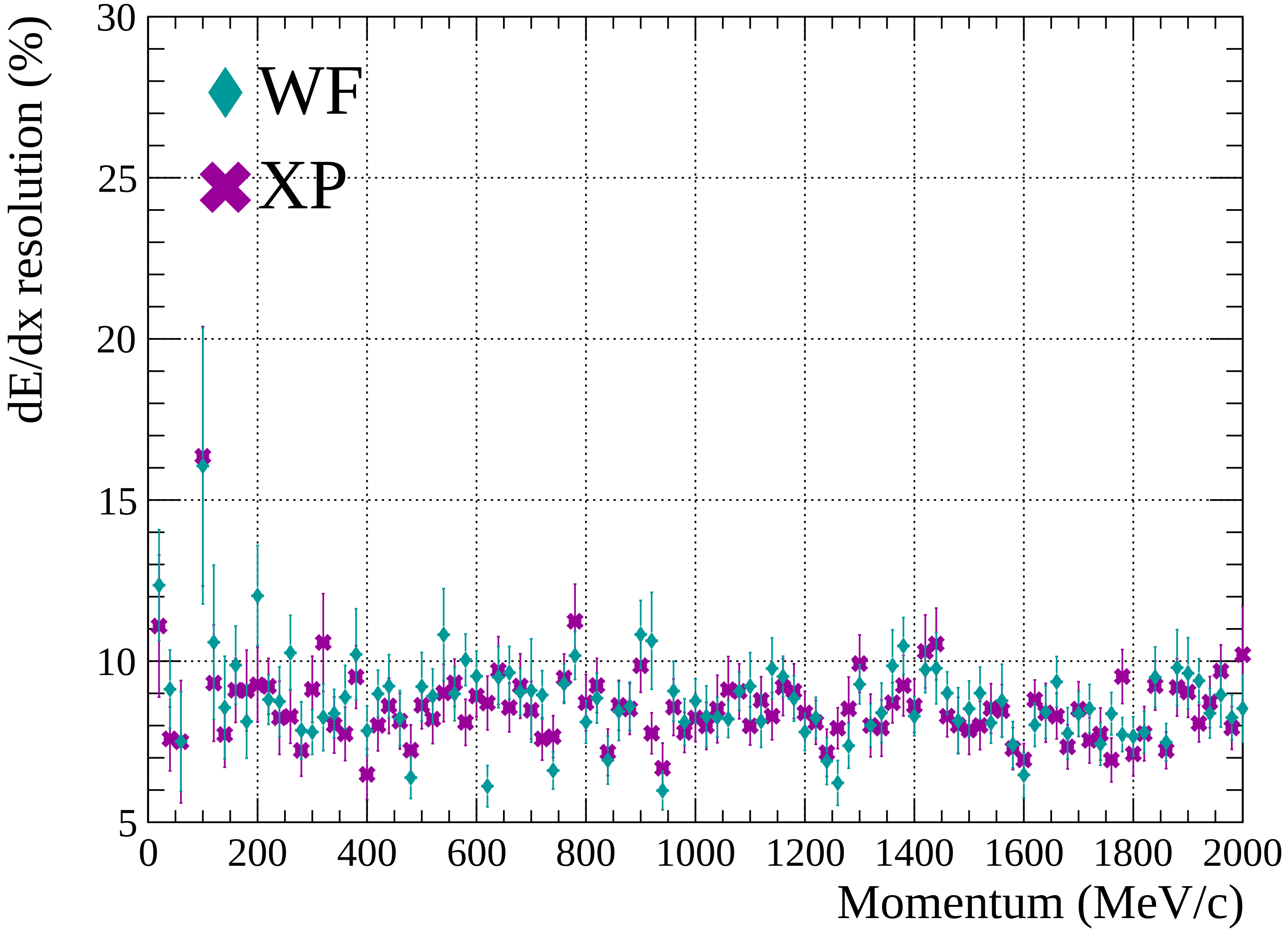
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
10
8
6
4
2
0



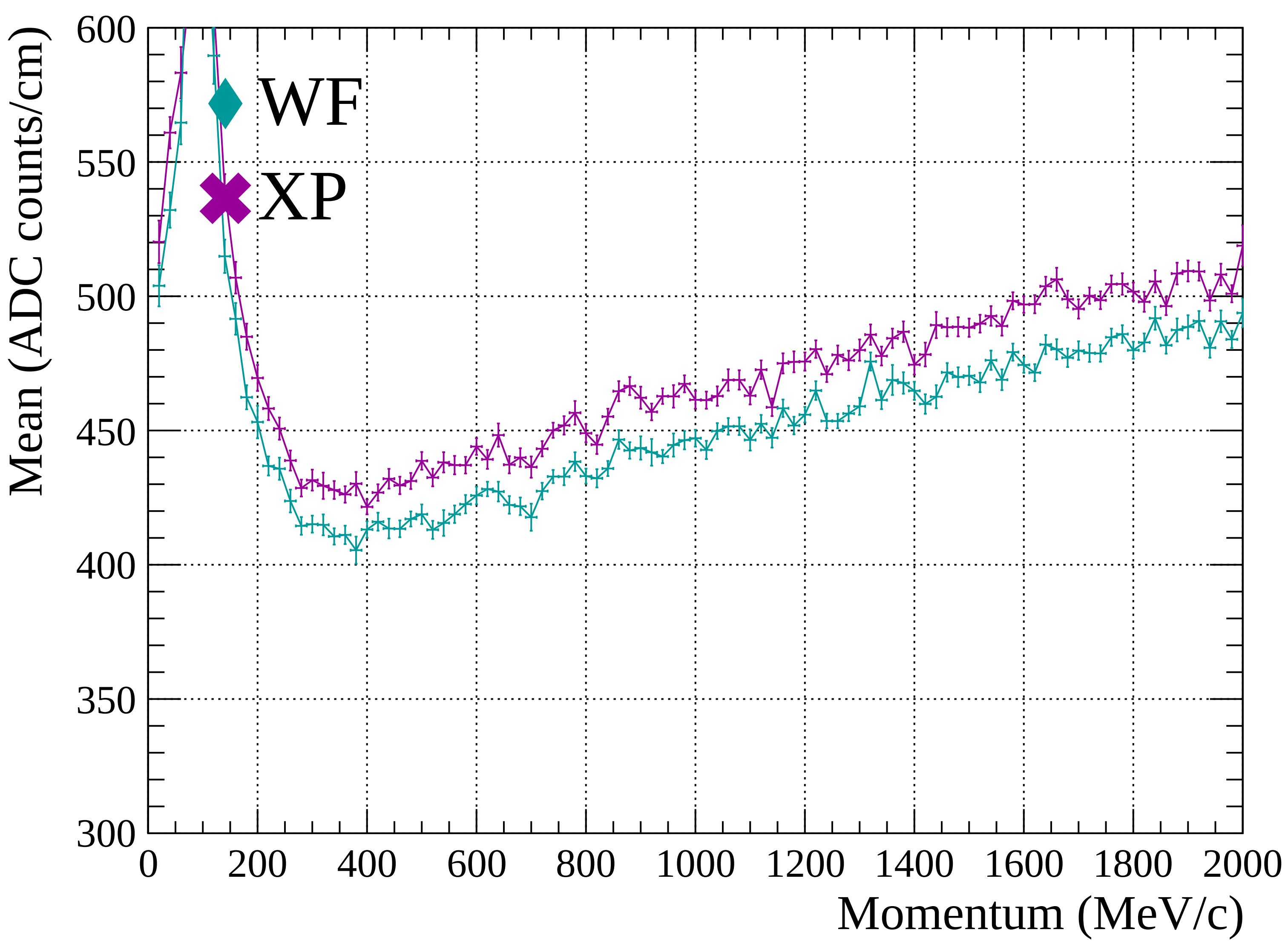
Count
2000
1800
1600
1400
1200
1000
800
600
400
200
0
−80
−60
−40
−20
0
20
40
60
80
θ angle



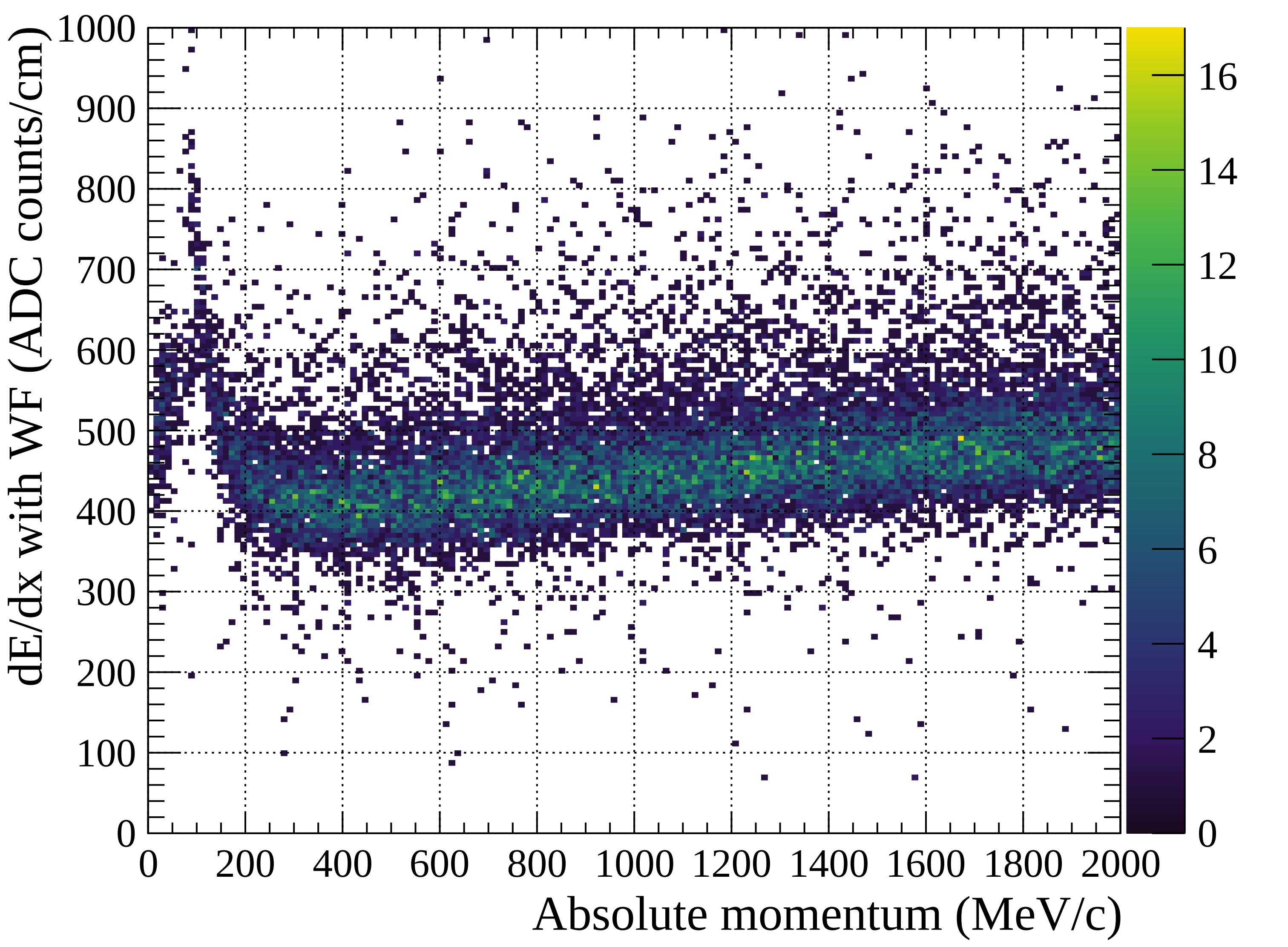
φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
100
80
60
40
20
0

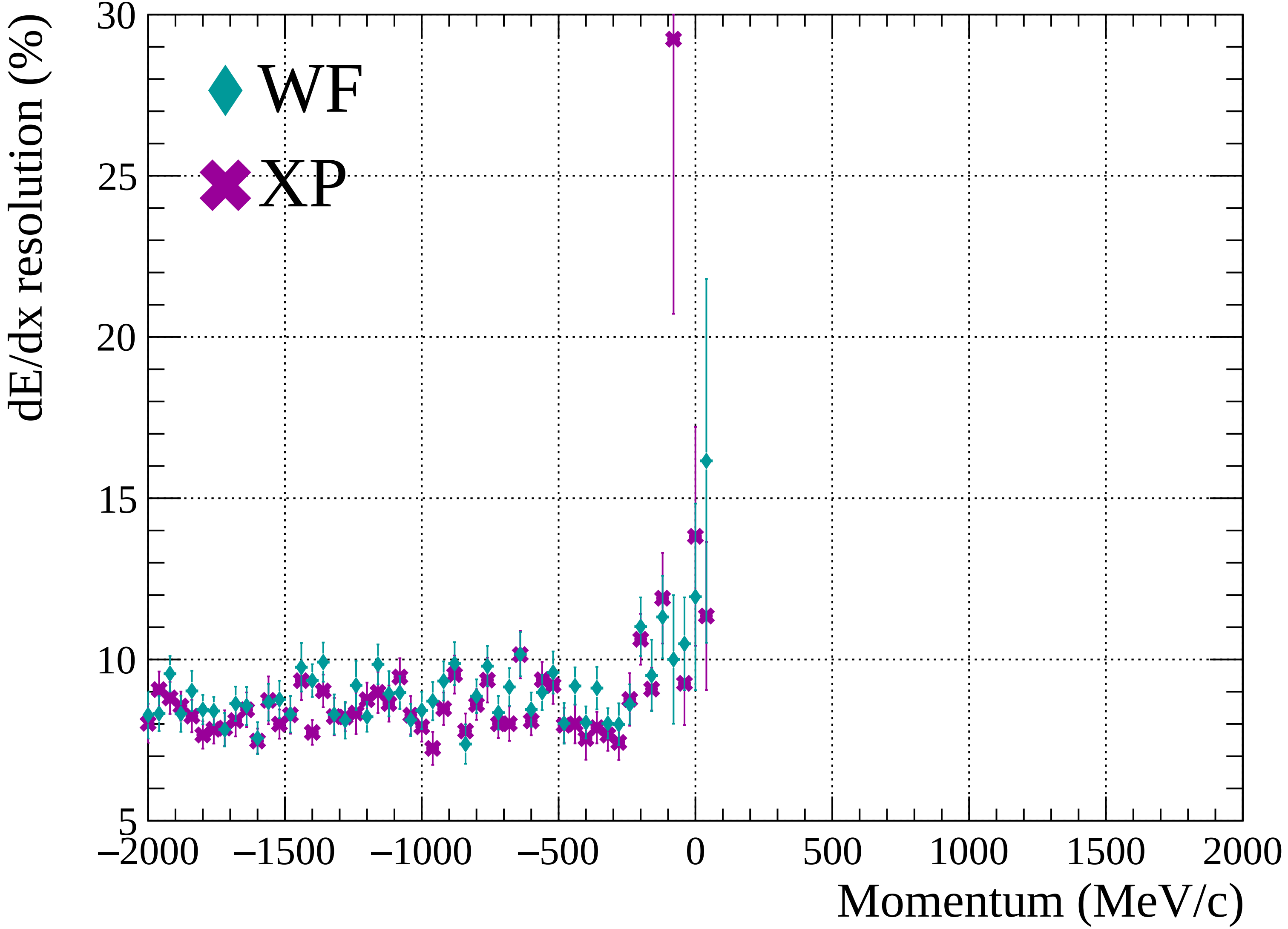


WF
XP
dE/dx resolution (%)
Momentum (MeV/c)
30
25
20
15
10
5
0
200
400
600
800
1000
1200
1400
1600
1800
2000

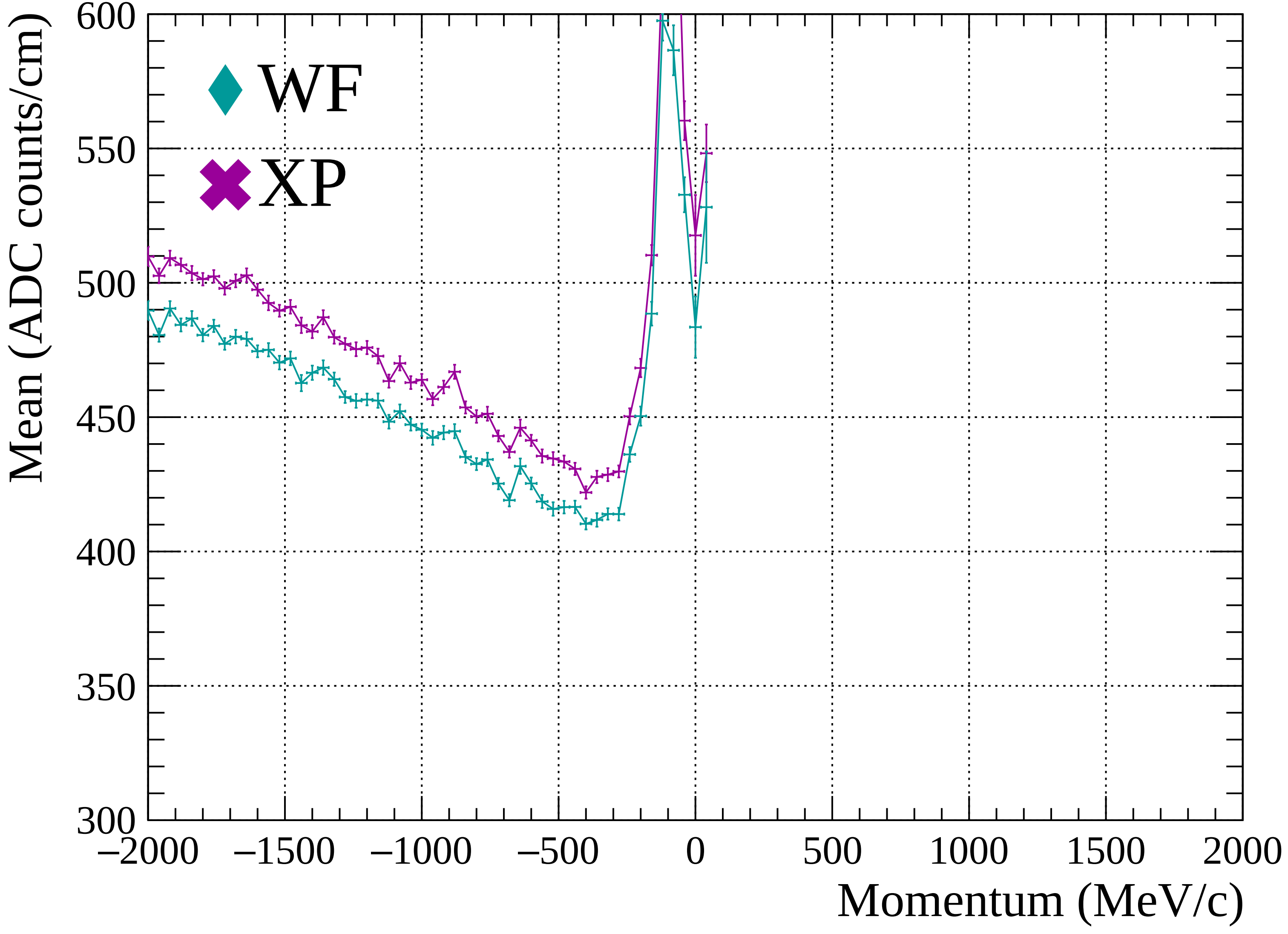


WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800
2000
Momentum (MeV/c)

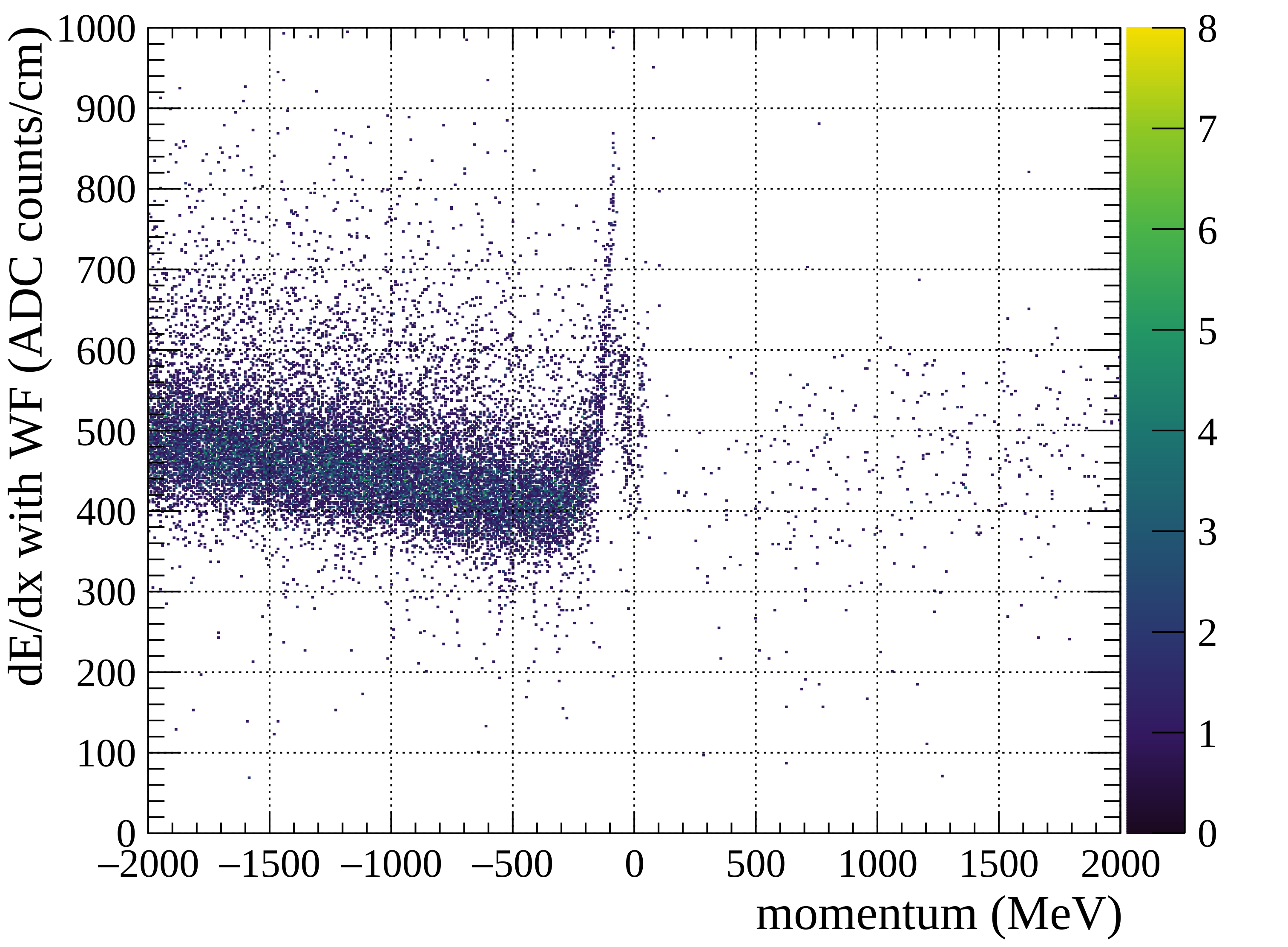

dE/dx with WF (ADC counts/cm)
Absolute momentum (MeV/c)
0
100
200
300
400
500
600
700
800
900
1000
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
2
4
6
8
10
12
14
16



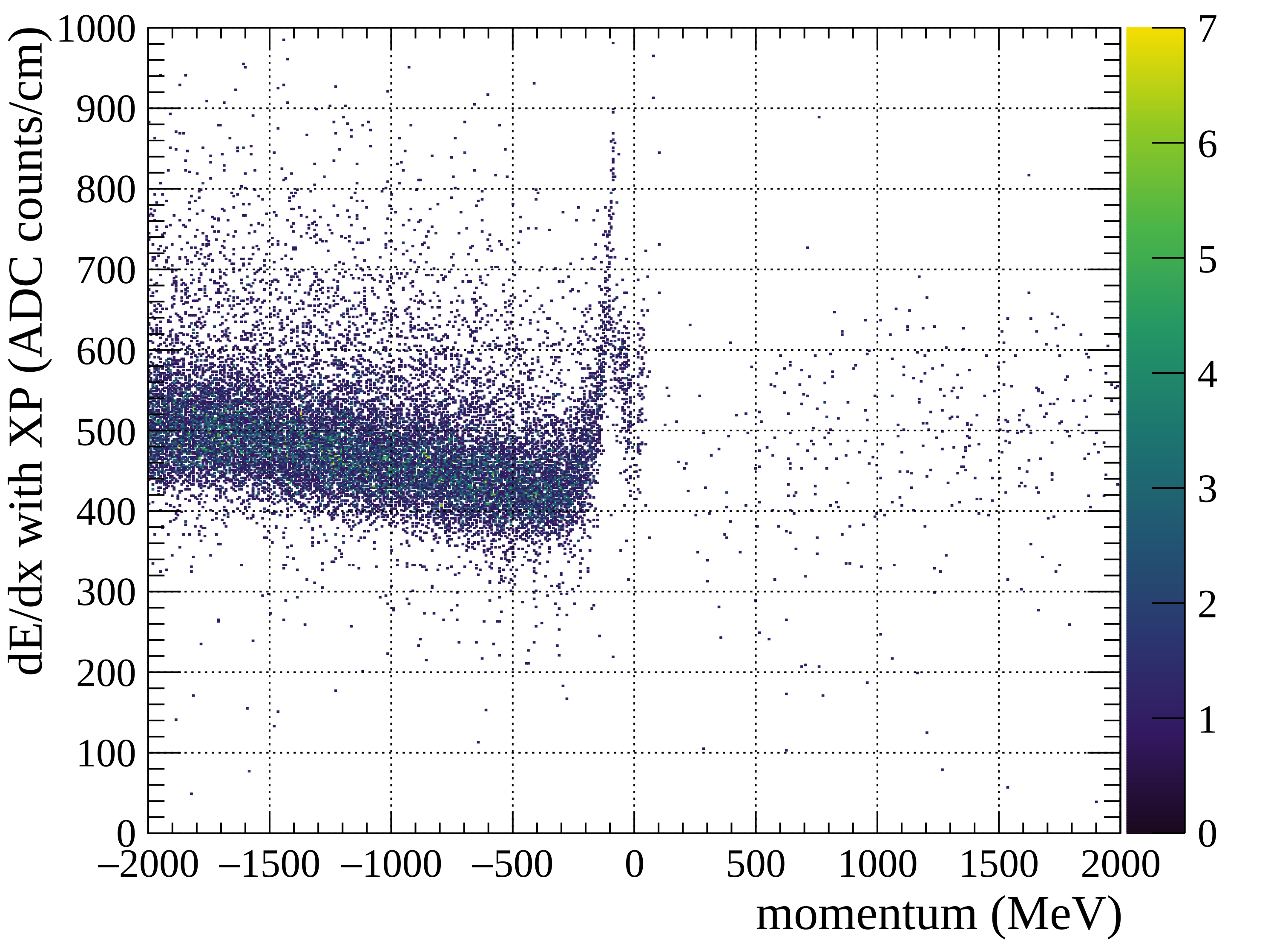
WF
XP
dE/dx resolution (%)
Momentum (MeV/c)
30
25
20
15
10
5
−2000
−1500
−1000
−500
0
500
1000
1500
2000



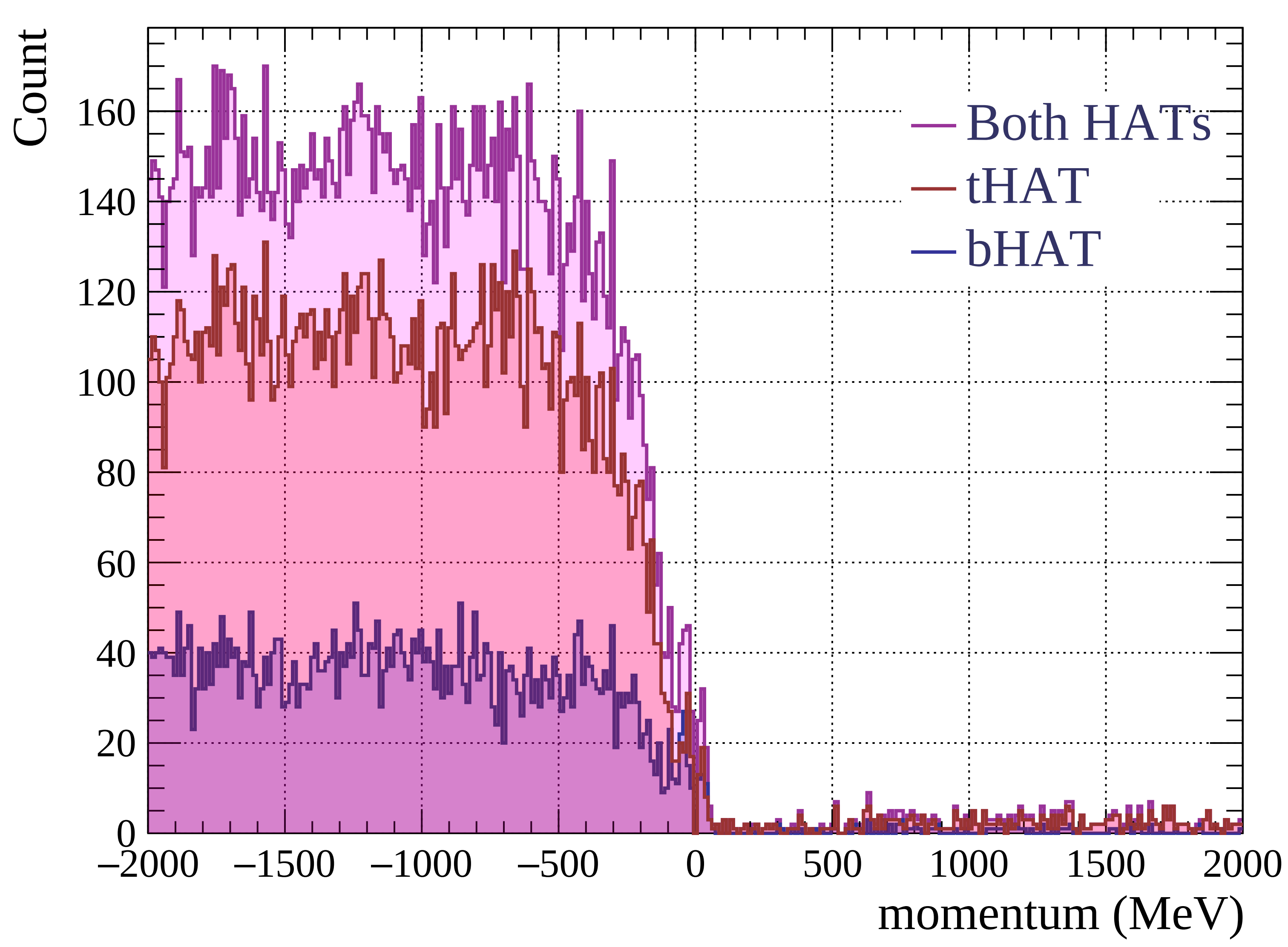
WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
−2000
−1500
−1000
−500
0
500
1000
1500
2000



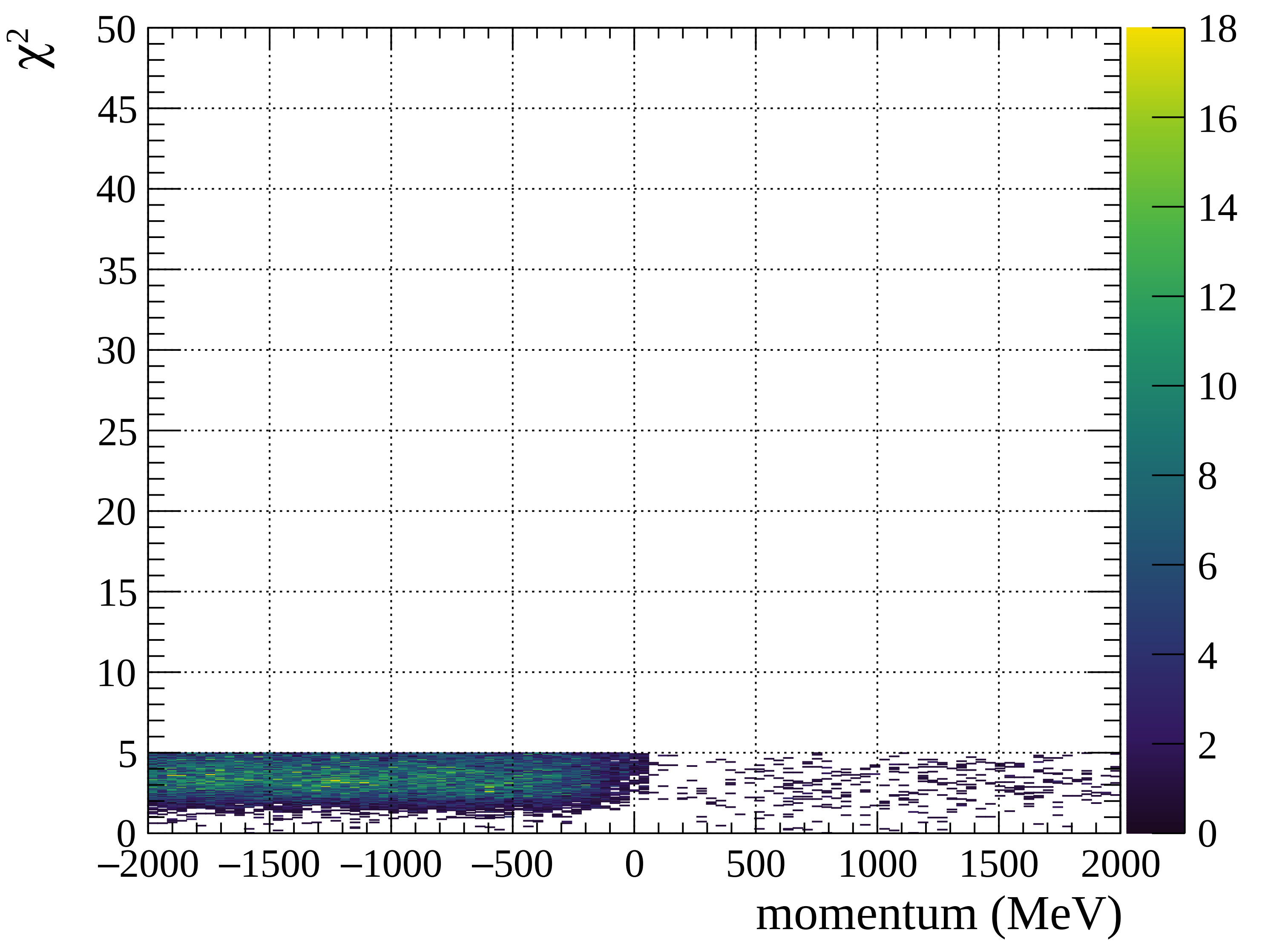
dE/dx with WF (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−2000
−1500
−1000
−500
0
500
1000
1500
2000
0
1
2
3
4
5
6
7
8



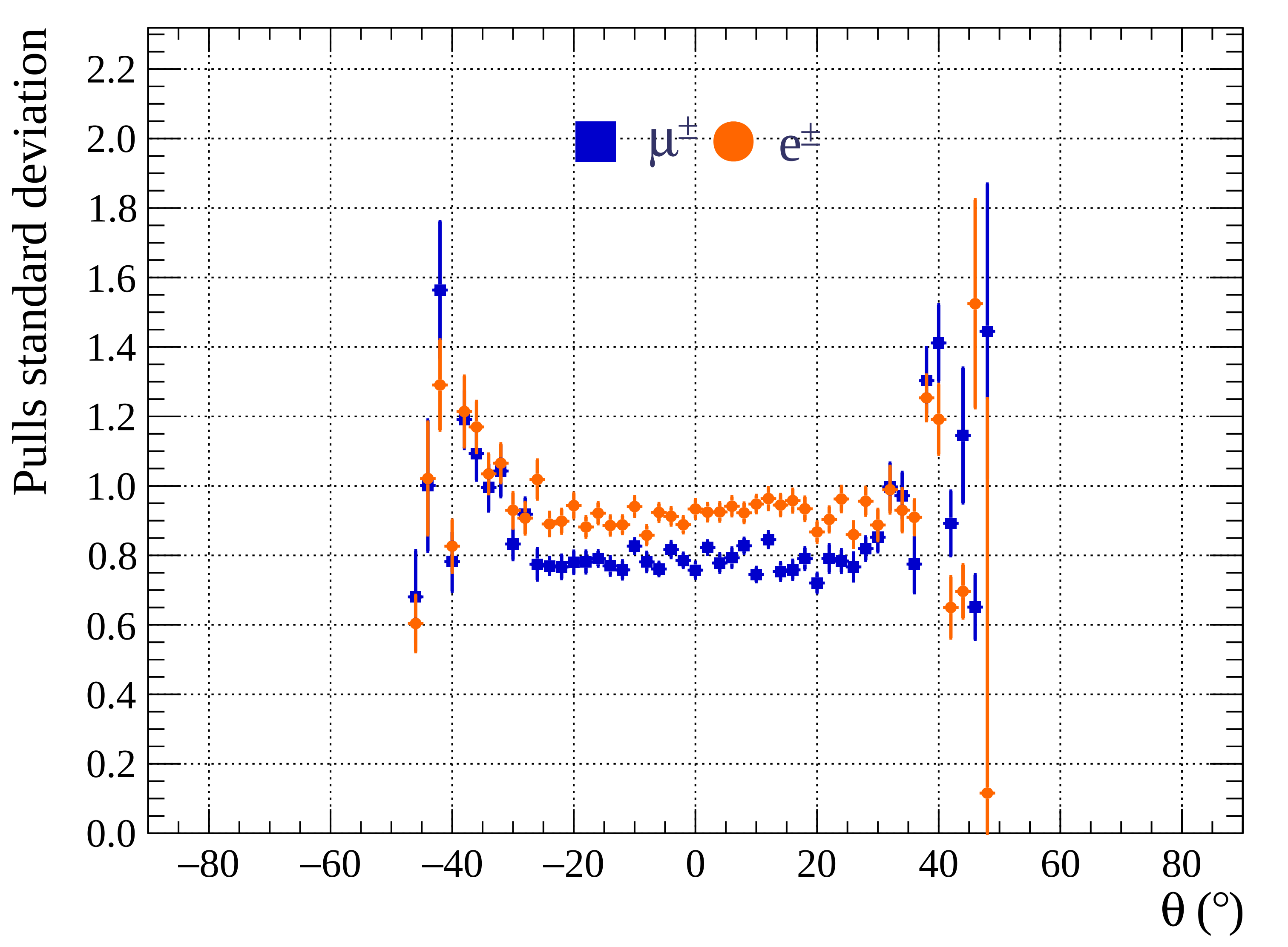
dE/dx with XP (ADC counts/cm)
momentum (MeV)
−2000
−1500
−1000
−500
0
500
1000
1500
2000
0
100
200
300
400
500
600
700
800
900
1000
0
1
2
3
4
5
6
7



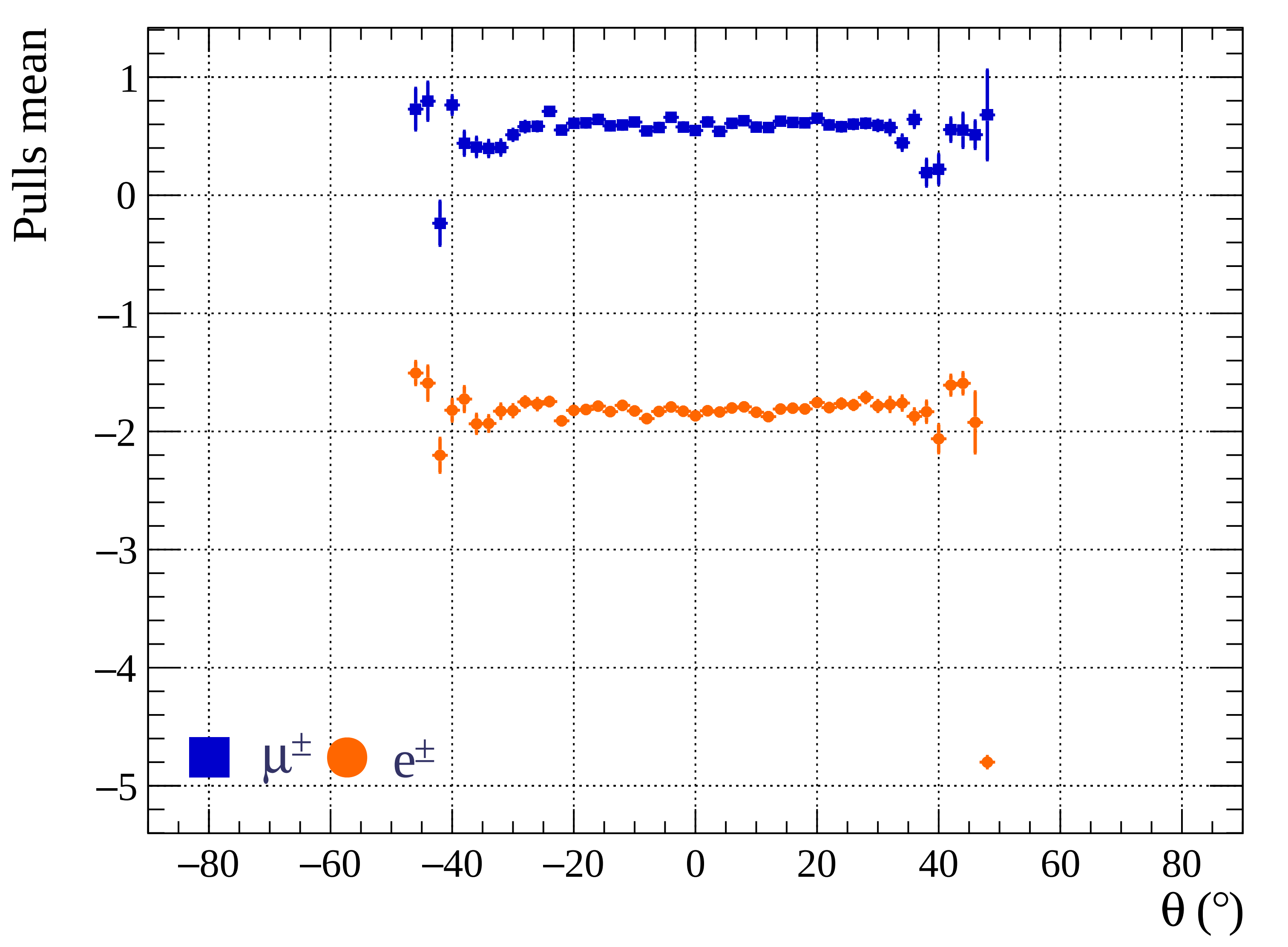
Count
160
140
120
100
80
60
40
20
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
Both HATs
tHAT
bHAT

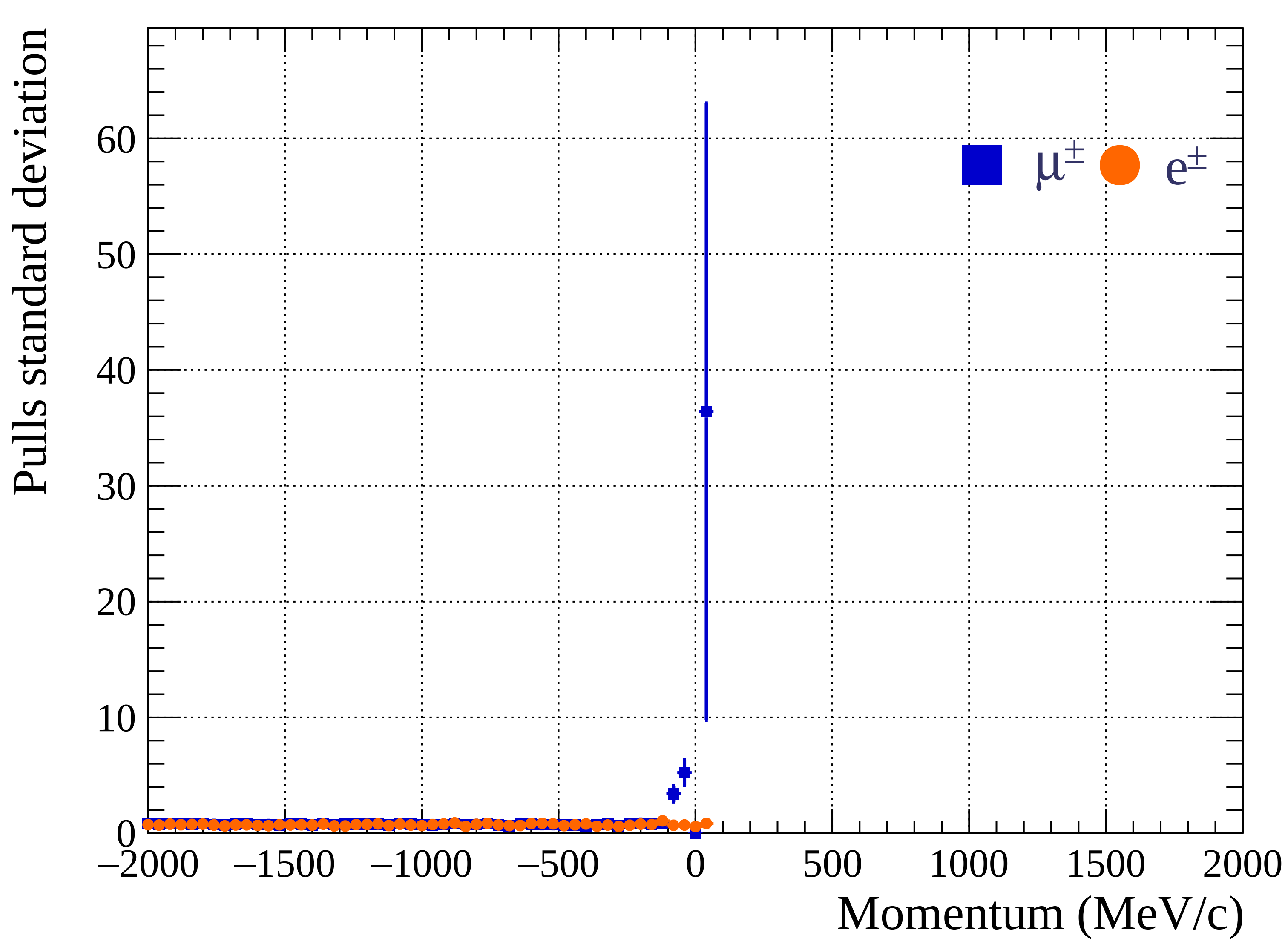


χ^2
50
45
40
35
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
18
16
14
12
10
8
6
4
2
0

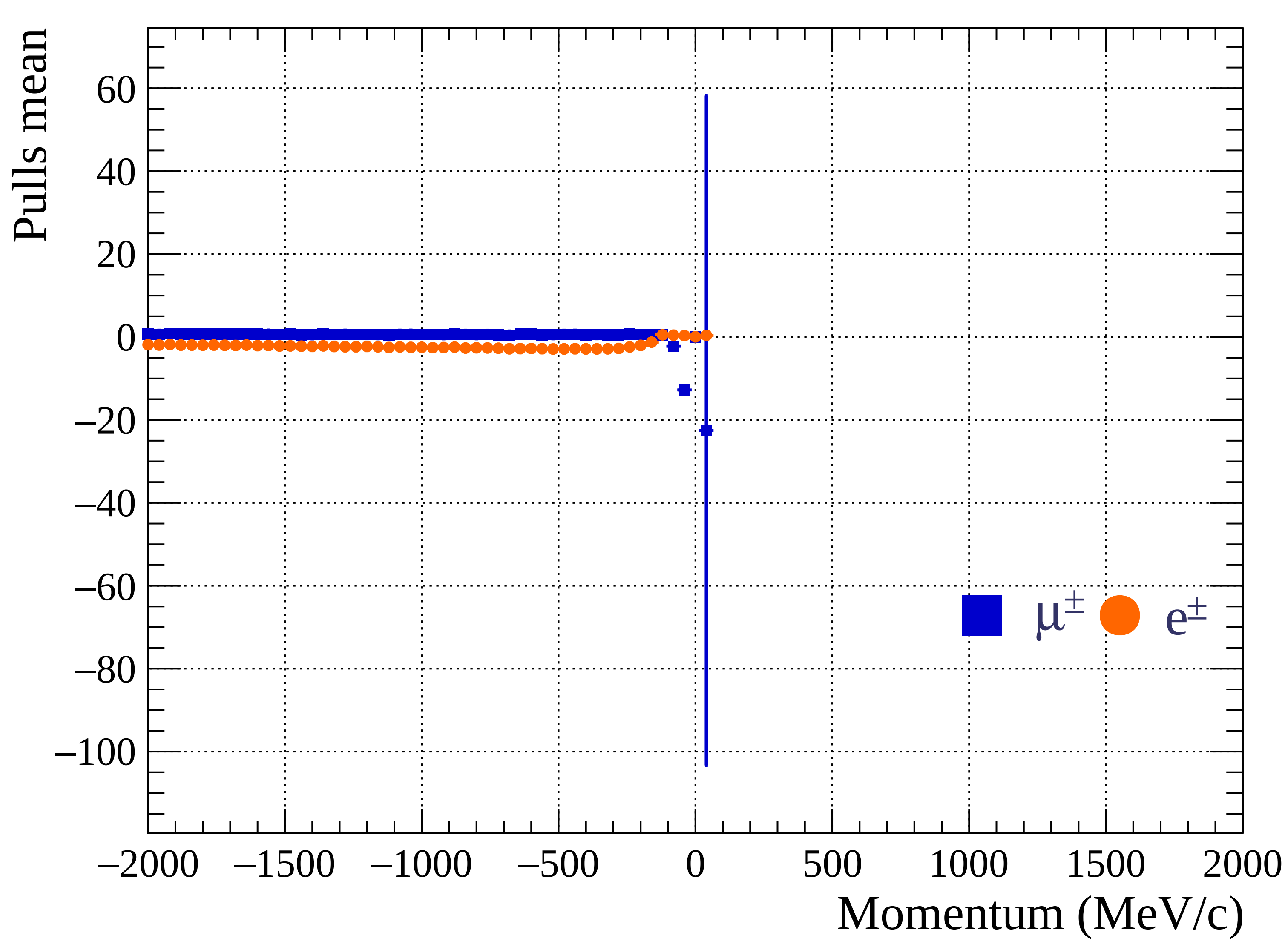


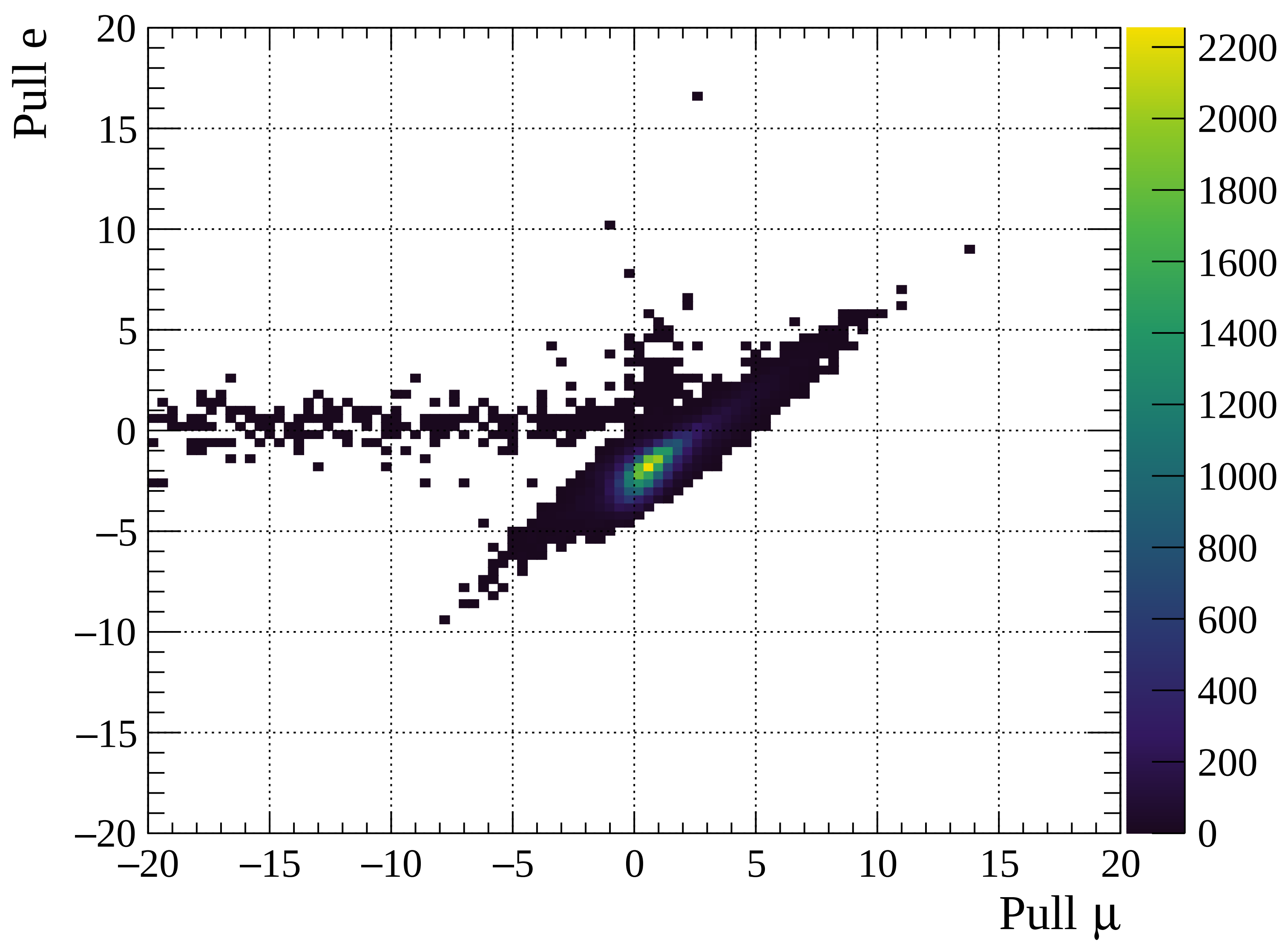
Pulls standard deviation
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±


Pulls mean
1
0
−1
−2
−3
−4
−5
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±

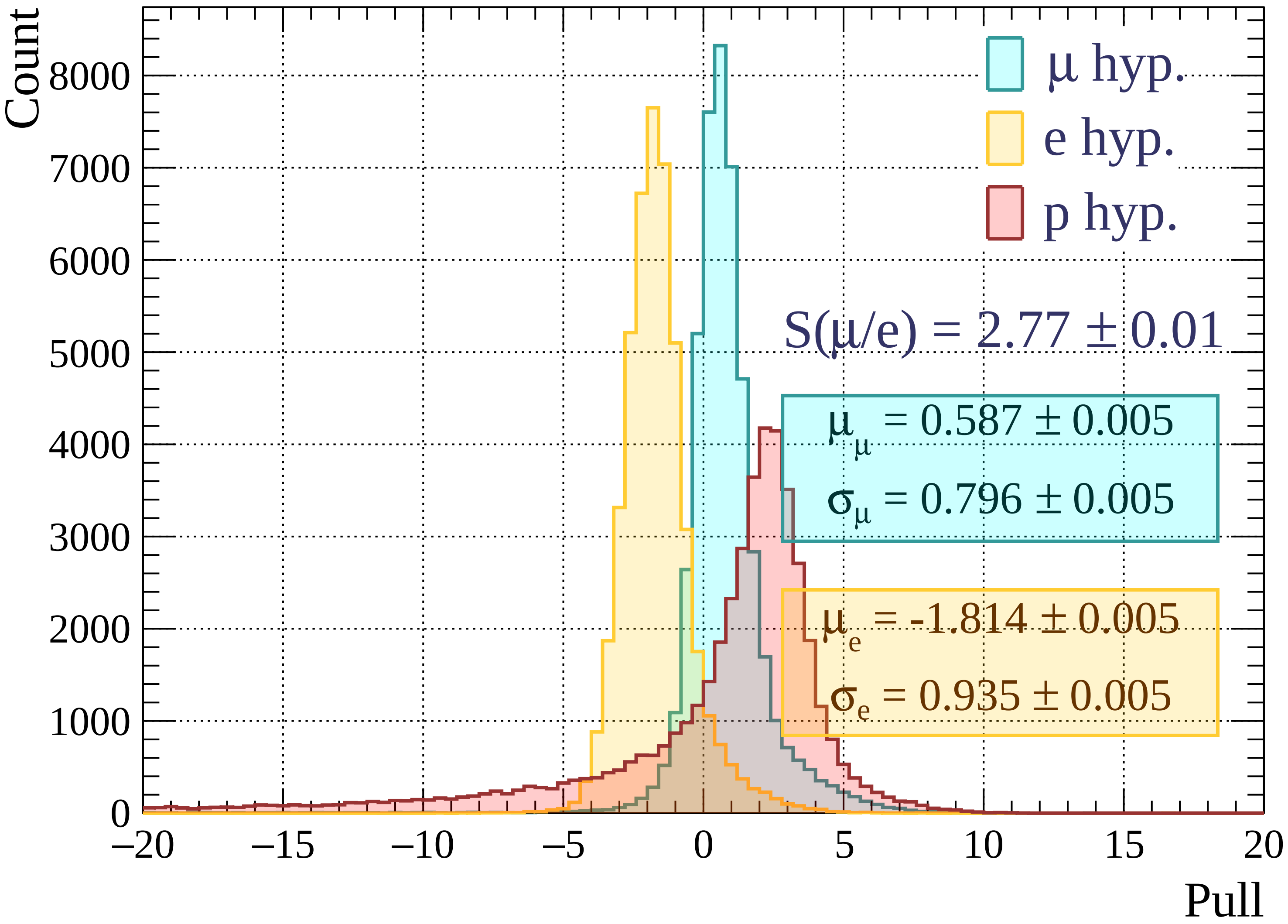


Pulls standard deviation
60
50
40
30
20
10
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$

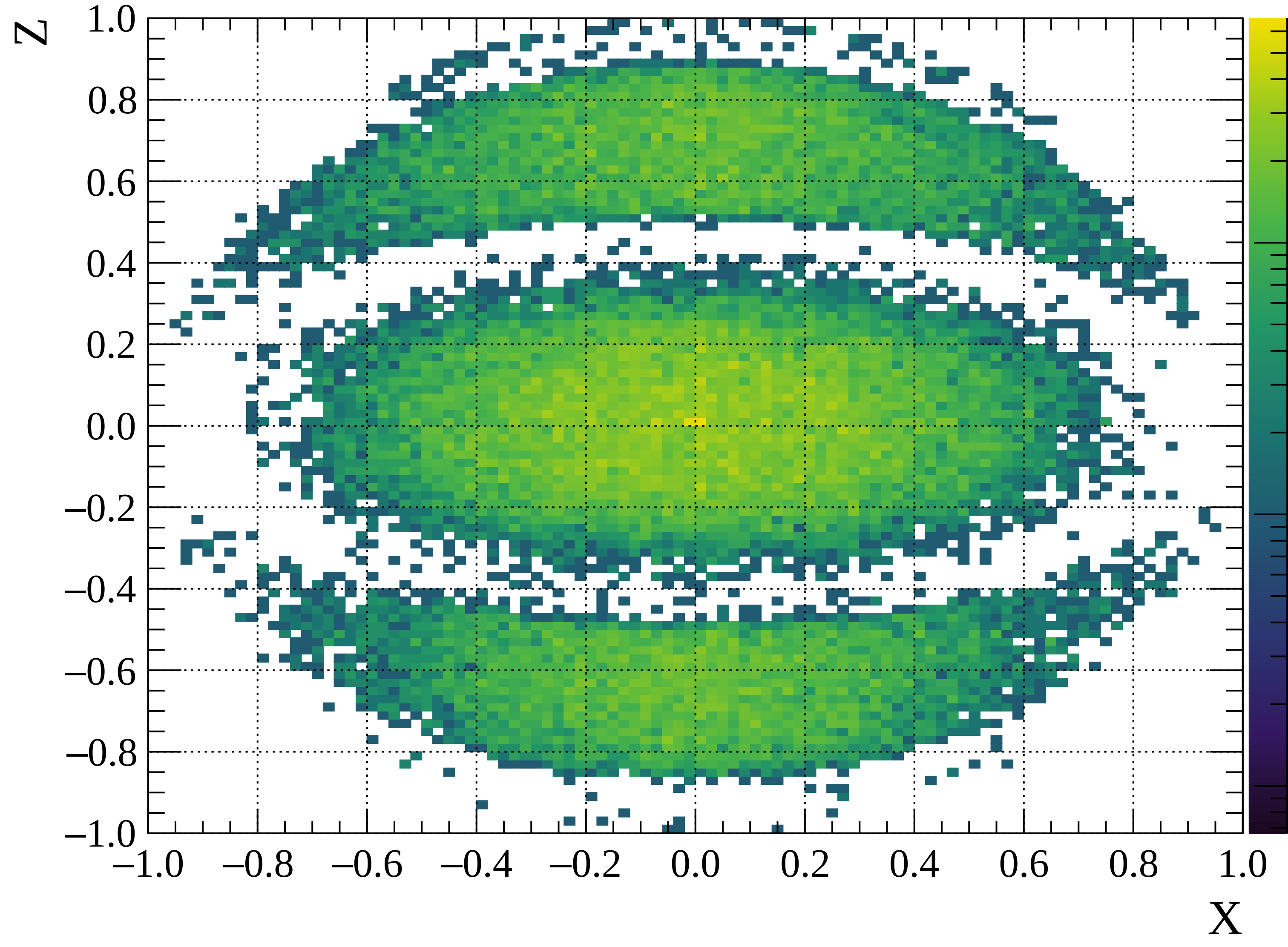

Pulls mean
60
40
20
0
−20
−40
−60
−80
−100
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
μ±
e±



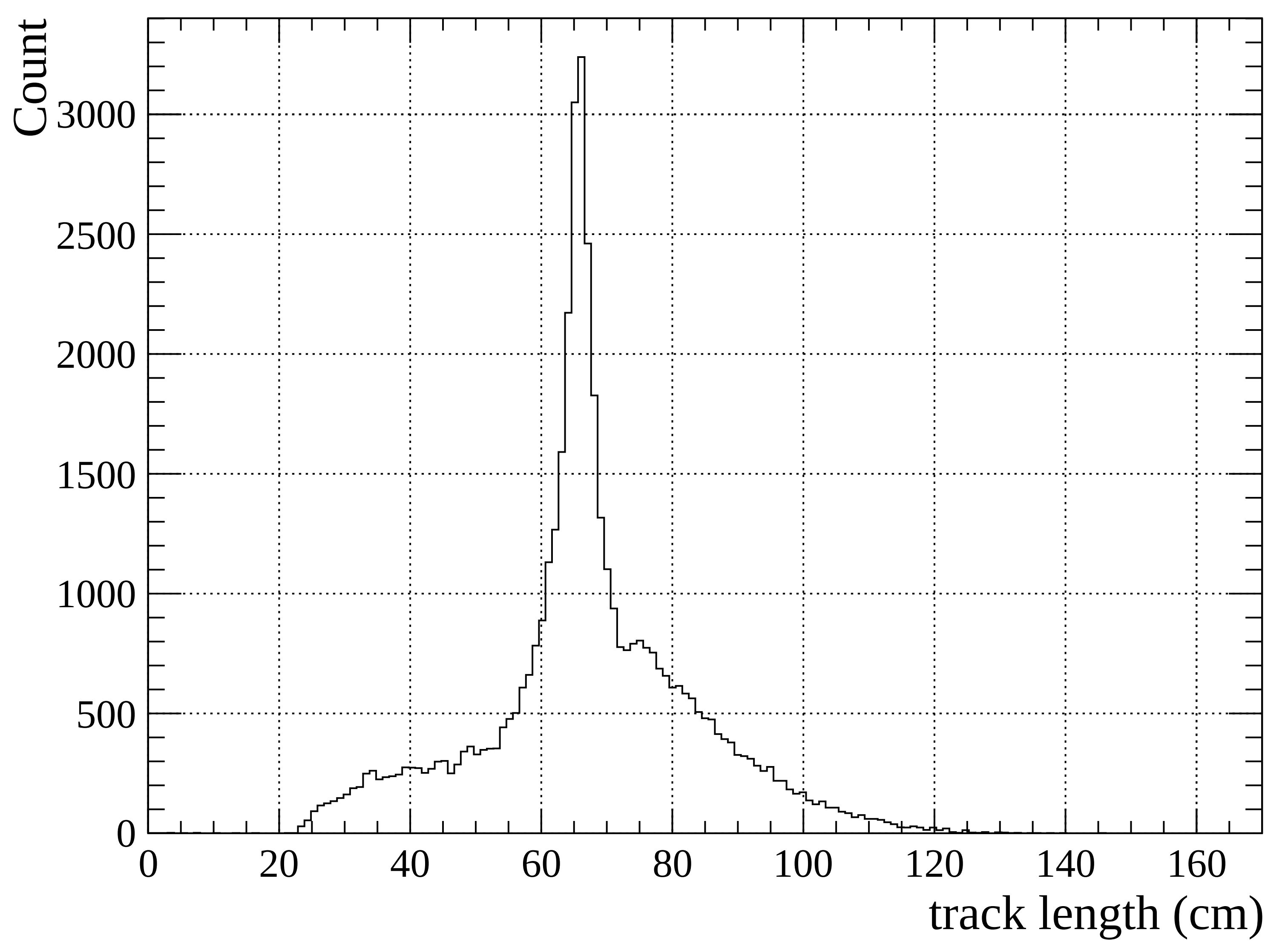
Pull e
Pull μ
20
15
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0



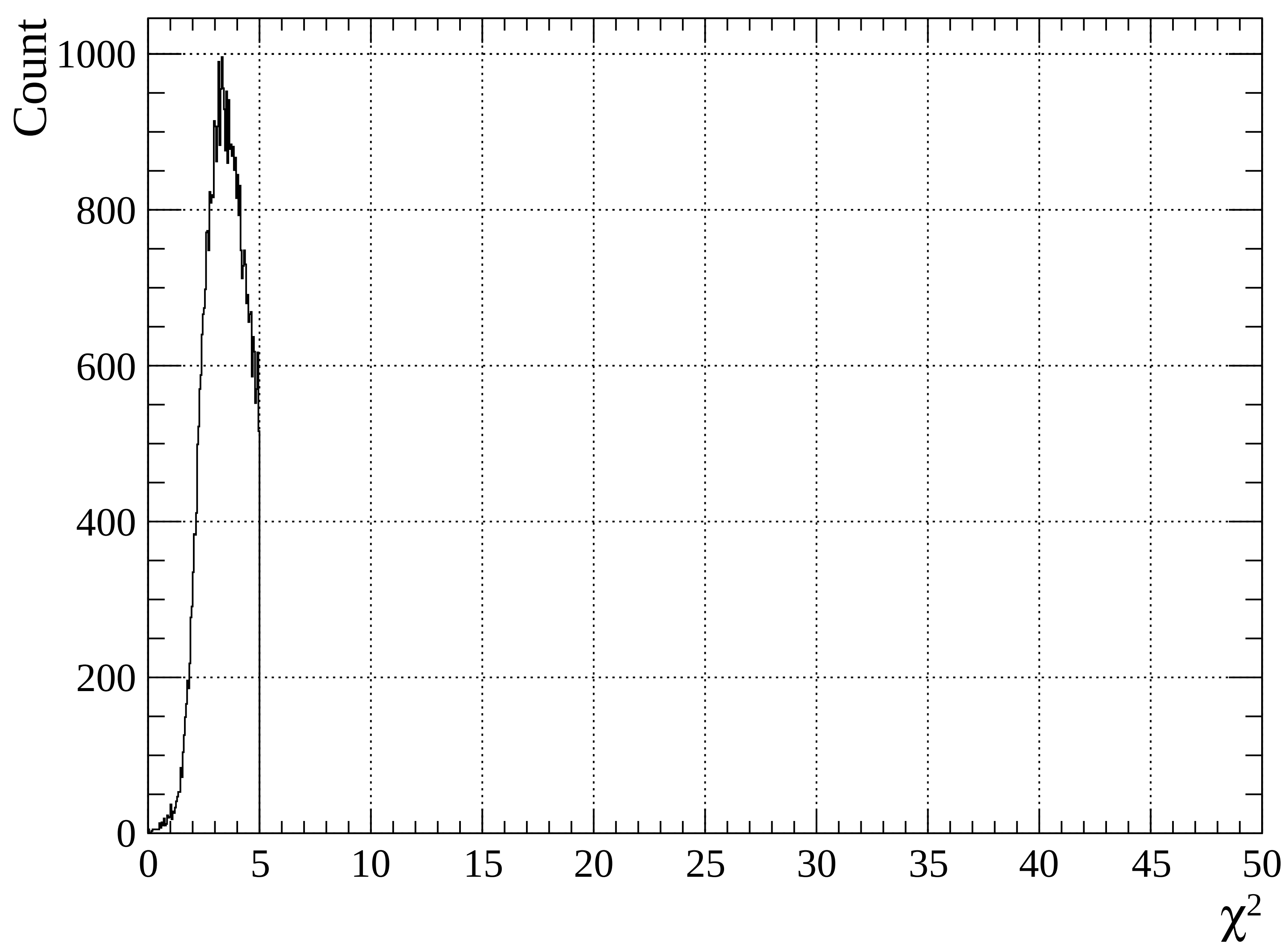
Count
8000
7000
6000
5000
4000
3000
2000
1000
0
−20
−15
−10
−5
0
5
10
15
20
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 2.77 ± 0.01
μ_μ = 0.587 ± 0.005
σ_μ = 0.796 ± 0.005
μ_e = -1.814 ± 0.005
σ_e = 0.935 ± 0.005



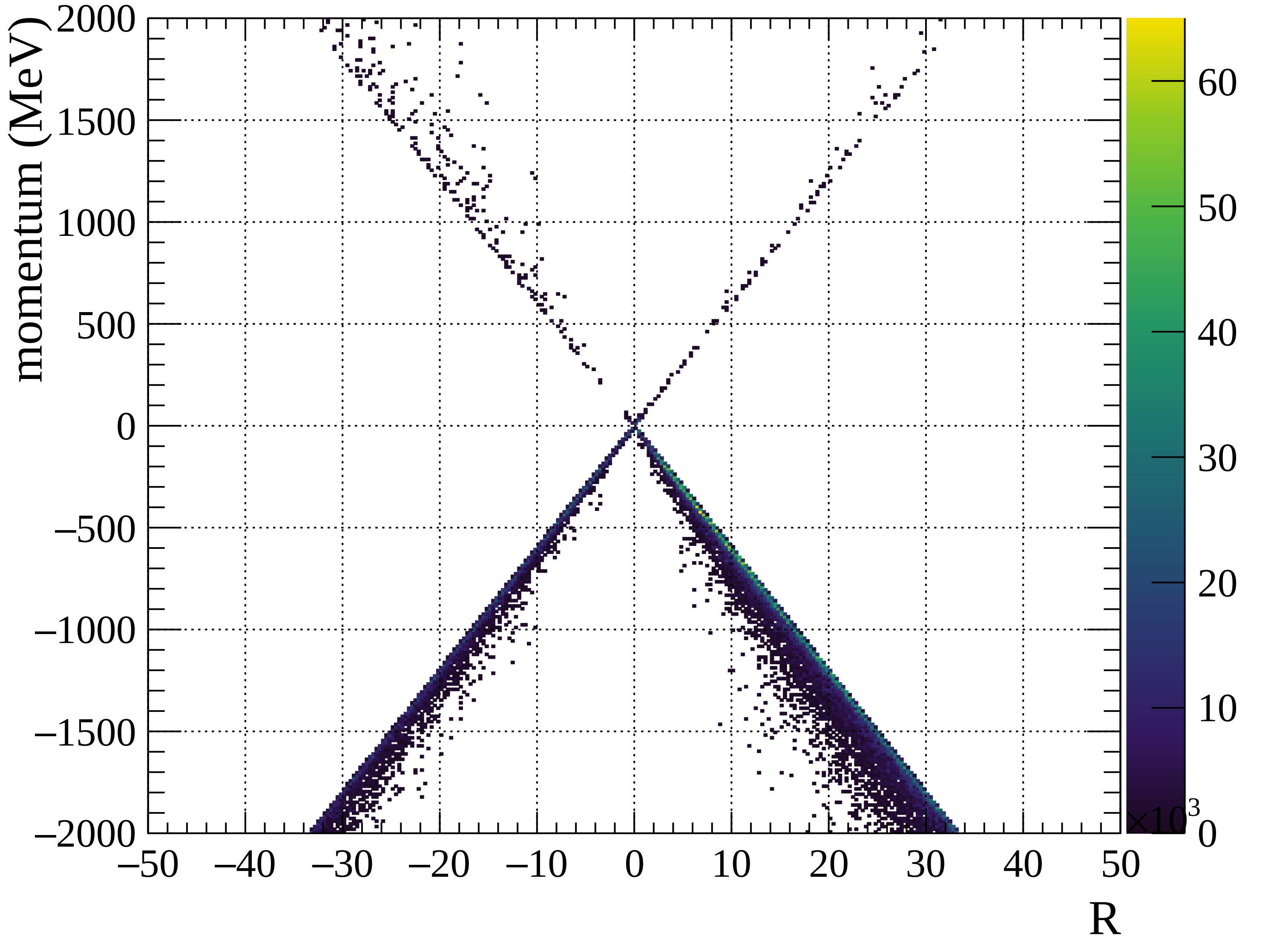
Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X



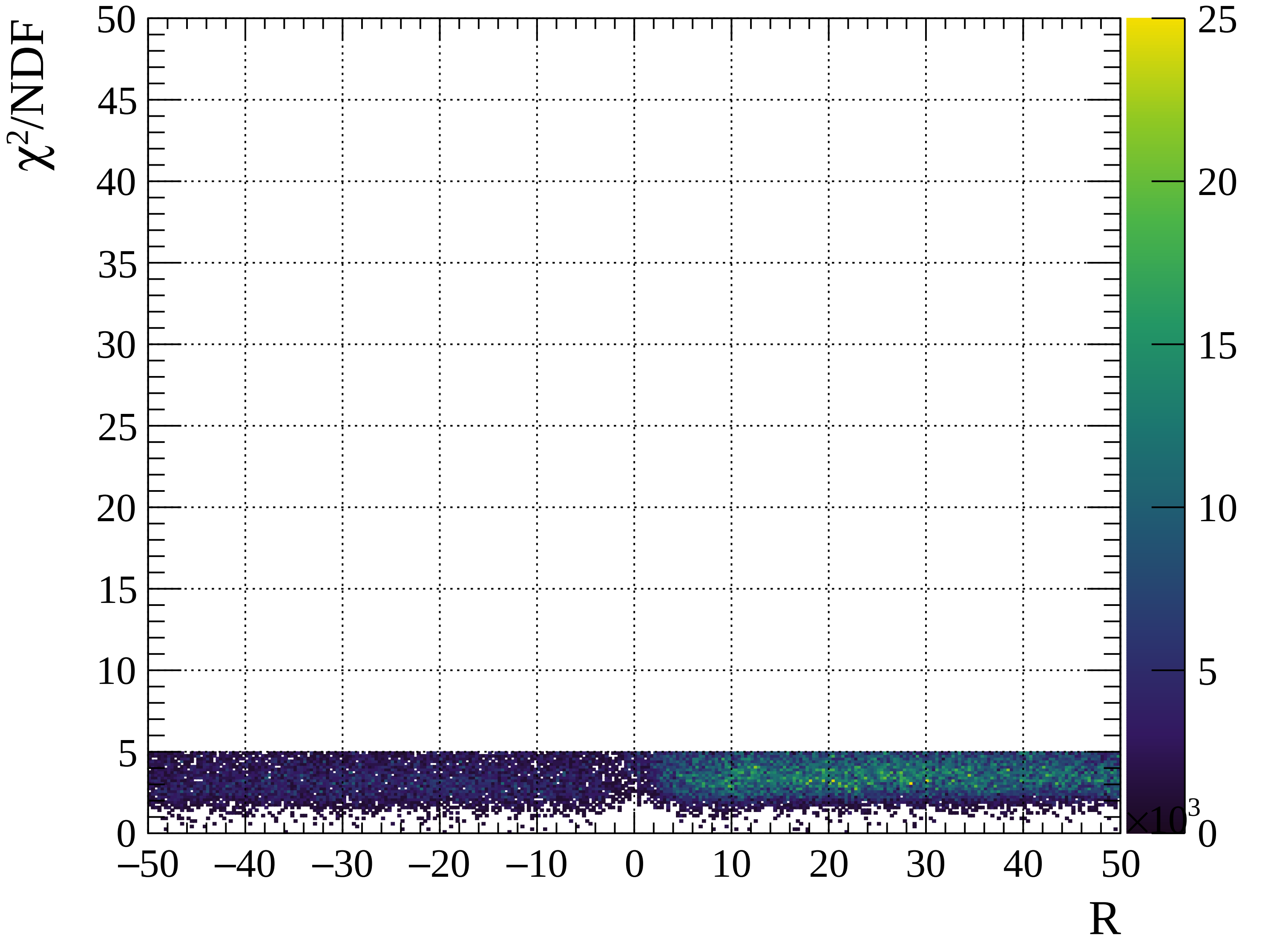
Count
3000
2500
2000
1500
1000
500
0
0
20
40
60
80
100
120
140
160
track length (cm)



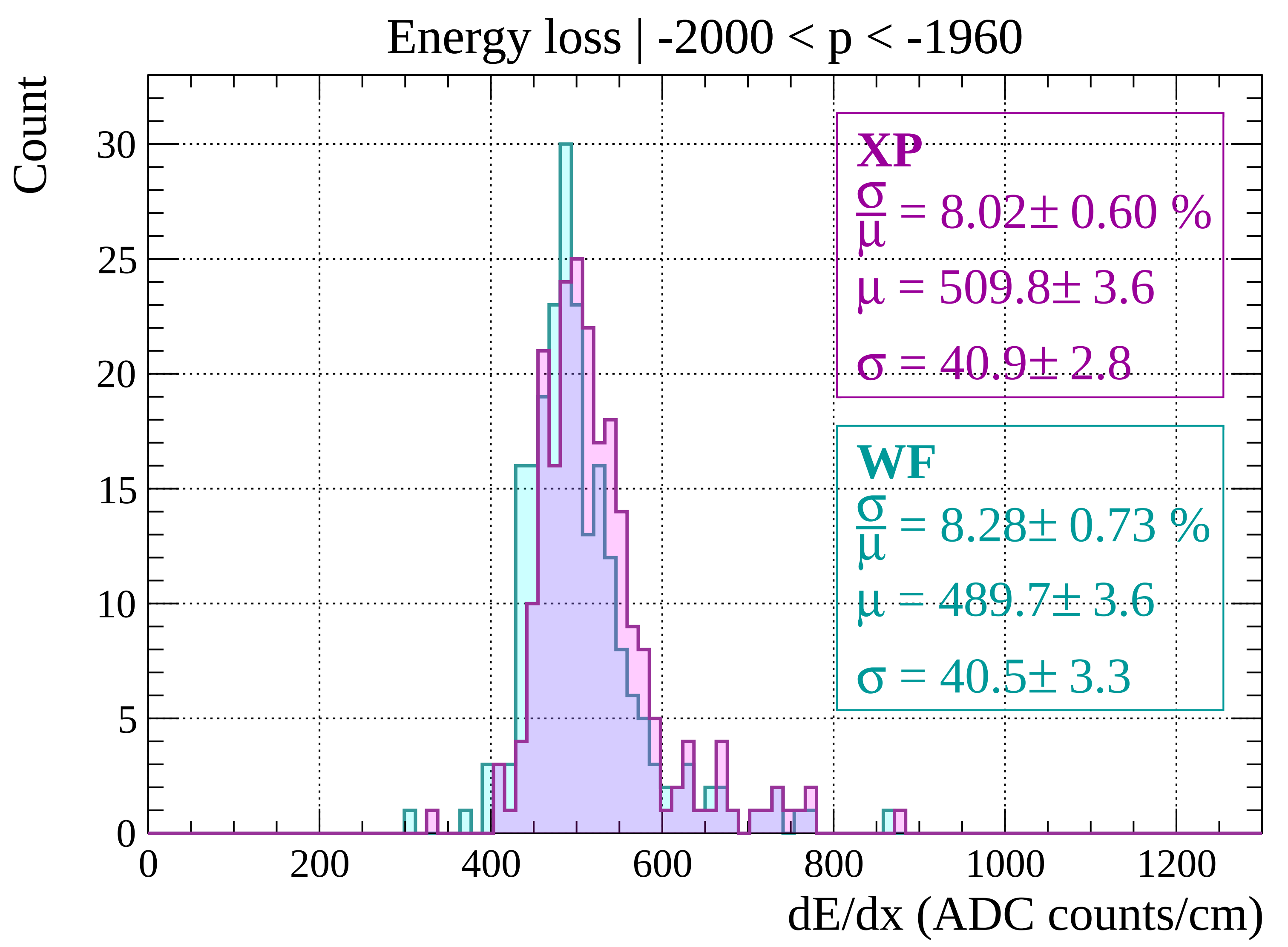
Count
1000
800
600
400
200
0
0
5
10
15
20
25
30
35
40
45
50
χ^2



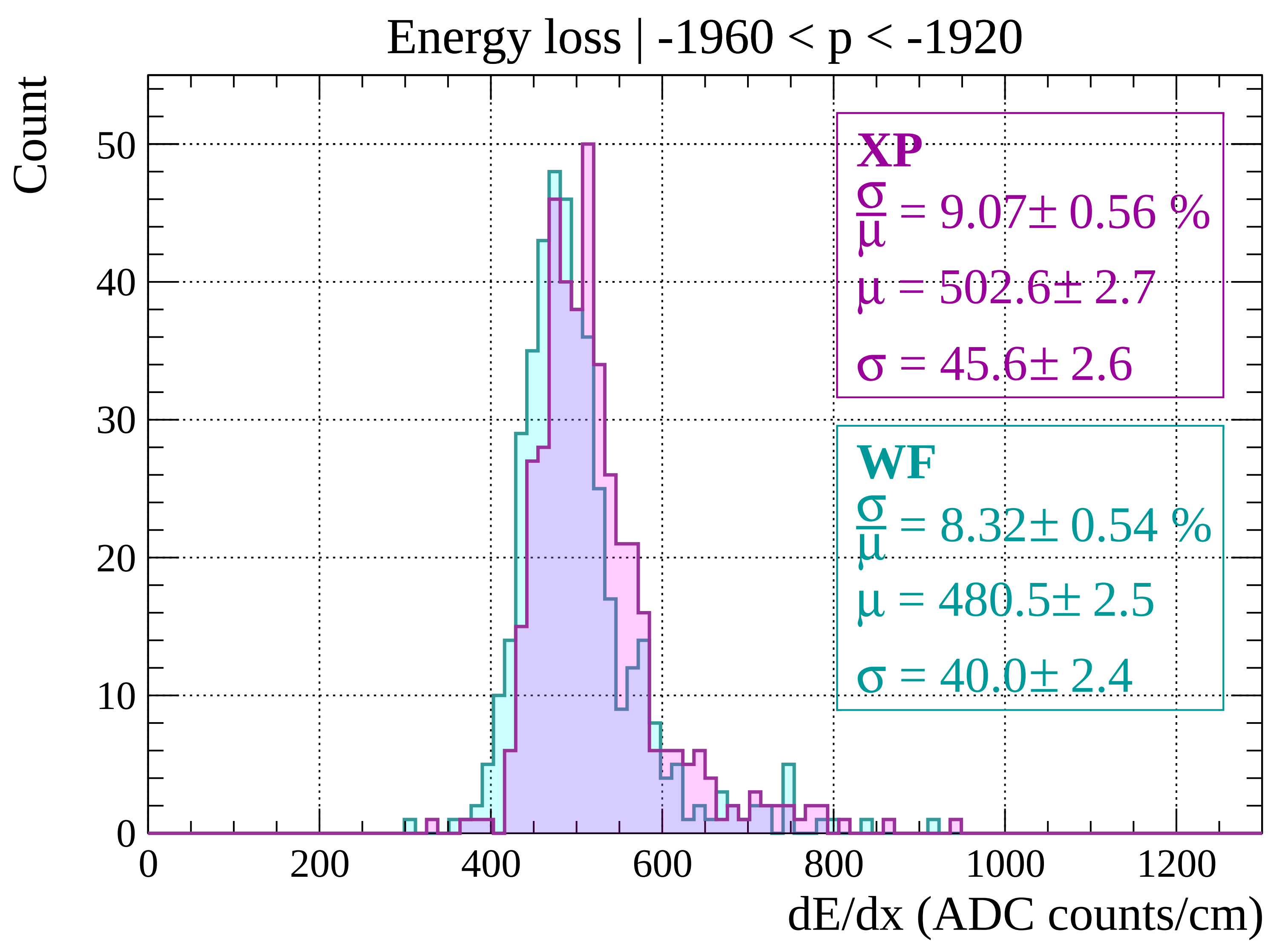
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
60
50
40
30
20
10
0
×10^3



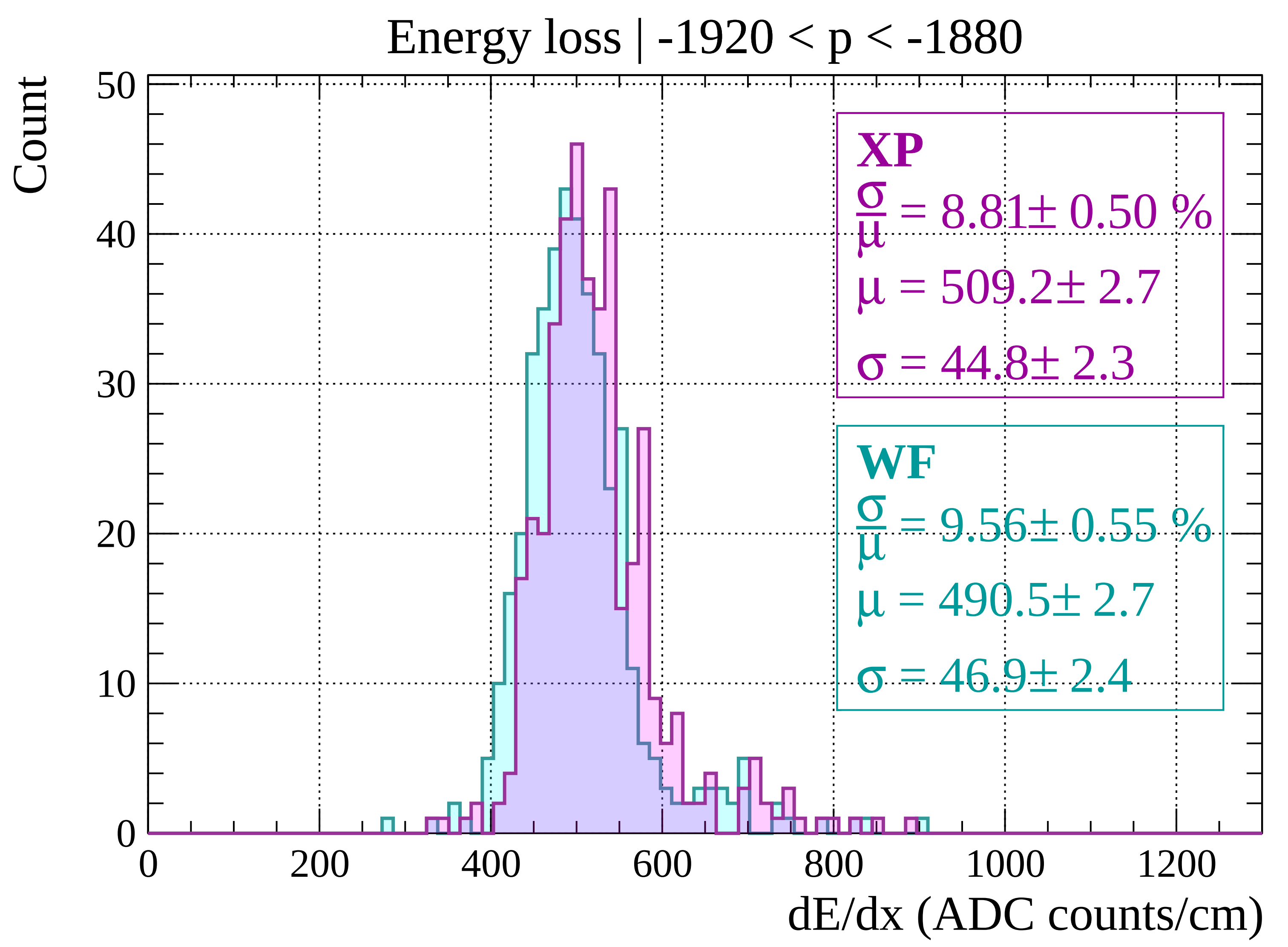
χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
25
20
15
10
5
0
$\times 10^3$



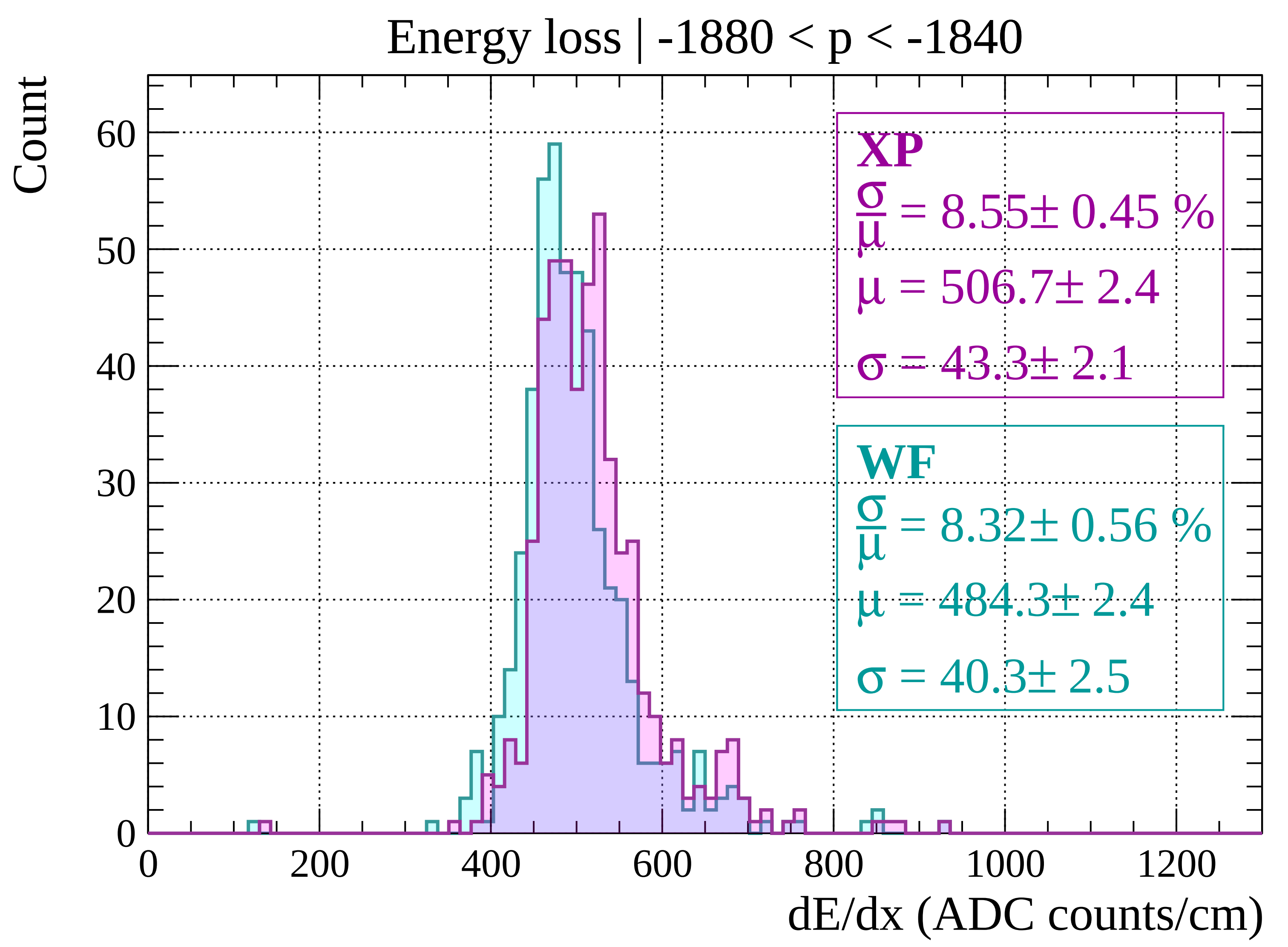
Energy loss | -2000 < p < -1960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
XP
σ/μ = 8.02± 0.60 %
μ = 509.8± 3.6
σ = 40.9± 2.8
WF
σ/μ = 8.28± 0.73 %
μ = 489.7± 3.6
σ = 40.5± 3.3



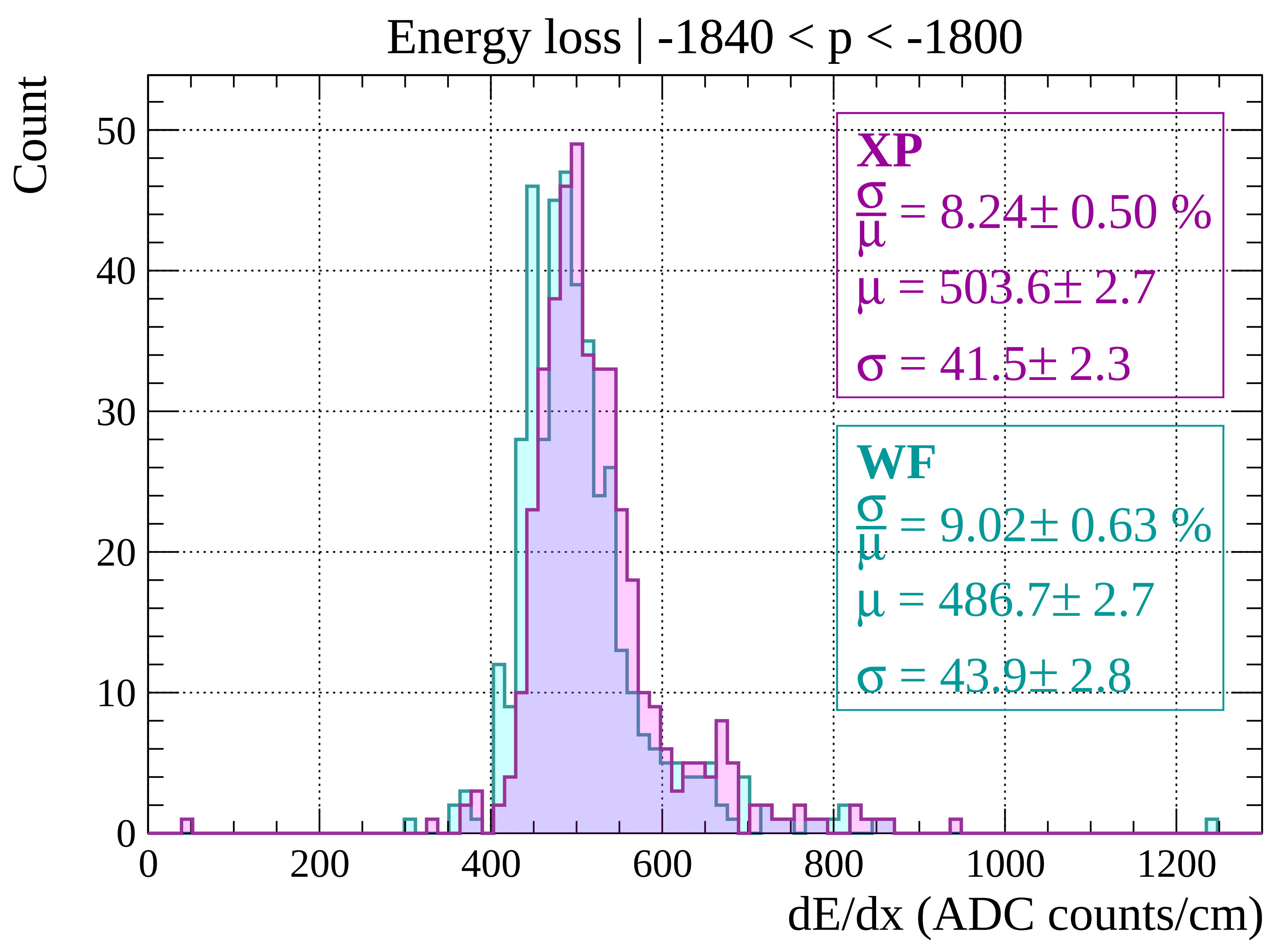
Energy loss | -1960 < p < -1920
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 9.07± 0.56 %
μ = 502.6± 2.7
σ = 45.6± 2.6
WF
σ/μ = 8.32± 0.54 %
μ = 480.5± 2.5
σ = 40.0± 2.4



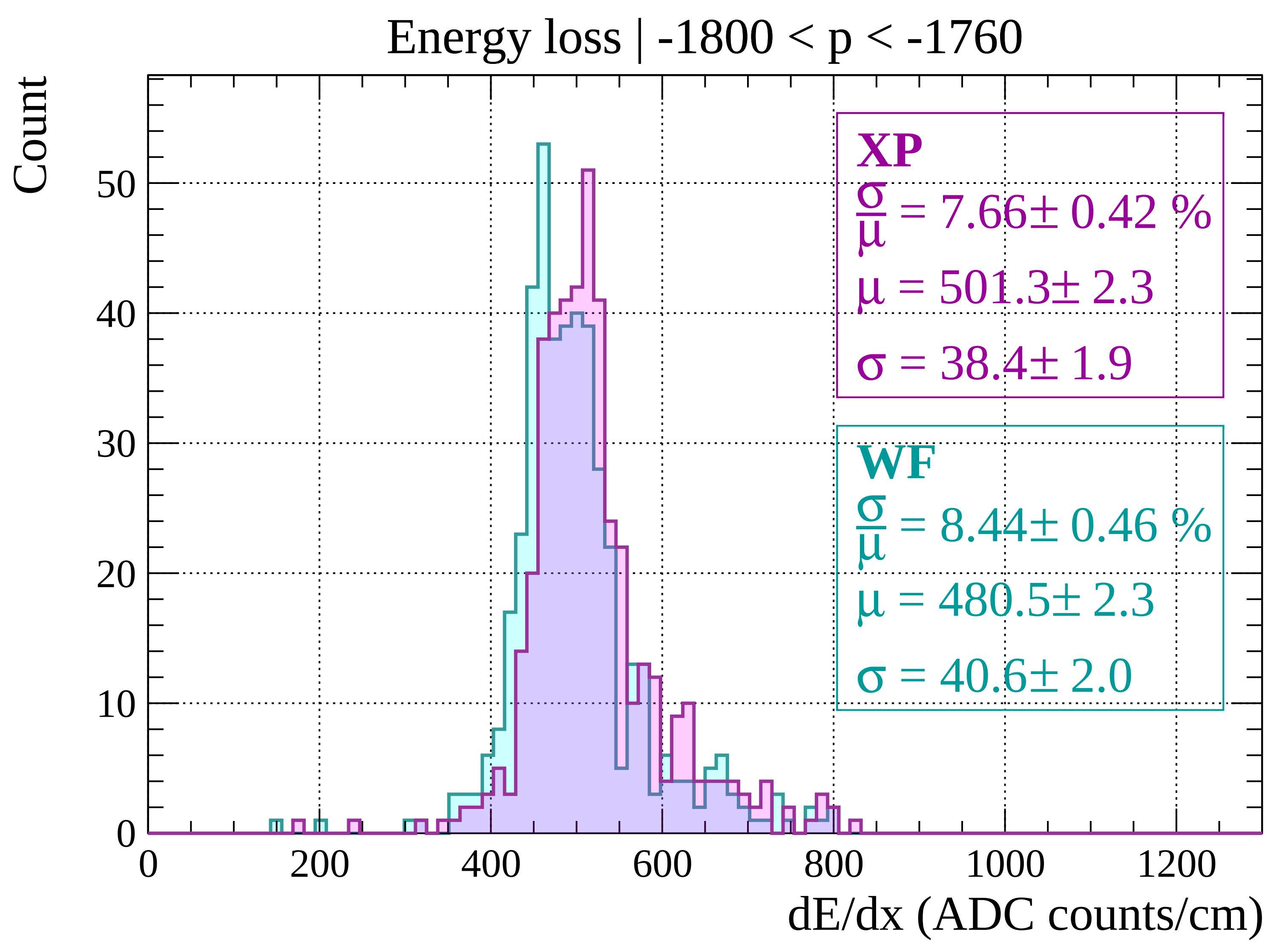
Energy loss | -1920 < p < -1880
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 8.81± 0.50 %
μ = 509.2± 2.7
σ = 44.8± 2.3
WF
σ/μ = 9.56± 0.55 %
μ = 490.5± 2.7
σ = 46.9± 2.4



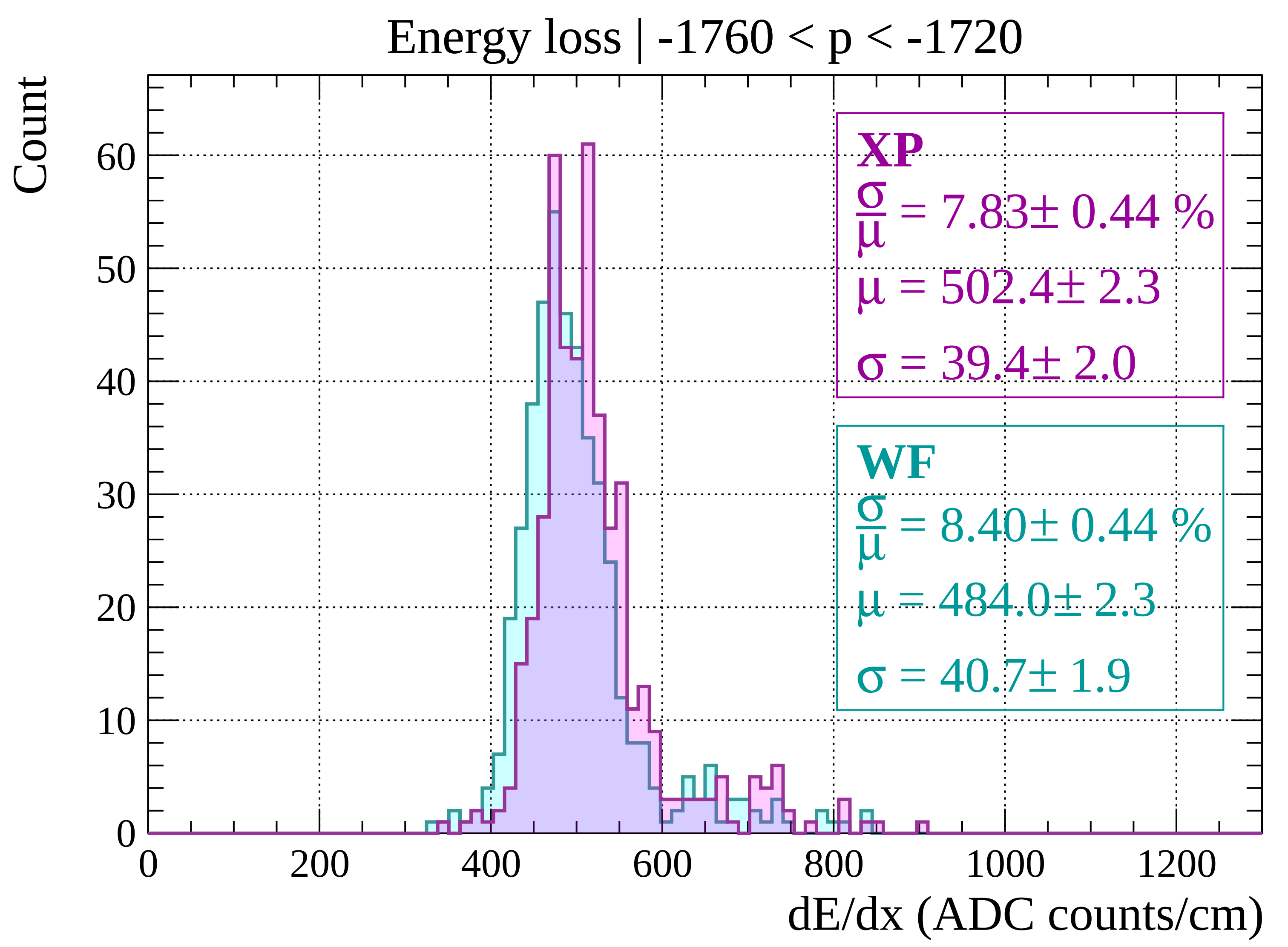
Energy loss | -1880 < p < -1840
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 8.55± 0.45 %
μ = 506.7± 2.4
σ = 43.3± 2.1
WF
σ/μ = 8.32± 0.56 %
μ = 484.3± 2.4
σ = 40.3± 2.5



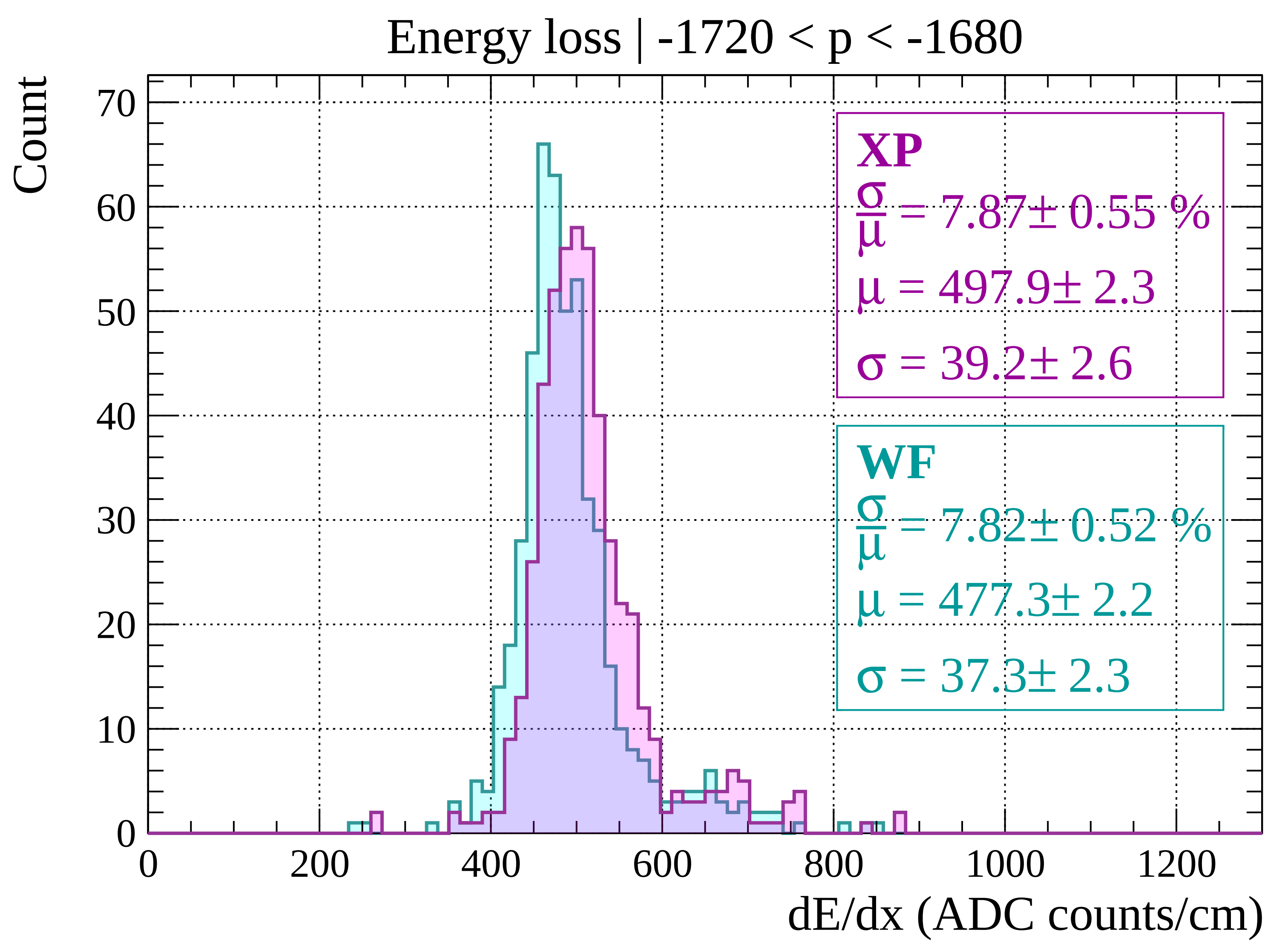
Energy loss | -1840 < p < -1800
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
XP
σ/μ = 8.24± 0.50 %
μ = 503.6± 2.7
σ = 41.5± 2.3
WF
σ/μ = 9.02± 0.63 %
μ = 486.7± 2.7
σ = 43.9± 2.8



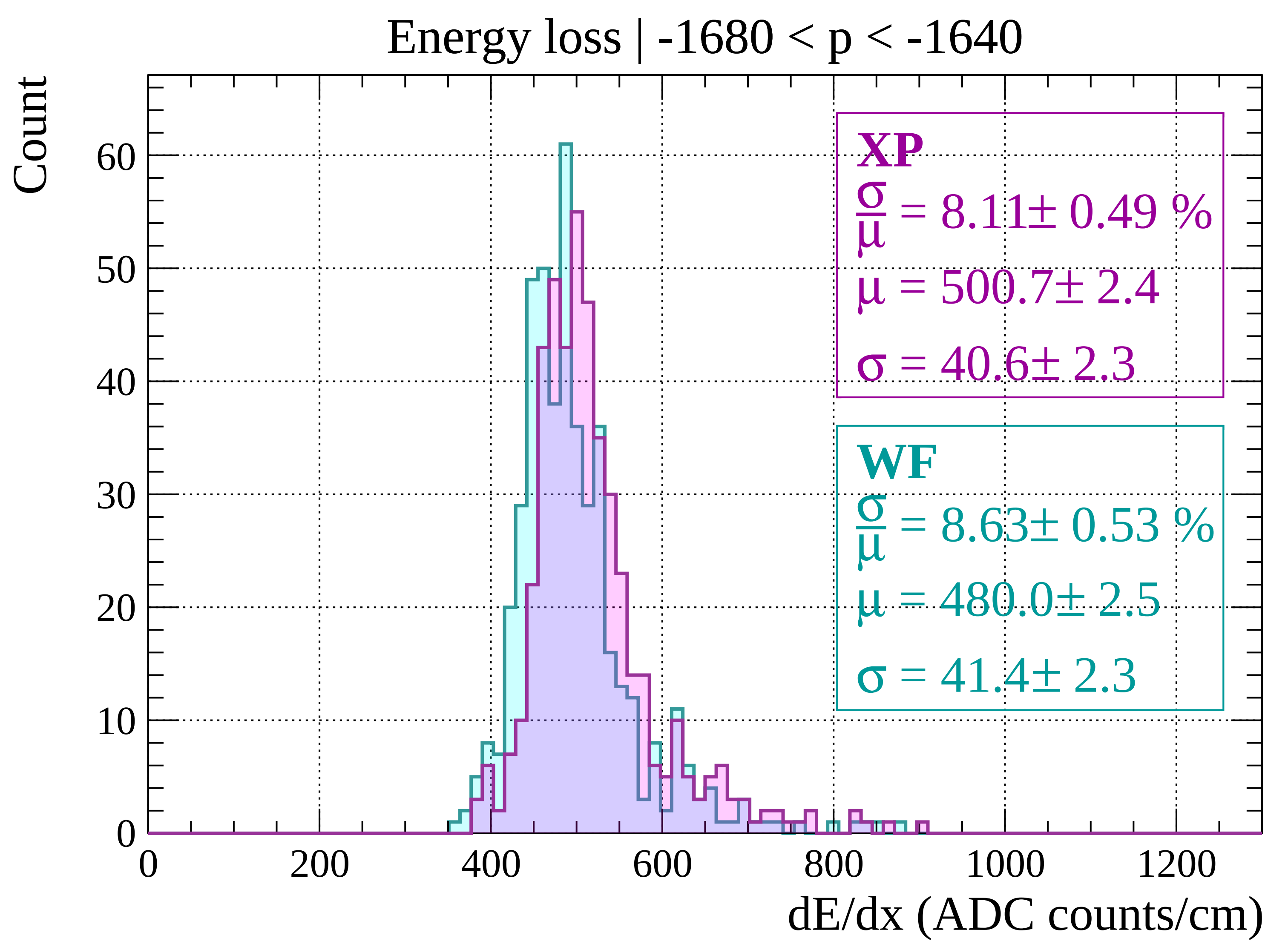
Energy loss | -1800 < p < -1760
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
XP
$\frac{\sigma}{\mu}$ = 7.66± 0.42 %
μ = 501.3± 2.3
σ = 38.4± 1.9
WF
$\frac{\sigma}{\mu}$ = 8.44± 0.46 %
μ = 480.5± 2.3
σ = 40.6± 2.0



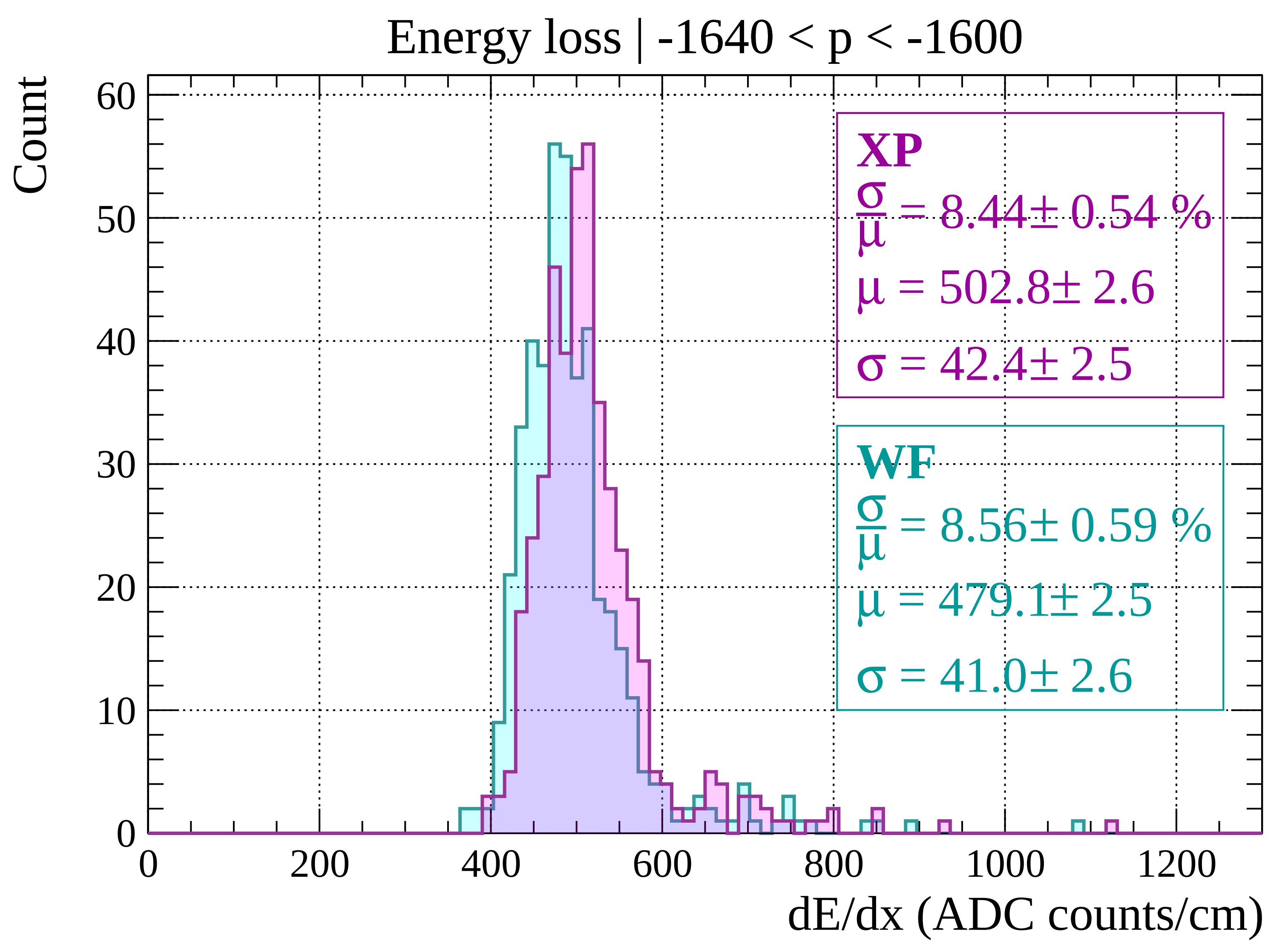
Energy loss | -1760 < p < -1720
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.83± 0.44 %
μ = 502.4± 2.3
σ = 39.4± 2.0
WF
σ/μ = 8.40± 0.44 %
μ = 484.0± 2.3
σ = 40.7± 1.9



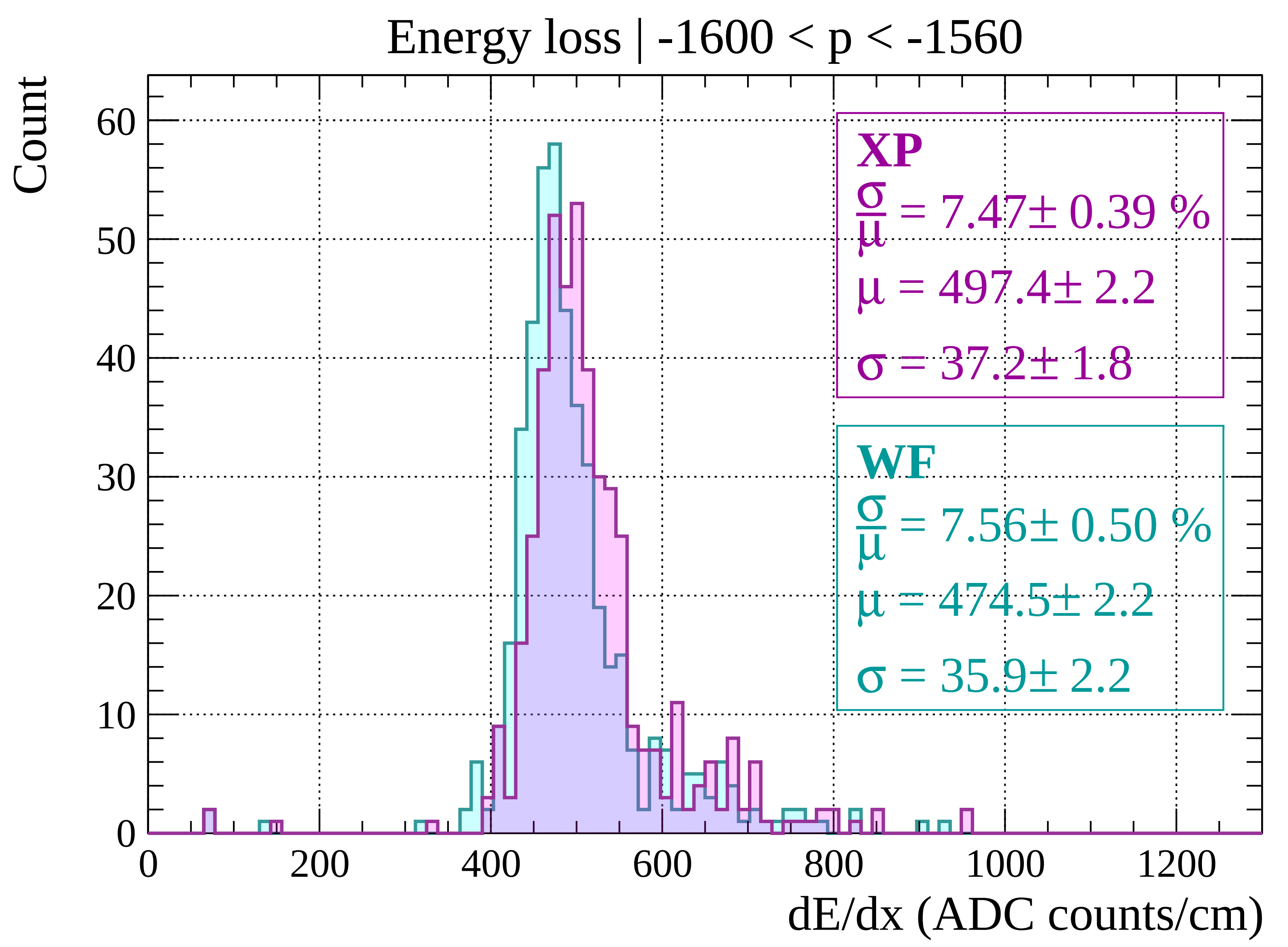
Energy loss | -1720 < p < -1680
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
XP
σ/μ = 7.87± 0.55 %
μ = 497.9± 2.3
σ = 39.2± 2.6
WF
σ/μ = 7.82± 0.52 %
μ = 477.3± 2.2
σ = 37.3± 2.3



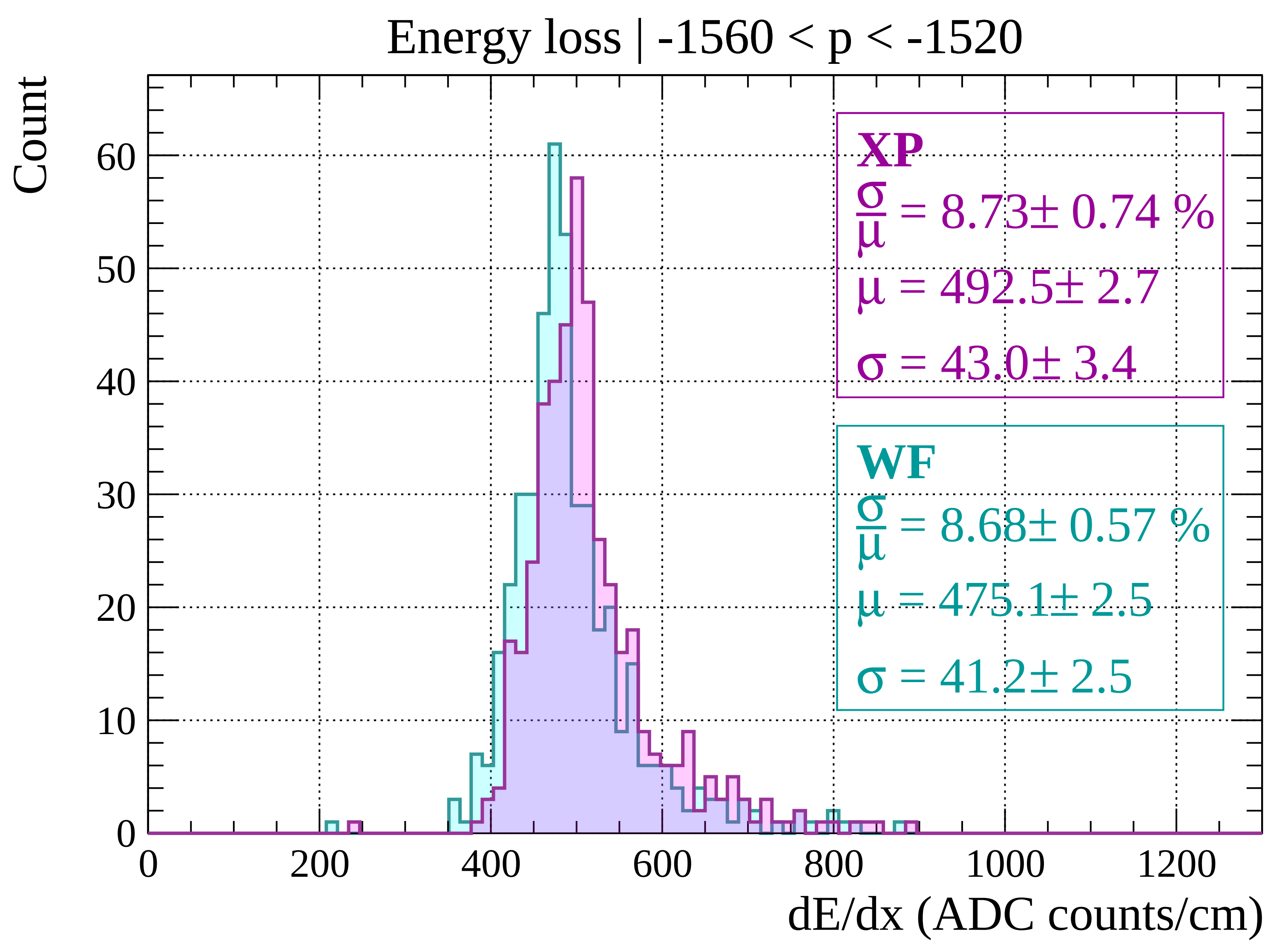
Energy loss | -1680 < p < -1640
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.11± 0.49 %
μ = 500.7± 2.4
σ = 40.6± 2.3
WF
$\frac{\sigma}{\mu}$ = 8.63± 0.53 %
μ = 480.0± 2.5
σ = 41.4± 2.3



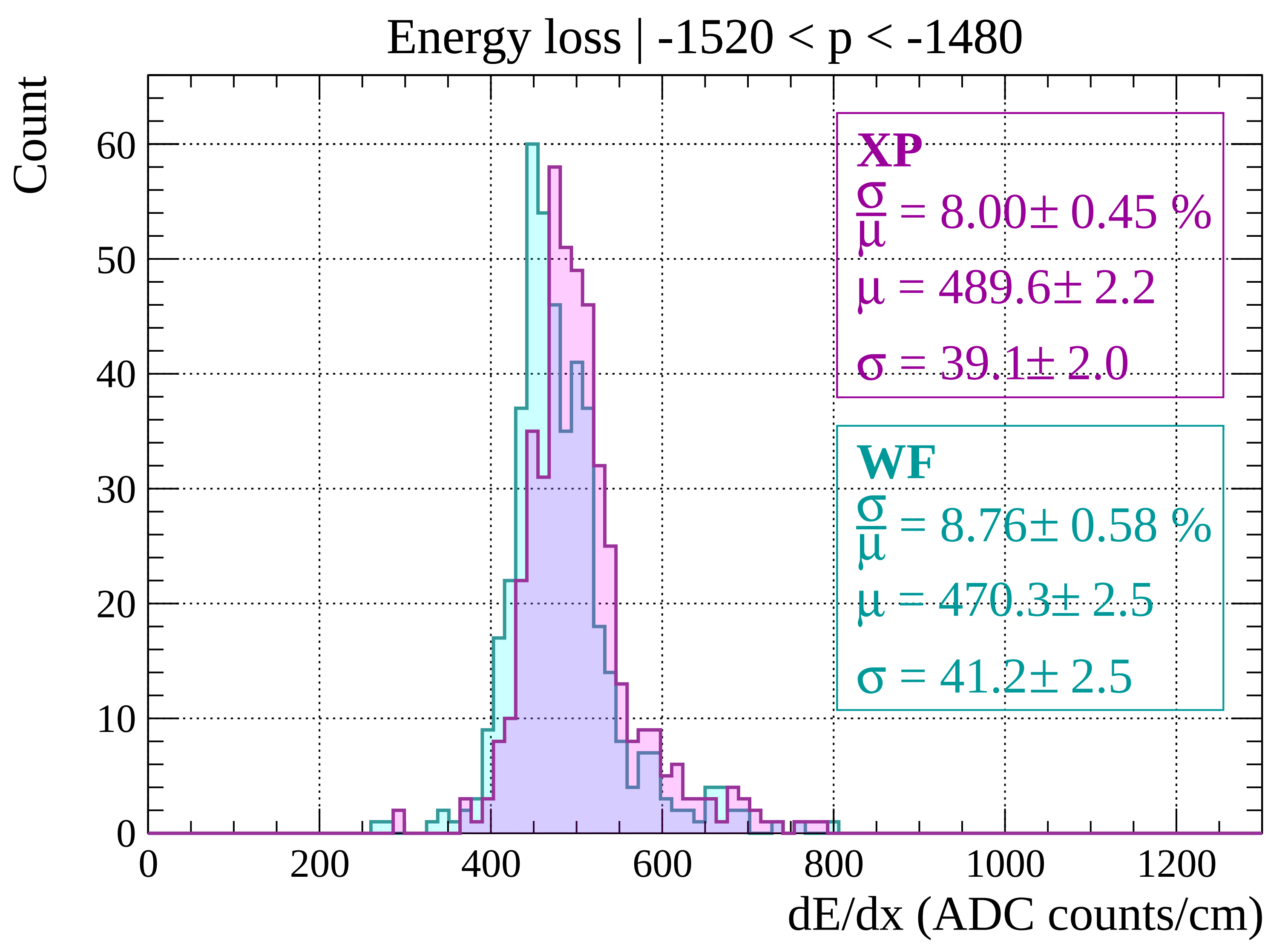
Energy loss | -1640 < p < -1600
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
XP
σ/μ = 8.44± 0.54 %
μ = 502.8± 2.6
σ = 42.4± 2.5
WF
σ/μ = 8.56± 0.59 %
μ = 479.1± 2.5
σ = 41.0± 2.6



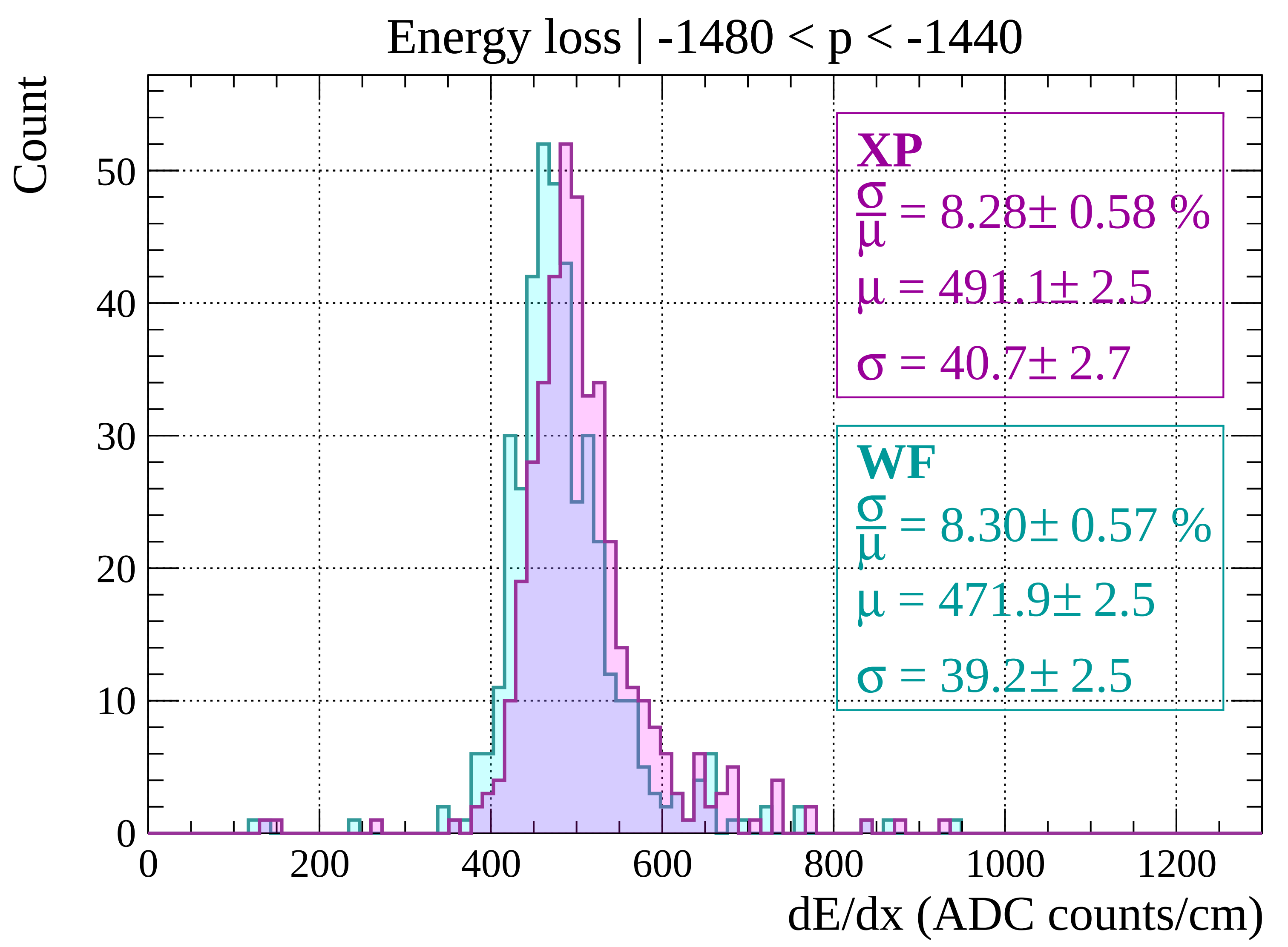
Energy loss | -1600 < p < -1560
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
XP
σ/μ = 7.47± 0.39 %
μ = 497.4± 2.2
σ = 37.2± 1.8
WF
σ/μ = 7.56± 0.50 %
μ = 474.5± 2.2
σ = 35.9± 2.2



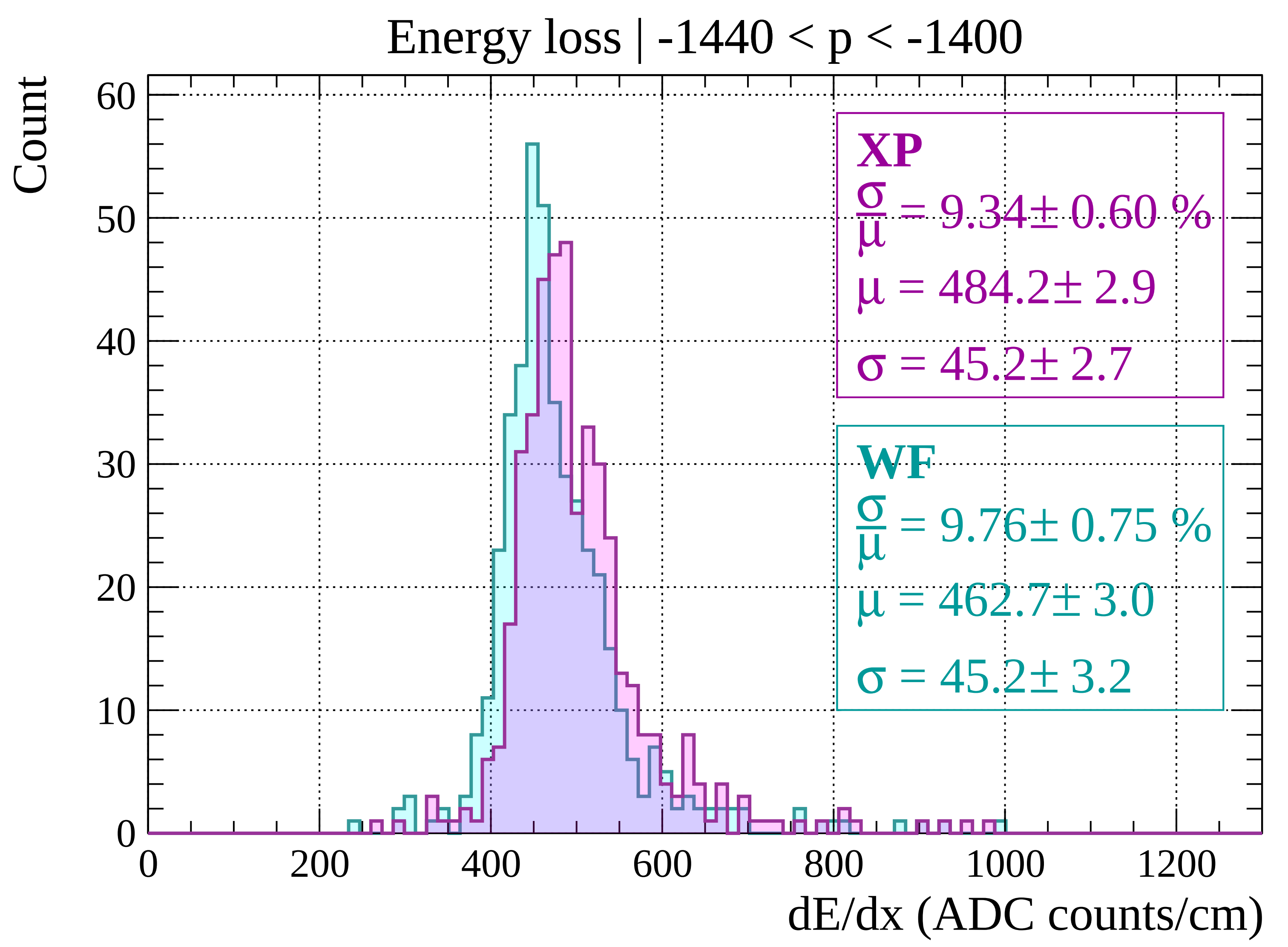
Energy loss | -1560 < p < -1520
Count
XP
$\frac{\sigma}{\mu}$ = 8.73± 0.74 %
μ = 492.5± 2.7
σ = 43.0± 3.4
WF
$\frac{\sigma}{\mu}$ = 8.68± 0.57 %
μ = 475.1± 2.5
σ = 41.2± 2.5
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



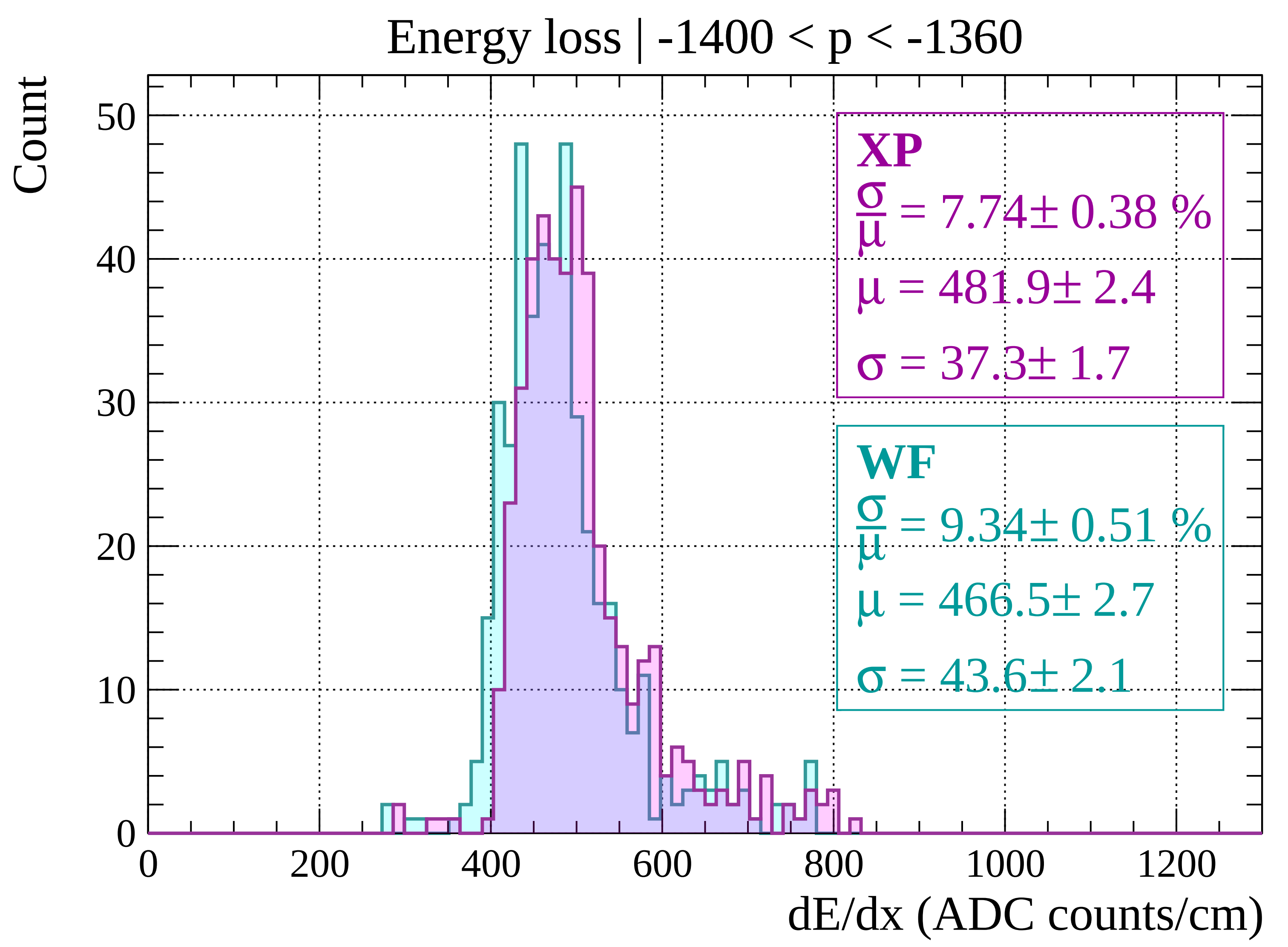
Energy loss | -1520 < p < -1480
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
XP
$\frac{\sigma}{\mu}$ = 8.00± 0.45 %
μ = 489.6± 2.2
σ = 39.1± 2.0
WF
$\frac{\sigma}{\mu}$ = 8.76± 0.58 %
μ = 470.3± 2.5
σ = 41.2± 2.5



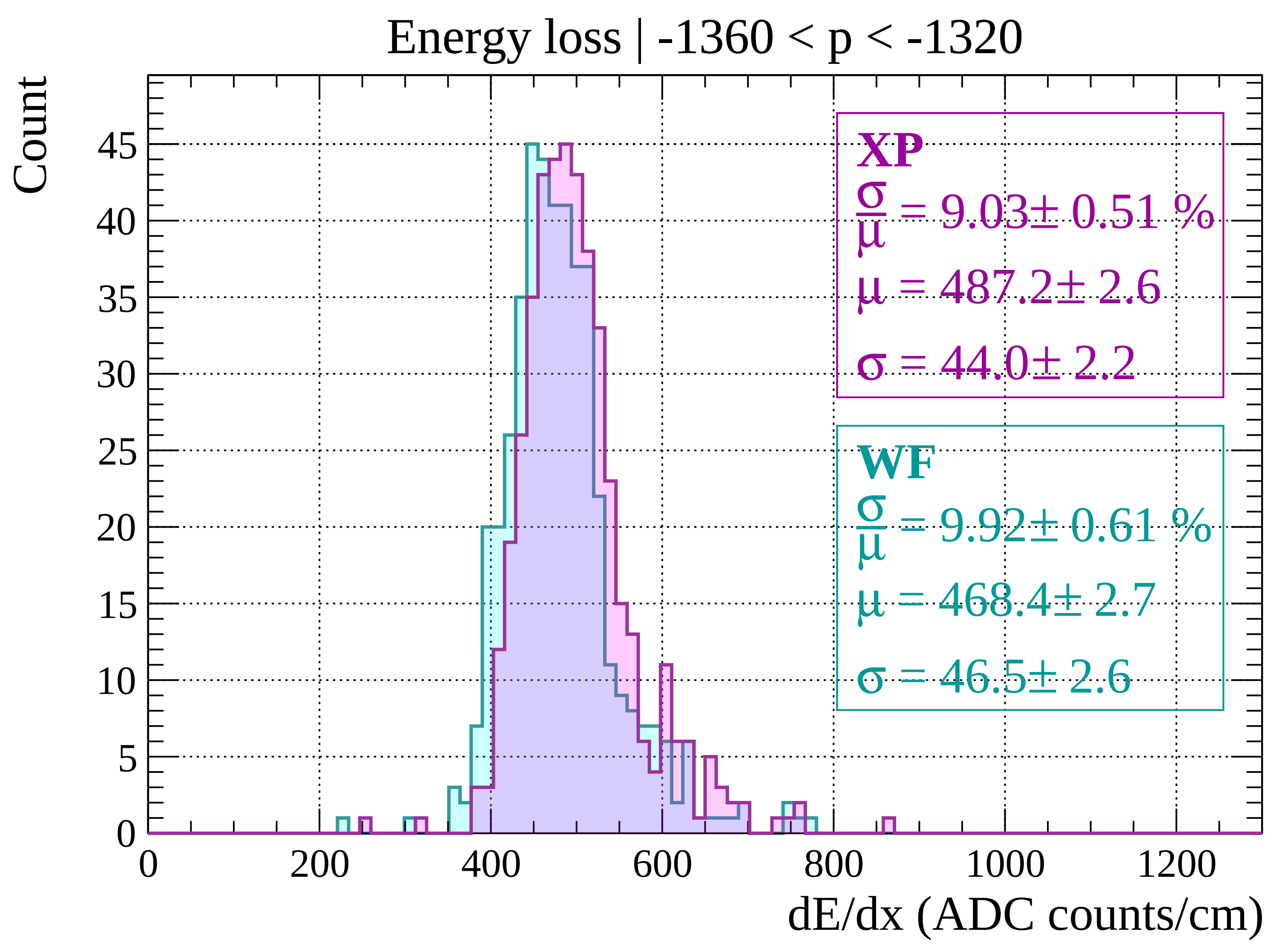
Energy loss | -1480 < p < -1440
Count
0
10
20
30
40
50
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.28± 0.58 %
μ = 491.1± 2.5
σ = 40.7± 2.7
WF
σ/μ = 8.30± 0.57 %
μ = 471.9± 2.5
σ = 39.2± 2.5

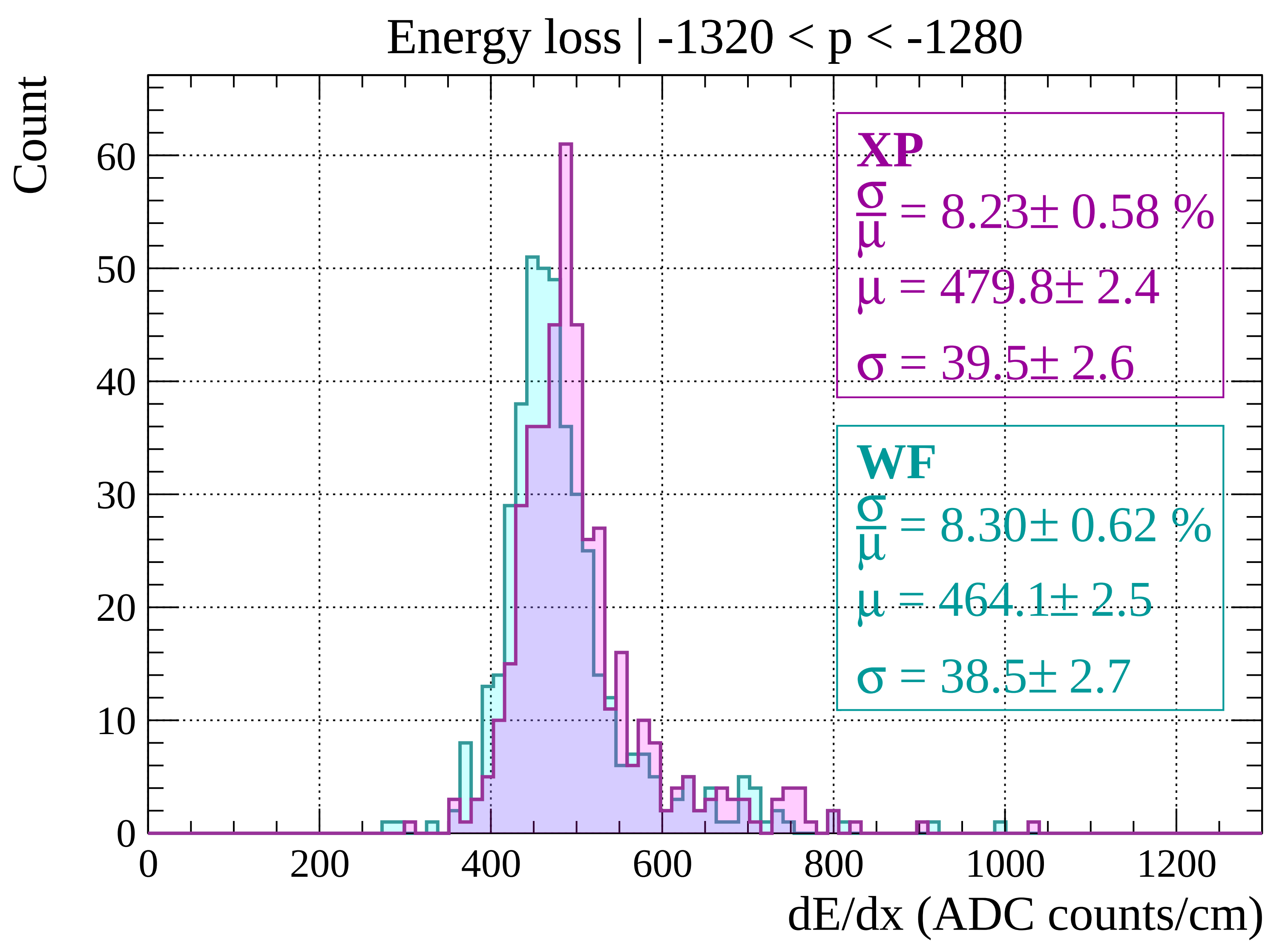


Energy loss | -1440 < p < -1400
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
XP
σ/μ = 9.34± 0.60 %
μ = 484.2± 2.9
σ = 45.2± 2.7
WF
σ/μ = 9.76± 0.75 %
μ = 462.7± 3.0
σ = 45.2± 3.2

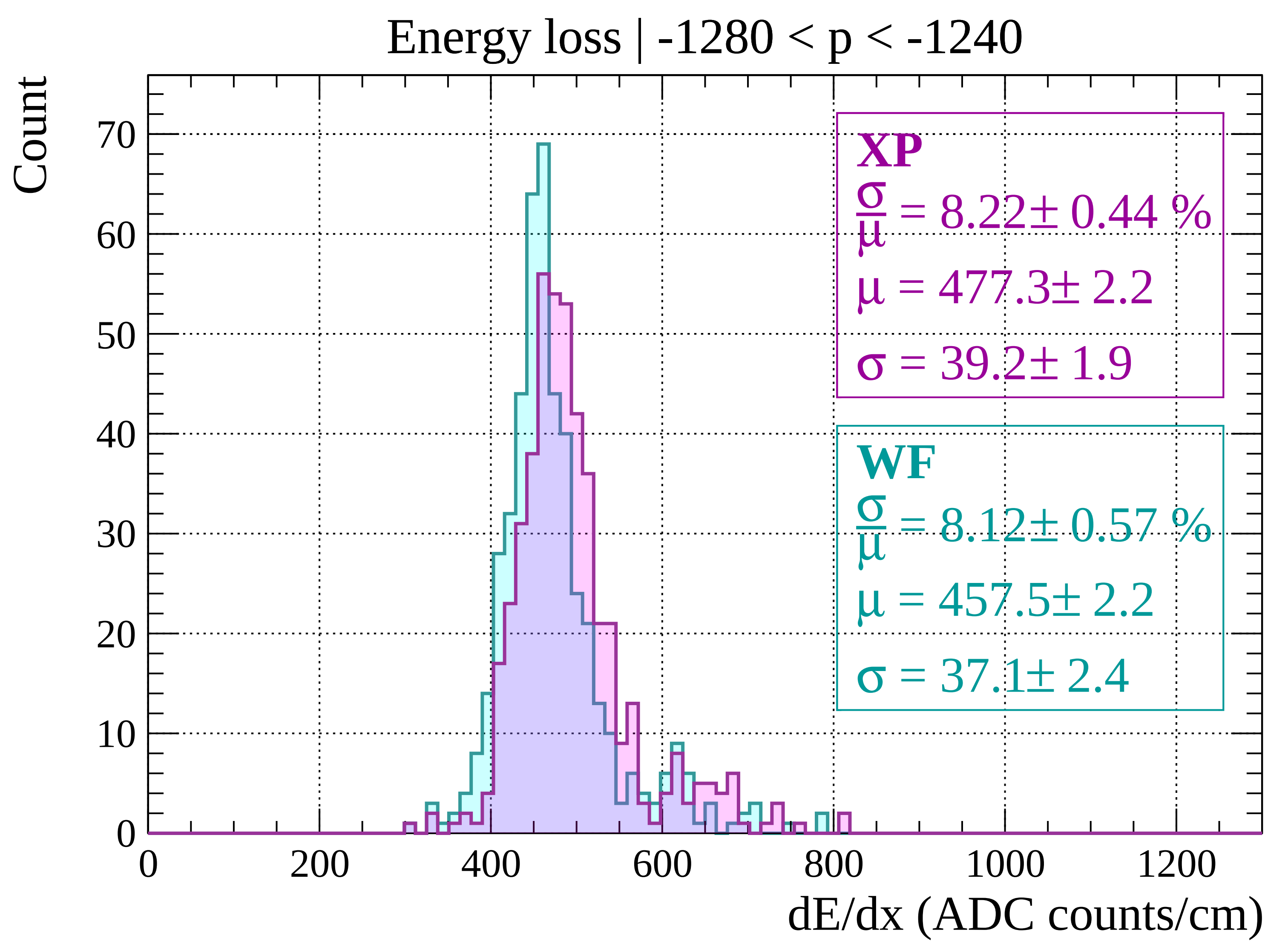


Energy loss | -1400 < p < -1360
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
XP
σ/μ = 7.74± 0.38 %
μ = 481.9± 2.4
σ = 37.3± 1.7
WF
σ/μ = 9.34± 0.51 %
μ = 466.5± 2.7
σ = 43.6± 2.1

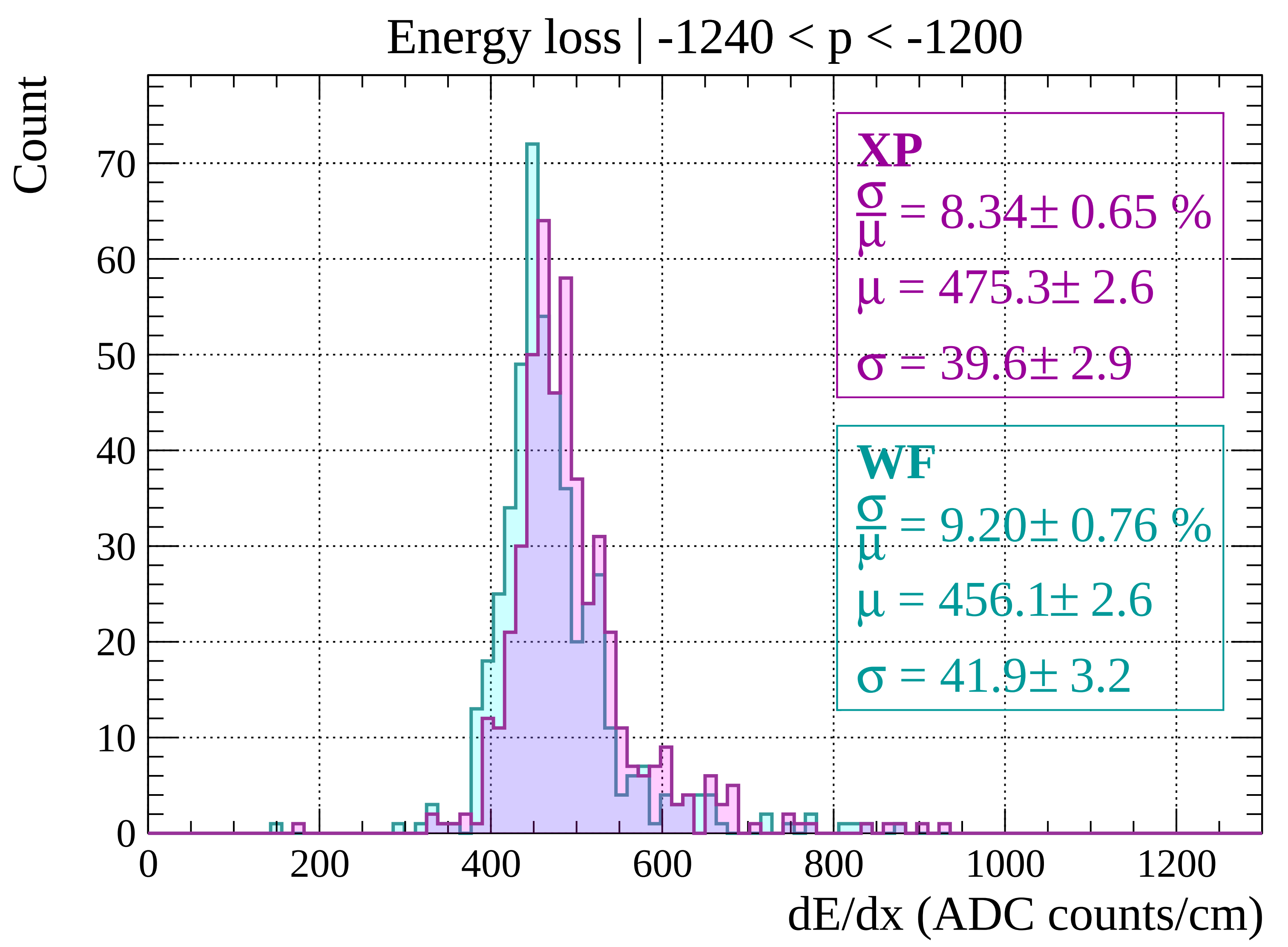

Energy loss | -1360 < p < -1320
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
40
45
XP
σ/μ = 9.03± 0.51 %
μ = 487.2± 2.6
σ = 44.0± 2.2
WF
σ/μ = 9.92± 0.61 %
μ = 468.4± 2.7
σ = 46.5± 2.6



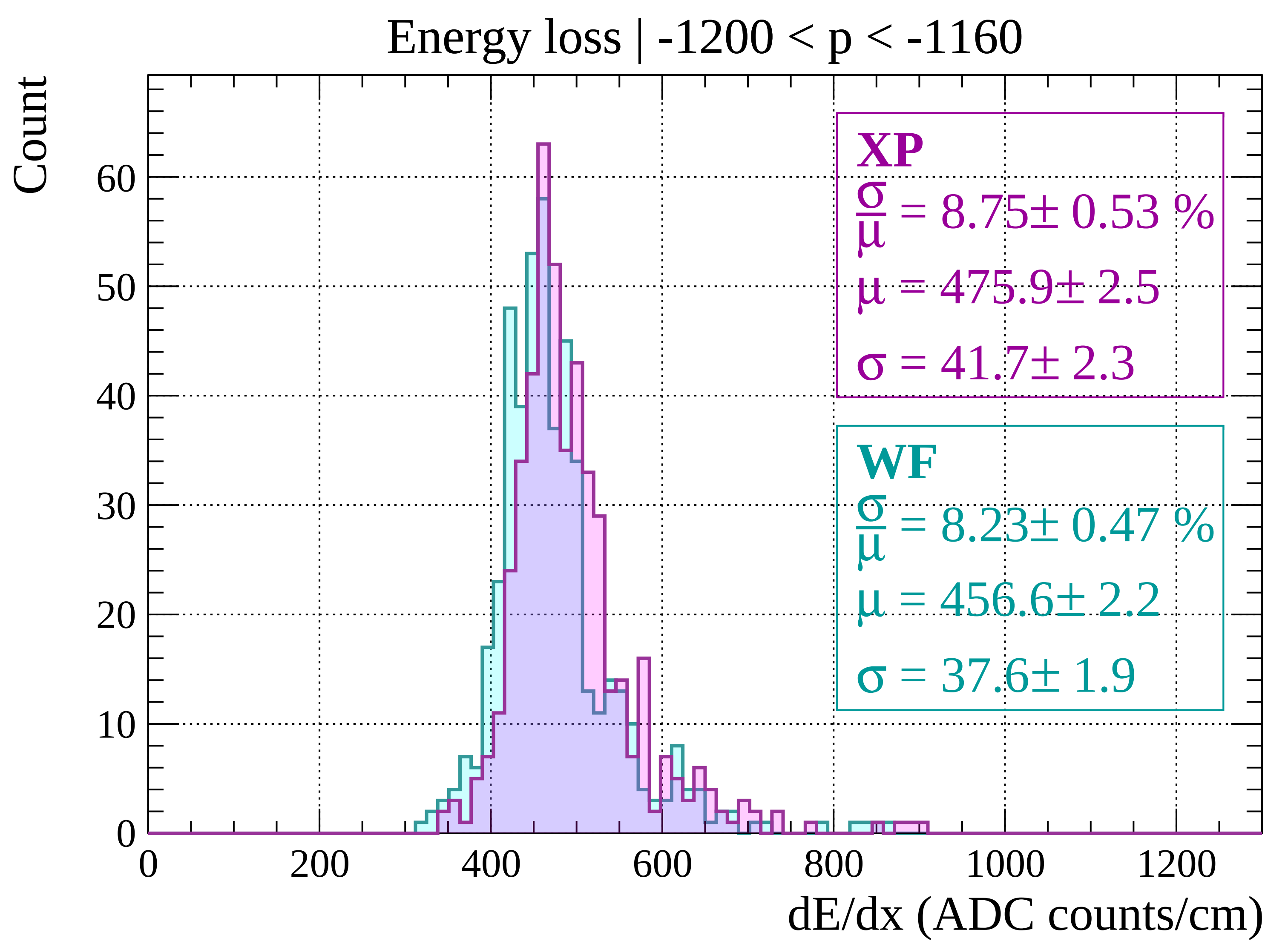
Energy loss | -1320 < p < -1280
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
XP
$\frac{\sigma}{\mu}$ = 8.23± 0.58 %
μ = 479.8± 2.4
σ = 39.5± 2.6
WF
$\frac{\sigma}{\mu}$ = 8.30± 0.62 %
μ = 464.1± 2.5
σ = 38.5± 2.7



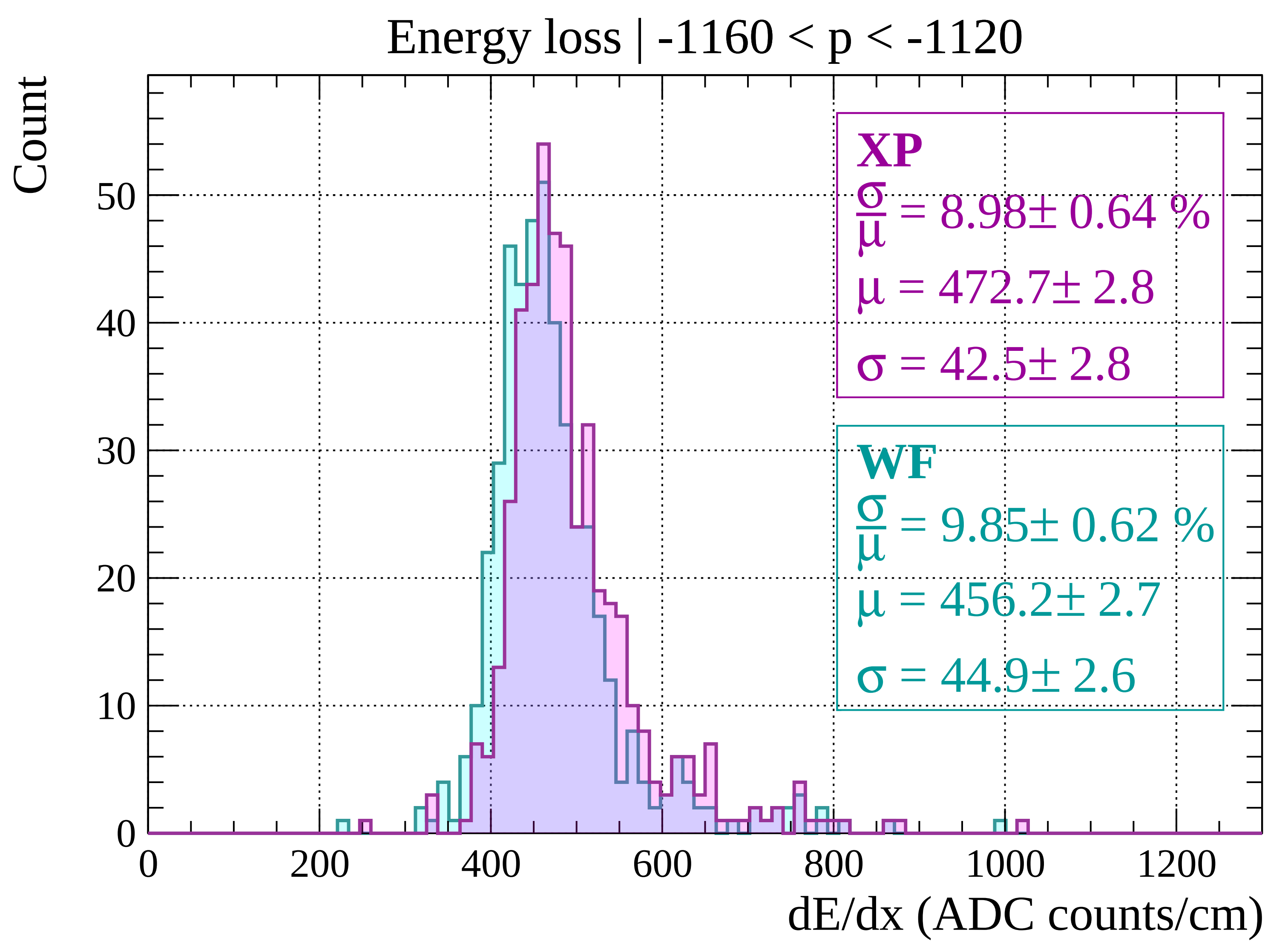
Energy loss | -1280 < p < -1240
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
XP
$\frac{\sigma}{\mu}$ = 8.22± 0.44 %
μ = 477.3± 2.2
σ = 39.2± 1.9
WF
$\frac{\sigma}{\mu}$ = 8.12± 0.57 %
μ = 457.5± 2.2
σ = 37.1± 2.4



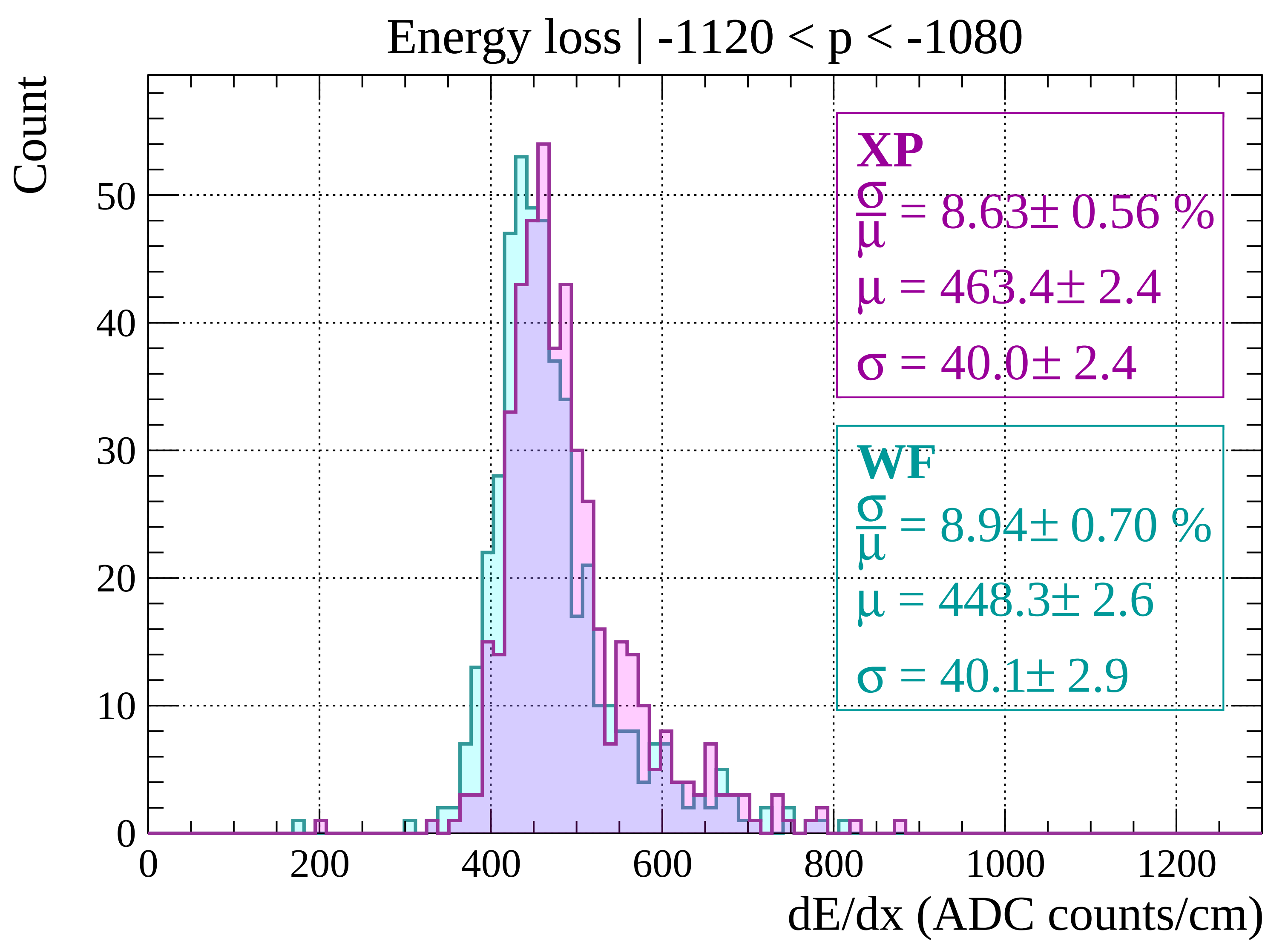
Energy loss | -1240 < p < -1200
Count
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.34± 0.65 %
μ = 475.3± 2.6
σ = 39.6± 2.9
WF
σ/μ = 9.20± 0.76 %
μ = 456.1± 2.6
σ = 41.9± 3.2

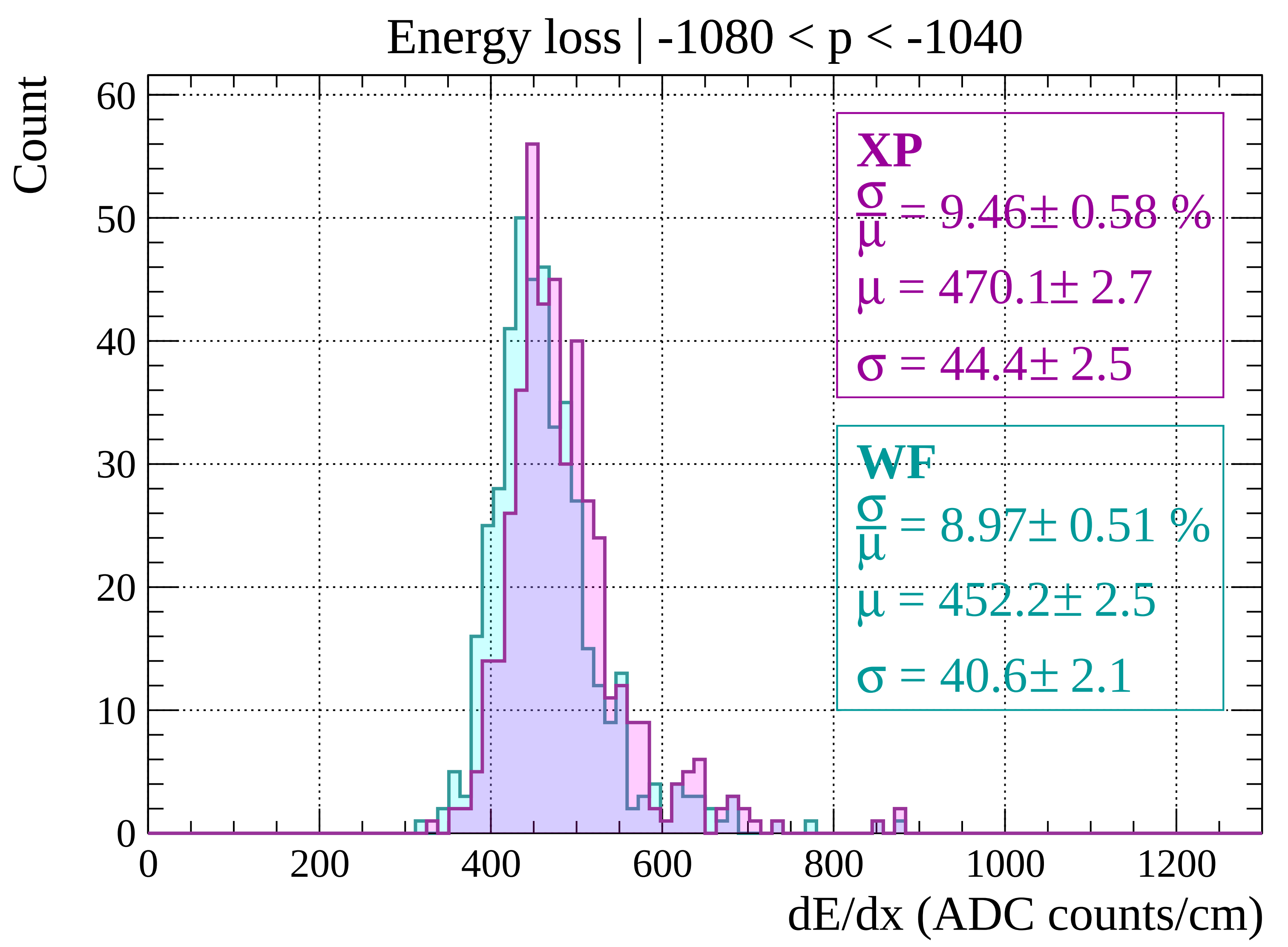


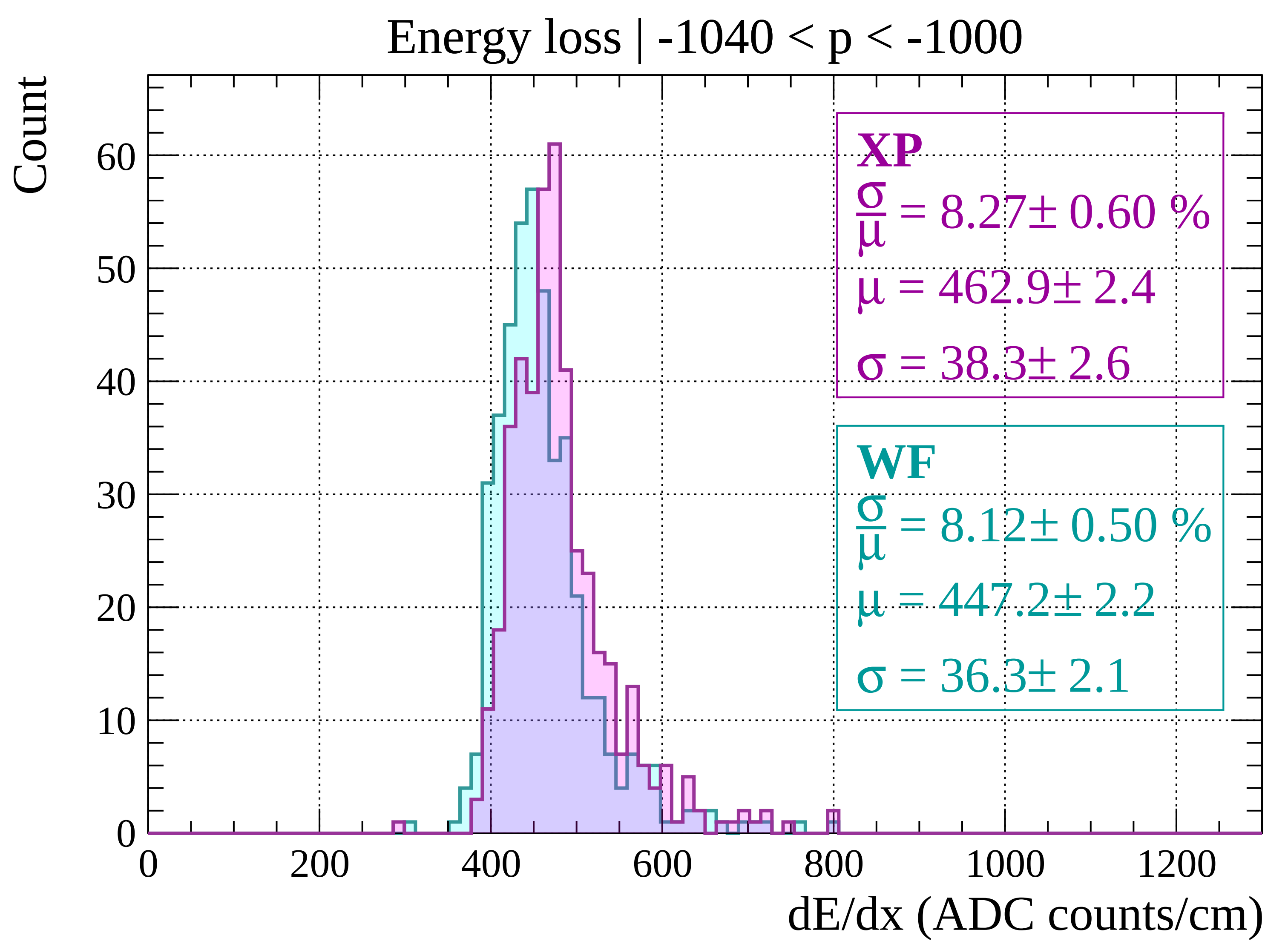
Energy loss | -1200 < p < -1160
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
XP
σ/μ = 8.75± 0.53 %
μ = 475.9± 2.5
σ = 41.7± 2.3
WF
σ/μ = 8.23± 0.47 %
μ = 456.6± 2.2
σ = 37.6± 1.9



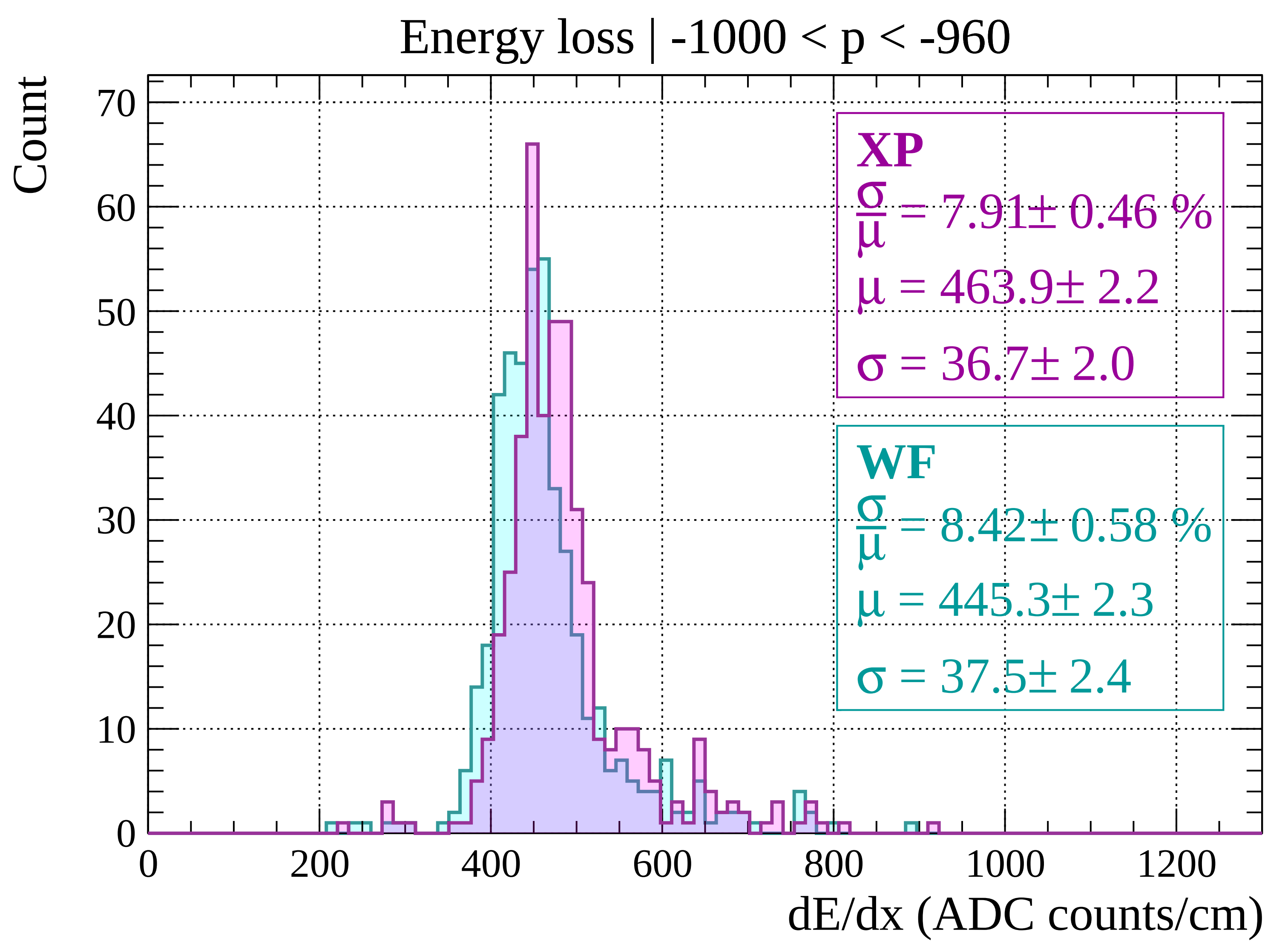
Energy loss | -1160 < p < -1120
Count
0
10
20
30
40
50
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.98± 0.64 %
μ = 472.7± 2.8
σ = 42.5± 2.8
WF
σ/μ = 9.85± 0.62 %
μ = 456.2± 2.7
σ = 44.9± 2.6

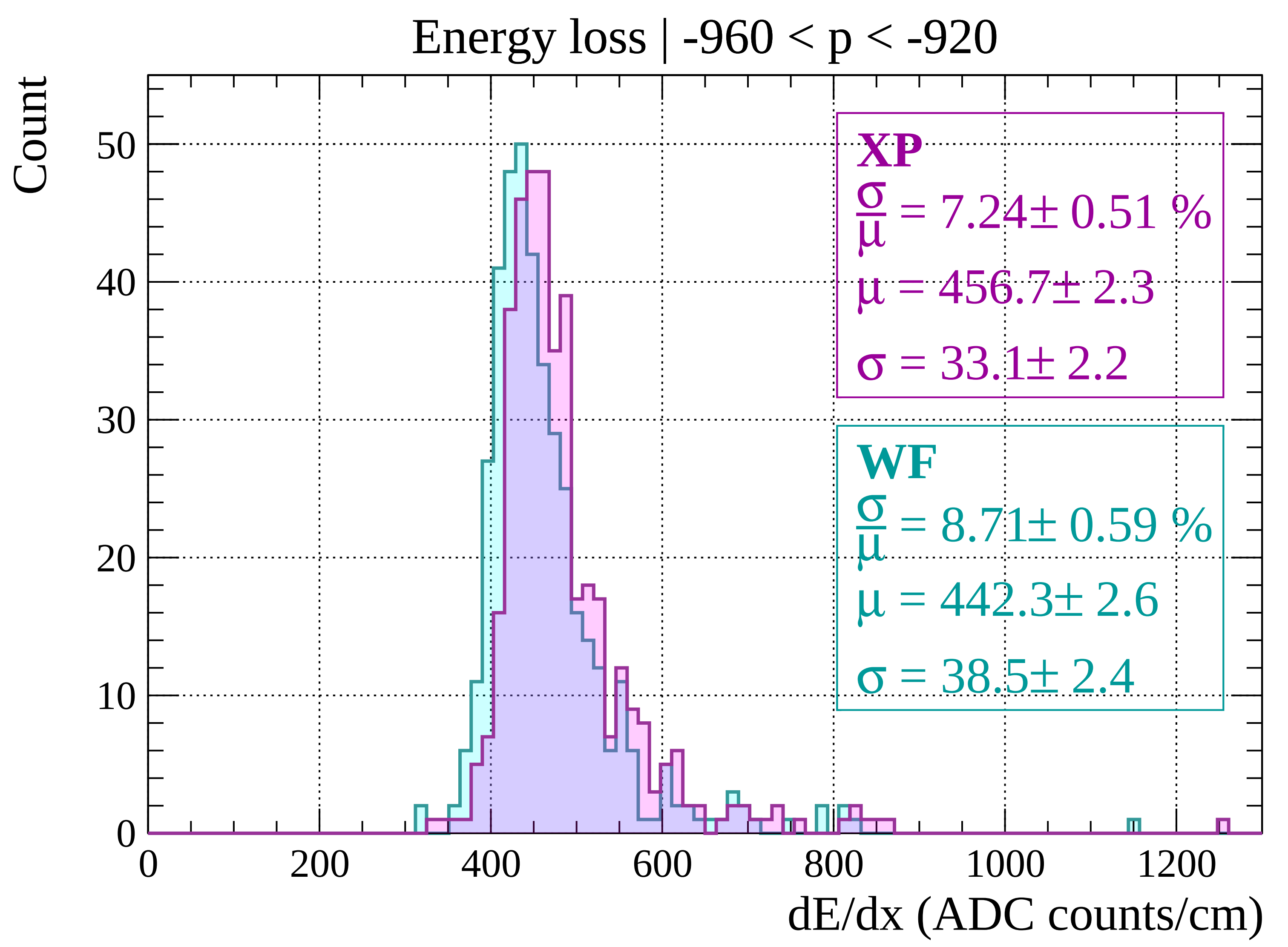

Energy loss | -1120 < p < -1080
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 8.63± 0.56 %
μ = 463.4± 2.4
σ = 40.0± 2.4
WF
σ/μ = 8.94± 0.70 %
μ = 448.3± 2.6
σ = 40.1± 2.9


Energy loss | -1080 < p < -1040
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.46± 0.58 %
μ = 470.1± 2.7
σ = 44.4± 2.5
WF
σ/μ = 8.97± 0.51 %
μ = 452.2± 2.5
σ = 40.6± 2.1

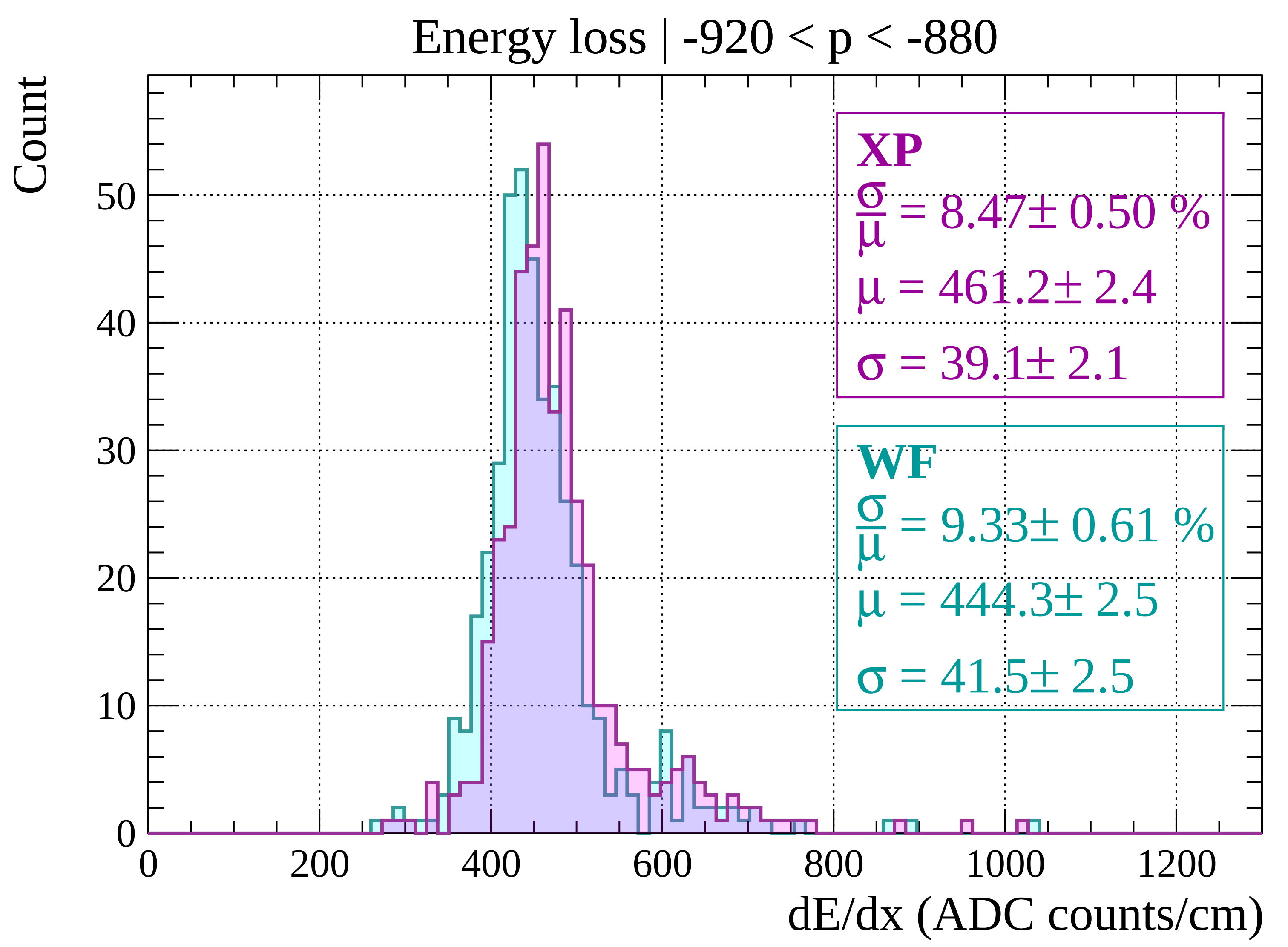


Energy loss | -1040 < p < -1000
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.27± 0.60 %
μ = 462.9± 2.4
σ = 38.3± 2.6
WF
σ/μ = 8.12± 0.50 %
μ = 447.2± 2.2
σ = 36.3± 2.1
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200

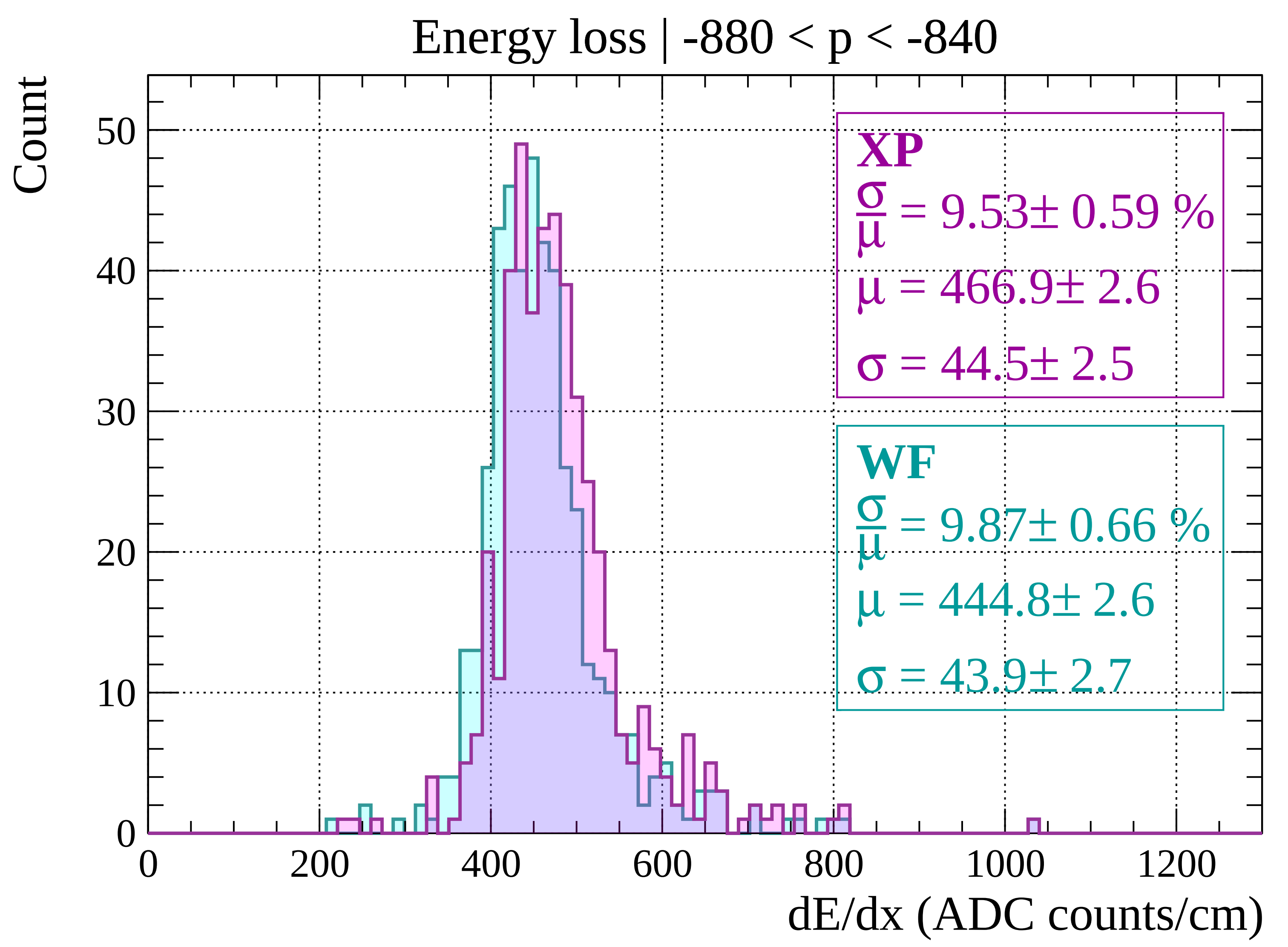

Energy loss | -1000 < p < -960
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP
σ/μ = 7.91± 0.46 %
μ = 463.9± 2.2
σ = 36.7± 2.0
WF
σ/μ = 8.42± 0.58 %
μ = 445.3± 2.3
σ = 37.5± 2.4



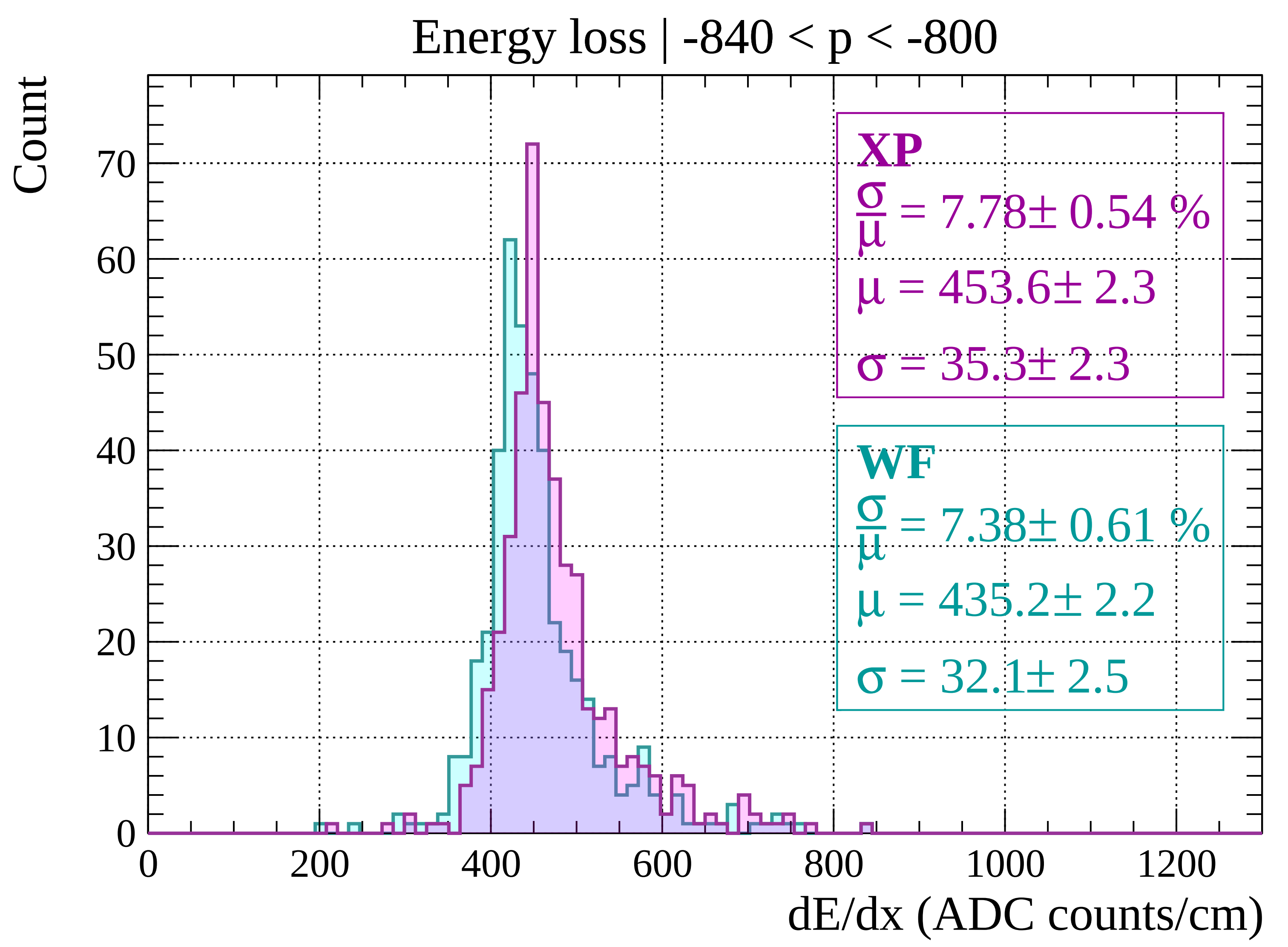
Energy loss | -960 < p < -920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
XP
σ/μ = 7.24± 0.51 %
μ = 456.7± 2.3
σ = 33.1± 2.2
WF
σ/μ = 8.71± 0.59 %
μ = 442.3± 2.6
σ = 38.5± 2.4

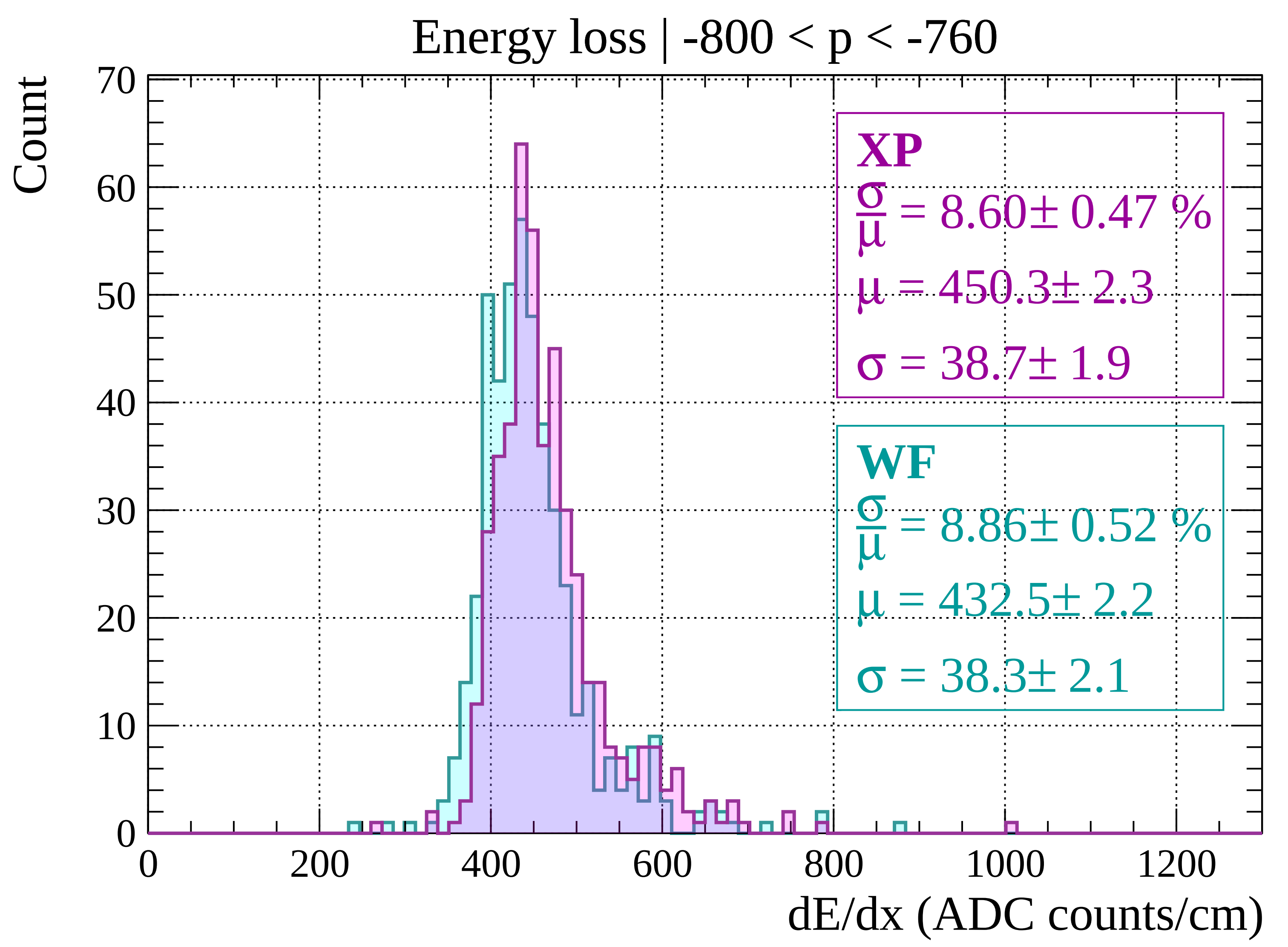


Energy loss | -920 < p < -880
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
XP
σ/μ = 8.47± 0.50 %
μ = 461.2± 2.4
σ = 39.1± 2.1
WF
σ/μ = 9.33± 0.61 %
μ = 444.3± 2.5
σ = 41.5± 2.5

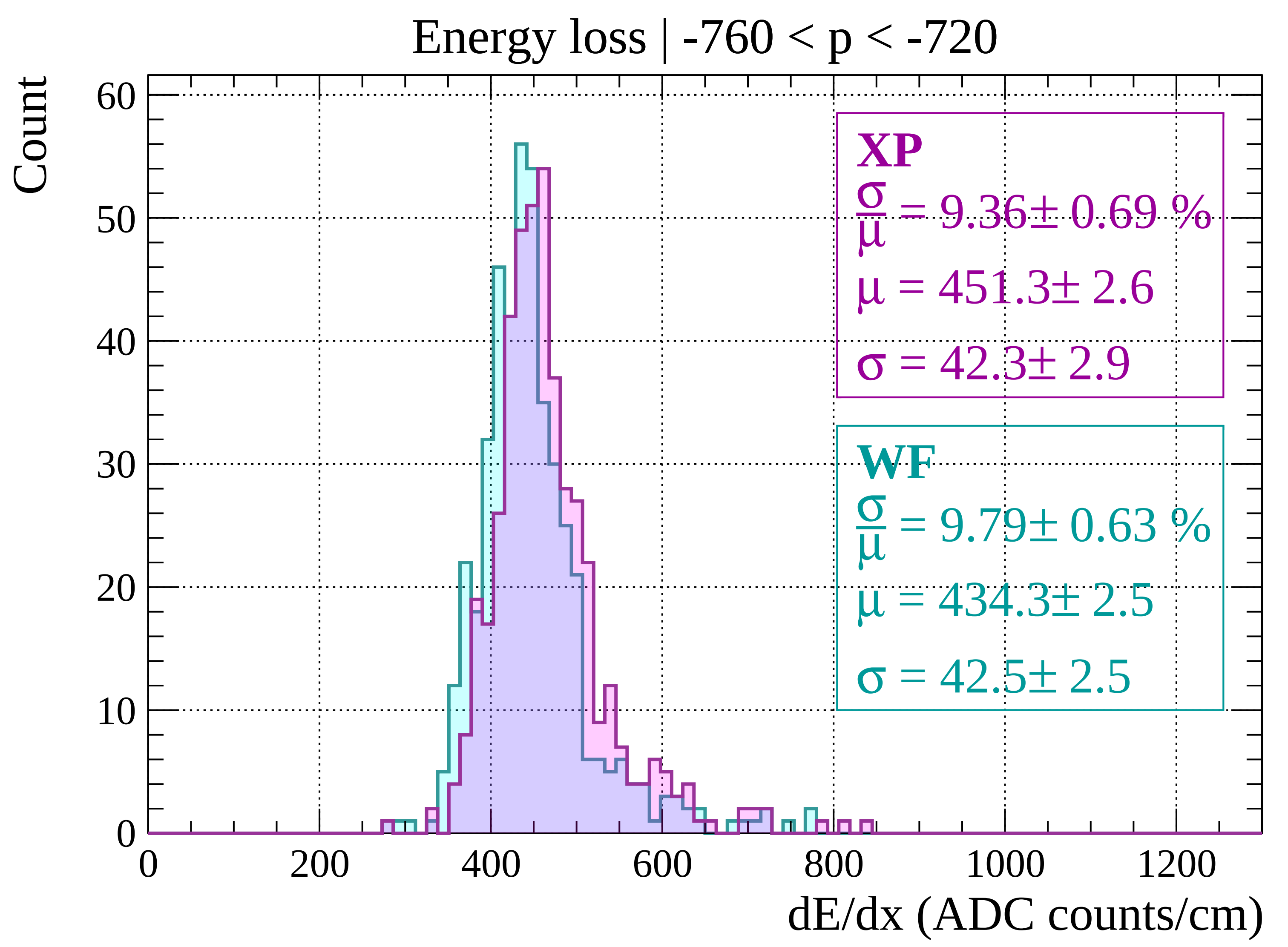


Energy loss | -880 < p < -840
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 9.53± 0.59 %
μ = 466.9± 2.6
σ = 44.5± 2.5
WF
σ/μ = 9.87± 0.66 %
μ = 444.8± 2.6
σ = 43.9± 2.7

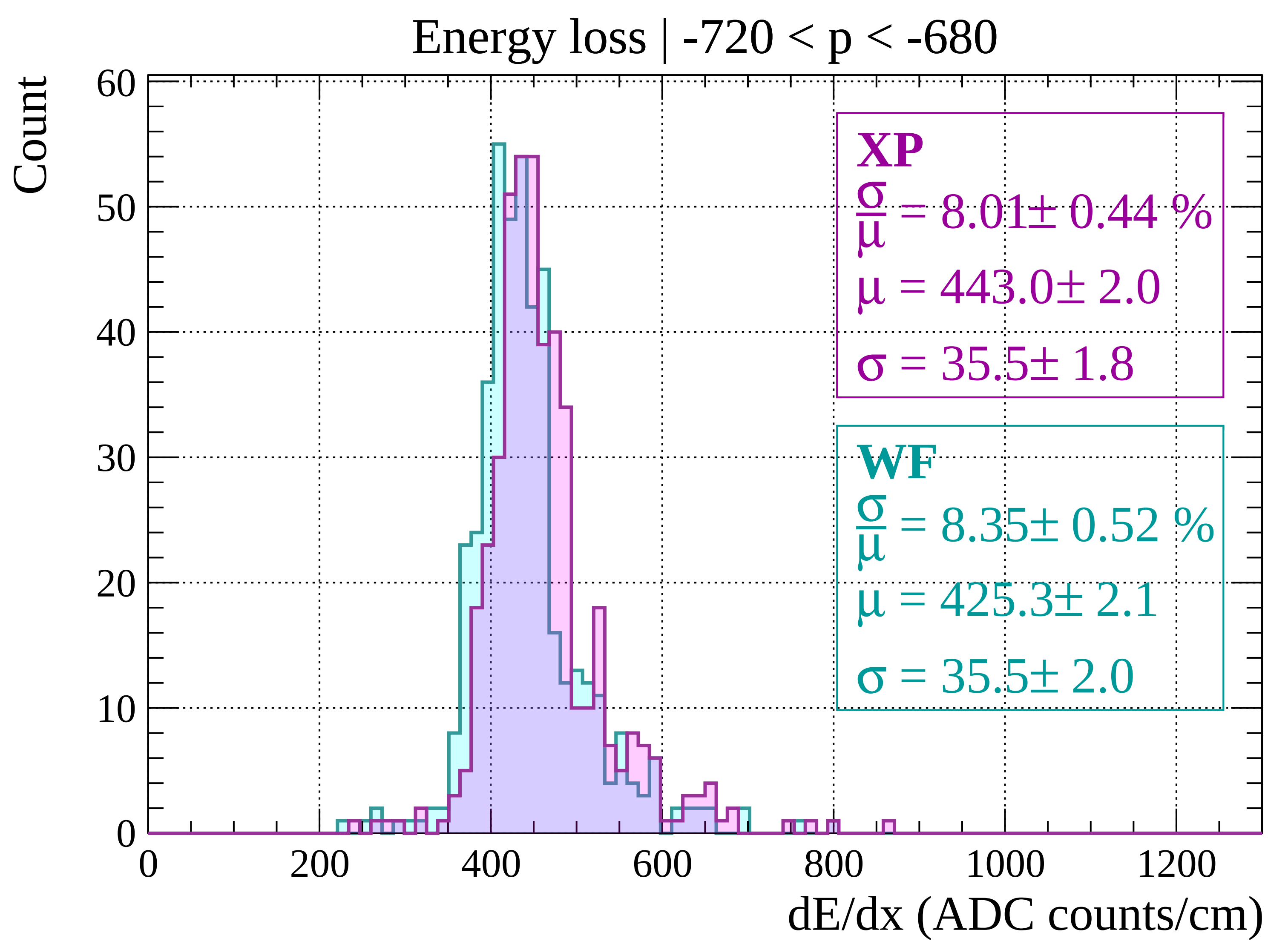

Energy loss | -840 < p < -800
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.78± 0.54 %
μ = 453.6± 2.3
σ = 35.3± 2.3
WF
σ/μ = 7.38± 0.61 %
μ = 435.2± 2.2
σ = 32.1± 2.5
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200



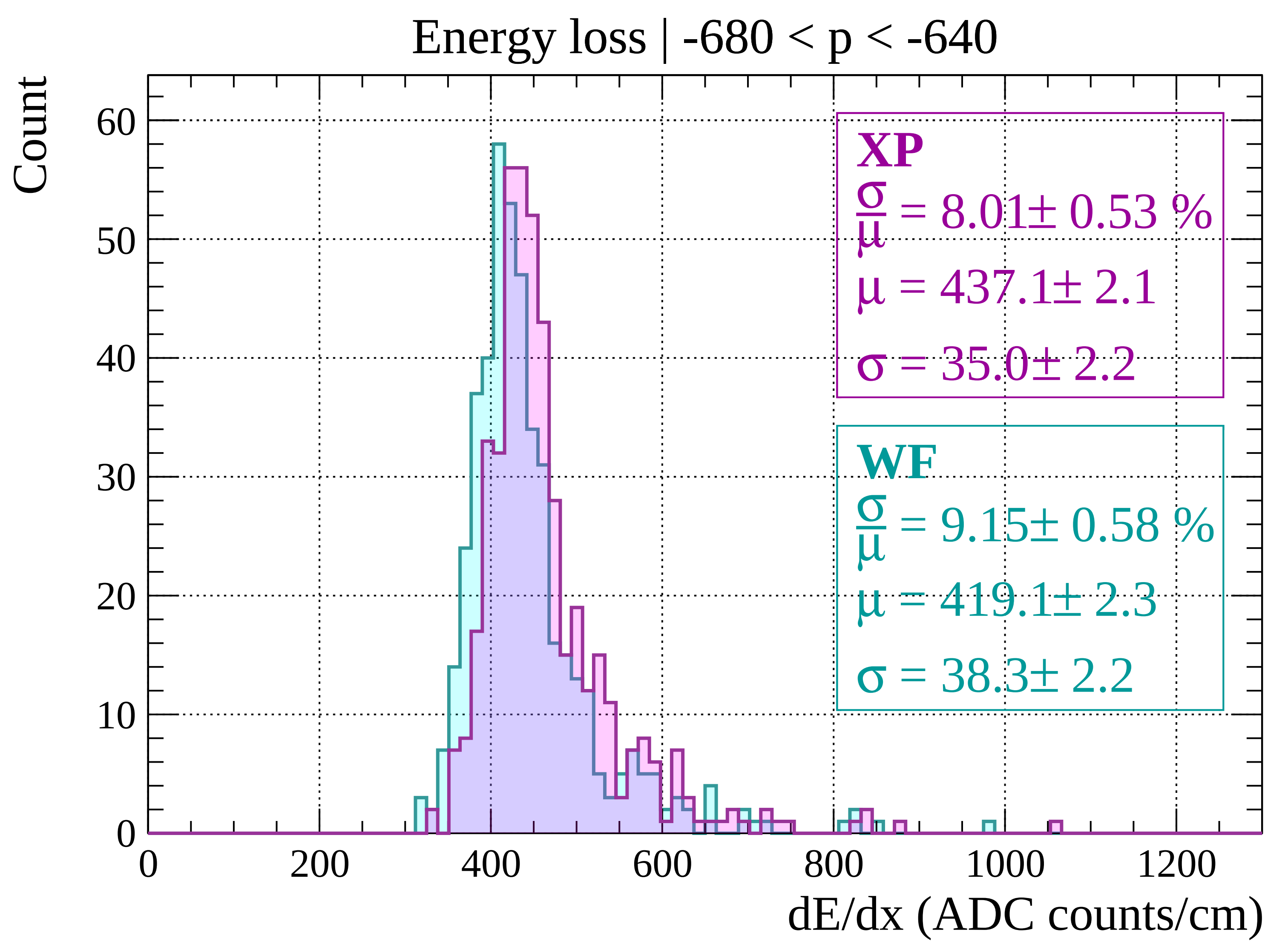
Energy loss | -800 < p < -760
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP
σ/μ = 8.60± 0.47 %
μ = 450.3± 2.3
σ = 38.7± 1.9
WF
σ/μ = 8.86± 0.52 %
μ = 432.5± 2.2
σ = 38.3± 2.1



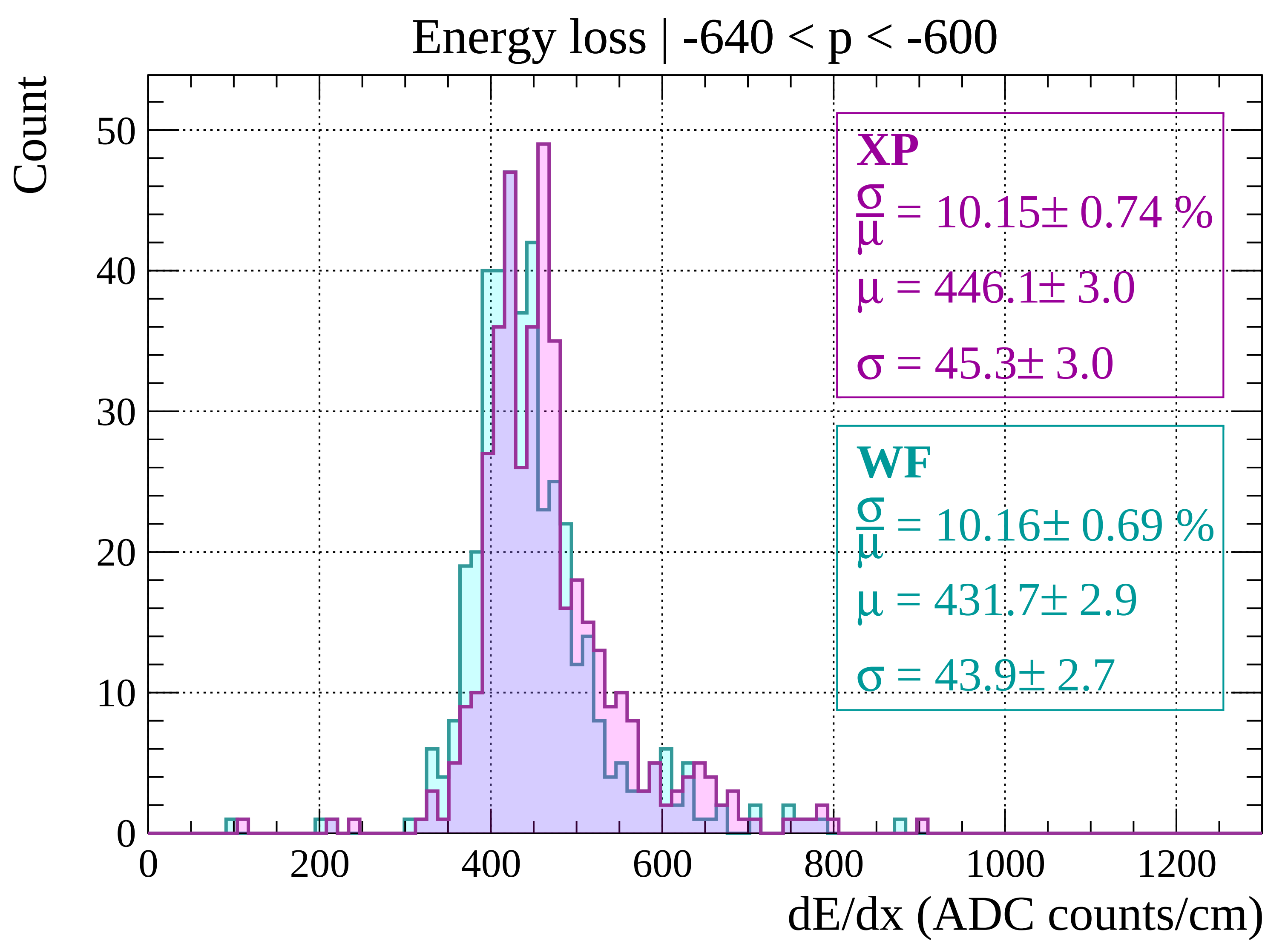
Energy loss | -760 < p < -720
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 9.36± 0.69 %
μ = 451.3± 2.6
σ = 42.3± 2.9
WF
σ/μ = 9.79± 0.63 %
μ = 434.3± 2.5
σ = 42.5± 2.5



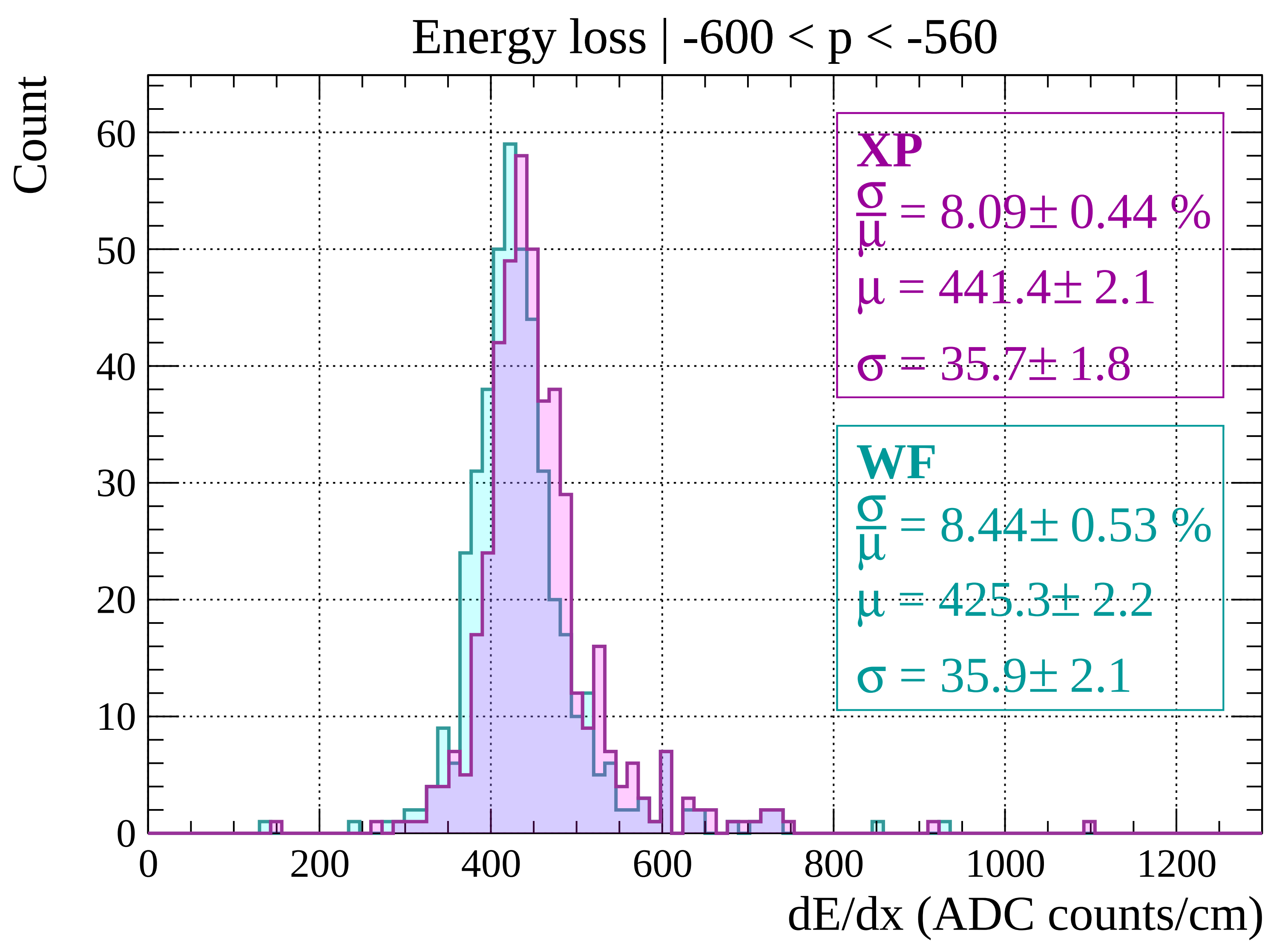
Energy loss | -720 < p < -680
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
XP
$\frac{\sigma}{\mu}$ = 8.01± 0.44 %
μ = 443.0± 2.0
σ = 35.5± 1.8
WF
$\frac{\sigma}{\mu}$ = 8.35± 0.52 %
μ = 425.3± 2.1
σ = 35.5± 2.0



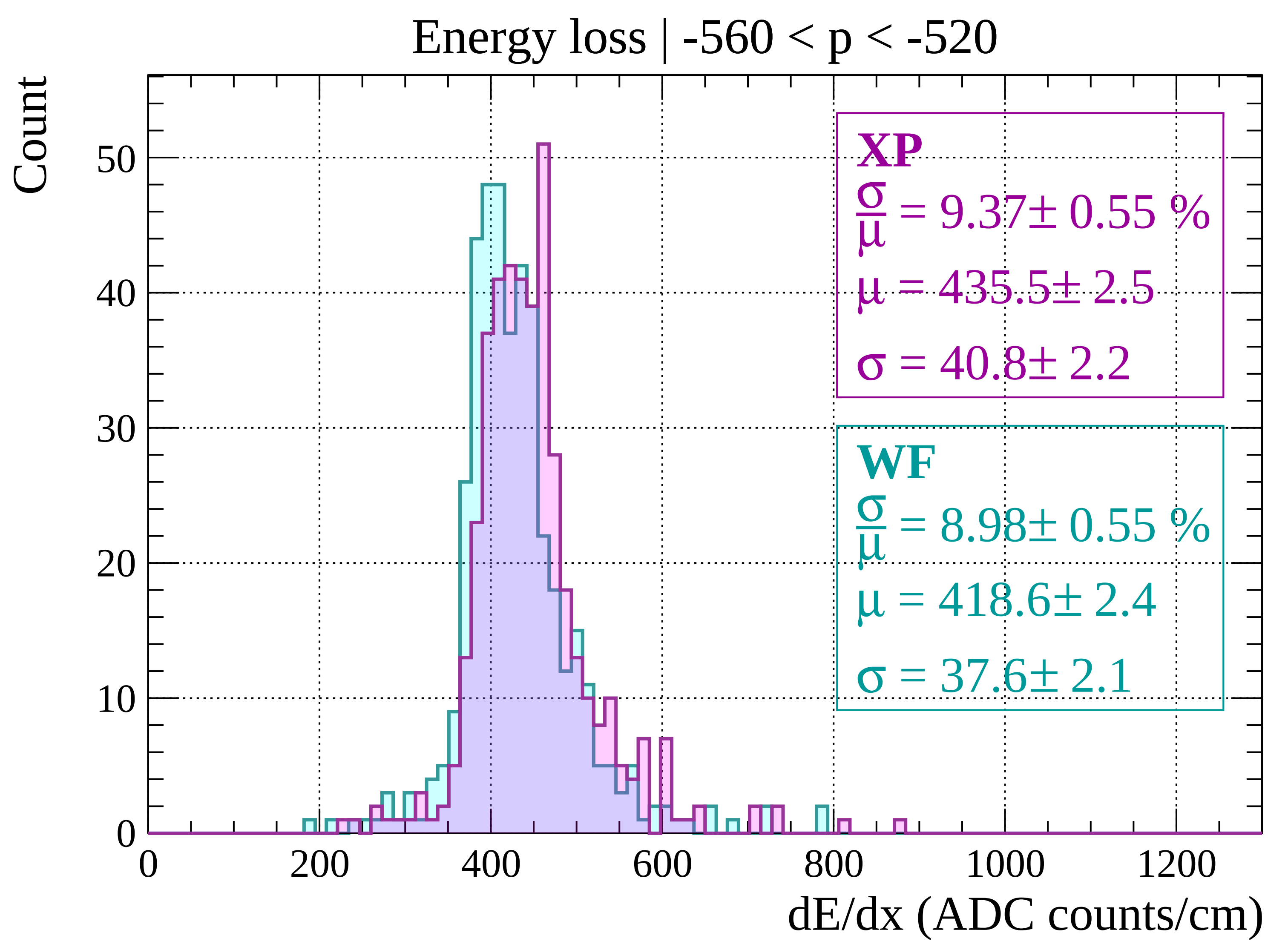
Energy loss | -680 < p < -640
Count
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.01± 0.53 %
μ = 437.1± 2.1
σ = 35.0± 2.2
WF
σ/μ = 9.15± 0.58 %
μ = 419.1± 2.3
σ = 38.3± 2.2



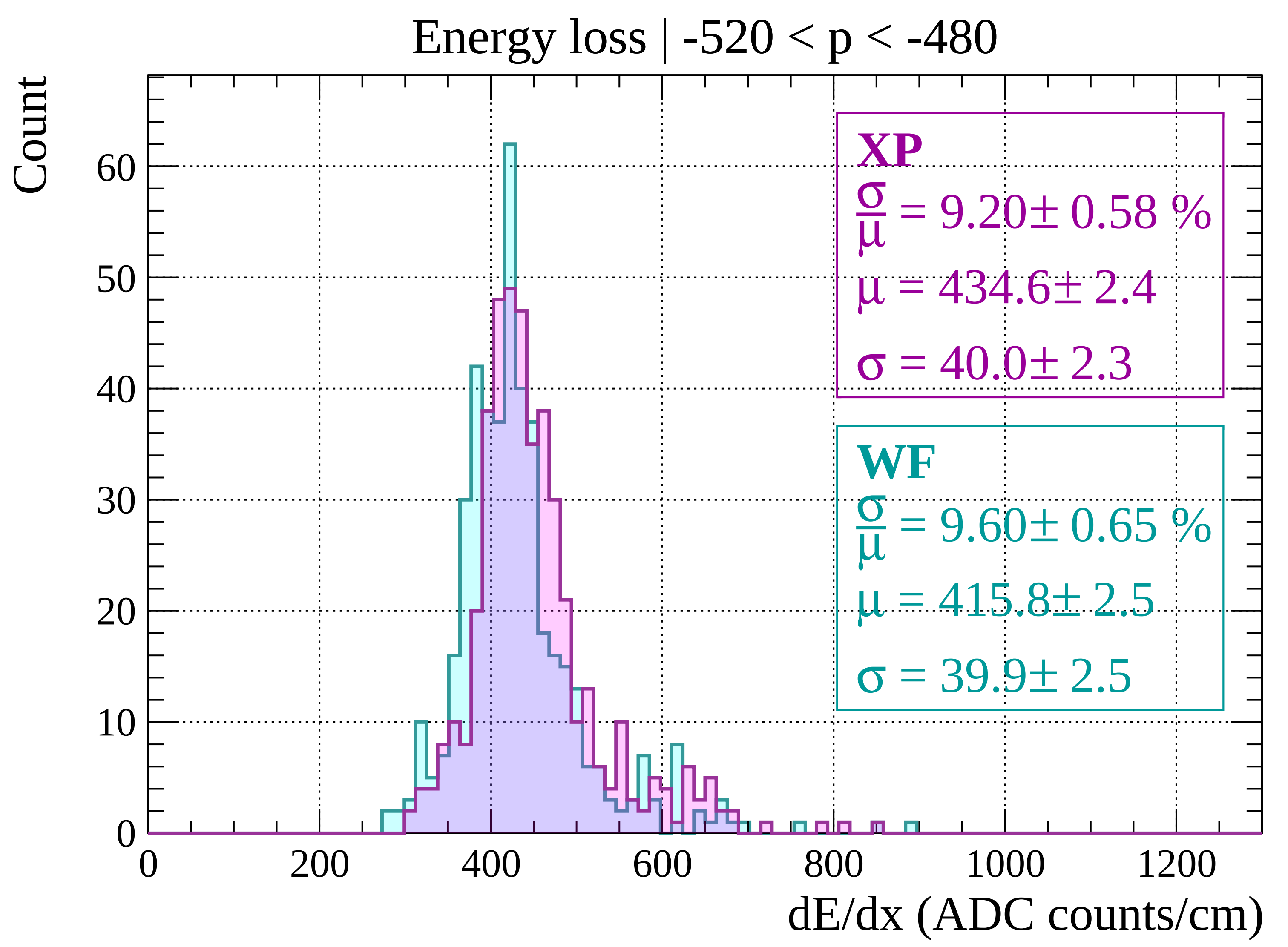
Energy loss | -640 < p < -600
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.15± 0.74 %
μ = 446.1± 3.0
σ = 45.3± 3.0
WF
$\frac{\sigma}{\mu}$ = 10.16± 0.69 %
μ = 431.7± 2.9
σ = 43.9± 2.7



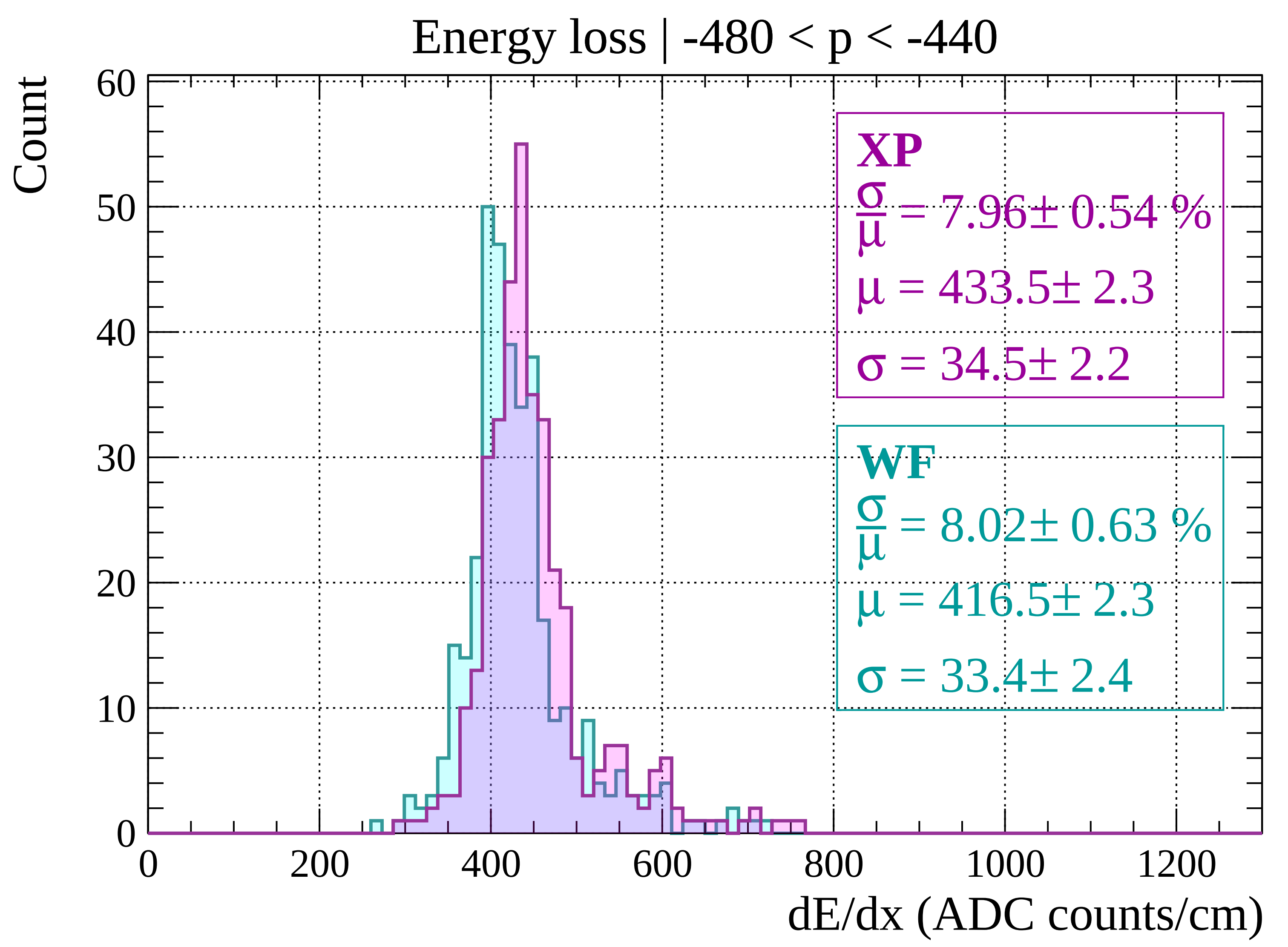
Energy loss | -600 < p < -560
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.09± 0.44 %
μ = 441.4± 2.1
σ = 35.7± 1.8
WF
$\frac{\sigma}{\mu}$ = 8.44± 0.53 %
μ = 425.3± 2.2
σ = 35.9± 2.1



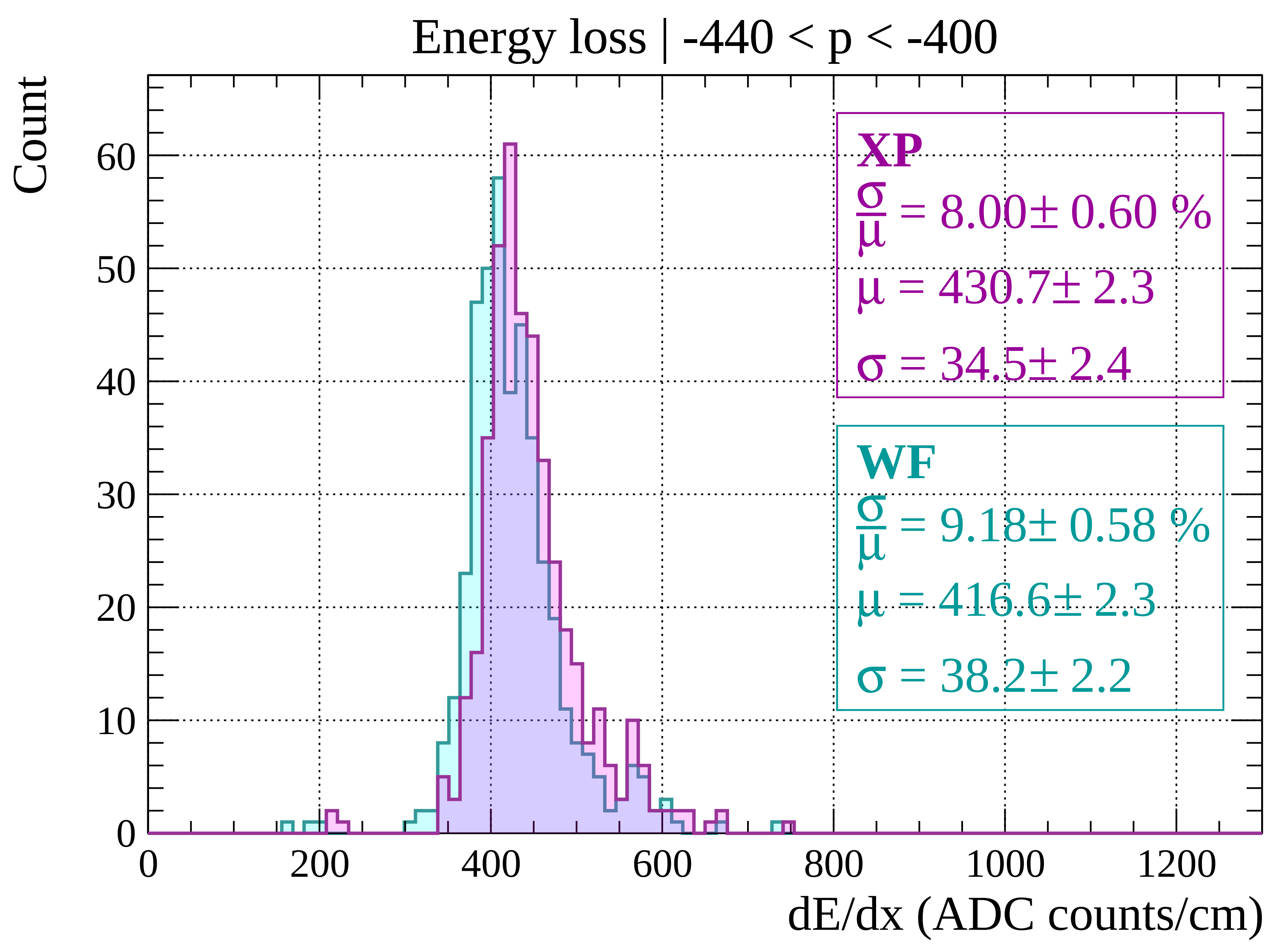
Energy loss | -560 < p < -520
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
XP
σ/μ = 9.37± 0.55 %
μ = 435.5± 2.5
σ = 40.8± 2.2
WF
σ/μ = 8.98± 0.55 %
μ = 418.6± 2.4
σ = 37.6± 2.1

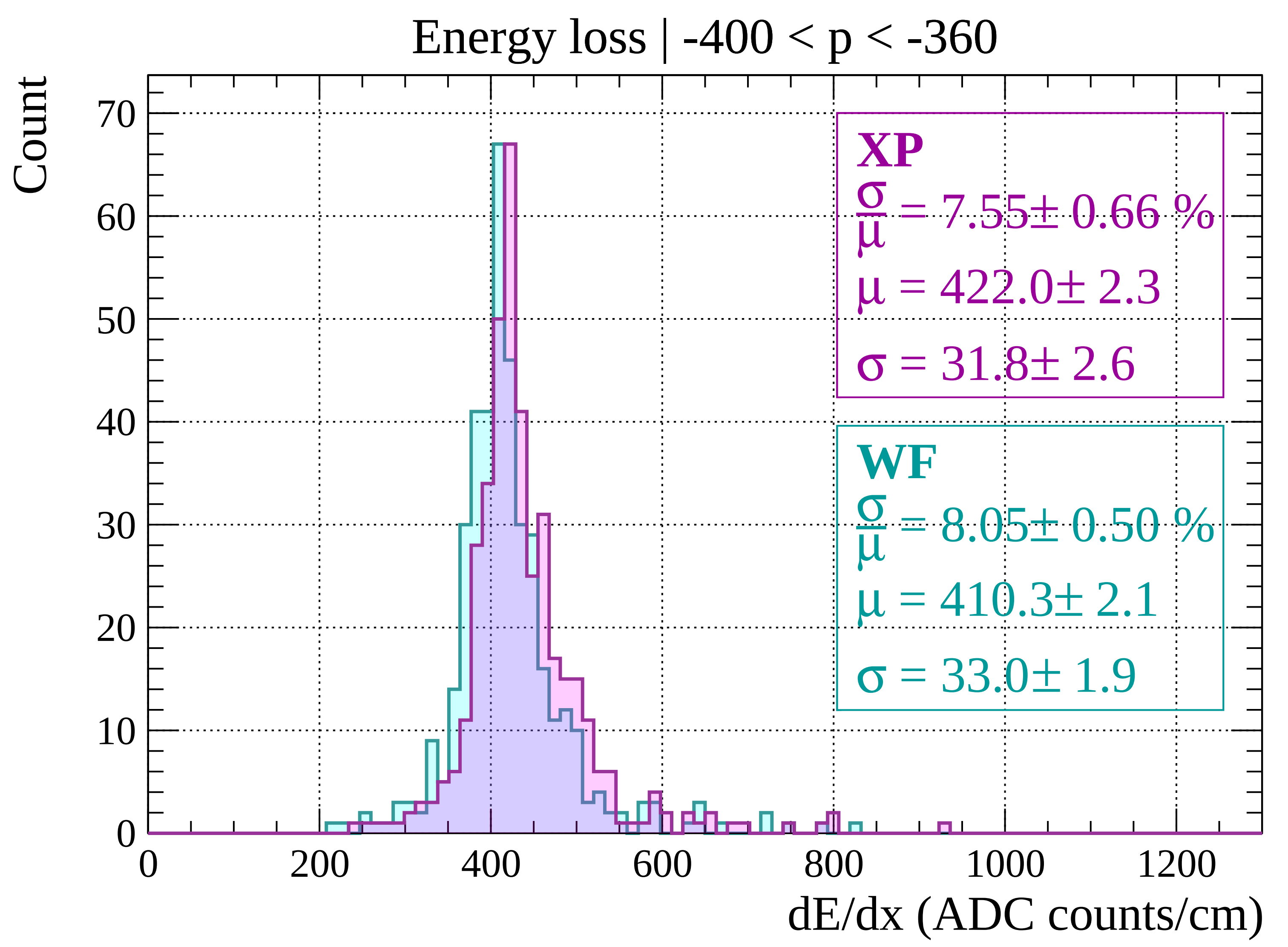


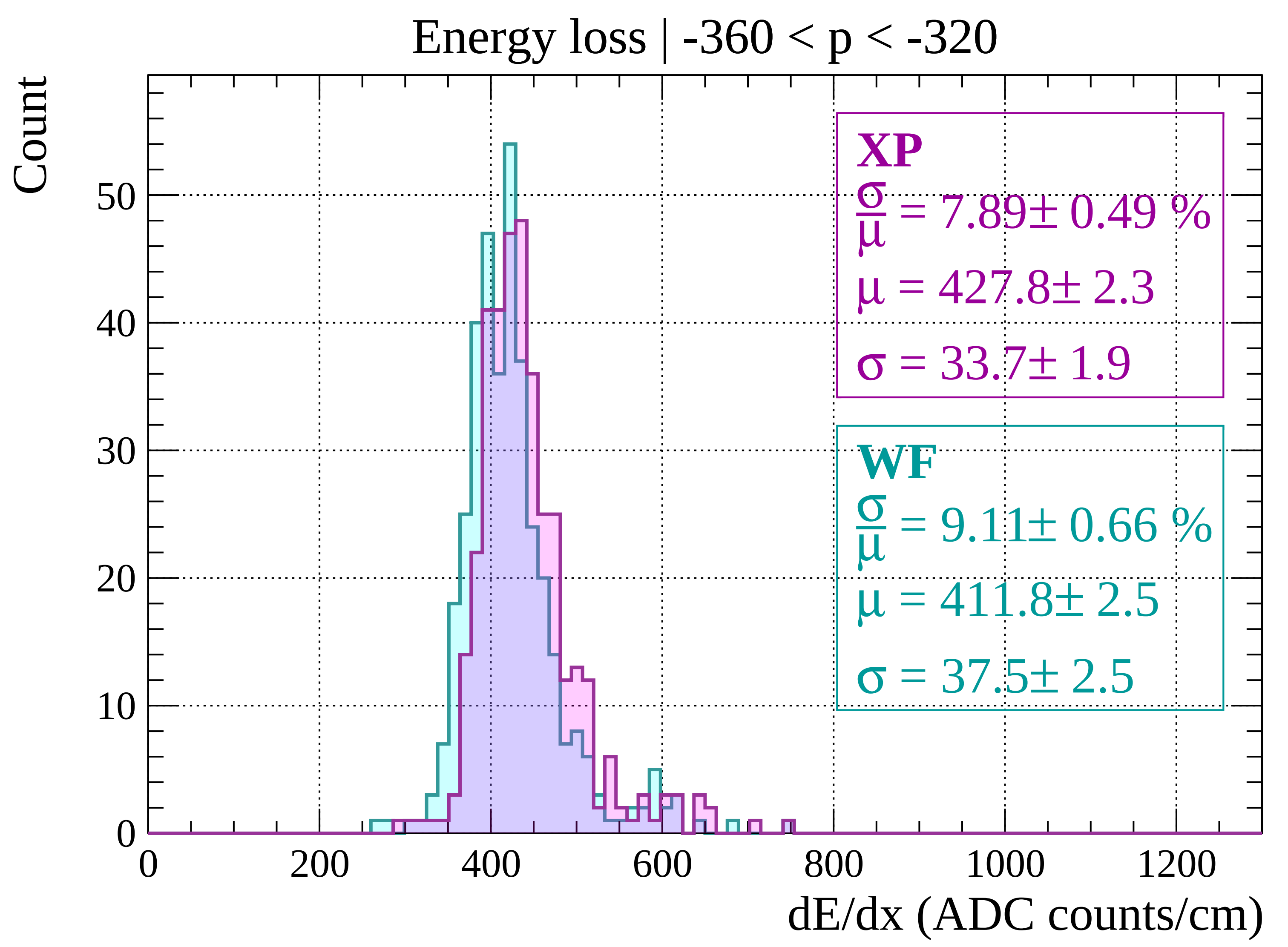
Energy loss | -520 < p < -480
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 9.20± 0.58 %
μ = 434.6± 2.4
σ = 40.0± 2.3
WF
σ/μ = 9.60± 0.65 %
μ = 415.8± 2.5
σ = 39.9± 2.5



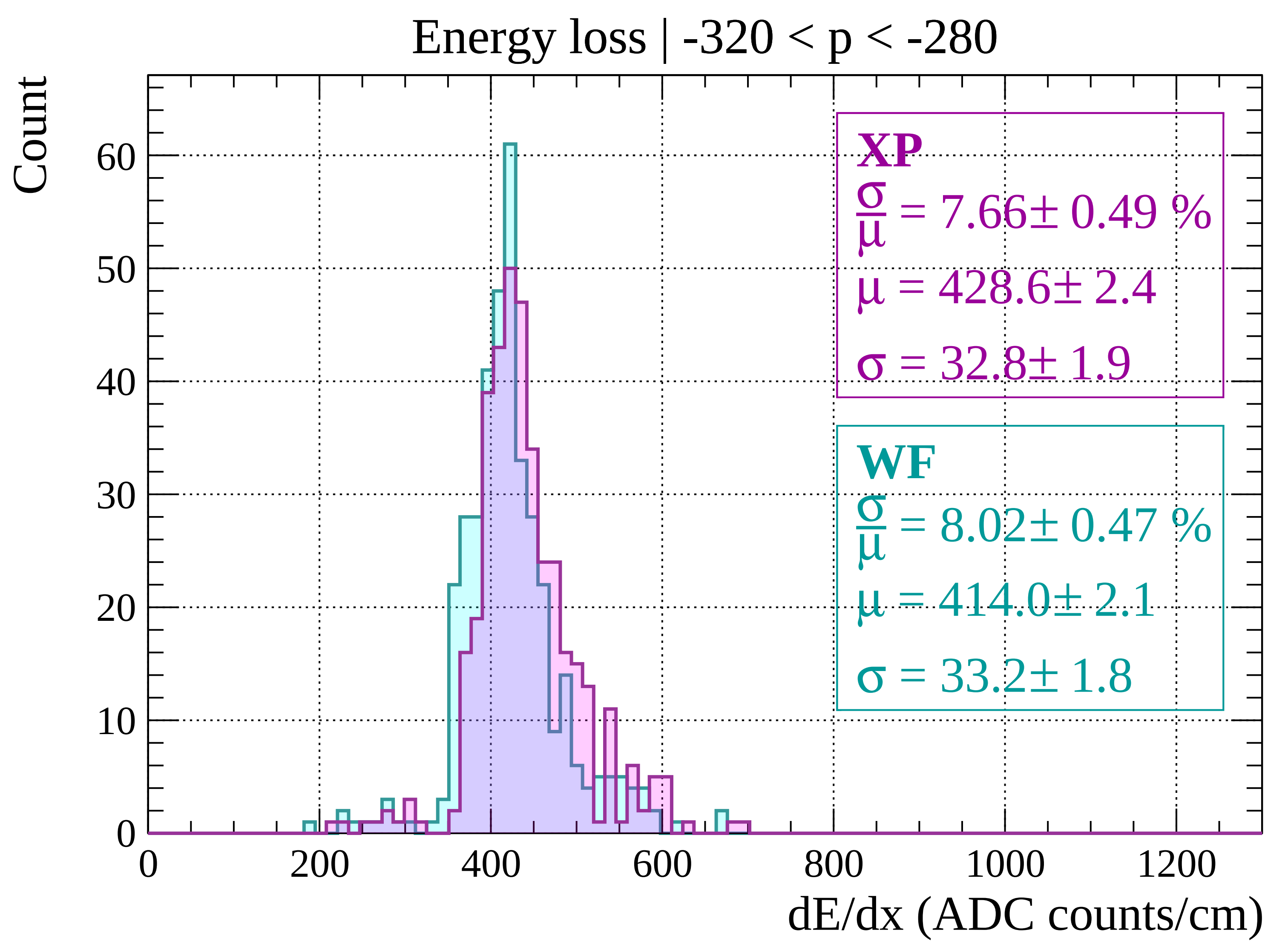
Energy loss | -480 < p < -440
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.96± 0.54 %
μ = 433.5± 2.3
σ = 34.5± 2.2
WF
$\frac{\sigma}{\mu}$ = 8.02± 0.63 %
μ = 416.5± 2.3
σ = 33.4± 2.4

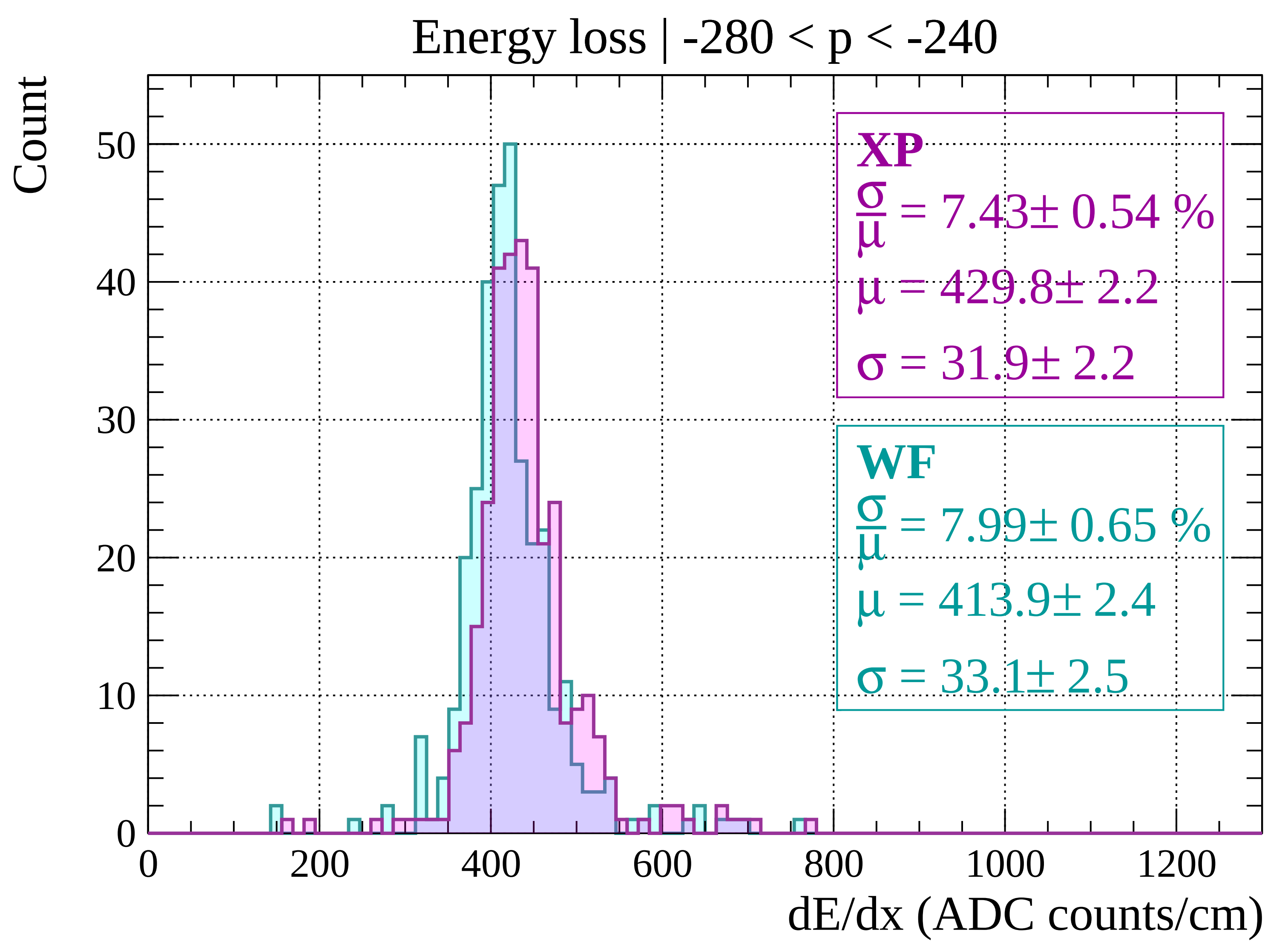

Energy loss | -440 < p < -400
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 8.00± 0.60 %
μ = 430.7± 2.3
σ = 34.5± 2.4
WF
σ/μ = 9.18± 0.58 %
μ = 416.6± 2.3
σ = 38.2± 2.2


Energy loss | -400 < p < -360
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.55± 0.66 %
μ = 422.0± 2.3
σ = 31.8± 2.6
WF
$\frac{\sigma}{\mu}$ = 8.05± 0.50 %
μ = 410.3± 2.1
σ = 33.0± 1.9
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200

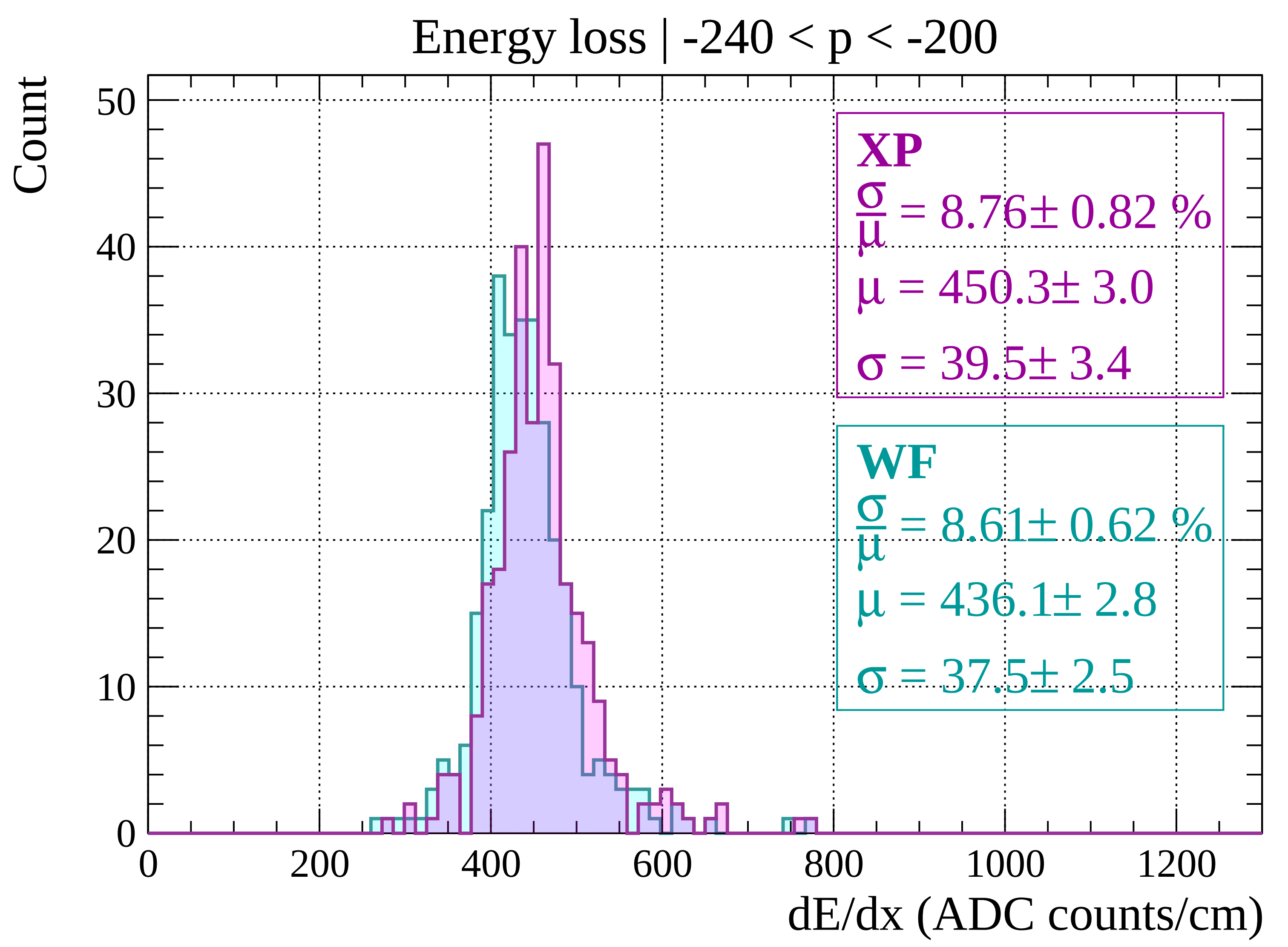


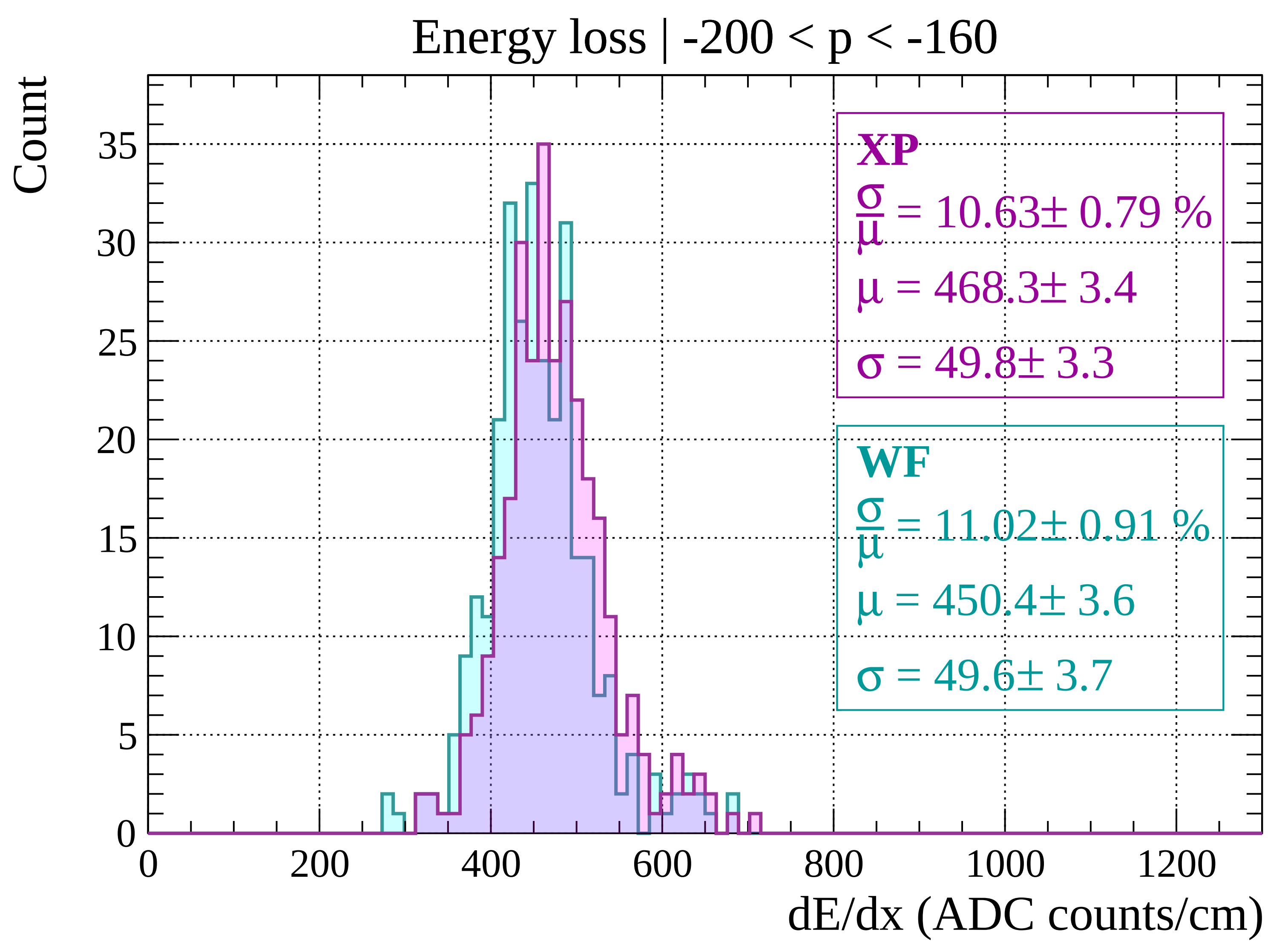
Energy loss | -360 < p < -320
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
XP
σ/μ = 7.89± 0.49 %
μ = 427.8± 2.3
σ = 33.7± 1.9
WF
σ/μ = 9.11± 0.66 %
μ = 411.8± 2.5
σ = 37.5± 2.5


Energy loss | -320 < p < -280
Count
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.66± 0.49 %
μ = 428.6± 2.4
σ = 32.8± 1.9
WF
σ/μ = 8.02± 0.47 %
μ = 414.0± 2.1
σ = 33.2± 1.8

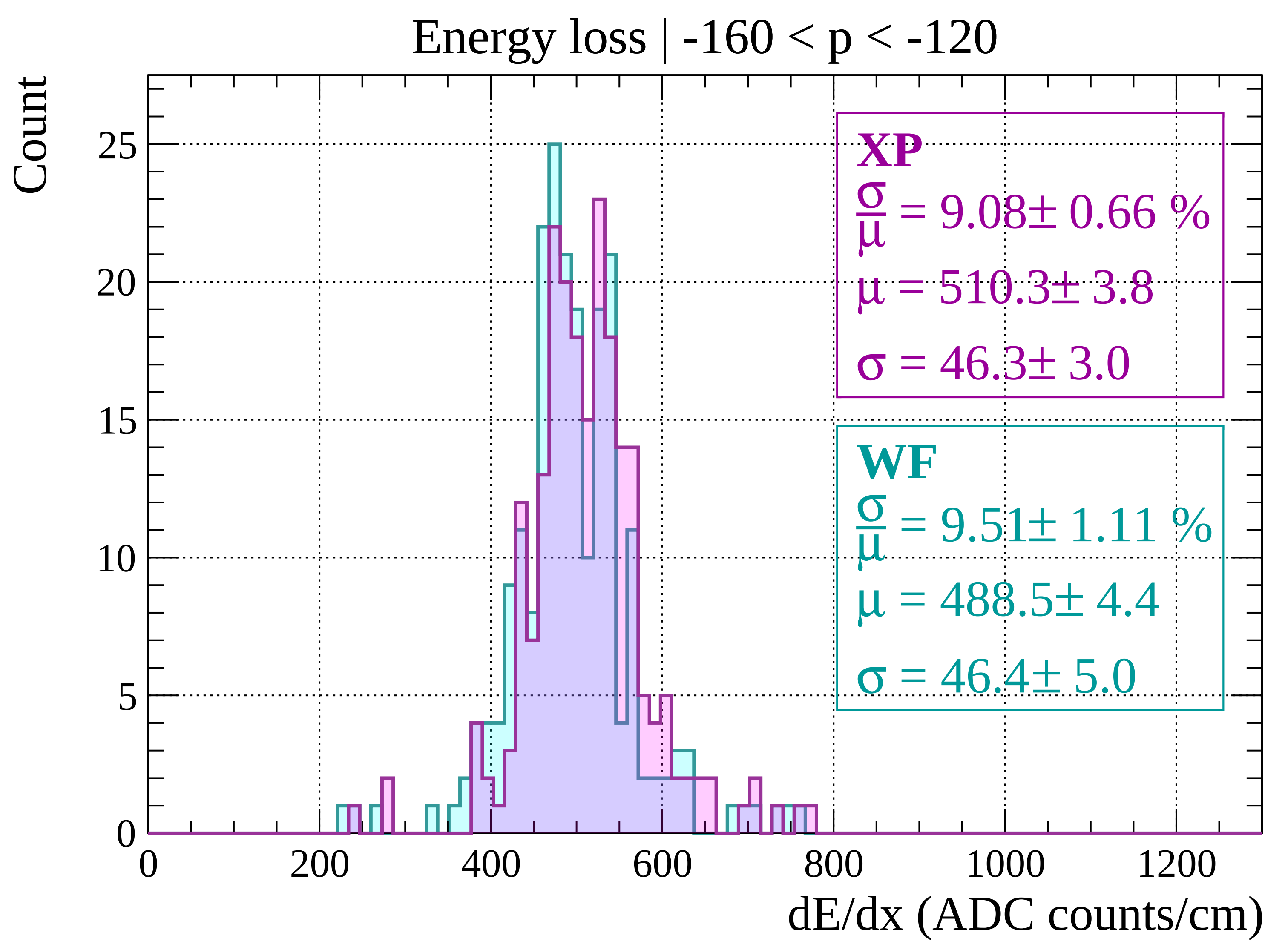


Energy loss | -280 < p < -240
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.43± 0.54 %
μ = 429.8± 2.2
σ = 31.9± 2.2
WF
σ/μ = 7.99± 0.65 %
μ = 413.9± 2.4
σ = 33.1± 2.5
0
10
20
30
40
50
0
200
400
600
800
1000
1200

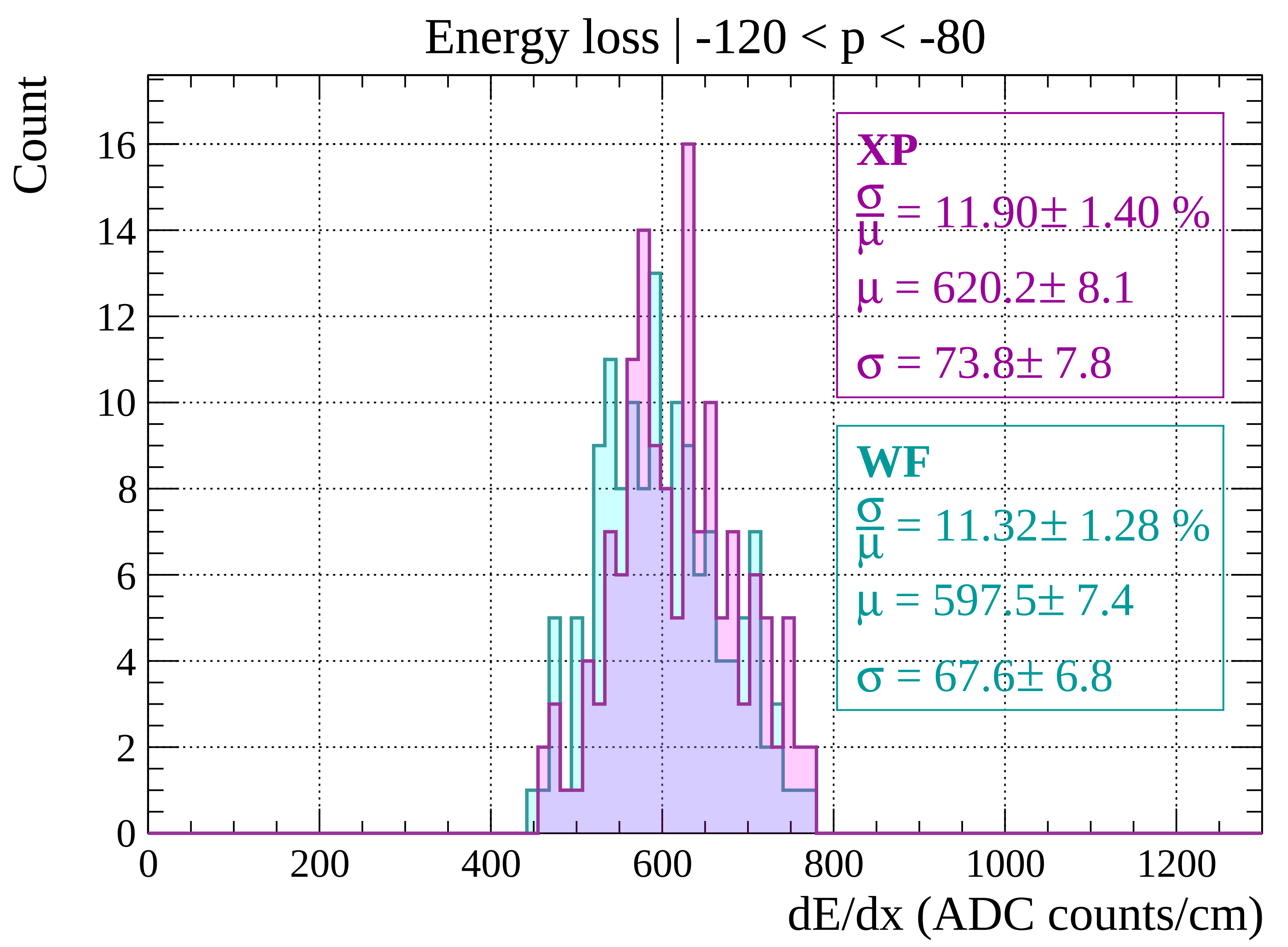

Energy loss | -240 < p < -200
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.76± 0.82 %
μ = 450.3± 3.0
σ = 39.5± 3.4
WF
σ/μ = 8.61± 0.62 %
μ = 436.1± 2.8
σ = 37.5± 2.5
0
10
20
30
40
50
0
200
400
600
800
1000
1200



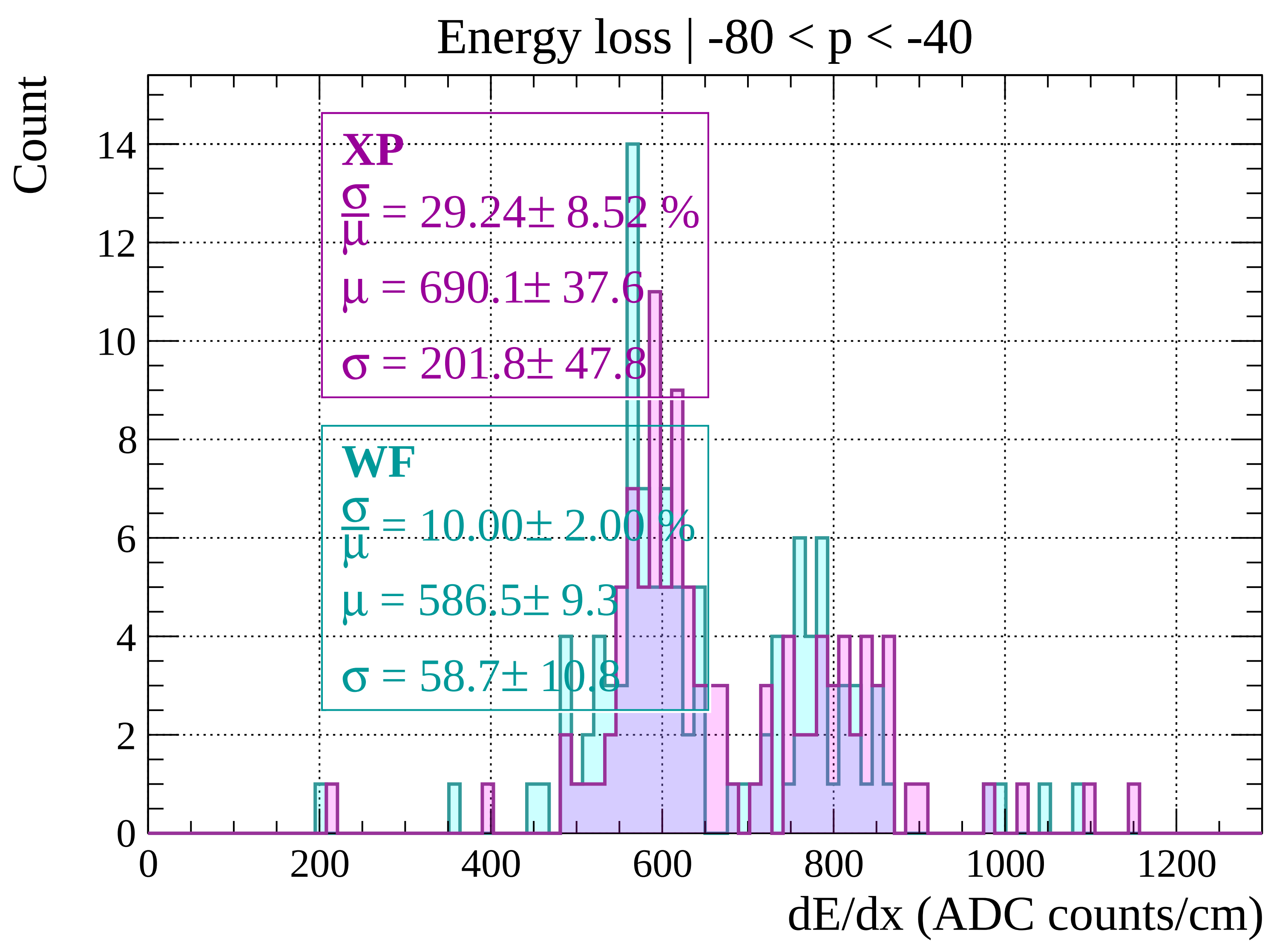
Energy loss | -200 < p < -160
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
XP
σ/μ = 10.63± 0.79 %
μ = 468.3± 3.4
σ = 49.8± 3.3
WF
σ/μ = 11.02± 0.91 %
μ = 450.4± 3.6
σ = 49.6± 3.7

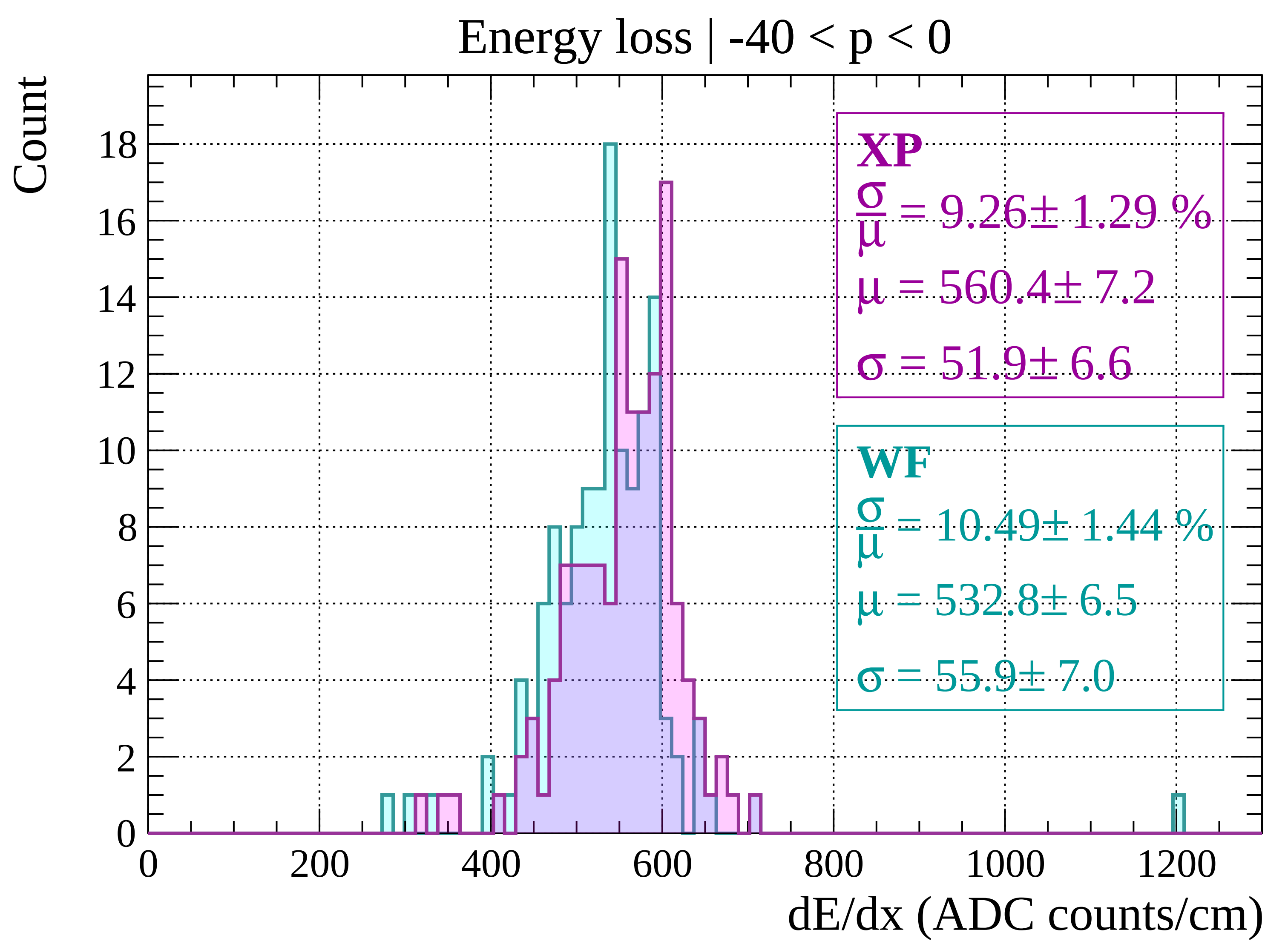


Energy loss | -160 < p < -120
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
XP
σ/μ = 9.08± 0.66 %
μ = 510.3± 3.8
σ = 46.3± 3.0
WF
σ/μ = 9.51± 1.11 %
μ = 488.5± 4.4
σ = 46.4± 5.0

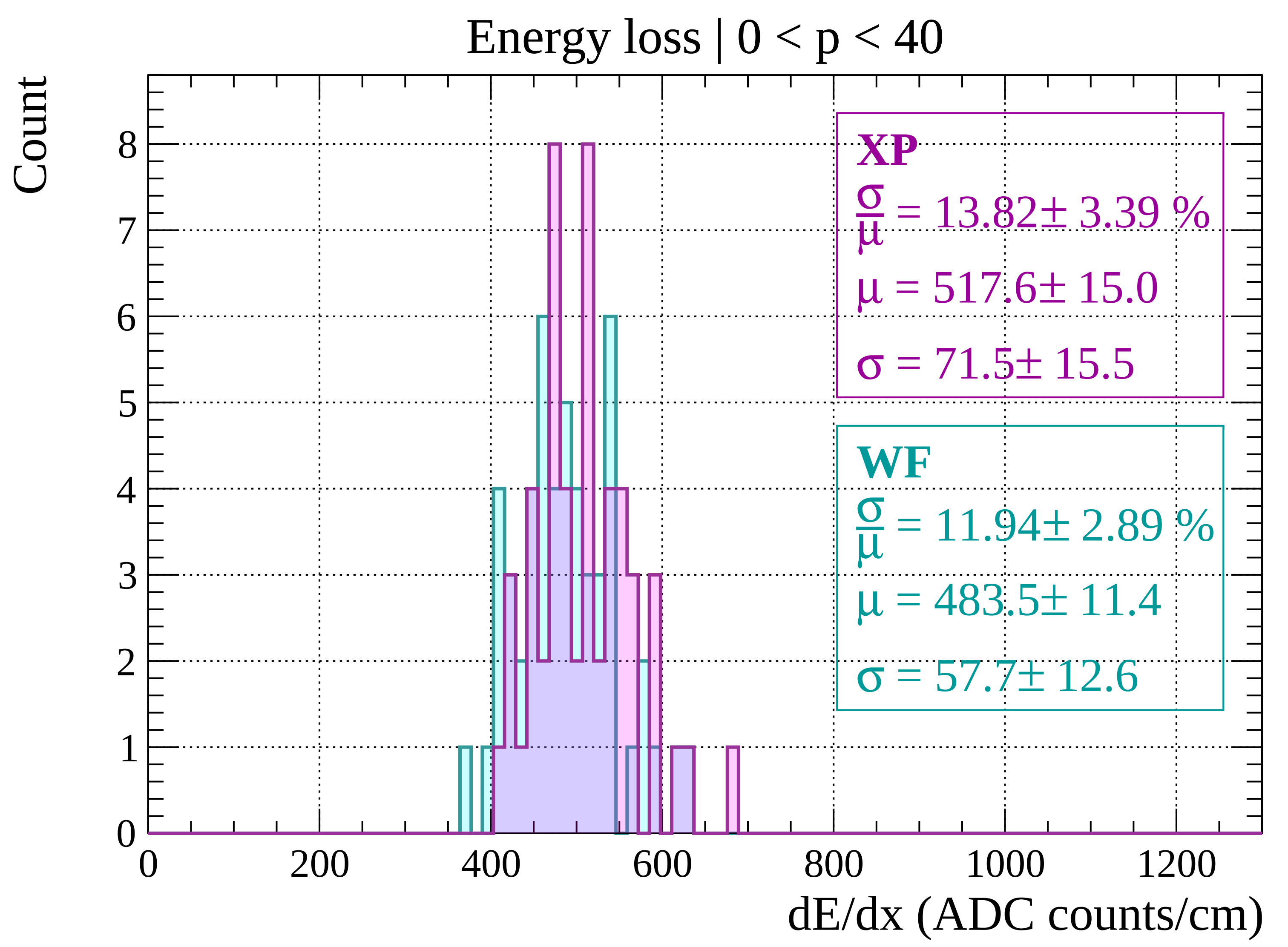


Energy loss | -120 < p < -80
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
0
200
400
600
800
1000
1200
XP
σ/μ = 11.90± 1.40 %
μ = 620.2± 8.1
σ = 73.8± 7.8
WF
σ/μ = 11.32± 1.28 %
μ = 597.5± 7.4
σ = 67.6± 6.8

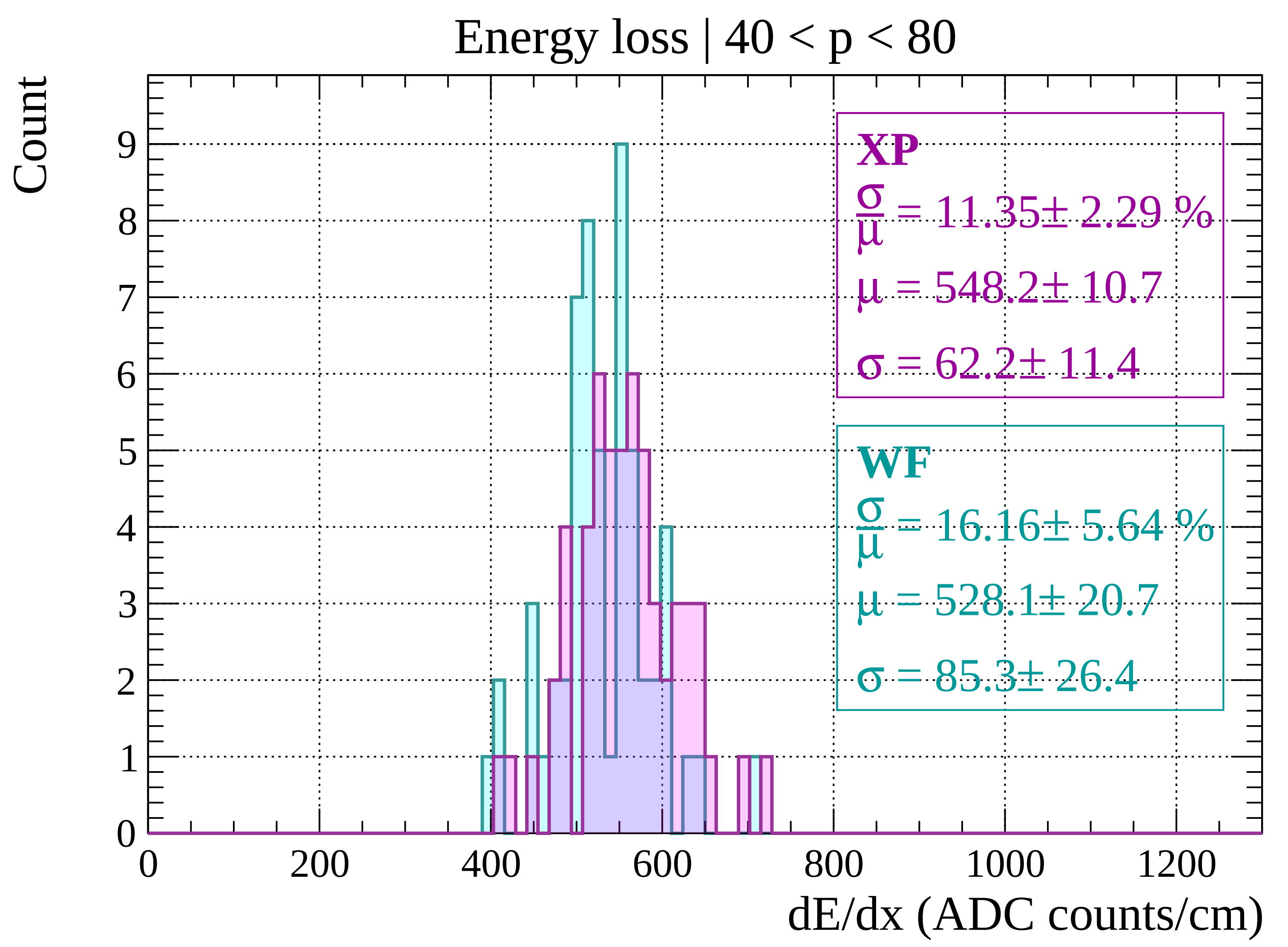

Energy loss | -80 < p < -40
XP
σ/μ = 29.24± 8.52 %
μ = 690.1± 37.6
σ = 201.8± 47.8
WF
σ/μ = 10.00± 2.00 %
μ = 586.5± 9.3
σ = 58.7± 10.8
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200



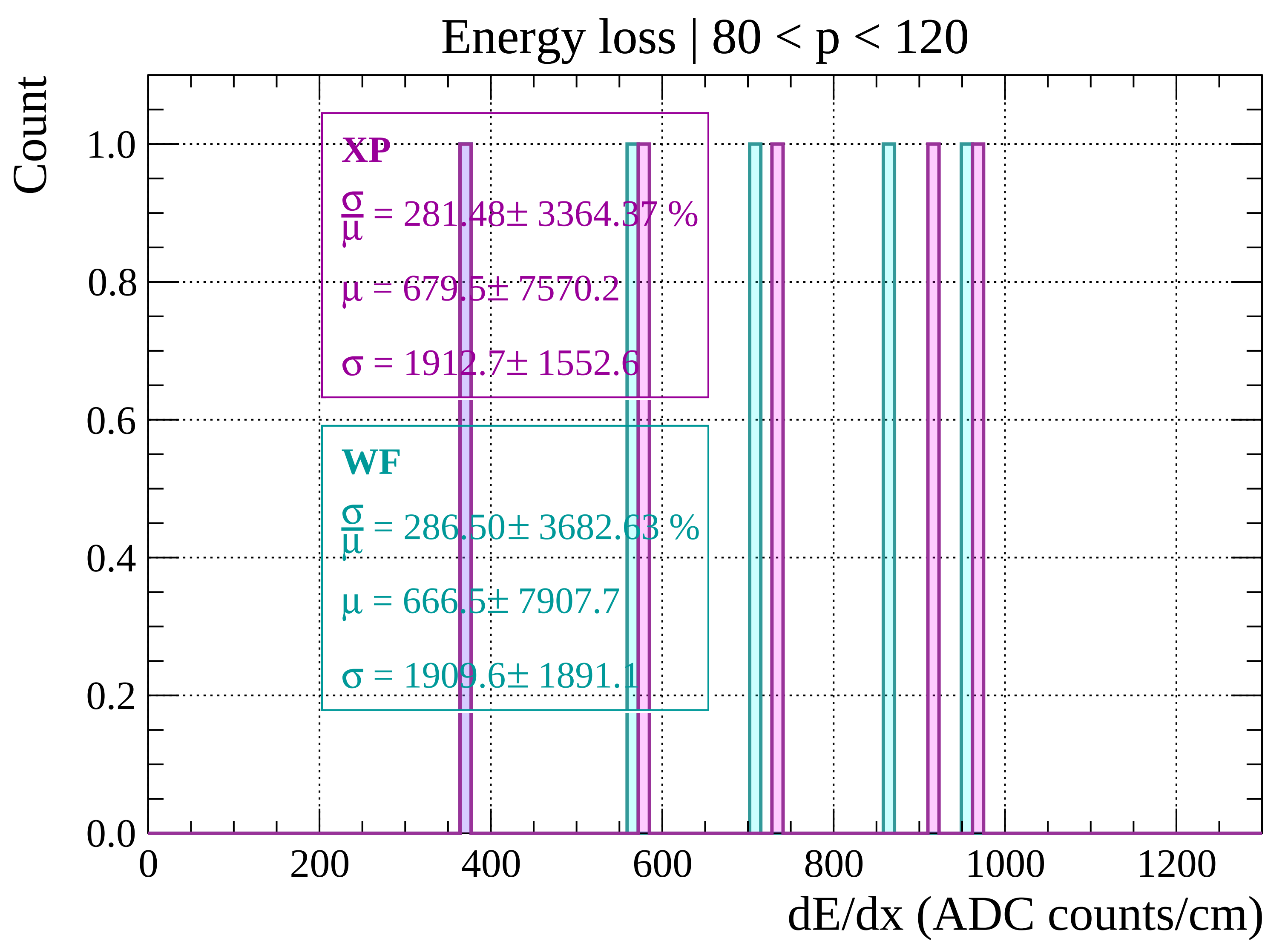
Energy loss | -40 < p < 0
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
18
XP
σ/μ = 9.26± 1.29 %
μ = 560.4± 7.2
σ = 51.9± 6.6
WF
σ/μ = 10.49± 1.44 %
μ = 532.8± 6.5
σ = 55.9± 7.0



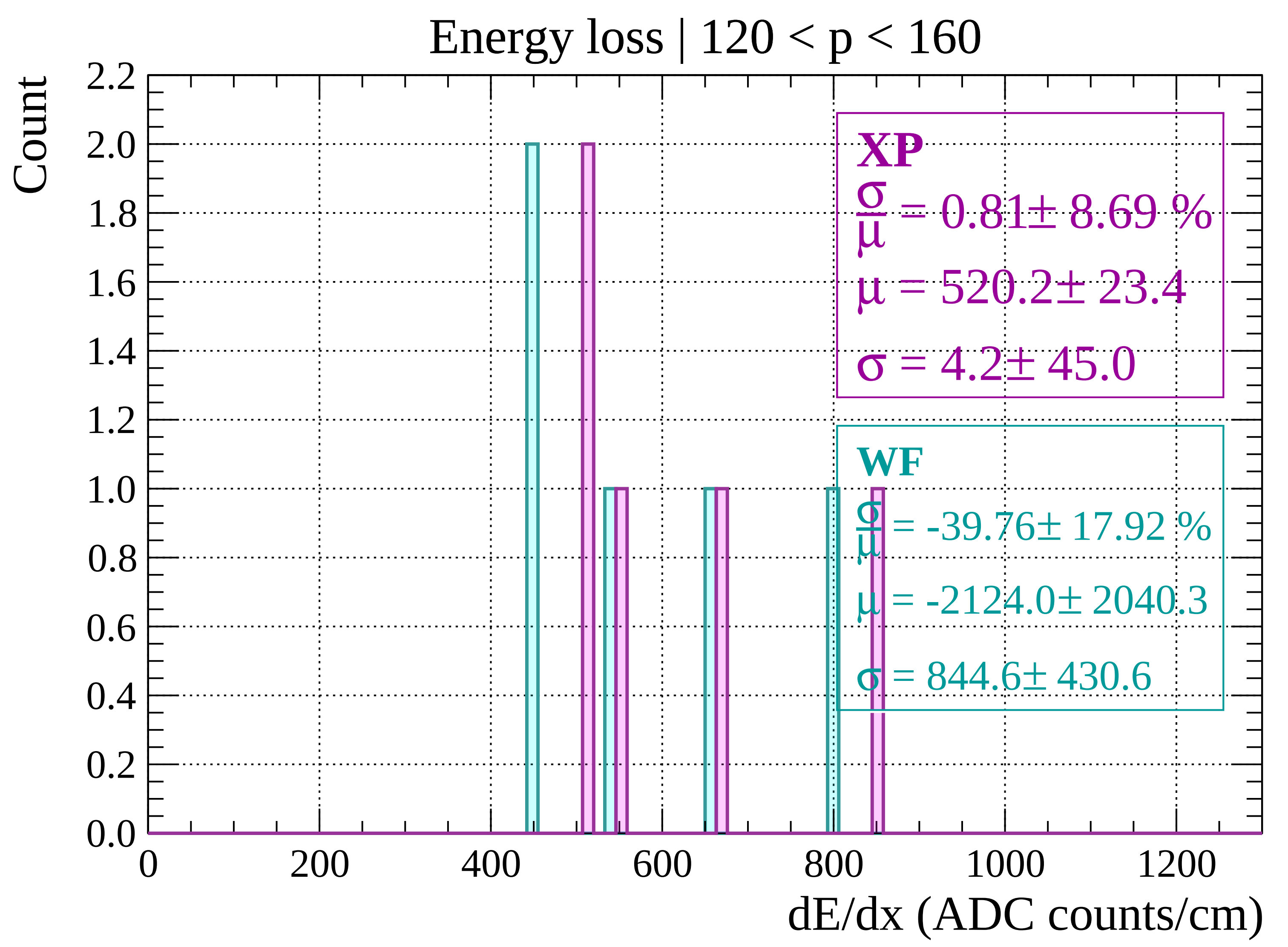
Energy loss | 0 < p < 40
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7
8
XP
σ/μ = 13.82± 3.39 %
μ = 517.6± 15.0
σ = 71.5± 15.5
WF
σ/μ = 11.94± 2.89 %
μ = 483.5± 11.4
σ = 57.7± 12.6



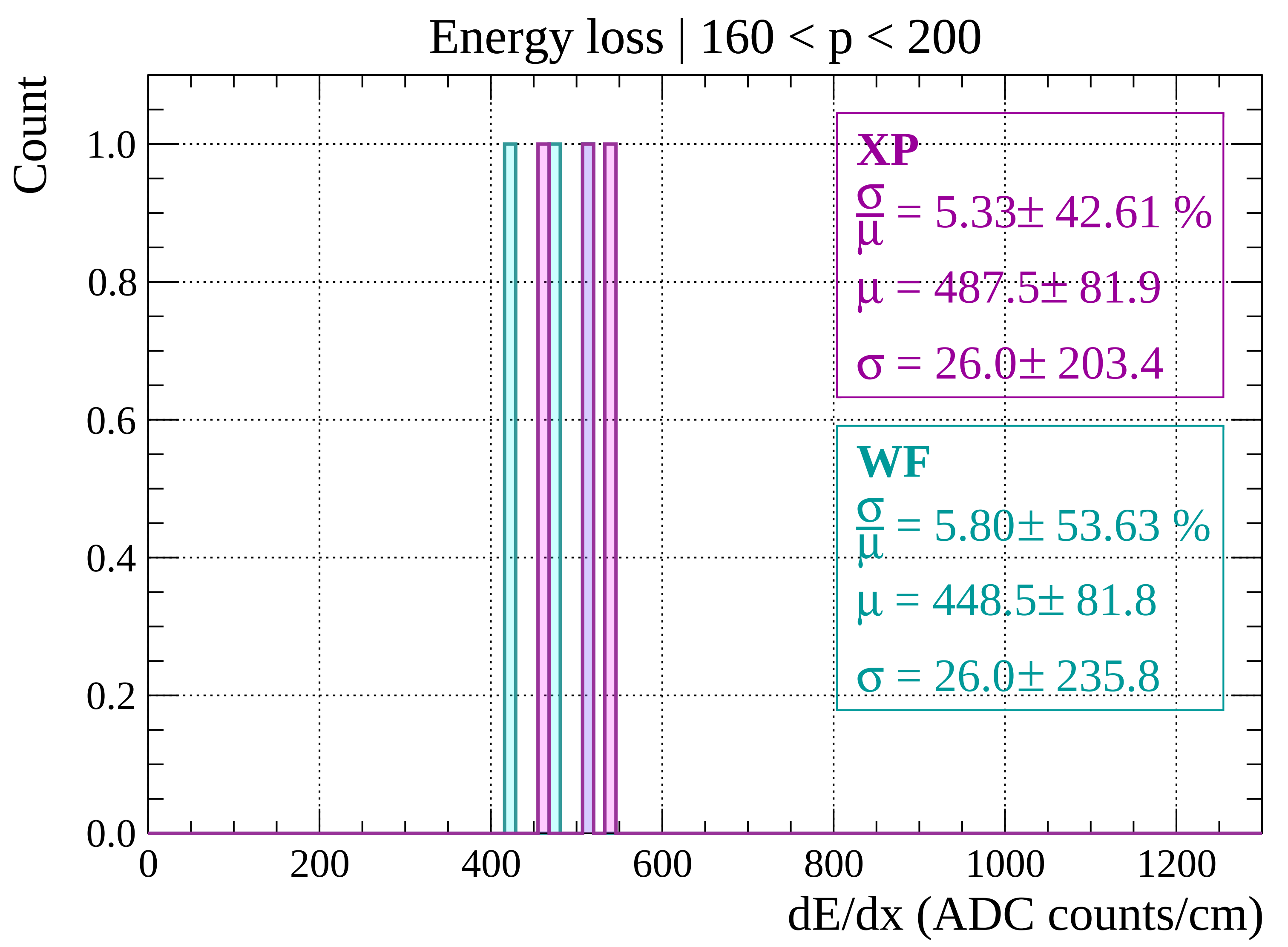
Energy loss | 40 < p < 80
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
9
XP
σ/μ = 11.35± 2.29 %
μ = 548.2± 10.7
σ = 62.2± 11.4
WF
σ/μ = 16.16± 5.64 %
μ = 528.1± 20.7
σ = 85.3± 26.4



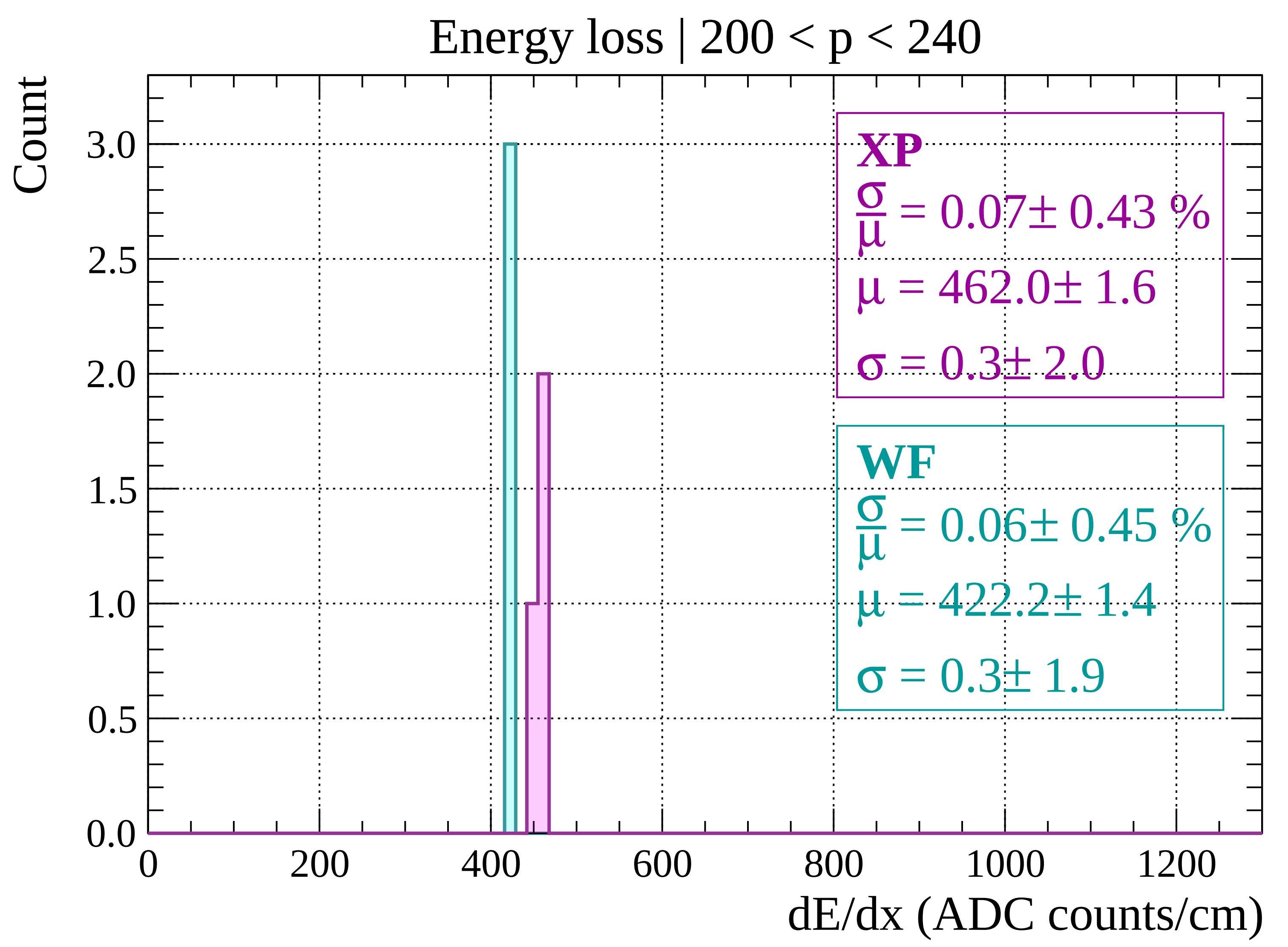
Energy loss | 80 < p < 120
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 281.48± 3364.37 %
μ = 679.5± 7570.2
σ = 1912.7± 1552.6
WF
$\frac{\sigma}{\mu}$ = 286.50± 3682.63 %
μ = 666.5± 7907.7
σ = 1909.6± 1891.1

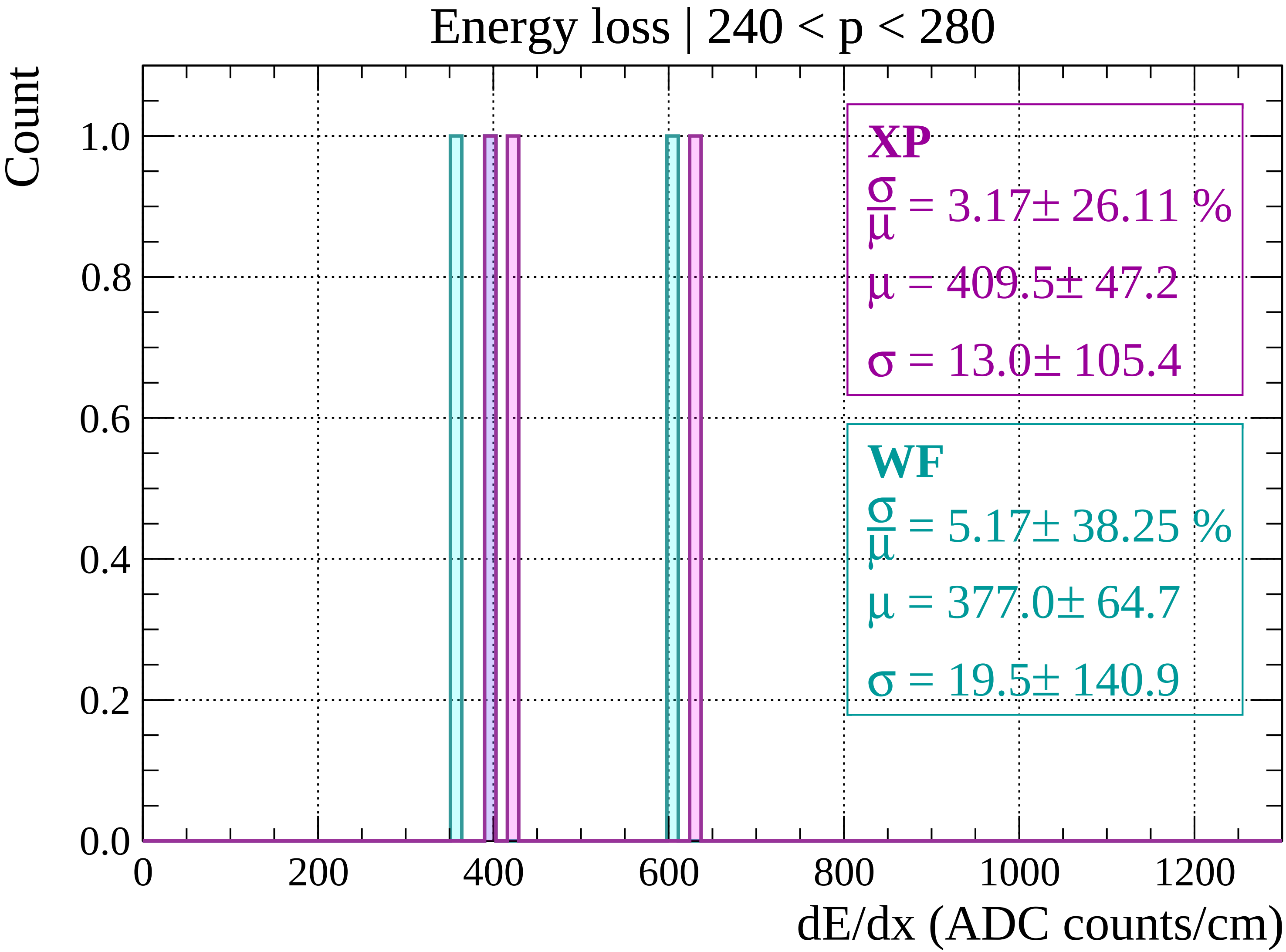


Energy loss | 120 < p < 160
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 0.81± 8.69 %
μ = 520.2± 23.4
σ = 4.2± 45.0
WF
$\frac{\sigma}{\mu}$ = -39.76± 17.92 %
μ = -2124.0± 2040.3
σ = 844.6± 430.6

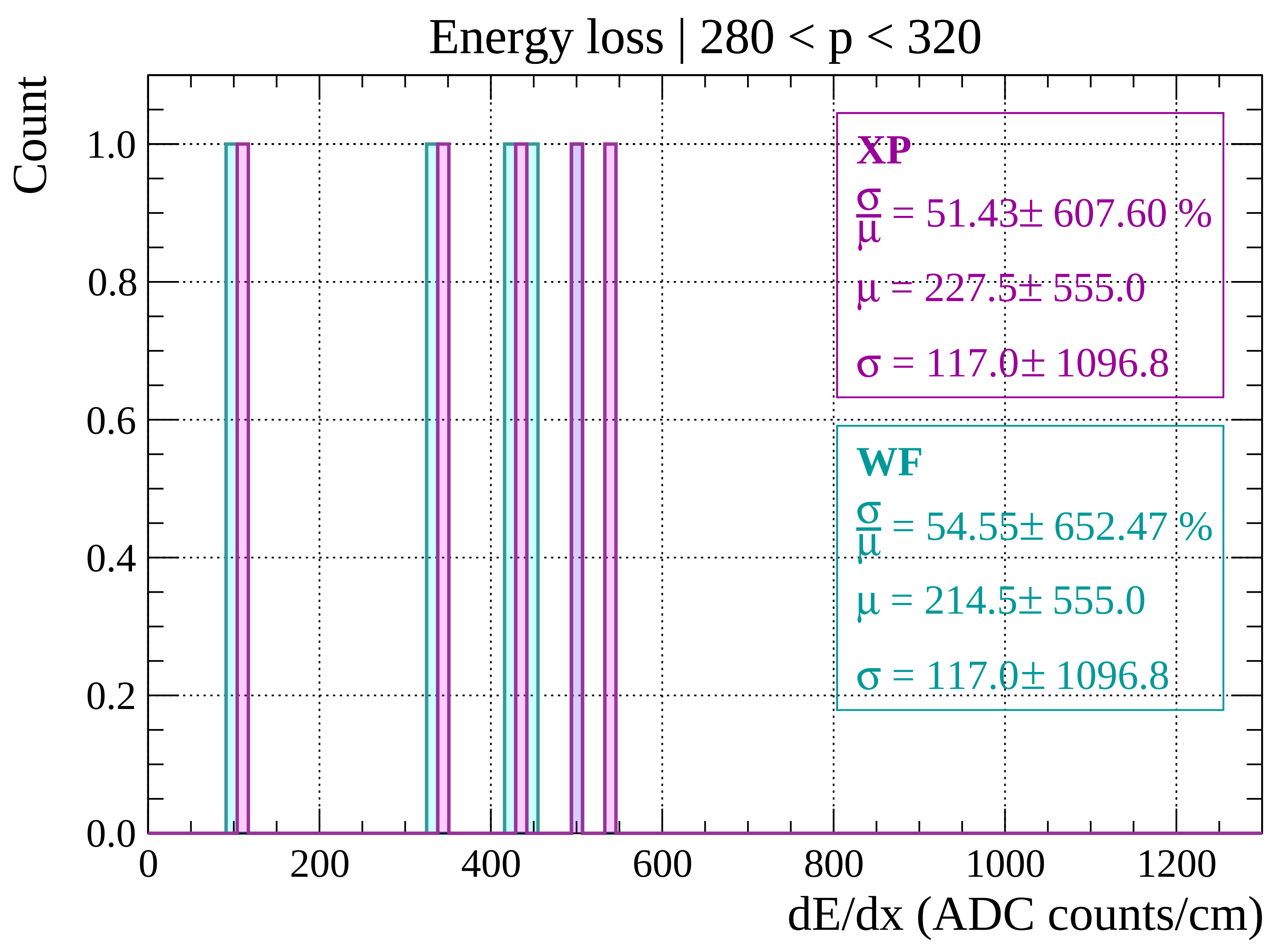


Energy loss | 160 < p < 200
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.33± 42.61 %
μ = 487.5± 81.9
σ = 26.0± 203.4
WF
σ/μ = 5.80± 53.63 %
μ = 448.5± 81.8
σ = 26.0± 235.8

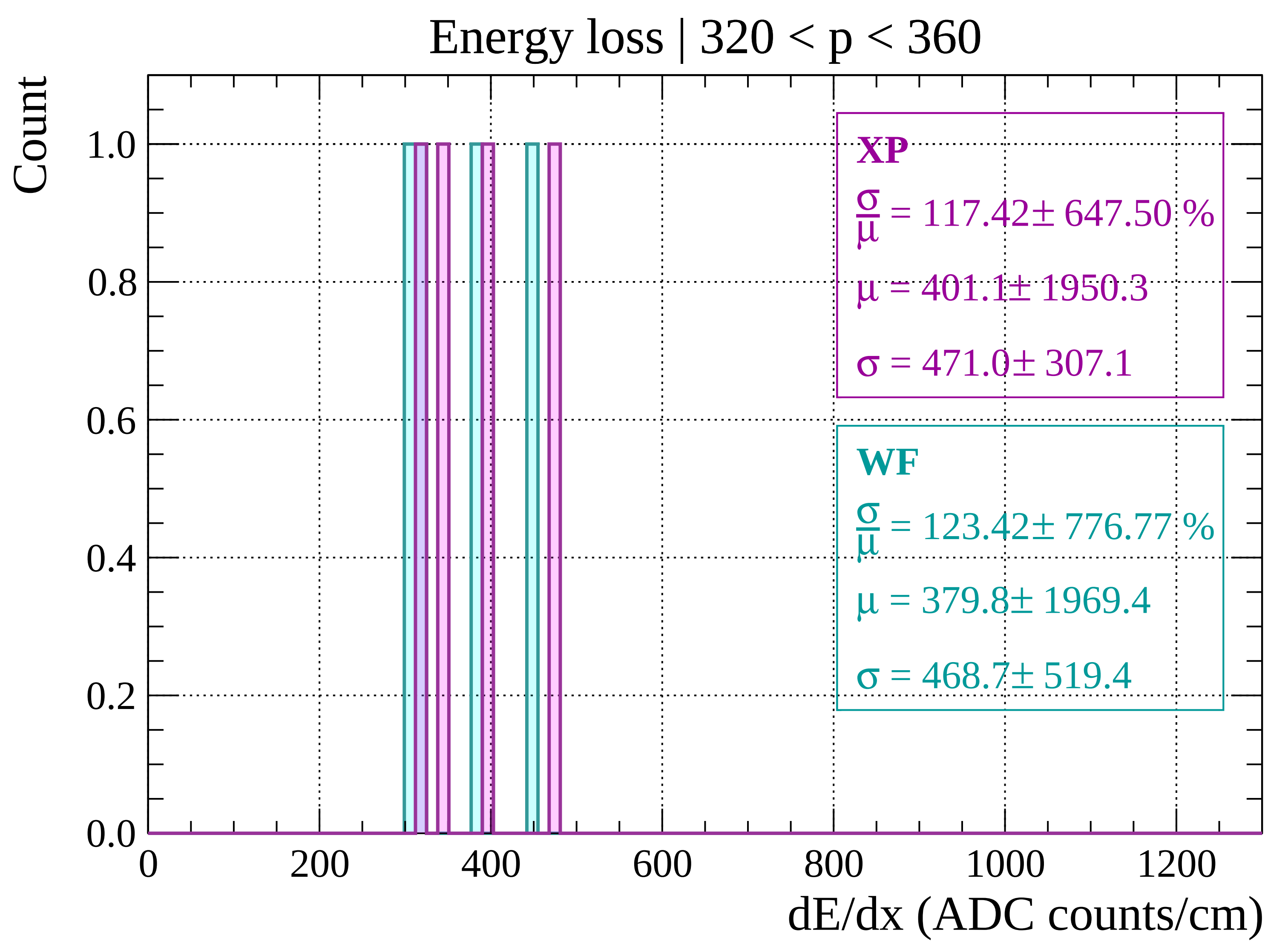

Energy loss | 200 < p < 240
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 0.07± 0.43 %
μ = 462.0± 1.6
σ = 0.3± 2.0
WF
$\frac{\sigma}{\mu}$ = 0.06± 0.45 %
μ = 422.2± 1.4
σ = 0.3± 1.9



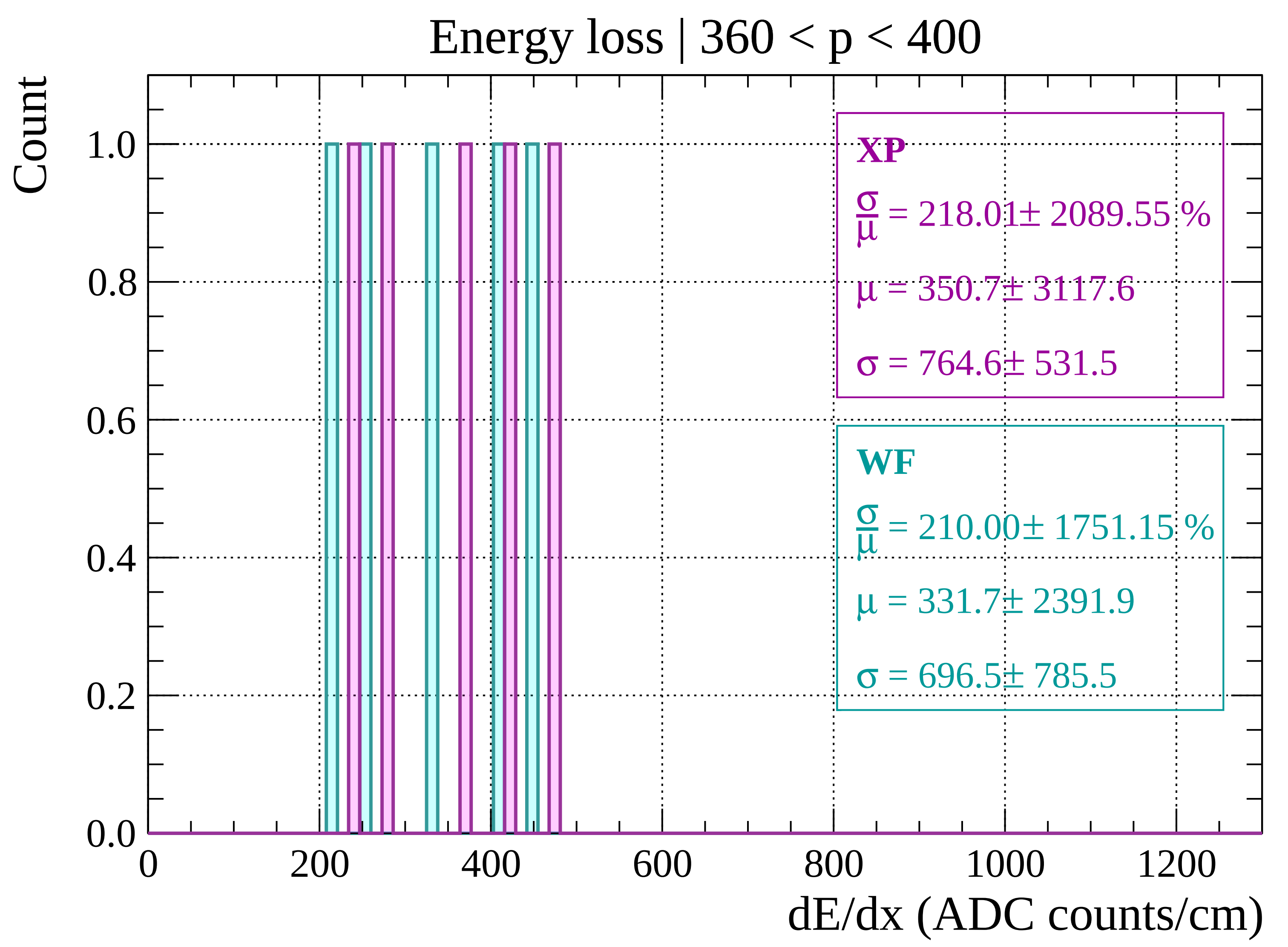
Energy loss | 240 < p < 280
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 3.17± 26.11 %
μ = 409.5± 47.2
σ = 13.0± 105.4
WF
$\frac{\sigma}{\mu}$ = 5.17± 38.25 %
μ = 377.0± 64.7
σ = 19.5± 140.9



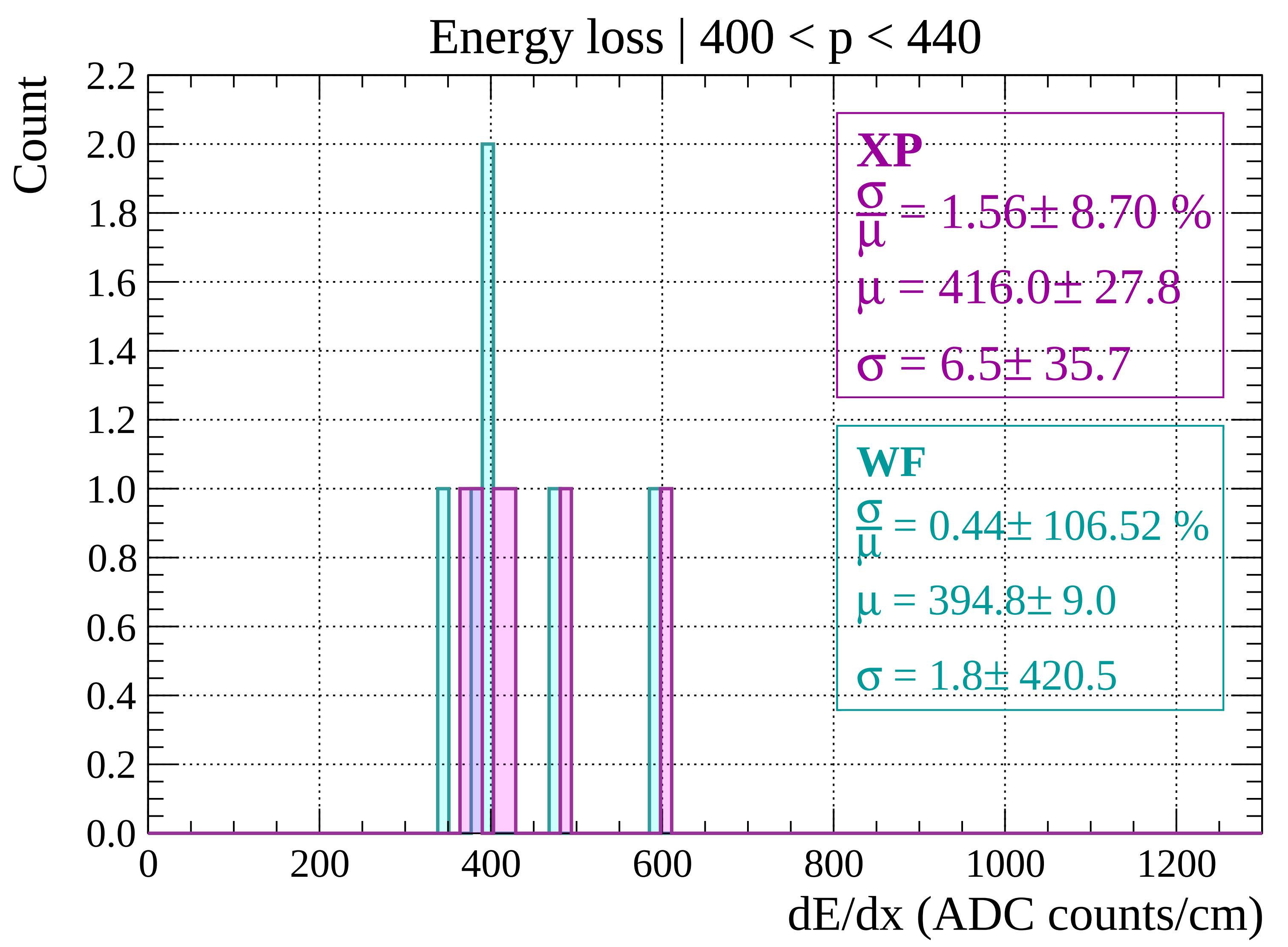
Energy loss | 280 < p < 320
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 51.43± 607.60 %
μ = 227.5± 555.0
σ = 117.0± 1096.8
WF
σ/μ = 54.55± 652.47 %
μ = 214.5± 555.0
σ = 117.0± 1096.8



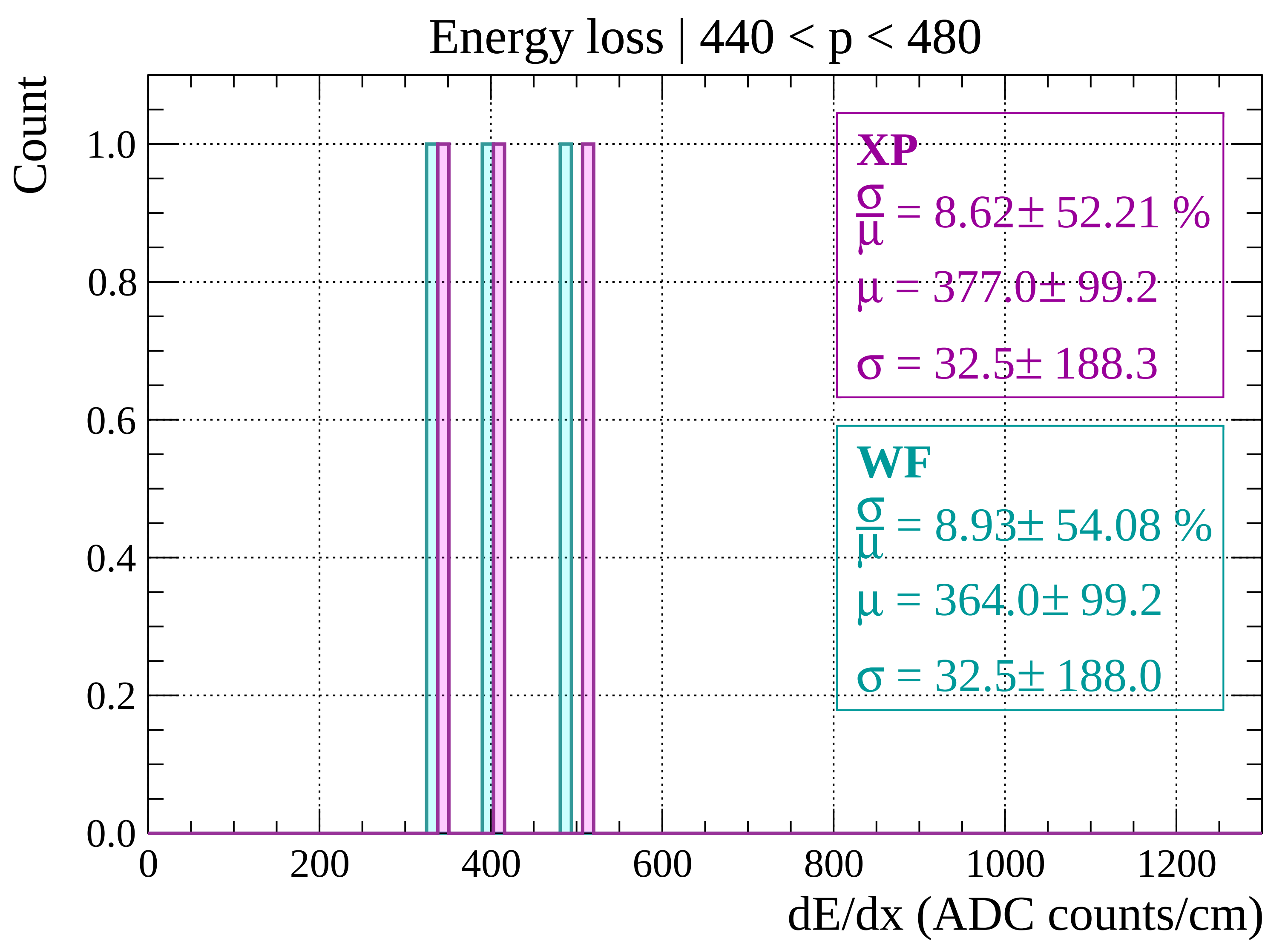
Energy loss | 320 < p < 360
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 117.42± 647.50 %
μ = 401.1± 1950.3
σ = 471.0± 307.1
WF
σ/μ = 123.42± 776.77 %
μ = 379.8± 1969.4
σ = 468.7± 519.4



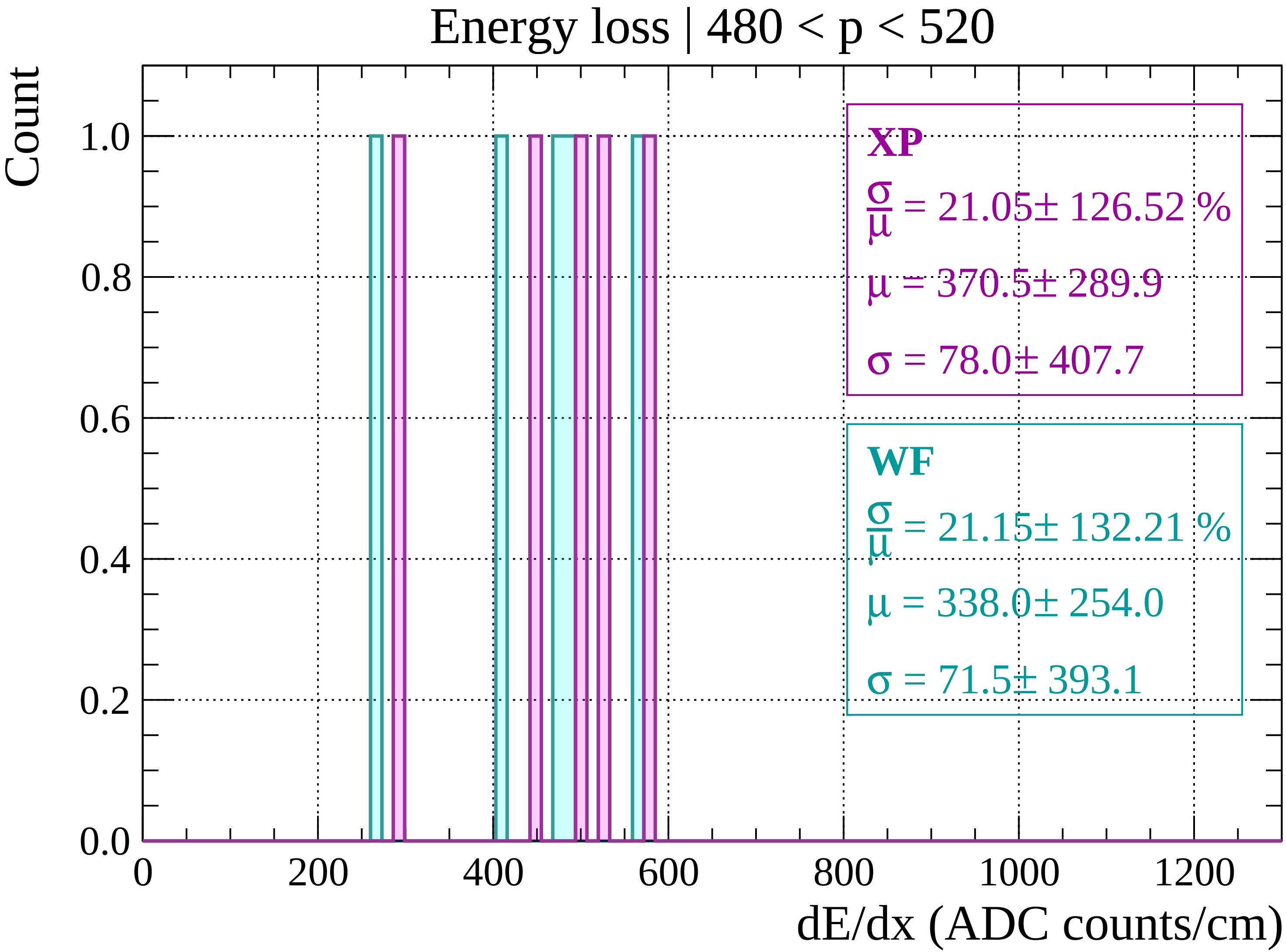
Energy loss | 360 < p < 400
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 218.01± 2089.55 %
μ = 350.7± 3117.6
σ = 764.6± 531.5
WF
$\frac{\sigma}{\mu}$ = 210.00± 1751.15 %
μ = 331.7± 2391.9
σ = 696.5± 785.5



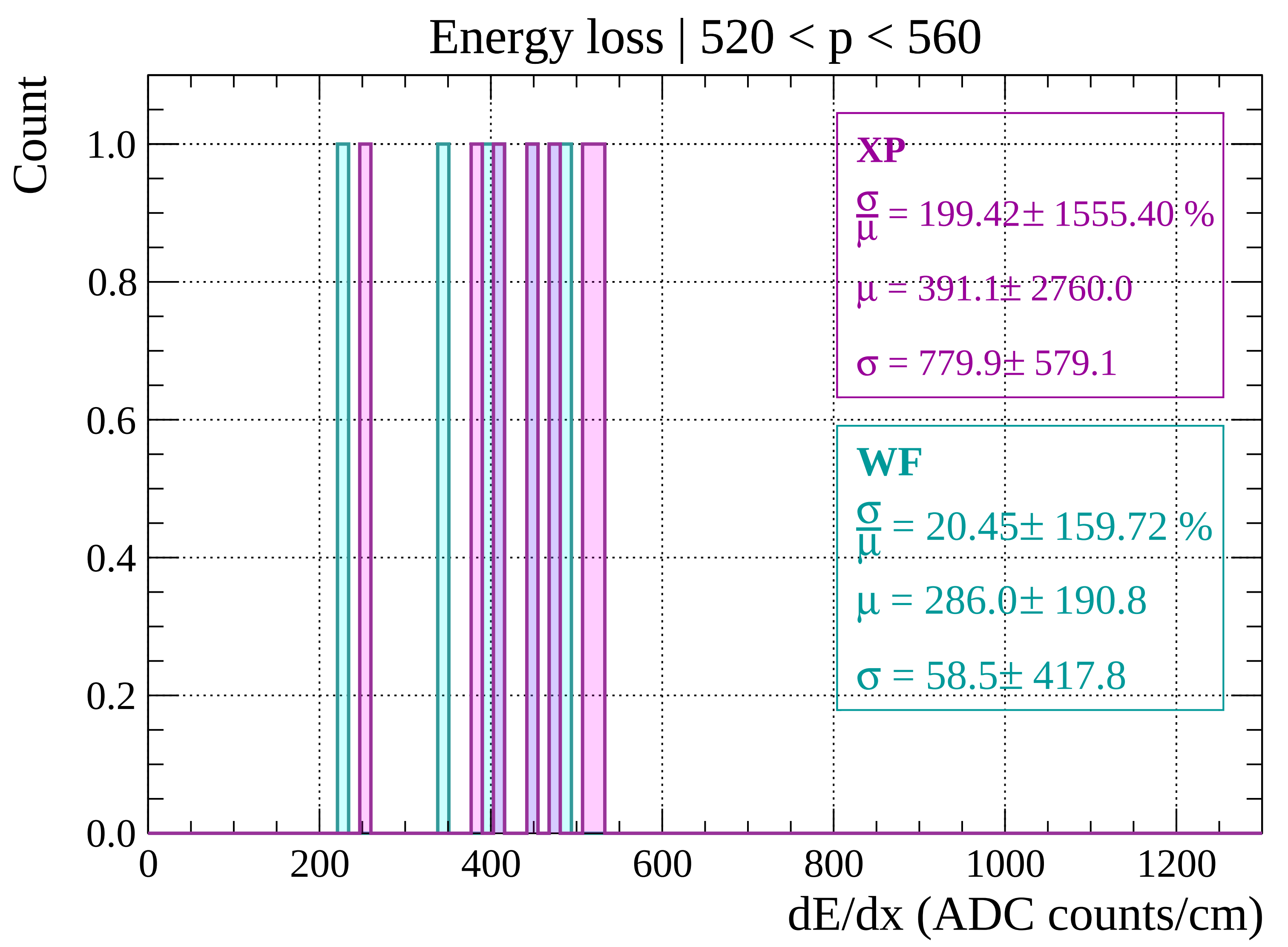
Energy loss | 400 < p < 440
Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 1.56± 8.70 %
μ = 416.0± 27.8
σ = 6.5± 35.7
WF
σ/μ = 0.44± 106.52 %
μ = 394.8± 9.0
σ = 1.8± 420.5



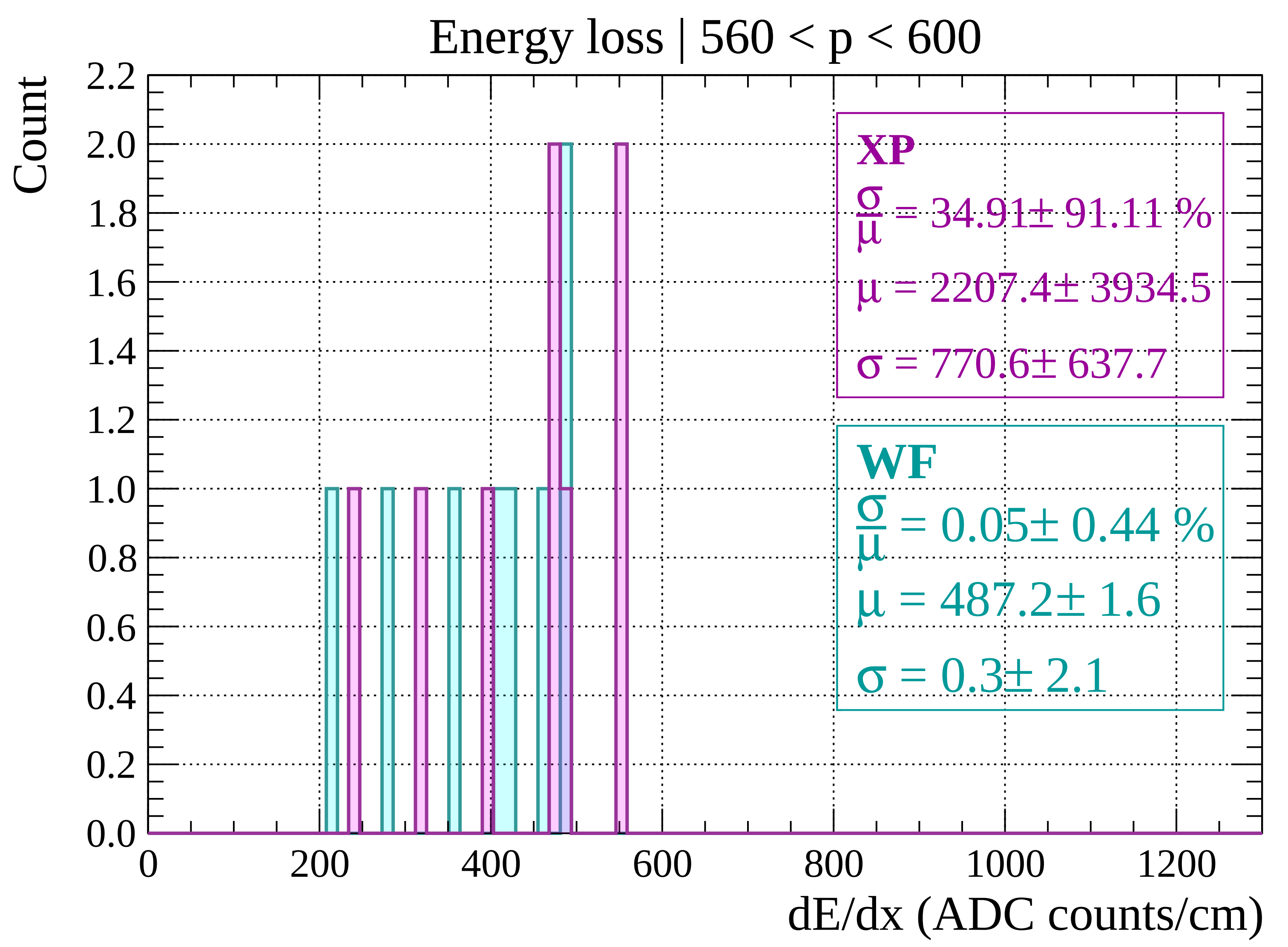
Energy loss | 440 < p < 480
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.62± 52.21 %
μ = 377.0± 99.2
σ = 32.5± 188.3
WF
σ/μ = 8.93± 54.08 %
μ = 364.0± 99.2
σ = 32.5± 188.0



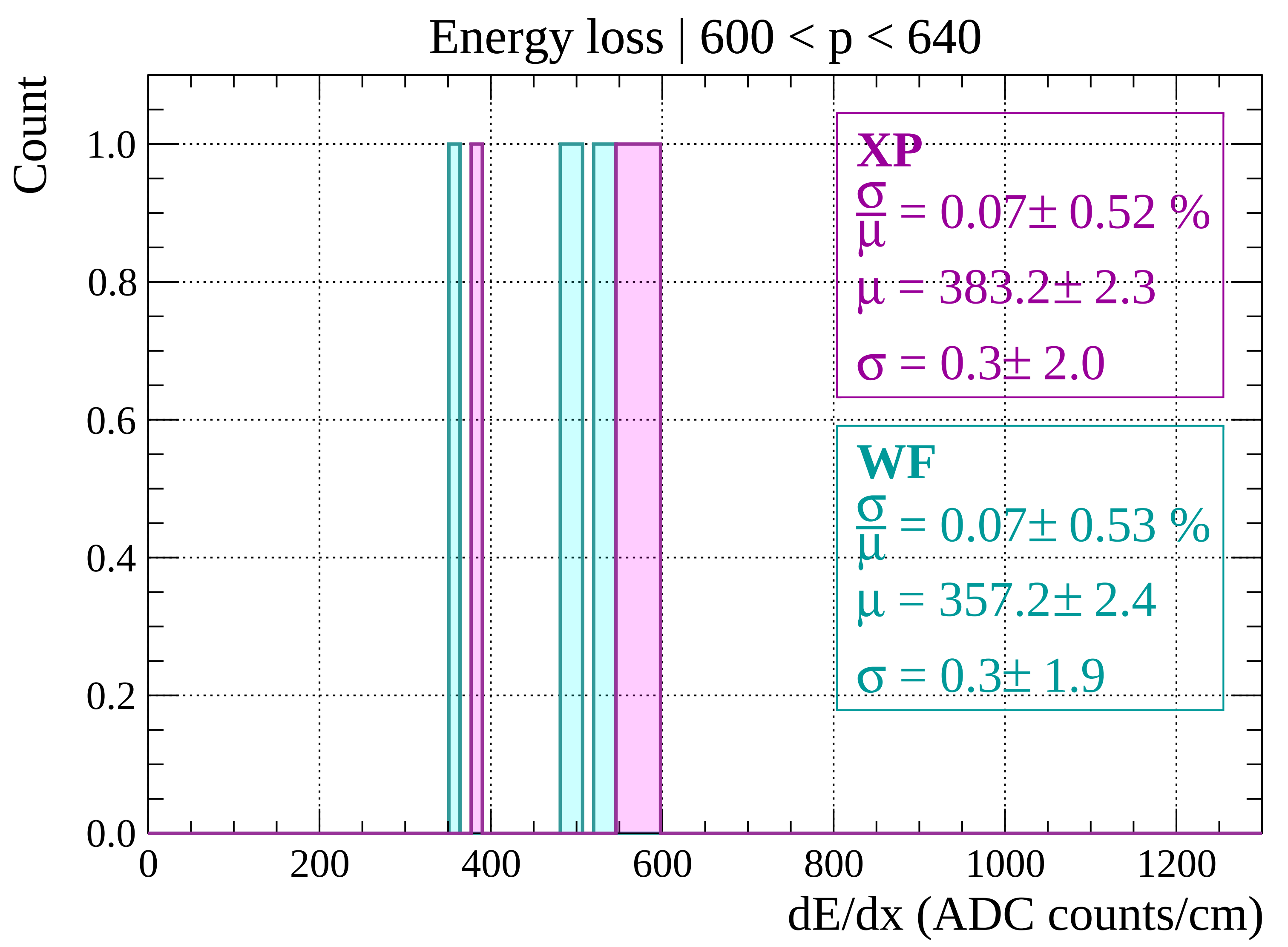
Energy loss | 480 < p < 520
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = 21.05± 126.52 %
μ = 370.5± 289.9
σ = 78.0± 407.7
WF
σ/μ = 21.15± 132.21 %
μ = 338.0± 254.0
σ = 71.5± 393.1



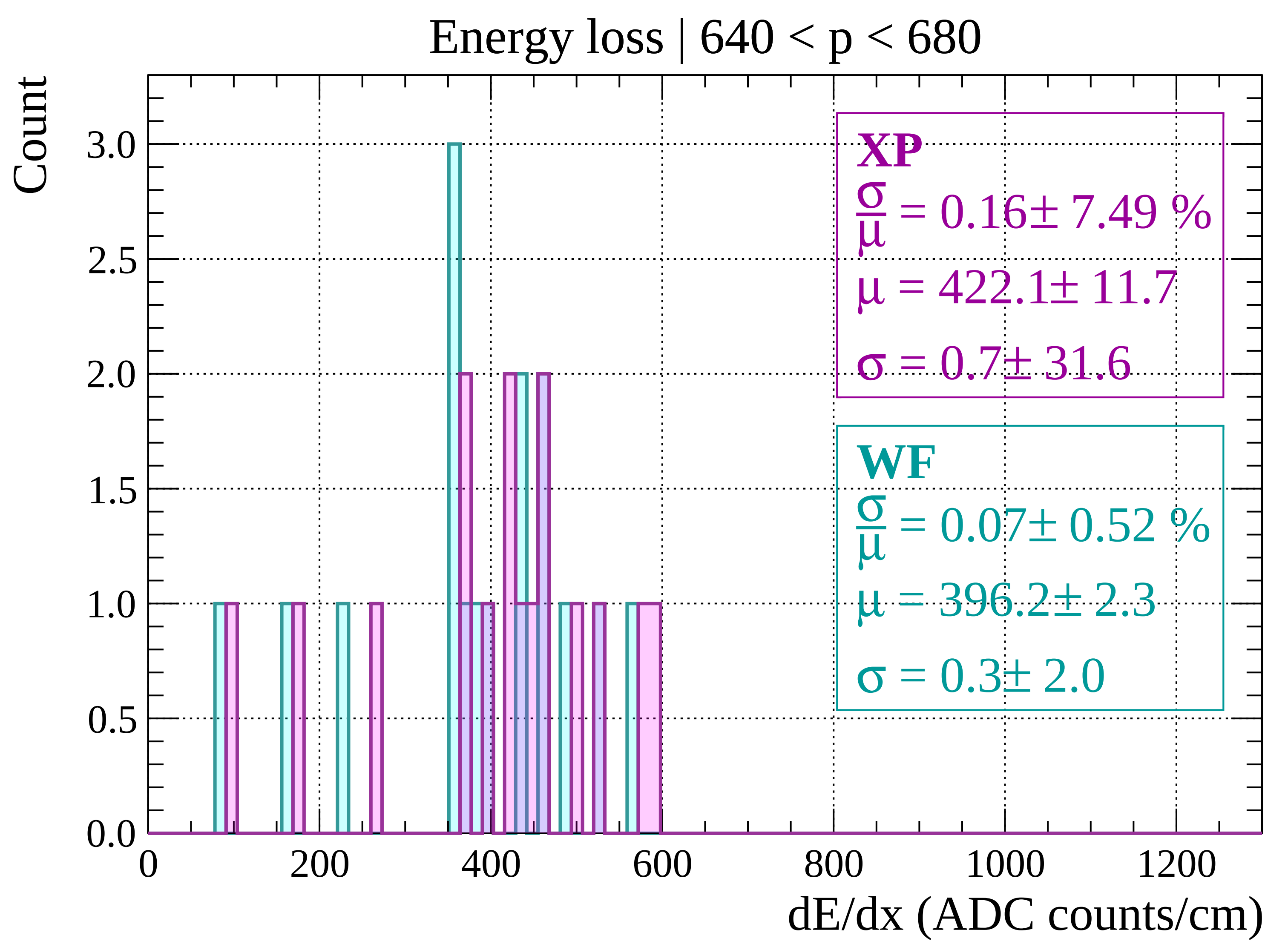
Energy loss | 520 < p < 560
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 199.42± 1555.40 %
μ = 391.1± 2760.0
σ = 779.9± 579.1
WF
σ/μ = 20.45± 159.72 %
μ = 286.0± 190.8
σ = 58.5± 417.8



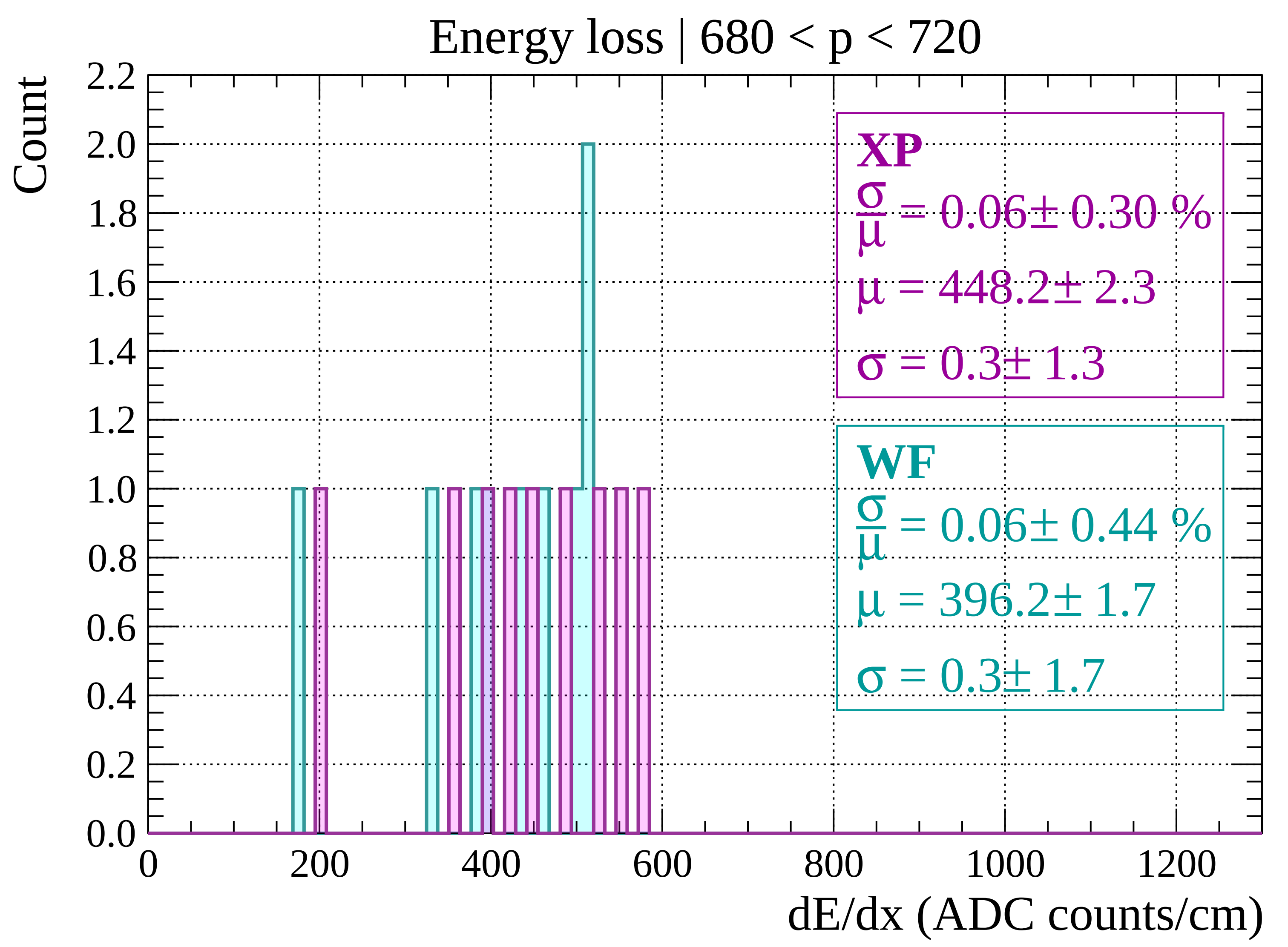
Energy loss | 560 < p < 600
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
σ/μ = 34.91± 91.11 %
μ = 2207.4± 3934.5
σ = 770.6± 637.7
WF
σ/μ = 0.05± 0.44 %
μ = 487.2± 1.6
σ = 0.3± 2.1



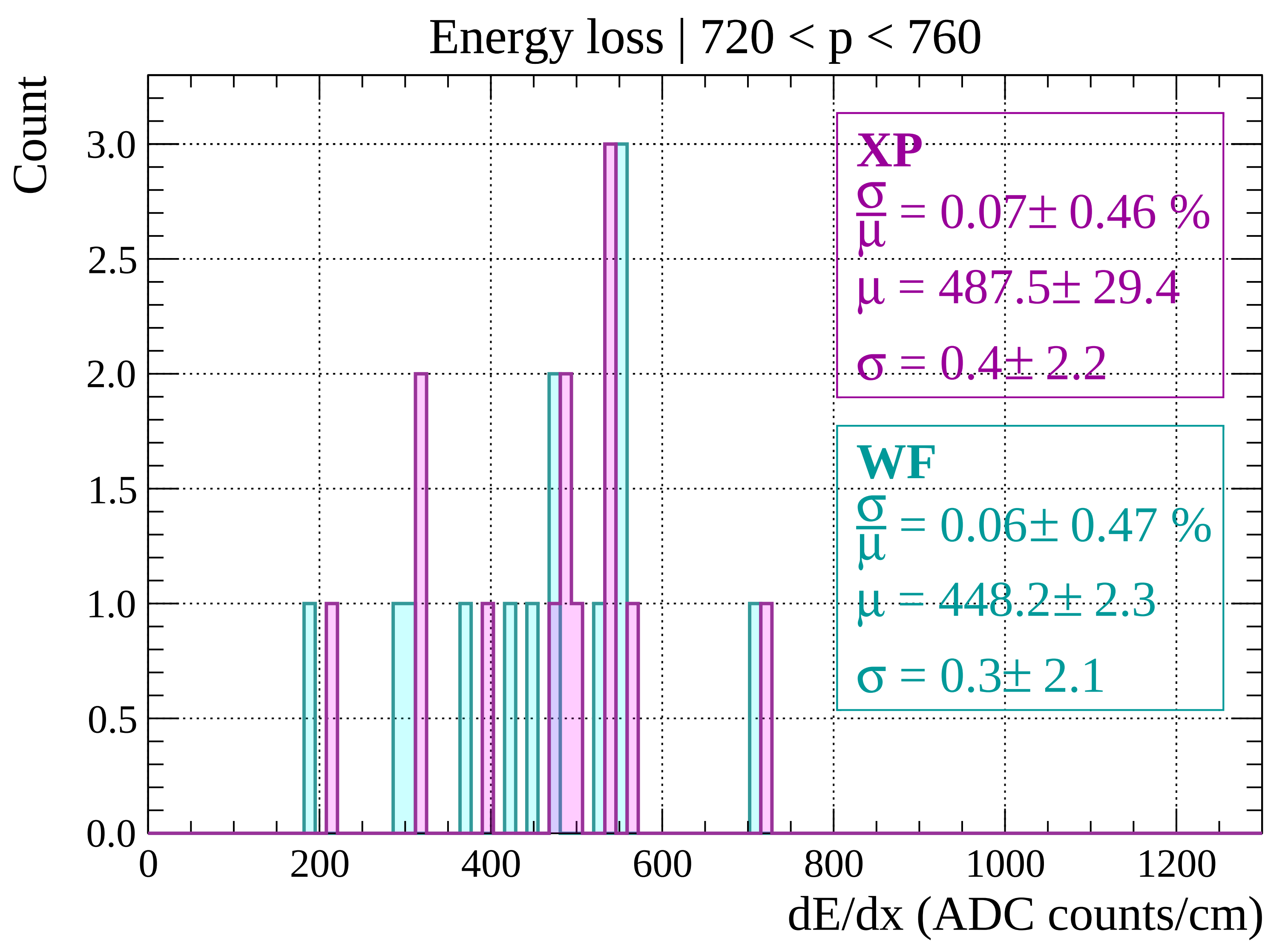
Energy loss | 600 < p < 640
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 0.07± 0.52 %
μ = 383.2± 2.3
σ = 0.3± 2.0
WF
σ/μ = 0.07± 0.53 %
μ = 357.2± 2.4
σ = 0.3± 1.9



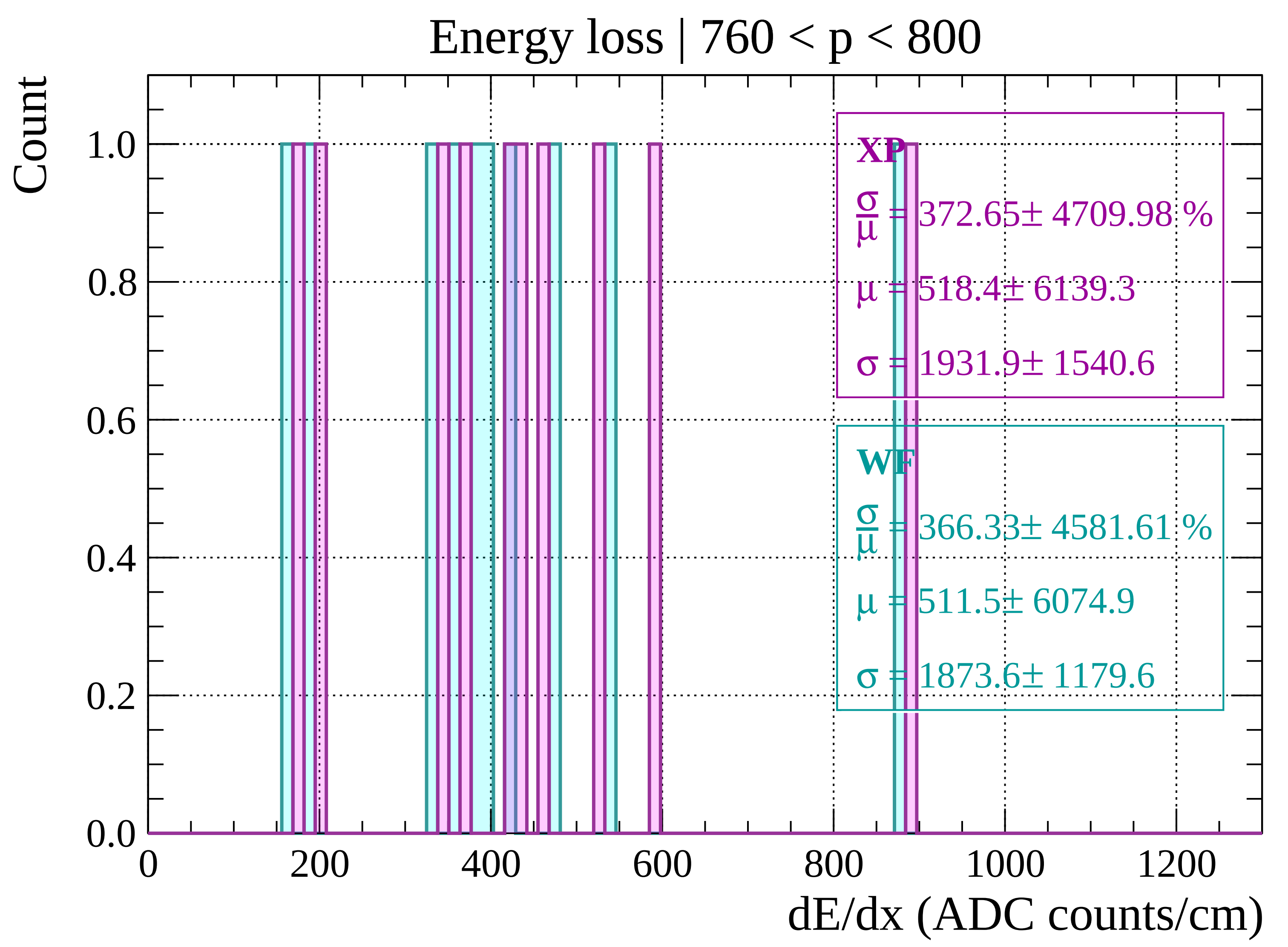
Energy loss | 640 < p < 680
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 0.16± 7.49 %
μ = 422.1± 11.7
σ = 0.7± 31.6
WF
$\frac{\sigma}{\mu}$ = 0.07± 0.52 %
μ = 396.2± 2.3
σ = 0.3± 2.0



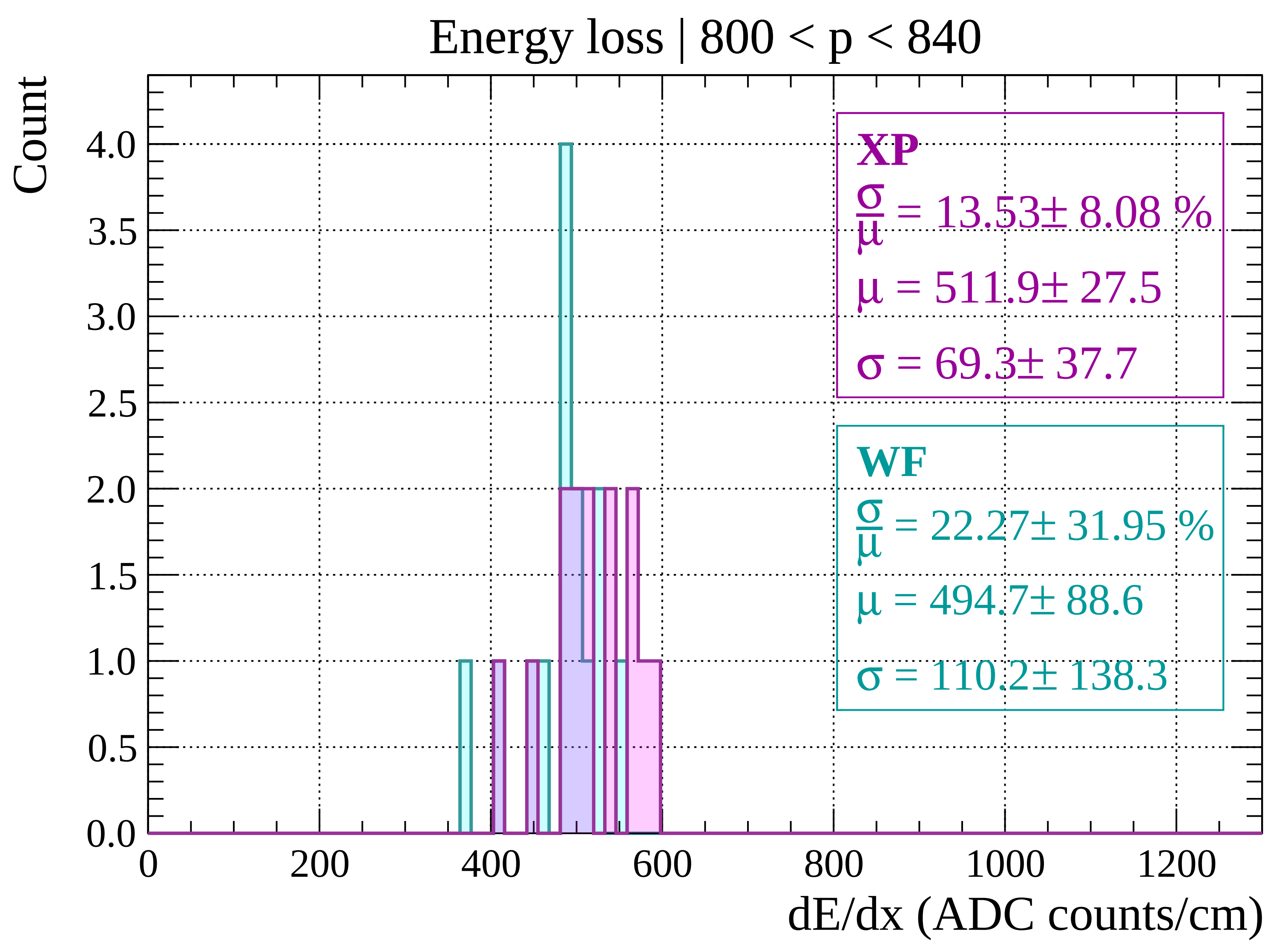
Energy loss | 680 < p < 720
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 0.06± 0.30 %
μ = 448.2± 2.3
σ = 0.3± 1.3
WF
σ/μ = 0.06± 0.44 %
μ = 396.2± 1.7
σ = 0.3± 1.7
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200



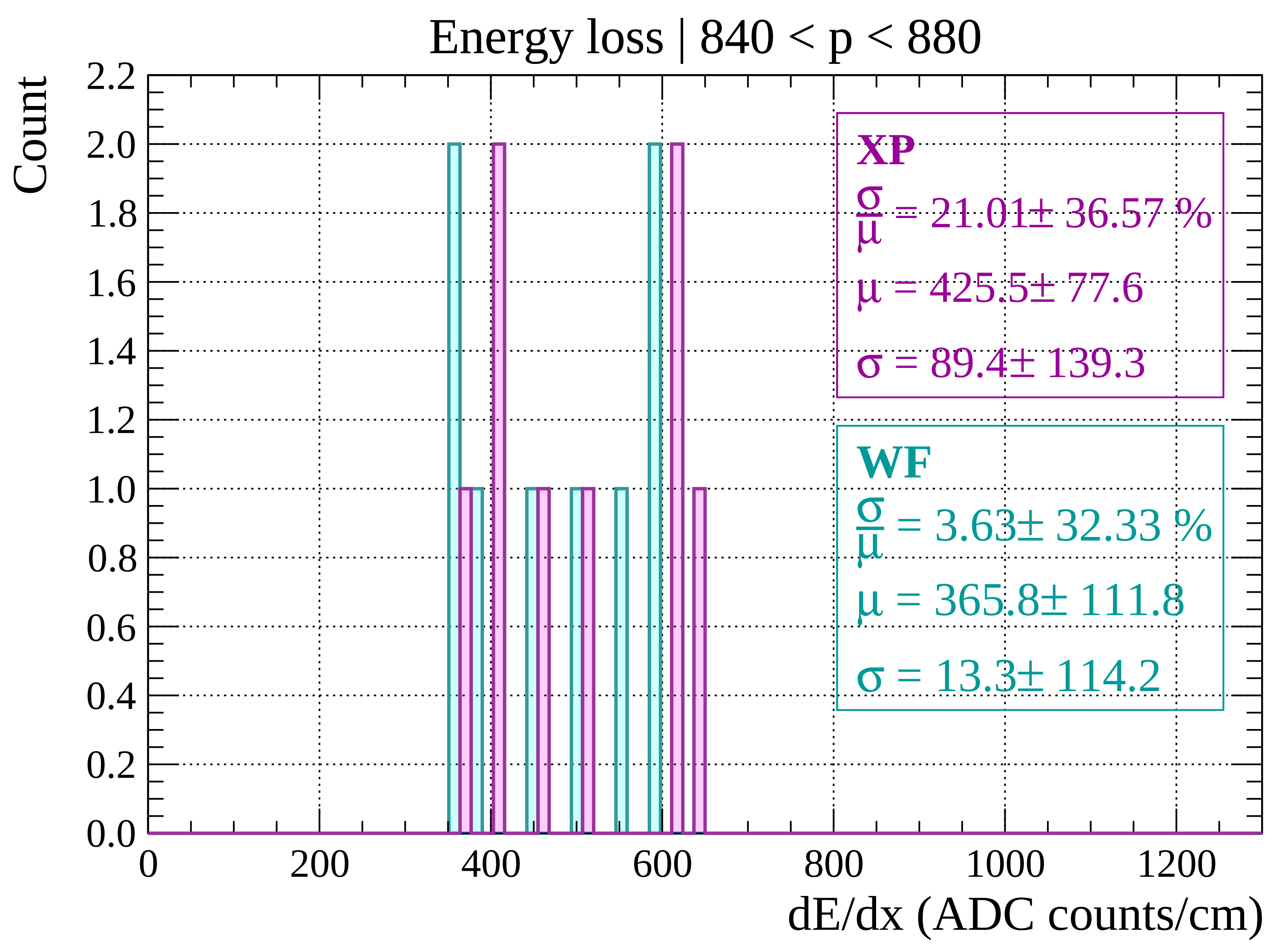
Energy loss | 720 < p < 760
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 0.07± 0.46 %
μ = 487.5± 29.4
σ = 0.4± 2.2
WF
$\frac{\sigma}{\mu}$ = 0.06± 0.47 %
μ = 448.2± 2.3
σ = 0.3± 2.1



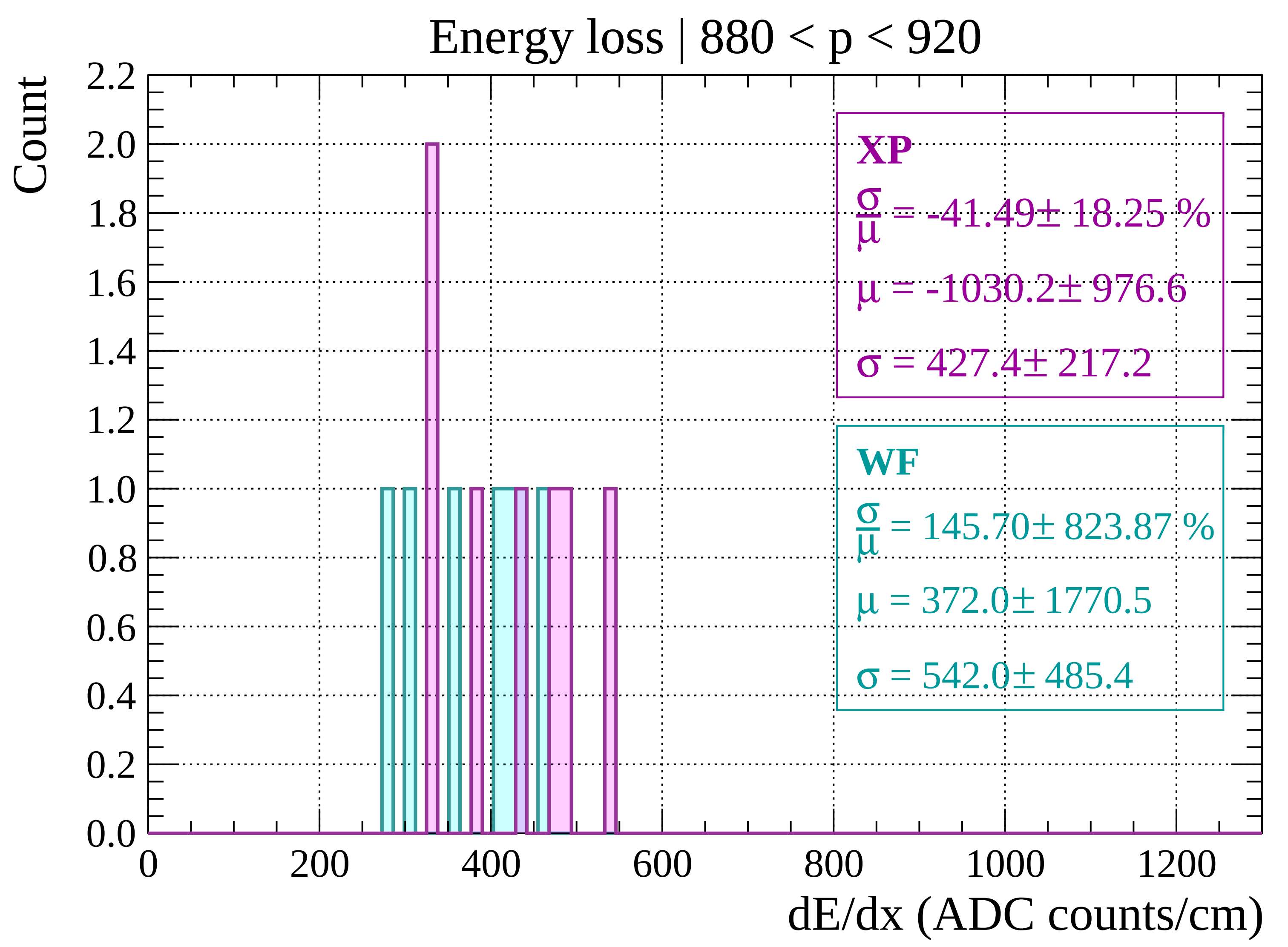
Energy loss | 760 < p < 800
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = 372.65± 4709.98 %
μ = 518.4± 6139.3
σ = 1931.9± 1540.6
WF
σ/μ = 366.33± 4581.61 %
μ = 511.5± 6074.9
σ = 1873.6± 1179.6



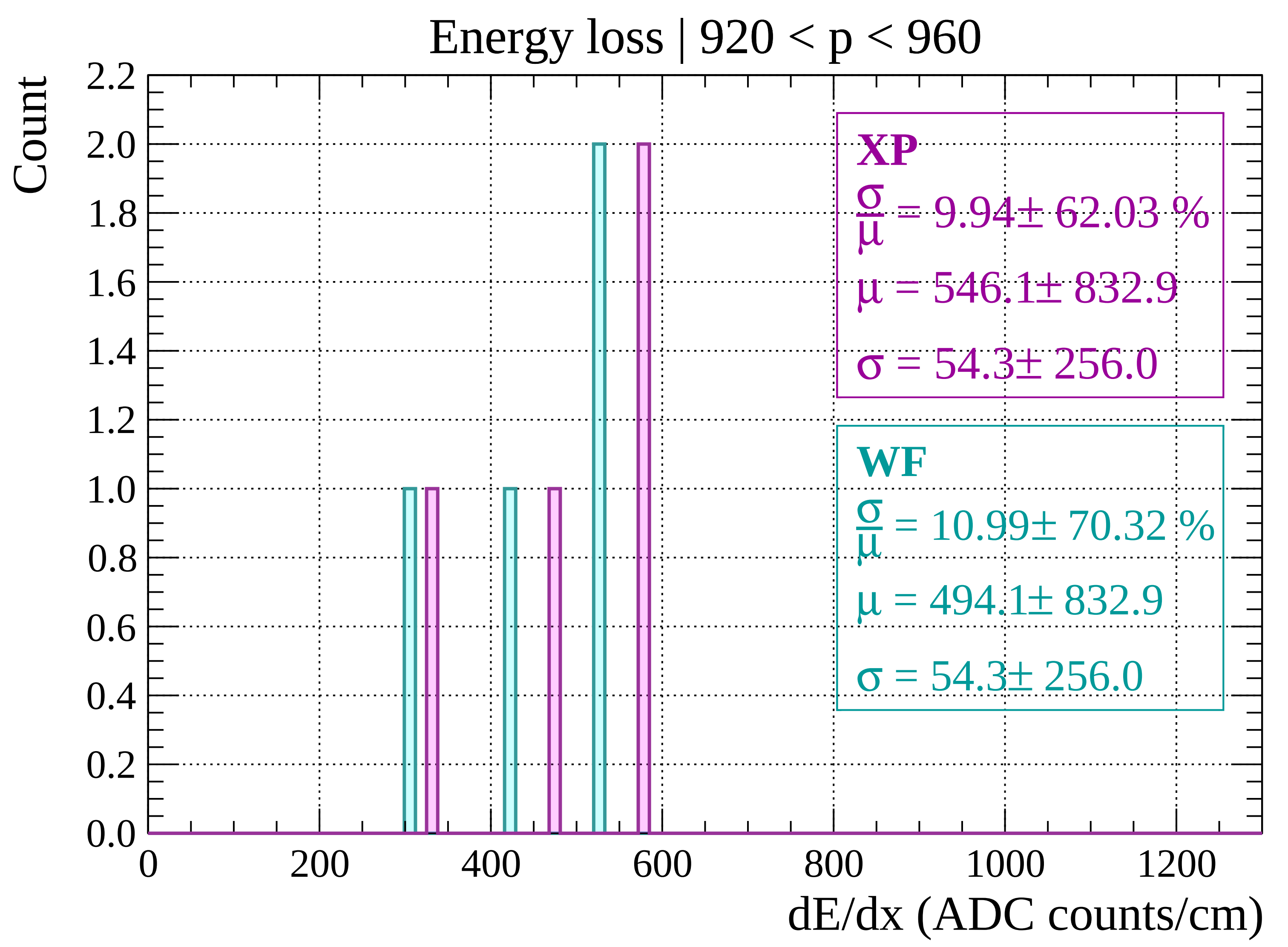
Energy loss | 800 < p < 840
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 13.53± 8.08 %
μ = 511.9± 27.5
σ = 69.3± 37.7
WF
σ/μ = 22.27± 31.95 %
μ = 494.7± 88.6
σ = 110.2± 138.3
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200



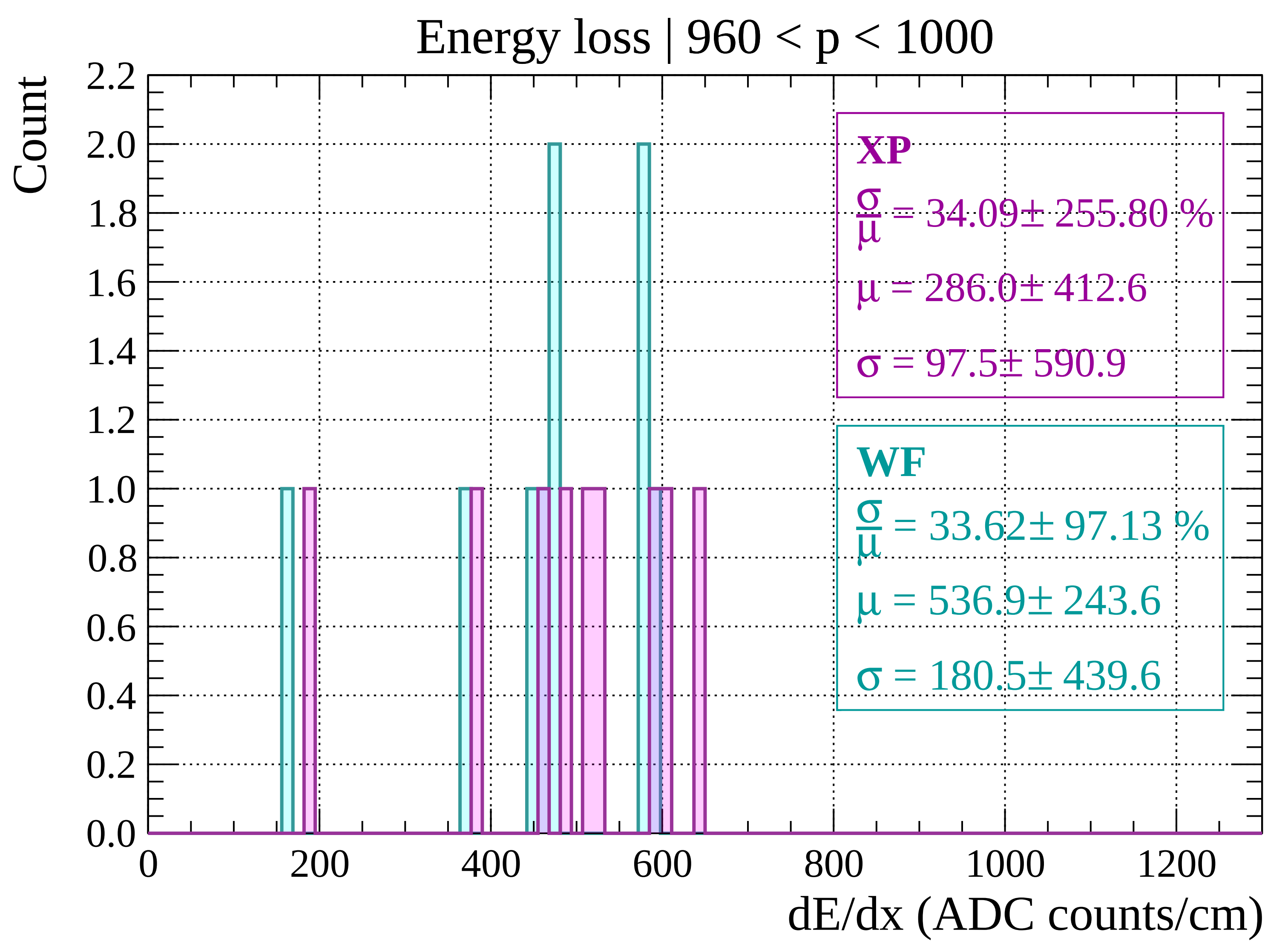
Energy loss | 840 < p < 880
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 21.01± 36.57 %
μ = 425.5± 77.6
σ = 89.4± 139.3
WF
σ/μ = 3.63± 32.33 %
μ = 365.8± 111.8
σ = 13.3± 114.2
0
200
400
600
800
1000
1200
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2



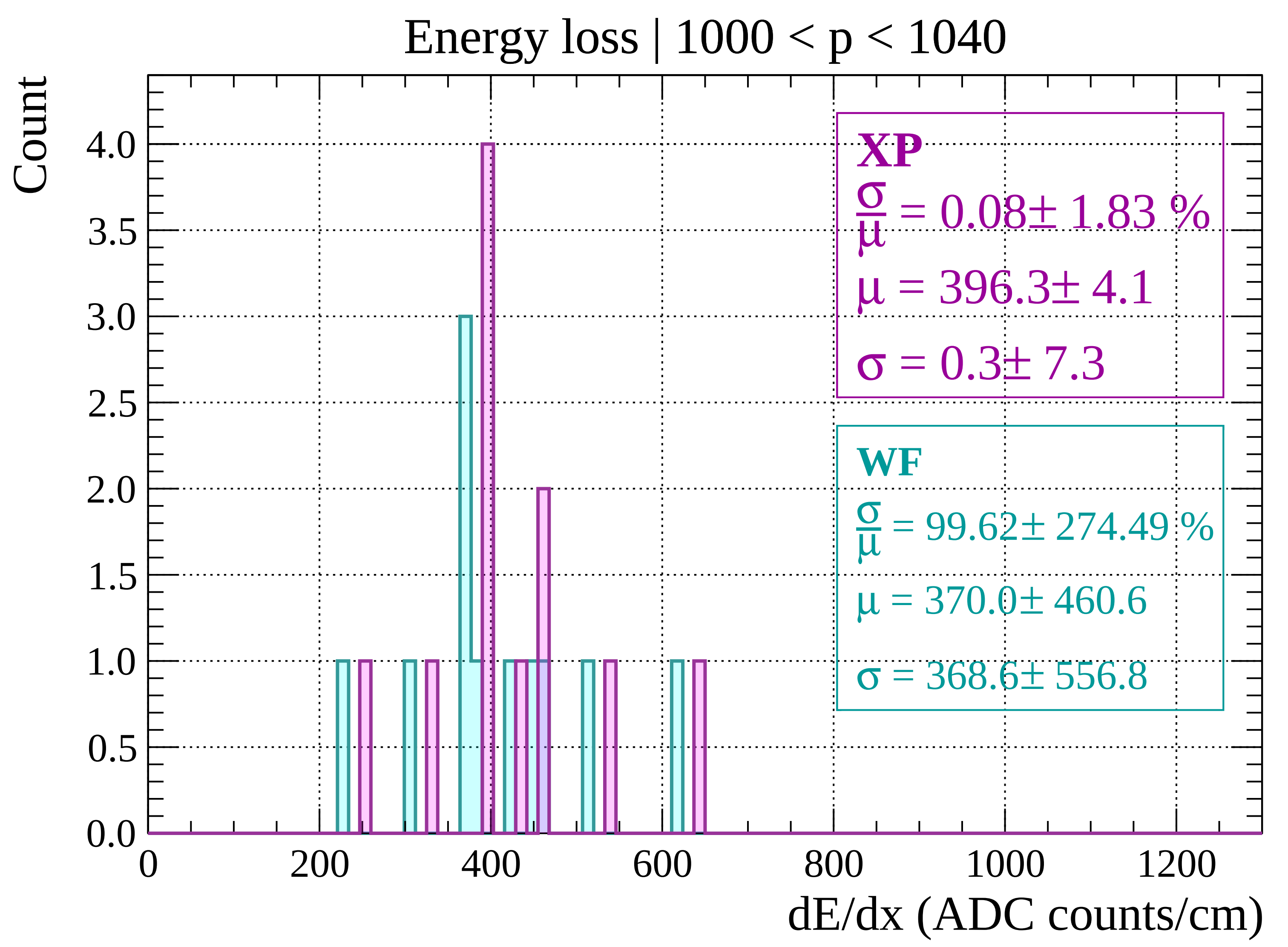
Energy loss | 880 < p < 920
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -41.49± 18.25 %
μ = -1030.2± 976.6
σ = 427.4± 217.2
WF
σ/μ = 145.70± 823.87 %
μ = 372.0± 1770.5
σ = 542.0± 485.4
0
200
400
600
800
1000
1200
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2



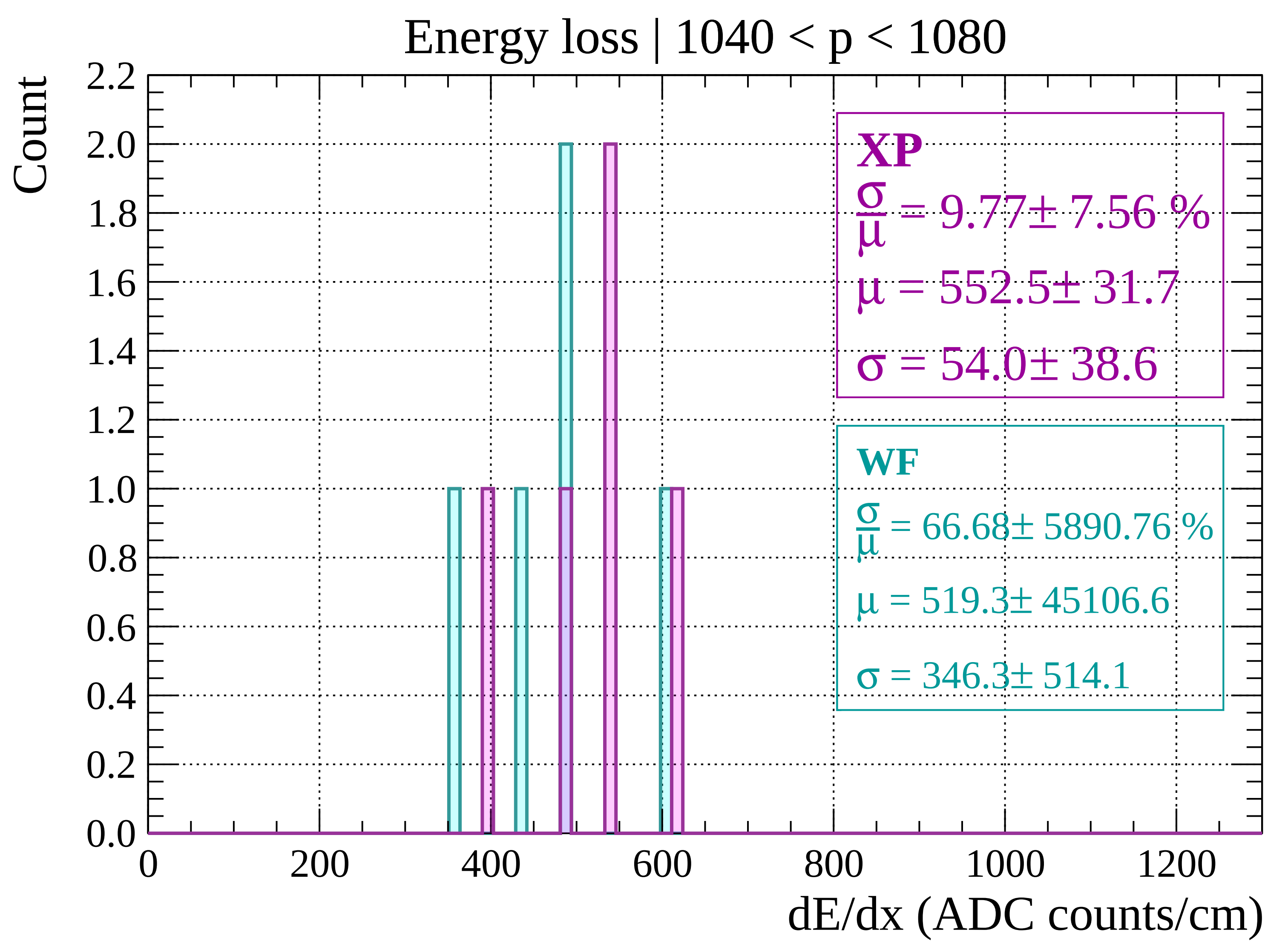
Energy loss | 920 < p < 960
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.94± 62.03 %
μ = 546.1± 832.9
σ = 54.3± 256.0
WF
$\frac{\sigma}{\mu}$ = 10.99± 70.32 %
μ = 494.1± 832.9
σ = 54.3± 256.0



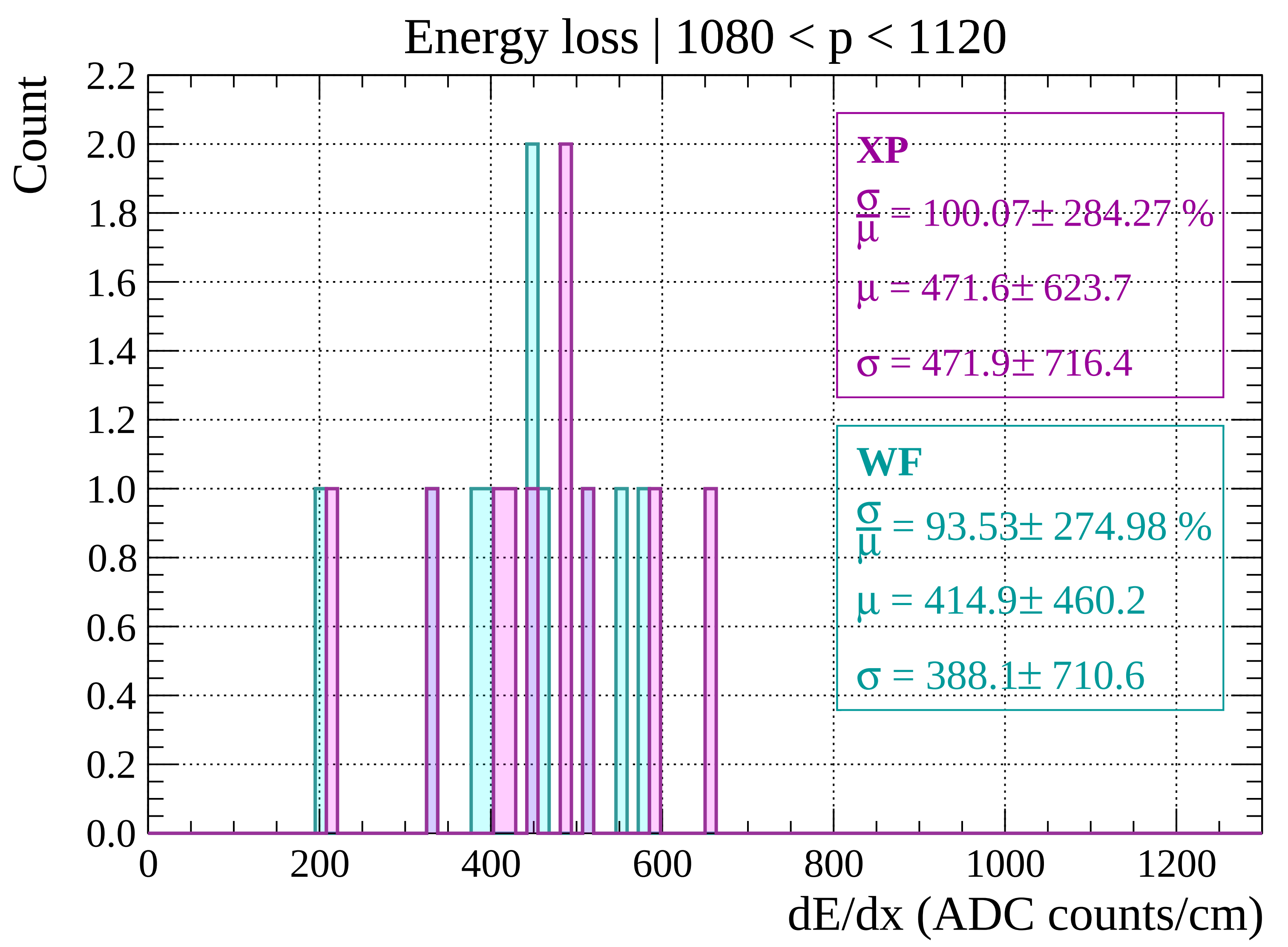
Energy loss | 960 < p < 1000
Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 34.09± 255.80 %
μ = 286.0± 412.6
σ = 97.5± 590.9
WF
$\frac{\sigma}{\mu}$ = 33.62± 97.13 %
μ = 536.9± 243.6
σ = 180.5± 439.6



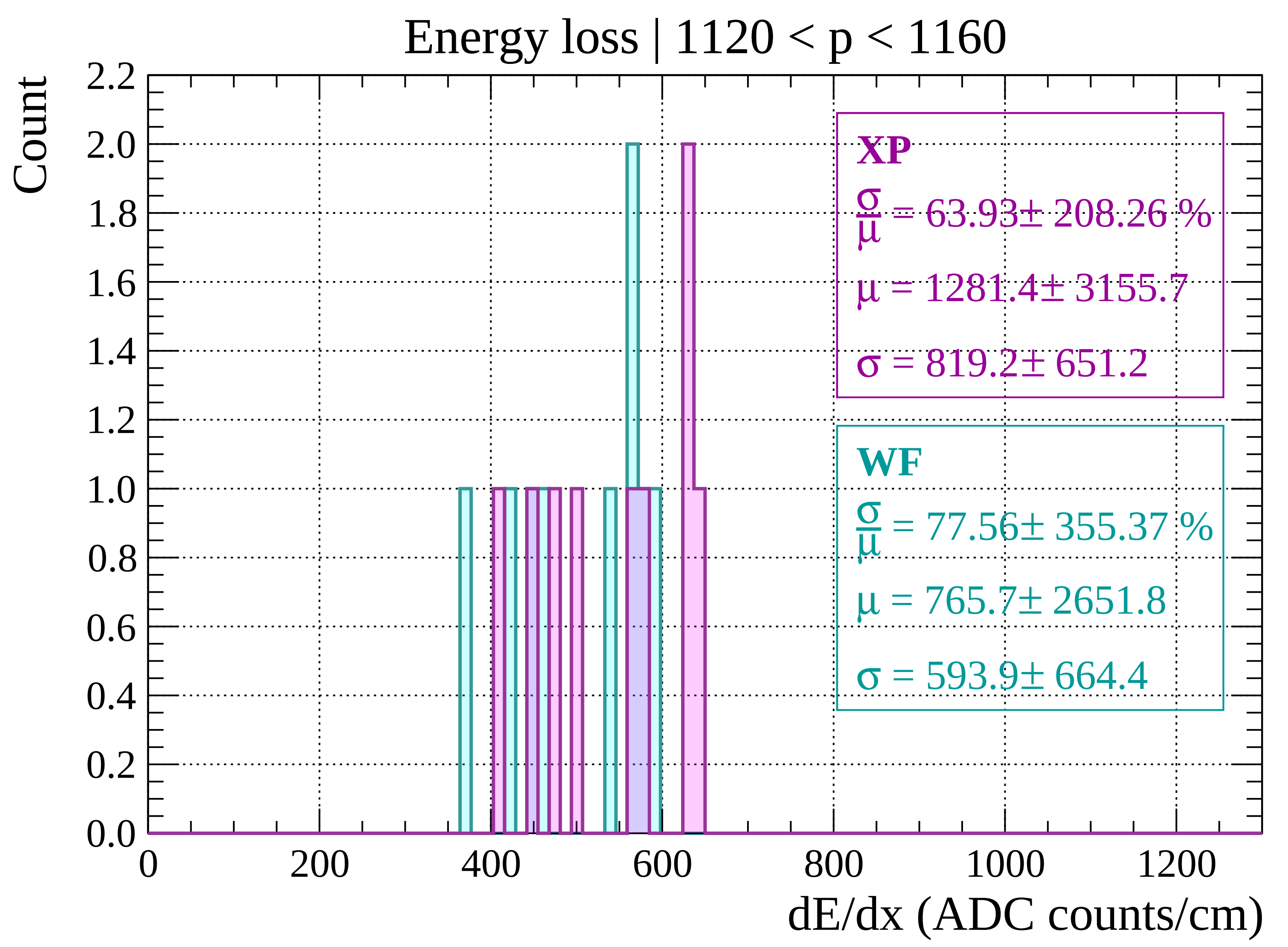
Energy loss | 1000 < p < 1040
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 0.08± 1.83 %
μ = 396.3± 4.1
σ = 0.3± 7.3
WF
σ/μ = 99.62± 274.49 %
μ = 370.0± 460.6
σ = 368.6± 556.8
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200



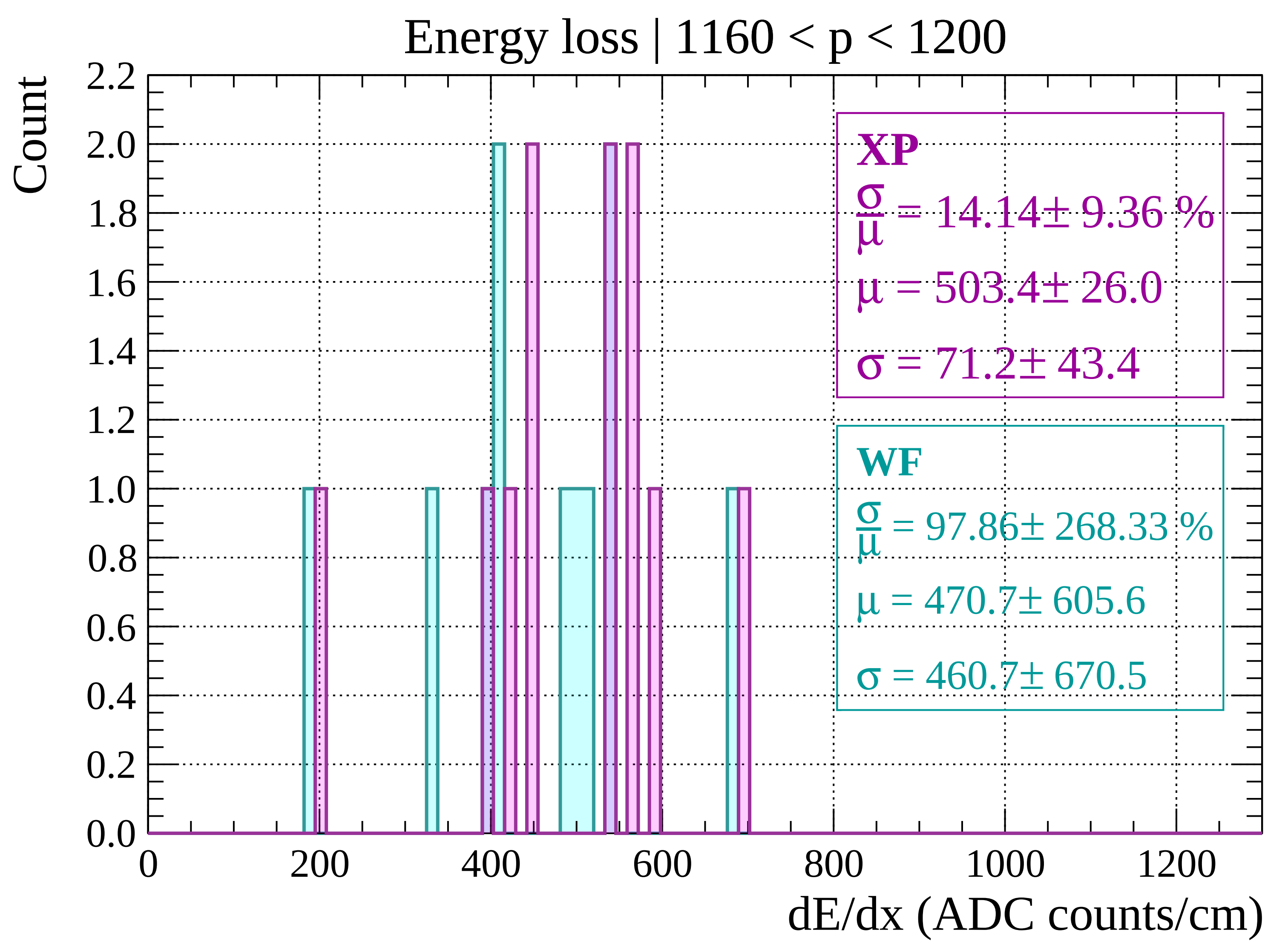
Energy loss | 1040 < p < 1080
Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.77± 7.56 %
μ = 552.5± 31.7
σ = 54.0± 38.6
WF
σ/μ = 66.68± 5890.76 %
μ = 519.3± 45106.6
σ = 346.3± 514.1



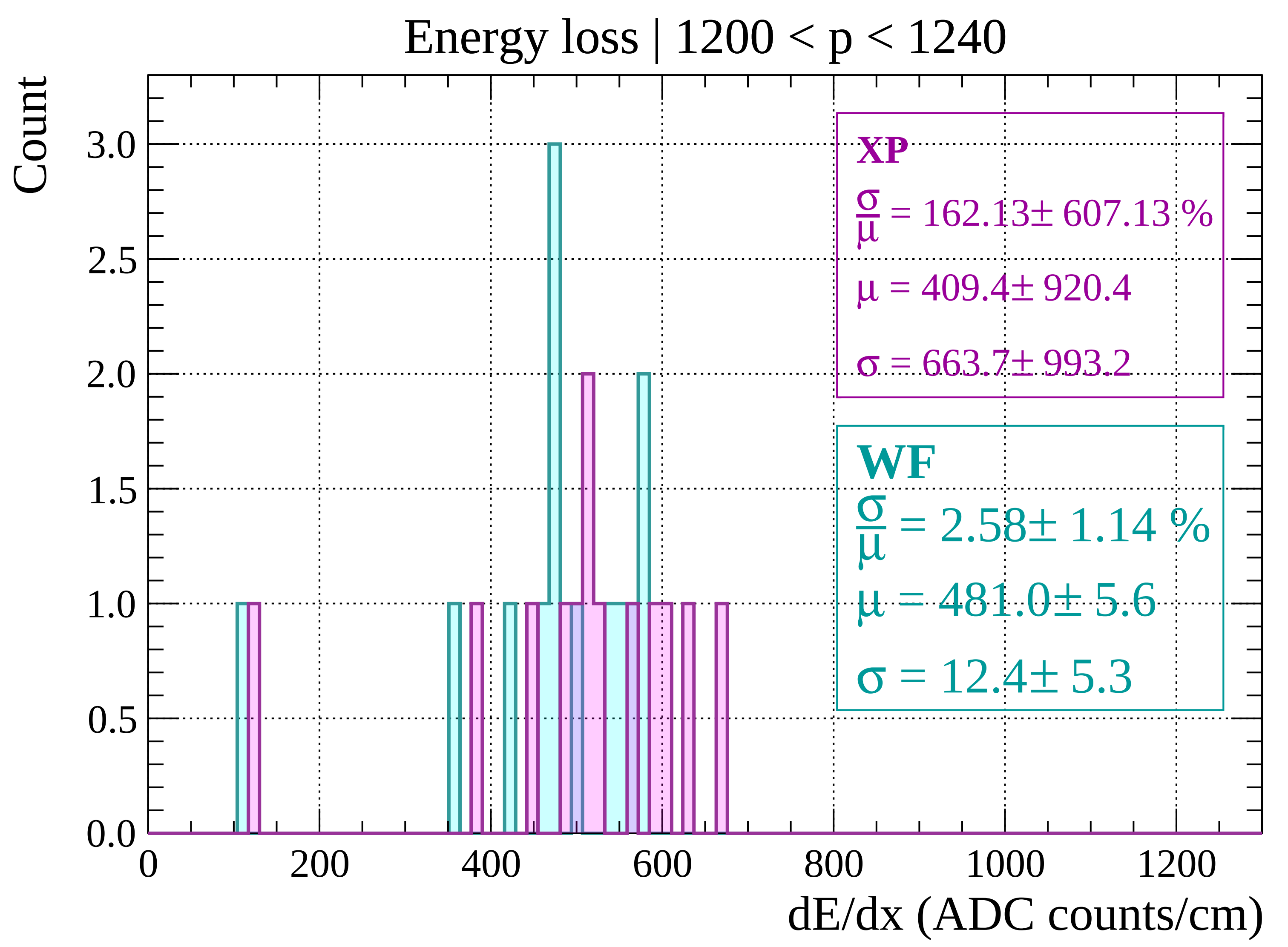
Energy loss | 1080 < p < 1120
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 100.07± 284.27 %
μ = 471.6± 623.7
σ = 471.9± 716.4
WF
σ/μ = 93.53± 274.98 %
μ = 414.9± 460.2
σ = 388.1± 710.6
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200



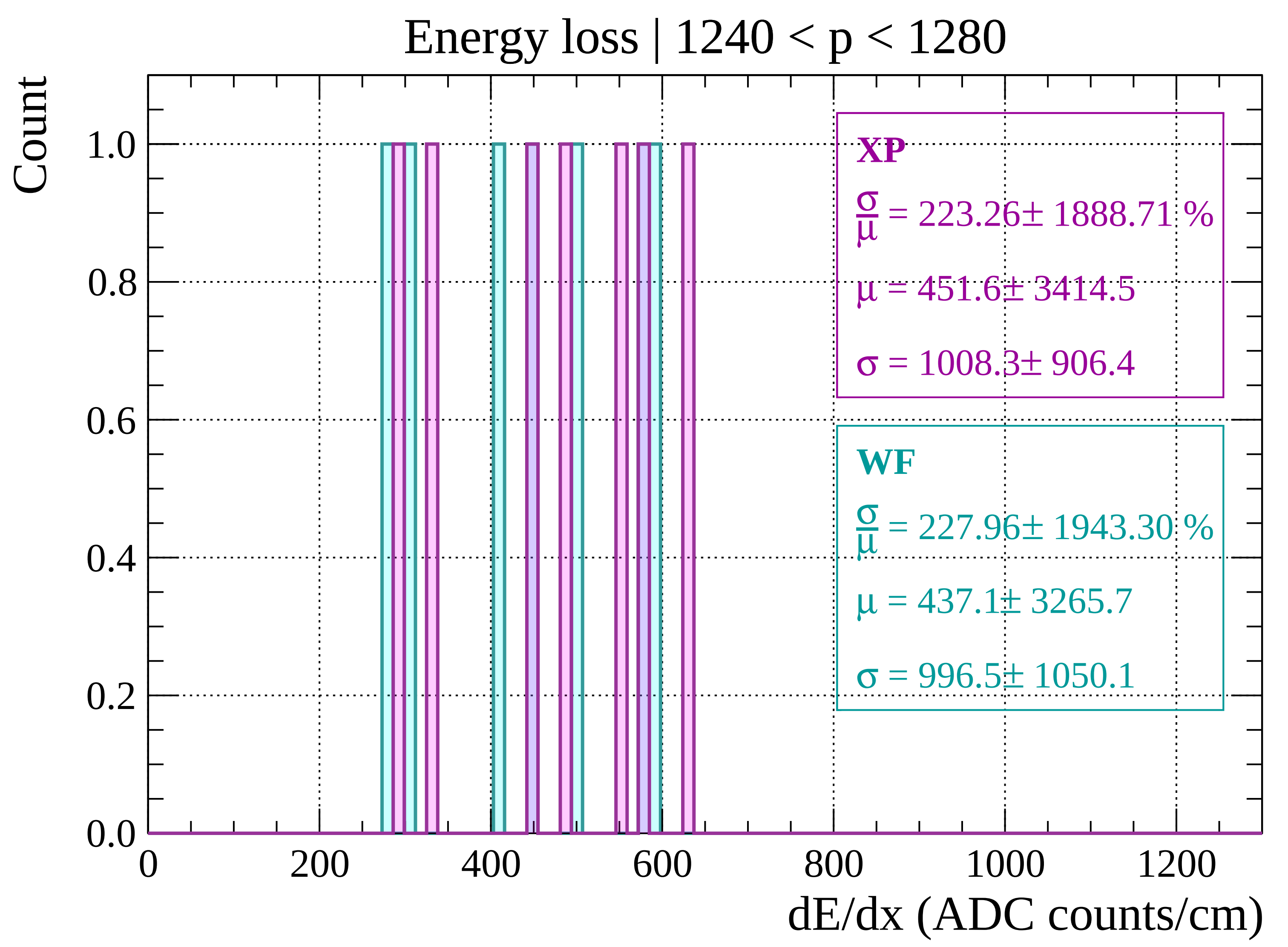
Energy loss | 1120 < p < 1160
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
σ/μ = 63.93± 208.26 %
μ = 1281.4± 3155.7
σ = 819.2± 651.2
WF
σ/μ = 77.56± 355.37 %
μ = 765.7± 2651.8
σ = 593.9± 664.4



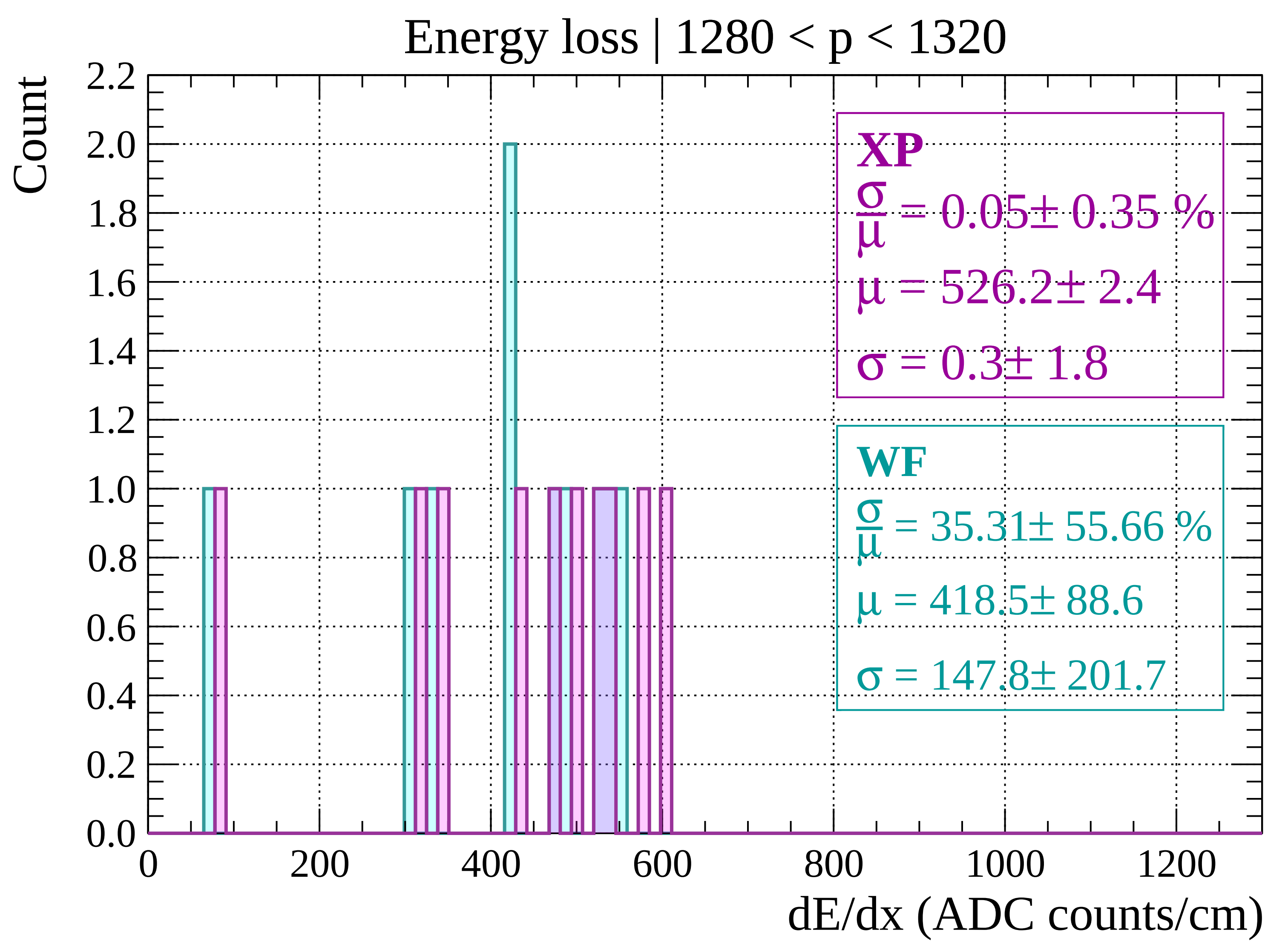
Energy loss | 1160 < p < 1200
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 14.14± 9.36 %
μ = 503.4± 26.0
σ = 71.2± 43.4
WF
$\frac{\sigma}{\mu}$ = 97.86± 268.33 %
μ = 470.7± 605.6
σ = 460.7± 670.5



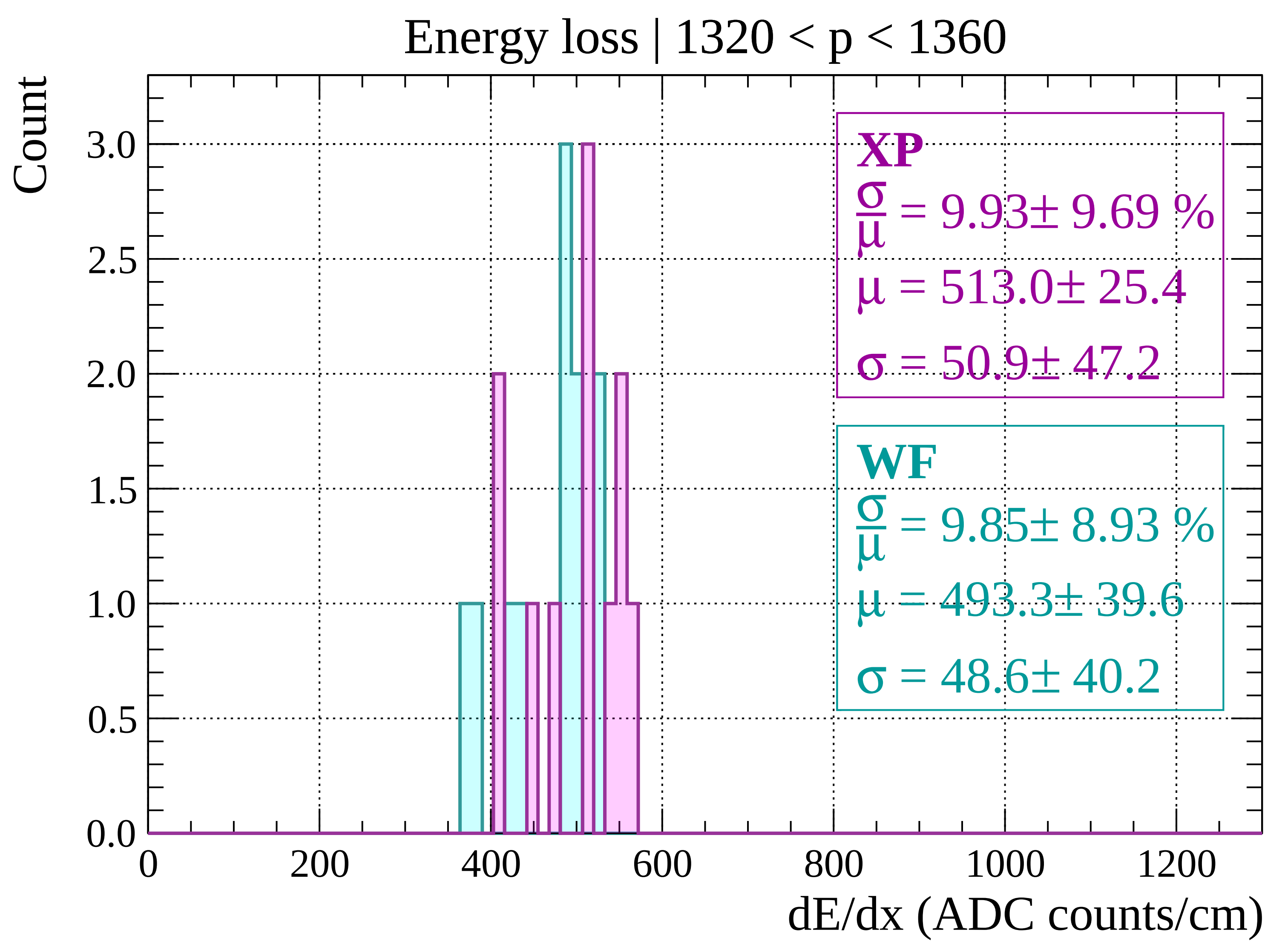
Energy loss | 1200 < p < 1240
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 162.13± 607.13 %
μ = 409.4± 920.4
σ = 663.7± 993.2
WF
σ/μ = 2.58± 1.14 %
μ = 481.0± 5.6
σ = 12.4± 5.3

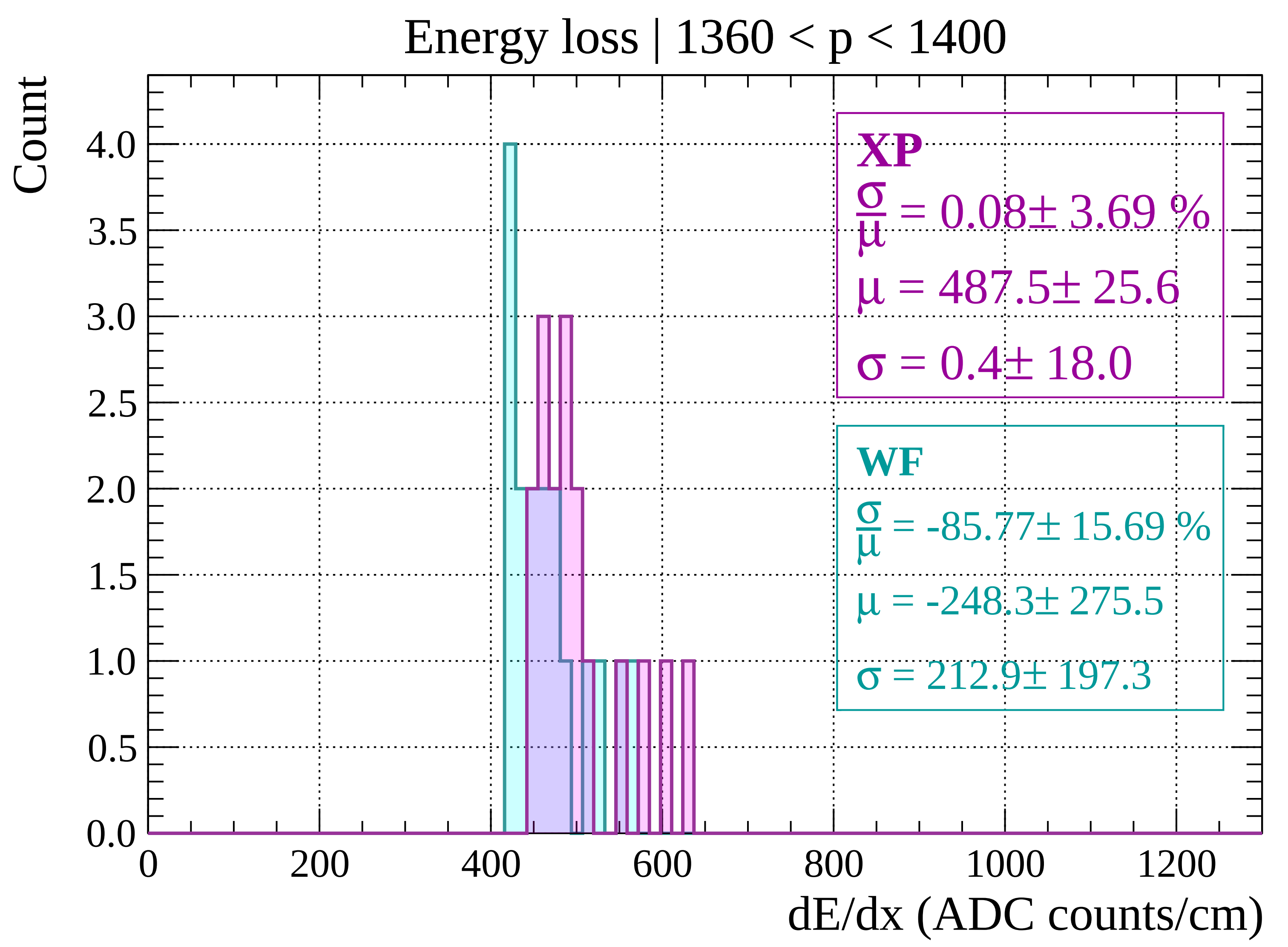


Energy loss | 1240 < p < 1280
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 223.26± 1888.71 %
μ = 451.6± 3414.5
σ = 1008.3± 906.4
WF
σ/μ = 227.96± 1943.30 %
μ = 437.1± 3265.7
σ = 996.5± 1050.1

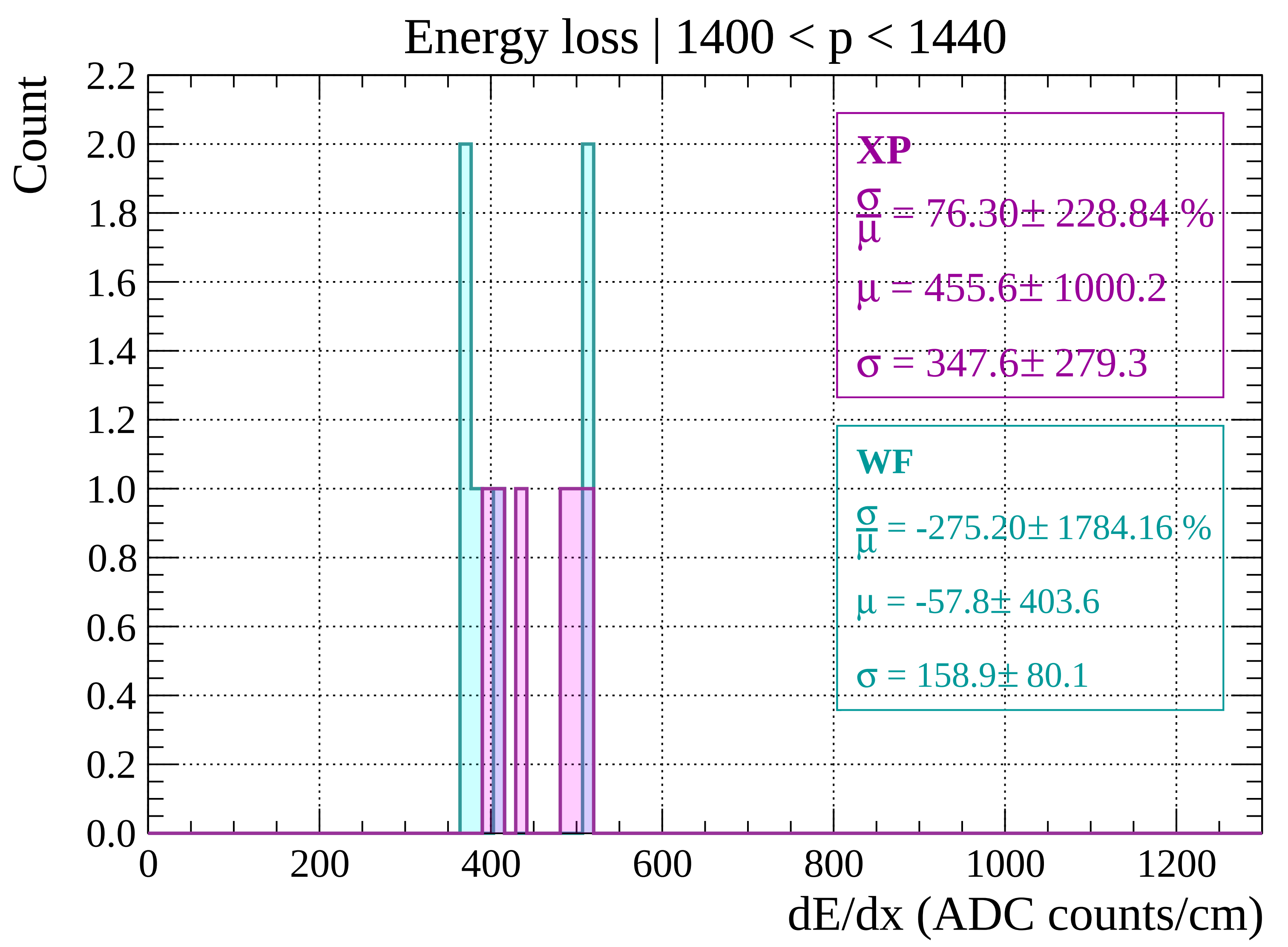


Energy loss | 1280 < p < 1320
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 0.05± 0.35 %
μ = 526.2± 2.4
σ = 0.3± 1.8
WF
$\frac{\sigma}{\mu}$ = 35.31± 55.66 %
μ = 418.5± 88.6
σ = 147.8± 201.7

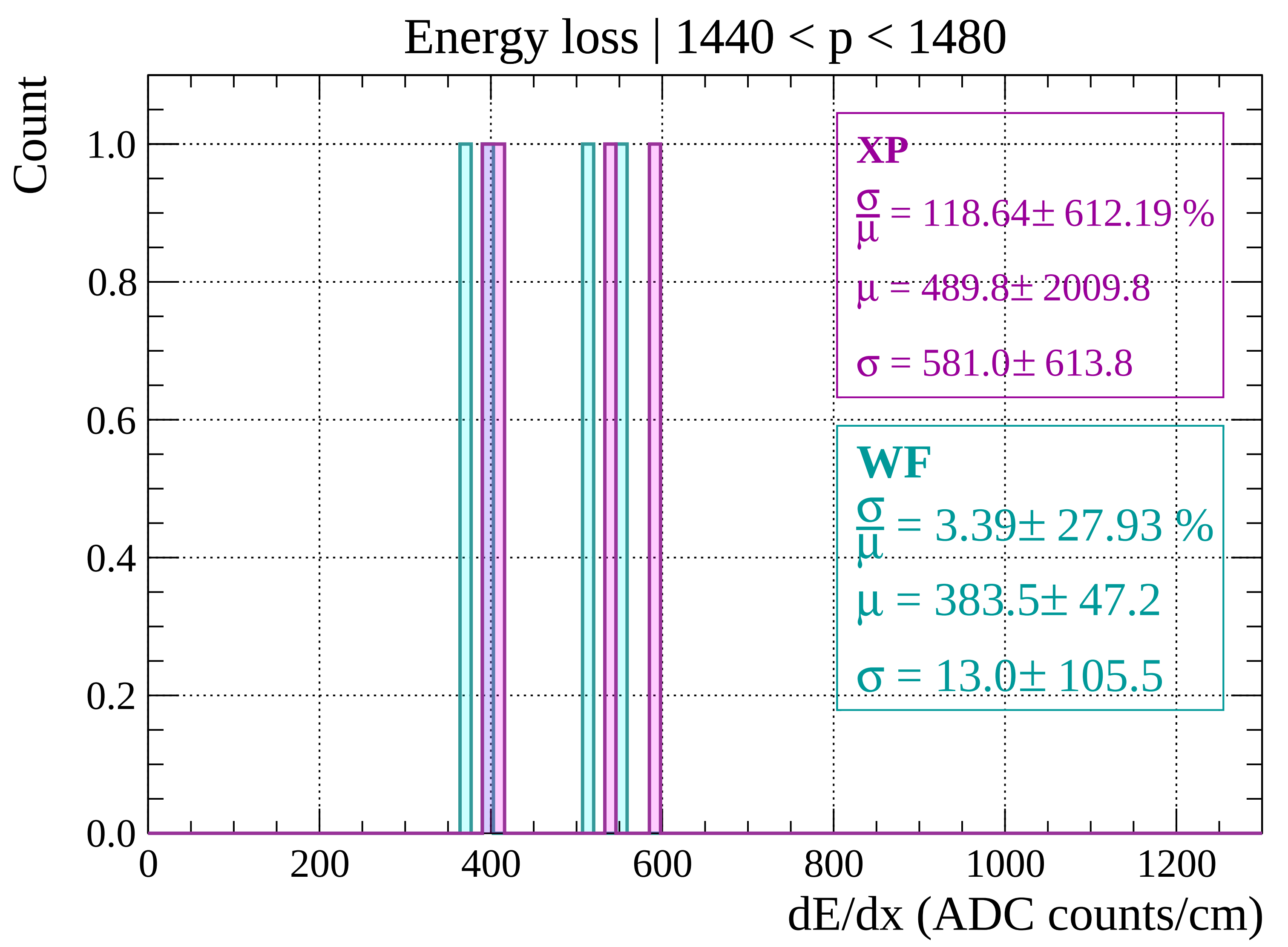

Energy loss | 1320 < p < 1360
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200
XP
σ/μ = 9.93± 9.69 %
μ = 513.0± 25.4
σ = 50.9± 47.2
WF
σ/μ = 9.85± 8.93 %
μ = 493.3± 39.6
σ = 48.6± 40.2



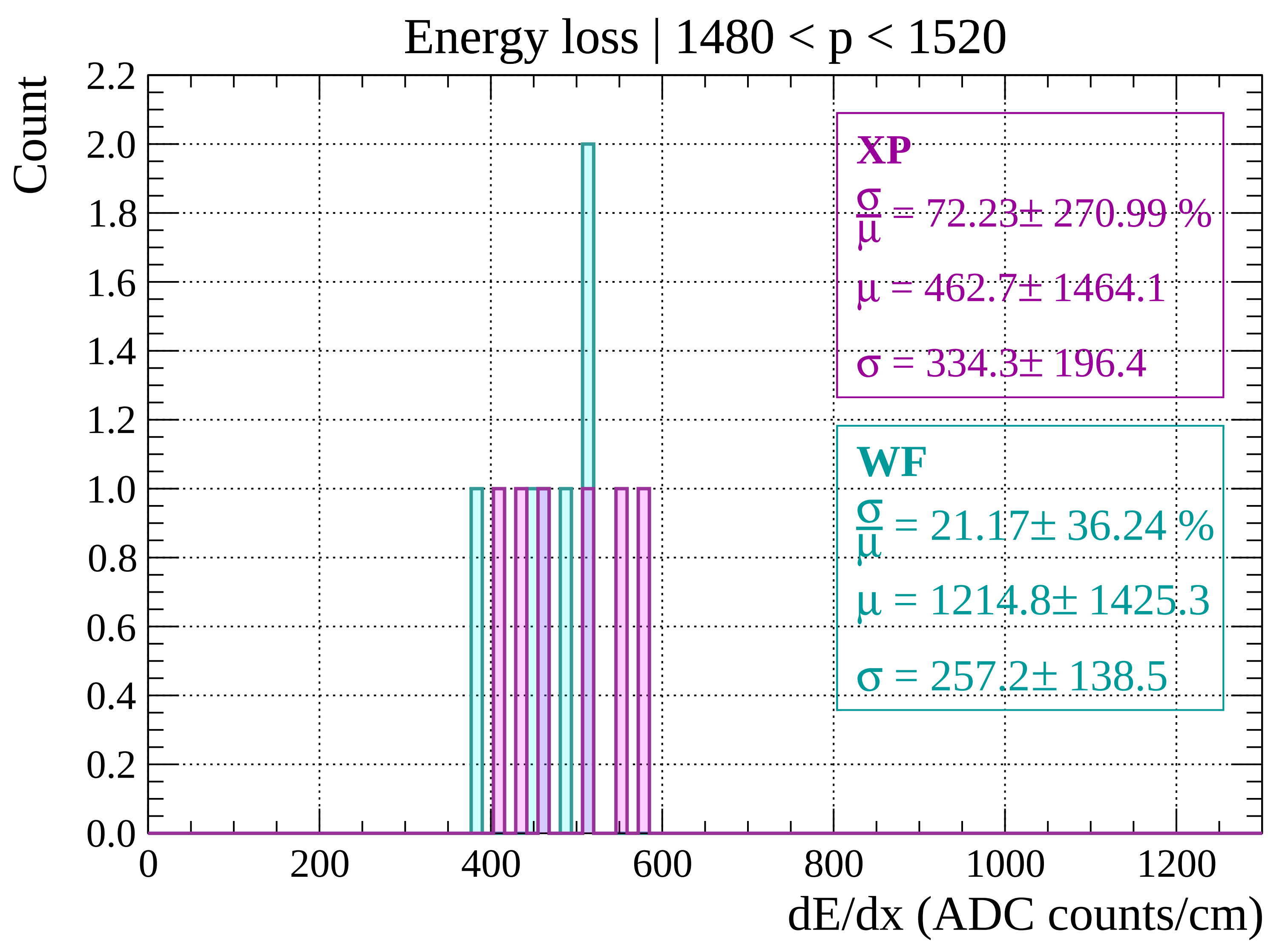
Energy loss | 1360 < p < 1400
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200
XP
σ/μ = 0.08± 3.69 %
μ = 487.5± 25.6
σ = 0.4± 18.0
WF
σ/μ = -85.77± 15.69 %
μ = -248.3± 275.5
σ = 212.9± 197.3

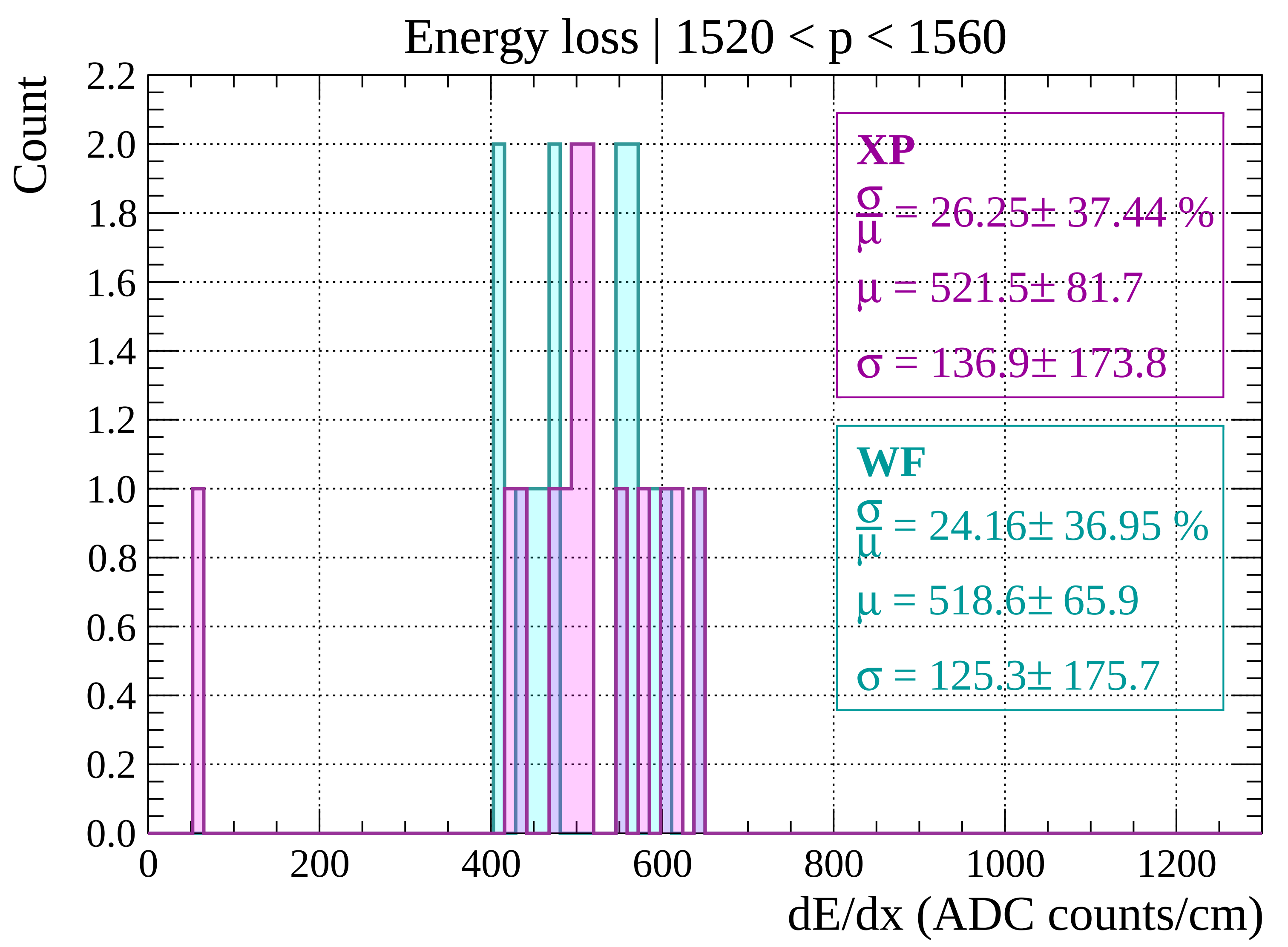


Energy loss | 1400 < p < 1440
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 76.30± 228.84 %
μ = 455.6± 1000.2
σ = 347.6± 279.3
WF
$\frac{\sigma}{\mu}$ = -275.20± 1784.16 %
μ = -57.8± 403.6
σ = 158.9± 80.1

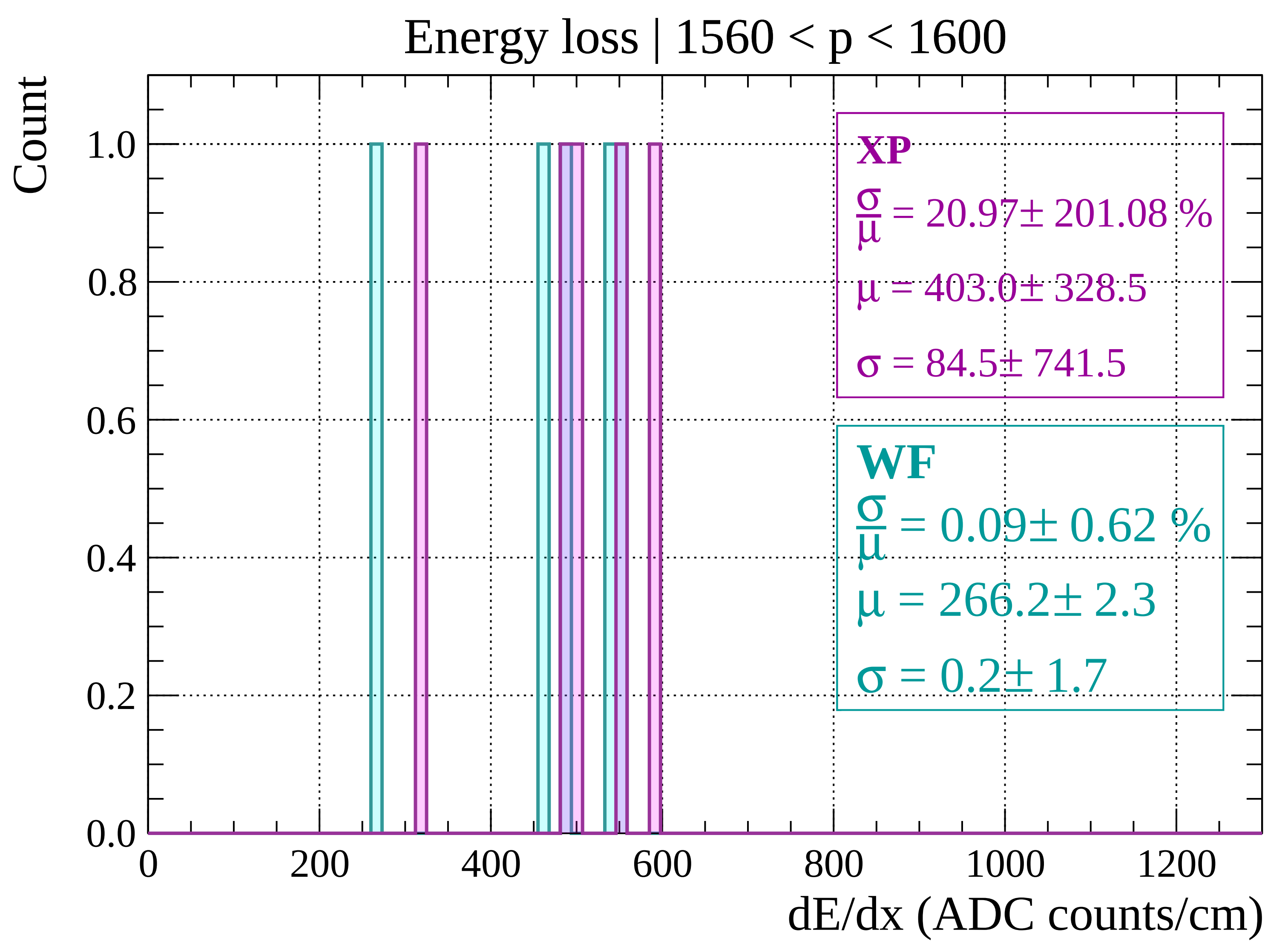


Energy loss | 1440 < p < 1480
Count
dE/dx (ADC counts/cm)
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 118.64± 612.19 %
μ = 489.8± 2009.8
σ = 581.0± 613.8
WF
$\frac{\sigma}{\mu}$ = 3.39± 27.93 %
μ = 383.5± 47.2
σ = 13.0± 105.5

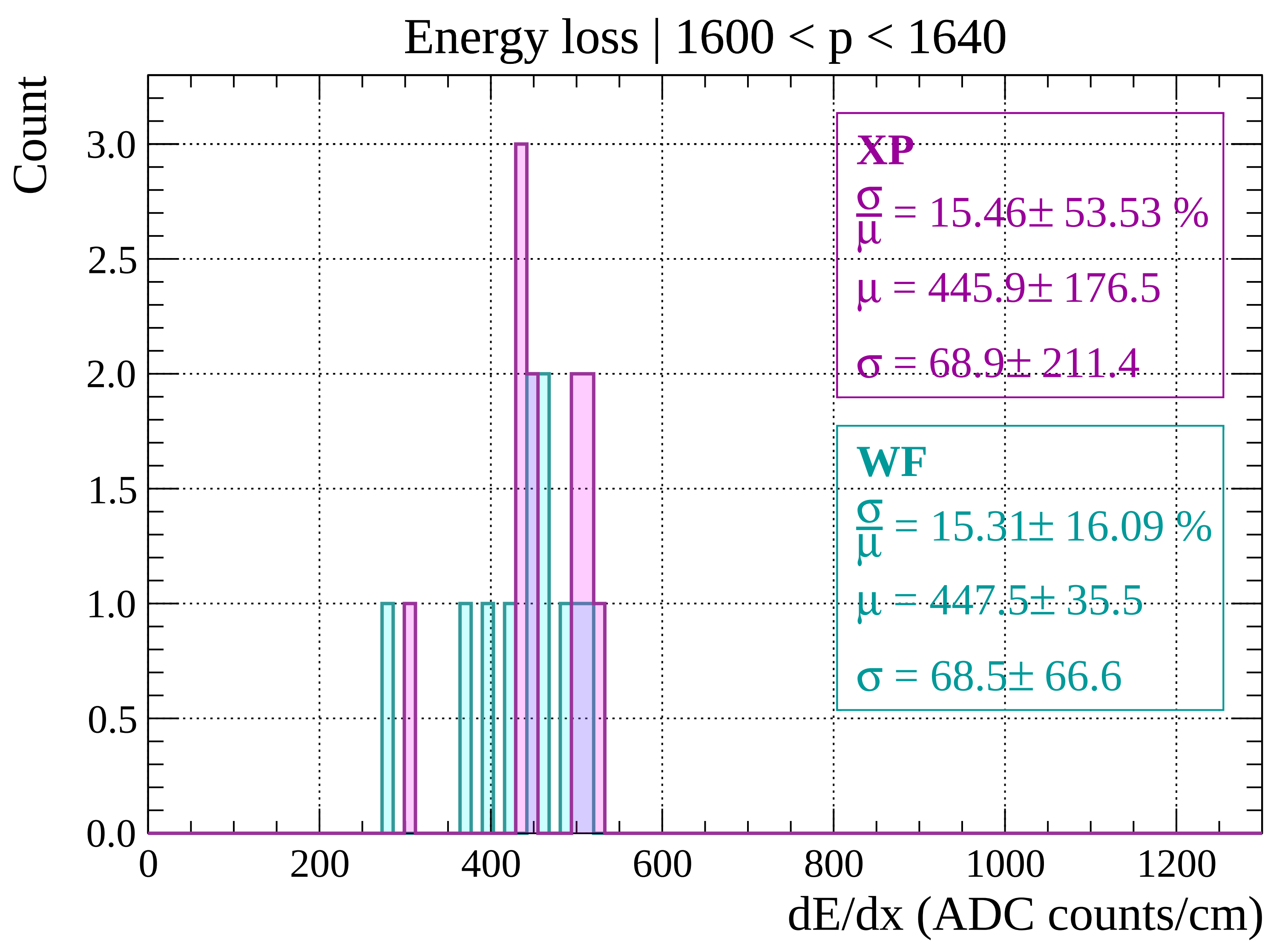

Energy loss | 1480 < p < 1520
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 72.23± 270.99 %
μ = 462.7± 1464.1
σ = 334.3± 196.4
WF
σ/μ = 21.17± 36.24 %
μ = 1214.8± 1425.3
σ = 257.2± 138.5



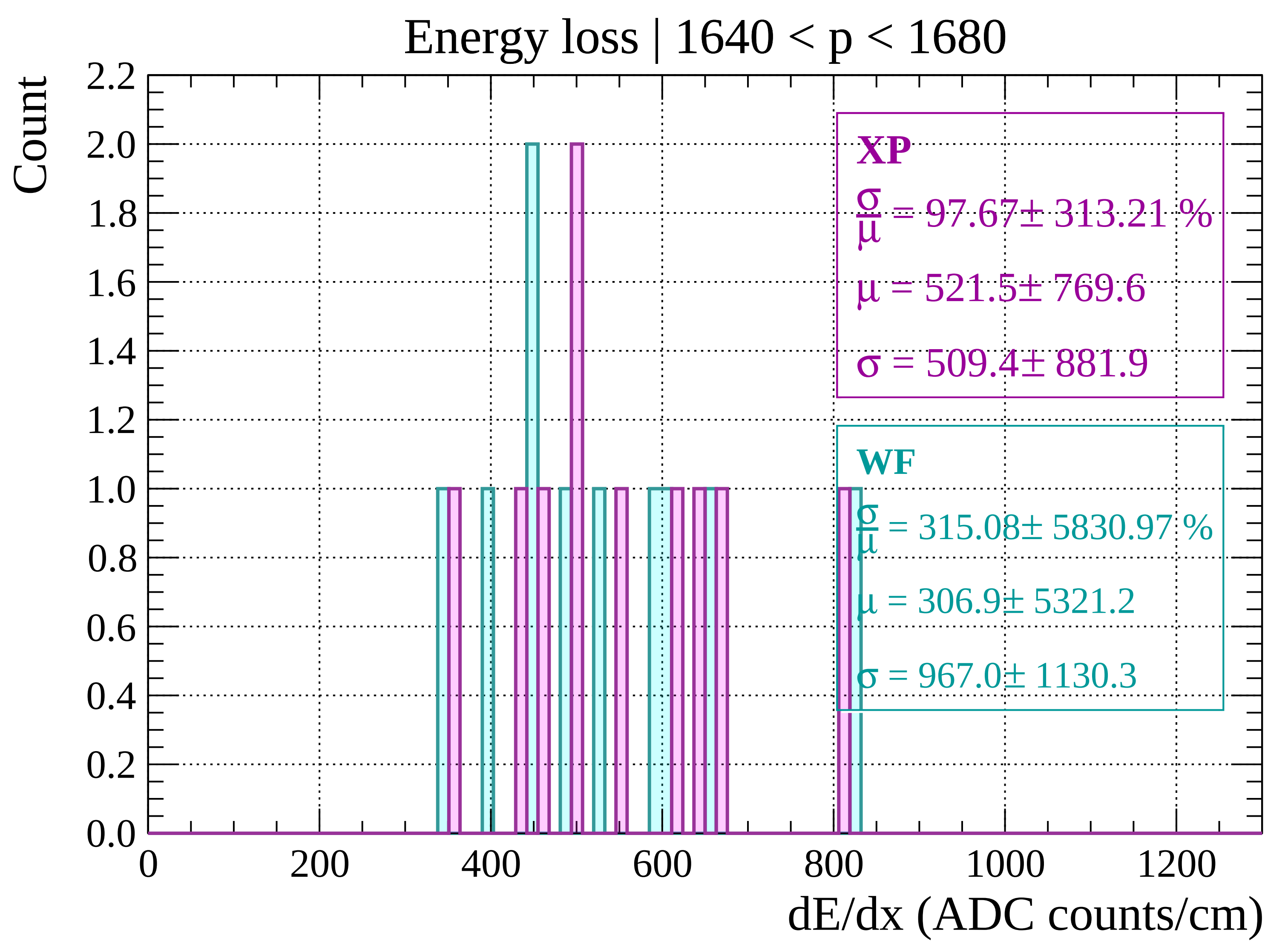
Energy loss | 1520 < p < 1560
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 26.25± 37.44 %
μ = 521.5± 81.7
σ = 136.9± 173.8
WF
$\frac{\sigma}{\mu}$ = 24.16± 36.95 %
μ = 518.6± 65.9
σ = 125.3± 175.7



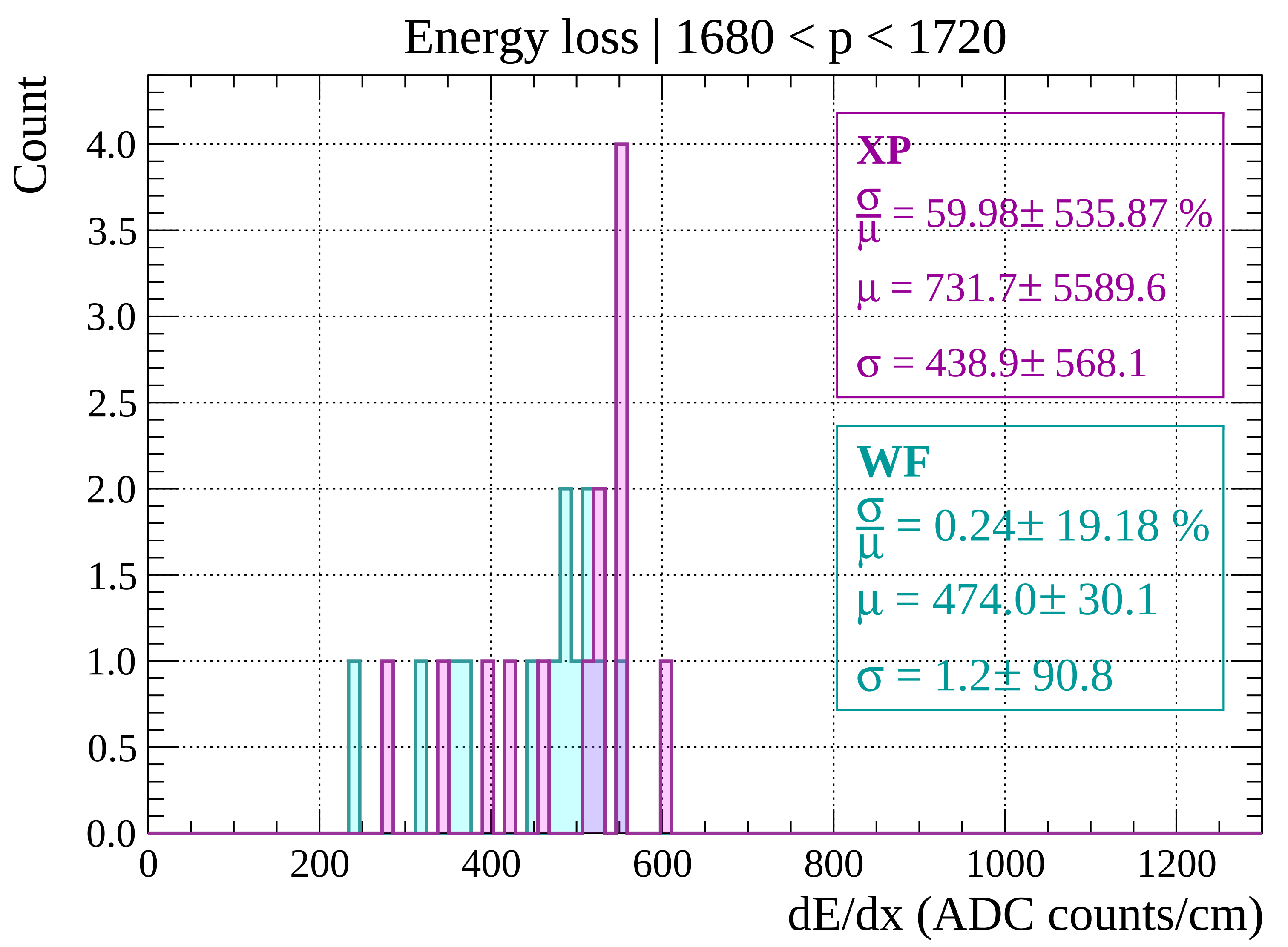
Energy loss | 1560 < p < 1600
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 20.97± 201.08 %
μ = 403.0± 328.5
σ = 84.5± 741.5
WF
$\frac{\sigma}{\mu}$ = 0.09± 0.62 %
μ = 266.2± 2.3
σ = 0.2± 1.7



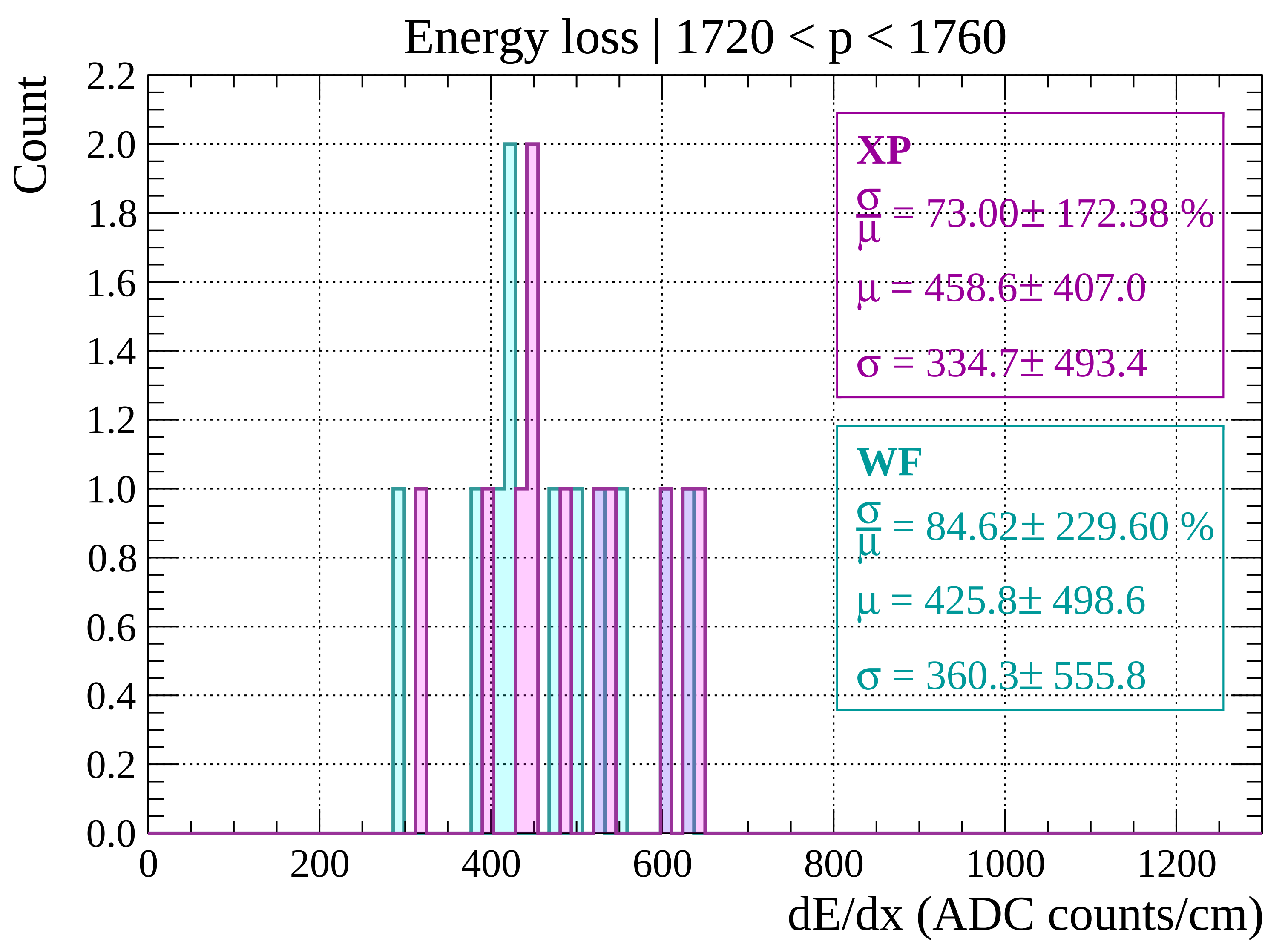
Energy loss | 1600 < p < 1640
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 15.46± 53.53 %
μ = 445.9± 176.5
σ = 68.9± 211.4
WF
$\frac{\sigma}{\mu}$ = 15.31± 16.09 %
μ = 447.5± 35.5
σ = 68.5± 66.6



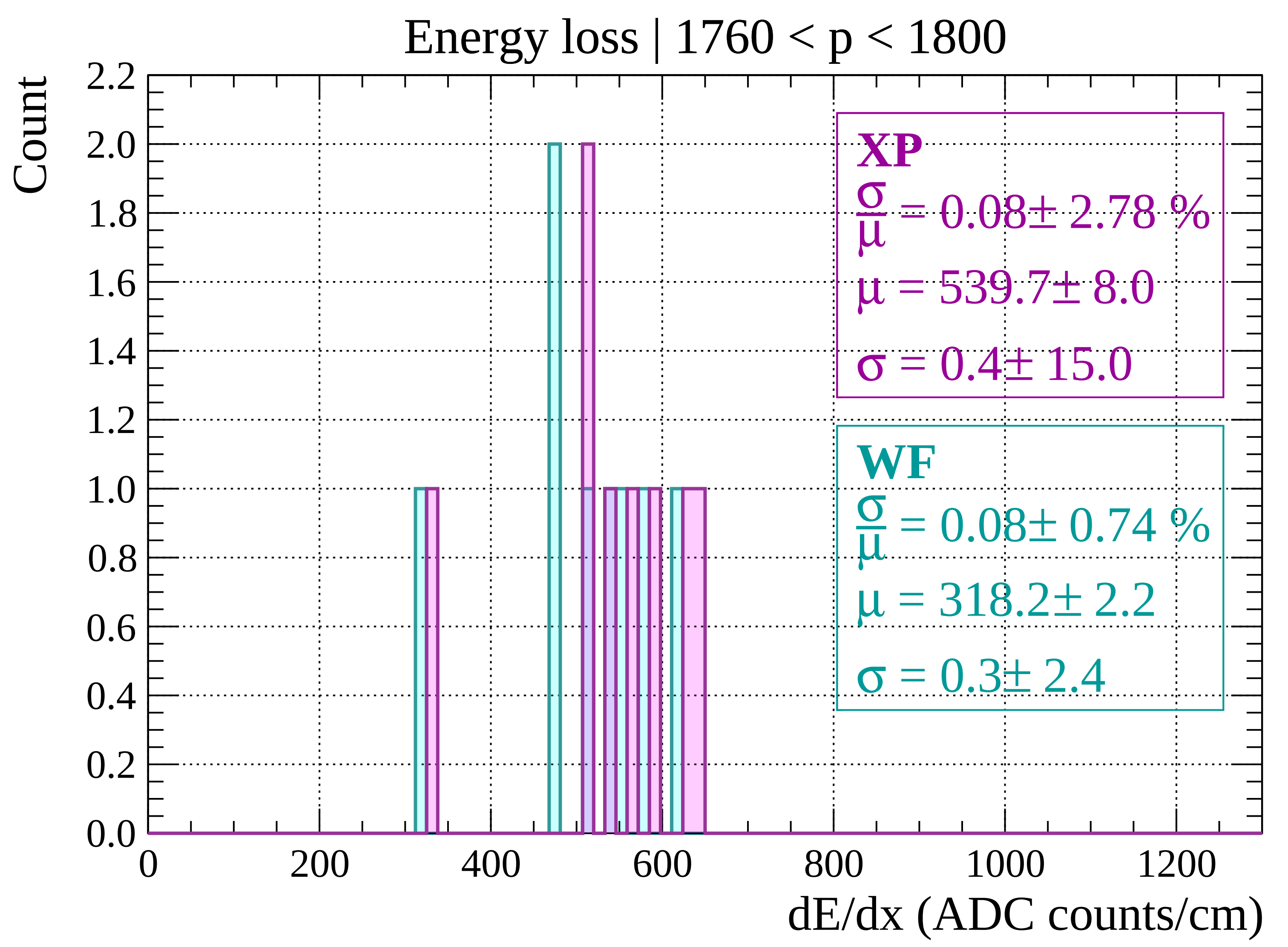
Energy loss | 1640 < p < 1680
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
σ/μ = 97.67± 313.21 %
μ = 521.5± 769.6
σ = 509.4± 881.9
WF
σ/μ = 315.08± 5830.97 %
μ = 306.9± 5321.2
σ = 967.0± 1130.3



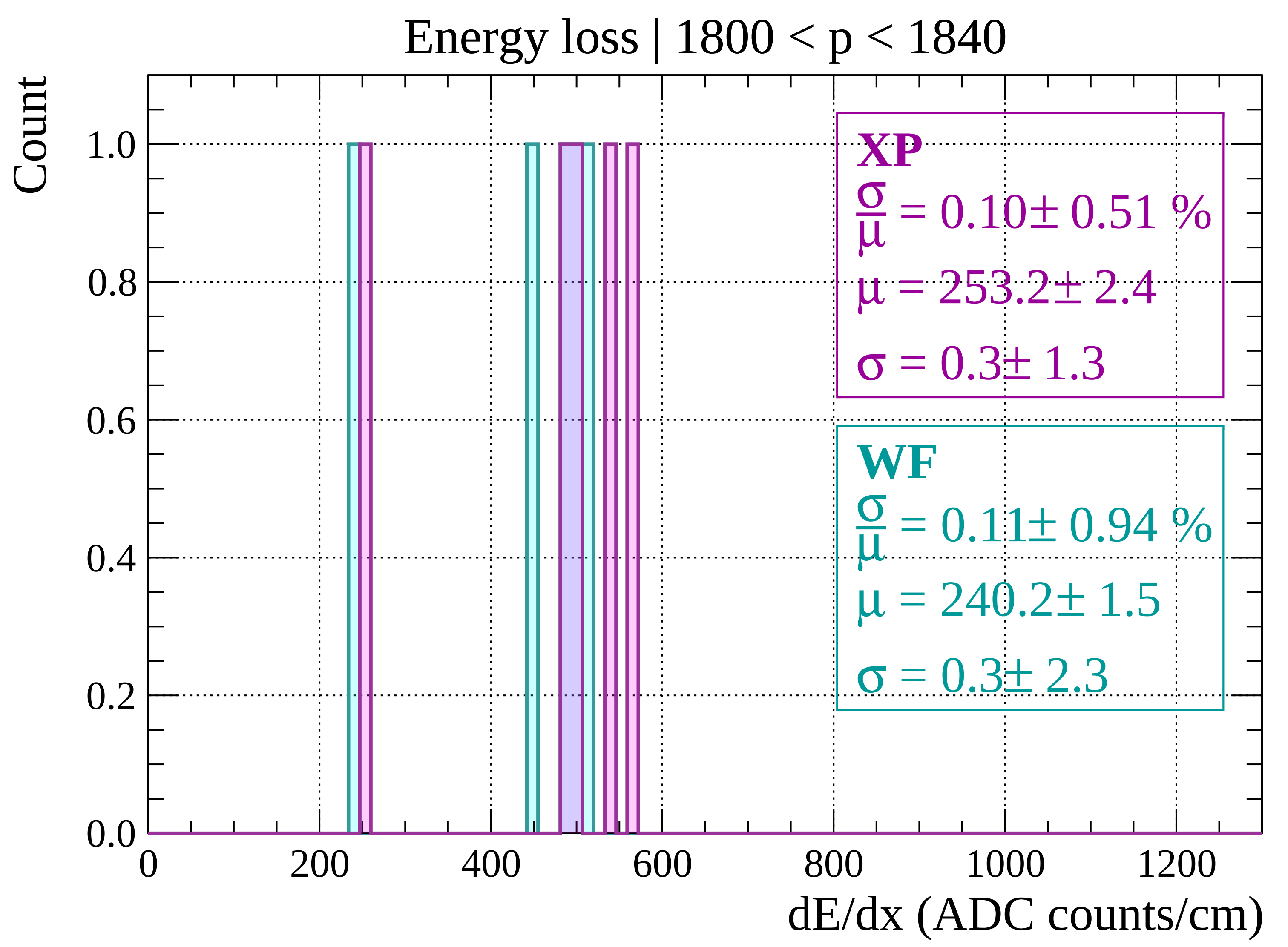
Energy loss | 1680 < p < 1720
Count
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 59.98± 535.87 %
μ = 731.7± 5589.6
σ = 438.9± 568.1
WF
σ/μ = 0.24± 19.18 %
μ = 474.0± 30.1
σ = 1.2± 90.8



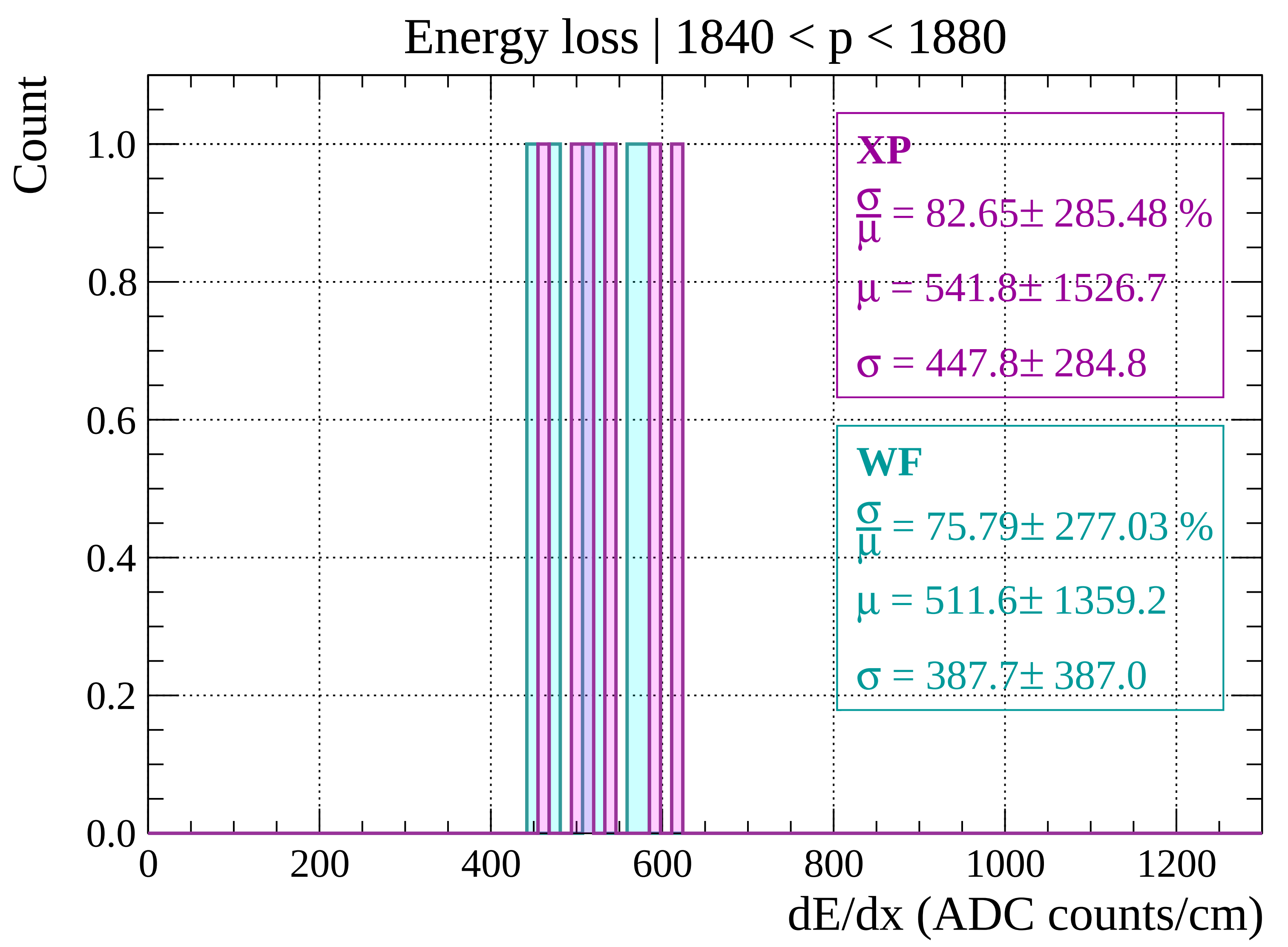
Energy loss | 1720 < p < 1760
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 73.00± 172.38 %
μ = 458.6± 407.0
σ = 334.7± 493.4
WF
$\frac{\sigma}{\mu}$ = 84.62± 229.60 %
μ = 425.8± 498.6
σ = 360.3± 555.8



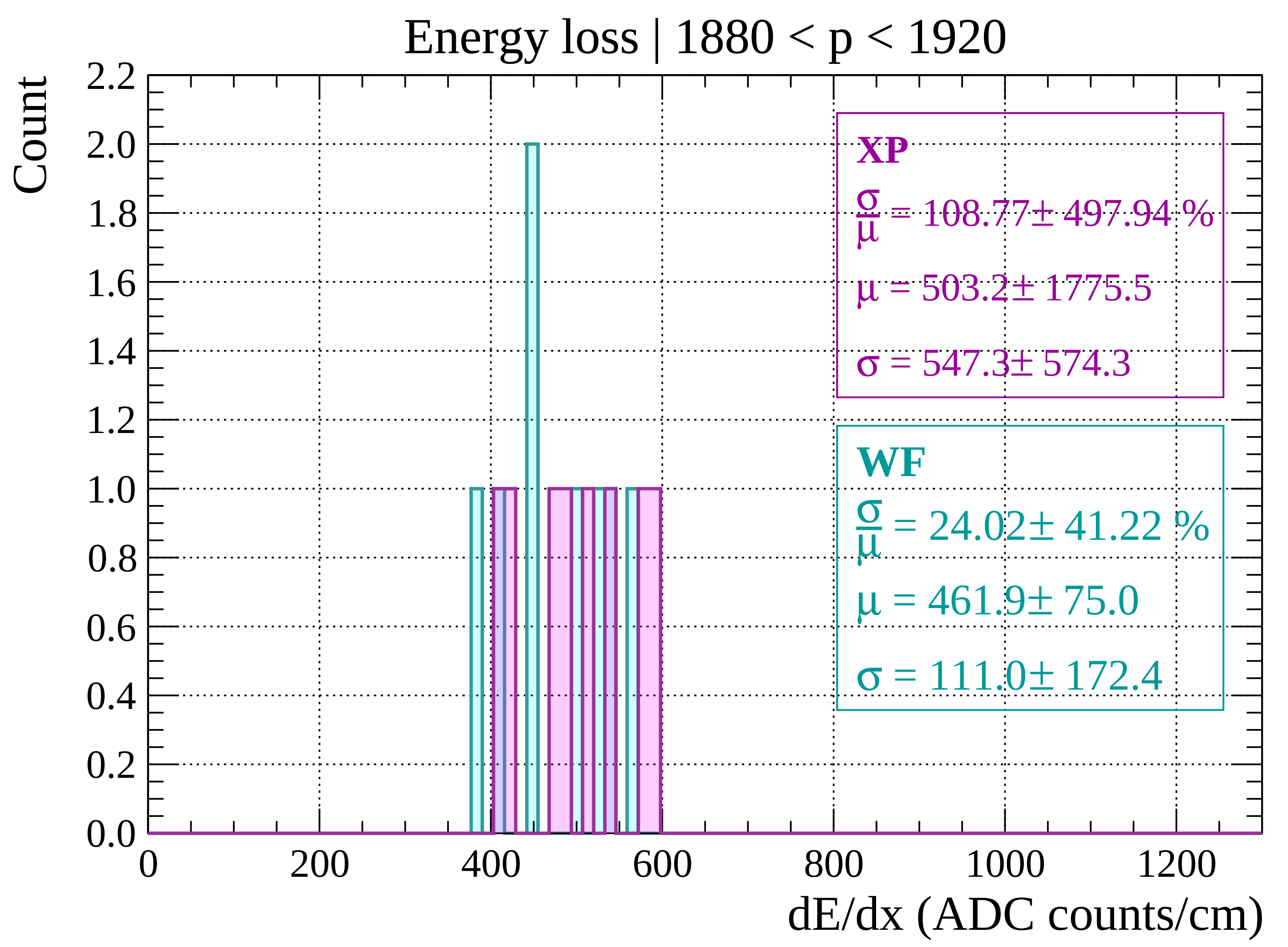
Energy loss | 1760 < p < 1800
Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 0.08± 2.78 %
μ = 539.7± 8.0
σ = 0.4± 15.0
WF
σ/μ = 0.08± 0.74 %
μ = 318.2± 2.2
σ = 0.3± 2.4



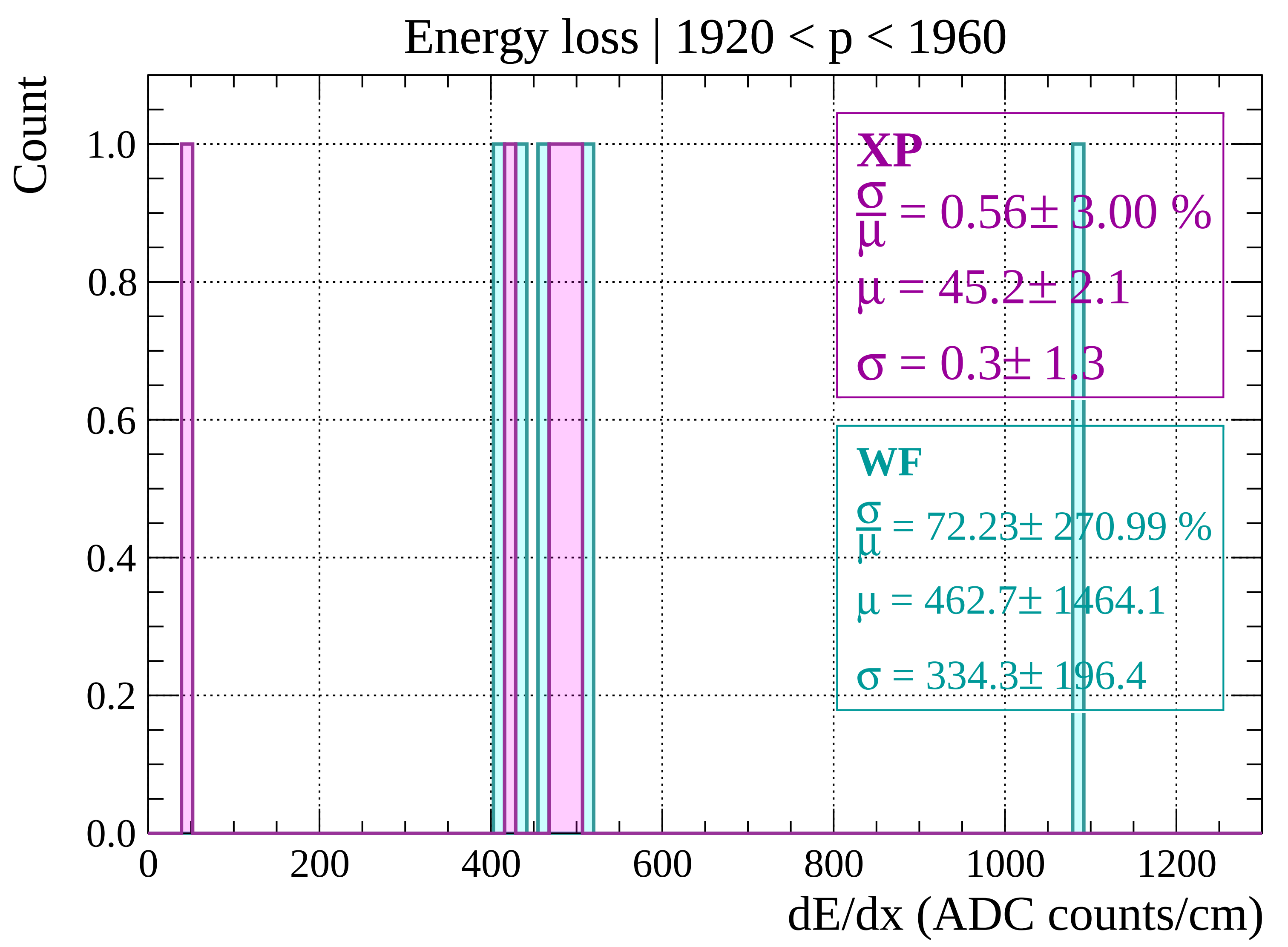
Energy loss | 1800 < p < 1840
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 0.10 \pm 0.51$ %
$\mu = 253.2 \pm 2.4$
$\sigma = 0.3 \pm 1.3$
WF
$\frac{\sigma}{\mu} = 0.11 \pm 0.94$ %
$\mu = 240.2 \pm 1.5$
$\sigma = 0.3 \pm 2.3$



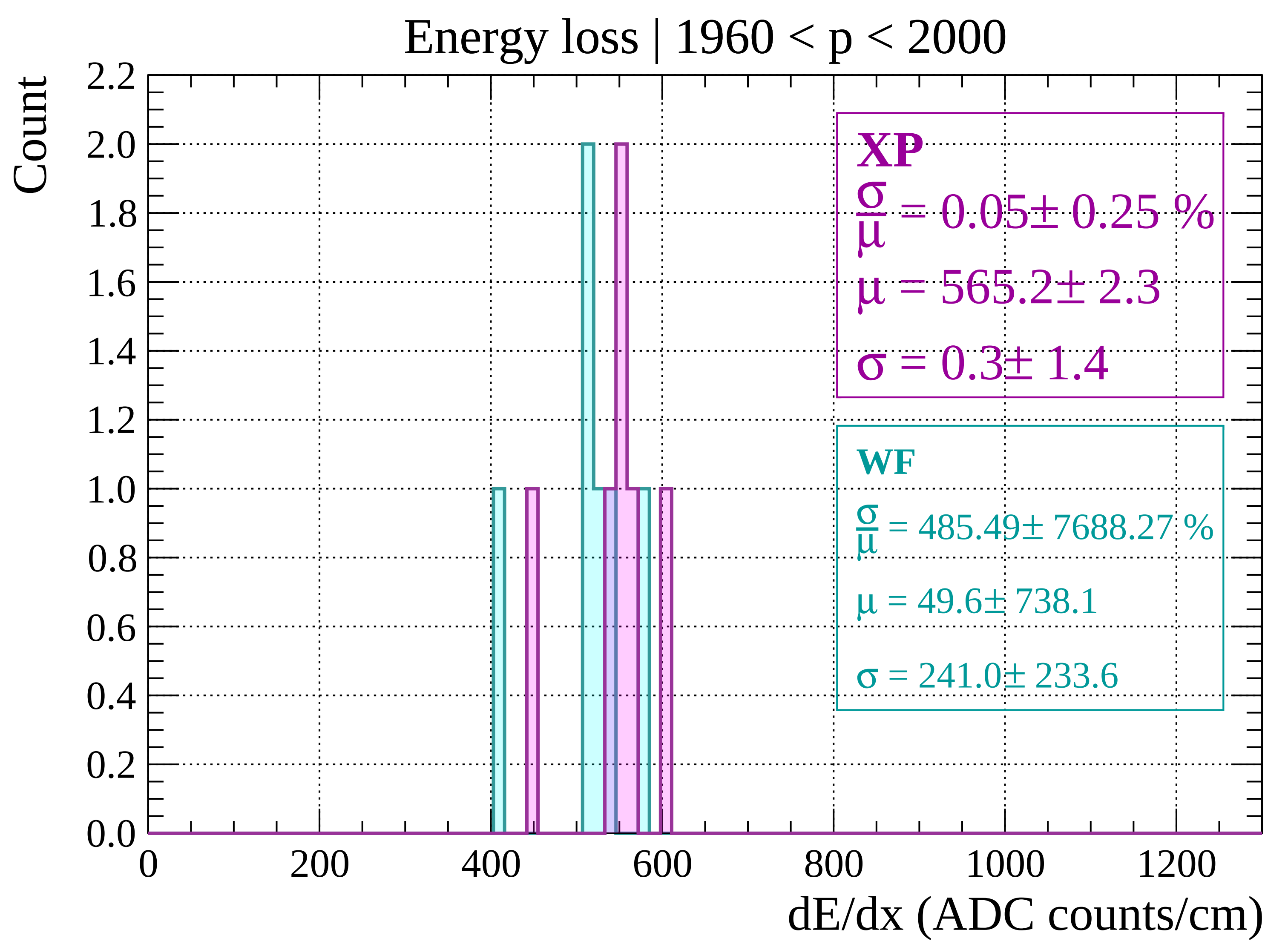
Energy loss | 1840 < p < 1880
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 82.65± 285.48 %
μ = 541.8± 1526.7
σ = 447.8± 284.8
WF
σ/μ = 75.79± 277.03 %
μ = 511.6± 1359.2
σ = 387.7± 387.0



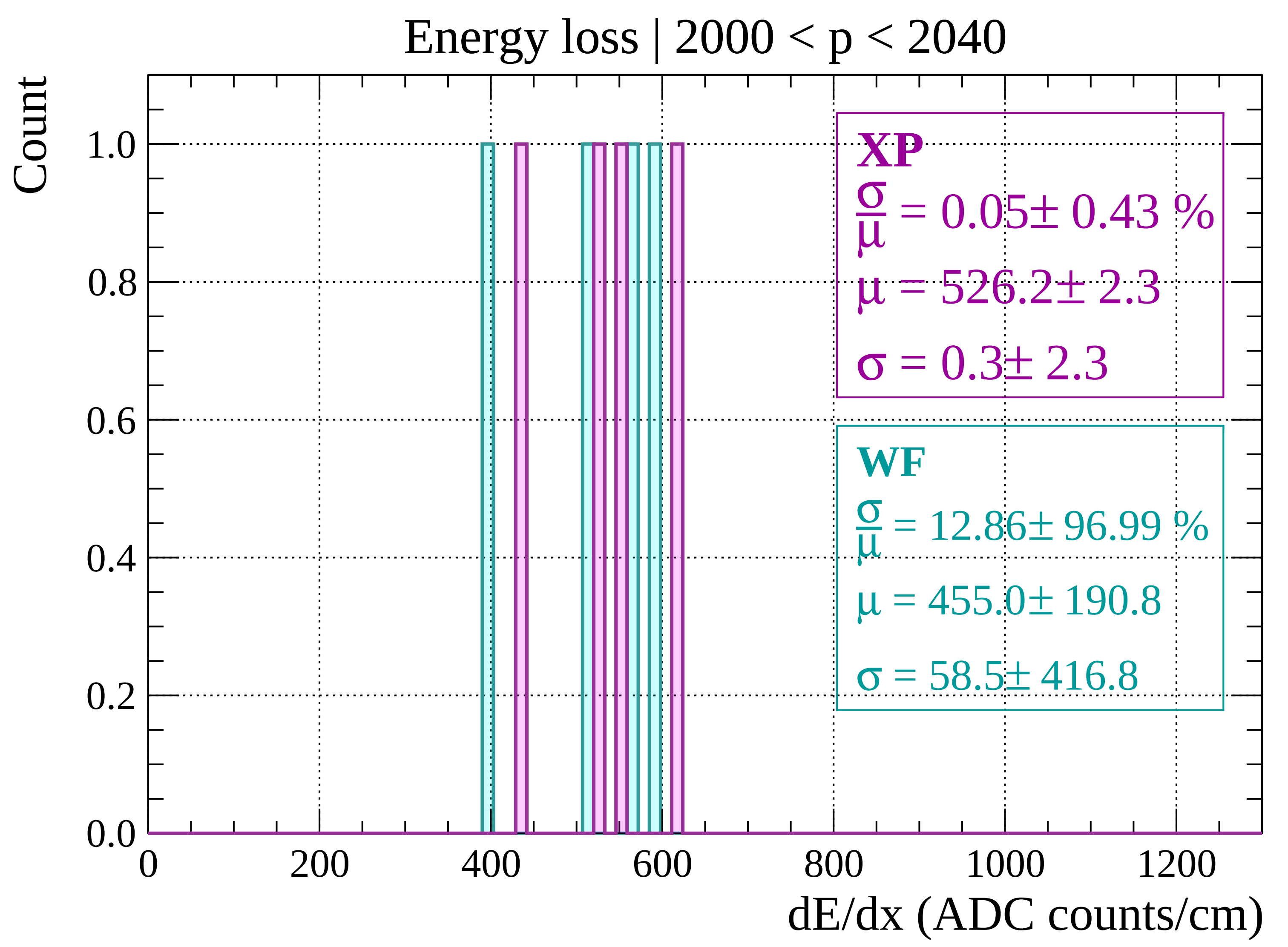
Energy loss | 1880 < p < 1920
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
σ/μ = 108.77± 497.94 %
μ = 503.2± 1775.5
σ = 547.3± 574.3
WF
σ/μ = 24.02± 41.22 %
μ = 461.9± 75.0
σ = 111.0± 172.4



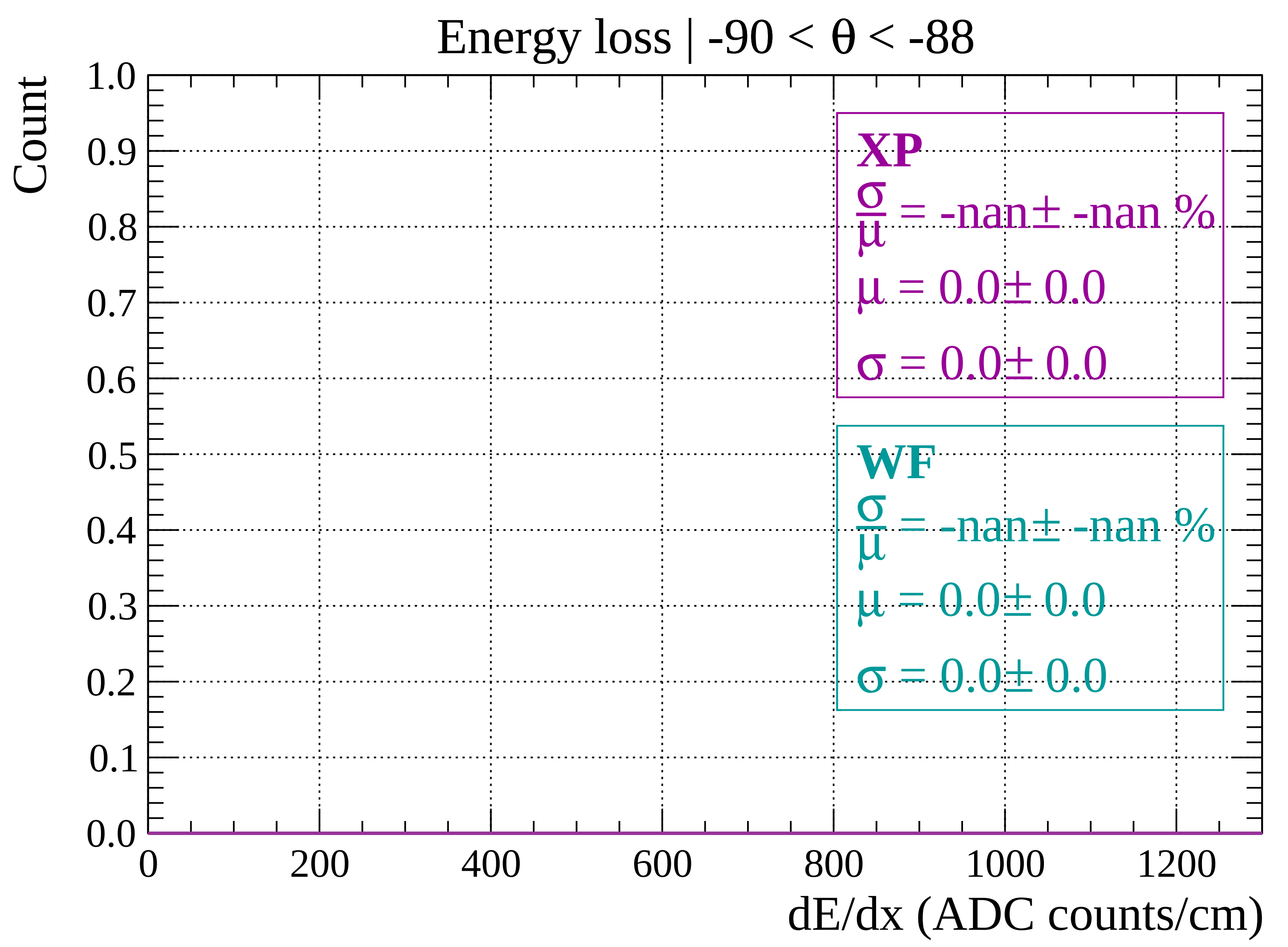
Energy loss | 1920 < p < 1960
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 0.56± 3.00 %
μ = 45.2± 2.1
σ = 0.3± 1.3
WF
$\frac{\sigma}{\mu}$ = 72.23± 270.99 %
μ = 462.7± 1464.1
σ = 334.3± 196.4

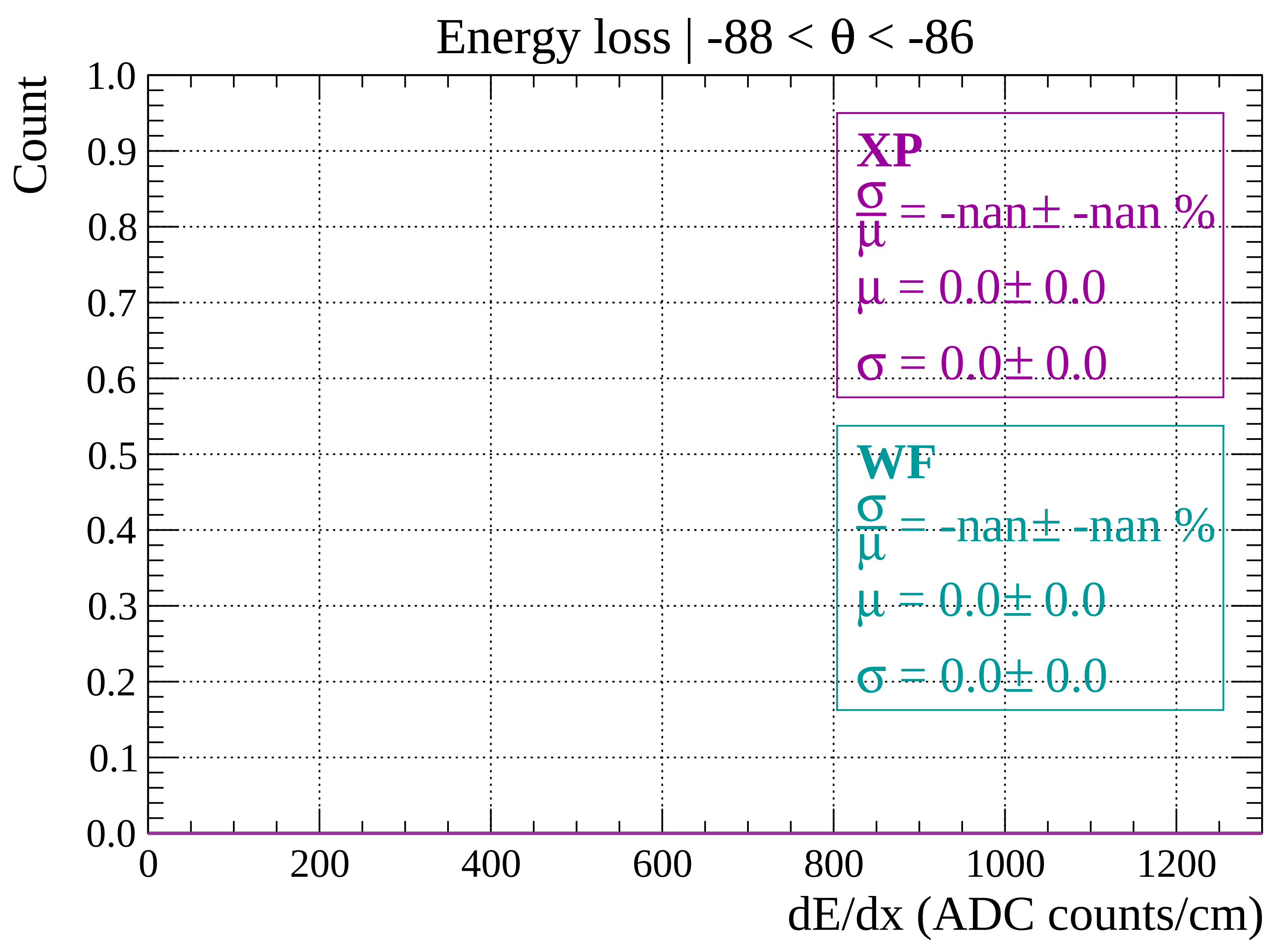


Energy loss | 1960 < p < 2000
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
XP
σ/μ = 0.05± 0.25 %
μ = 565.2± 2.3
σ = 0.3± 1.4
WF
σ/μ = 485.49± 7688.27 %
μ = 49.6± 738.1
σ = 241.0± 233.6

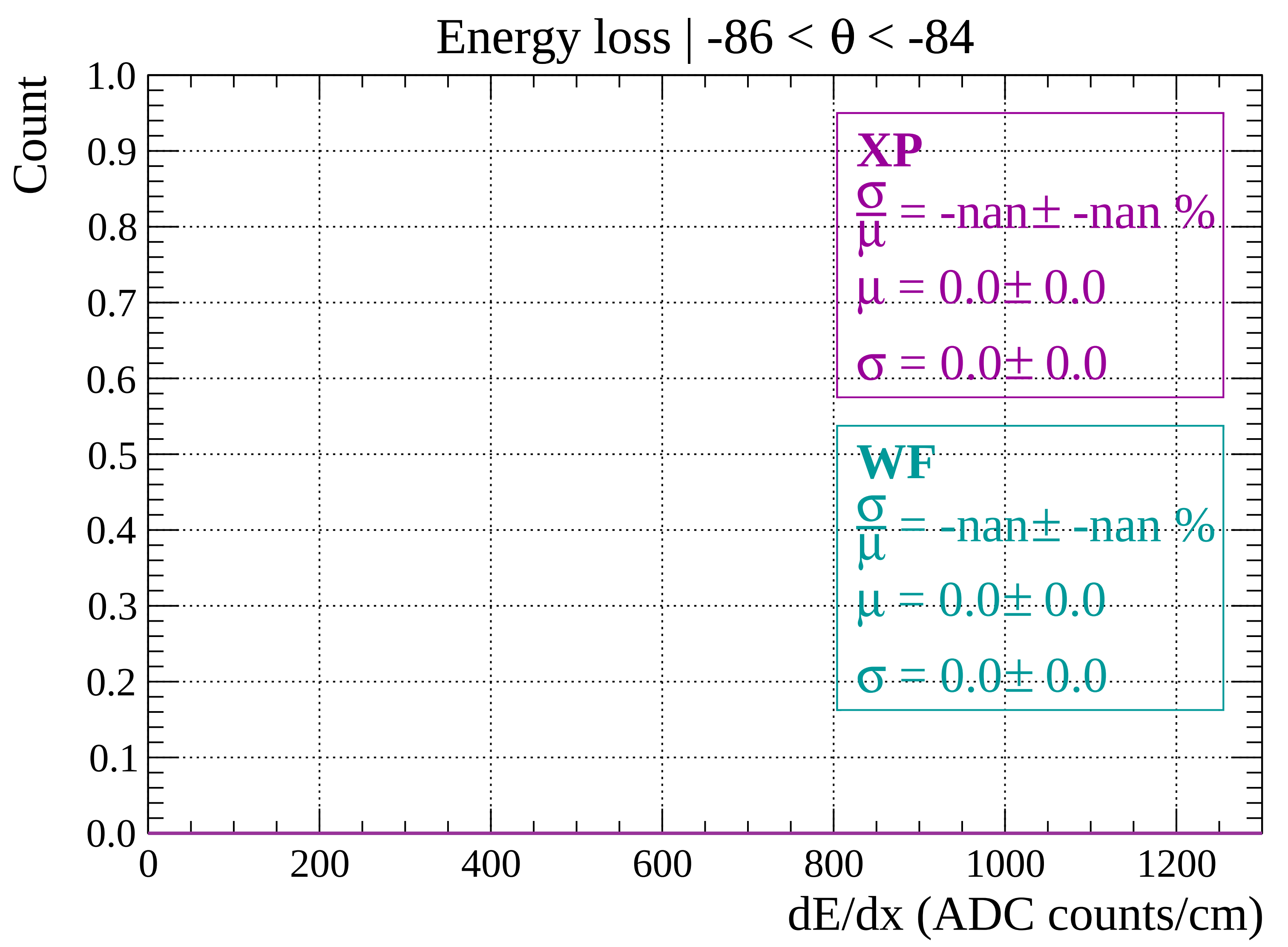


Energy loss | 2000 < p < 2040
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = 0.05± 0.43 %
μ = 526.2± 2.3
σ = 0.3± 2.3
WF
σ/μ = 12.86± 96.99 %
μ = 455.0± 190.8
σ = 58.5± 416.8

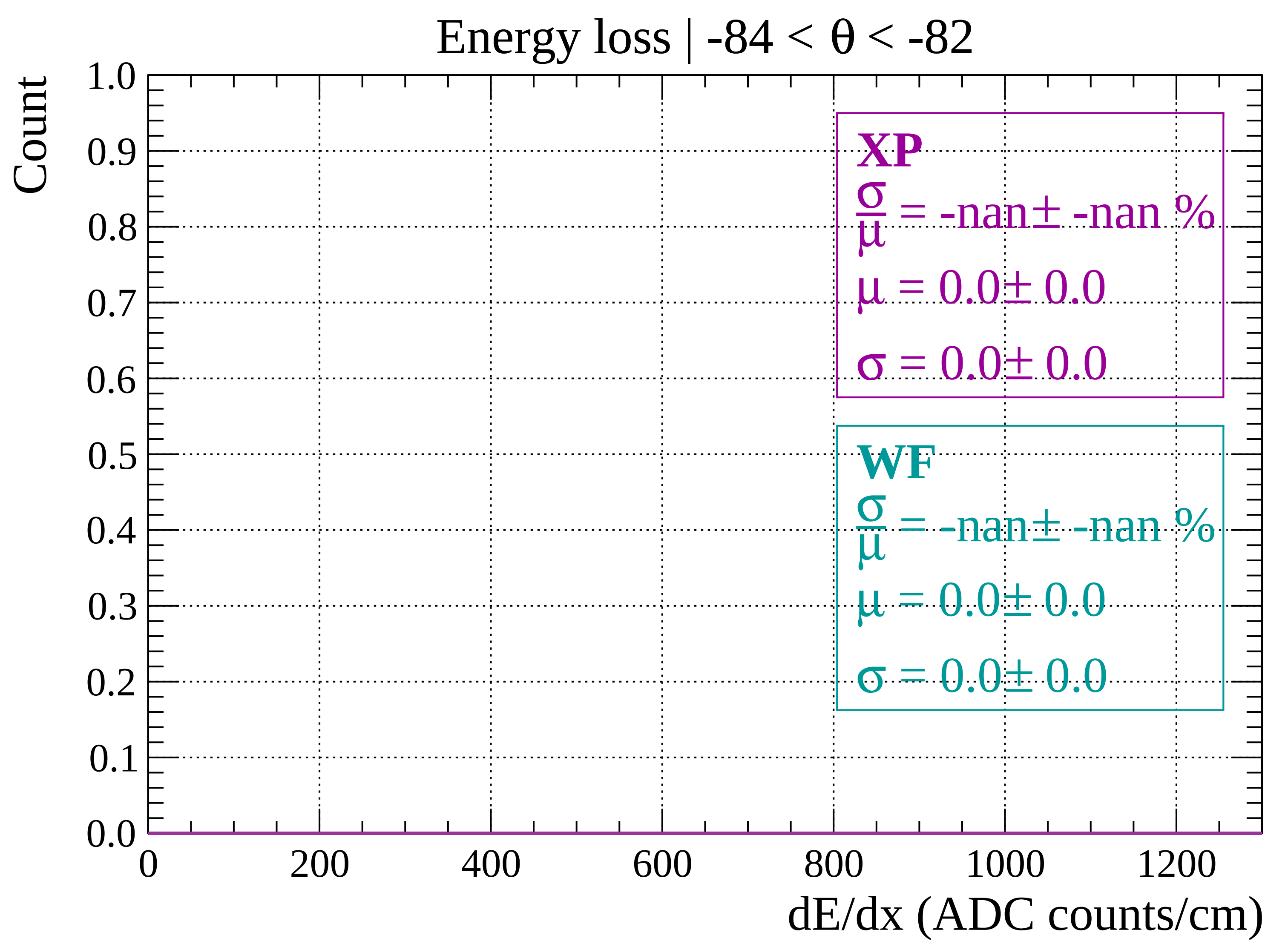

Energy loss | -90 < θ < -88
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



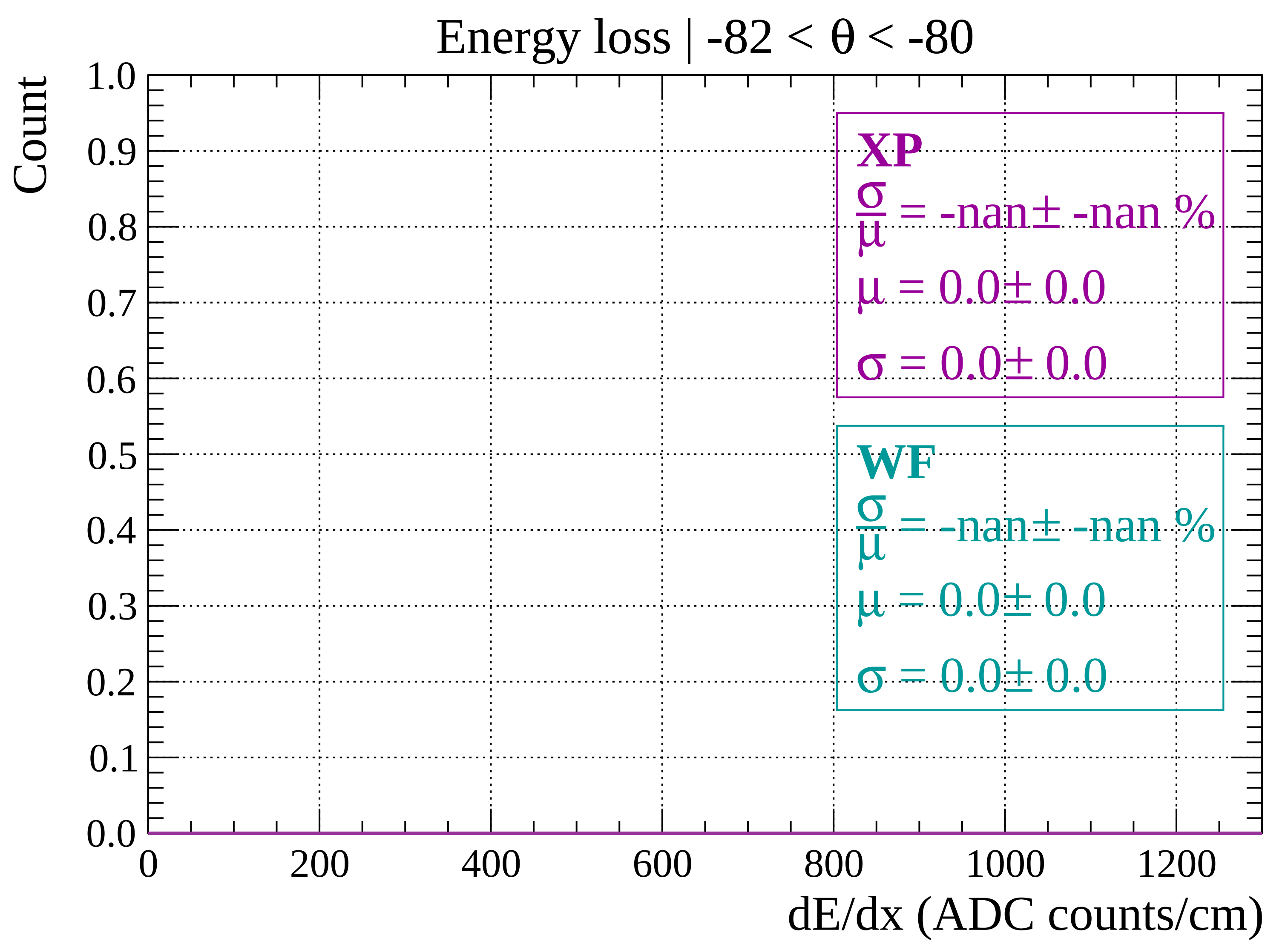
Energy loss | -88 < θ < -86
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



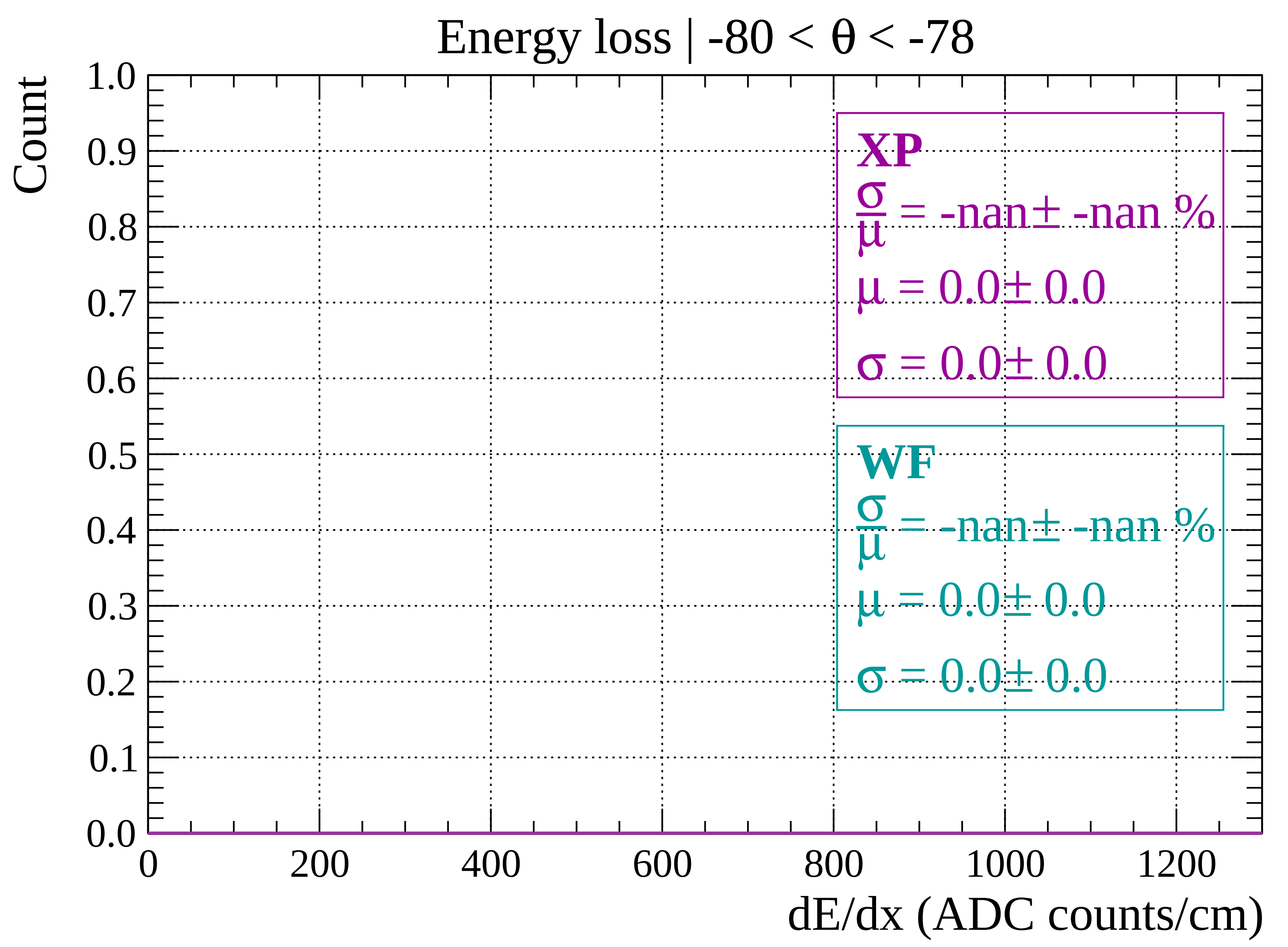
Energy loss | -86 < θ < -84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



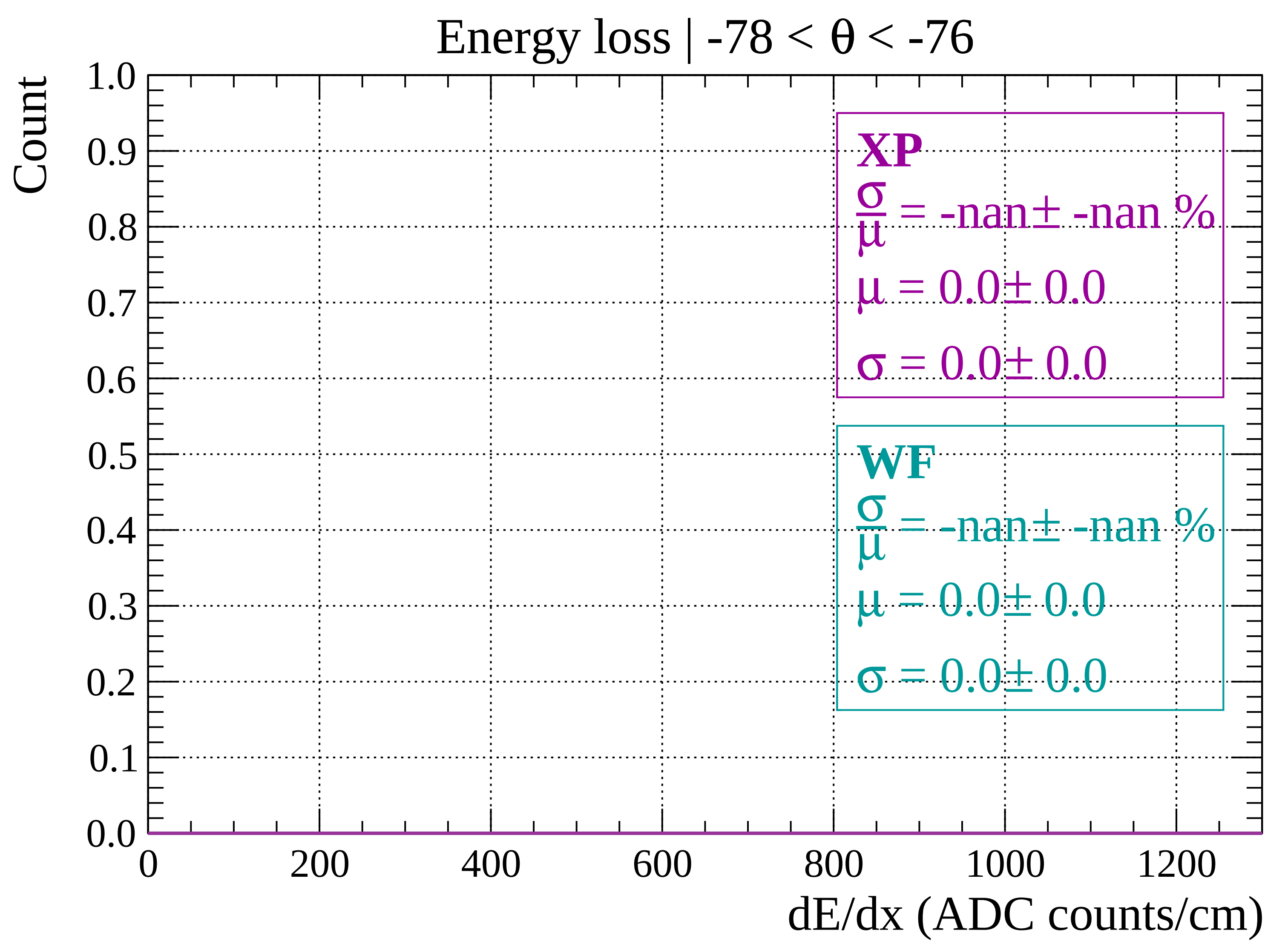
Energy loss | -84 < θ < -82
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

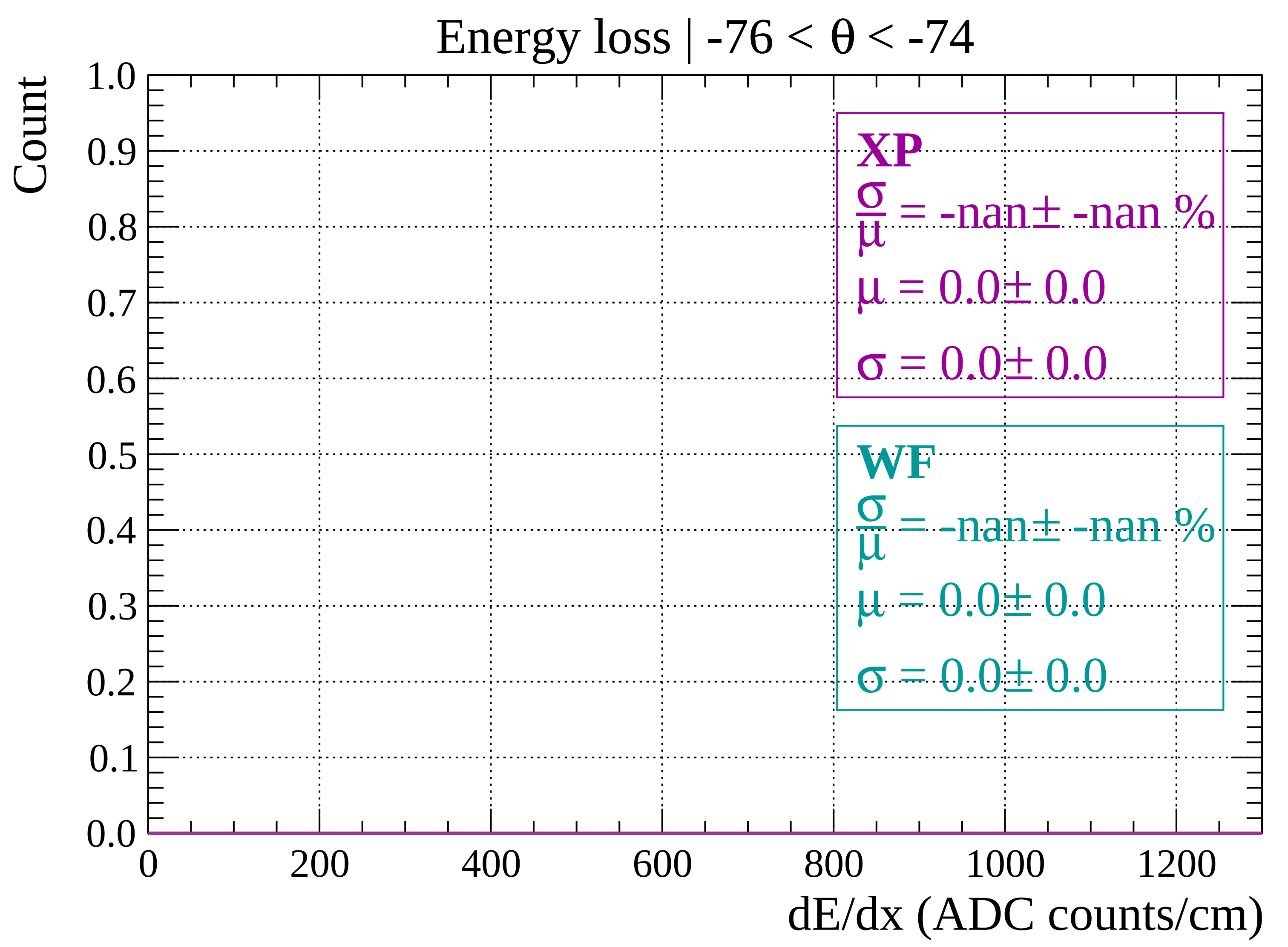


Energy loss | -82 < θ < -80
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

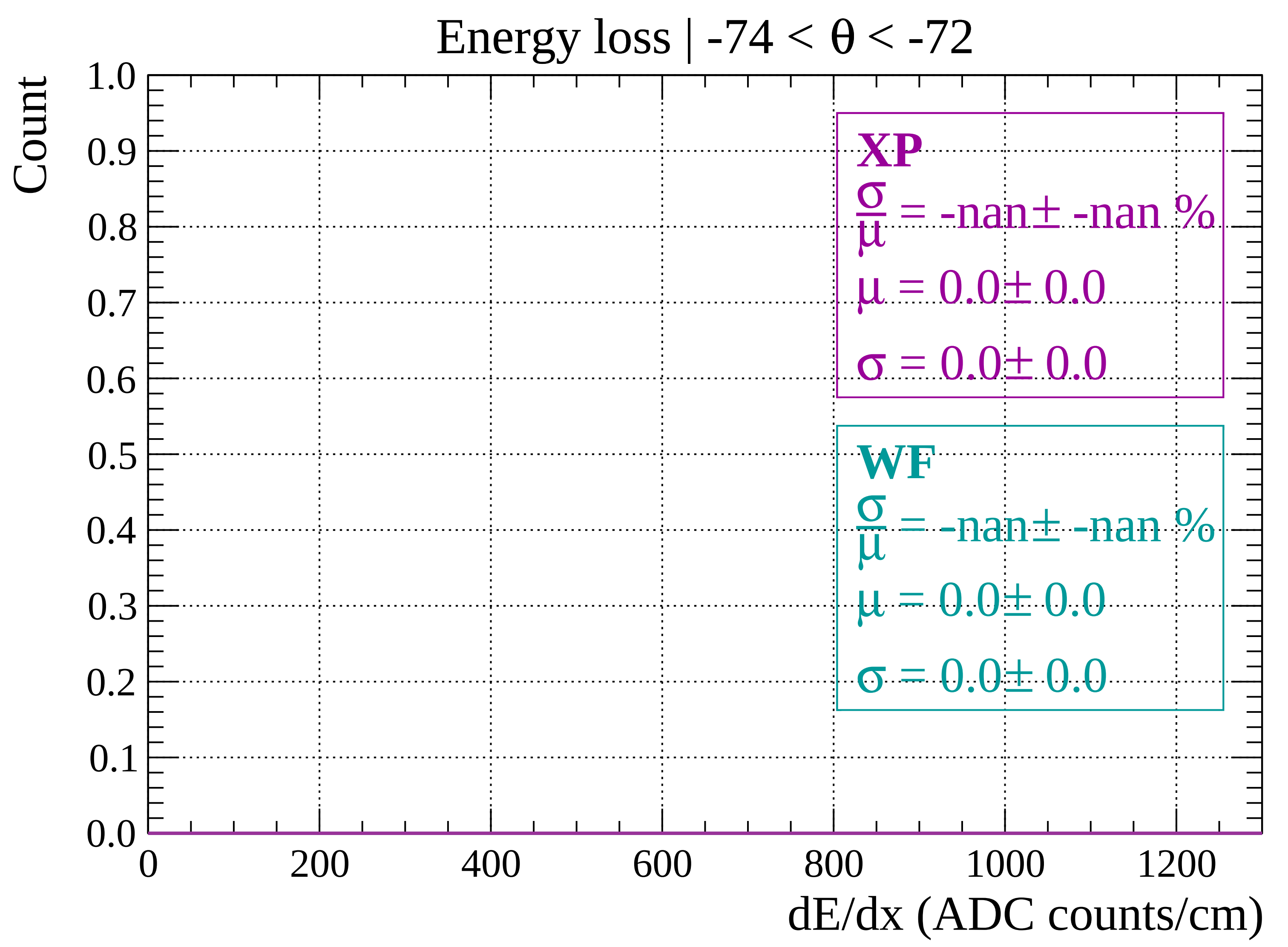


Energy loss | -80 < θ < -78
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

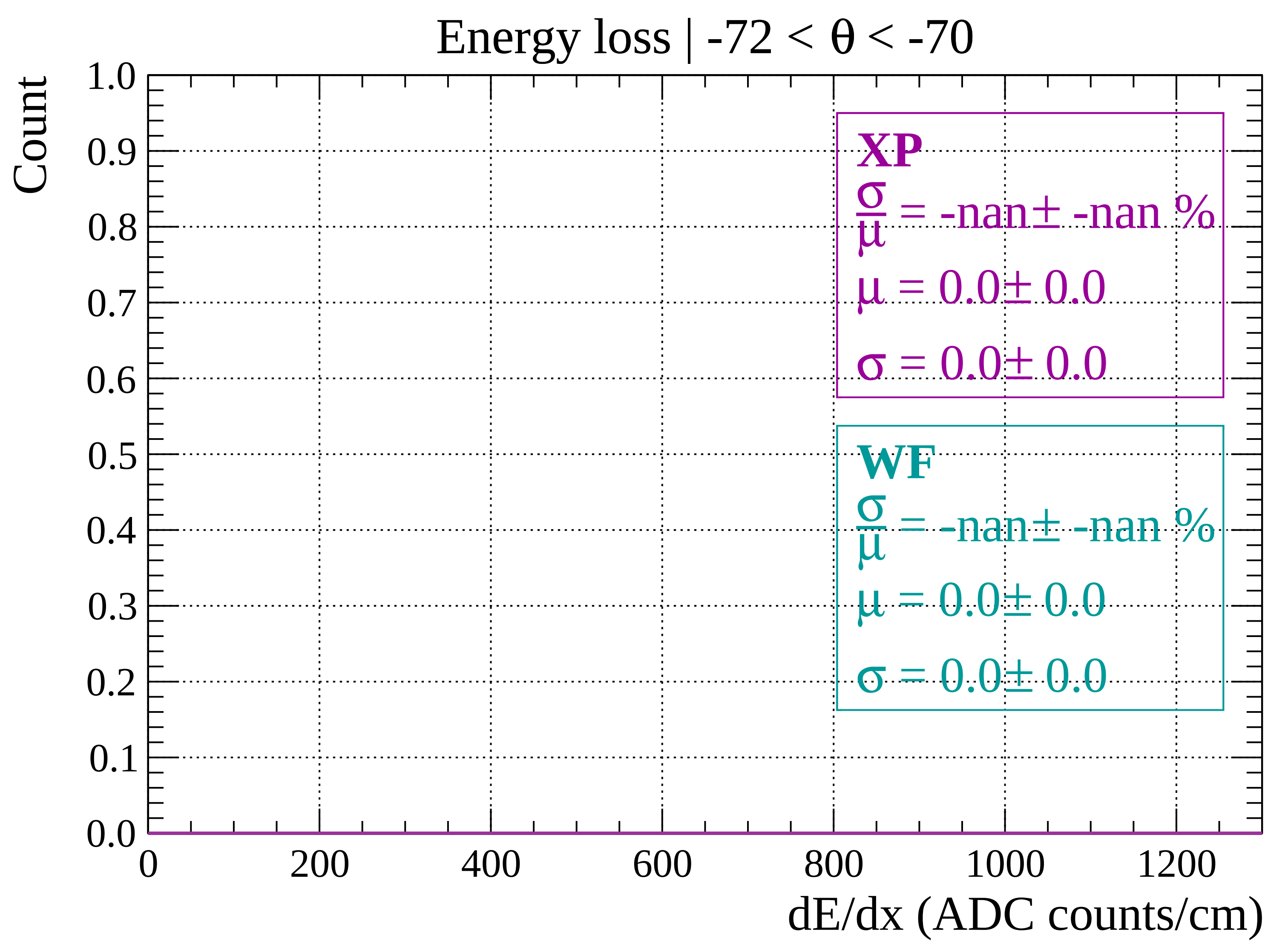

Energy loss | -78 < θ < -76
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

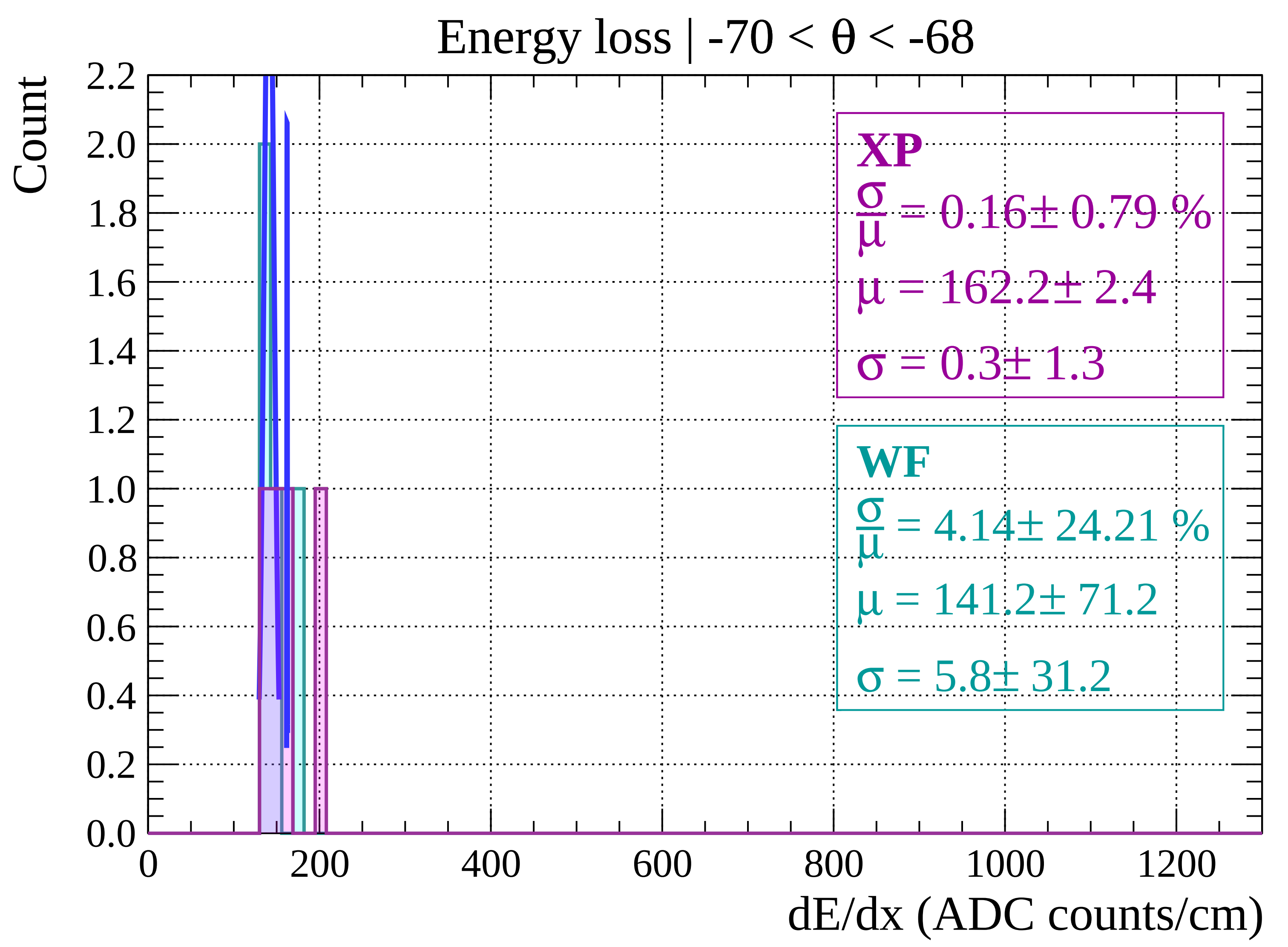


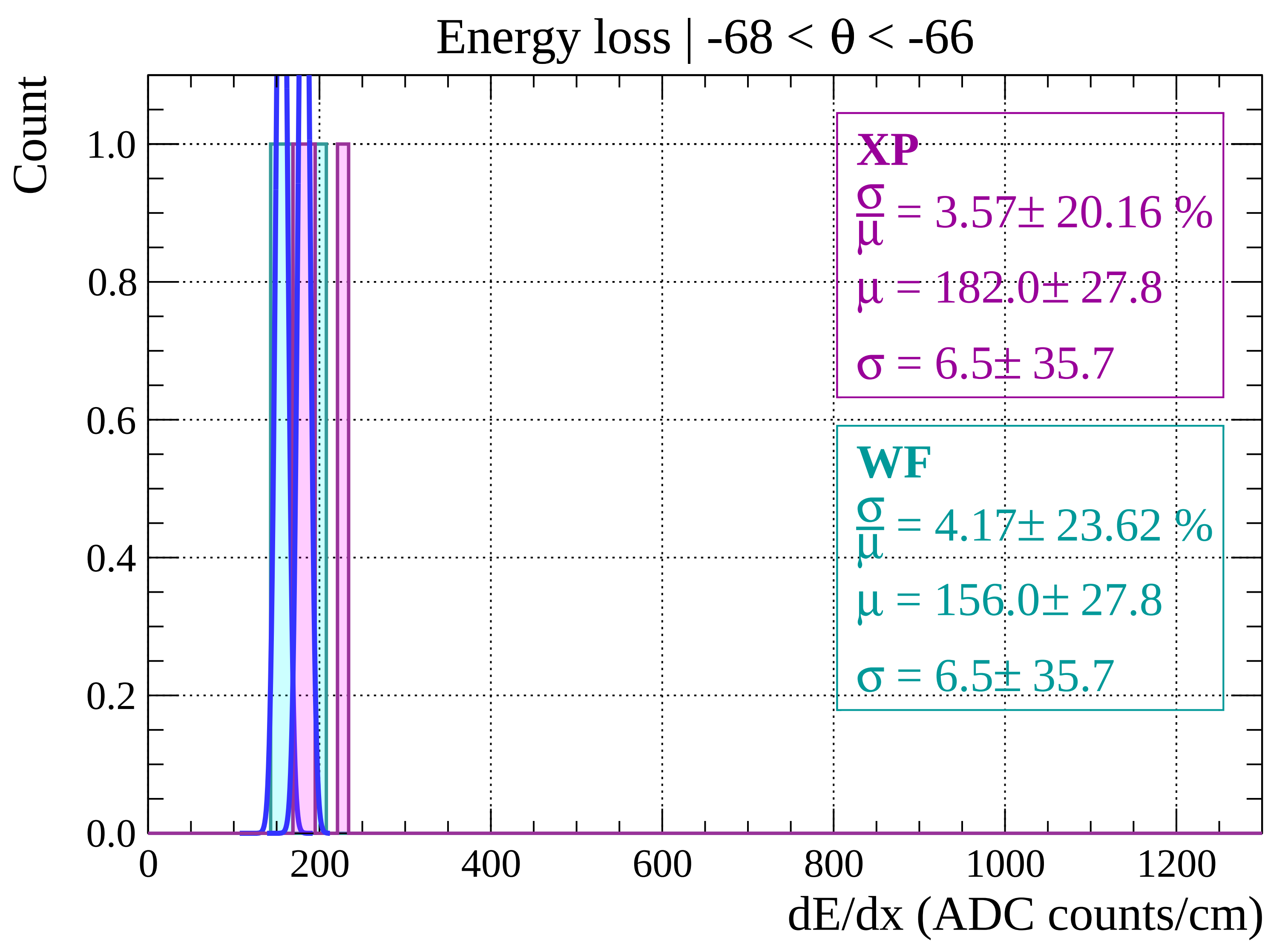
Energy loss | -76 < θ < -74
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



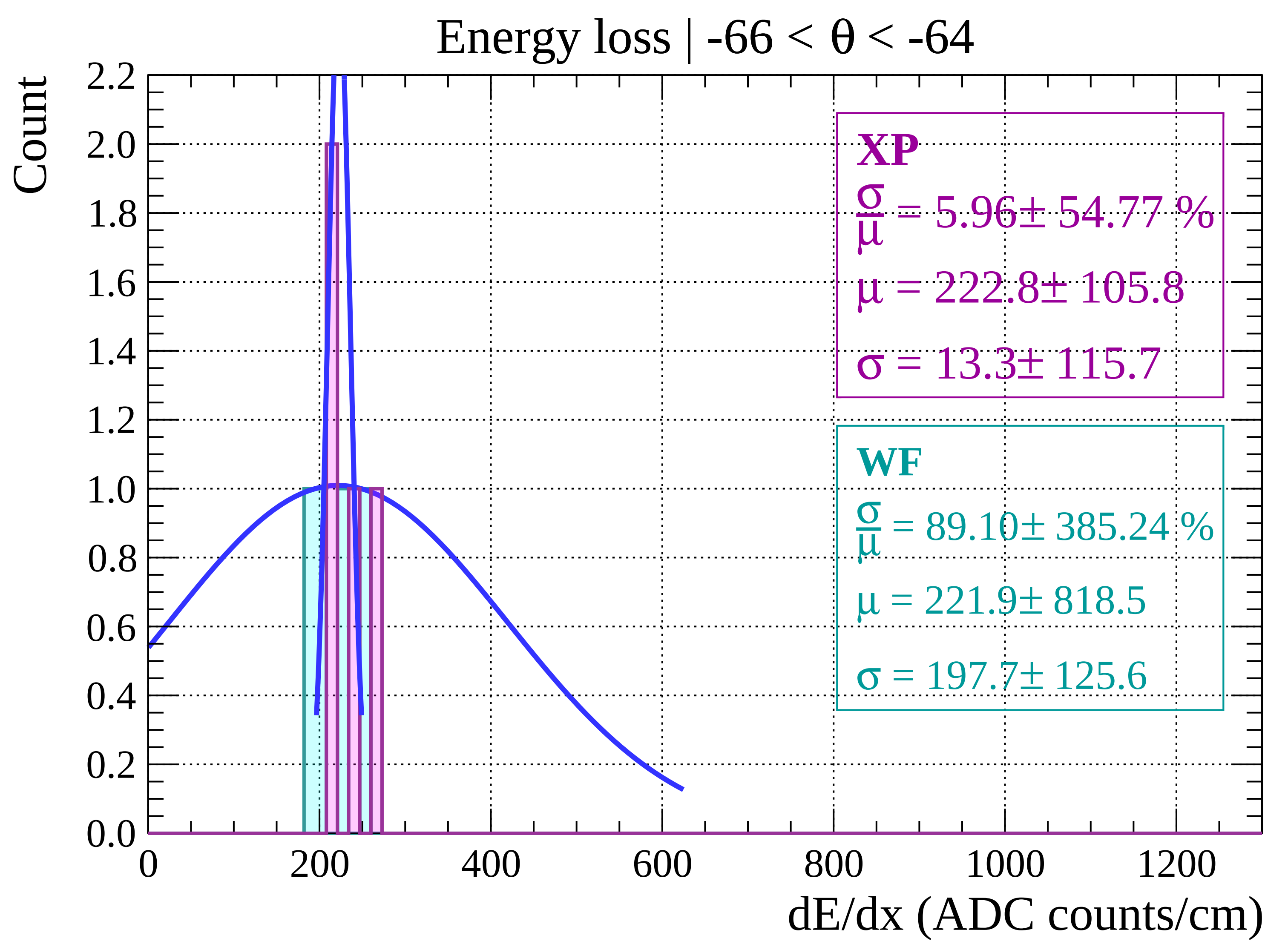
Energy loss | -74 < θ < -72
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | -72 < θ < -70
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

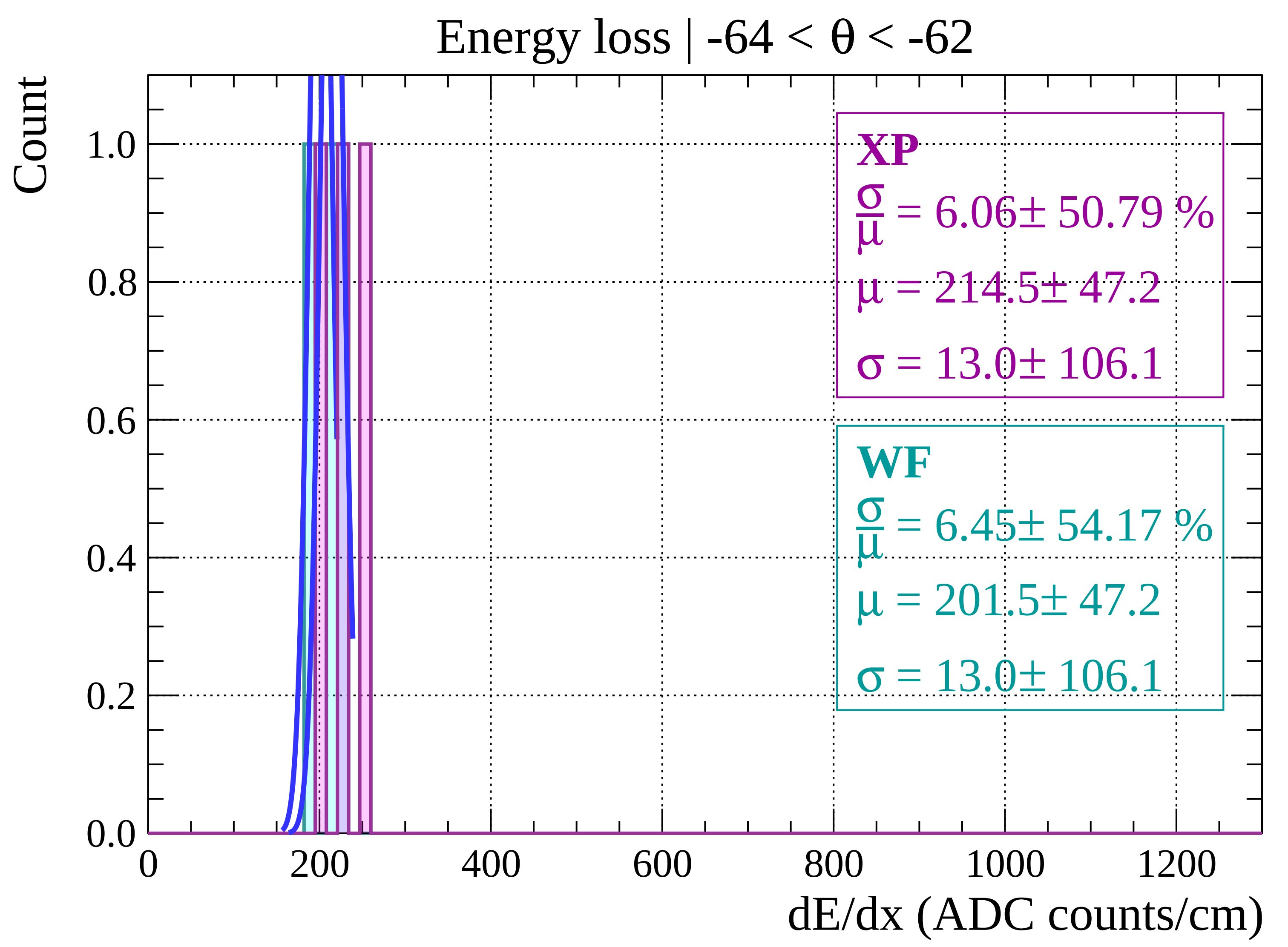

Energy loss | -70 < θ < -68
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
σ/μ = 0.16± 0.79 %
μ = 162.2± 2.4
σ = 0.3± 1.3
WF
σ/μ = 4.14± 24.21 %
μ = 141.2± 71.2
σ = 5.8± 31.2



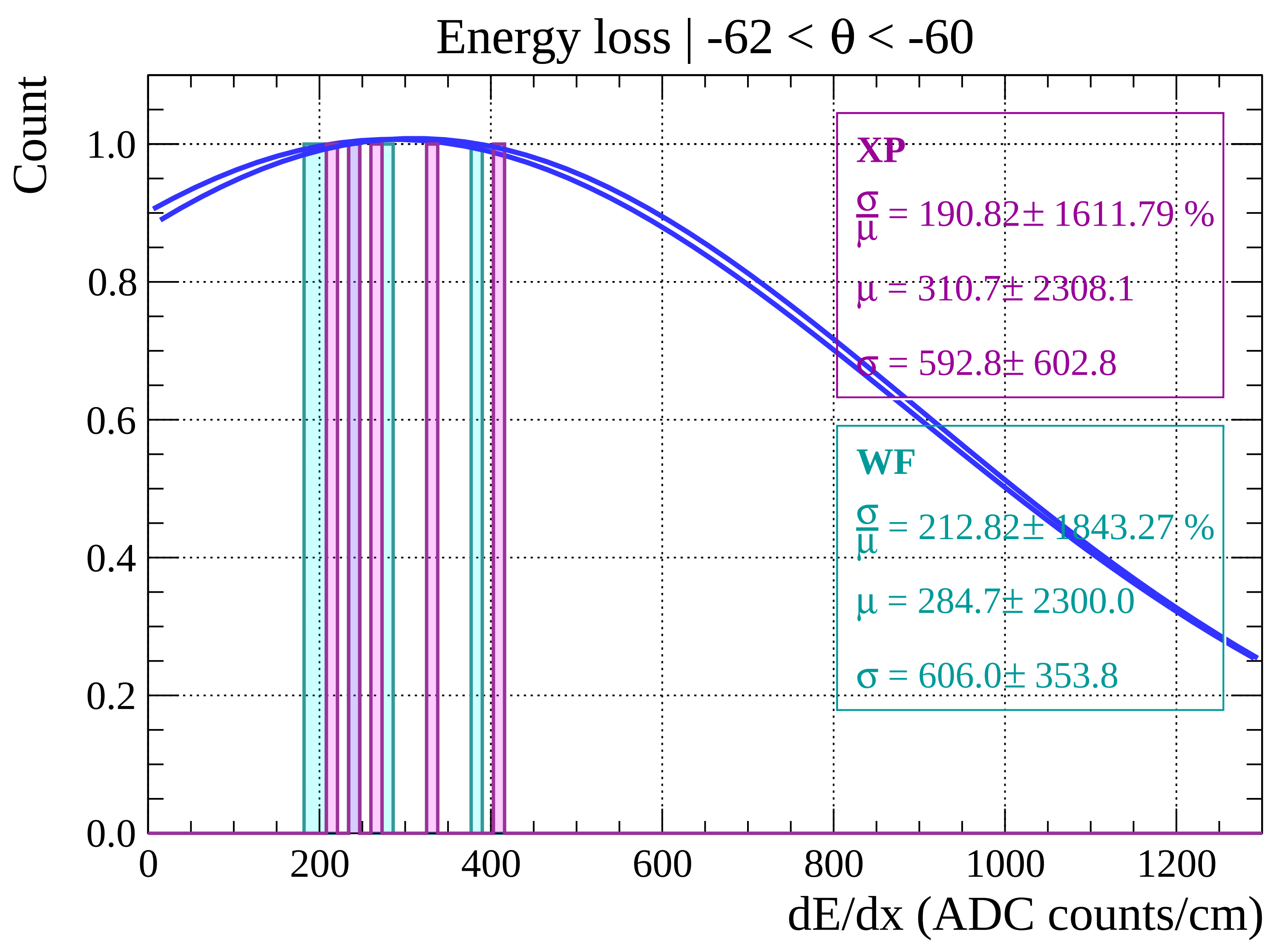
Energy loss | -68 < θ < -66
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 3.57± 20.16 %
μ = 182.0± 27.8
σ = 6.5± 35.7
WF
σ/μ = 4.17± 23.62 %
μ = 156.0± 27.8
σ = 6.5± 35.7



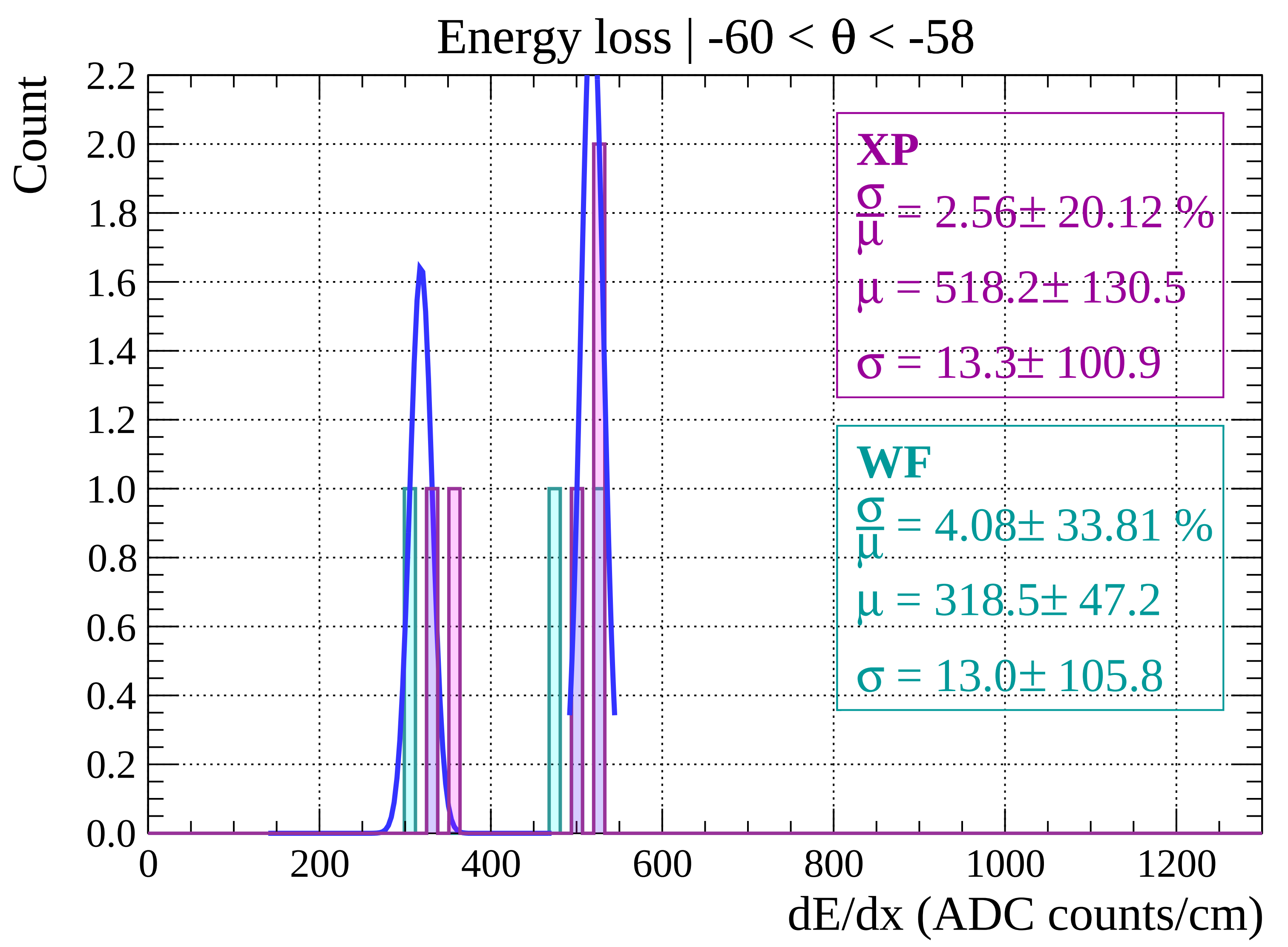
Energy loss | -66 < θ < -64
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
σ/μ = 5.96± 54.77 %
μ = 222.8± 105.8
σ = 13.3± 115.7
WF
σ/μ = 89.10± 385.24 %
μ = 221.9± 818.5
σ = 197.7± 125.6



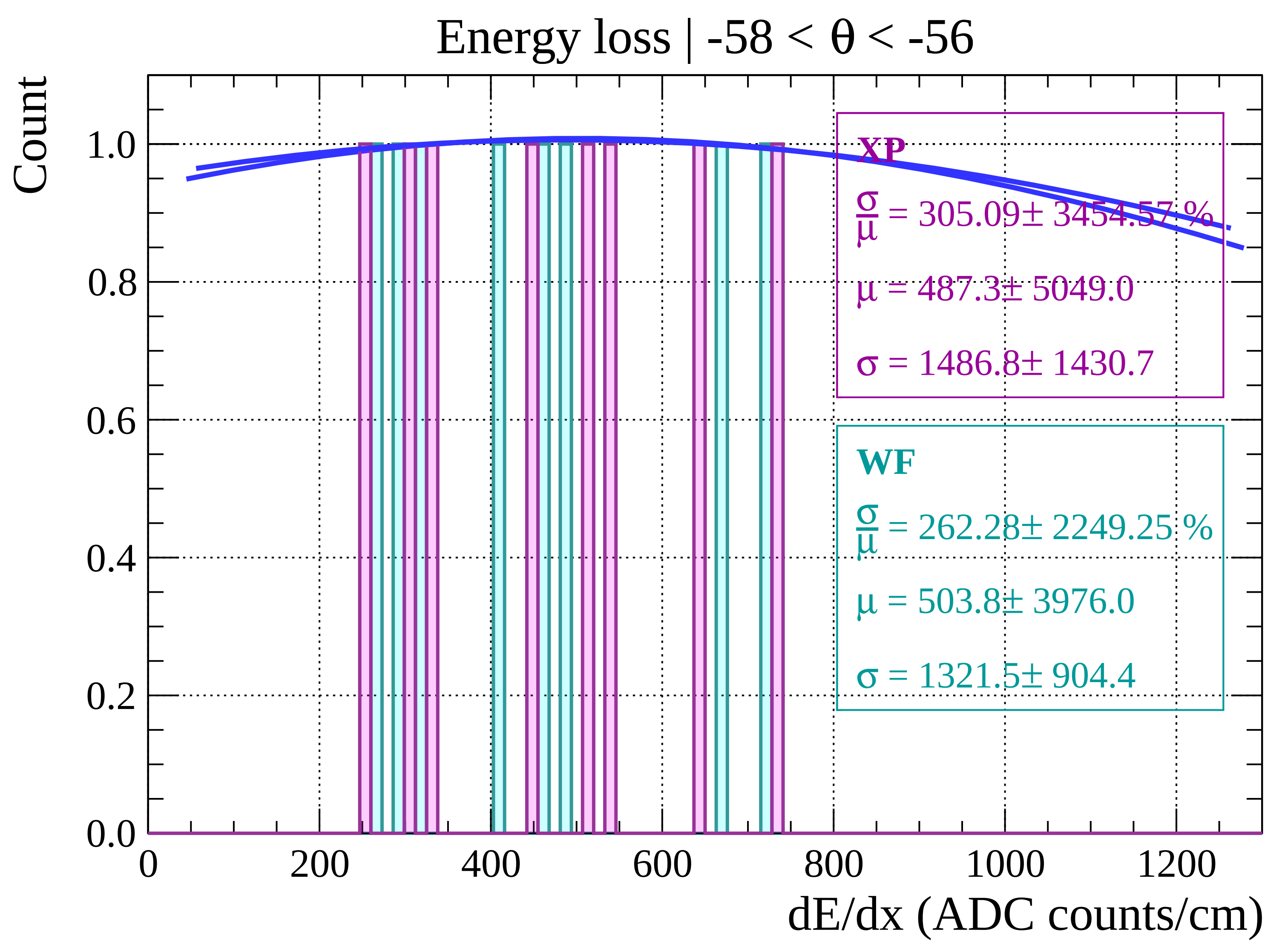
Energy loss | -64 < θ < -62
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 6.06± 50.79 %
μ = 214.5± 47.2
σ = 13.0± 106.1
WF
σ/μ = 6.45± 54.17 %
μ = 201.5± 47.2
σ = 13.0± 106.1



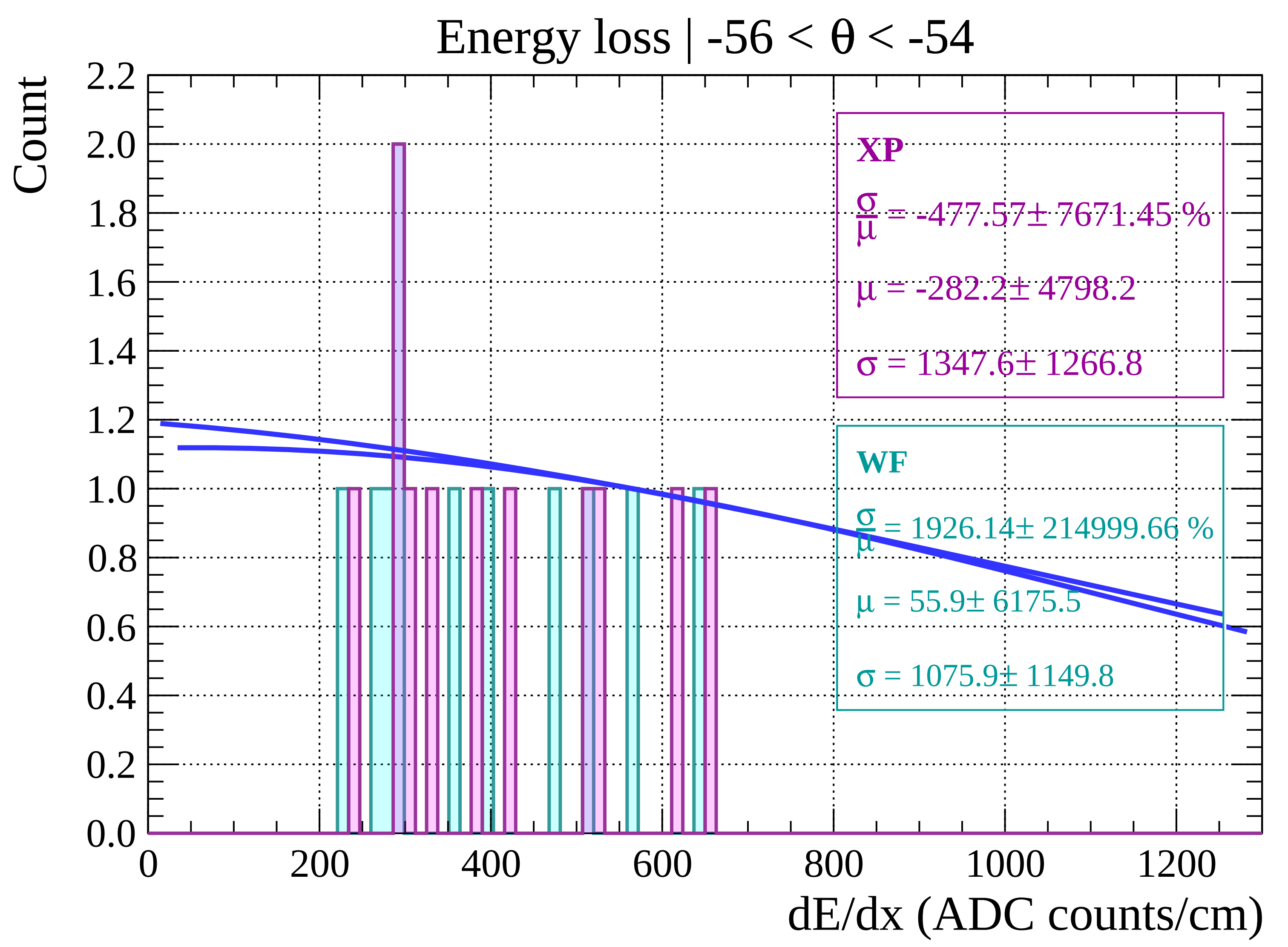
Energy loss | -62 < θ < -60
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 190.82± 1611.79 %
μ = 310.7± 2308.1
σ = 592.8± 602.8
WF
$\frac{\sigma}{\mu}$ = 212.82± 1843.27 %
μ = 284.7± 2300.0
σ = 606.0± 353.8

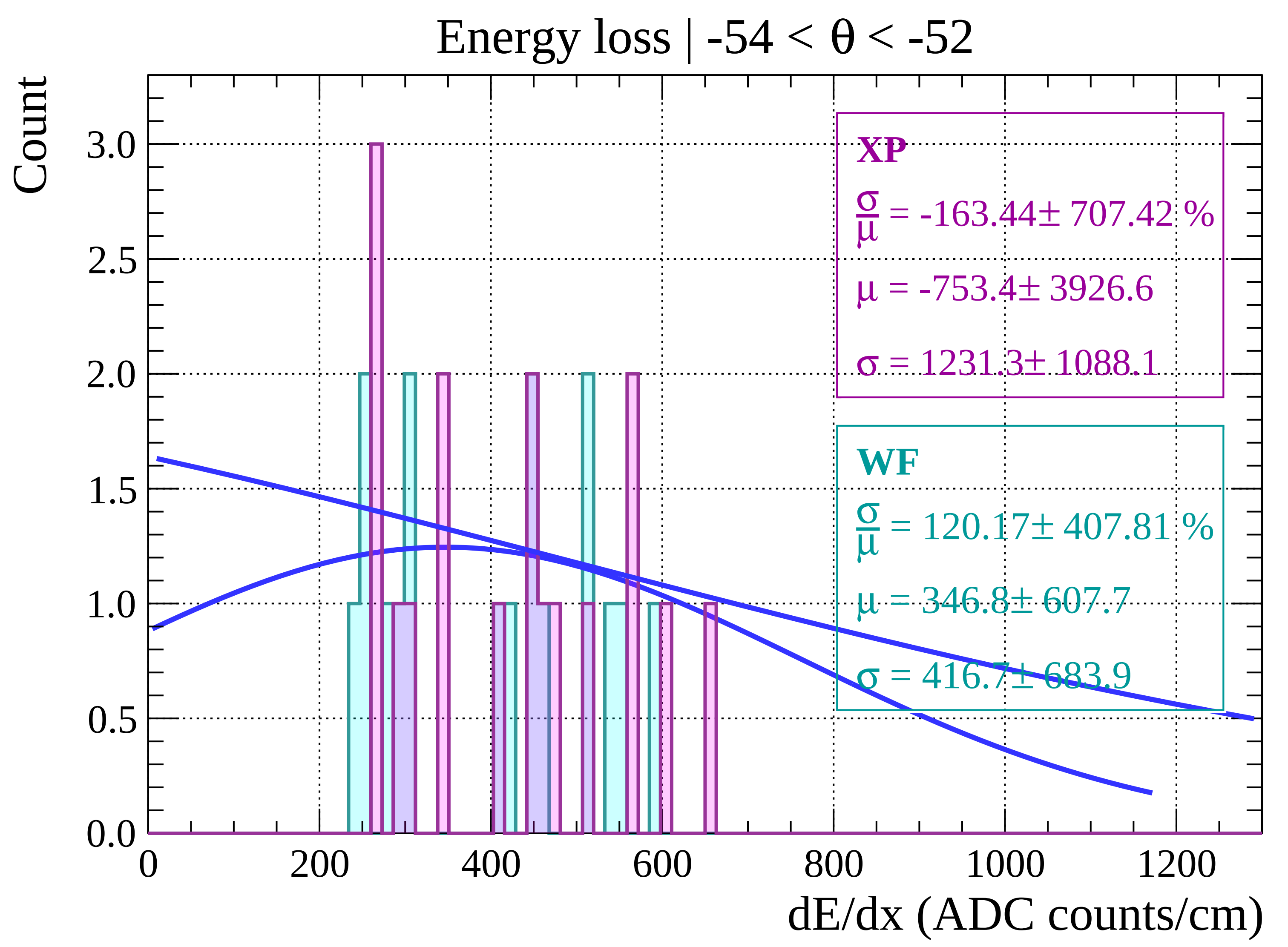


Energy loss | -60 < θ < -58
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
σ/μ = 2.56± 20.12 %
μ = 518.2± 130.5
σ = 13.3± 100.9
WF
σ/μ = 4.08± 33.81 %
μ = 318.5± 47.2
σ = 13.0± 105.8

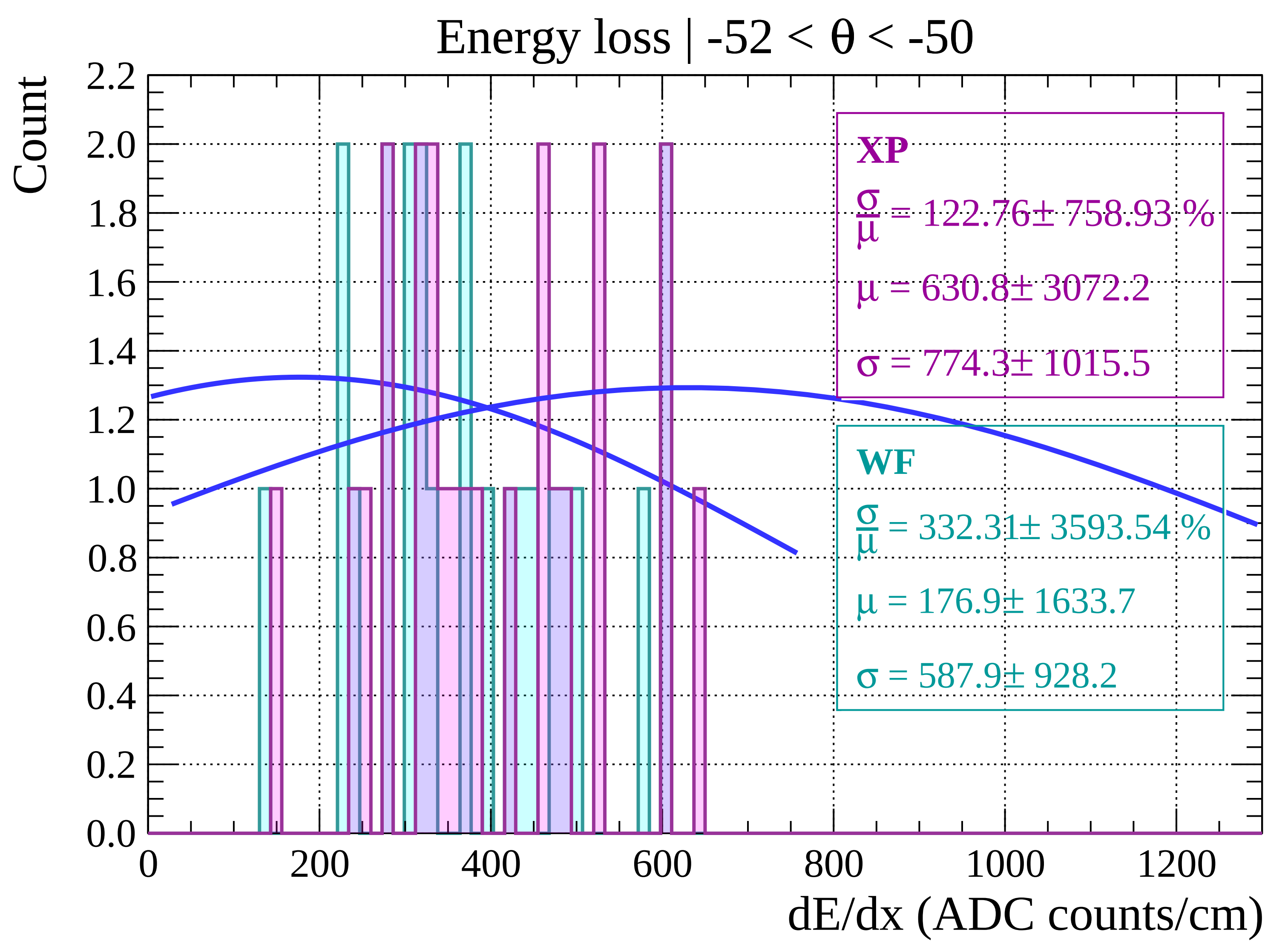


Energy loss | -58 < θ < -56
Count
0.0
0.2
0.4
0.6
0.8
1.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 305.09± 3454.57 %
μ = 487.3± 5049.0
σ = 1486.8± 1430.7
WF
σ/μ = 262.28± 2249.25 %
μ = 503.8± 3976.0
σ = 1321.5± 904.4

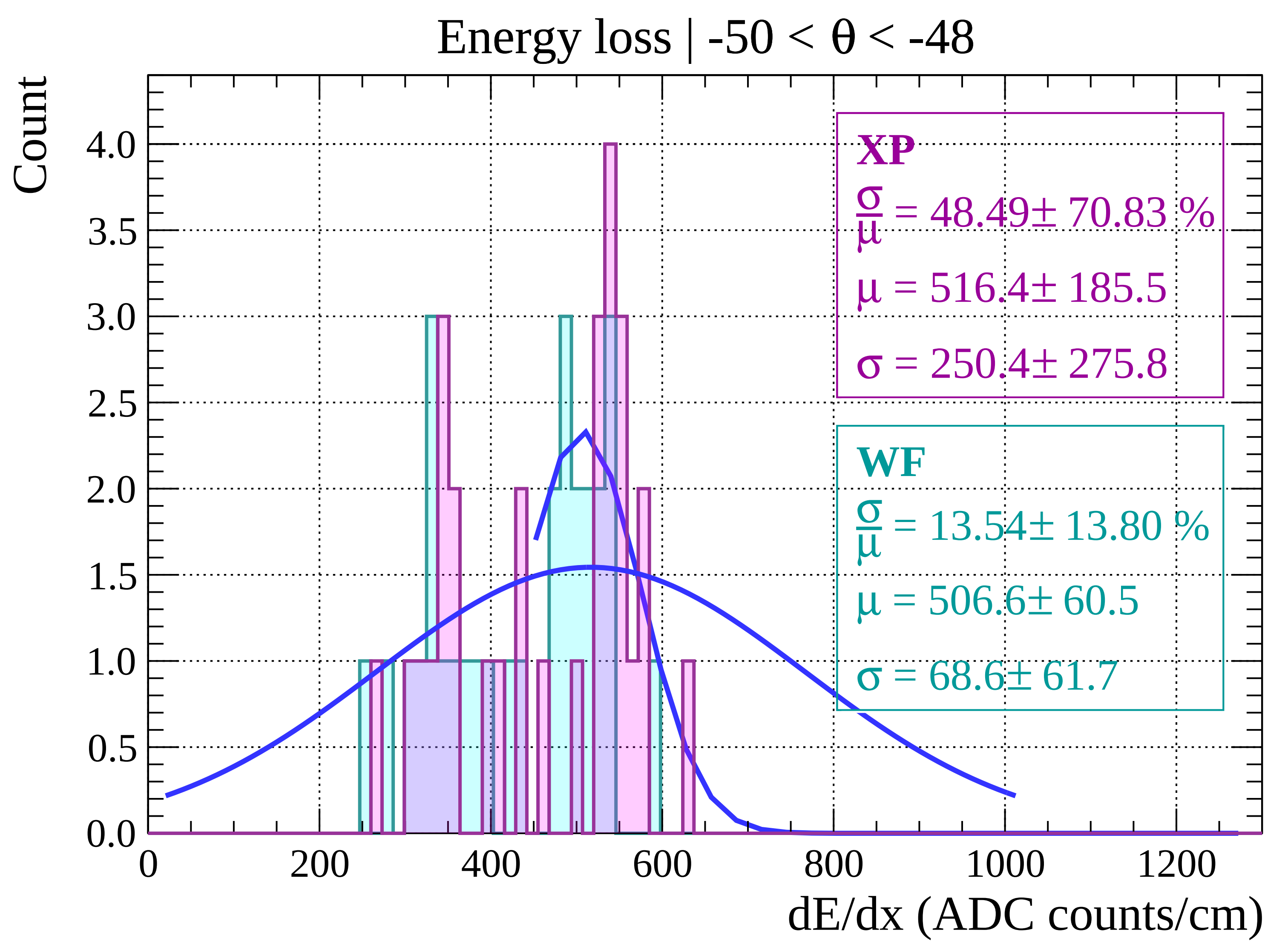

Energy loss | -56 < θ < -54
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
σ/μ = -477.57± 7671.45 %
μ = -282.2± 4798.2
σ = 1347.6± 1266.8
WF
σ/μ = 1926.14± 214999.66 %
μ = 55.9± 6175.5
σ = 1075.9± 1149.8



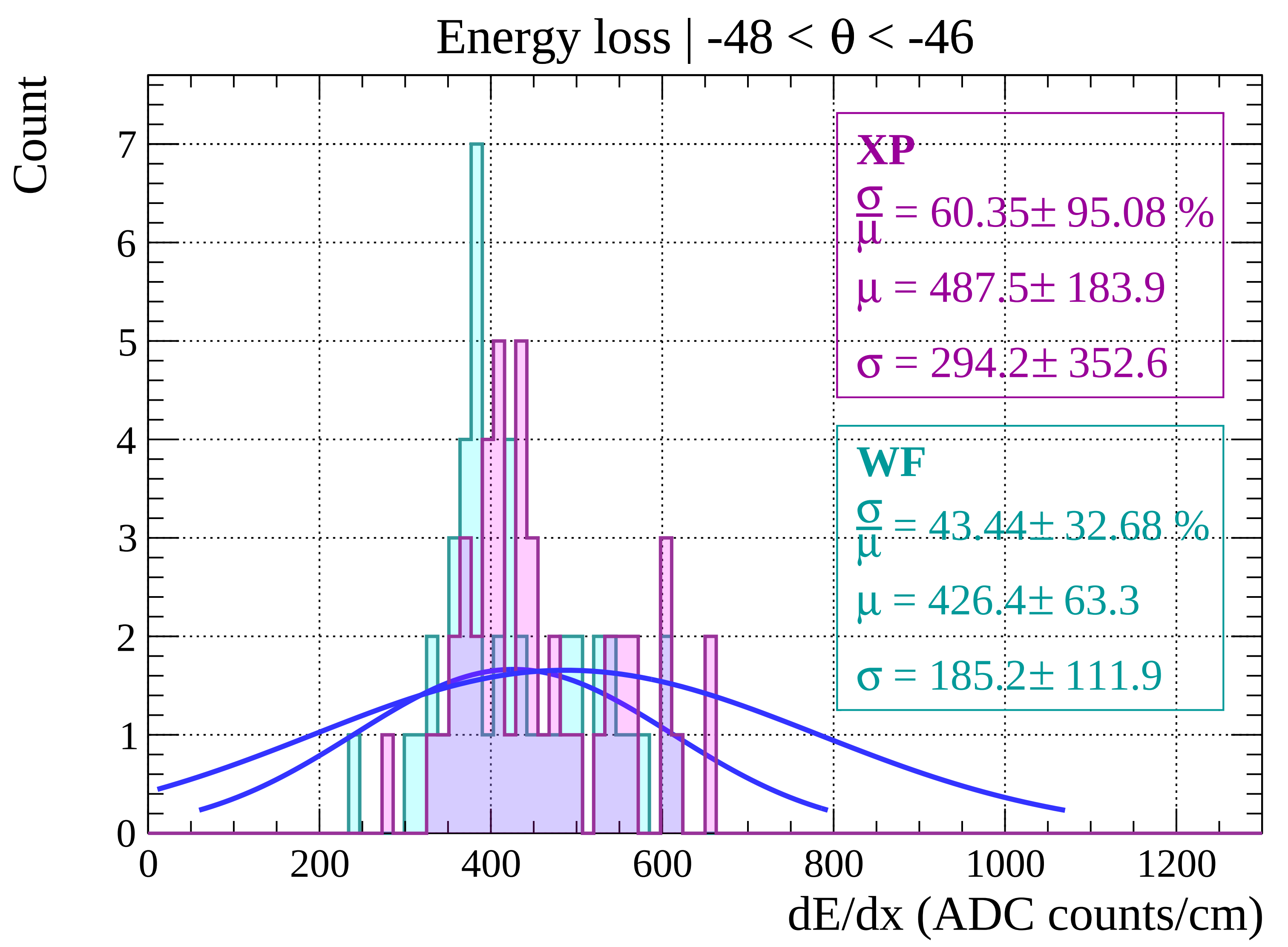
Energy loss | -54 < θ < -52
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -163.44± 707.42 %
μ = -753.4± 3926.6
σ = 1231.3± 1088.1
WF
σ/μ = 120.17± 407.81 %
μ = 346.8± 607.7
σ = 416.7± 683.9



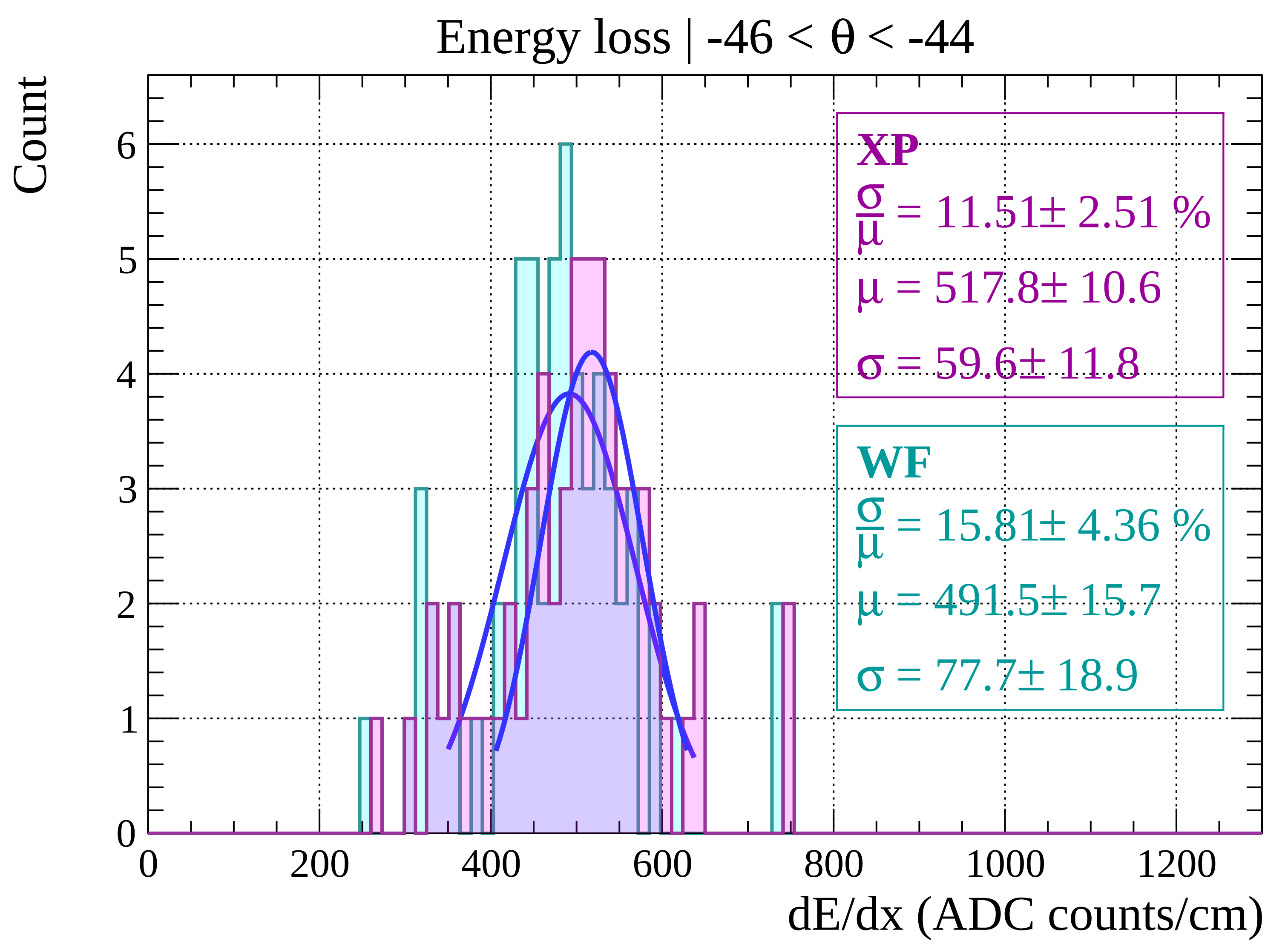
Energy loss | -52 < θ < -50
Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 122.76± 758.93 %
μ = 630.8± 3072.2
σ = 774.3± 1015.5
WF
σ/μ = 332.31± 3593.54 %
μ = 176.9± 1633.7
σ = 587.9± 928.2



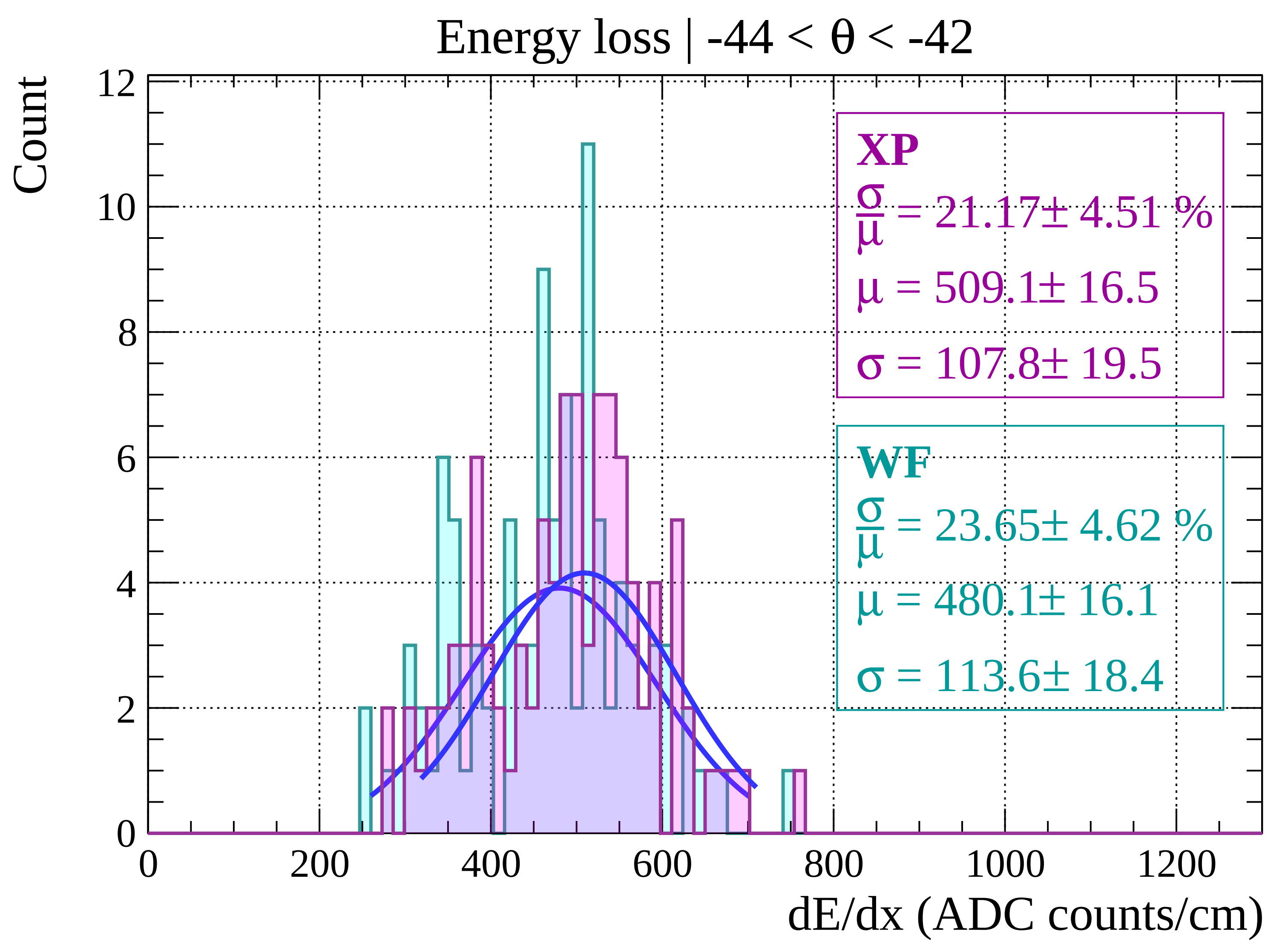
Energy loss | -50 < θ < -48
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 48.49± 70.83 %
μ = 516.4± 185.5
σ = 250.4± 275.8
WF
σ/μ = 13.54± 13.80 %
μ = 506.6± 60.5
σ = 68.6± 61.7



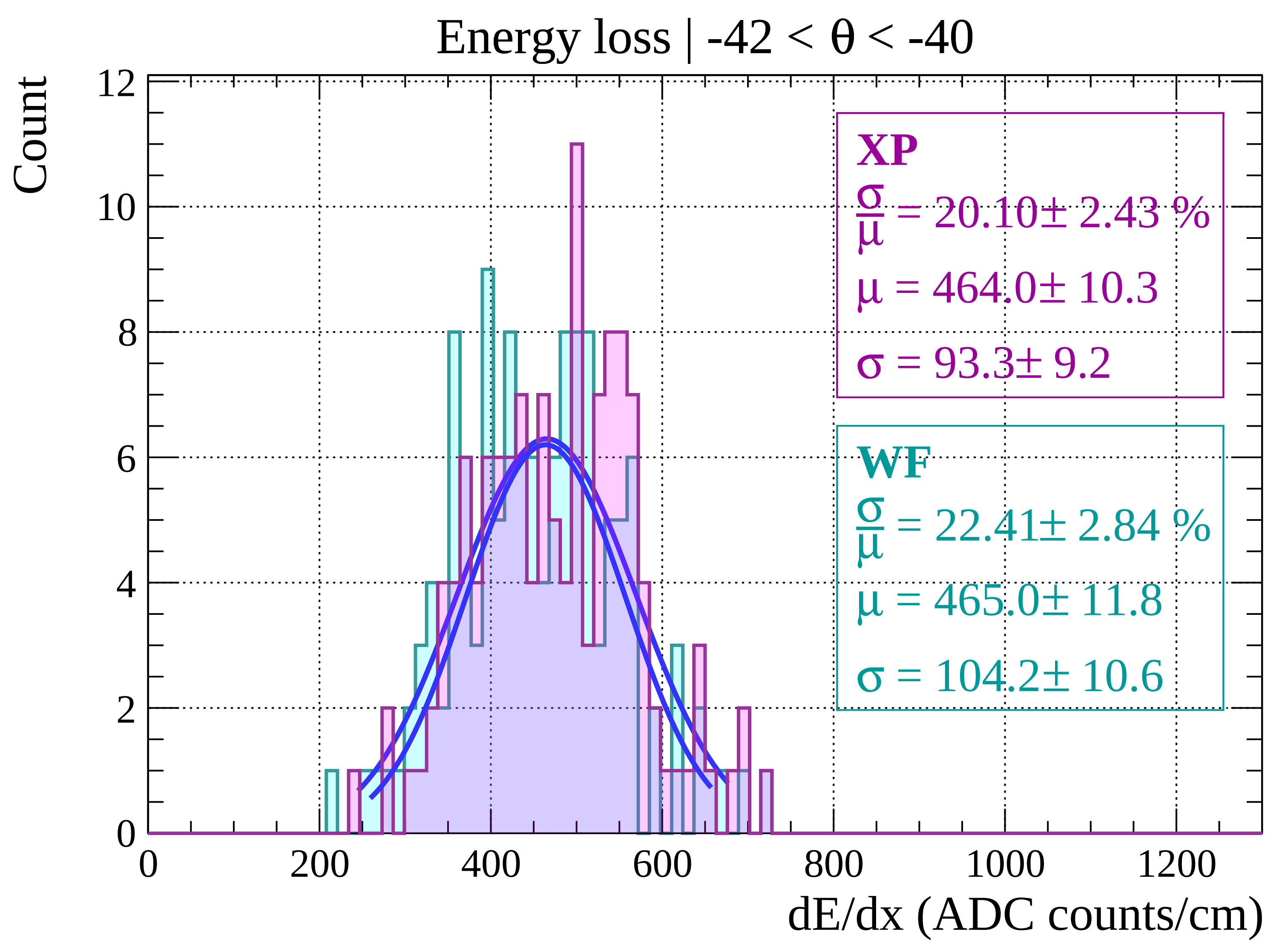
Energy loss | -48 < θ < -46
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7
XP
σ/μ = 60.35± 95.08 %
μ = 487.5± 183.9
σ = 294.2± 352.6
WF
σ/μ = 43.44± 32.68 %
μ = 426.4± 63.3
σ = 185.2± 111.9

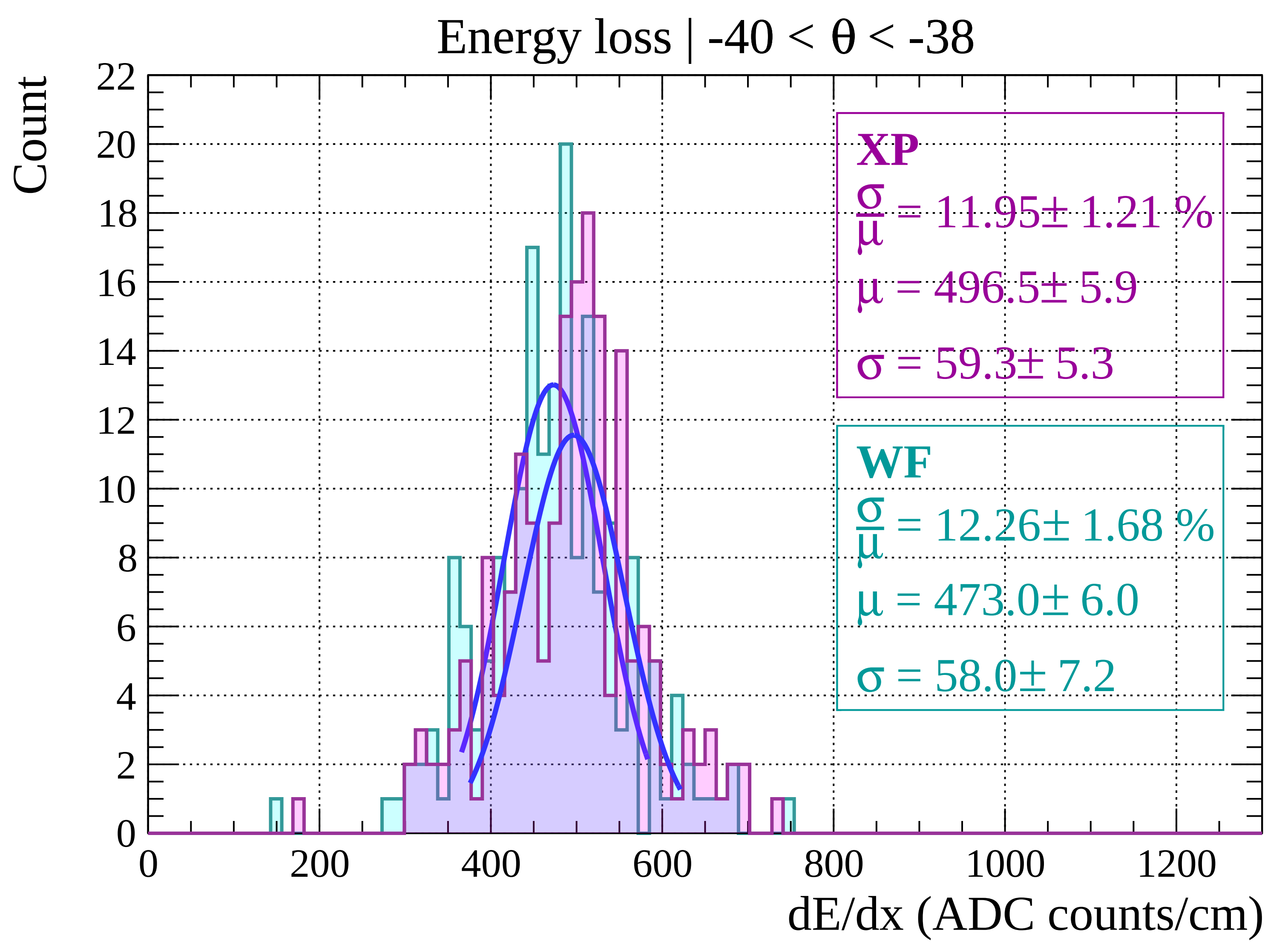


Energy loss | -46 < θ < -44
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.51± 2.51 %
μ = 517.8± 10.6
σ = 59.6± 11.8
WF
σ/μ = 15.81± 4.36 %
μ = 491.5± 15.7
σ = 77.7± 18.9

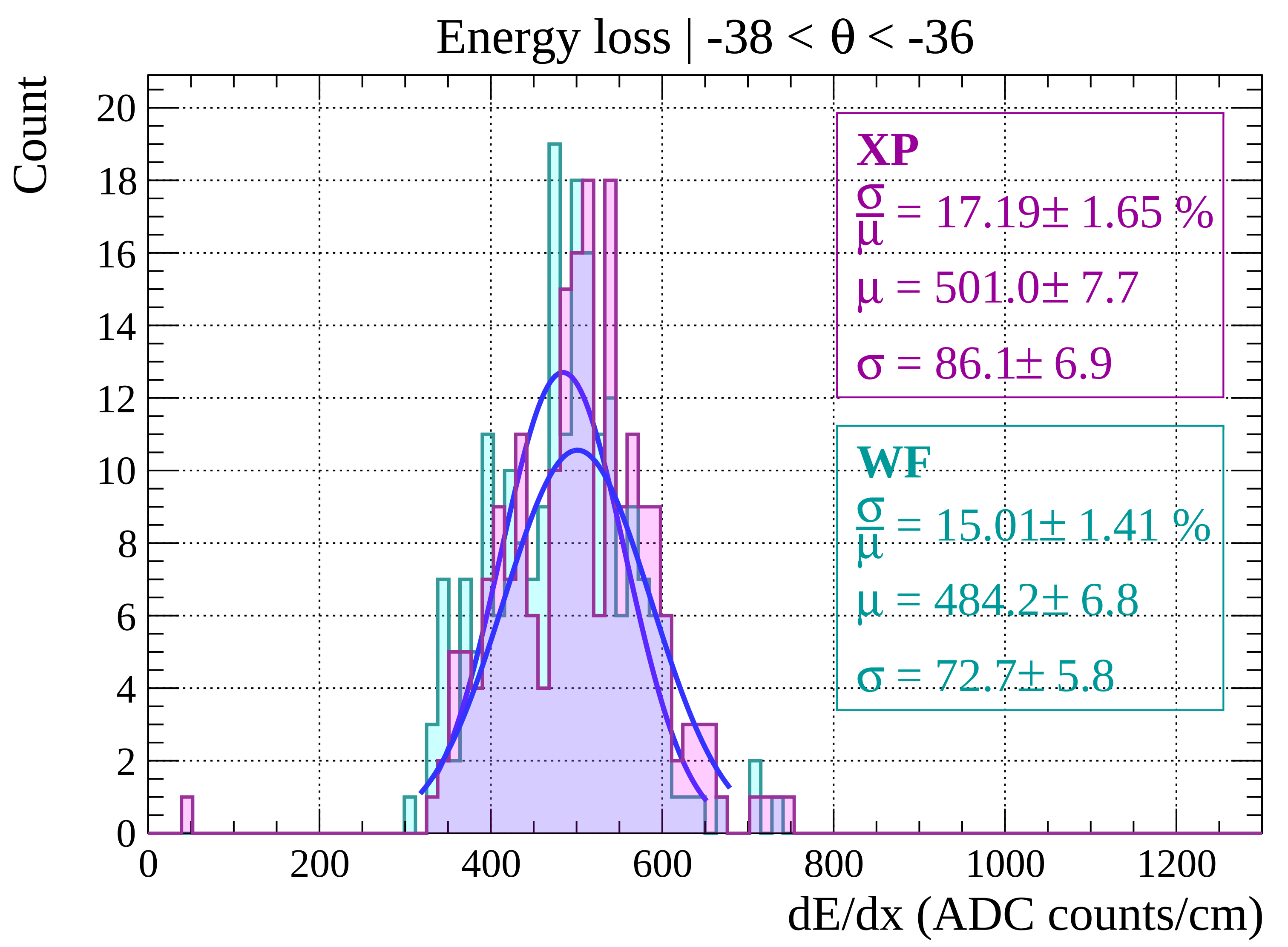


Energy loss | -44 < θ < -42
Count
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 21.17± 4.51 %
μ = 509.1± 16.5
σ = 107.8± 19.5
WF
σ/μ = 23.65± 4.62 %
μ = 480.1± 16.1
σ = 113.6± 18.4

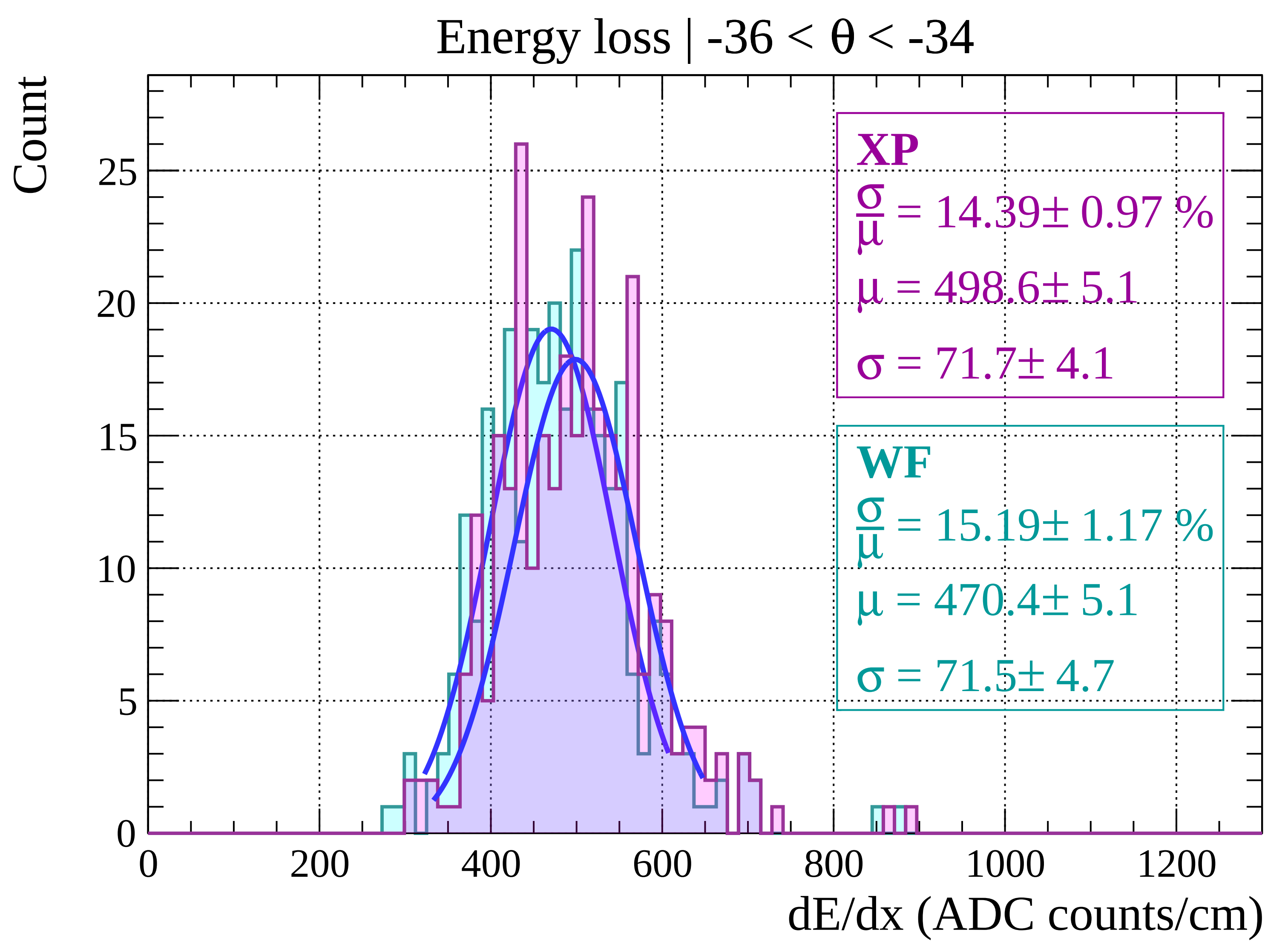

Energy loss | -42 < θ < -40
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 20.10± 2.43 %
μ = 464.0± 10.3
σ = 93.3± 9.2
WF
σ/μ = 22.41± 2.84 %
μ = 465.0± 11.8
σ = 104.2± 10.6
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200



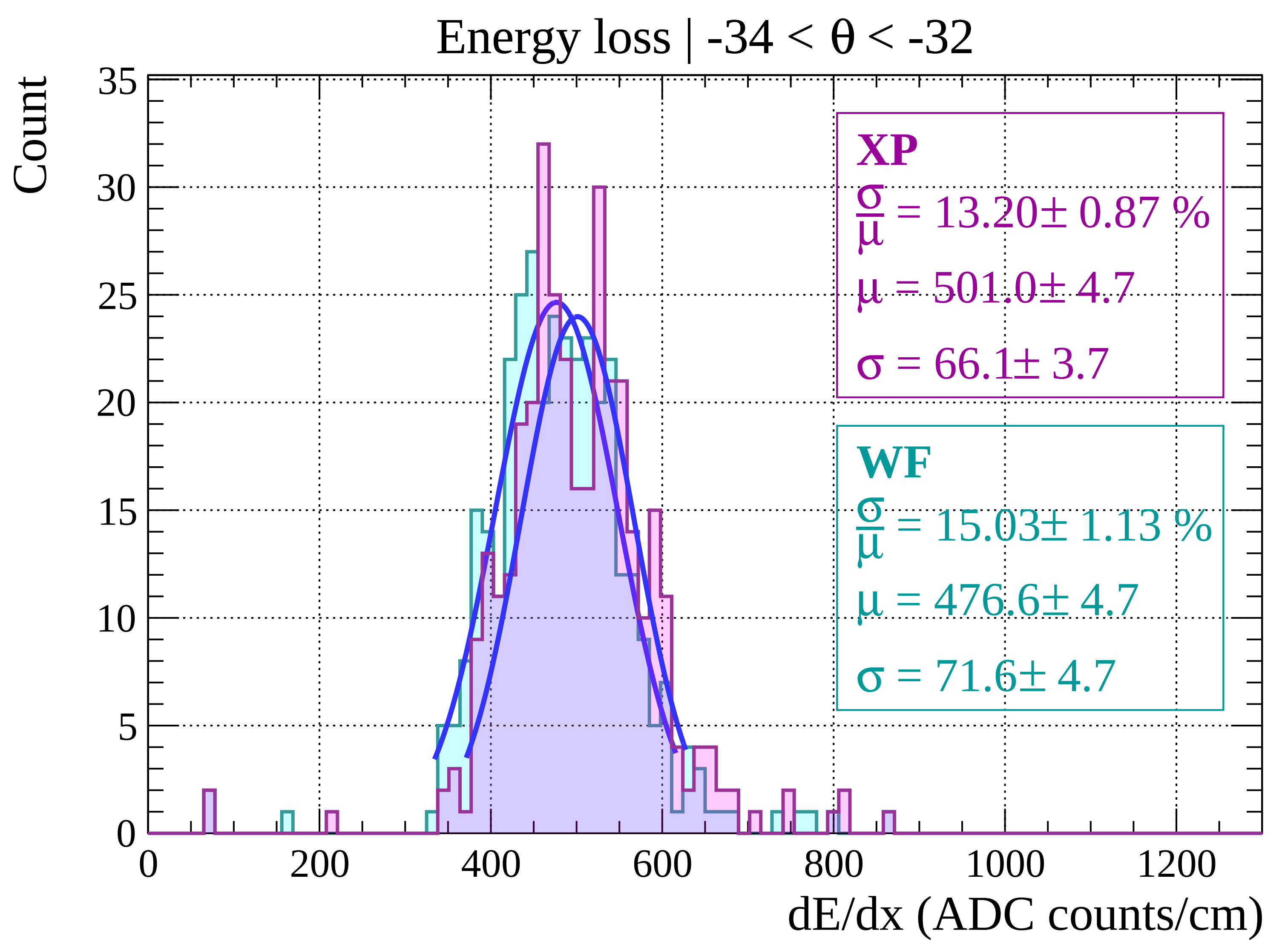
Energy loss | -40 < θ < -38
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.95± 1.21 %
μ = 496.5± 5.9
σ = 59.3± 5.3
WF
σ/μ = 12.26± 1.68 %
μ = 473.0± 6.0
σ = 58.0± 7.2
0
2
4
6
8
10
12
14
16
18
20
22
0
200
400
600
800
1000
1200

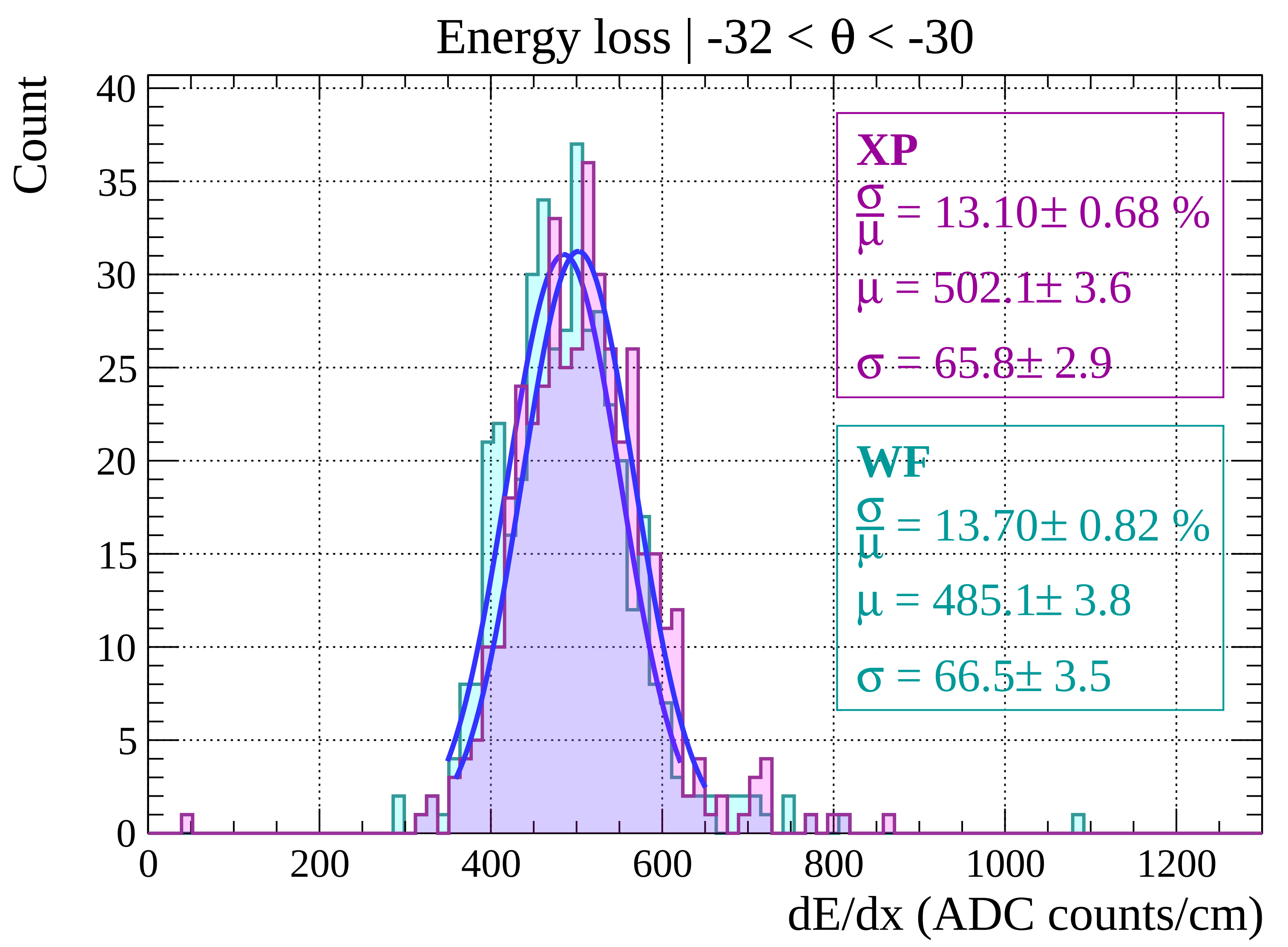


Energy loss | -38 < θ < -36
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
20
0
200
400
600
800
1000
1200
XP
σ/μ = 17.19± 1.65 %
μ = 501.0± 7.7
σ = 86.1± 6.9
WF
σ/μ = 15.01± 1.41 %
μ = 484.2± 6.8
σ = 72.7± 5.8

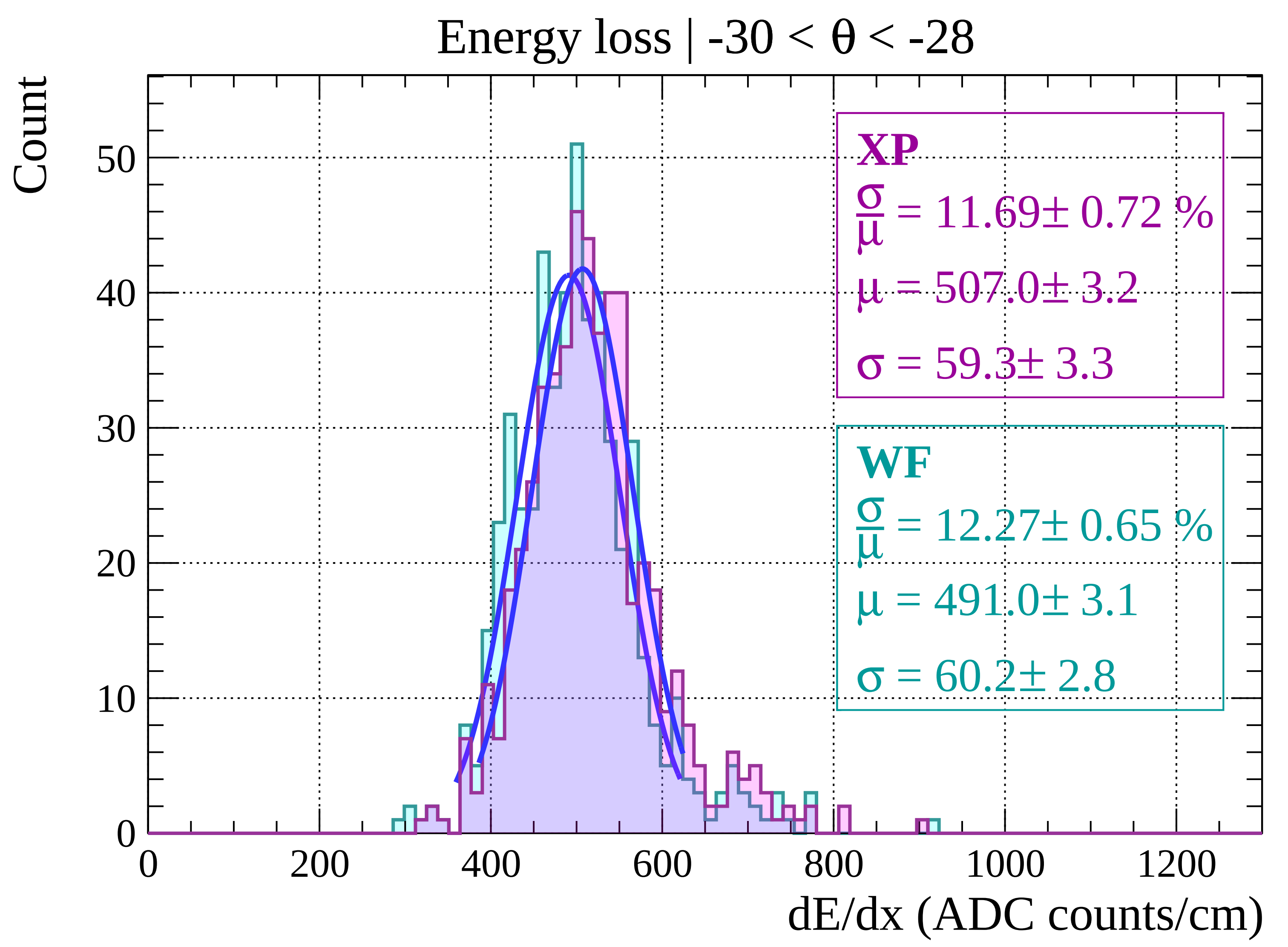


Energy loss | -36 < θ < -34
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 14.39± 0.97 %
μ = 498.6± 5.1
σ = 71.7± 4.1
WF
σ/μ = 15.19± 1.17 %
μ = 470.4± 5.1
σ = 71.5± 4.7

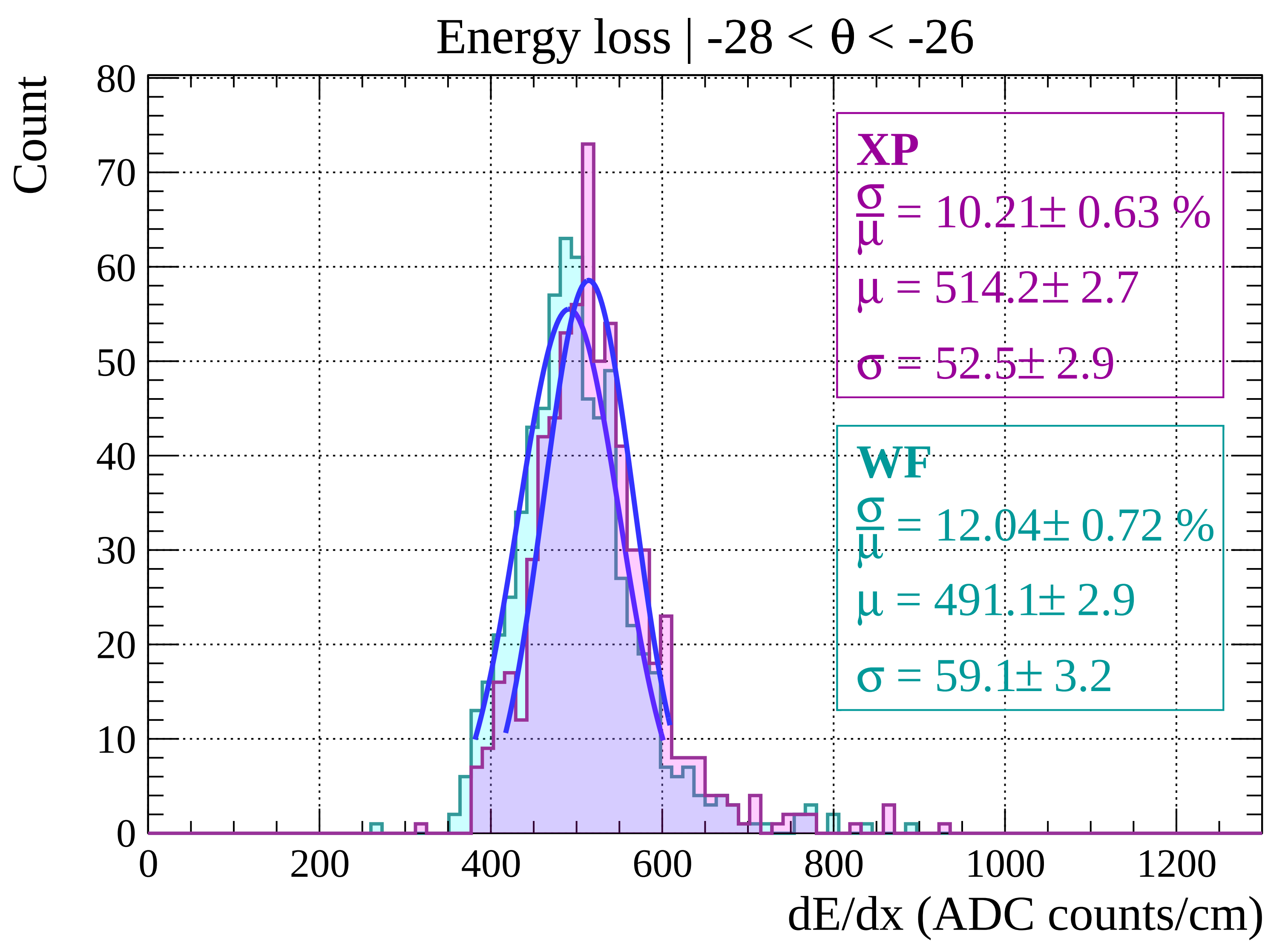

Energy loss | -34 < θ < -32
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
XP
σ/μ = 13.20± 0.87 %
μ = 501.0± 4.7
σ = 66.1± 3.7
WF
σ/μ = 15.03± 1.13 %
μ = 476.6± 4.7
σ = 71.6± 4.7



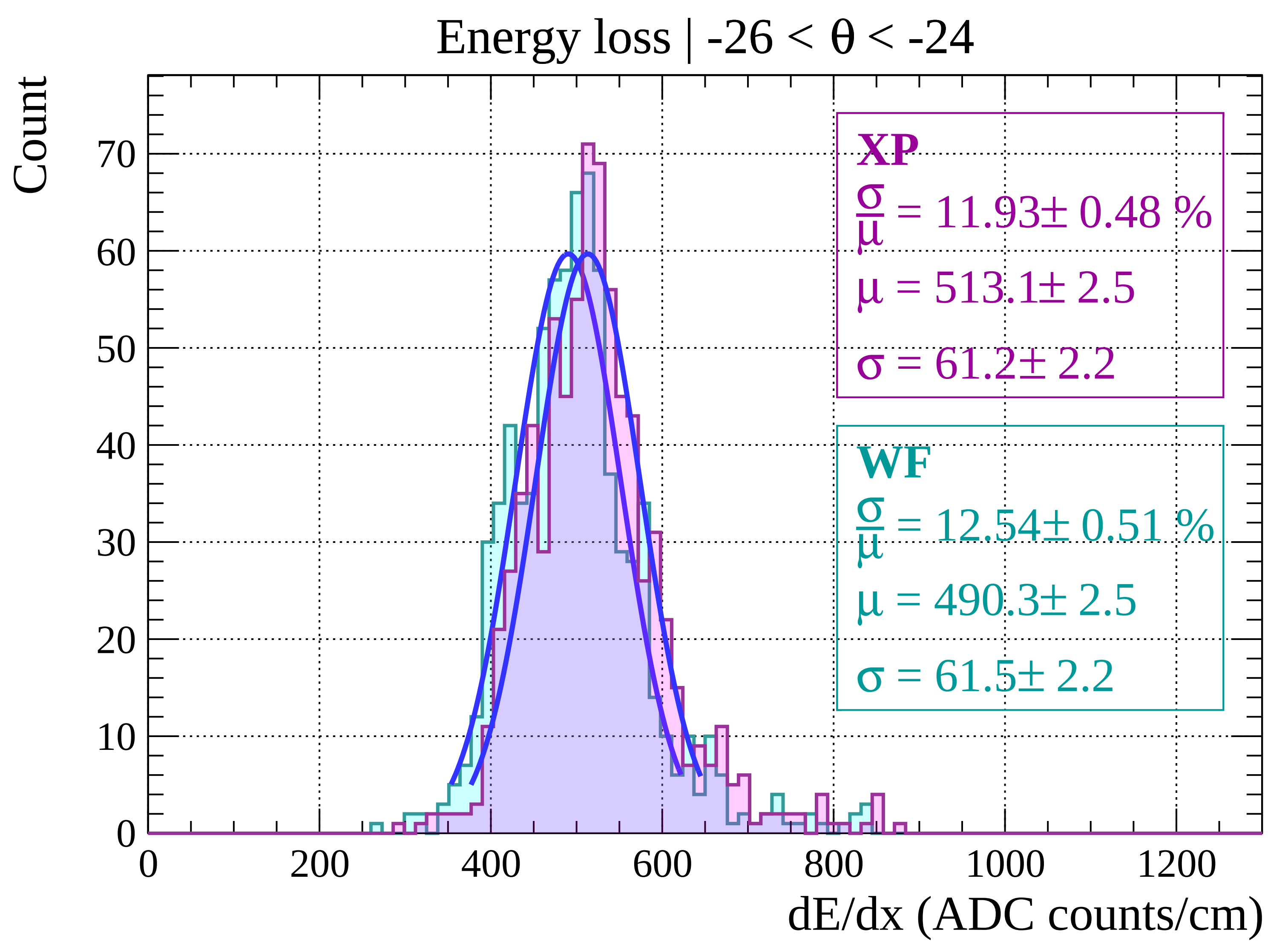
Energy loss | -32 < θ < -30
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
40
XP
σ/μ = 13.10± 0.68 %
μ = 502.1± 3.6
σ = 65.8± 2.9
WF
σ/μ = 13.70± 0.82 %
μ = 485.1± 3.8
σ = 66.5± 3.5



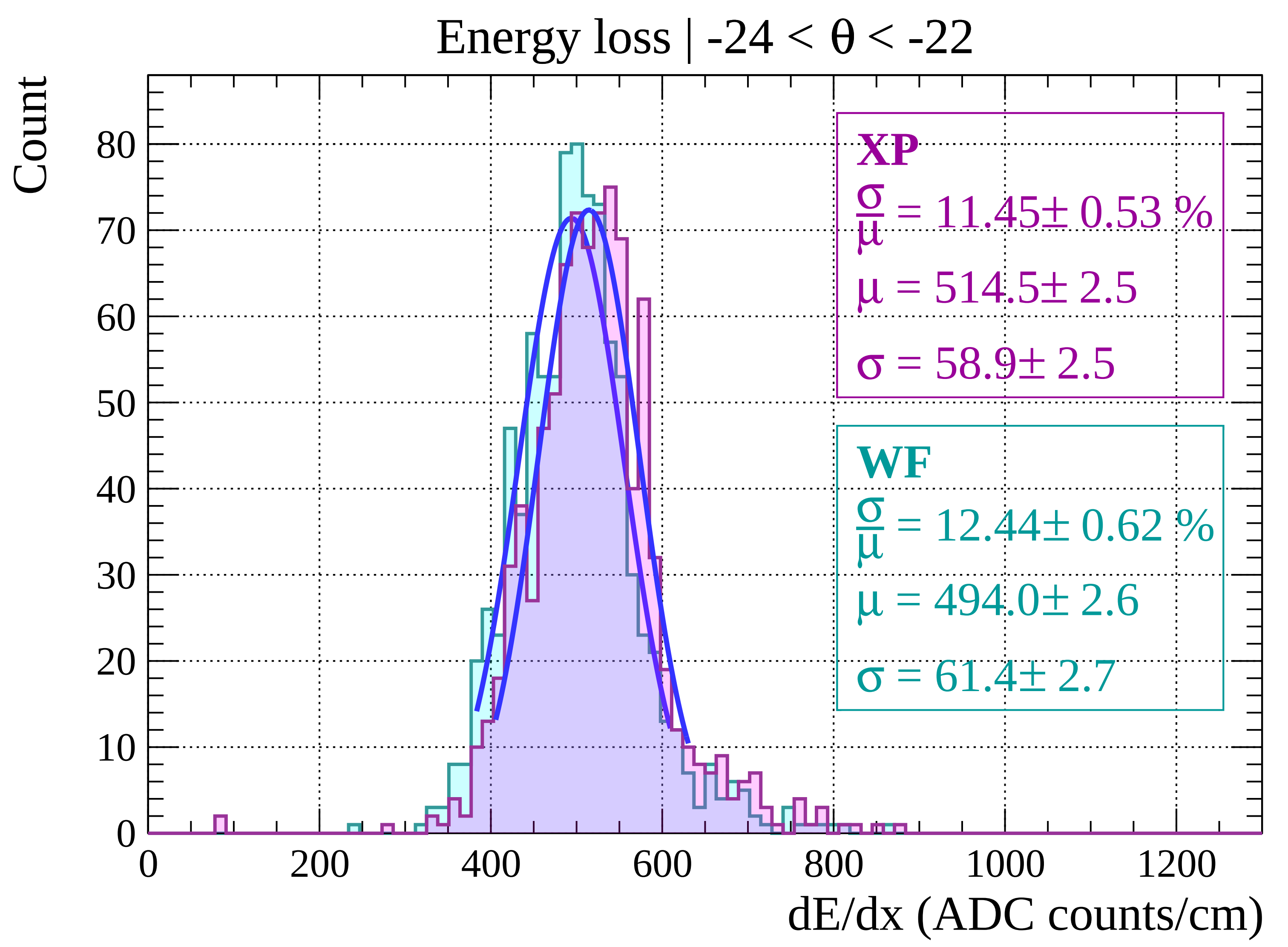
Energy loss | -30 < θ < -28
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 11.69± 0.72 %
μ = 507.0± 3.2
σ = 59.3± 3.3
WF
$\frac{\sigma}{\mu}$ = 12.27± 0.65 %
μ = 491.0± 3.1
σ = 60.2± 2.8



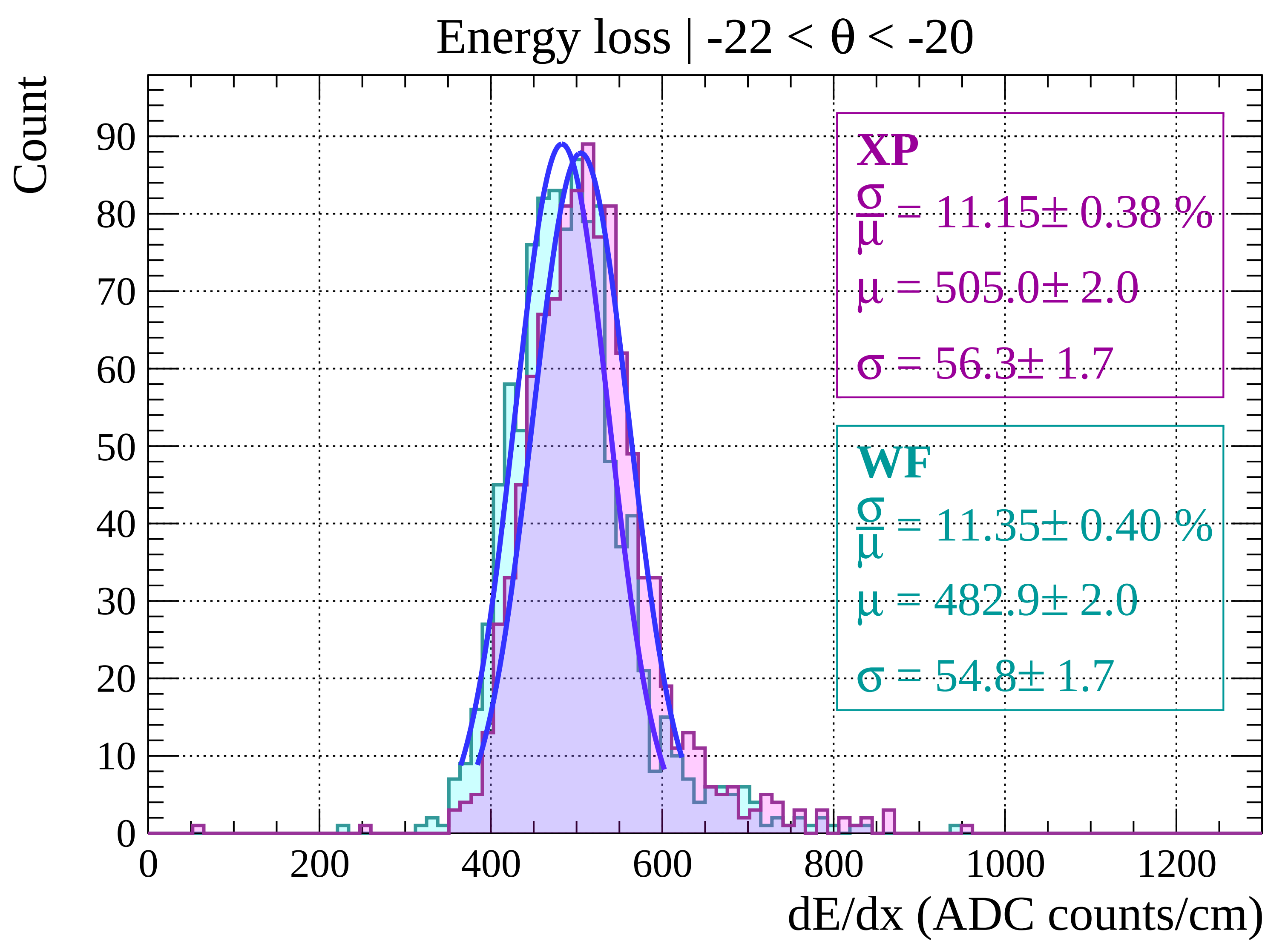
Energy loss | -28 < θ < -26
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
XP
σ/μ = 10.21± 0.63 %
μ = 514.2± 2.7
σ = 52.5± 2.9
WF
σ/μ = 12.04± 0.72 %
μ = 491.1± 2.9
σ = 59.1± 3.2

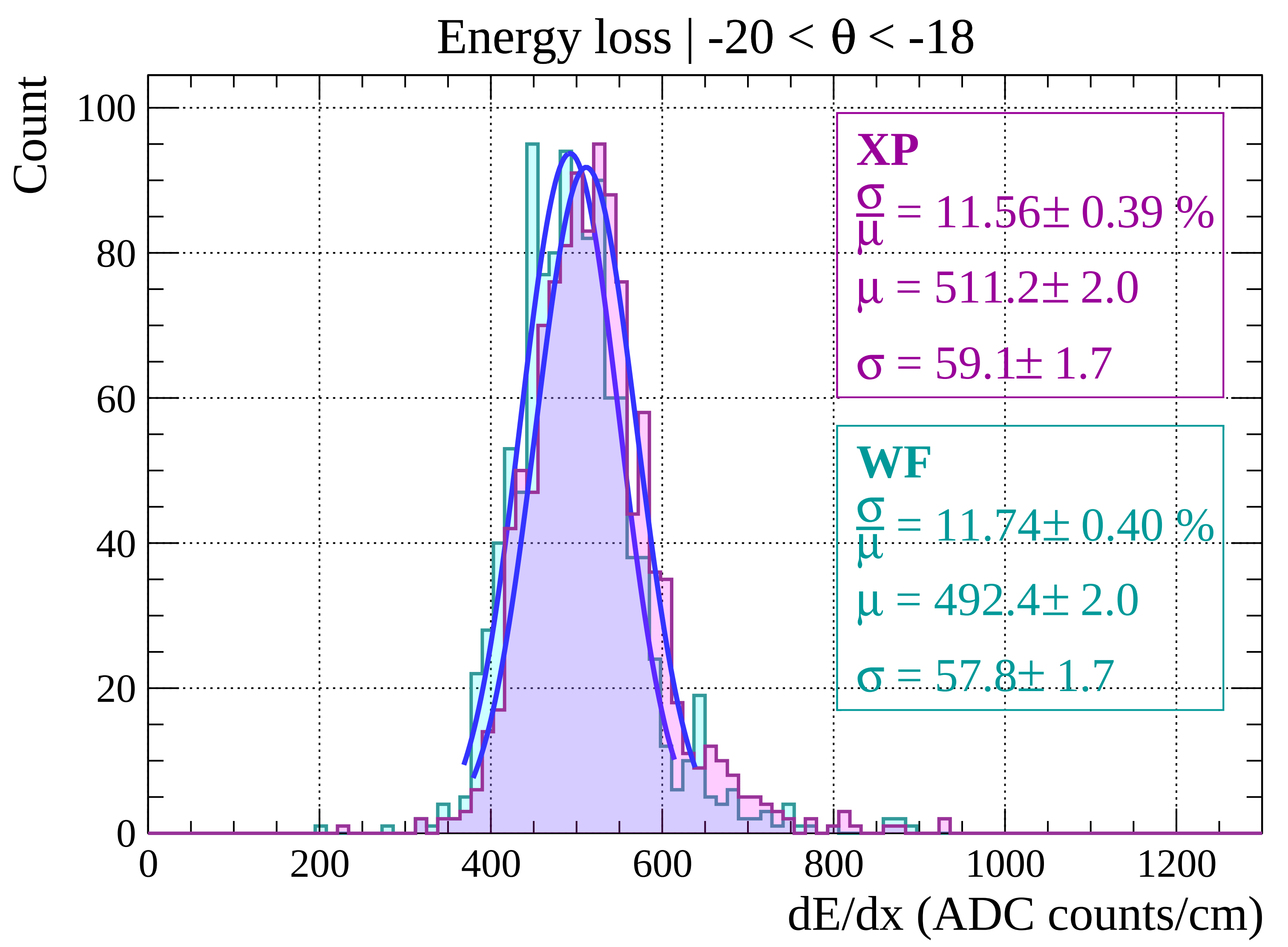


Energy loss | -26 < θ < -24
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP
σ/μ = 11.93± 0.48 %
μ = 513.1± 2.5
σ = 61.2± 2.2
WF
σ/μ = 12.54± 0.51 %
μ = 490.3± 2.5
σ = 61.5± 2.2

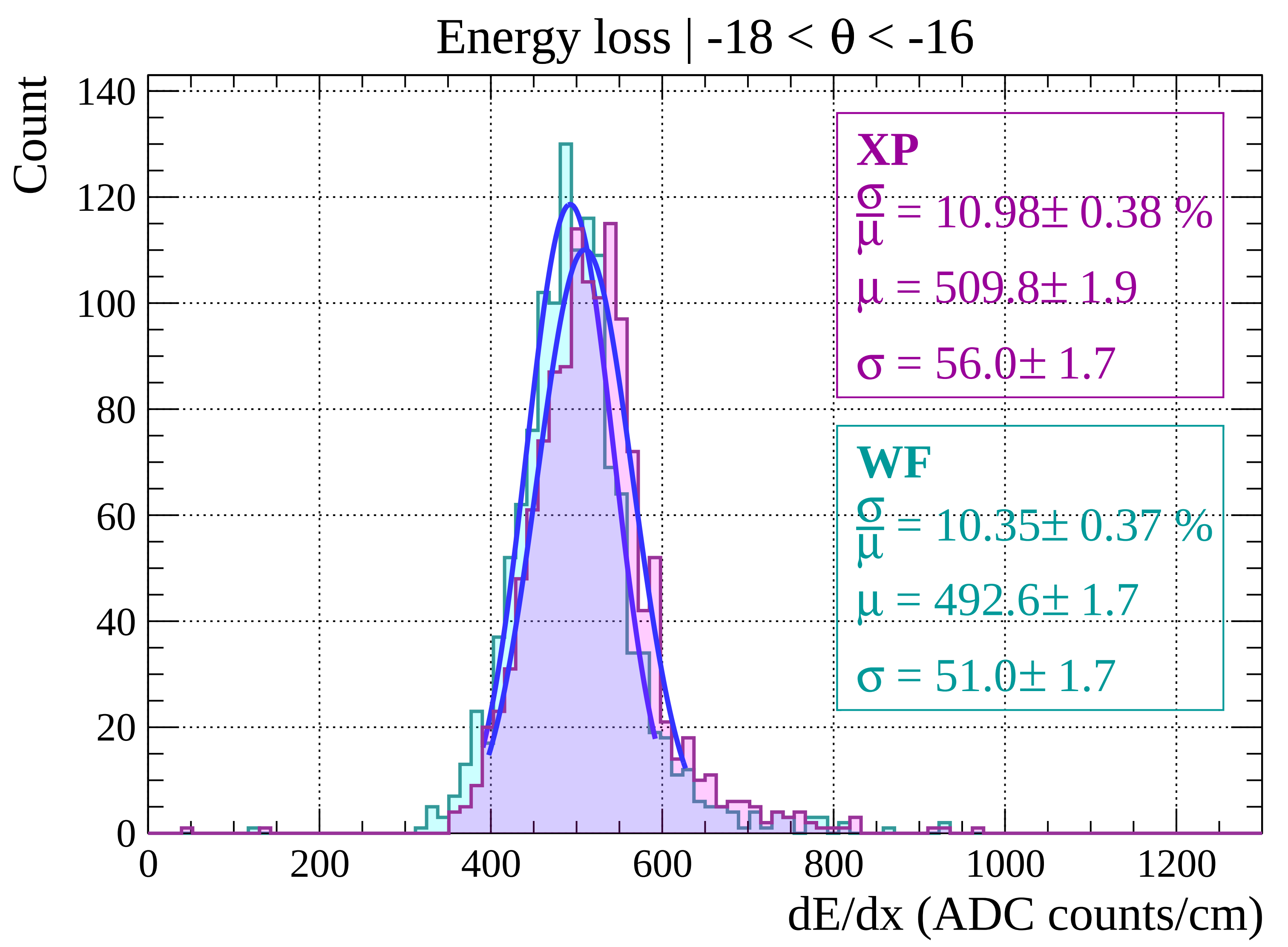


Energy loss | -24 < θ < -22
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
XP
σ/μ = 11.45± 0.53 %
μ = 514.5± 2.5
σ = 58.9± 2.5
WF
σ/μ = 12.44± 0.62 %
μ = 494.0± 2.6
σ = 61.4± 2.7

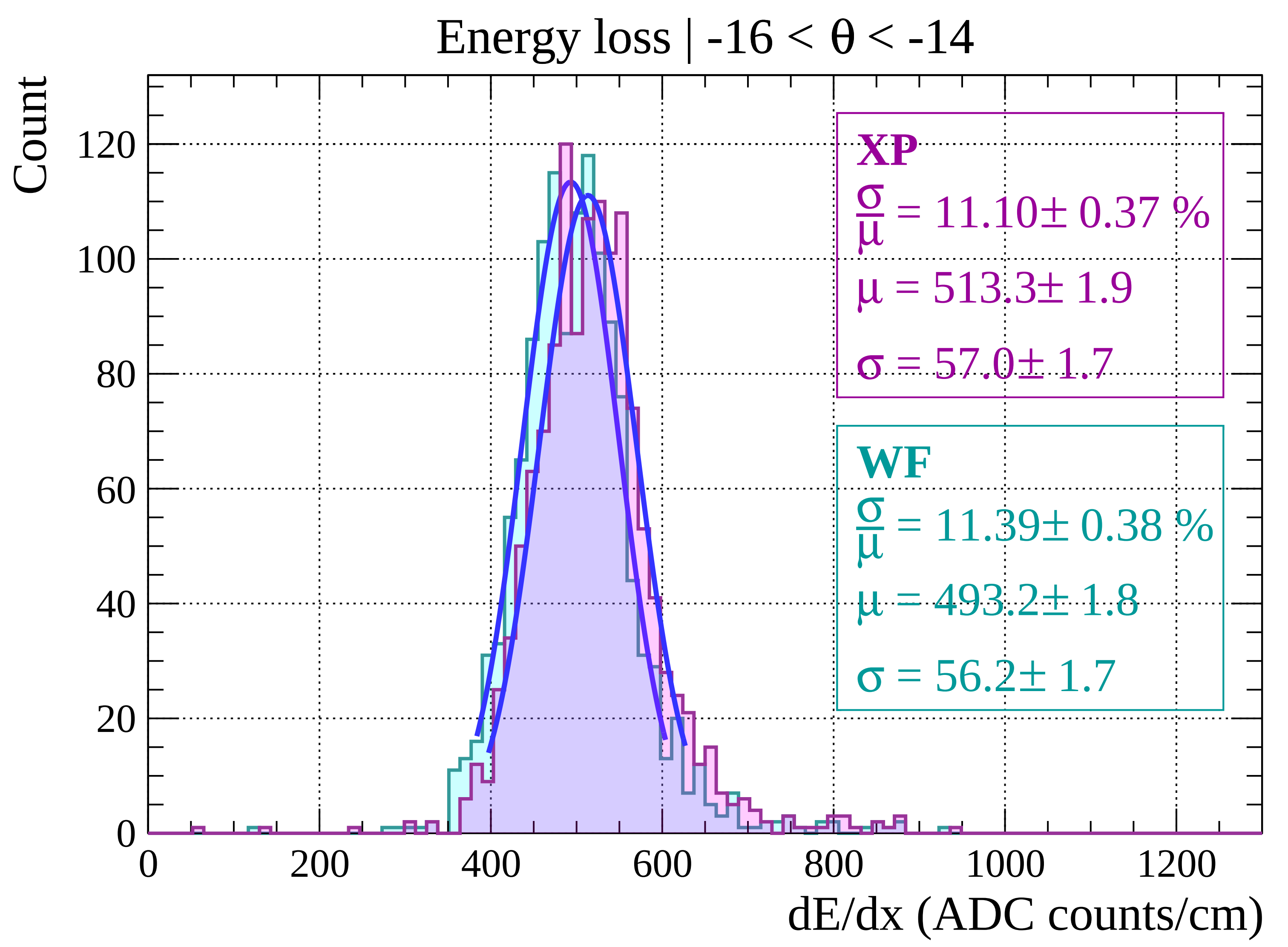

Energy loss | -22 < θ < -20
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.15± 0.38 %
μ = 505.0± 2.0
σ = 56.3± 1.7
WF
σ/μ = 11.35± 0.40 %
μ = 482.9± 2.0
σ = 54.8± 1.7



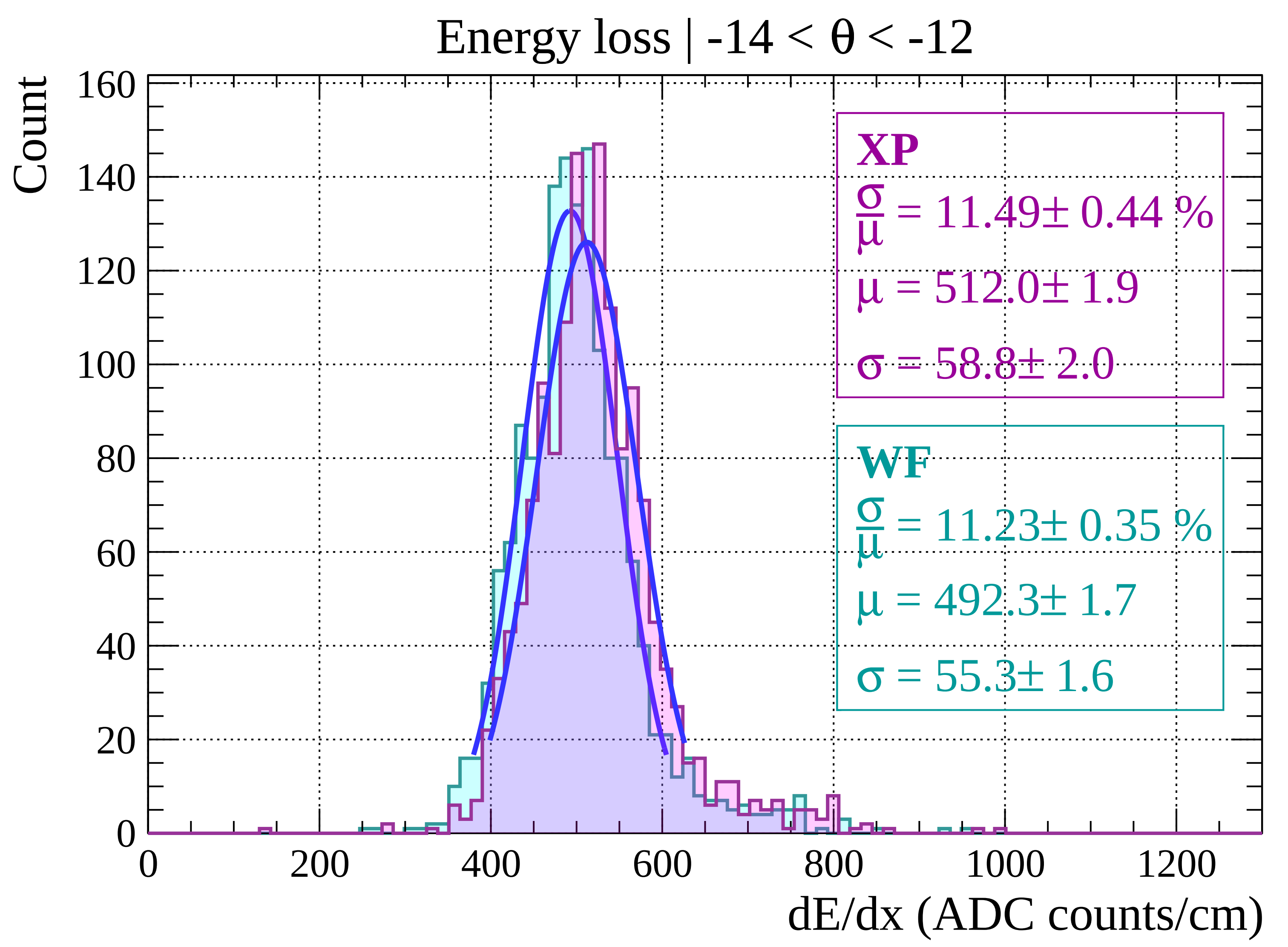
Energy loss | -20 < θ < -18
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.56± 0.39 %
μ = 511.2± 2.0
σ = 59.1± 1.7
WF
σ/μ = 11.74± 0.40 %
μ = 492.4± 2.0
σ = 57.8± 1.7



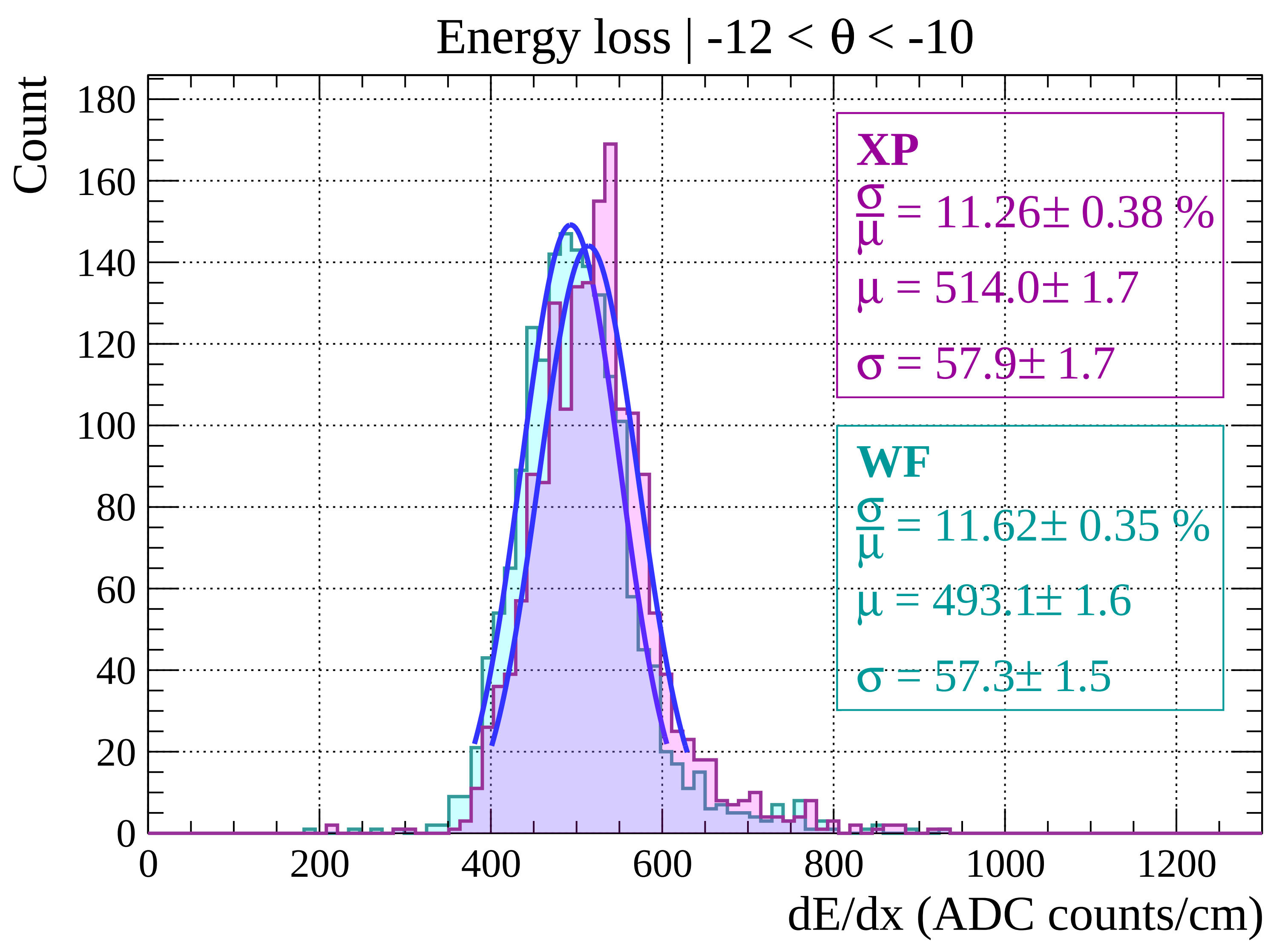
Energy loss | -18 < θ < -16
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.98± 0.38 %
μ = 509.8± 1.9
σ = 56.0± 1.7
WF
σ/μ = 10.35± 0.37 %
μ = 492.6± 1.7
σ = 51.0± 1.7
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200



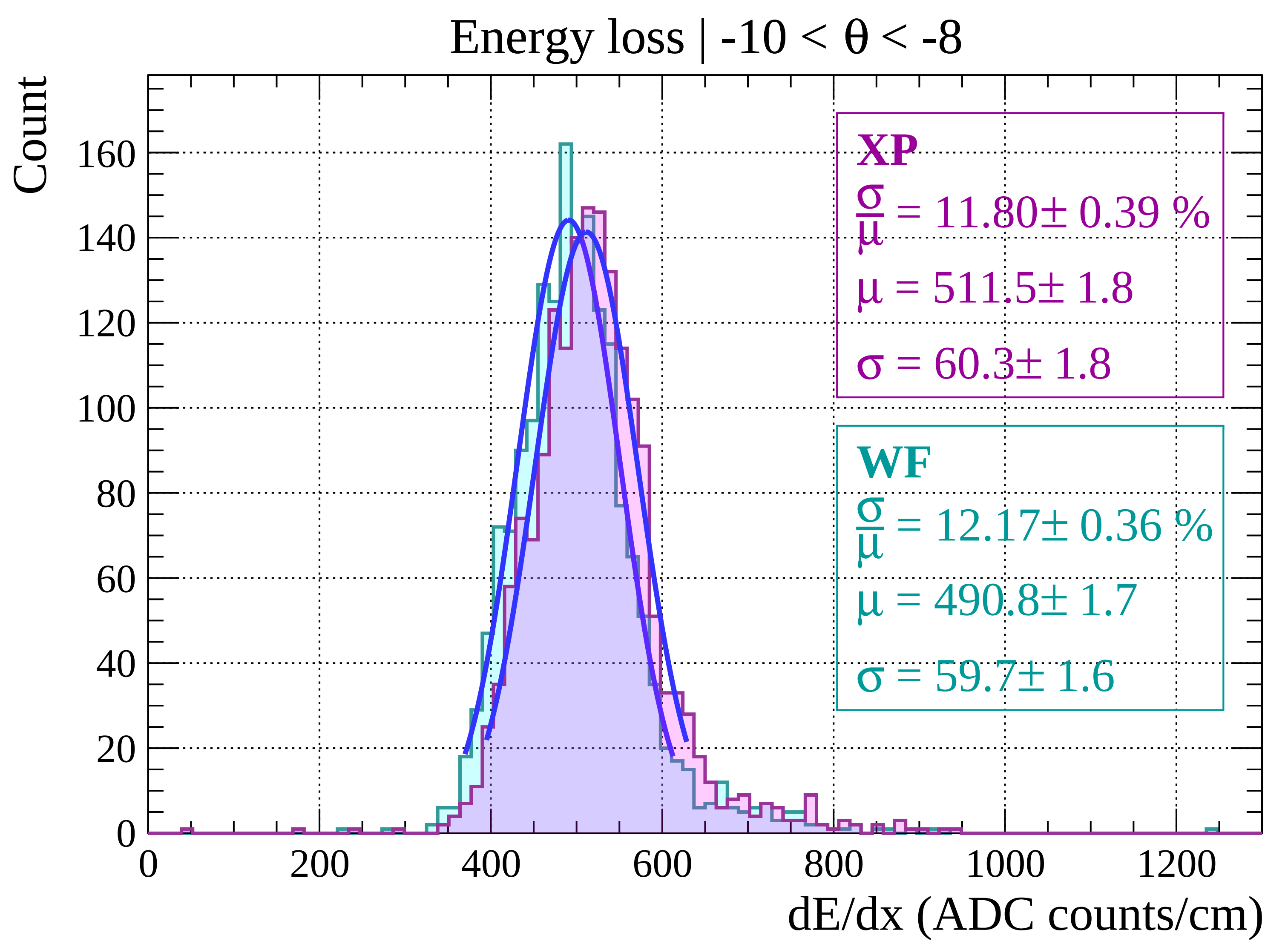
Energy loss | -16 < θ < -14
Count
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.10± 0.37 %
μ = 513.3± 1.9
σ = 57.0± 1.7
WF
σ/μ = 11.39± 0.38 %
μ = 493.2± 1.8
σ = 56.2± 1.7

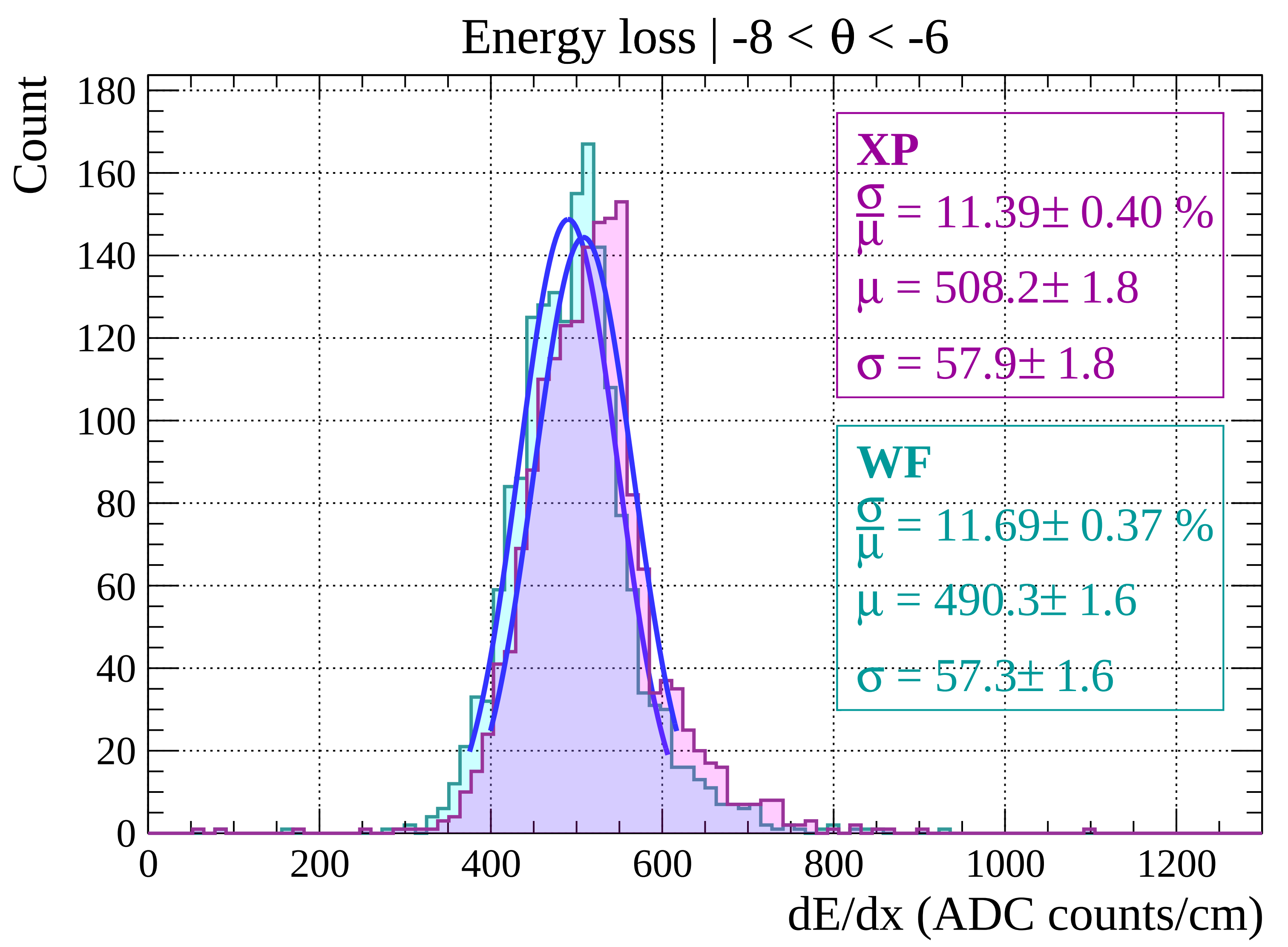


Energy loss | -14 < θ < -12
Count
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.49± 0.44 %
μ = 512.0± 1.9
σ = 58.8± 2.0
WF
σ/μ = 11.23± 0.35 %
μ = 492.3± 1.7
σ = 55.3± 1.6

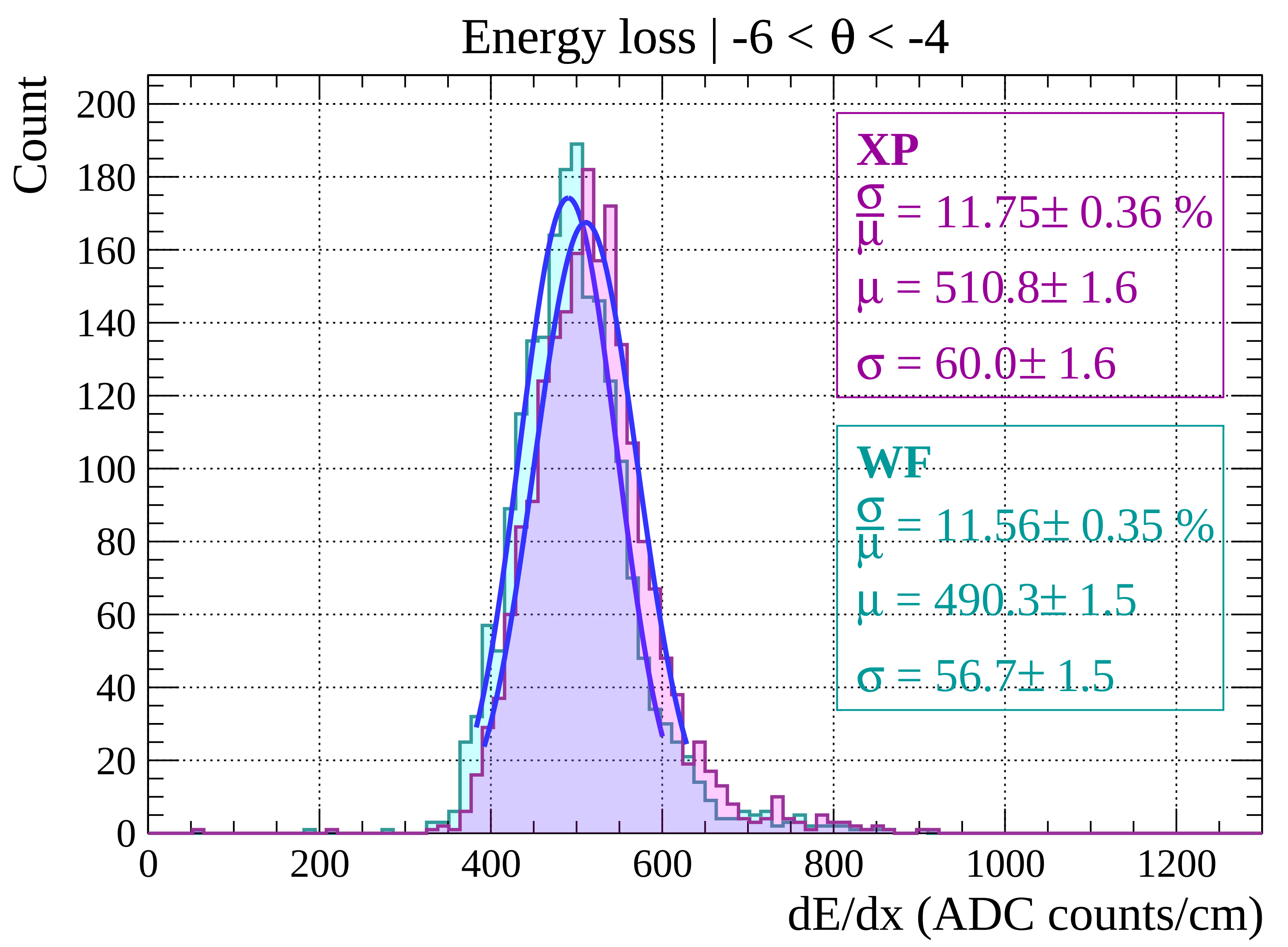


Energy loss | -12 < θ < -10
Count
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.26± 0.38 %
μ = 514.0± 1.7
σ = 57.9± 1.7
WF
σ/μ = 11.62± 0.35 %
μ = 493.1± 1.6
σ = 57.3± 1.5

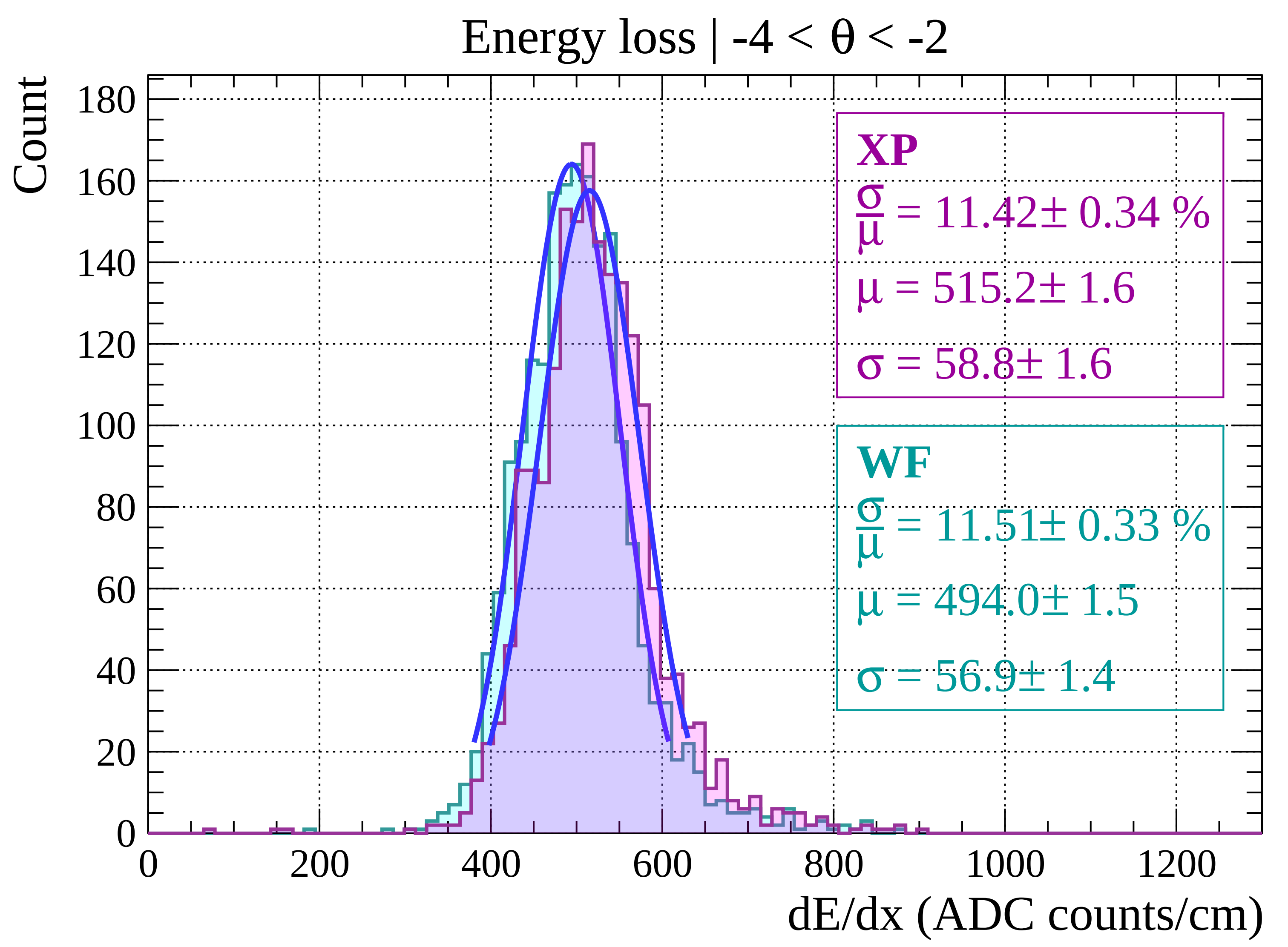

Energy loss | -10 < θ < -8
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.80± 0.39 %
μ = 511.5± 1.8
σ = 60.3± 1.8
WF
σ/μ = 12.17± 0.36 %
μ = 490.8± 1.7
σ = 59.7± 1.6
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200



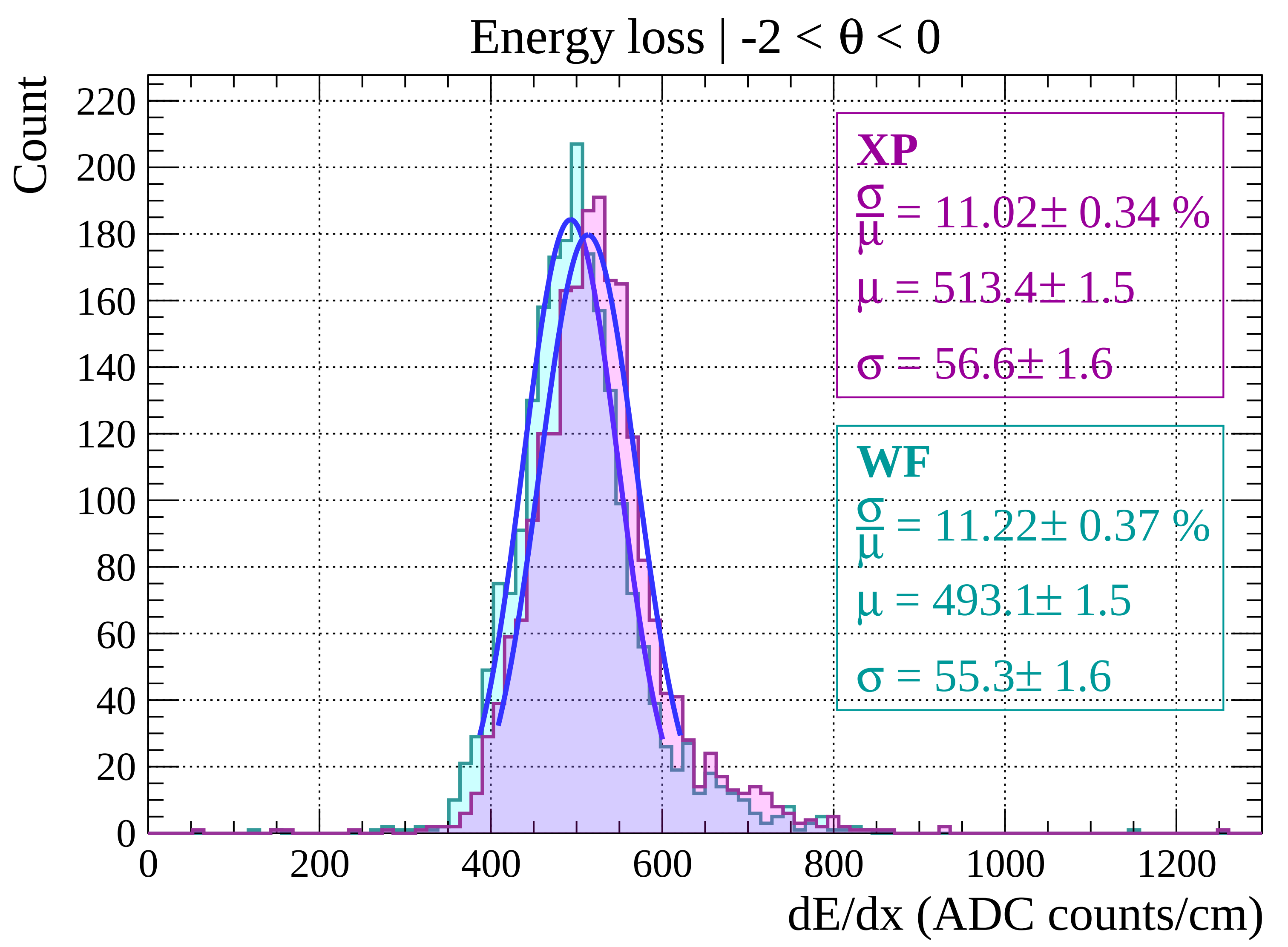
Energy loss | -8 < θ < -6
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.39± 0.40 %
μ = 508.2± 1.8
σ = 57.9± 1.8
WF
σ/μ = 11.69± 0.37 %
μ = 490.3± 1.6
σ = 57.3± 1.6
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200



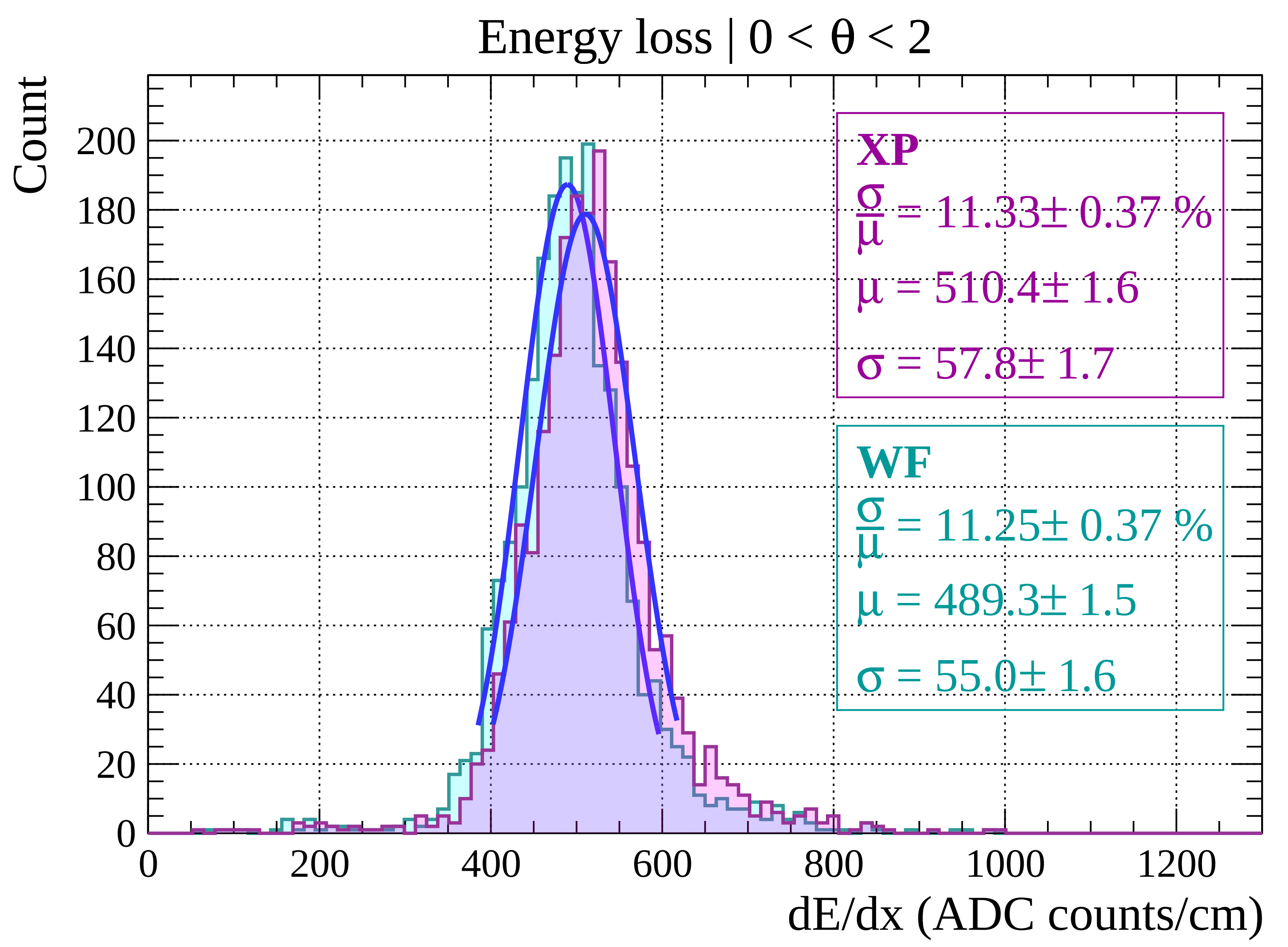
Energy loss | -6 < θ < -4
Count
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.75± 0.36 %
μ = 510.8± 1.6
σ = 60.0± 1.6
WF
σ/μ = 11.56± 0.35 %
μ = 490.3± 1.5
σ = 56.7± 1.5



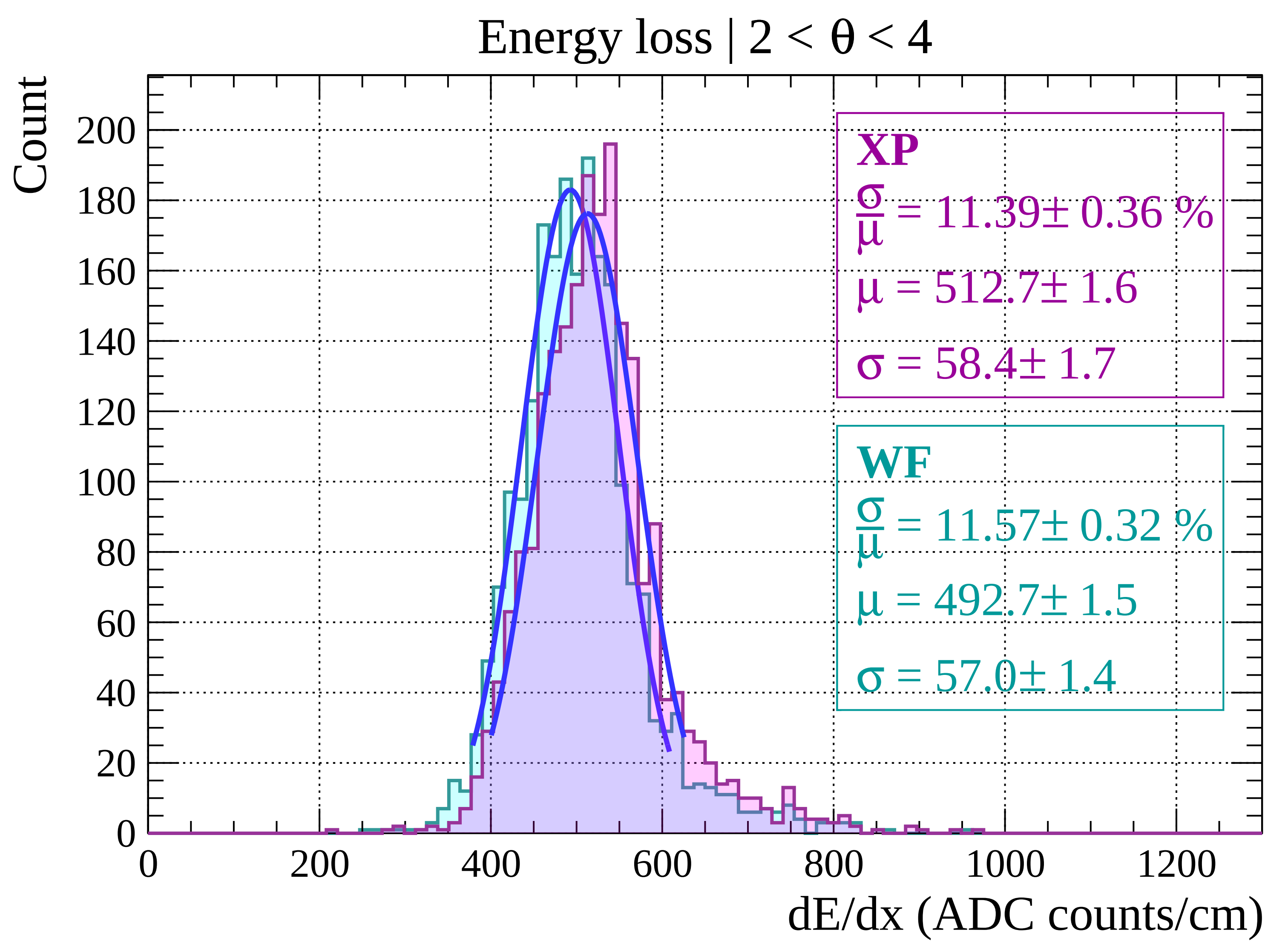
Energy loss | -4 < θ < -2
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP
σ/μ = 11.42± 0.34 %
μ = 515.2± 1.6
σ = 58.8± 1.6
WF
σ/μ = 11.51± 0.33 %
μ = 494.0± 1.5
σ = 56.9± 1.4



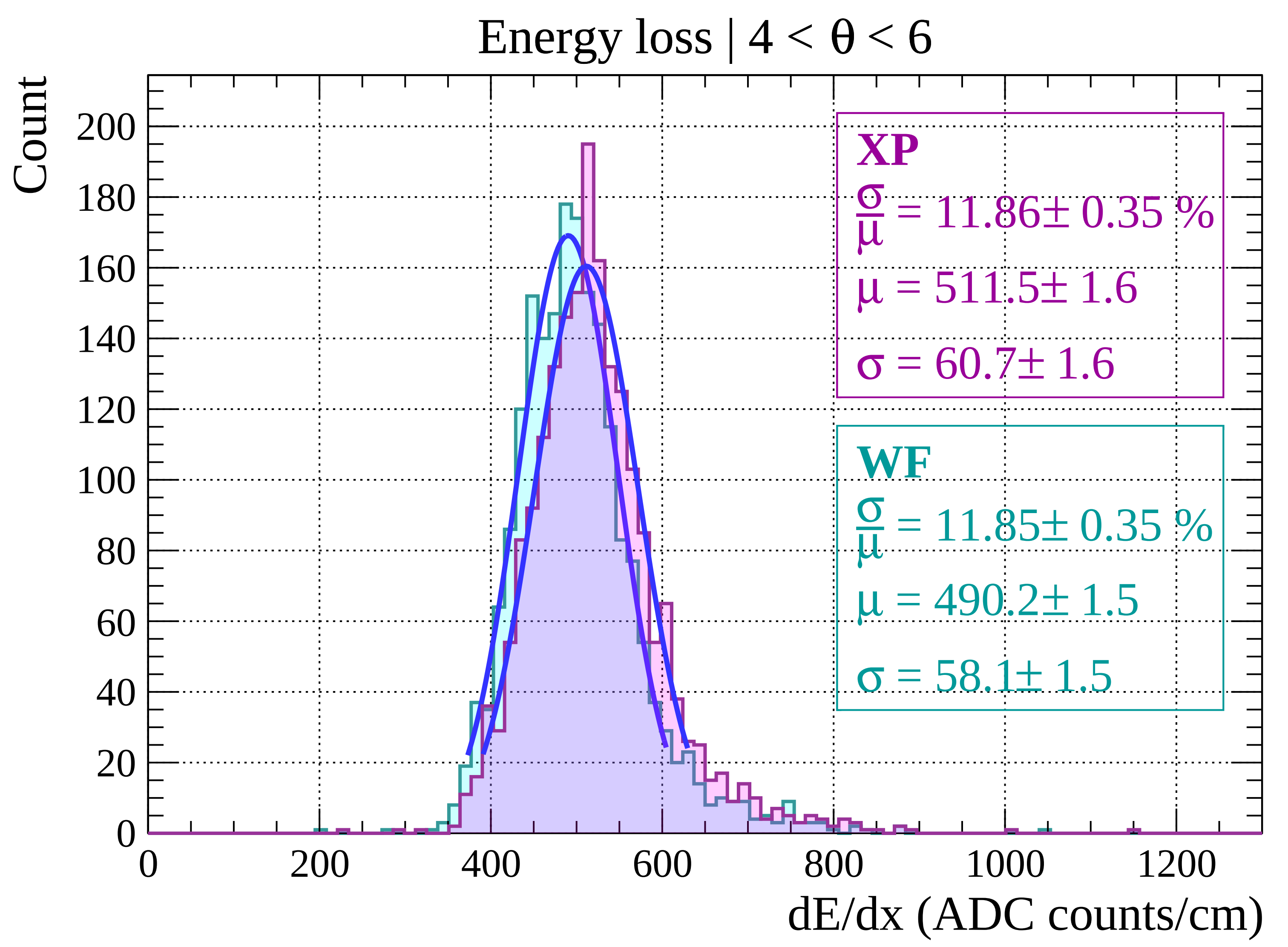
Energy loss | -2 < θ < 0
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP
σ/μ = 11.02± 0.34 %
μ = 513.4± 1.5
σ = 56.6± 1.6
WF
σ/μ = 11.22± 0.37 %
μ = 493.1± 1.5
σ = 55.3± 1.6



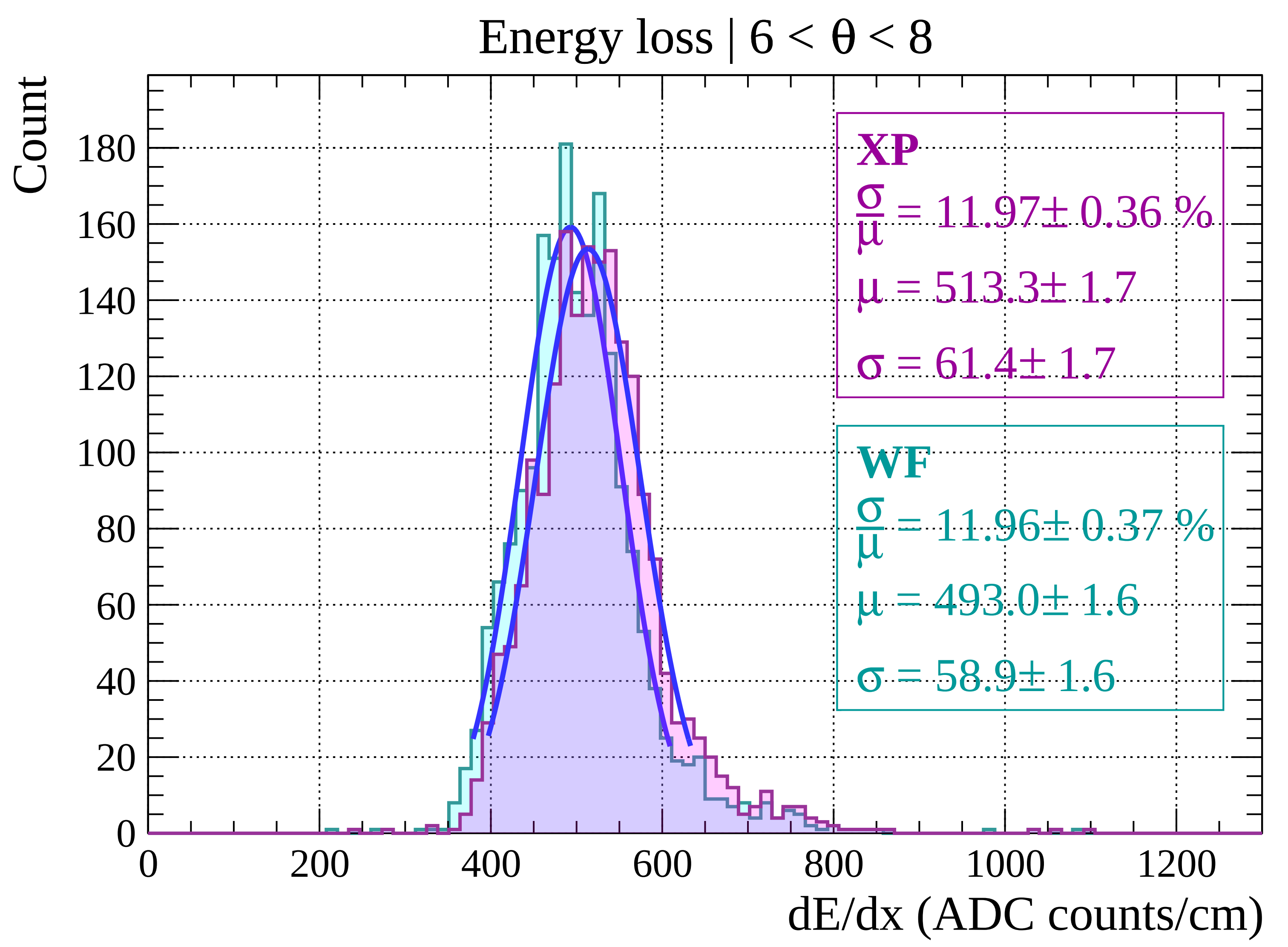
Energy loss | 0 < θ < 2
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
σ/μ = 11.33± 0.37 %
μ = 510.4± 1.6
σ = 57.8± 1.7
WF
σ/μ = 11.25± 0.37 %
μ = 489.3± 1.5
σ = 55.0± 1.6



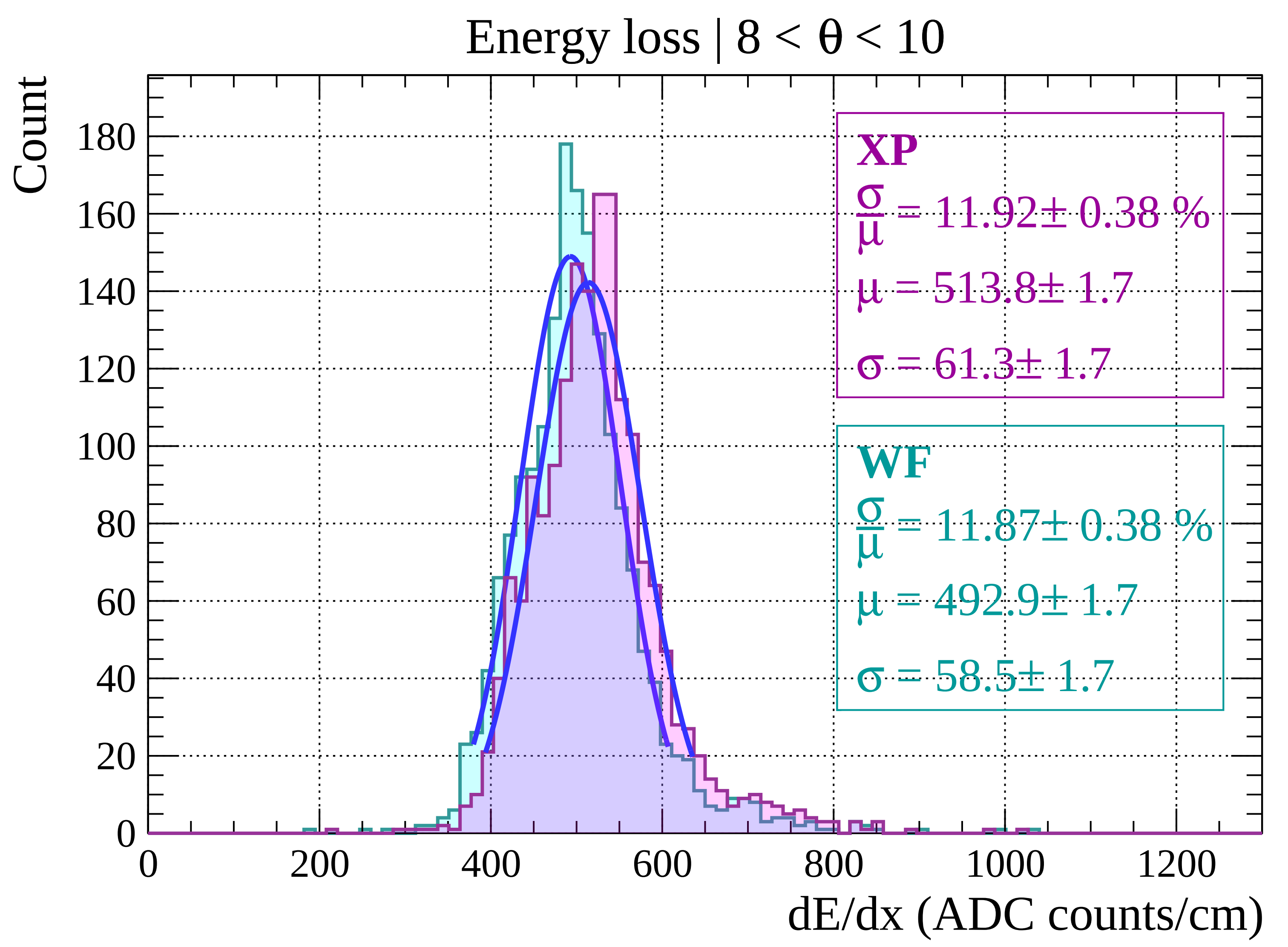
Energy loss | 2 < θ < 4
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP
σ/μ = 11.39± 0.36 %
μ = 512.7± 1.6
σ = 58.4± 1.7
WF
σ/μ = 11.57± 0.32 %
μ = 492.7± 1.5
σ = 57.0± 1.4



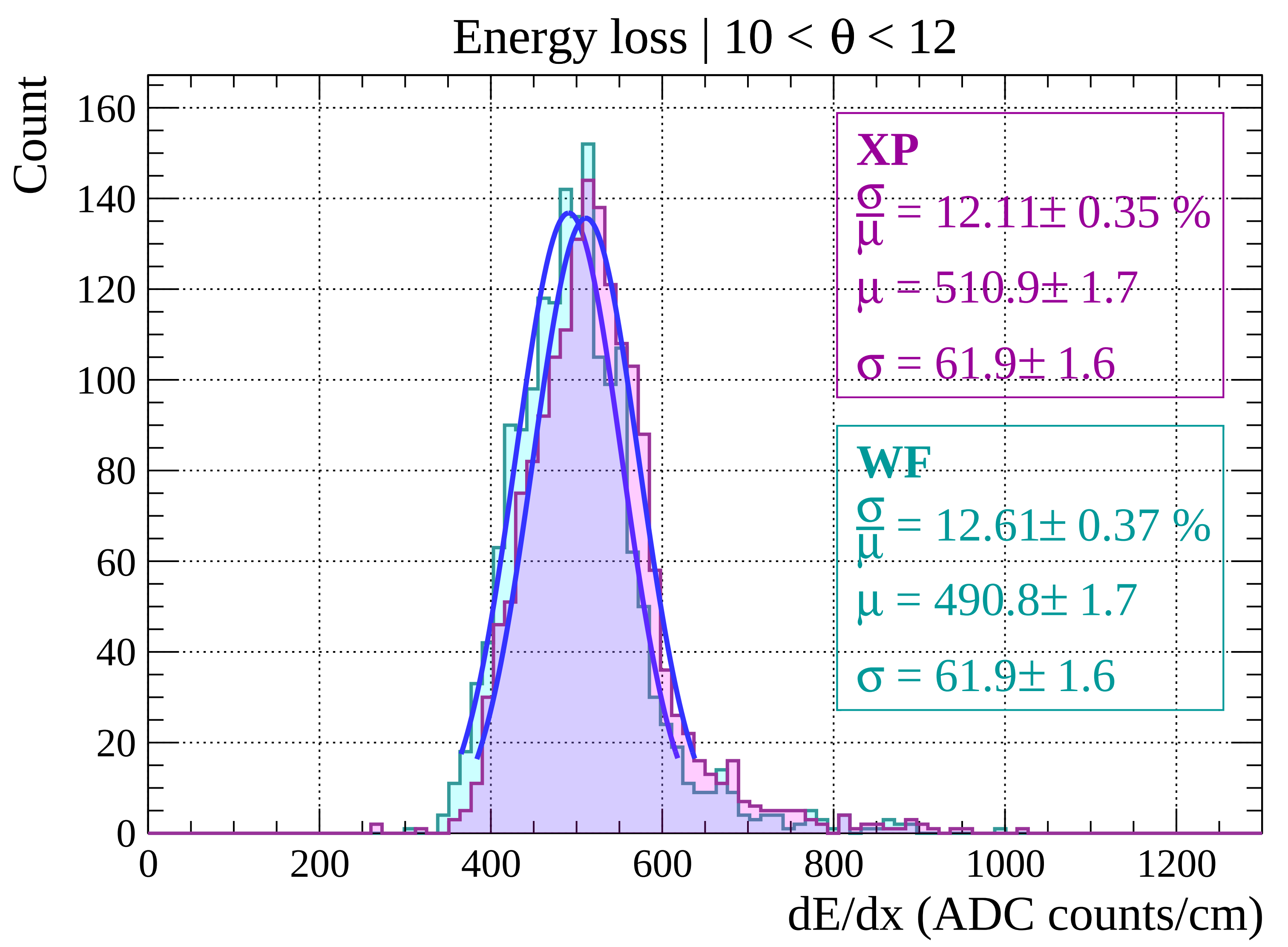
Energy loss | 4 < θ < 6
Count
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.86± 0.35 %
μ = 511.5± 1.6
σ = 60.7± 1.6
WF
σ/μ = 11.85± 0.35 %
μ = 490.2± 1.5
σ = 58.1± 1.5



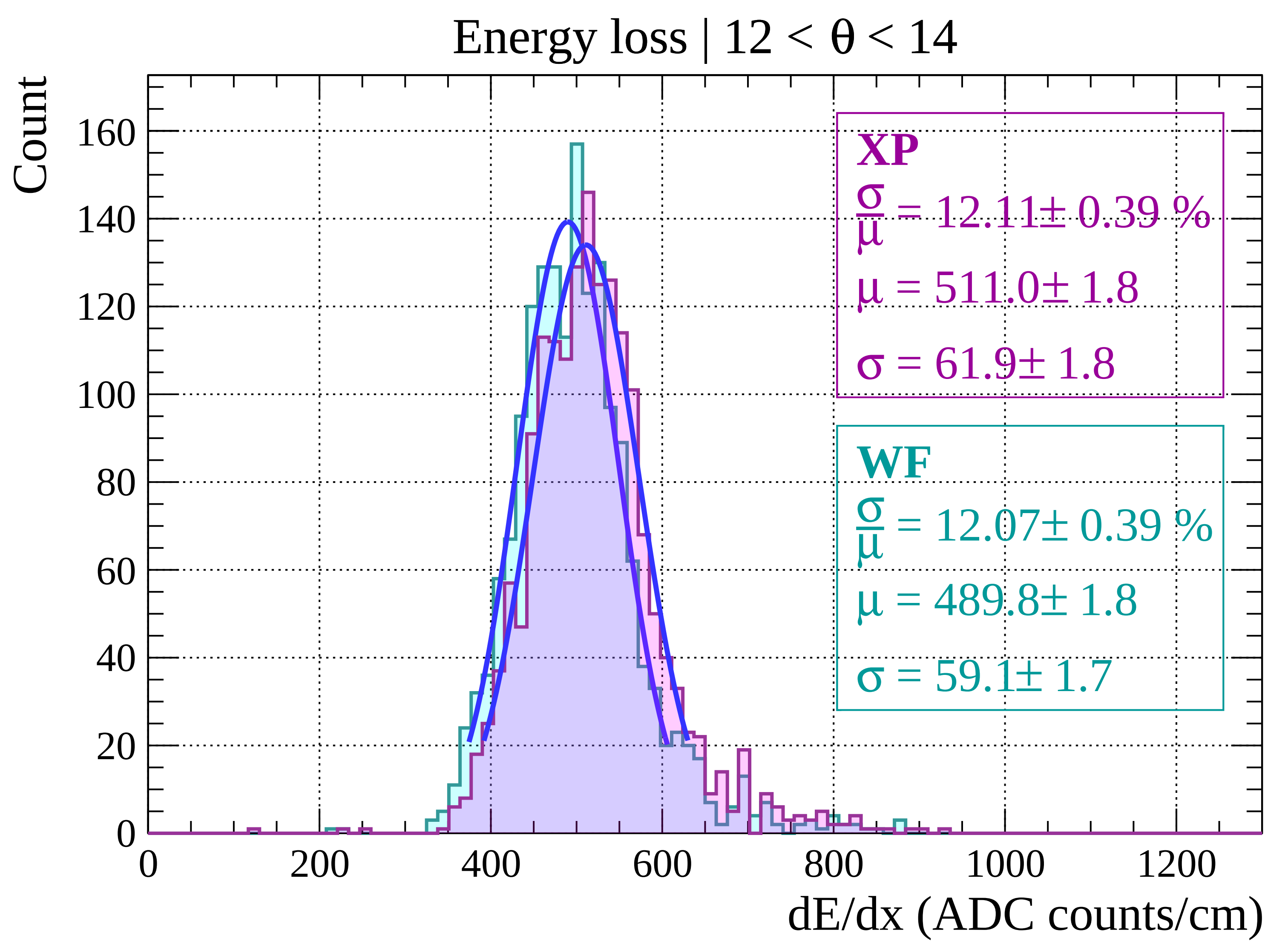
Energy loss | 6 < θ < 8
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP
σ/μ = 11.97± 0.36 %
μ = 513.3± 1.7
σ = 61.4± 1.7
WF
σ/μ = 11.96± 0.37 %
μ = 493.0± 1.6
σ = 58.9± 1.6



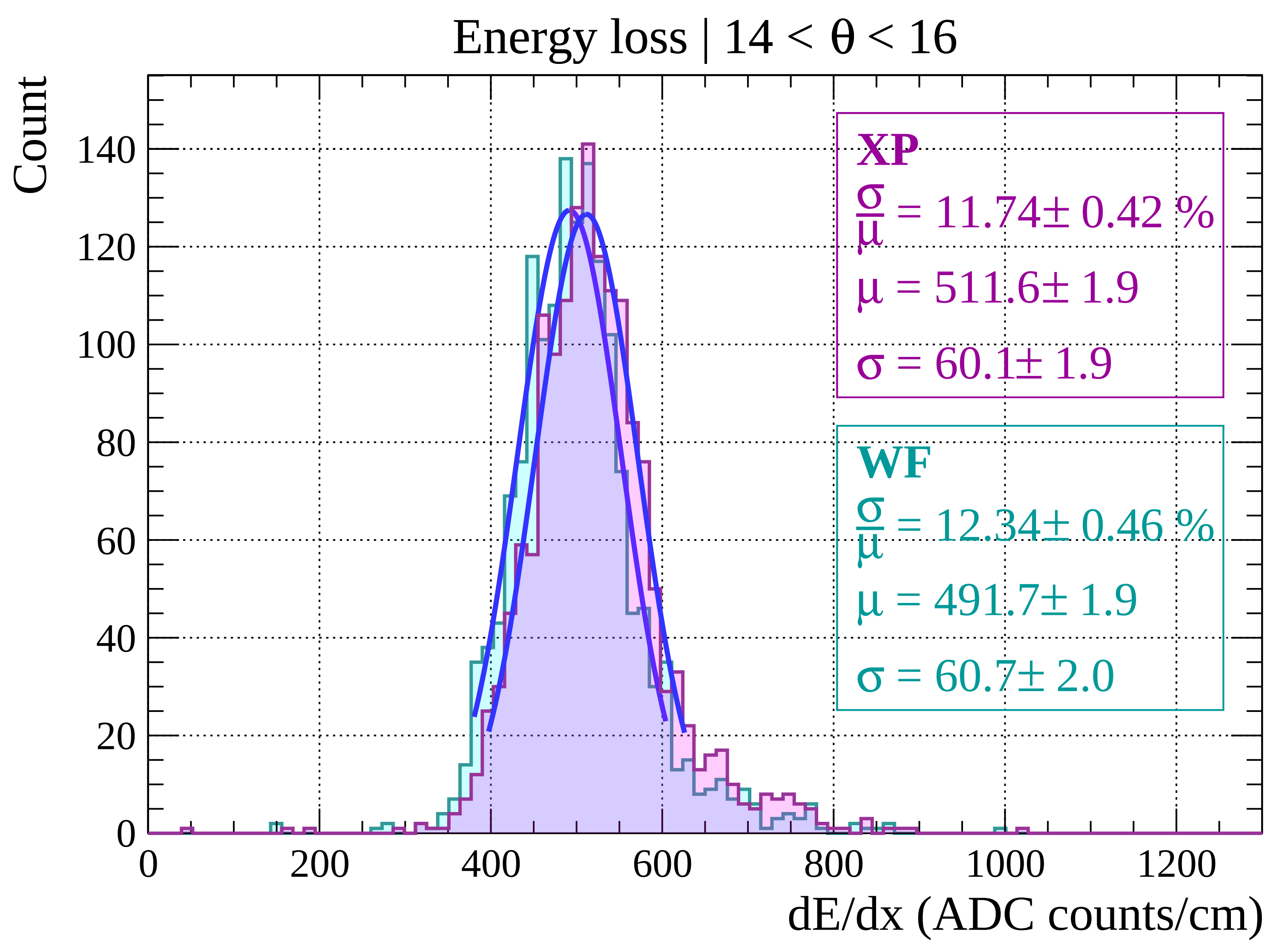
Energy loss | 8 < θ < 10
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 11.92± 0.38 %
μ = 513.8± 1.7
σ = 61.3± 1.7
WF
$\frac{\sigma}{\mu}$ = 11.87± 0.38 %
μ = 492.9± 1.7
σ = 58.5± 1.7



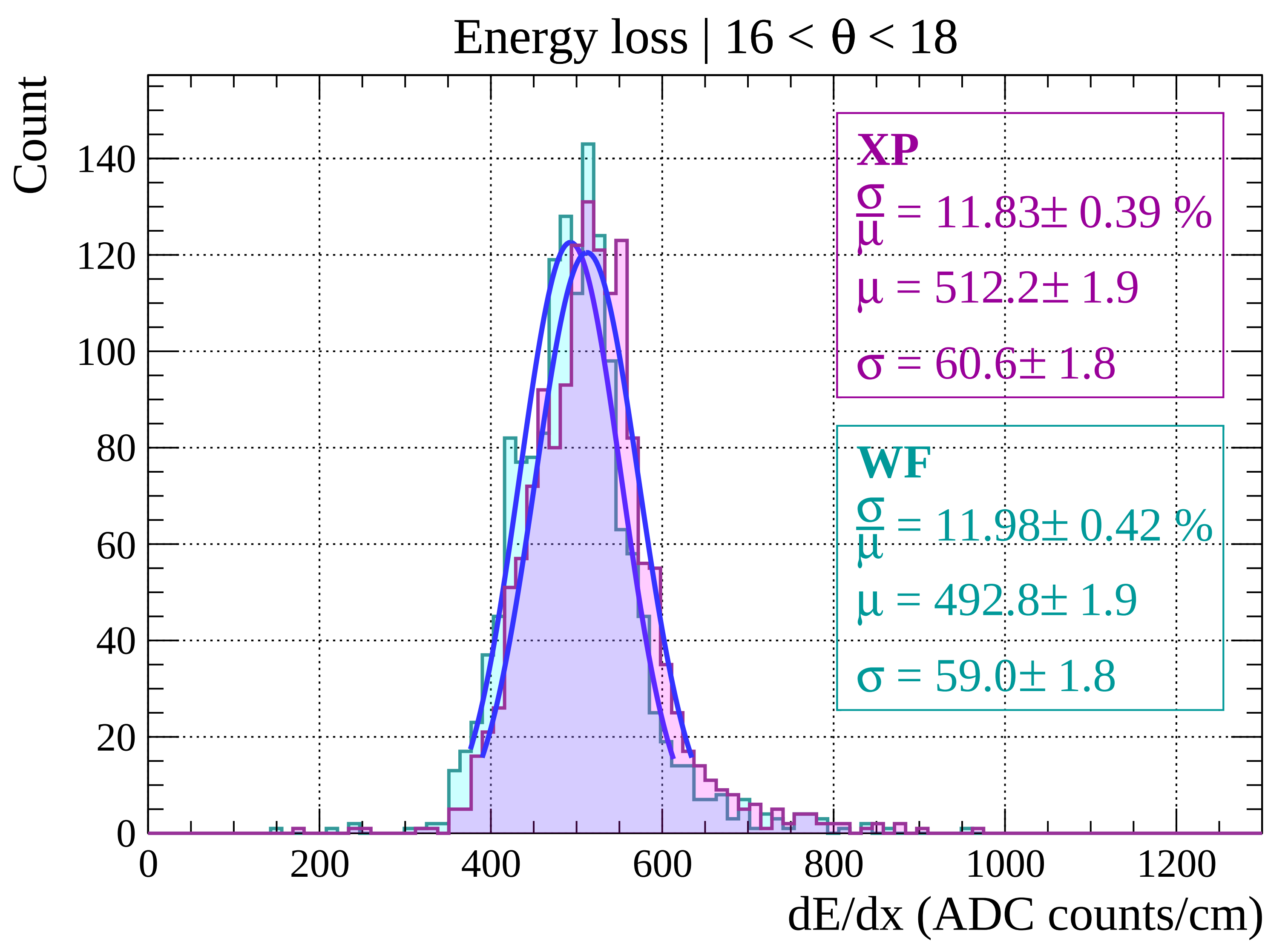
Energy loss | 10 < θ < 12
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 12.11± 0.35 %
μ = 510.9± 1.7
σ = 61.9± 1.6
WF
σ/μ = 12.61± 0.37 %
μ = 490.8± 1.7
σ = 61.9± 1.6



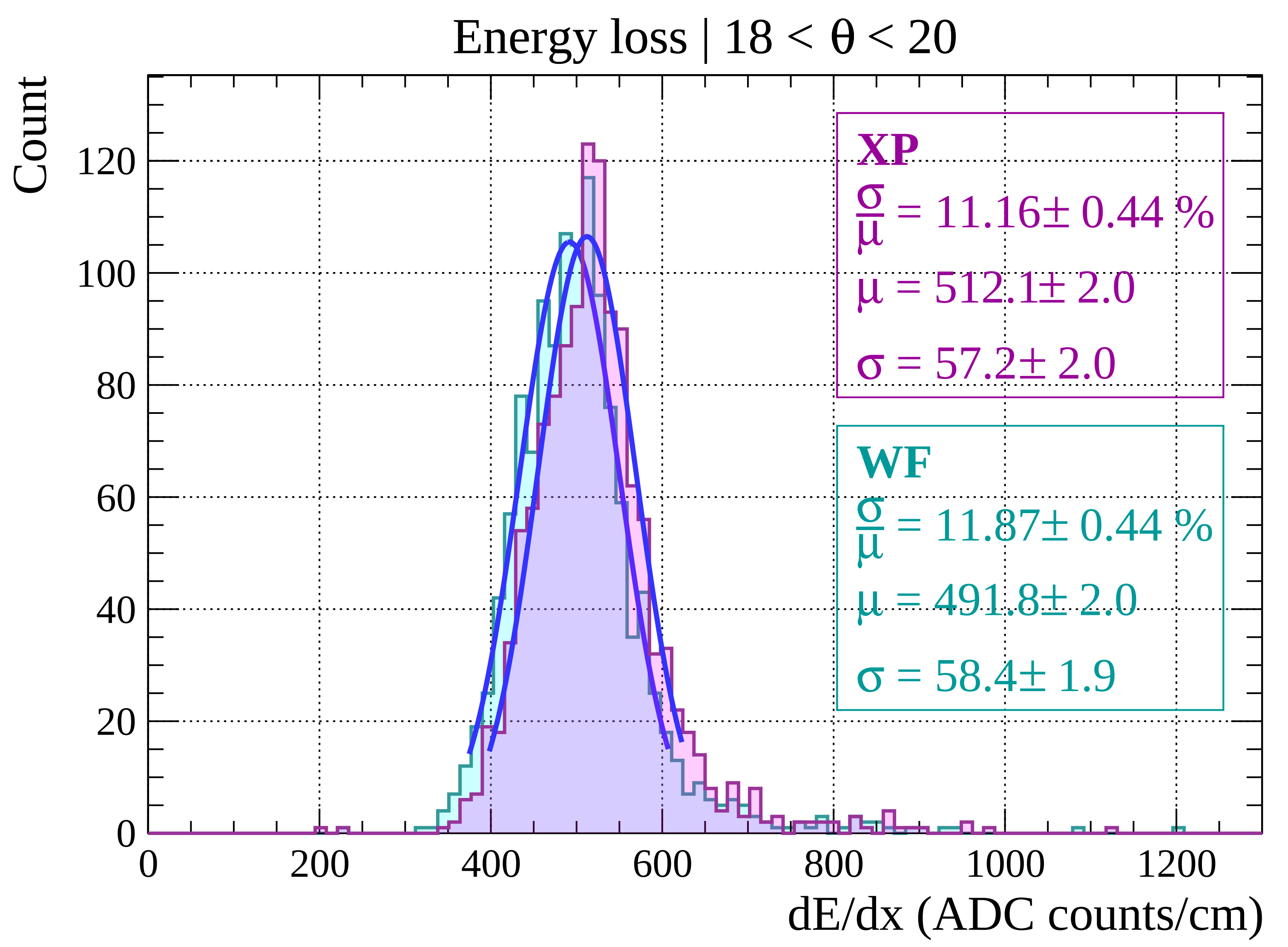
Energy loss | 12 < θ < 14
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 12.11± 0.39 %
μ = 511.0± 1.8
σ = 61.9± 1.8
WF
σ/μ = 12.07± 0.39 %
μ = 489.8± 1.8
σ = 59.1± 1.7
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200



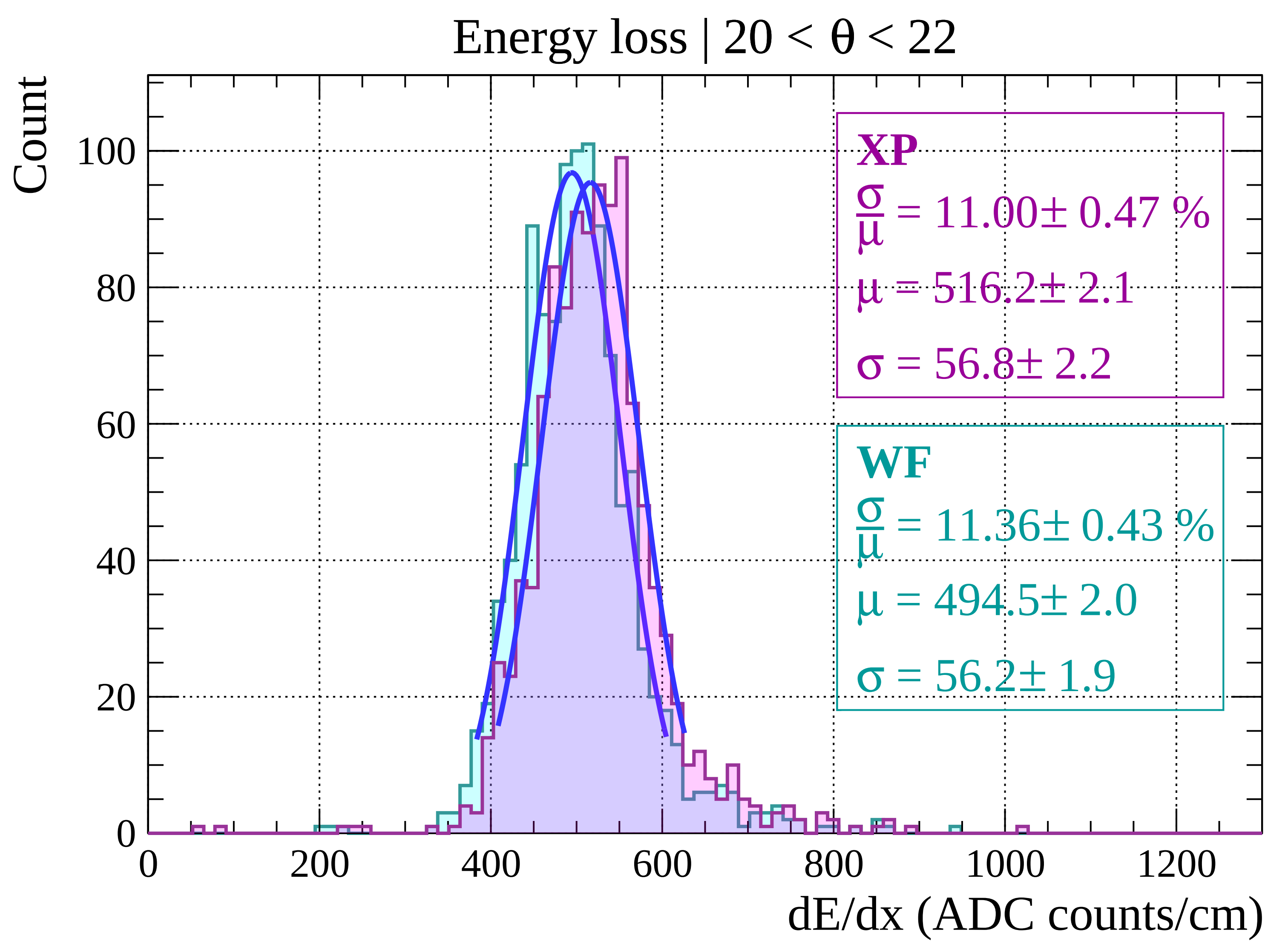
Energy loss | 14 < θ < 16
Count
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 11.74± 0.42 %
μ = 511.6± 1.9
σ = 60.1± 1.9
WF
$\frac{\sigma}{\mu}$ = 12.34± 0.46 %
μ = 491.7± 1.9
σ = 60.7± 2.0



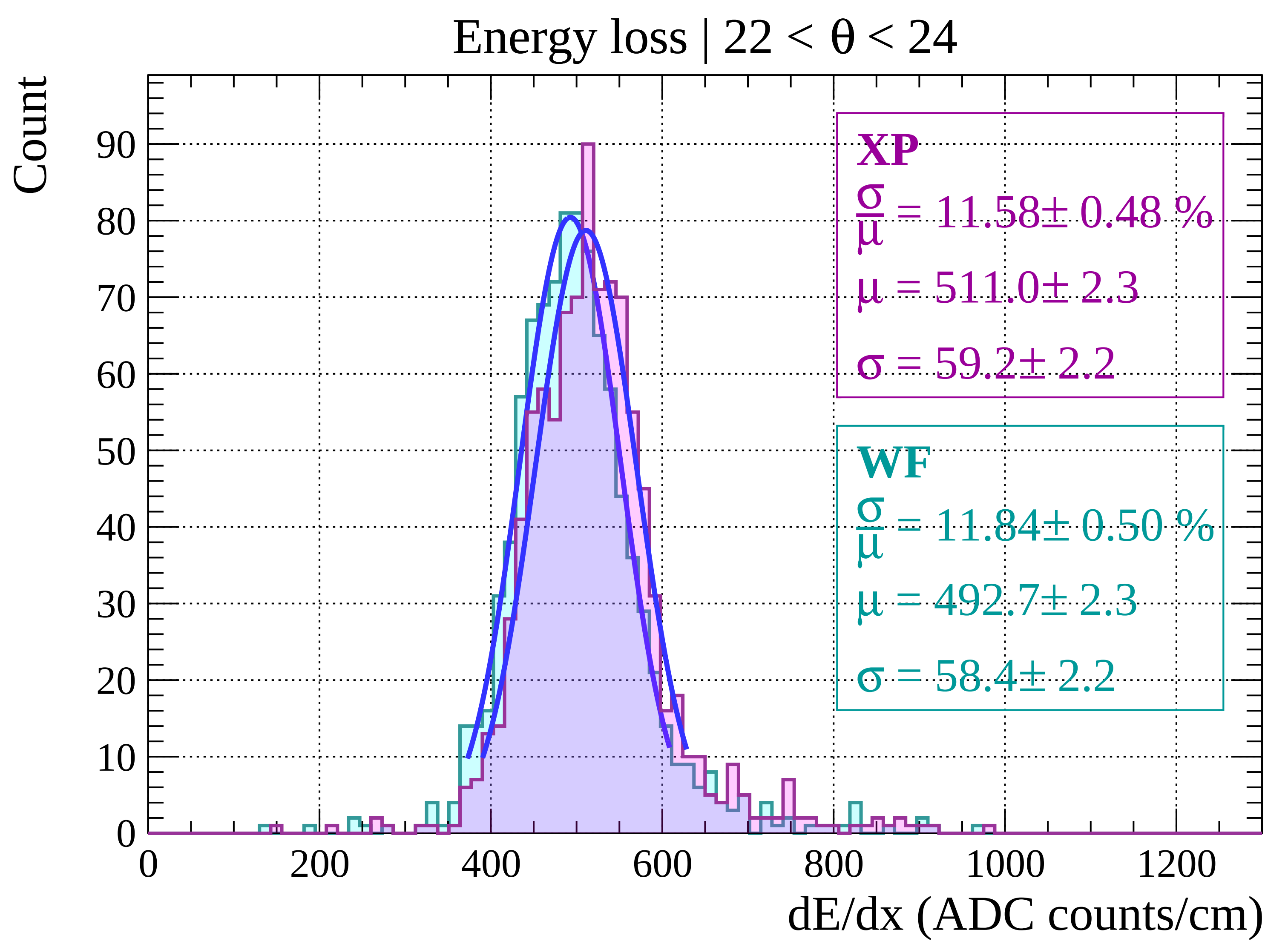
Energy loss | 16 < θ < 18
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.83± 0.39 %
μ = 512.2± 1.9
σ = 60.6± 1.8
WF
σ/μ = 11.98± 0.42 %
μ = 492.8± 1.9
σ = 59.0± 1.8
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200



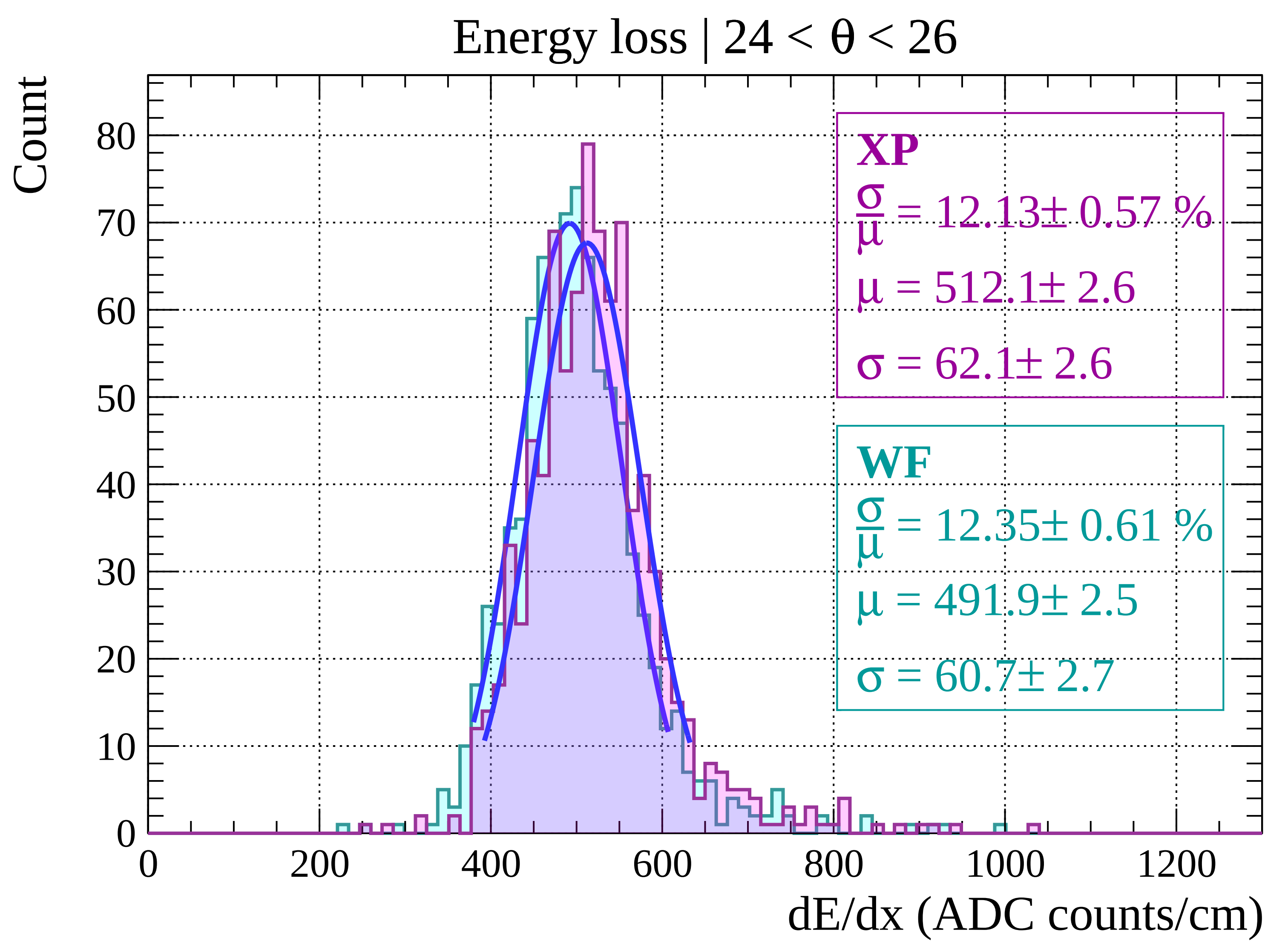
Energy loss | 18 < θ < 20
Count
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.16± 0.44 %
μ = 512.1± 2.0
σ = 57.2± 2.0
WF
σ/μ = 11.87± 0.44 %
μ = 491.8± 2.0
σ = 58.4± 1.9



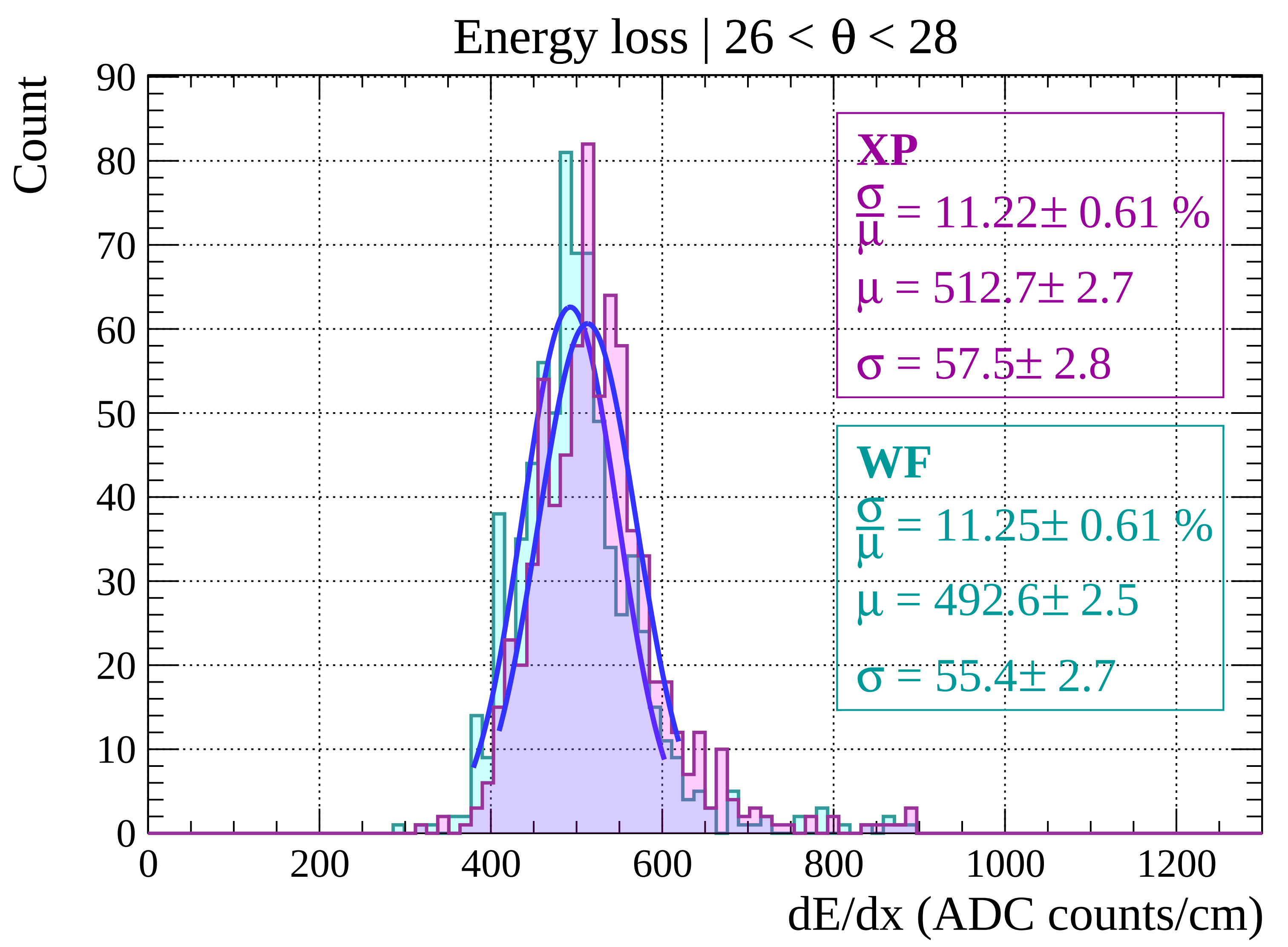
Energy loss | 20 < θ < 22
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.00± 0.47 %
μ = 516.2± 2.1
σ = 56.8± 2.2
WF
σ/μ = 11.36± 0.43 %
μ = 494.5± 2.0
σ = 56.2± 1.9
0
20
40
60
80
100
0
200
400
600
800
1000
1200



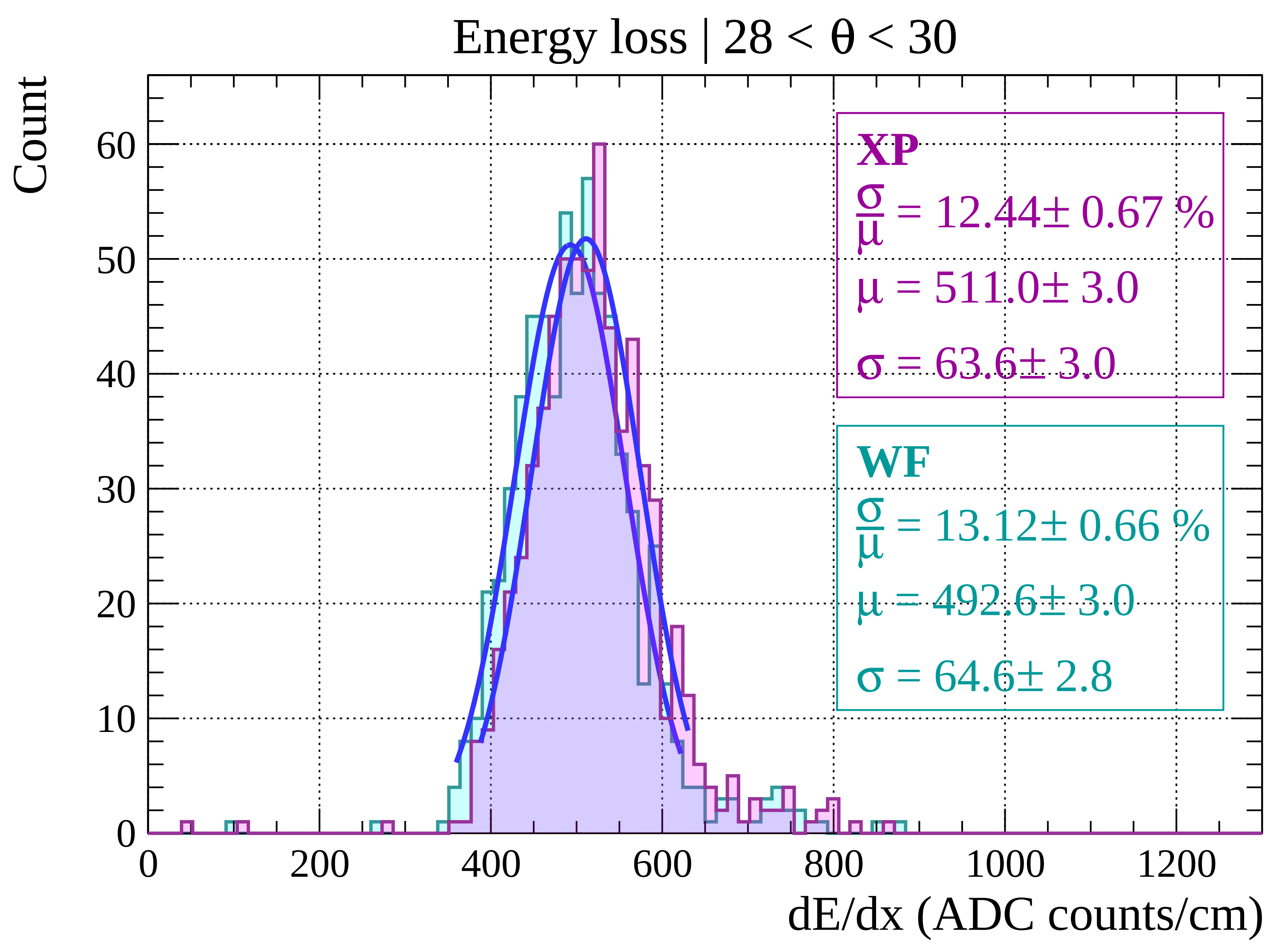
Energy loss | 22 < θ < 24
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP
σ/μ = 11.58± 0.48 %
μ = 511.0± 2.3
σ = 59.2± 2.2
WF
σ/μ = 11.84± 0.50 %
μ = 492.7± 2.3
σ = 58.4± 2.2



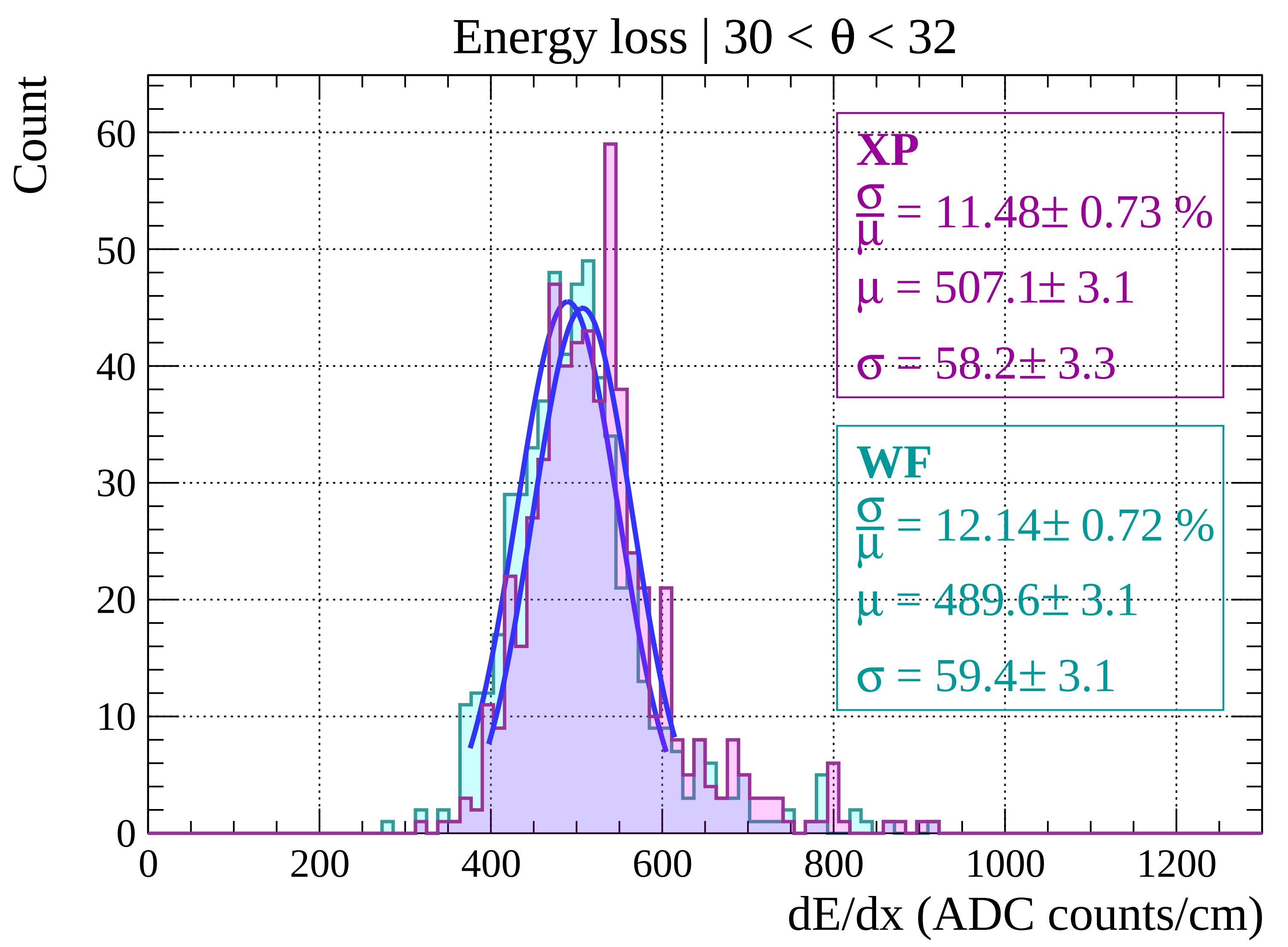
Energy loss | 24 < θ < 26
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
XP
σ/μ = 12.13± 0.57 %
μ = 512.1± 2.6
σ = 62.1± 2.6
WF
σ/μ = 12.35± 0.61 %
μ = 491.9± 2.5
σ = 60.7± 2.7



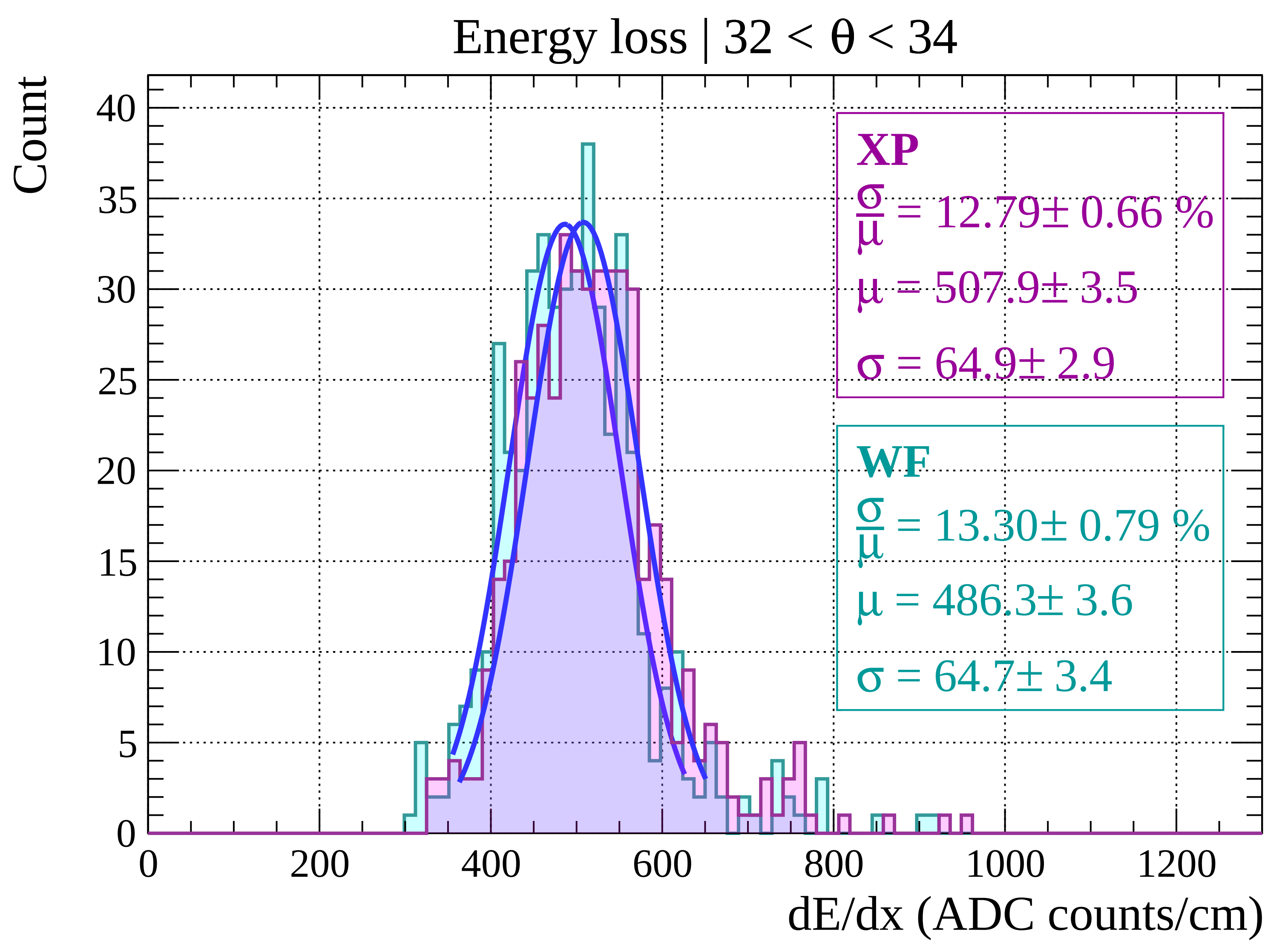
Energy loss | 26 < θ < 28
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
90
XP
σ/μ = 11.22± 0.61 %
μ = 512.7± 2.7
σ = 57.5± 2.8
WF
σ/μ = 11.25± 0.61 %
μ = 492.6± 2.5
σ = 55.4± 2.7

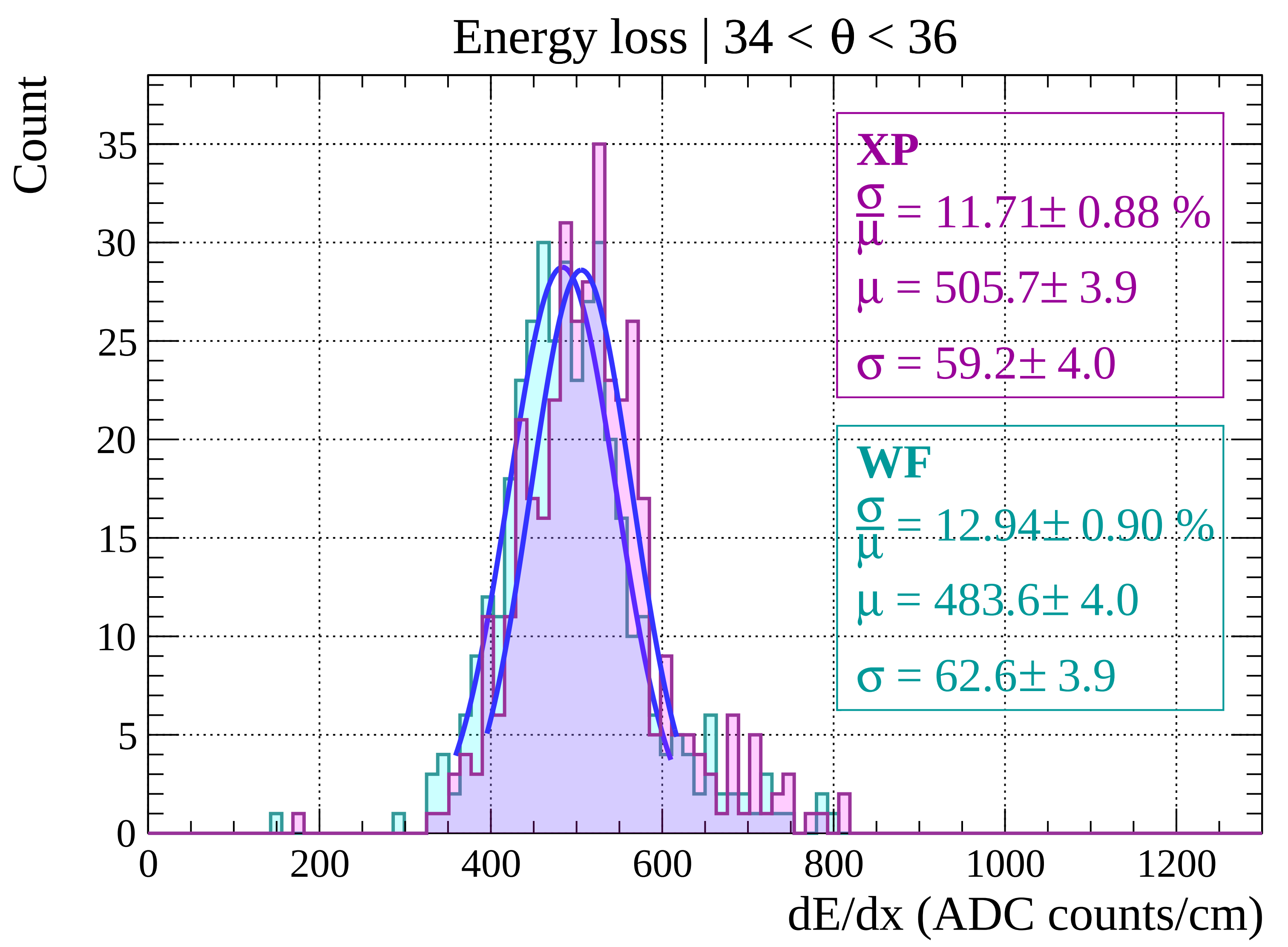


Energy loss | 28 < θ < 30
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 12.44± 0.67 %
μ = 511.0± 3.0
σ = 63.6± 3.0
WF
σ/μ = 13.12± 0.66 %
μ = 492.6± 3.0
σ = 64.6± 2.8

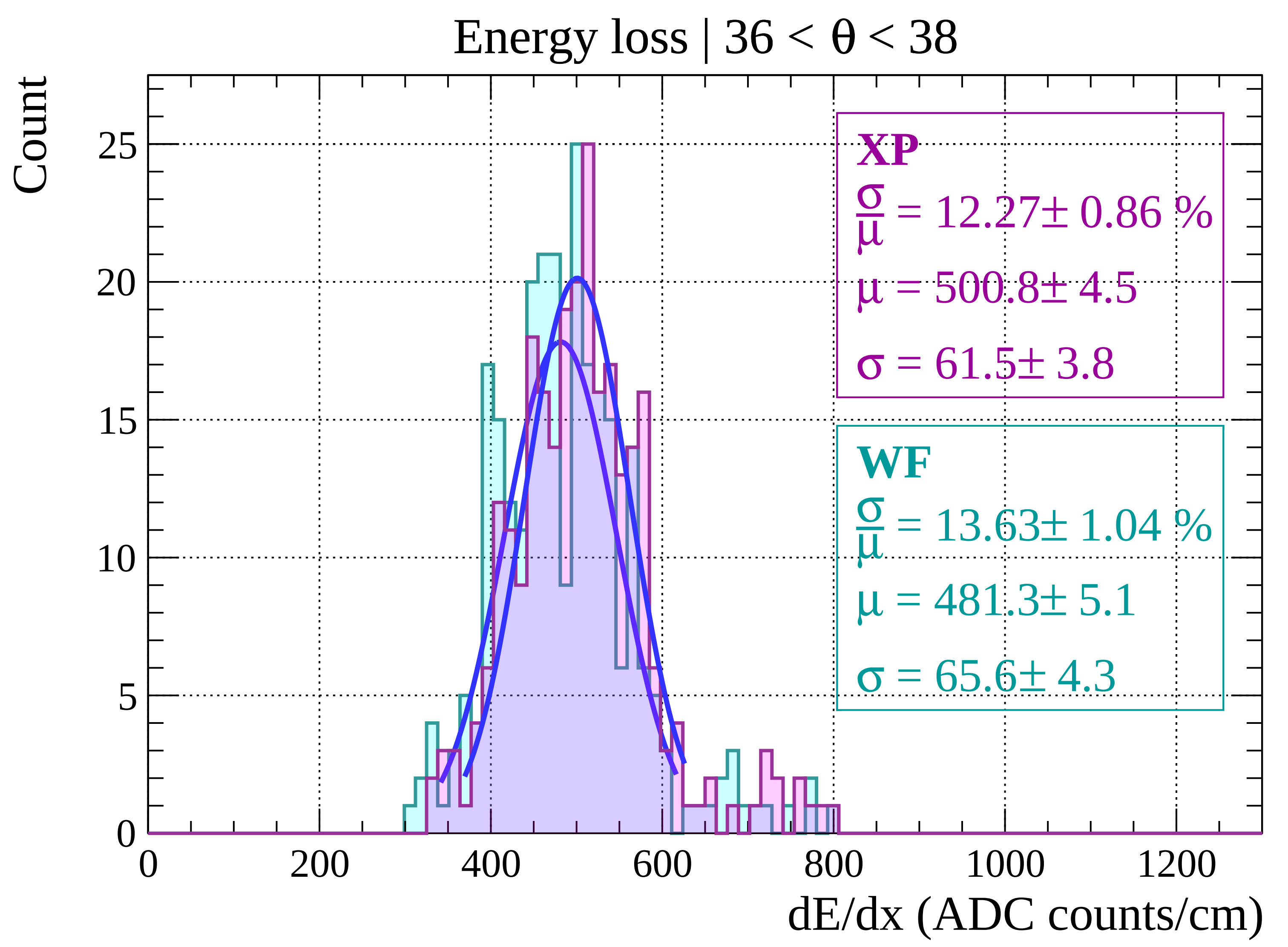


Energy loss | 30 < θ < 32
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
XP
σ/μ = 11.48± 0.73 %
μ = 507.1± 3.1
σ = 58.2± 3.3
WF
σ/μ = 12.14± 0.72 %
μ = 489.6± 3.1
σ = 59.4± 3.1

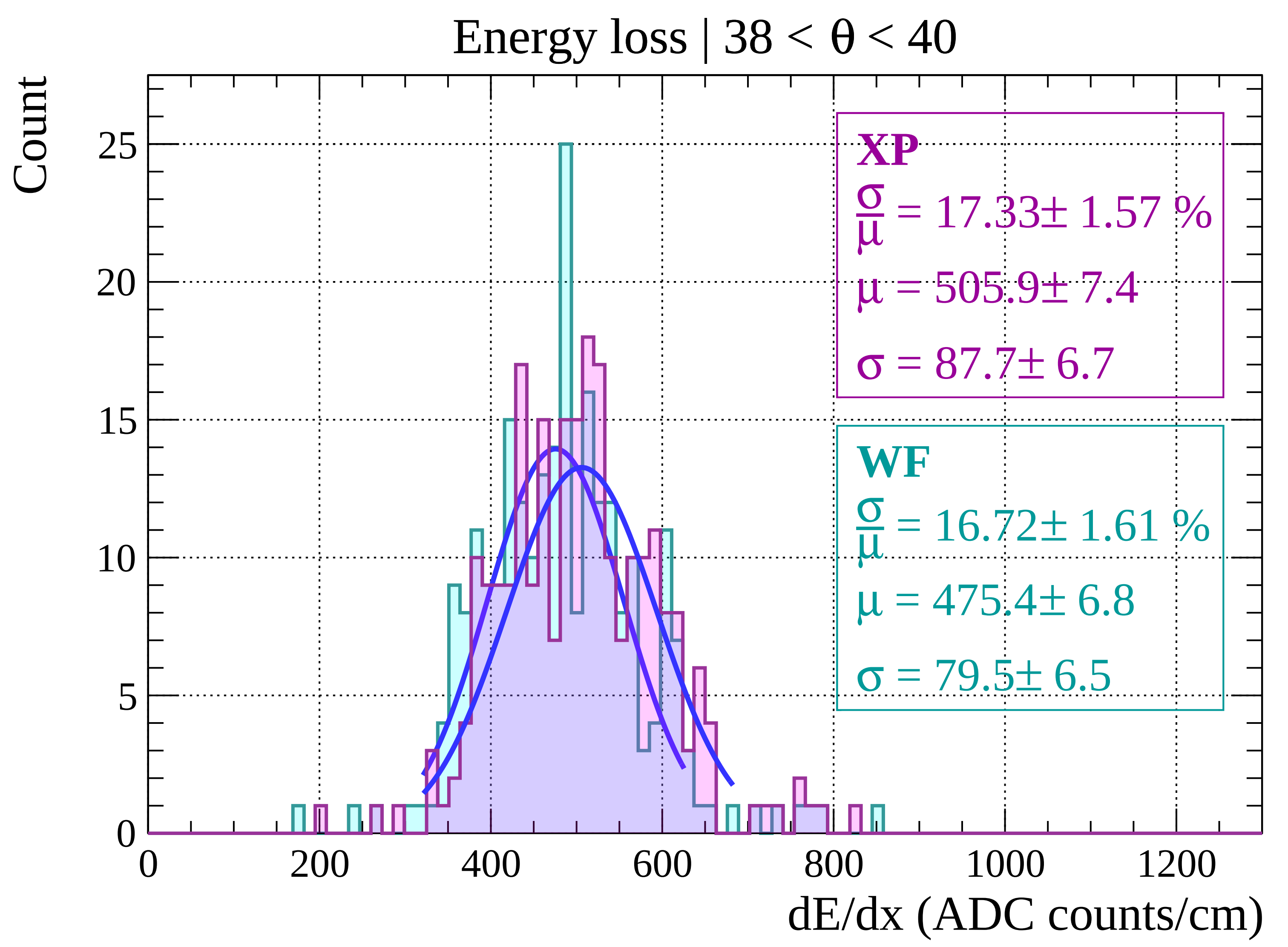

Energy loss | 32 < θ < 34
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 12.79± 0.66 %
μ = 507.9± 3.5
σ = 64.9± 2.9
WF
σ/μ = 13.30± 0.79 %
μ = 486.3± 3.6
σ = 64.7± 3.4
0
200
400
600
800
1000
1200
5
10
15
20
25
30
35
40



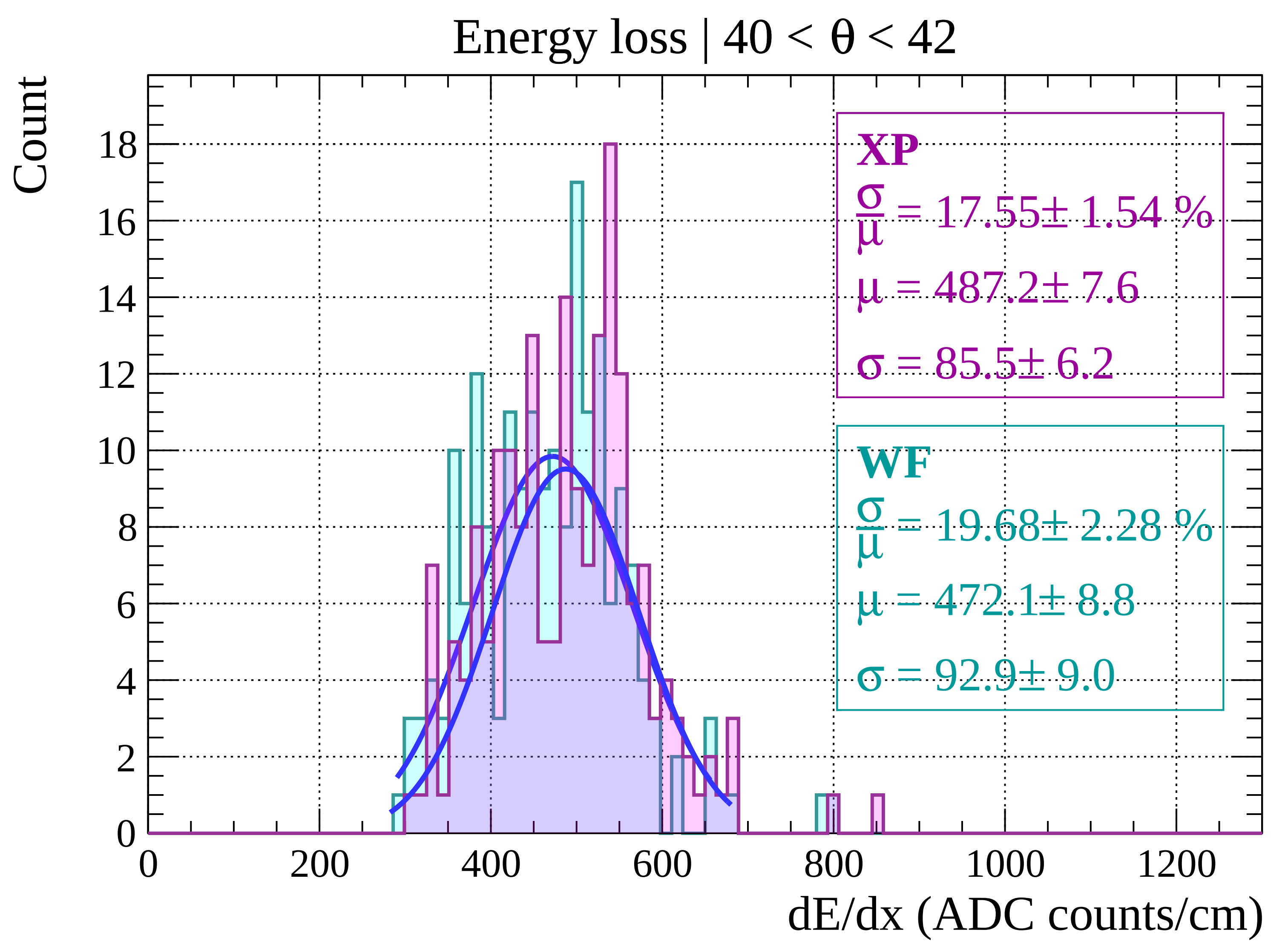
Energy loss | 34 < θ < 36
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
XP
σ/μ = 11.71± 0.88 %
μ = 505.7± 3.9
σ = 59.2± 4.0
WF
σ/μ = 12.94± 0.90 %
μ = 483.6± 4.0
σ = 62.6± 3.9

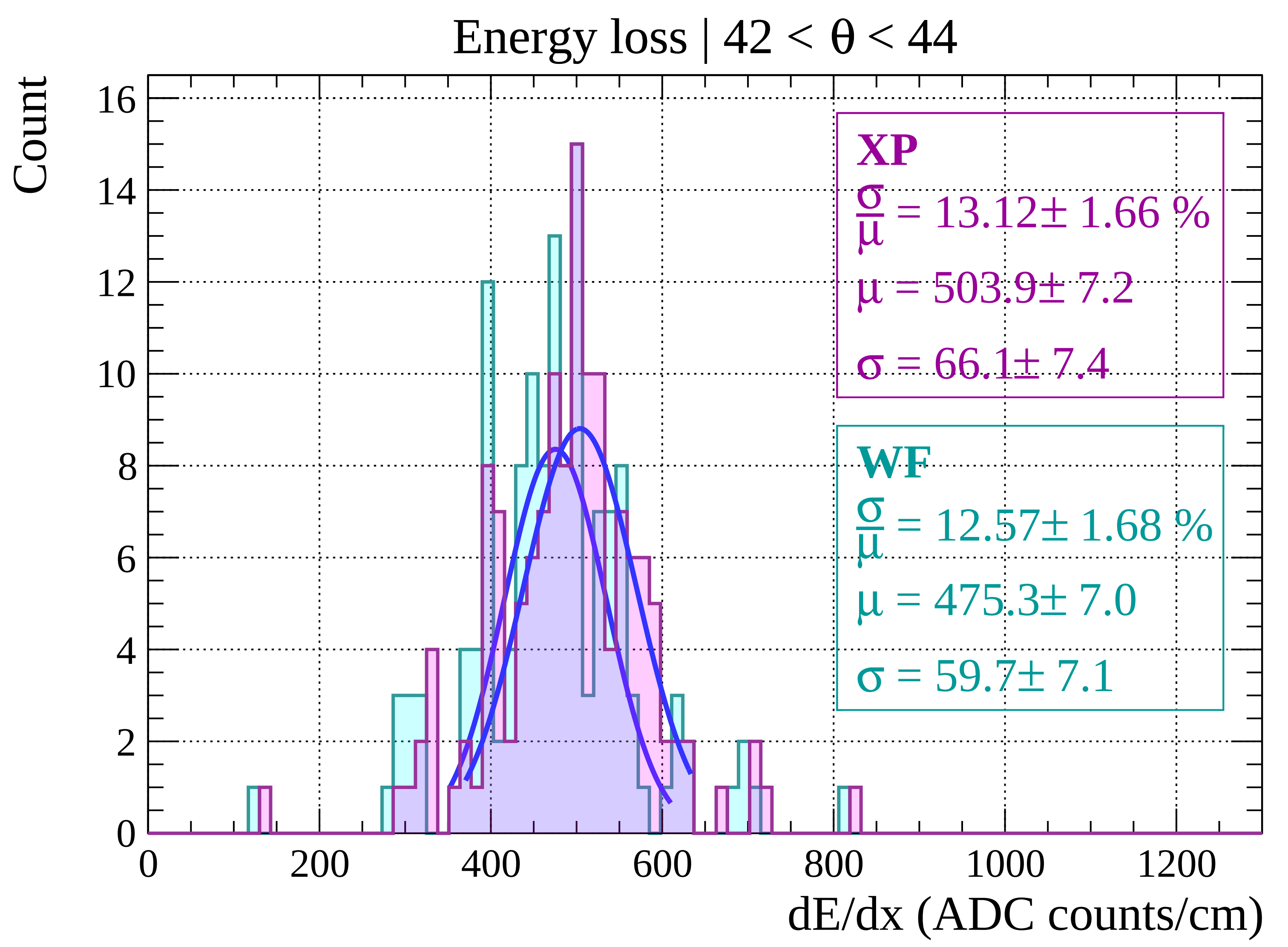


Energy loss | 36 < θ < 38
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 12.27± 0.86 %
μ = 500.8± 4.5
σ = 61.5± 3.8
WF
σ/μ = 13.63± 1.04 %
μ = 481.3± 5.1
σ = 65.6± 4.3

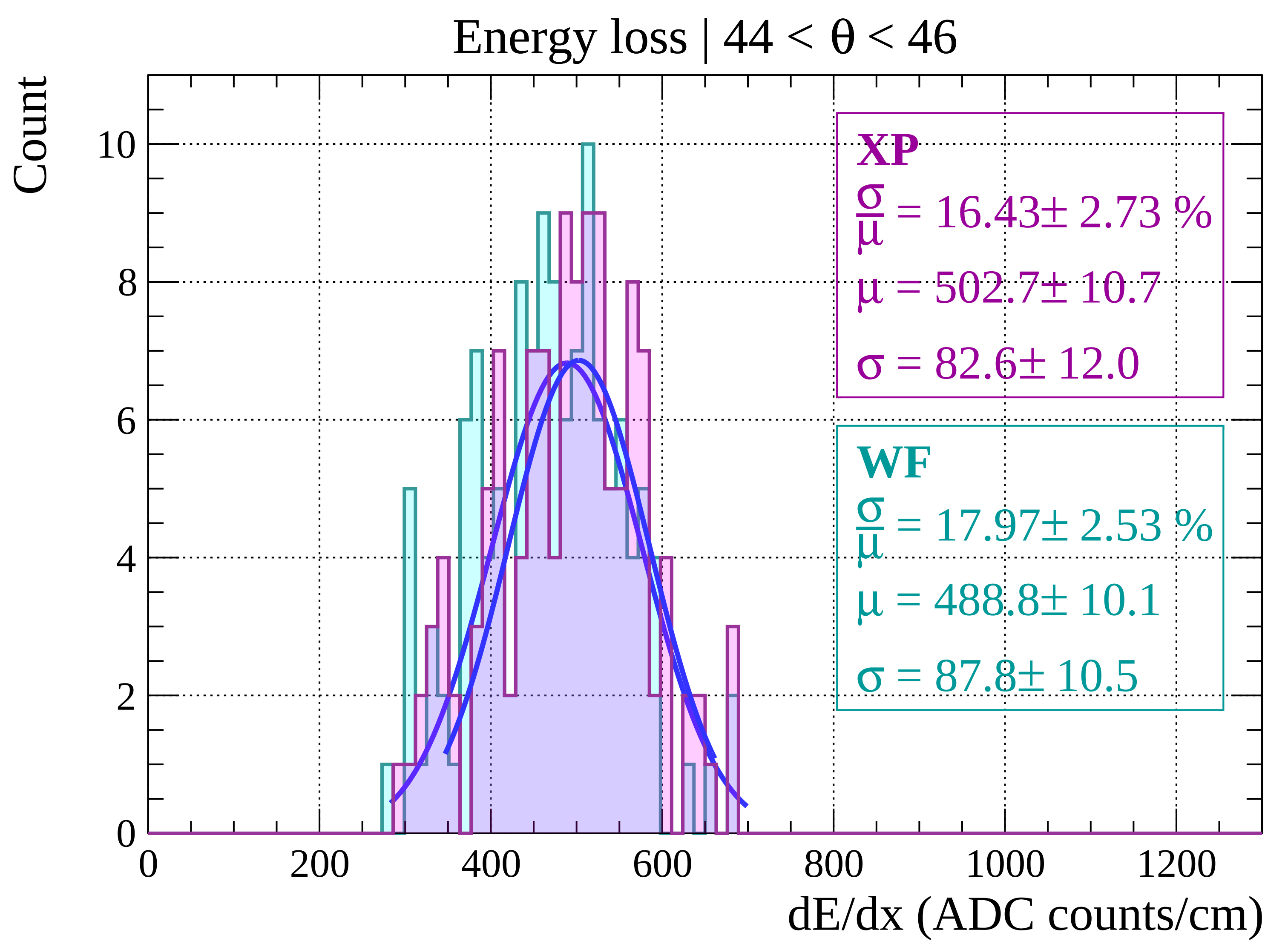


Energy loss | 38 < θ < 40
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
XP
σ/μ = 17.33± 1.57 %
μ = 505.9± 7.4
σ = 87.7± 6.7
WF
σ/μ = 16.72± 1.61 %
μ = 475.4± 6.8
σ = 79.5± 6.5

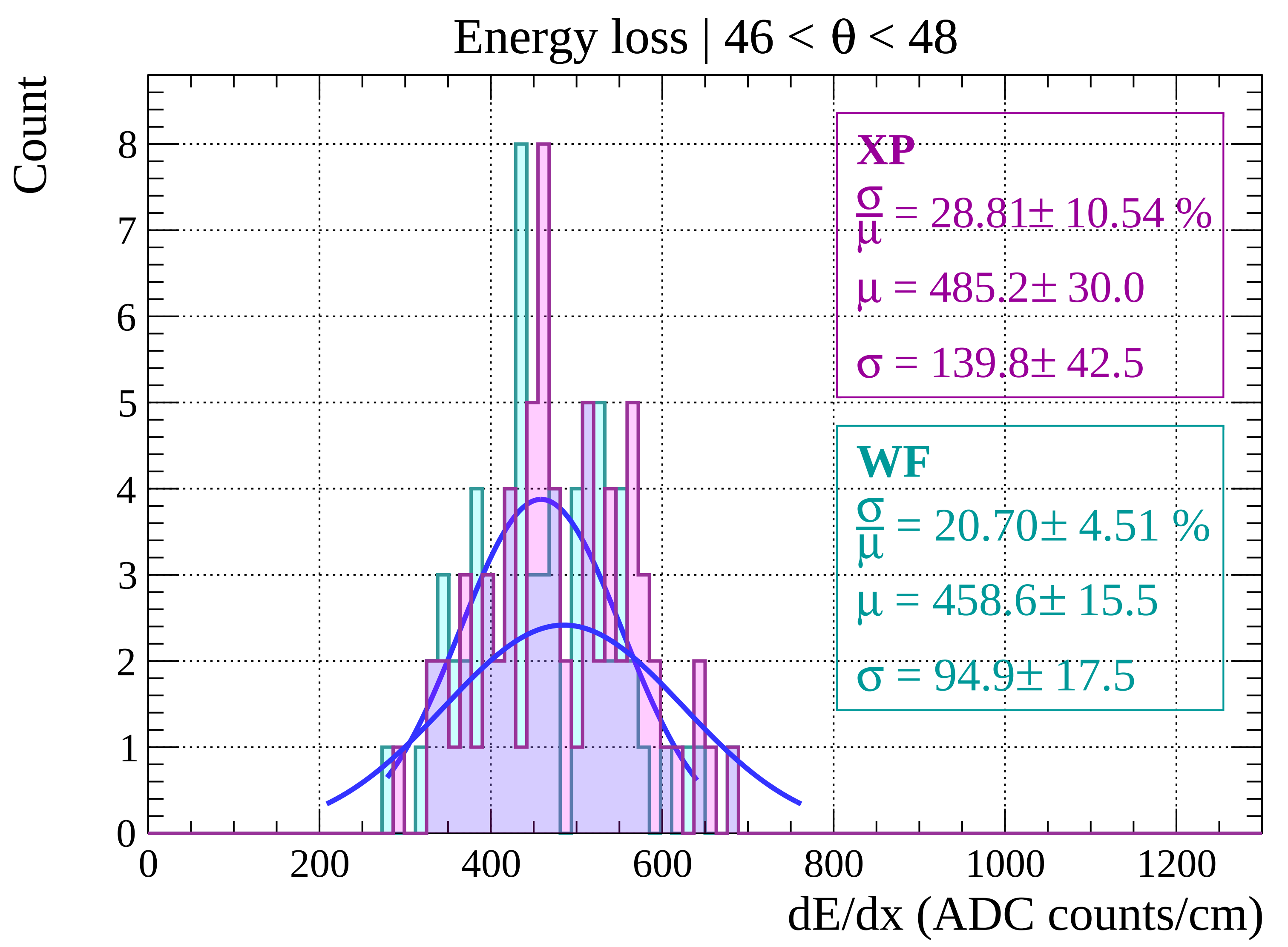

Energy loss | 40 < θ < 42
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
18
XP
$\frac{\sigma}{\mu}$ = 17.55± 1.54 %
μ = 487.2± 7.6
σ = 85.5± 6.2
WF
$\frac{\sigma}{\mu}$ = 19.68± 2.28 %
μ = 472.1± 8.8
σ = 92.9± 9.0



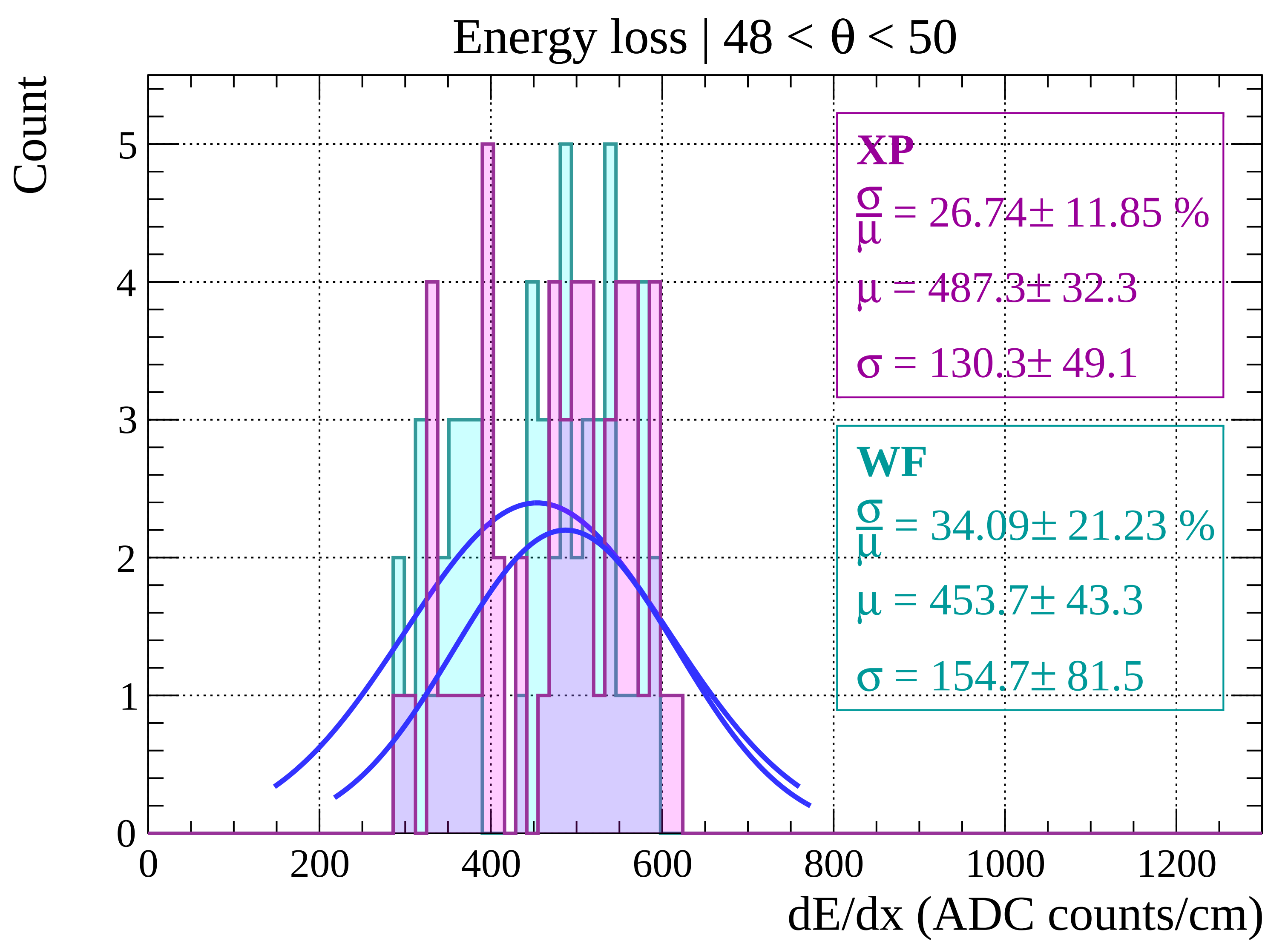
Energy loss | 42 < θ < 44
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
0
200
400
600
800
1000
1200
XP
σ/μ = 13.12± 1.66 %
μ = 503.9± 7.2
σ = 66.1± 7.4
WF
σ/μ = 12.57± 1.68 %
μ = 475.3± 7.0
σ = 59.7± 7.1



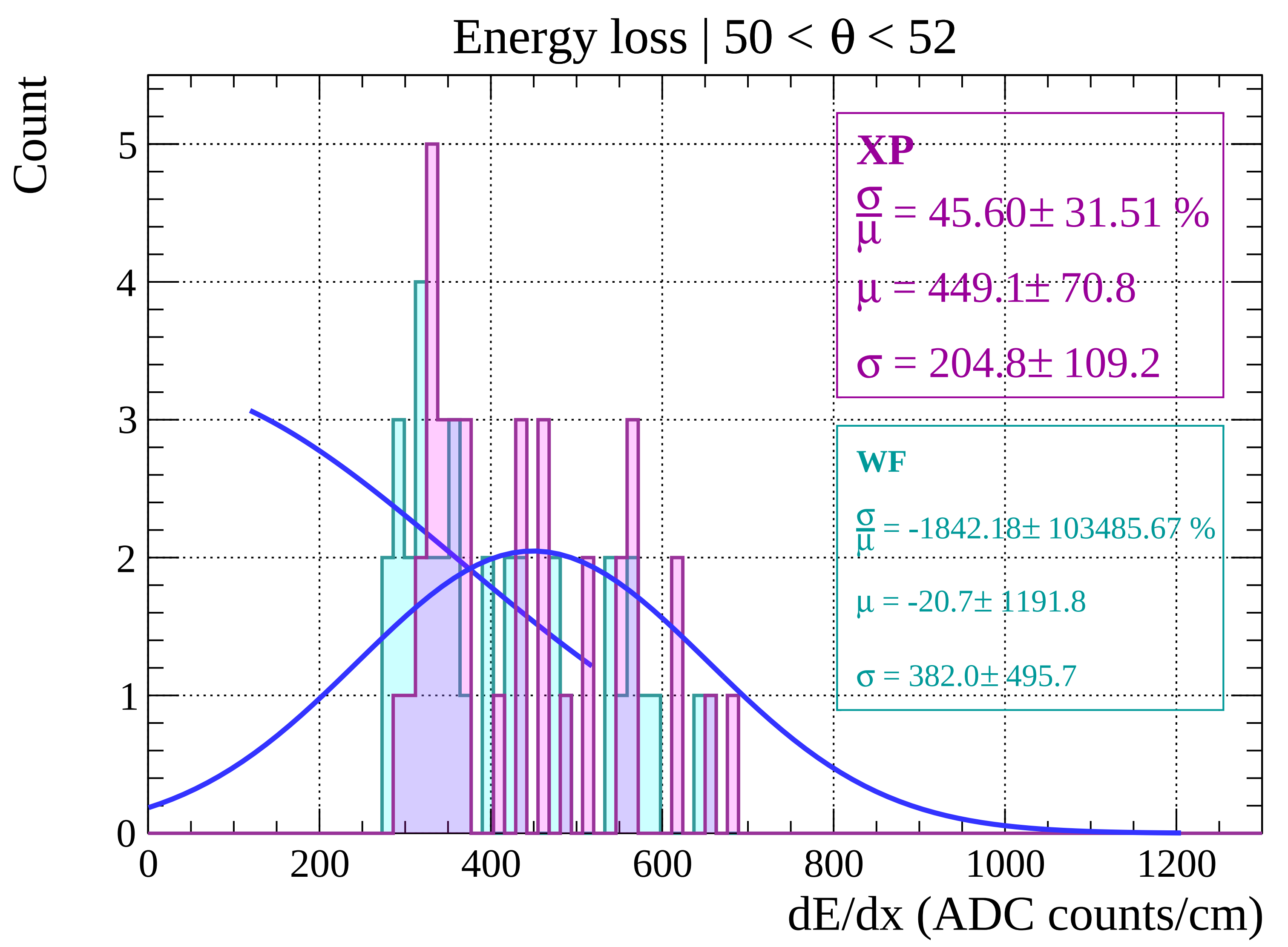
Energy loss | 44 < θ < 46
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.43± 2.73 %
μ = 502.7± 10.7
σ = 82.6± 12.0
WF
σ/μ = 17.97± 2.53 %
μ = 488.8± 10.1
σ = 87.8± 10.5



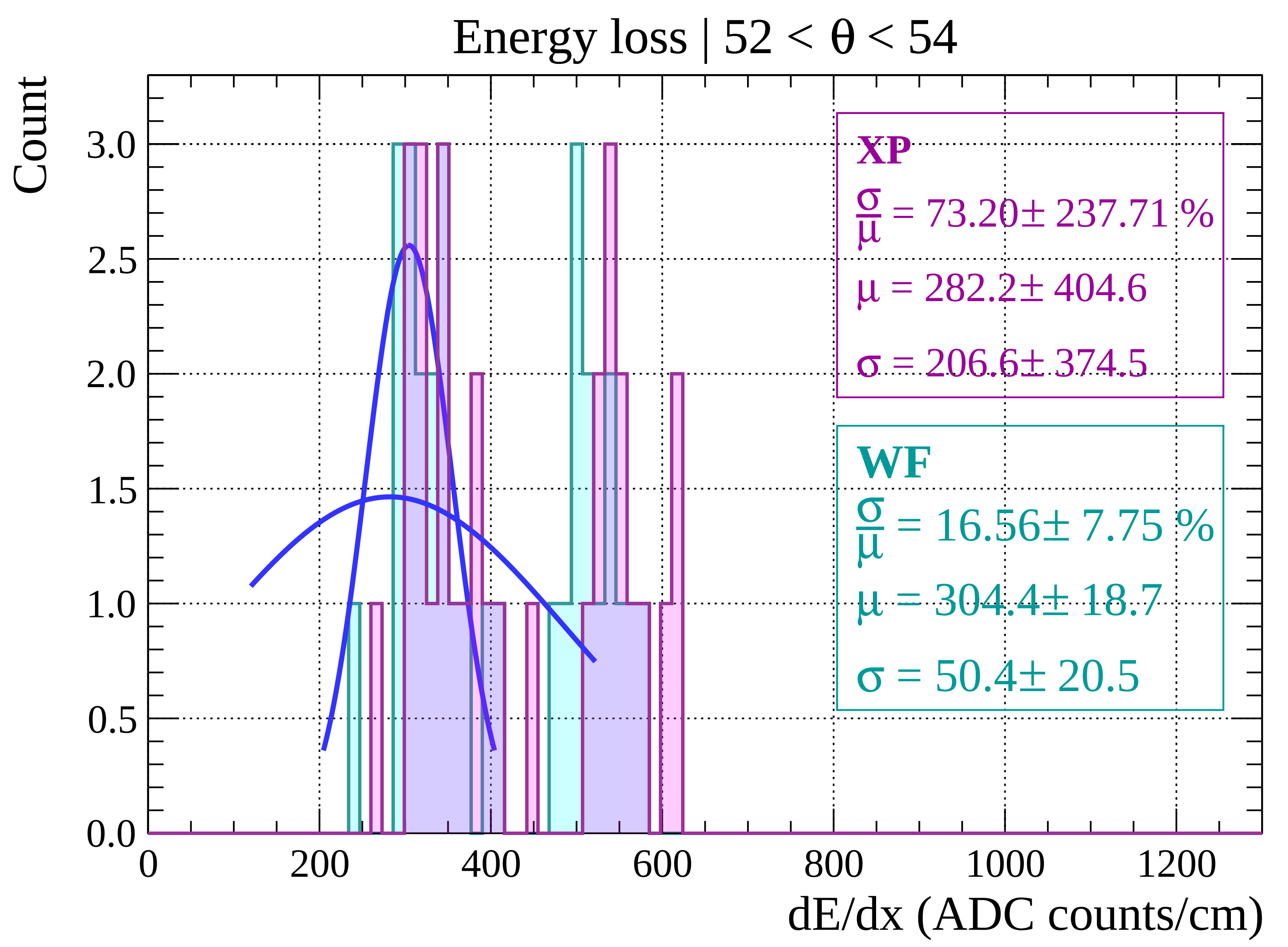
Energy loss | 46 < θ < 48
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
XP
σ/μ = 28.81± 10.54 %
μ = 485.2± 30.0
σ = 139.8± 42.5
WF
σ/μ = 20.70± 4.51 %
μ = 458.6± 15.5
σ = 94.9± 17.5



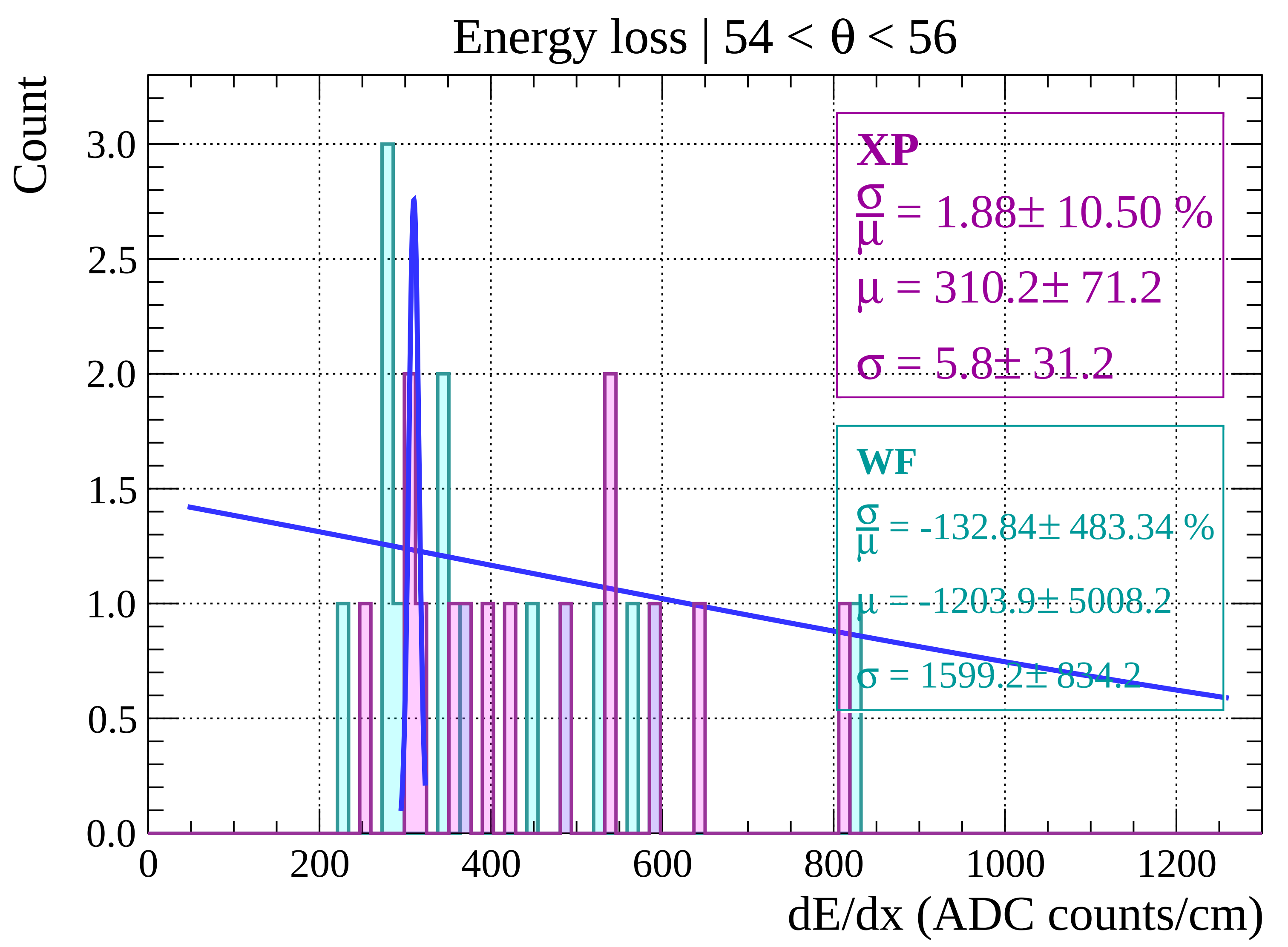
Energy loss | 48 < θ < 50
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
XP
σ/μ = 26.74± 11.85 %
μ = 487.3± 32.3
σ = 130.3± 49.1
WF
σ/μ = 34.09± 21.23 %
μ = 453.7± 43.3
σ = 154.7± 81.5



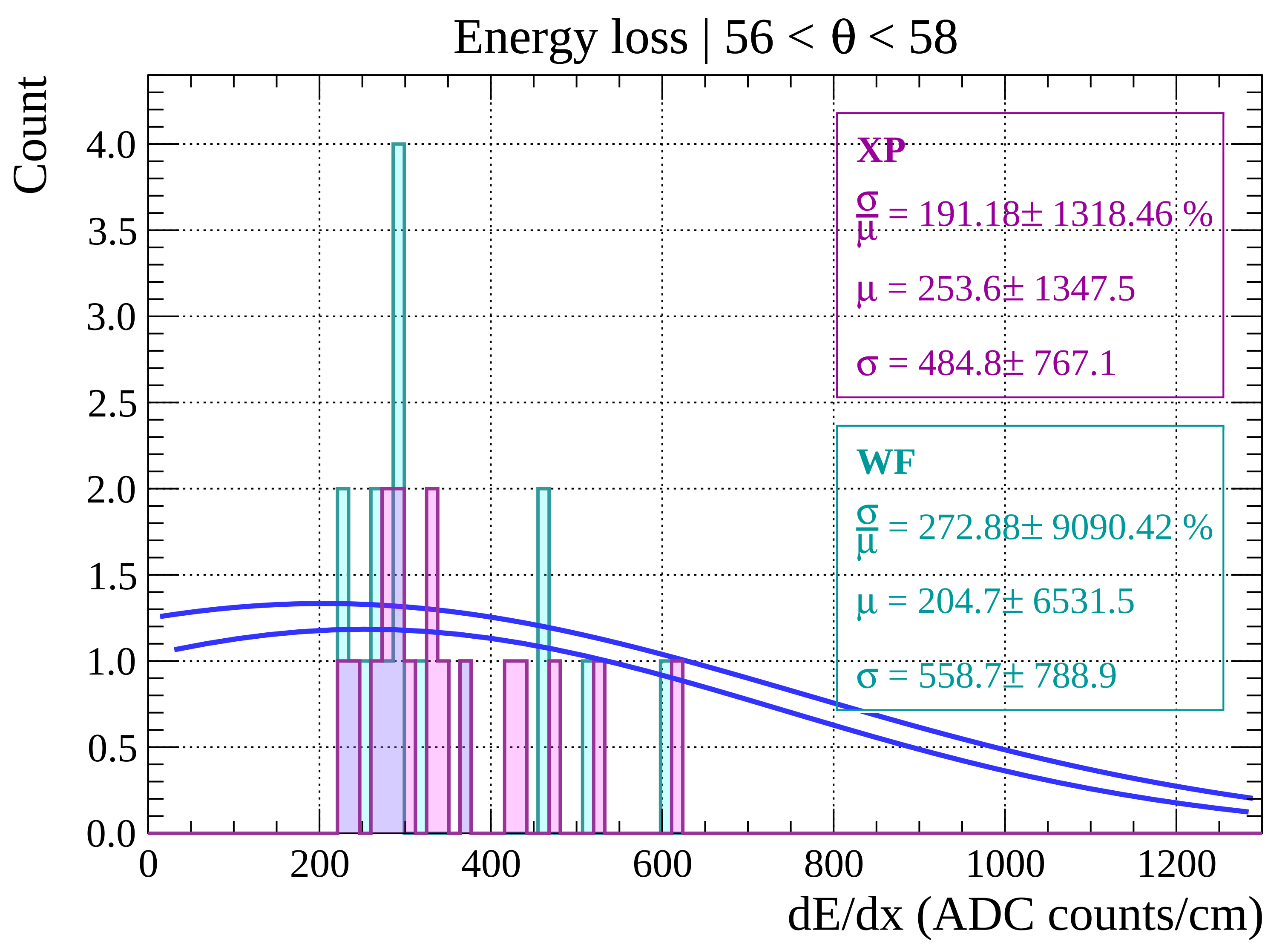
Energy loss | 50 < θ < 52
Count
5
4
3
2
1
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 45.60± 31.51 %
μ = 449.1± 70.8
σ = 204.8± 109.2
WF
σ/μ = -1842.18± 103485.67 %
μ = -20.7± 1191.8
σ = 382.0± 495.7

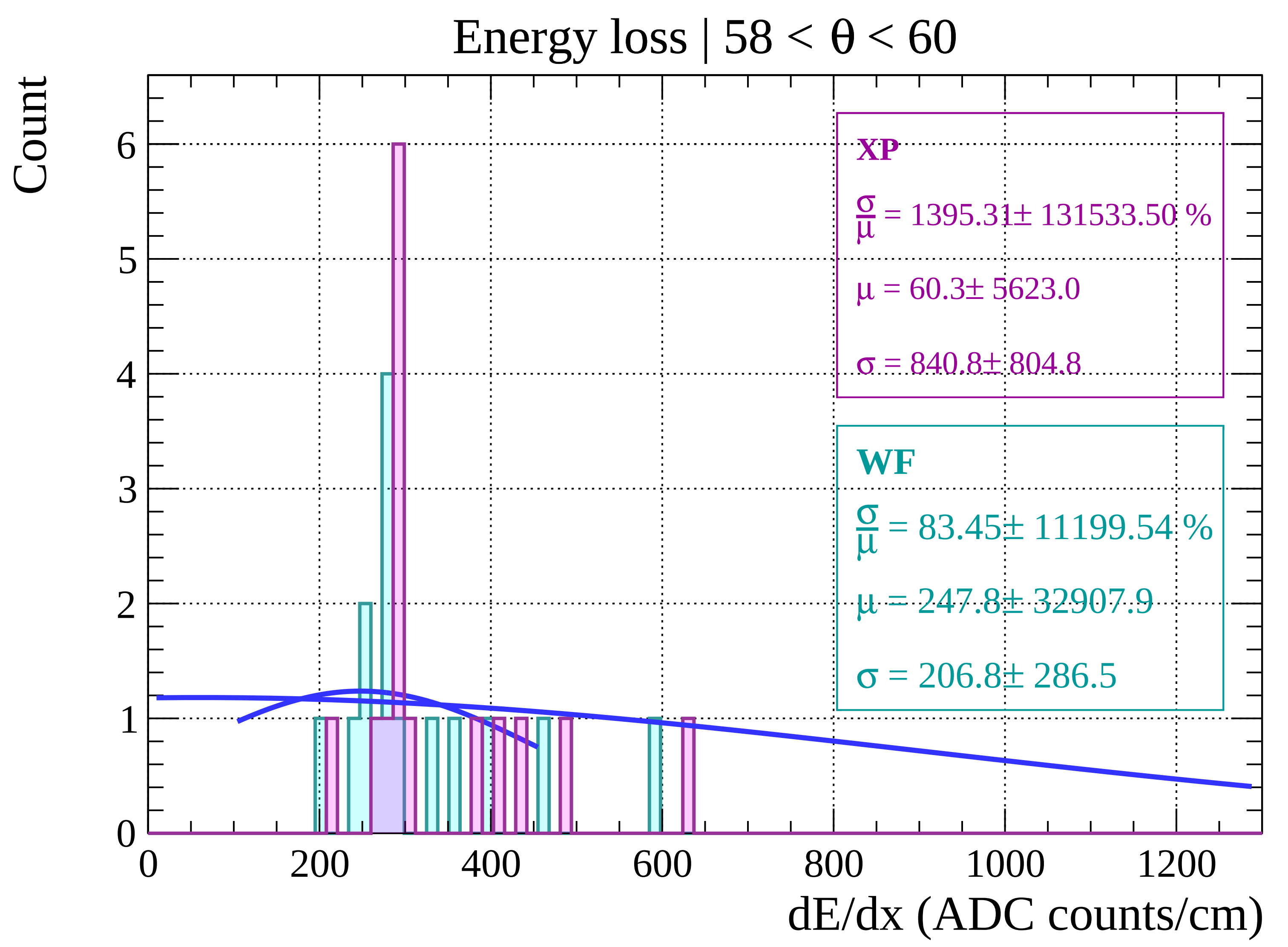


Energy loss | 52 < θ < 54
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 73.20± 237.71 %
μ = 282.2± 404.6
σ = 206.6± 374.5
WF
σ/μ = 16.56± 7.75 %
μ = 304.4± 18.7
σ = 50.4± 20.5

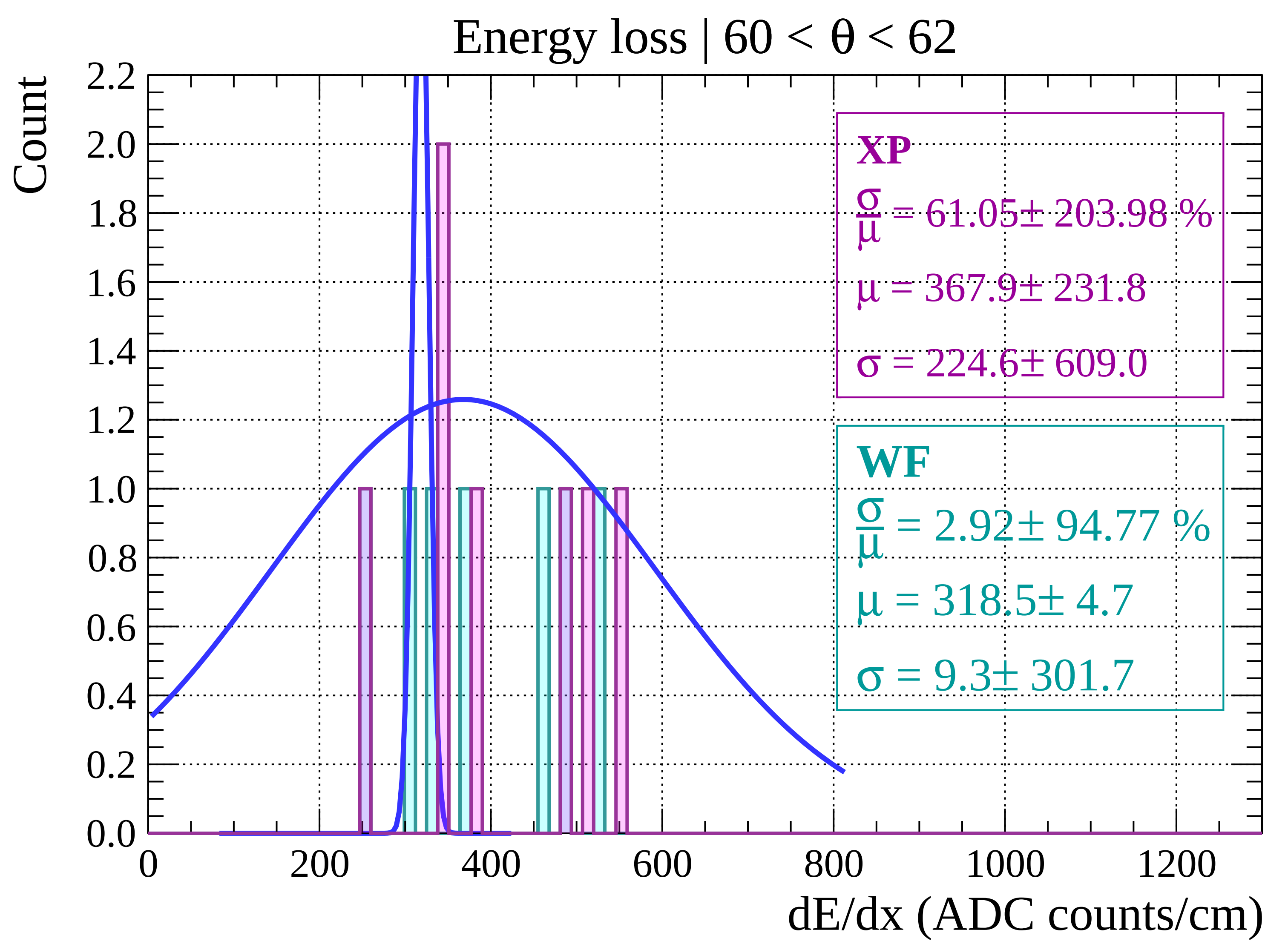


Energy loss | 54 < θ < 56
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 1.88± 10.50 %
μ = 310.2± 71.2
σ = 5.8± 31.2
WF
σ/μ = -132.84± 483.34 %
μ = -1203.9± 5008.2
σ = 1599.2± 834.2

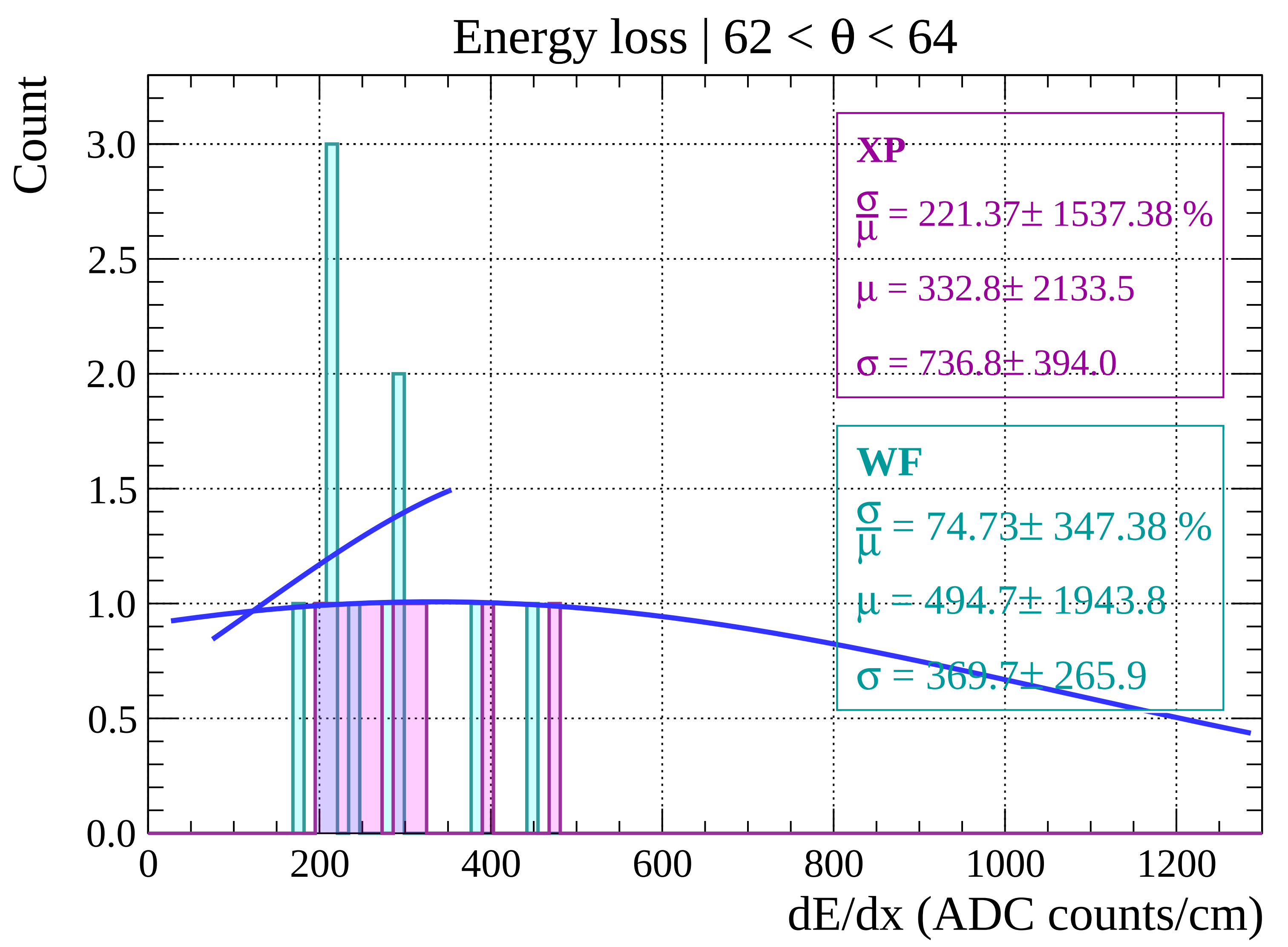

Energy loss | 56 < θ < 58
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 191.18± 1318.46 %
μ = 253.6± 1347.5
σ = 484.8± 767.1
WF
σ/μ = 272.88± 9090.42 %
μ = 204.7± 6531.5
σ = 558.7± 788.9
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200



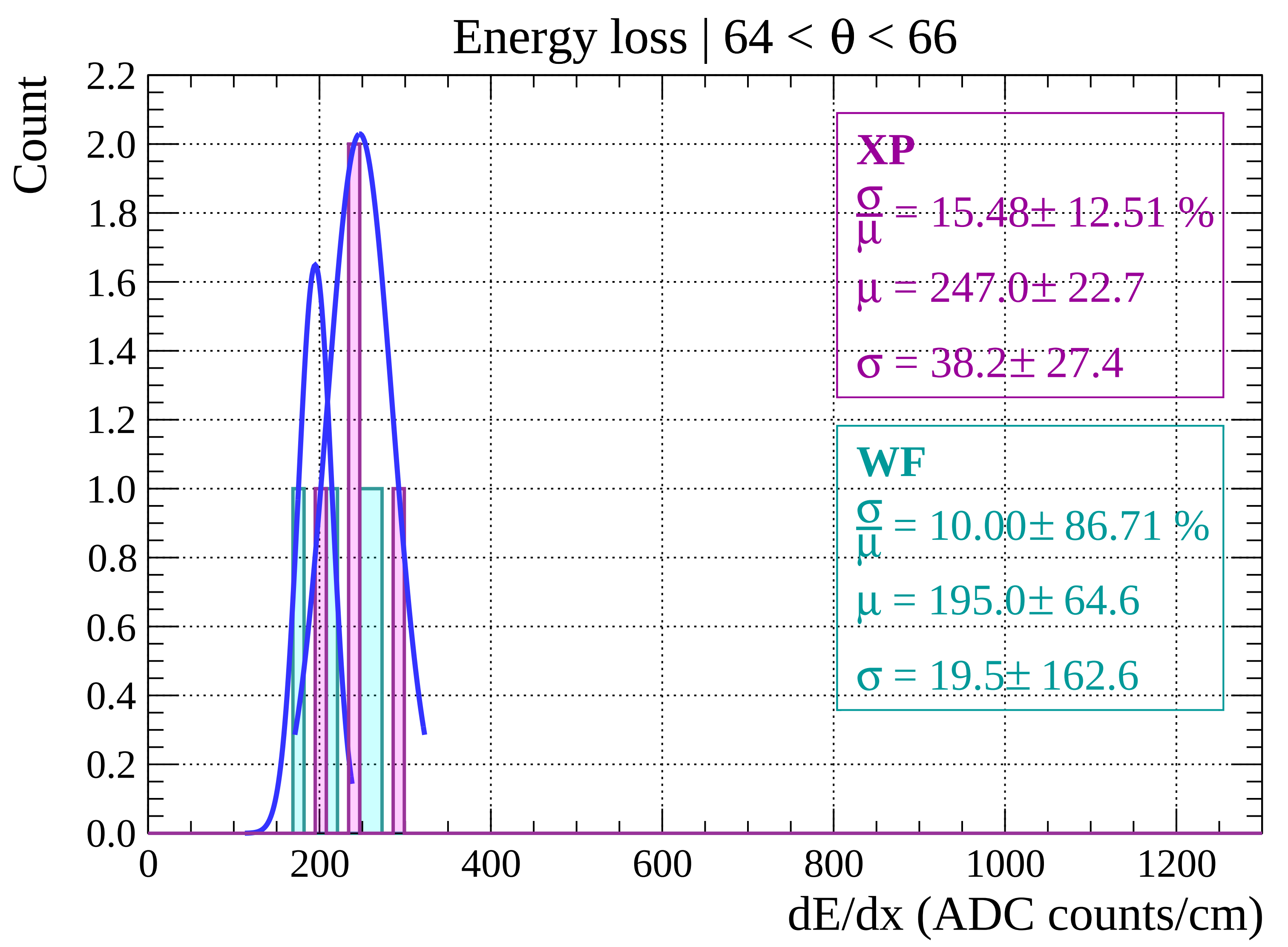
Energy loss | 58 < θ < 60
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200
XP
σ/μ = 1395.31± 131533.50 %
μ = 60.3± 5623.0
σ = 840.8± 804.8
WF
σ/μ = 83.45± 11199.54 %
μ = 247.8± 32907.9
σ = 206.8± 286.5



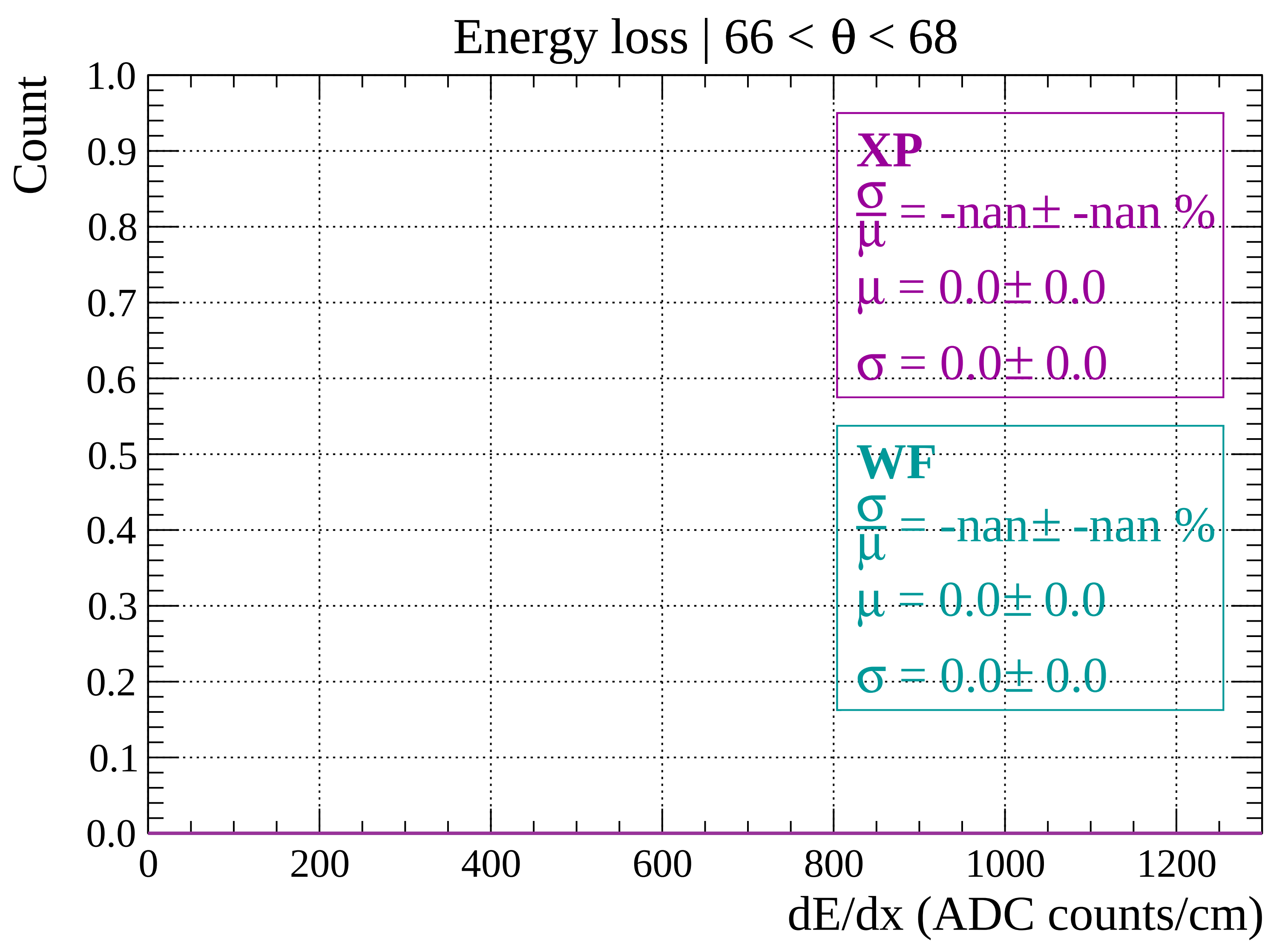
Energy loss | 60 < θ < 62
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
σ/μ = 61.05± 203.98 %
μ = 367.9± 231.8
σ = 224.6± 609.0
WF
σ/μ = 2.92± 94.77 %
μ = 318.5± 4.7
σ = 9.3± 301.7

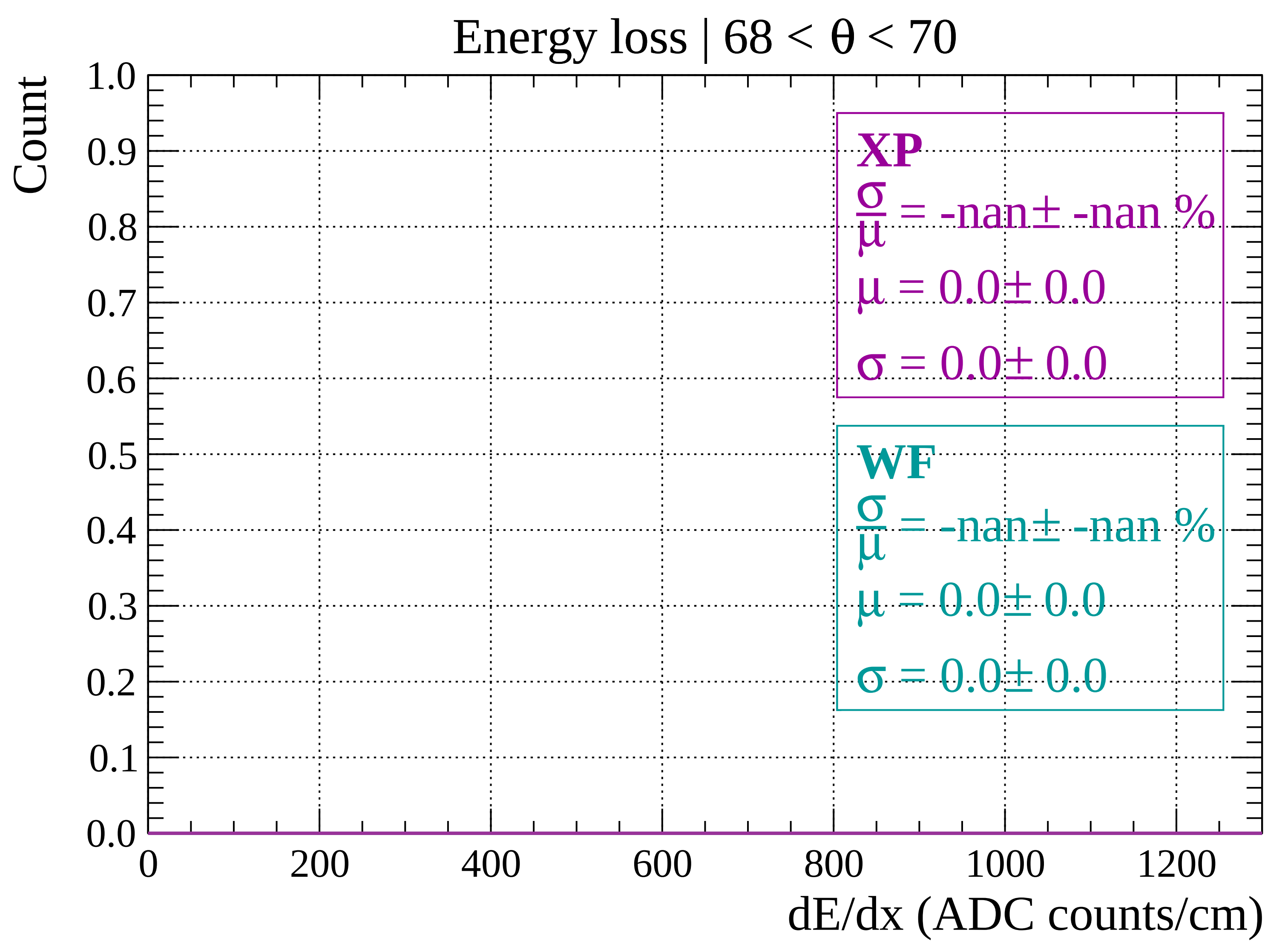


Energy loss | 62 < θ < 64
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 221.37± 1537.38 %
μ = 332.8± 2133.5
σ = 736.8± 394.0
WF
σ/μ = 74.73± 347.38 %
μ = 494.7± 1943.8
σ = 369.7± 265.9

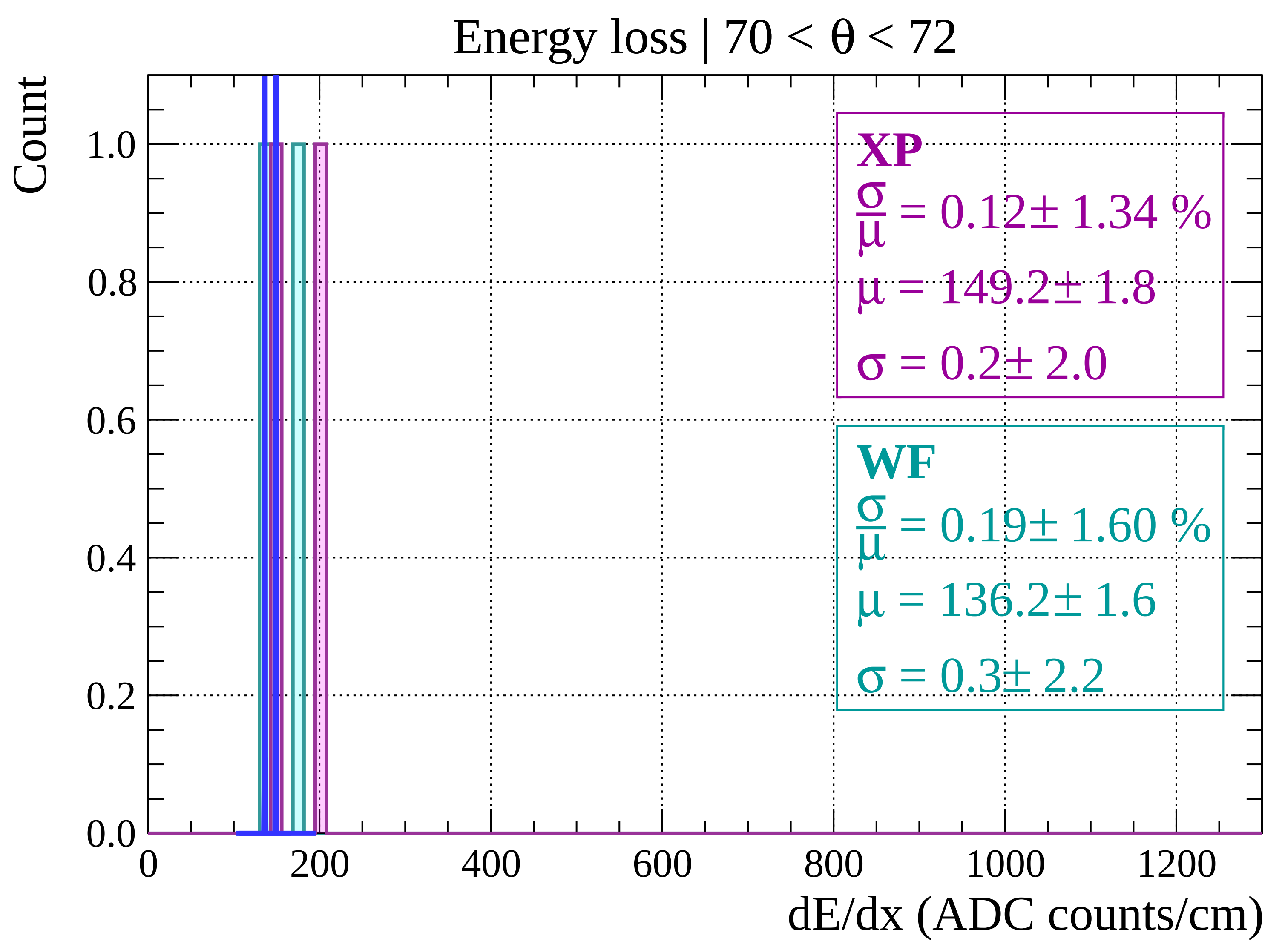


Energy loss | 64 < θ < 66
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.48± 12.51 %
μ = 247.0± 22.7
σ = 38.2± 27.4
WF
σ/μ = 10.00± 86.71 %
μ = 195.0± 64.6
σ = 19.5± 162.6
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200

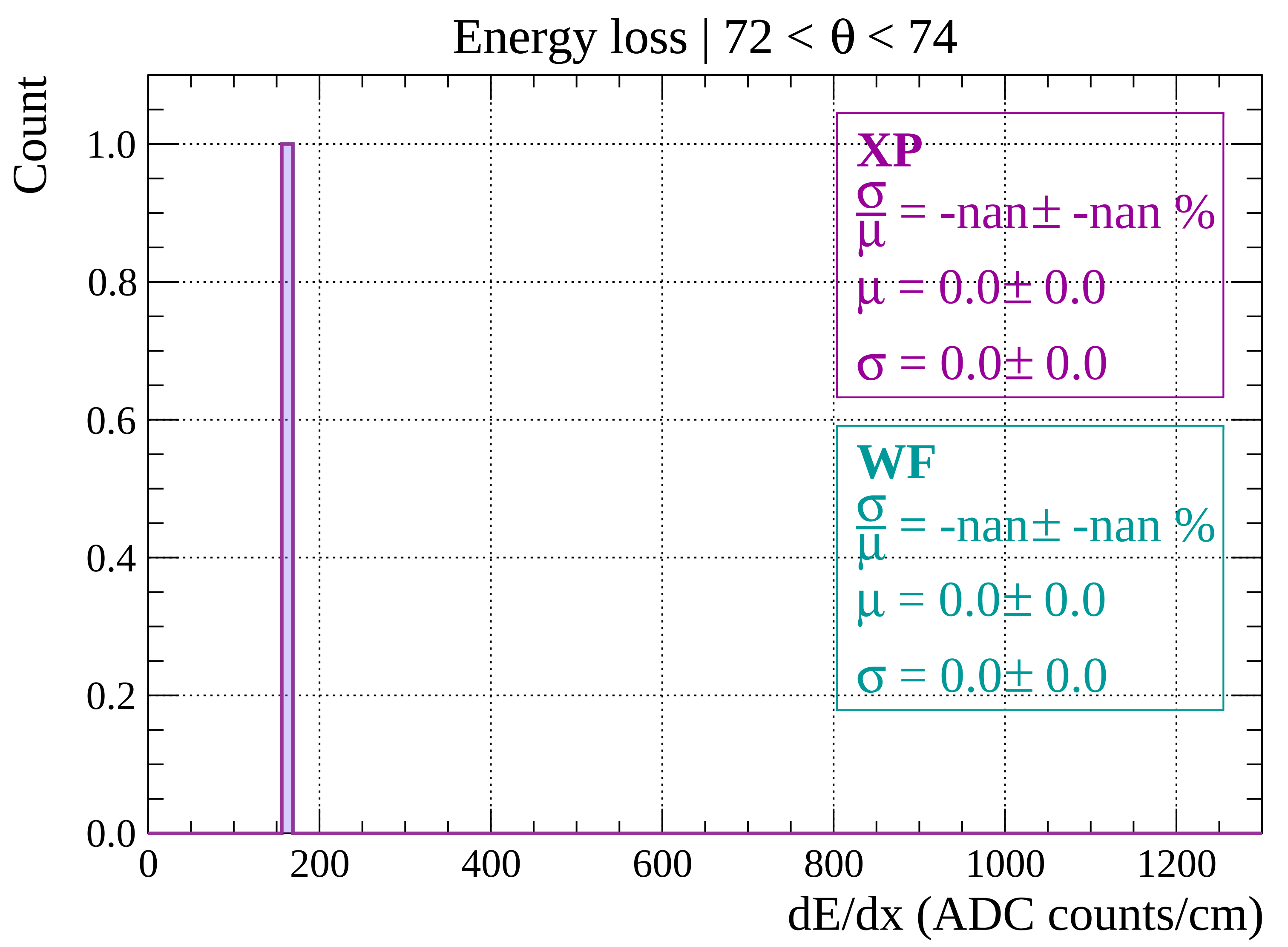

Energy loss | 66 < θ < 68
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



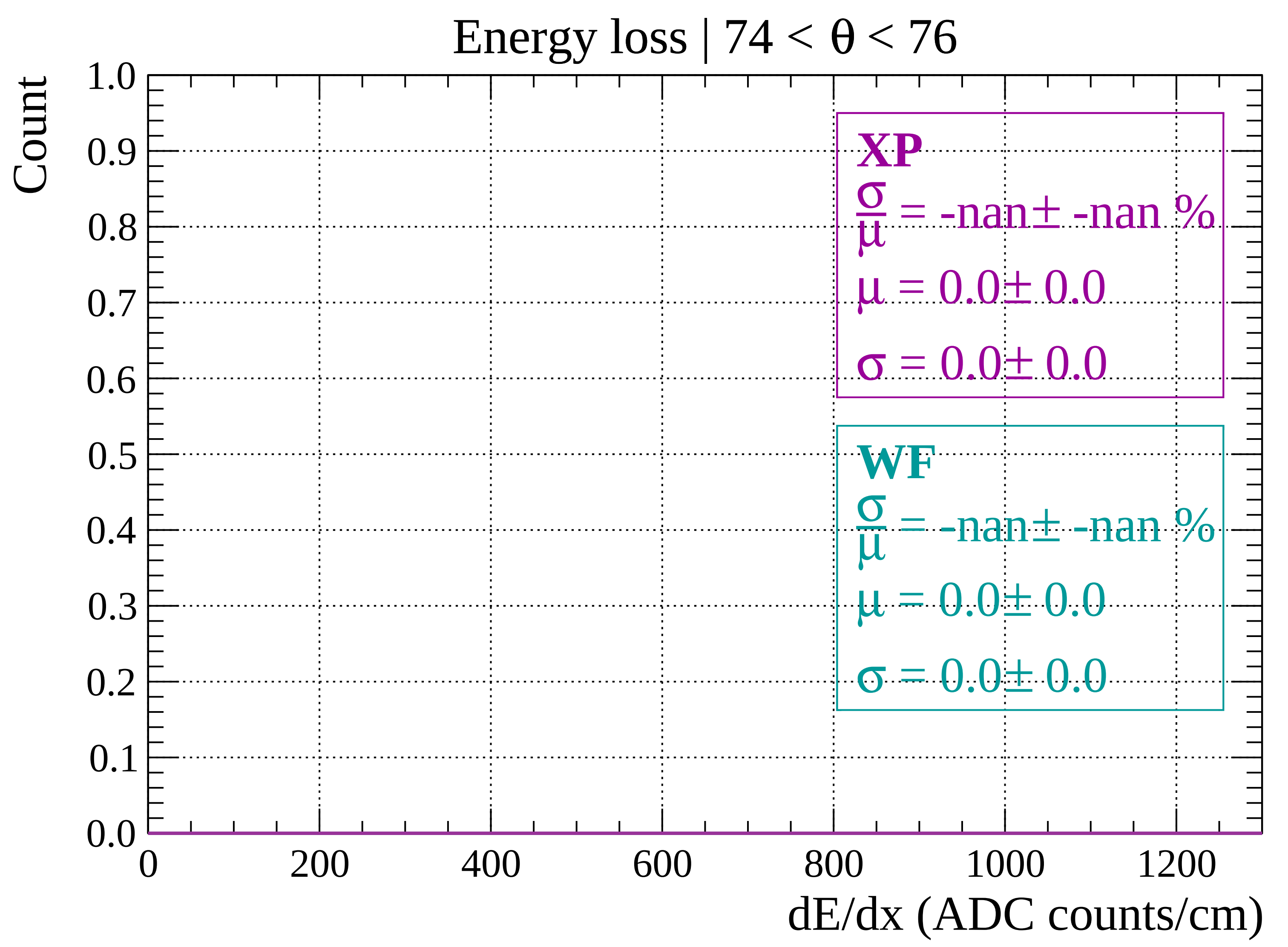
Energy loss | 68 < θ < 70
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



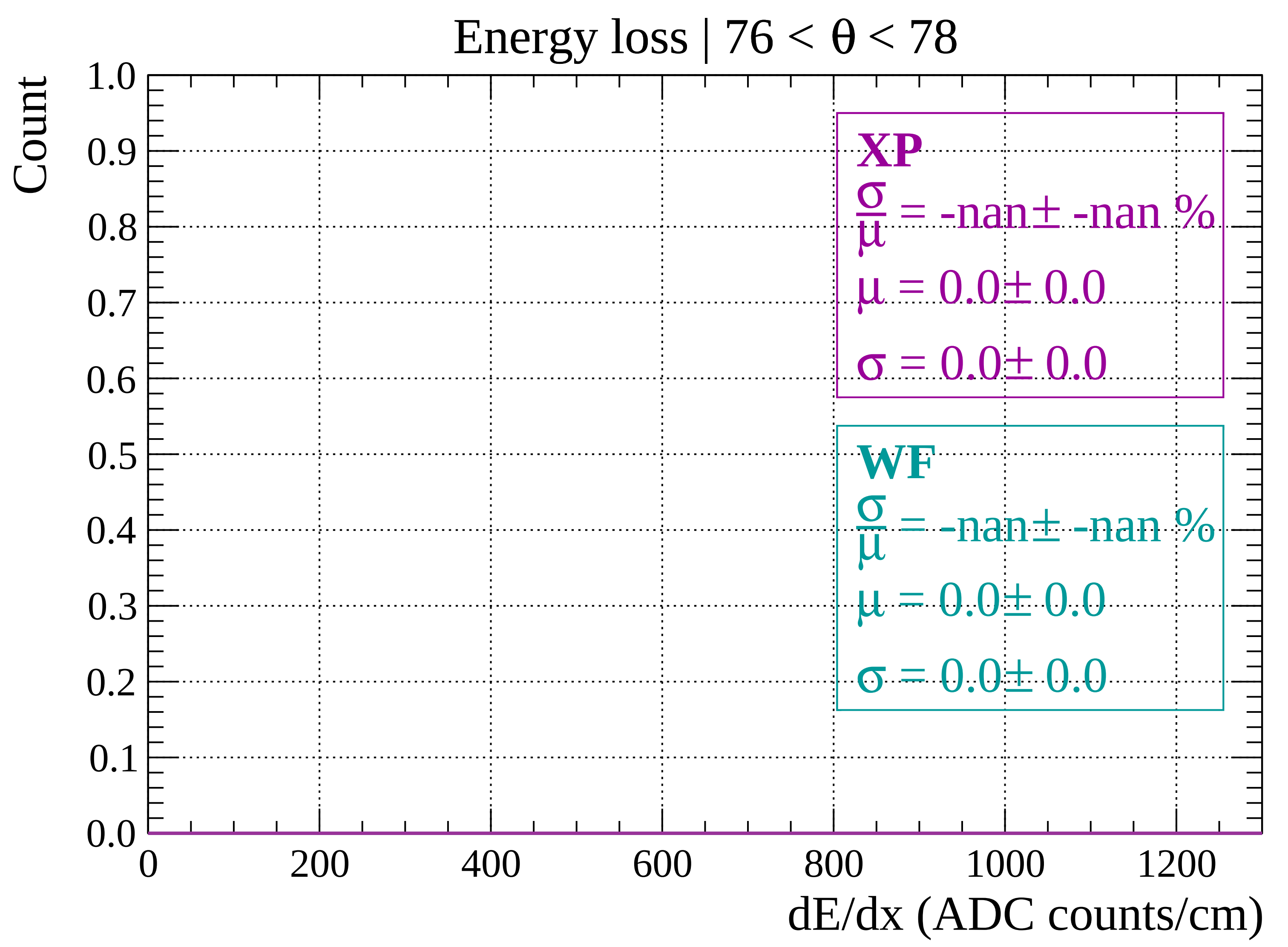
Energy loss | 70 < θ < 72
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 0.12± 1.34 %
μ = 149.2± 1.8
σ = 0.2± 2.0
WF
σ/μ = 0.19± 1.60 %
μ = 136.2± 1.6
σ = 0.3± 2.2

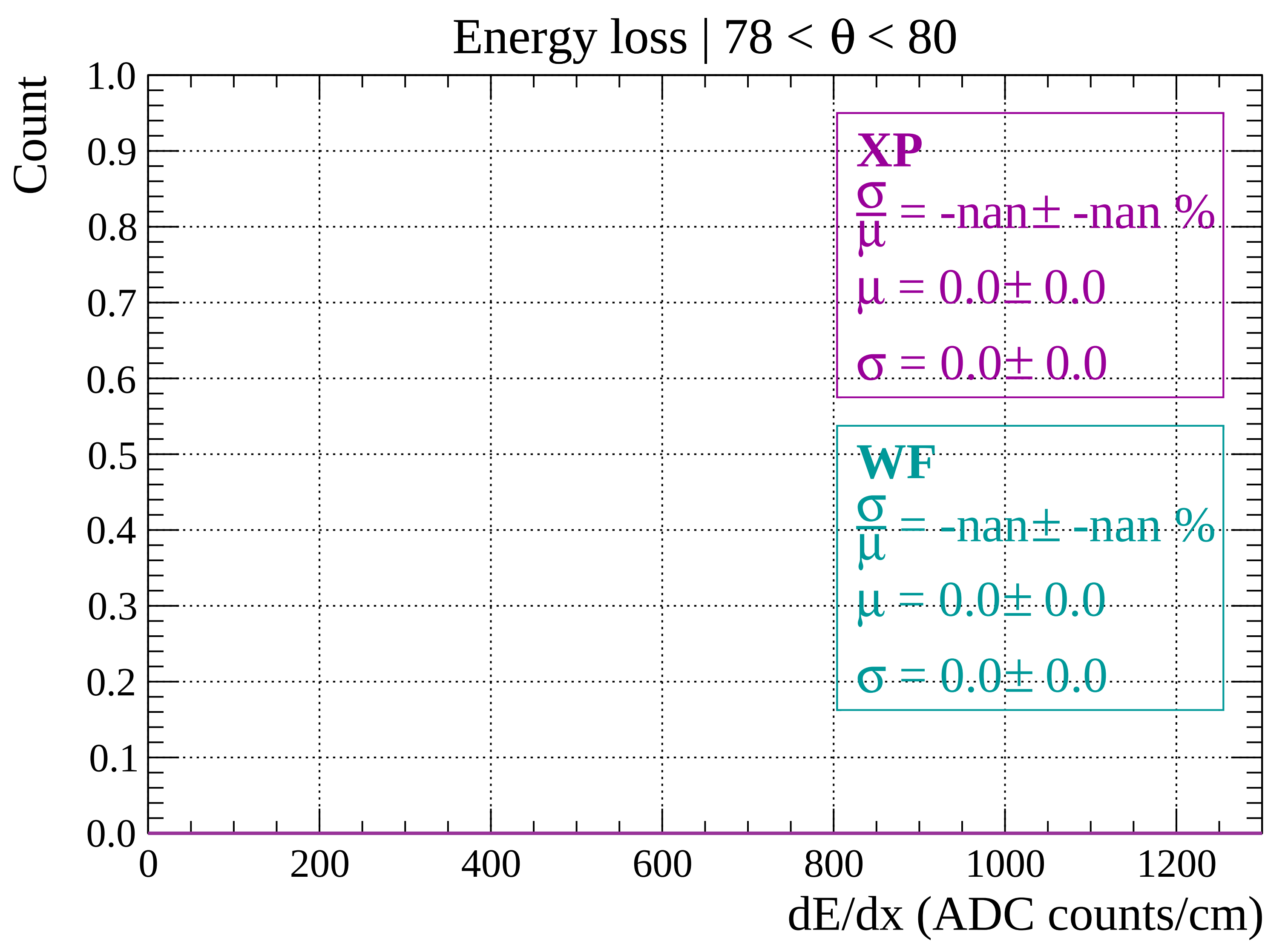


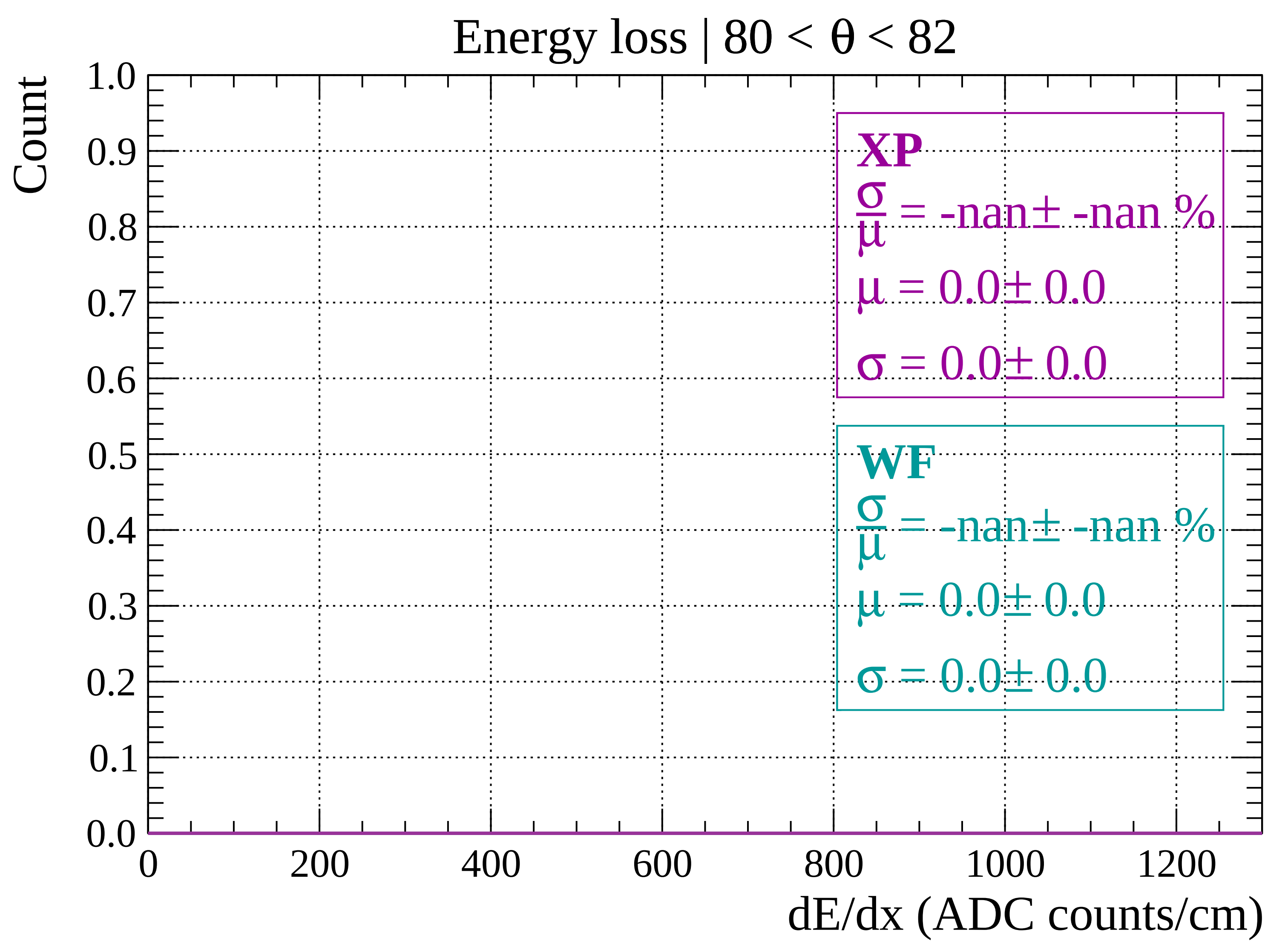
Energy loss | 72 < θ < 74
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



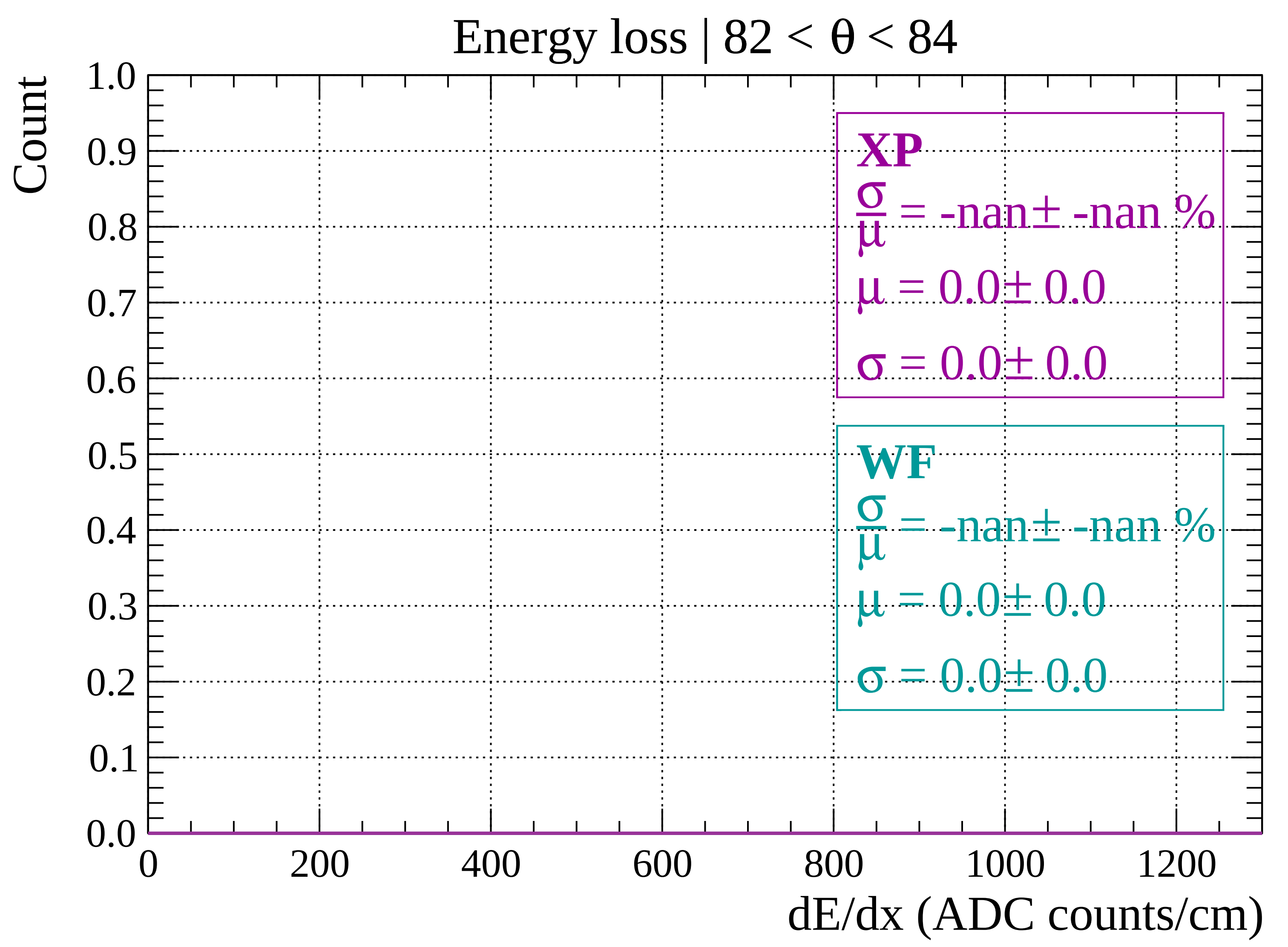
Energy loss | 74 < θ < 76
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

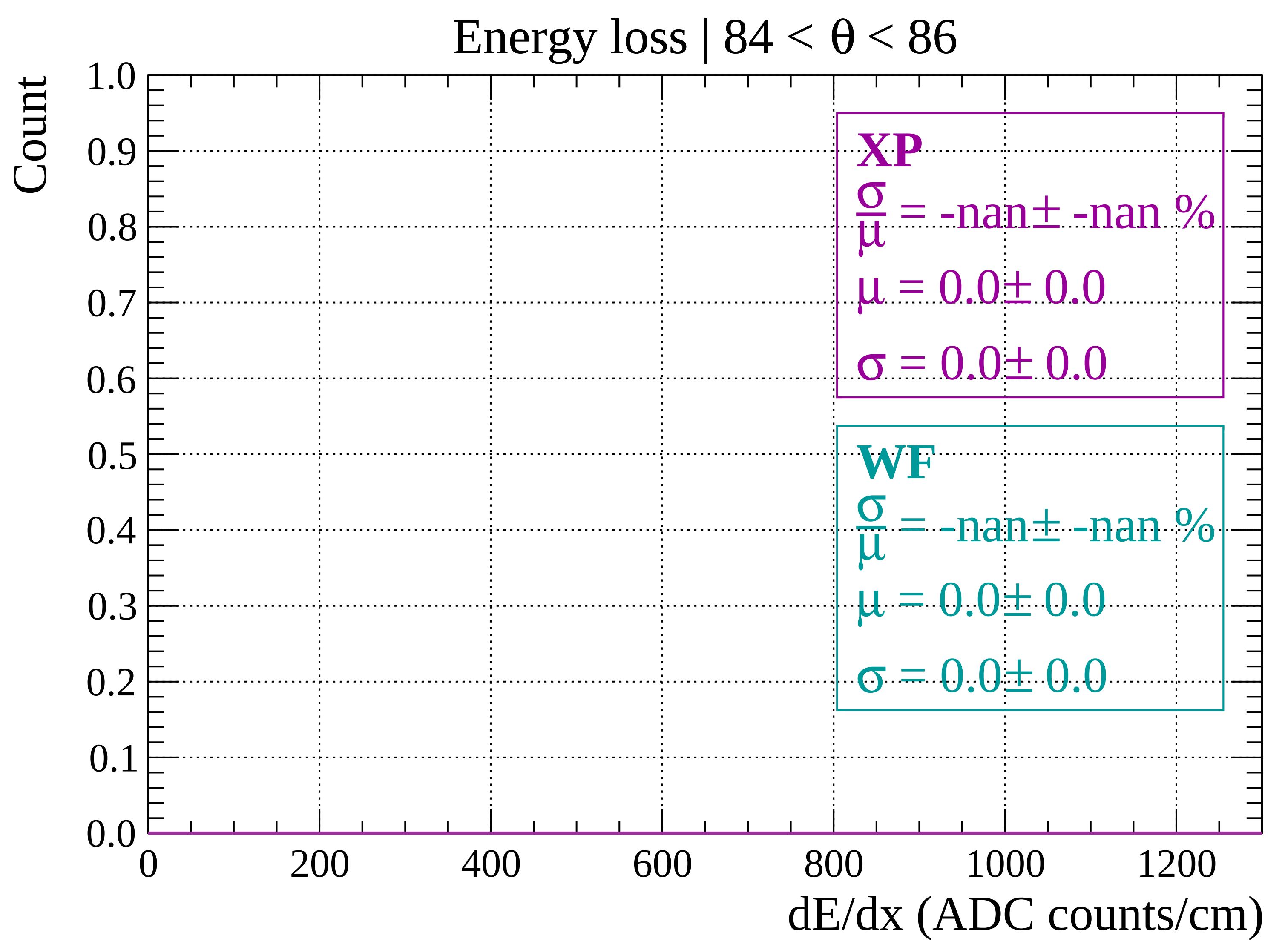

Energy loss | 76 < θ < 78
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 78 < θ < 80
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

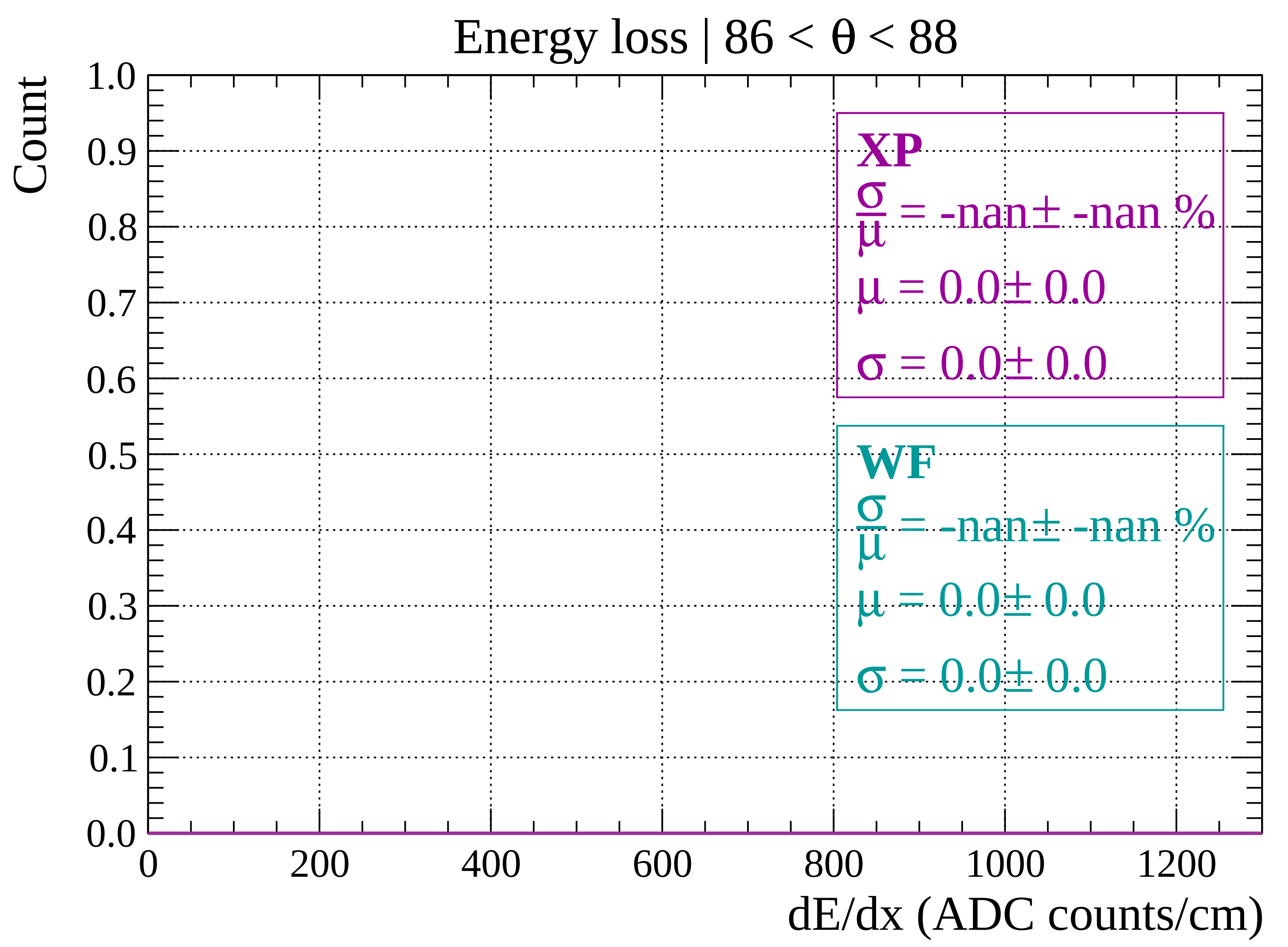


Energy loss | 80 < θ < 82
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

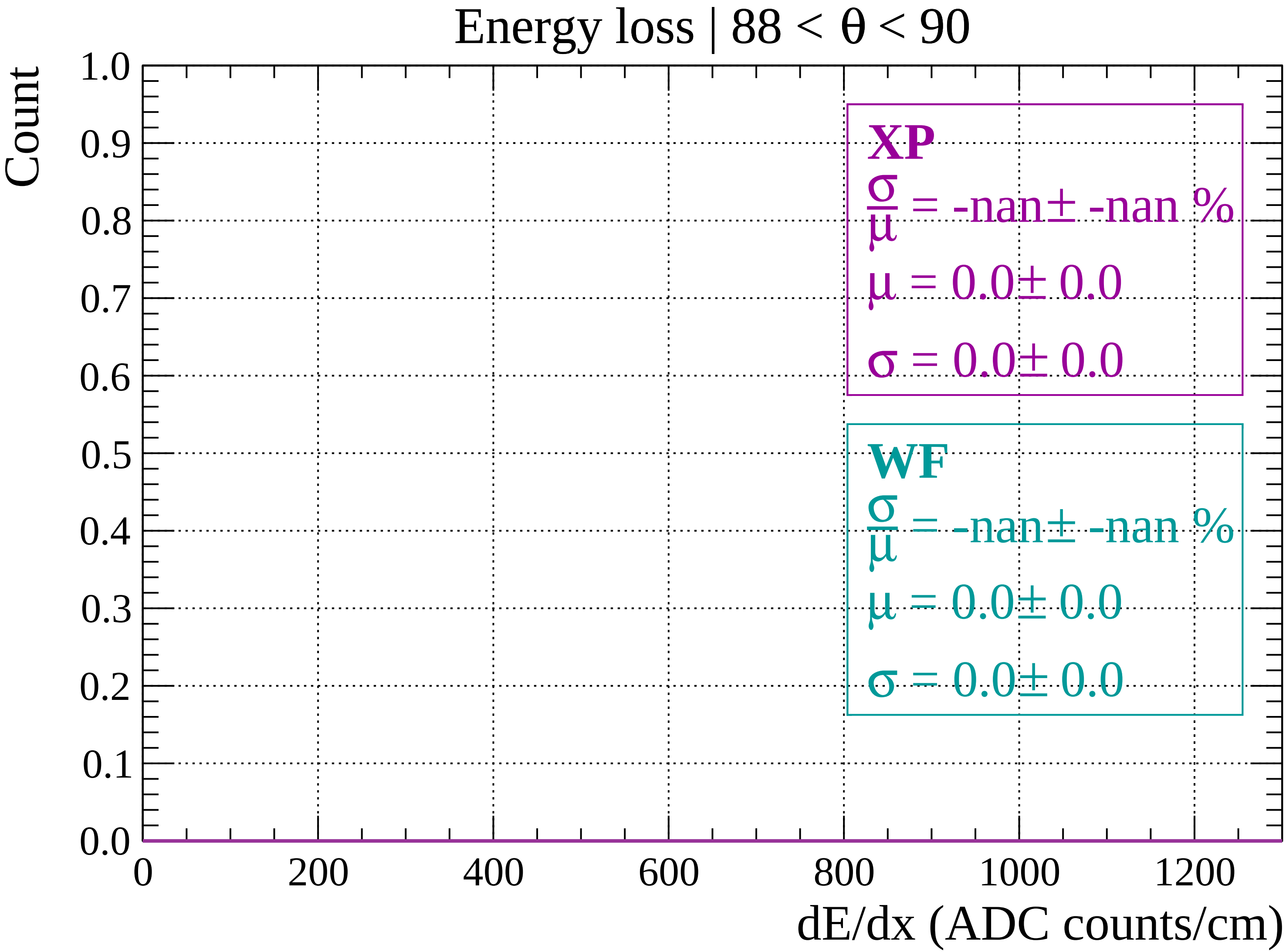

Energy loss | 82 < θ < 84
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



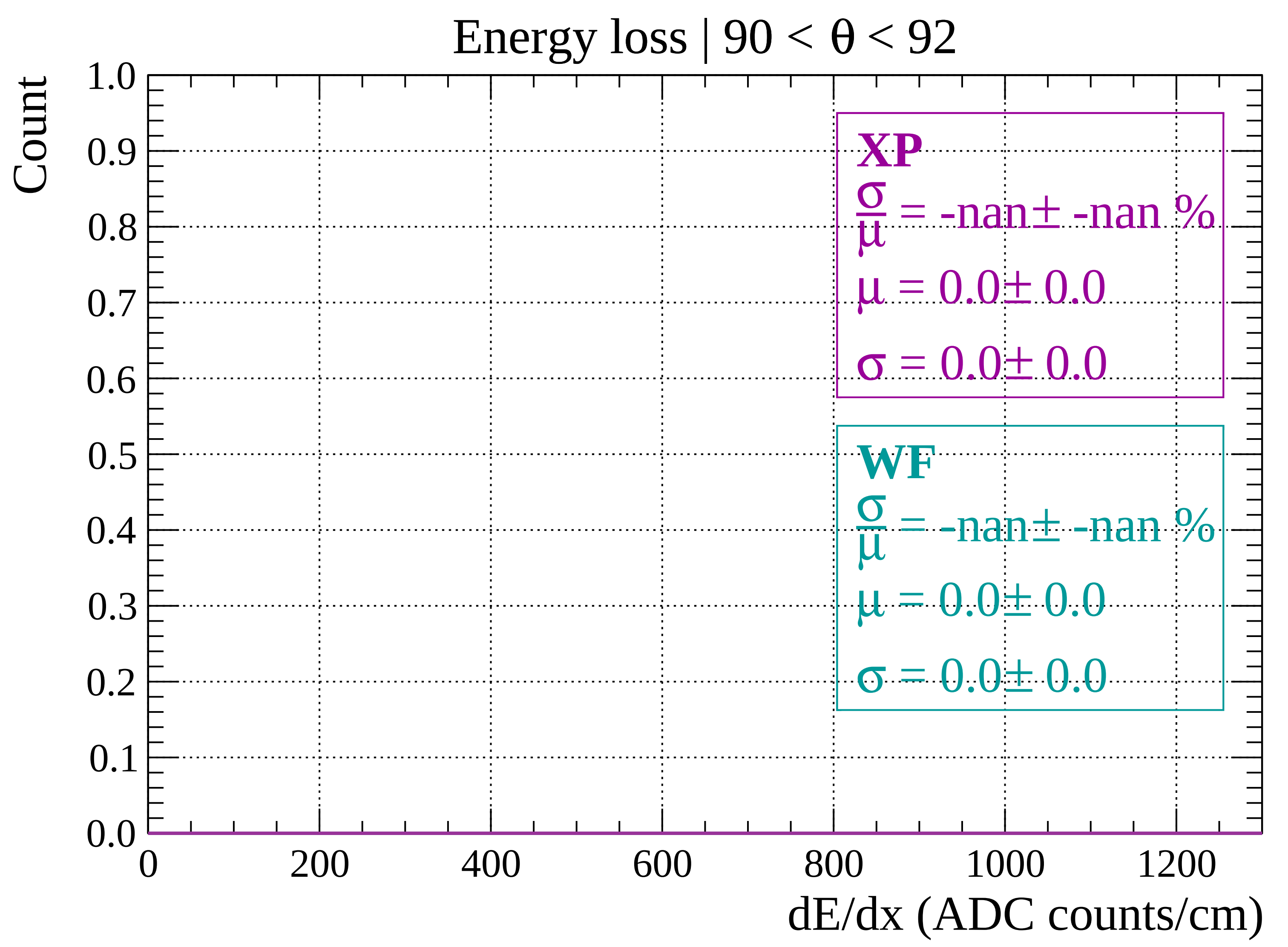
Energy loss | 84 < θ < 86
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



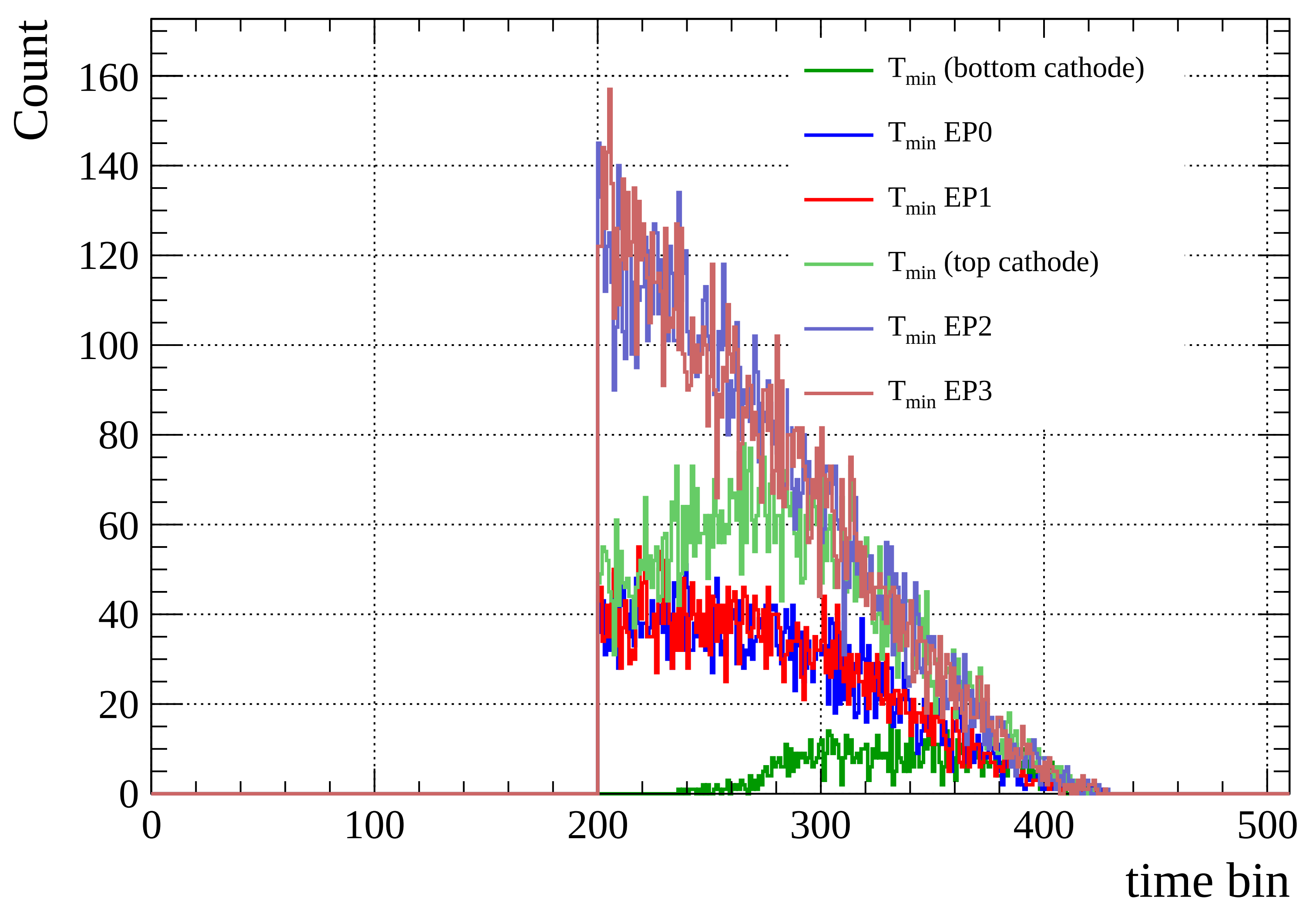
Energy loss | 86 < θ < 88
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



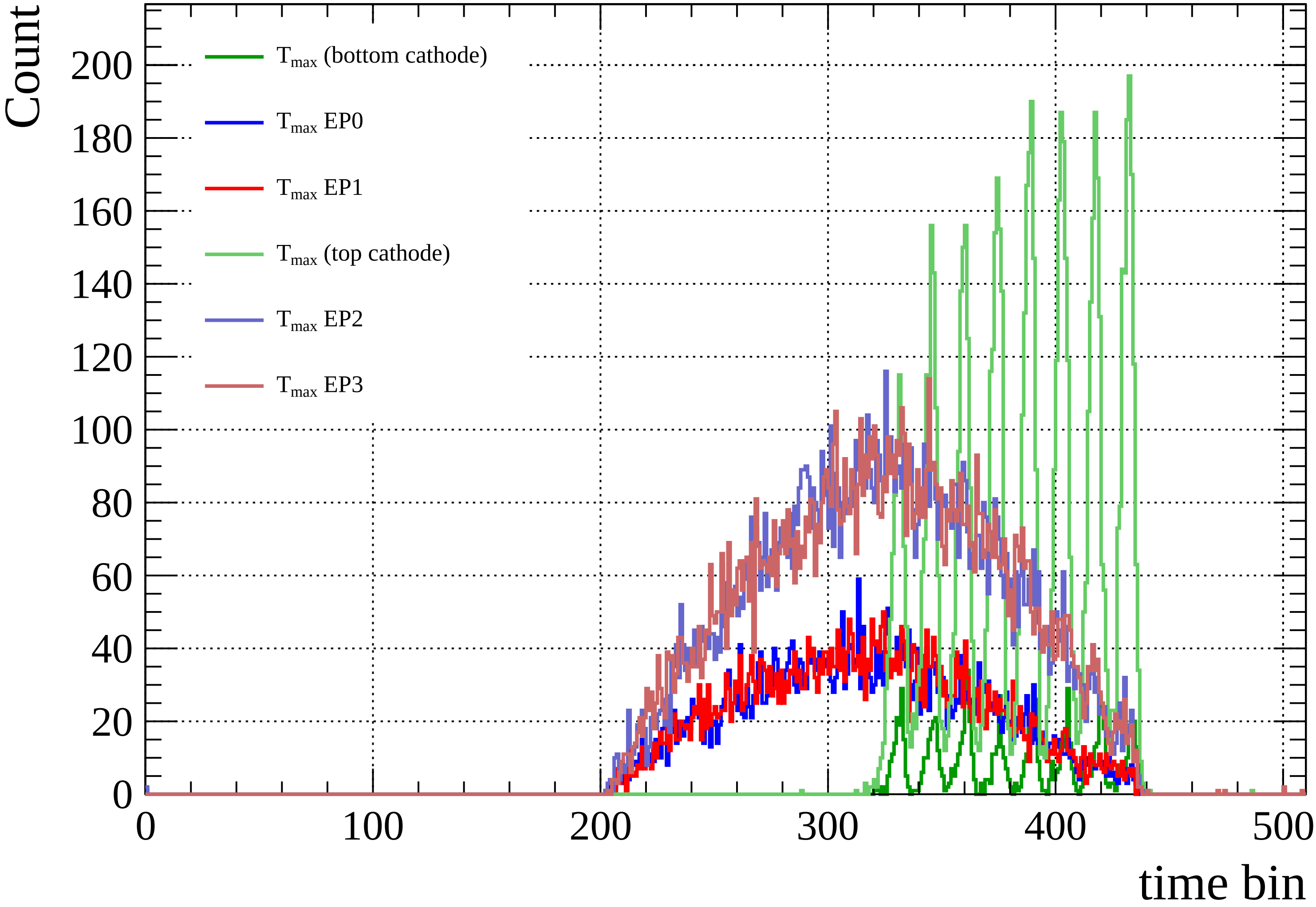
Energy loss | 88 < θ < 90
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Energy loss | 90 < θ < 92
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
time bin
0
20
40
60
80
100
120
140
160
0
100
200
300
400
500
T_min (bottom cathode)
T_min EP0
T_min EP1
T_min (top cathode)
T_min EP2
T_min EP3



Count
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3
0
20
40
60
80
100
120
140
160
180
200
0
100
200
300
400
500
time bin